# UNIVERSAL PHYSICS

# Universal Physics

*A Map of Why a Final Theory Is Open — and What Would Close It*

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universal-physics.khe.money

First edition, June 2026.

universal-physics.khe.money · github.com/HKTITAN/universal-physics

Research and content: Harshit Khemani + Claude.

Design inspired by Dan Hollick's *Making Software* (makingsoftware.com).

Typeset with Paged.js and KaTeX from the project's markdown sources. Epistemic tags [ESTABLISHED] [INFERENCE] [SPECULATIVE] [OPEN] [CONTESTED] are reproduced verbatim from the source wiki.

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# Preface

This book is a map of a question, not an answer to it. The question is the oldest and most abused in physics: is there a single universal theory underneath everything, and if so, what would it take to find it and to know we had found it? Shelves sag under books that answer yes and then sell you a candidate. This one does something different. It maps, as carefully as we could manage, the best current understanding of fundamental physics; it locates exactly where that understanding breaks; it sorts the field's assumptions into bedrock and historical accident; it surveys the candidate roads to unification without favoritism; and it delivers a verdict about what the evidence actually supports — which is neither a theory of everything nor a proof that none can exist.

The verdict, stated up front because you should not have to wait for it: a universal theory is **not forbidden and not delivered**. The two pillars of modern physics, quantum field theory and general relativity, are mutually consistent everywhere we can currently test them. None of the seventeen inter-framework clashes catalogued in Part III is a proven logical contradiction; they are domain mismatches and unsolved-but-consistent problems, concentrated where both theories are extrapolated past their tested regimes. A consistent unified framework is therefore very plausibly possible. But the most developed candidate direction — the wager that causal, algebraic, and entanglement structure is more fundamental than metric geometry — turns out, under sustained adversarial pressure, to *encode* geometry without *generating* it, and to have no distinguishing experimental test even in principle. By this project's own standard, that makes it a sharp and coherent research strategy, not yet physics. Part VI states this verdict in full, with its three explicit hedges. It is a stable conclusion, not a final one: the same chapter specifies precisely what would reopen it.

A word on how the book was made, because the method is unusual and the reader deserves to be able to discount for it. The text began as a research wiki built and refined over six iterations by a multi-agent process: teams of AI research agents, directed by a human author, each iteration deploying dozens of agents on focused questions — and, critically, every new claim passed through an adversarial referee step whose job was to break it. Iteration 2 stress-tested the central wager against the recent literature. Iteration 3 stopped surveying and started *attempting* — trying to derive the results the wager would need, and red-teaming the project's own posture. Iterations 4 and 5 executed the remaining designated verdict-flipping moves (they did not flip) and declared analytical saturation. Iteration 6 injected two more years of new literature, with four tracks explicitly tasked to attack the standing verdict; it survived, tightened. Citations were web-verified against the actual papers, not recalled from memory; the bibliography contains real references only. The project's anti-crank protocol was simple and absolute: never let a speculative idea wear the clothes of an established fact, prefer a negative result to a flattering one, and treat overclaiming as a defect rather than a flourish. Where the investigation corrected itself — and it did, more than once — the correction is recorded, not erased.

The visible instrument of that protocol is the system of epistemic tags you will meet on nearly every page: `[ESTABLISHED]`, `[INFERENCE]`, `[SPECULATIVE]`, `[OPEN]`, and `[CONTESTED]`. They are defined precisely in Chapter 2, but the spirit is this: `[ESTABLISHED]` means experimentally confirmed or rigorously proven, with broad expert consensus; `[INFERENCE]` means a strong, standard bet that is not yet a fact; `[SPECULATIVE]` means principled but undecided; `[OPEN]` means the honest answer is *we do not know*; `[CONTESTED]` means competent experts actively disagree. The tags are not decoration. They are the book's load-bearing honesty, and reading them is reading the book. A claim's tag tells you how hard to lean on it, and the most important sentences here are often the ones tagged weakest.

Be clear, too, about what the verdict is not. It is not a proof that unification is impossible — we explicitly decline the fashionable Gödelian overreach. It is not a dismissal of the emergent-spacetime program, which the investigation found to be the most rigorous and self-aware road on the map; the diagnosis "encodes, does not generate" was reached by taking that program more seriously than its popularizations do. And it is not a theorem: it rests on named hedges, graded at medium confidence, which Part VI sets out without embarrassment. What it is, we hope, is the most precise available account of *why* the question remains open — and a usable specification of the analytical construction or experimental result that would close it.

On authorship. This book was directed, scoped, and edited by a human, Harshit Khemani, and researched and written in collaboration with Claude, an AI system built by Anthropic; the research labor — the literature sweeps, the derivation attempts, the refereeing — was substantially the machine's, under human direction and judgment. We list both names because both did the work, and because pretending otherwise would violate the book's own first rule. Whatever errors survive the process are failures of that process, and we ask readers to report them: the living version of this text, with its full working notes and change log, remains public at universal-physics.khe.money and github.com/HKTITAN/universal-physics, licensed for reuse.

The question is open. Here is the map.

*Harshit Khemani & Claude — June 2026*

PART I

# THE QUESTION AND THE METHOD

Every attempt at a final theory begins with a temptation: to start building before deciding what would count as success. This part exists to remove that temptation before the physics begins.

Chapter 1 states the project. Not an argument for a pet theory of everything, but a structured, self-critical investigation with two acceptable outcomes: a candidate framework with real backing, or a precise account of why the evidence does not yet support one. It also gives the honest one-paragraph status of the field — two spectacularly confirmed pillars, mutually consistent everywhere we can test, both known to be incomplete — which is the ground truth everything later must respect.

Chapter 2 is the method, and it is the most important chapter in the book. It defines the five epistemic tags — ESTABLISHED, INFERENCE, SPECULATIVE, OPEN, CONTESTED — that mark nearly every claim in the chapters that follow, along with the cardinal rule: never let a speculative idea wear the clothes of an established fact. Overclaiming is the characteristic failure mode of unification work, and this book treats it as a defect, not a flourish.

What this part establishes is modest and essential: a question worth asking, a standard of evidence strict enough to be trusted, and a vocabulary for uncertainty. Read Chapter 2 carefully. The rest of the book is written in its language.

CHAPTER 1

# The Project: A Universal Theory, Taken Seriously

> An LLM-maintained, adversarially-refereed research wiki driving toward — and rigorously bounding — a universal theory of physics.

site universal-physics.khe.money

book PDF

verdict PARTIAL · not-yet-physics

iteration 15 · saturated

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## What this is

This repository is a serious, cumulative, self-critical knowledge base — an **LLM-maintained research wiki** built by **Harshit Khemani** (research direction) and **Claude** (Anthropic; research labor, via multi-agent workflows) — whose single goal is to map the best current understanding of fundamental physics, find exactly where it breaks, separate genuinely fundamental assumptions from historical artifacts, search for deeper unifying principles, and evaluate **honestly** whether a universal theory of physics is possible. It is **not** an essay arguing for a pet "theory of everything"; every claim is epistemically tagged, every hypothesis is run through an adversarial anti-crank referee, and the explicit, equally-valuable success condition is *either* a candidate framework with real backing *or* a precise, reasoned account of why the evidence does not yet support one. Knowledge is **compiled once** into interlinked markdown that compounds, rather than re-derived on every query.

## The standing verdict

After fifteen iterations, the verdict has been stable for **14 consecutive iterations** and the analytical program has reached **terminal analytical saturation** — continued reasoning now mostly re-derives the same map; only new external input can move the result.

**A single universal theory of physics is *not forbidden and not delivered.***

- **PARTIAL coherence.** There is not one unified program but two internally-coherent ones — *geometry-from-algebra* and *is-the-quantum-framework-final* — joined only by thin bridges. The tempting "it's all information" fusion is refused.
- **Encodes, does not generate.** The most developed direction — *causal/algebraic/entanglement structure precedes metric geometry* — **encodes** geometry without yet **generating** it. Every load-bearing result is conceded by its own architects to presuppose a fixed background with a Killing/conformal symmetry.
- **Not yet physics.** The central wager has **no distinguishing experimental test even in principle**. By this wiki's own anti-crank standard it is a sharp, coherent research *strategy*, not a result.
- **Not forecastable.** The object-level question — *is there a final theory, and is it this one?* — is gated less by the ~15-order-of-magnitude energy desert than by the absence of any experiment that bears on the central wager.

**The single open gate** is the **carrier problem**: derive an algebra-compatible, *localized* (fiberwise) indefinite `(n,n)` pairing from modular data alone — installing no symmetry, η-form, foliation, twist, indefinite-pairing axiom, or signature-valued template by hand. Five independent routes converge on the same closing step. A construction meeting the full spec would flip the headline from "encodes" to "generates" and reopen the capstone. As of iteration 15, no carrier closes, no reverse-Weinberg–Witten witness exists, and the external channels (notably Connes' bicentralizer problem, re-confirmed open in general) remain dry.

➡️ **Read the full reasoning in *Chapter 38 (p. 705)* — the capstone verdict.**

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## How it is built — the llm-wiki structure

This wiki follows the persistent **LLM-maintained wiki** pattern (Karpathy, *llm-wiki.md*): a three-layer system.

| Layer | What it is | Where it lives |
|---|---|---|
| **Raw sources** | The literature — textbooks, papers, data. Immutable; only cited. | the Bibliography (p. 776) |
| **The wiki** | Interlinked markdown that *compiles* understanding once and reuses it. | this repository |
| **The schema** | The rules of structure, labeling, and workflow. | `AGENTS.md` (in the repository) + *Chapter 2 (p. 8)* |

Two disciplines hold the whole thing together:

- **Epistemic tags.** Every nontrivial claim wears exactly one of `[ESTABLISHED]` · `[INFERENCE]` · `[SPECULATIVE]` · `[OPEN]` · `[CONTESTED]`. Speculation never wears established clothes. See *Chapter 2 (p. 8)*.
- **The anti-crank referee protocol** (`AGENTS.md` (in the repository) §4). Before any hypothesis is "kept", it is stated precisely (with its math), red-teamed against the relevant no-go theorems (Weinberg–Witten, Coleman–Mandula/HLS, Haag, Bell/PBR/Kochen–Specker, Reeh–Schlieder, the singularity theorems, unitarity/causality bounds) and against this wiki's own closures, then assigned a severity (none/minor/major/fatal — `fatal` ⇒ rejected, with the corpse and the cause of death kept). A candidate that *explains everything and predicts nothing* fails by construction.

Work proceeds through three operations — **Ingest** (add knowledge, enrich in place, tag, cross-link, log), **Query** (search the wiki before the web; answers inherit their weakest premise's tag), and **Lint** (hunt contradictions, stale claims, orphans, overclaiming, fabrication, ID integrity).

## Repository map

### Start here

| Page | What it is |
|---|---|
| *Chapter 38 (p. 705)* | **The capstone verdict.** Read first. |
| *Chapter 21 (p. 497)* | Running executive synthesis, iteration by iteration. |
| *Chapter 3 (p. 15)* | Every framework, its math, domain of validity, reductions. |
| *Chapter 14 (p. 191)* | Where the frameworks clash, with the technical root of each. |
| *Chapter 39 (p. 725)* | The empirical channels that gate the object-level question. |

### Core pages

| Page | Role |
|---|---|
| `AGENTS.md` (in the repository) | Behavioral schema — how to read, extend, and maintain the wiki |
| *Chapter 2 (p. 8)* | Evidence tags and labeling rules — **binding** |
| *Chapter 15 (p. 233)* | Open-problems registry (`OP-n`) |
| *Chapter 19 (p. 392)* | Candidate unifying directions, refereed (`Hn`) |
| *Chapter 18 (p. 349)* | Critical survey of unification / quantum-gravity programs |
| *Chapter 17 (p. 325)* | Candidate deep principles |
| *Chapter 20 (p. 458)* | Assumptions: fundamental ↔ artifact (`A-n`) |

| | |
|---|---|
| *Chapter 16 (p. 308)* | Constants, units, scales, free parameters, naturalness |
| the Glossary (p. 747) · the Bibliography (p. 776) | Definitions · real references only |
| *Chapter 40 (p. 739)* · the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)) | Research agenda · append-only iteration log |

### Directories

| PATH | CONTENTS |
|---|---|
| *the domain chapters (Part II, Chapters 4–13)* | Deep dossiers: quantum mechanics, QFT, general relativity, classical mechanics, thermodynamics, statistical mechanics, cosmology, particle physics, information theory, mathematics |
| `notes/` (notes index — repo: `notes/README.md`) | Forward-research iteration logs (the adversarial attack tracks, iteration by iteration) |
| `book/` | Source for the typeset PDF book |
| `site/` | Source for the website |

## Read it elsewhere

- **Website** — universal-physics.khe.money: the full wiki as a technical reference manual (built from `site/`).
- **Book (PDF)** — universal-physics-book.pdf: *Universal Physics*, the properly typeset edition (6×9 in; typeset through iteration 15; built from `book/`).
- **Vault** — open this repo directly in Obsidian: frontmatter, graph-view color groups, the `index.md` (in the repository) catalog, and `MAP.canvas` (in the repository) are configured.

## How to read this

- **Want the bottom line?** *Chapter 38 (p. 705)*, then *Chapter 21 (p. 497)*.

- **Want the map of physics?** *Chapter 3 (p. 15)* and the *the domain chapters (Part II, Chapters 4–13)* dossiers.
- **Want the open frontier?** *Chapter 15 (p. 233)*, *Chapter 19 (p. 392)*, *Chapter 39 (p. 725)*.
- **Want to see the work?** The iteration logs under *`notes/`* (notes index — repo: `notes/README.md`).

Throughout, trust the tags: `[ESTABLISHED]` is load-bearing fact; `[SPECULATIVE]` and `[OPEN]` are flagged as such on purpose.

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## How to contribute

Contributions are invited — from typo fixes to full attack tracks on the open problems. **The bar is proper backing, not credentials.** See **`CONTRIBUTING.md` (in the repository)** for the full anti-crank protocol; in short:

1. **Tag every nontrivial claim** with exactly one epistemic tag.
2. **Never fabricate a citation, number, theorem, or arXiv ID** — a missing reference is acceptable, a fake one is disqualifying.
3. **Red-team your own claim** against the relevant no-go theorems and this wiki's own closures.
4. **Enrich pages in place**, cross-link, and log every change in the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)).

The analytical program is saturated; the verdict moves only on **new input**. The single open gate is posed as a self-contained, formally-precise **open-problem dossier for external solvers** — *Chapter 35 (p. 674)*. Help is most valuable there (the net-naturality / carrier theorem — the verdict-flipper), and on watchlist maintenance (new bounds in any channel, with citations).

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**Authors:** Harshit Khemani + Claude (Anthropic) · **License:** *CC BY-SA 4.0 (content) + MIT (code)* (repo: `LICENSE.md`) · **Site design**

inspired by Dan Hollick's Making Software

CHAPTER 2

# Epistemics: Standards of Evidence and Labeling

This page defines *how strongly* any statement in this wiki is claimed to be true. It is the most important meta-page: a research program toward a universal theory lives or dies on its honesty about what is known, what is inferred, and what is guessed. Every other page is required to use the conventions below.

> **The cardinal rule:** Never let a *speculative* idea wear the clothes of an *established* fact. Overclaiming is the characteristic failure mode of "theory of everything" work, and it is treated here as a defect, not a flourish.

## 1. The five epistemic tags

Every nontrivial claim carries exactly one inline tag.

| Tag | Meaning | Test for using it |
|---|---|---|
| `[ESTABLISHED]` | Experimentally confirmed to high significance, **or** rigorously proven inside an accepted framework; broad expert consensus. | "Would the overwhelming majority of working physicists stake their reputation on this?" |
| `[INFERENCE]` | Well-motivated, widely accepted reasoning, not yet *directly* confirmed. A strong bet, not a fact. | "Is this the standard conclusion of careful reasoning, while still logically defeasible by future data?" |
| `[SPECULATIVE]` | Plausible and principled, but lacking decisive evidence; a minority or research-frontier position. | "Is there a real argument for it, but no experiment that forces it?" |

| | | |
|---|---|---|
| `[OPEN]` | Genuinely unresolved; the honest answer is *we do not know*. | "Is there active, undecided disagreement or simply no answer?" |
| `[CONTESTED]` | Competent experts actively disagree, with no consensus. | "Do serious camps hold incompatible views?" |

Tags may be applied to a sentence, a clause, or a whole paragraph (stated once at the top of the paragraph). When a single sentence mixes a fact and an inference, split it.

### Examples (calibration)

- General relativity predicts gravitational waves, now directly detected. `[ESTABLISHED]`
- Dark matter is a particle (rather than a modification of gravity). `[INFERENCE]` — the *gravitational* evidence for missing mass is `[ESTABLISHED]`; that it is a *particle* is the leading inference, not yet confirmed.
- Spacetime geometry is emergent from entanglement. `[SPECULATIVE]`
- The correct interpretation of quantum mechanics. `[CONTESTED]`
- The mechanism that sets the neutrino mass scale. `[OPEN]`

---

## 2. Confidence (optional secondary signal)

Where useful, attach a coarse confidence to `[INFERENCE]`/`[SPECULATIVE]` items: **high / medium / low / very-low**. This is a subjective Bayesian credence, not a frequentist statistic. Use it sparingly and never to dress up a guess — a "high-confidence speculation" is still a speculation.

---

## 3. Assumption classification (used in `ASSUMPTIONS_LEDGER.md`)

Assumptions are graded on a separate axis — not *how true* but *how load-bearing and how revisable*:

| Class | Meaning |
|---|---|
| `fundamental` | Appears irreducible; relaxing it breaks the theory's coherence or contradicts robust data. |
| `plausibly-fundamental` | Looks basic, but no proof it could not be derived from something deeper. |
| `historical-artifact` | Adopted for historical/pedagogical reasons; may be a contingent choice, not a necessity. |
| `mathematical-convenience` | A simplifying technical choice (smoothness, separability, a particular gauge) that buys tractability. |
| `untested-empirically` | Assumed because it has never been probed in the relevant regime, not because it is confirmed. |

A central research activity of this wiki is moving assumptions *down* this list — turning supposed bedrock into a derivable or contingent feature.

---

# 4. Sourcing rules

1. **No fabrication.** Never invent a citation, an experimental number, a theorem, or a date. A missing reference is acceptable; a fake one is a disqualifying error.
2. **Real sources only.** Cite canonical textbooks and genuinely landmark results. If unsure of exact bibliographic detail, give author + title + approximate year and mark `[unverified]`.
3. **Numbers carry context.** Quote measured quantities with their nature (e.g., that the "10^120" cosmological-constant figure is *regularization-dependent*, see *Chapter 16 (p. 308)*). A number without its caveat is a half-truth.
4. **Folklore is flagged.** Widely repeated claims that are actually subtle or wrong (e.g., loose statements of Haag's theorem, "energy is not conserved in GR") must be stated carefully, not parroted.

---

## 5. Distinguishing four things that get conflated

A recurring discipline here, especially in *Chapter 14 (p. 191)*:

- **Logical inconsistency** — two accepted statements cannot both be true. (Rare. Most "contradictions" are not this.)
- **Domain-of-validity mismatch** — two frameworks disagree only where neither is trusted (e.g., GR vs QFT at the Planck scale). Not a paradox; a frontier.
- **Unsolved-but-consistent problem** — no known inconsistency, just no derivation/solution yet (e.g., the value of a parameter).
- **Conceptual tension** — the frameworks are formally compatible but rest on jarringly different worldviews (e.g., block-universe GR vs the apparent dynamical collapse in QM).

Mislabeling a frontier as a paradox is itself an epistemic error.

---

## 6. How claims change status

Status is not permanent. A claim moves between tags as evidence and argument accumulate. Every promotion/demotion is logged in the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)) with a one-line reason. The wiki is **cumulative**: we revise pages in place and record *why* in the changelog, rather than starting over.

---

PART II

# THE ESTABLISHED MAP

Before asking where physics breaks, you must know what physics *is* — not as folklore, but as a precise inventory. This part is that inventory: the established map any universal theory would have to reproduce, and the largest body of ESTABLISHED-tagged material in the book.

Chapter 3 draws the map whole: every major framework, its mathematical home, its domain of validity, and — crucially — the reduction relations between them, the limits in which one theory hands off to another. Those handoffs are where unification claims are ultimately cashed out, so the map is drawn with the seams showing.

The ten chapters that follow are domain dossiers, ordered roughly along the historical and logical grain: classical mechanics, thermodynamics, and statistical mechanics; then the quantum sequence — quantum mechanics, quantum field theory, and the Standard Model of particle physics; then general relativity and the cosmology built on it; and finally two connective disciplines, information theory and the mathematics that underwrites all of the above. Each dossier records what the framework asserts, what has been confirmed and to what precision, what it assumes, and where its edges lie.

What this part establishes is the non-negotiable core: the results — Noether's theorem, the Born rule's confirmed form, the anomaly-free Standard Model, general relativity at gravitational-wave precision, the fluctuation theorems, ΛCDM — that constrain every candidate road in Part IV. It is long because the truth is long. It is also the foundation everything after it stands on.

CHAPTER 3

# The Theory Map: How the Frameworks Fit Together

---

This page is the **atlas** of the wiki: it situates every physical framework relative to every other, exhibits the *reduction structure* (which theory is a limit of which), and lays out the *scale ladder* from the Planck scale to the cosmological horizon. It is deliberately structural rather than exhaustive; per-framework depth lives on the linked `domains/` (Chapter 1 (p. 1)) pages, and the *failures* of the map — the places where frameworks clash rather than nest — are catalogued in *Chapter 14 (p. 191)*. Epistemic markers follow *Chapter 2 (p. 8)*; load-bearing modeling assumptions are tracked in *Chapter 20 (p. 458)*.

A guiding caveat: the words "limit," "reduction," and "emergence" below denote *controlled approximation in a specified regime*, **not** logical derivation in general. Several of the most important limits are mathematically **singular** (e.g. $\hbar \to 0$), so "X reduces to Y" almost always means "Y is the leading term of an asymptotic expansion of X, valid where a dimensionless ratio is small." `[INFERENCE]`

---

# 1. Master table of frameworks

| DOMAIN | DESCRIBES | CORE MATH OBJECT | KEY ASSUMPTIONS | VALIDITY WINDOW | KNOWN BREAKDOWN |
|---|---|---|---|---|---|
| **Classical mechanics** | Deterministic motion of finitely many bodies | Symplectic manifold $(M, \omega)$; Hamiltonian flow; Poisson algebra | Absolute time; smooth deterministic ODEs; $S \gg \hbar$, $v \ll c$, weak gravity | Celestial mechanics, engineering; verified to extreme precision in regime | $v \to c$ (SR); $S \sim \hbar$ (QM); strong gravity (GR); non-Lipschitz/$N$-body singularities; chaos |
| **Quantum mechanics (NRQM)** | Non-relativistic quantum systems, finite dof | Rays in complex separable Hilbert space; self-adjoint operators; PVM/POVM | Superposition; Born rule; tensor-product composition; external classical time | Atoms, chemistry, condensed matter, quantum info — no confirmed deviation | $v \to c$ / pair creation (QFT); dynamical gravity (problem of time); measurement cut |
| **Quantum field theory** | Relativistic quantum fields; SM substrate | Operator-valued distributions; path integral $\int \mathcal{D}\phi\, e^{iS/\hbar}$; local algebras (type III) | Microcausality; Poincaré invariance; unitarity; fixed background | QED $g-2$ to ~12 sig figs; LHC to ~few TeV; lattice QCD | Planck scale (gravity non-renorm.); $\Lambda$ problem; rigorous 4D construction; Haag's theorem |
| **Particle physics / SM** | EM, weak, strong interactions | Chiral gauge theory $SU(3)\times SU(2)\times U(1)$ + Higgs | Local Lorentz-invariant QFT; anomaly cancellation; 3 generations | sub-eV → ~TeV directly, higher indirectly; most-tested theory | $\nu$ mass (confirmed BSM); no DM/DE; hierarchy & strong-CP; baryogenesis |
| **General relativity** | Classical gravitation = spacetime geometry | Lorentzian manifold $(M, g)$; $G_{\mu\nu}+\Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ | Equivalence principle; diffeo invariance; smooth metric; classical $g$ | mm tests → $\sim 10^{26}$ m; pulsars, GWs, $\Lambda$ CDM | Singularities; Planck scale (perturbative non-renorm.); $\Lambda$ problem; info paradox |

| | | | | | |
|---|---|---|---|---|---|
| **Thermodynamics** | Macroscopic energy/heat/entropy at equilibrium | State functions; Legendre transforms; $dU = TdS - pdV + \mu dN$ | Equilibrium exists; extensivity; short-range forces; $N \sim N_A$ | Exact for macroscopic short-range systems | Smal $N$ (fl theor gravi syste time |
| **Statistical mechanics** | Micro → macro bridge | Ensembles; partition function $Z$; $S = k_B \ln W$ | Equal a priori probability; thermodynamic limit; molecular chaos | Equilibrium + short-range; controls fluctuations | Long range glass integ (ETH of tin |
| **Cosmology** | Dynamics & history of the universe | FLRW metric; Friedmann eqns; $\Lambda$ CDM | Cosmological principle; GR on horizon scales; perfect-fluid matter | BBN ($t \sim 1$ s) → today, scales $\gtrsim 100$ Mpc | Initia $\Lambda$ pro secto tensi |
| **Information theory** | Quantification & limits of information | Shannon $H$ ; von Neumann $S(\rho)$; RT surfaces | Observer-independent measure; unitarity; (in gravity) holography | Layer-dependent: exact math → AdS/CFT-only holography | Type (no lo info p outsi |
| **Mathematics for physics** | The rigorous substrate | Hilbert spaces; operator algebras; bundles; measures | Self-adjointness; smooth manifolds; well-defined measures | Rigorous in low-$d$ QFT, finite-dof QM, classical geometry | Loren integ 4D Y quan go |

A cross-cutting principle: **the action principle** $\delta \int L\, dt = 0$ / $\int \mathcal{D}\phi\, e^{iS/\hbar}$ is the single organizing structure shared from classical mechanics through QFT [ESTABLISHED]; the **symplectic/Poisson structure** survives (deformed) into QM via $\{\,,\} \to [\,,]/i\hbar$ [ESTABLISHED]; the **renormalization group** unifies QFT with critical phenomena in statistical mechanics [ESTABLISHED]. See *Chapter 17 (p. 325).*

## 2. The tower of effective theories (reduction structure)

Each arrow reads **"A reduces to B in the indicated limit"** — B is the effective/low-energy/coarse-grained description, A the more fundamental one. All limits are flagged for singularity.

- **QFT** $\xrightarrow{c\to\infty,\ \text{fixed } N}$ **NRQM.** Non-relativistic QM is the fixed-particle-number, low-energy sector of QFT `[ESTABLISHED]`. QFT *supplies what NRQM postulates*: spin–statistics, antiparticles, and the resolution of one-particle relativistic pathologies. The reduction is **not** representation-preserving: the Stone–von Neumann uniqueness that NRQM relies on **fails** for infinitely many dof (Haag's theorem, inequivalent vacua) `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.
- **QM** $\xrightarrow{\hbar\to 0\ (S\gg\hbar)}$ **classical mechanics.** Recovered via Ehrenfest, WKB/stationary phase, and the Wigner–Moyal map (Moyal bracket $\to$ Poisson bracket) `[ESTABLISHED]`. The limit is **singular and non-uniform**: $e^{iS/\hbar}$ oscillates without bound, tunneling $\sim e^{-S/\hbar}$ is non-analytic, and for chaotic systems the Ehrenfest time $t_E \sim \lambda^{-1}\ln(S/\hbar)$ truncates correspondence `[ESTABLISHED]`. Decoherence supplies the *effective* pointer basis; a fully general derivation of classicality is `[OPEN]`. Definite trajectories and noncontextual values are excluded by Bell / Kochen–Specker `[ESTABLISHED]`.
- **SR / relativistic dynamics** $\xrightarrow{c\to\infty}$ **Galilean mechanics.** The relativistic free Lagrangian $-mc^2\sqrt{1-v^2/c^2} \to \text{const} + \frac{1}{2}mv^2$ `[ESTABLISHED]`. Absolute time is the $c\to\infty$ contraction of Lorentzian causal structure.
- **GR** $\xrightarrow{\text{weak field, slow motion}}$ **Newtonian gravity.** $g_{00} \approx -(1+2\Phi/c^2)$ , geodesics $\to \ddot{\mathbf{x}} = -\nabla\Phi$, and the field equation $\to \nabla^2\Phi = 4\pi G\rho$; this limit fixes the coupling $8\pi G/c^4$ `[ESTABLISHED]`.
- **GR (+ matter)** $\xrightarrow{E\ll M_{\text{Pl}}}$ **GR as an effective field theory.** GR quantized perturbatively is a valid EFT at low energy but

perturbatively non-renormalizable at $M_{\rm Pl}$ `[ESTABLISHED]`. What UV-completes it is `[OPEN]` — the central problem of *Chapter 15 (p. 233)*.

- **Statistical mechanics $\xrightarrow{N\to\infty,\ \text{coarse-grain}}$ thermodynamics.** Stat-mech *derives* thermodynamics: $F = -k_B T \ln Z$ reproduces the potentials, $S = k_B \ln W$ grounds entropy `[ESTABLISHED]`. The reduction has a *subtle residue*: the strict second law is the probabilistic statement's $N \to \infty$ shadow, and irreversibility requires the extra **Past Hypothesis** input that thermodynamics merely posits `[INFERENCE/OPEN]`. See *Chapter 6 (p. 56)* and *Chapter 14 (p. 191)*.
- **Hamiltonian mechanics $\xrightarrow{+\ \text{Liouville measure}}$ statistical mechanics.** Hamiltonian phase-space flow plus Liouville's theorem is the microscopic foundation of the microcanonical ensemble `[ESTABLISHED]`; the tension is that reversible, volume-preserving microdynamics cannot *alone* produce monotone entropy increase (Loschmidt, Zermelo) `[ESTABLISHED tension]`.
- **Euclidean QFT $\cong$ classical statistical field theory** (Wick rotation, Osterwalder–Schrader). Not a reduction but a *formal isomorphism*: the path integral *is* a partition function, and Wilsonian RG is literally shared `[ESTABLISHED, where reflection positivity holds]`. See *Chapter 13 (p. 172)*.
- **SM $\xrightarrow{\text{leading operators}}$ SMEFT / low-energy EFTs** (chiral perturbation theory, Fermi theory). Renormalizability is *not* a fundamental axiom but an emergent low-energy property: irrelevant operators are suppressed by powers of $E/\Lambda$ `[ESTABLISHED, Wilsonian reinterpretation]`.

**Diagrammatic note.** The tower is **not** a single linear chain. Three deformation parameters act quasi-independently — $\hbar$ (quantum), $1/c$ (relativistic), $GM/rc^2$ (gravitational) — plus a coarse-graining/$N$ axis. The "missing corner" where all of $\hbar$, $1/c$, and $G$ are simultaneously $O(1)$ is **quantum gravity**, which no current framework occupies `[OPEN]`. See *Chapter 18 (p. 349)*.

# 3. Relationship diagram

```
graph TD
    QG["Quantum Gravity<br/>(missing: ℏ, 1/c, G all O(1))<br/>[OPEN]"]

    QG -->|low energy, classical g| GR["General Relativity"]
    QG -->|fixed background| QFT["Quantum Field Theory"]

    GR -->|weak field, v≪c| NG["Newtonian Gravity"]
    GR -->|flat metric, local frame| SR["Special Relativity"]
    GR -->|FLRW symmetry + fluid| COSMO["Cosmology (ΛCDM)"]

    QFT -->|c→∞, fixed N| QM["Quantum Mechanics (NRQM)"]
    QFT -->|gauge group + reps| SM["Standard Model"]
    QFT -.Wick rotation.- STAT["Statistical Mechanics"]

    QM -->|ℏ→0, S≫ℏ| CM["Classical Mechanics"]
    QM -->|density matrices, ETH| STAT
    SR -->|c→∞| CM

    CM -->|+ Liouville measure| STAT
    STAT -->|N→∞, coarse-grain| THERMO["Thermodynamics"]

    QM -.operational substrate.- INFO["Information Theory"]
    GR -.Bekenstein–Hawking S=A/4.- THERMO
    GR -.holography, RT.- INFO
    THERMO -.Landauer, Maxwell demon.- INFO

    SM --> COSMO
    QFT -.cosmological constant problem.- COSMO

    classDef open fill:#fdd,stroke:#900;
    class QG open;
```

**Solid arrows** = controlled reduction (limit/coarse-graining). **Dashed lines** = structural correspondence or shared formalism, not a reduction. The triple point **GR** ∩ **QFT/QM** ∩ **thermodynamics** (black-hole thermodynamics, the information paradox, $S = A/4$) is the densest cluster of `[OPEN]`/`[CONTESTED]` problems in physics. See *Chapter 14 (p. 191)* and *Chapter 12 (p. 155)*.

# 4. The scale ladder: from Planck to the horizon

Energies in GeV, lengths in metres; values are standard constants, not invented. The "ruling framework" column states which theory is *operationally correct* at that scale; arrows indicate where one framework hands off to another.

| **Scale (energy / length)** | **Regime** | **Ruling framework** | **Notes** |
|---|---|---|---|
| $E_{\rm Pl} \approx 1.22 \times 10^{19}$ GeV; $\ell_{\rm Pl} \approx 1.6 \times 10^{-35}$ m | Quantum gravity | **None established** `[OPEN]` | Curvature $\sim \ell_{\rm Pl}^{-2}$; metric fluctuations $O(1)$; GR-as-EFT and QFT both fail |
| $\sim 10^{16}$ GeV | GUT / proton decay | SM as EFT; GUTs `[SPECULATIVE]` | Gauge couplings near-unify (better with SUSY); $\tau_p > 10^{34}$ yr excludes minimal $SU(5)$ |
| $\sim 10^9$–$10^{15}$ GeV | Seesaw / inflation / baryogenesis | QFT + cosmology `[INFERENCE]` | $\nu$-mass seesaw scale; inflaton sector; leptogenesis — none directly probed |
| $\sim 246$ GeV ($v$ ); $\sim 100$ GeV–few TeV | Electroweak; collider frontier | **Standard Model** `[ESTABLISHED]` | Higgs mechanism: $M_W = \frac{1}{2}gv$, $M_Z = \frac{1}{2}\sqrt{g^2 + g'^2}\,v$; LHC reach |
| $\sim 0.2$ GeV ( $\Lambda_{\rm QCD}$) | Confinement / hadrons | Nonperturbative QCD (lattice) `[ESTABLISHED]` | Quark–gluon → hadron description switch; perturbation theory fails |
| $\sim$ MeV–keV | Nuclear, atomic; BBN | QM + QED; QFT for precision | $g{-}2$ to ~12 sig figs; BBN at $T \sim 1$ MeV, $t \sim 1$ s |
| $\sim$ eV and below | Chemistry, condensed matter | **NRQM** `[ESTABLISHED]` | Classical mechanics emerges where $S \gg \hbar$ and decoherence acts |

| | | | |
|---|---|---|---|
| Macroscopic ( $N \sim N_A$) | Bulk matter, engines | Classical mechanics; thermodynamics | Stat-mech bridges to micro; fluctuations $\sim N^{-1/2}$ |
| Solar system → galactic ($\sim 10^{20}$ m) | Gravitational dynamics | **General relativity / Newtonian** [ESTABLISHED] | PPN tests; pulsars; GW sources; rotation curves require dark matter |
| $\gtrsim$ 100 Mpc → $\sim 10^{26}$ m (horizon) | Cosmological | **GR + ΛCDM** [ESTABLISHED] | Cosmological principle holds statistically; $H_0$/ $S_8$ tensions [CONTESTED] |
| Cosmological horizon / de Sitter | Global structure | GR; holography [SPECULATIVE outside AdS] | de Sitter entropy by analogy with BH; RT/holography unestablished here |

Numerical anchors for the dimensionless ratios that switch frameworks live in *Chapter 16 (p. 308)*: $\hbar$, $c$, $G$, $k_B$, $M_{\rm Pl} = \sqrt{\hbar c/G}$, $\alpha \approx 1/137$, $\Lambda \sim 10^{-52}\ \mathrm{m}^{-2}$, and the Bekenstein–Hawking entropy $S_{\rm BH} = k_B c^3 A/4G\hbar$.

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## 5. Where the map tears (pointers, not content)

The reductions above are the *successes*. The wiki's value concentrates in the *failures* — pairs of frameworks that do not nest cleanly. These are catalogued in full in *Chapter 14 (p. 191)*; a structural index:

- **QM/QFT ∩ GR — quantum gravity & the problem of time.** QM/QFT presuppose a fixed classical background and external time; GR makes spacetime dynamical (Wheeler–DeWitt $H|\Psi\rangle = 0$ is timeless). Perturbative quantization is non-renormalizable. The sharpest clash in physics `[ESTABLISHED that it is unresolved]`. See *Chapter 15 (p. 233)*.

- **QFT ∩ GR ∩ cosmology — the cosmological constant problem.** Naive QFT vacuum energy exceeds observed $\rho_\Lambda$ by $\sim 10^{60}$–$10^{120}$ `[OPEN]`. A domain-clash, possibly the deepest quantitative mismatch known.
- **Reversible microdynamics ∩ thermodynamics — the arrow of time.** Time-symmetric mechanics + thermodynamic arrow forces a low-entropy boundary condition (Past Hypothesis), whose origin is cosmological and `[OPEN]`.
- **The measurement problem.** Unitary evolution (A5) and the projection postulate (A6) are jointly applied with no rule demarcating their jurisdictions `[OPEN/CONTESTED]`. A genuine incompleteness, not a domain mismatch — distinct in kind. See *Chapter 7 (p. 71)*.
- **Black-hole thermodynamics & information.** $S_{\mathrm{BH}} = A/4$ and Hawking radiation place thermodynamics, QM (unitarity), and GR at a triple point; the information paradox and microstate counting are `[OPEN/CONTESTED]`. See *Chapter 12 (p. 155)* and *Chapter 5 (p. 40)*.
- **Rigor gap.** No physically realistic interacting 4D QFT is rigorously constructed (Yang–Mills mass gap, a Clay problem); the Lorentzian path-integral measure does not exist as a measure `[OPEN]`. See *Chapter 13 (p. 172)*.

A discipline of classification (per *Chapter 2 (p. 8)*): distinguish a **true logical inconsistency** (none confirmed between accepted frameworks), a **domain-of-validity mismatch** (most "clashes" above — e.g. QFT-on-fixed-background vs dynamical GR), and an **unsolved-but-consistent problem** ($\Lambda$CDM dark sector, confinement proof, glass transition).

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## References

See the Bibliography (p. 776) for canonical sources underlying each framework (e.g. von Neumann and Reed–Simon for QM foundations; Weinberg, Haag, and Streater–Wightman for QFT; Wald and Misner–Thorne–Wheeler for GR; Callen and Jaynes for thermodynamics/statistical mechanics; Arnold for classical mechanics; Nielsen–Chuang for quantum information). Per wiki policy (`AGENTS.md` (in the repository)), only real, standard sources are cited; specific landmark results (Gleason, Stone–von Neumann, Haag, Lovelock, Coleman–Mandula, Penrose–Hawking, Aizenman–Duminil-Copin, Ryu–Takayanagi) are named on the relevant domain pages.

CHAPTER 4

# Classical Mechanics

Classical mechanics is the deterministic theory of the dynamics of systems with finitely many degrees of freedom — point particles, rigid bodies, and their continuum generalizations — in the regime of low velocity ($v \ll c$), large action ($S \gg \hbar$), and weak gravity. It is at once the historical taproot of physics and, in its mature geometric form, the structural skeleton that survives (deformed, not discarded) into quantum mechanics, statistical mechanics, and field theory. This page treats the three equivalent formulations, their geometry, the symmetry–conservation correspondence, integrability and chaos, and the precise boundaries where the framework yields.

## Scope

In scope: Newtonian, Lagrangian, and Hamiltonian dynamics; their geometric reformulation on configuration manifolds, tangent and cotangent bundles, and symplectic manifolds; Noether's theorem and the moment map; canonical/symplectic transformations and Hamilton–Jacobi theory; Liouville–Arnold integrability, KAM theory, and deterministic chaos; holonomic, nonholonomic, and gauge (singular-Lagrangian) constraints; and the structural relations to quantum mechanics via canonical quantization, deformation quantization, and the path integral.

Out of scope (treated as boundaries, not contents): special/general relativistic dynamics, the statistical mechanics of large ensembles, and quantum field theory — referenced here only insofar as they bound the domain of validity. See *Chapter 3 (p. 15)* for the global placement of this domain.

# Core formalism

Classical mechanics admits three logically distinct but — for a large class of regular systems — equivalent formulations, increasing in geometric abstraction and structural power.

## I. Newtonian (vectorial) mechanics

State: position vectors $\mathbf{r}_i(t) \in \mathbb{E}^3$ for $i = 1, \dots, N$. Dynamics is a second-order ODE,

$$m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i(\mathbf{r}_1, \dots, \mathbf{r}_N, \dot{\mathbf{r}}_1, \dots, \dot{\mathbf{r}}_N, t).$$

The three laws are the empirical backbone `[ESTABLISHED in their domain]`: (1) inertia, asserting the *existence* of inertial frames; (2) $\mathbf{F} = d\mathbf{p}/dt$ with $\mathbf{p} = m\mathbf{v}$; (3) action–reaction $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$, which for central forces yields conservation of total momentum and, with collinearity, of angular momentum. The physical content is not the equation alone but the claim that forces are *specified independently* of the kinematics (gravity, Lorentz, contact) and superpose linearly [INFERENCE — a modeling commitment, not a theorem; see the assumptions ledger]. Cartesian inertial coordinates are natural here; constraints and curvilinear coordinates motivate the analytical formulations.

## II. Lagrangian mechanics (configuration space, variational)

Let $Q$ be the configuration manifold of dimension $n$ (the number of degrees of freedom after holonomic constraints are solved). The dynamical arena is the tangent bundle $TQ$ with coordinates $(q^i, \dot{q}^i)$. A Lagrangian $L : TQ \times \mathbb{R} \to \mathbb{R}$, typically $L = T - V$, defines the action functional

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt.$$

**Hamilton's principle** `[INFERENCE – see below on its derived status]`: the physical trajectory makes $S$ stationary among paths with fixed endpoints, $\delta q(t_1) = \delta q(t_2) = 0$. Stationarity yields the **Euler–Lagrange equations**

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = 0,$$

which are manifestly form-invariant under arbitrary point transformations of $Q$. For $L = \frac{1}{2}g_{ij}(q)\dot{q}^i\dot{q}^j - V(q)$ they reduce exactly to the geodesic equation of the metric $g_{ij}$ perturbed by $-\partial V$: the Christoffel terms reproduce "fictitious" inertial forces automatically `[ESTABLISHED]`. For *regularL*, the velocity Hessian $\det\left(\partial^2 L/\partial \dot{q}^i \partial \dot{q}^j\right) \neq 0$, and the dynamics is a well-defined second-order vector field (SODE) on $TQ$. Singular Lagrangians (gauge systems) require Dirac–Bergmann constraint analysis. The genuinely first principle for constrained and nonconservative systems is **D'Alembert's principle**, $\sum_i (\mathbf{F}_i - m_i \mathbf{a}_i) \cdot \delta \mathbf{r}_i = 0$, from which both Newton's and the Euler–Lagrange equations descend `[ESTABLISHED]`; velocity-dependent potentials (e.g. the magnetic term $q\,\mathbf{A} \cdot \mathbf{v}$) and Rayleigh dissipation fit within or just beyond this scheme.

## III. Hamiltonian mechanics (phase space, symplectic)

Pass from $TQ$ to the cotangent bundle $T^*Q$ via the Legendre transform: conjugate momenta $p_i = \partial L/\partial \dot{q}^i$ and

$$H(q, p, t) = p_i \dot{q}^i - L,$$

well-defined when $L$ is regular. The dynamics is the first-order flow

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i}.$$

$T^*Q$ carries the canonical (tautological) 1-form $\theta = p_i\, dq^i$ and the canonical symplectic 2-form

$$\omega = -d\theta = dq^i \wedge dp_i,$$

which is closed ($d\omega = 0$) and nondegenerate. The dynamics is intrinsically the flow of the Hamiltonian vector field $X_H$ defined by

$\iota_{X_H}\omega = dH,$

giving coordinate-free Hamiltonian mechanics as a triple $(M, \omega, H)$ on a $2n$-dimensional symplectic manifold. The **Poisson bracket**

$$\{f, g\} = \frac{\partial f}{\partial q^i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q^i} = \omega(X_f, X_g)$$

makes $C^\infty(M)$ a Lie algebra (antisymmetry, Jacobi identity) and a Poisson algebra (Leibniz). Observables evolve by $df/dt = \{f, H\} + \partial f/\partial t$, and the fundamental brackets $\{q^i, q^j\} = 0$, $\{p_i, p_j\} = 0$, $\{q^i, p_j\} = \delta^i_j$ encode the entire structure.

**Liouville's theorem** `[ESTABLISHED]`: the Hamiltonian flow preserves $\omega$ and hence the phase-space volume $\omega^n/n!$; each map $\phi_t$ is a symplectomorphism ($\phi_t^*\omega = \omega$). This is the geometric root of incompressibility of the probability fluid in statistical mechanics and forbids attractors in the strict Hamiltonian setting. **Canonical (symplectic) transformations** are maps preserving $\omega$; a type-2 generating function $F_2(q, P, t)$ gives $p_i = \partial F_2/\partial q^i$, $Q^i = \partial F_2/\partial P_i$, $K = H + \partial F_2/\partial t$ — the freedom to choose canonical coordinates is the engine of integration.

**Noether's theorem** `[ESTABLISHED]`: every continuous one-parameter symmetry leaving $L$ invariant up to a total derivative $dF/dt$ under $q \to q + \epsilon\,\xi(q)$ yields a conserved

$$J = \frac{\partial L}{\partial \dot{q}^i}\,\xi^i - F, \qquad \frac{dJ}{dt} = 0.$$

In Hamiltonian form the statement is an iff: $J$ generates a symmetry of $H$ exactly when $\{J, H\} = 0$, in which case $J$ is conserved (and conversely). Time-translation $\to$ energy; spatial translation $\to$ momentum; rotation $\to$ angular momentum; Galilean boost $\to$ uniform center-of-mass motion; the Laplace–Runge–Lenz vector of the $1/r$ potential reflects a hidden $SO(4)$ symmetry. The deep statement: symmetries and conservation laws are two faces of the **moment map**$J : M \to \mathfrak{g}^*$ for a Lie group $G$ acting by symplectomorphisms — see *Chapter 17 (p. 325)*.

## IV. Hamilton–Jacobi theory

Seek a canonical transformation to coordinates in which the new Hamiltonian vanishes, so all new coordinates and momenta are constant. Hamilton's principal function $S(q, \alpha, t)$ satisfies the **Hamilton–Jacobi PDE**

$$H\left(q, \frac{\partial S}{\partial q}, t\right) + \frac{\partial S}{\partial t} = 0.$$

For time-independent $H$, write $S = W(q, \alpha) - Et$ with $H(q, \partial W/\partial q) = E$; a *complete integral* (with $n$ nontrivial constants $\alpha$) solves the dynamics by quadratures. Here $S = \int L\, dt$ evaluated on the actual trajectory as a function of endpoints, with $\partial S/\partial q^i = p_i$ , $\partial S/\partial t = -H$. Hamilton–Jacobi is the classical shadow of the Schrödinger equation in the eikonal/WKB limit: writing $\psi = \exp(iS/\hbar)$ and taking $\hbar \to 0$ reduces Schrödinger to Hamilton–Jacobi at leading order, the next order giving the amplitude transport equation `[ESTABLISHED as the standard WKB result]`. See *Chapter 7 (p. 71).*

## V. Integrability, action–angle variables, and chaos

**Liouville–Arnold theorem** `[ESTABLISHED]`: an $n$-degree-of-freedom system with $n$ functionally independent integrals $F_1 = H, F_2, \ldots, F_n$ in mutual involution ($\{F_i, F_j\} = 0$) is *completely integrable*. On a compact connected level set the motion is confined to an $n$-torus $T^n$, and there exist action–angle variables $(I_i, \phi_i)$ with

$$\dot{\phi}_i = \frac{\partial H}{\partial I_i} = \omega_i(I) = \text{const}, \qquad \dot{I}_i = 0,$$

so the motion is conditionally periodic (quasiperiodic) winding on the torus. Examples: harmonic oscillators, the Kepler problem, the free rigid body (Euler top), the Toda lattice.

**KAM theorem** (Kolmogorov–Arnold–Moser) `[ESTABLISHED]`: for a small perturbation $H = H_0(I) + \epsilon H_1(I, \phi)$ of a nondegenerate integrable $H_0$ ($\det(\partial^2 H_0/\partial I\, \partial I) \neq 0$), most invariant tori survive —

precisely those whose frequency vectors are sufficiently irrational (Diophantine),

$$|k \cdot \omega(I)| \geq \gamma \, |k|^{-\tau} \quad \forall \, k \in \mathbb{Z}^n \setminus \{0\}.$$

Resonant tori are destroyed, breaking into alternating elliptic/hyperbolic structures (Poincaré–Birkhoff) and seeding a fractal web of chaotic layers (with Arnold diffusion possible for $n \geq 3$).

**Deterministic chaos** `[ESTABLISHED]`: generic nonintegrable systems exhibit sensitive dependence on initial conditions,

$$|\delta x(t)| \sim |\delta x(0)| \, e^{\lambda t}, \quad \lambda > 0,$$

with $\lambda$ the largest Lyapunov exponent. Determinism (unique solution from initial data) coexists with practical unpredictability because finite-precision data loses information at rate $\lambda$ (Kolmogorov–Sinai entropy). The three-body problem is the historical seed (Poincaré, 1890s). See *Chapter 13 (p. 172)* for the dynamical-systems machinery.

## VI. Constraints

Holonomic constraints $f(q, t) = 0$ cut $Q$ down to a submanifold and are eliminated by adapted coordinates. Nonholonomic constraints — non-integrable velocity constraints $a_i(q) \, \dot{q}^i = 0$, e.g. rolling without slipping — cannot be integrated to position constraints and are handled by the Lagrange–d'Alembert principle. Crucially, the resulting equations are **not** the Euler–Lagrange equations of any action in general `[ESTABLISHED]` — a genuine structural subtlety placing nonholonomic systems outside the pure variational/Hamiltonian framework. Singular Lagrangians and gauge constraints invoke the Dirac constraint algorithm (primary/secondary, first/second class, Dirac brackets), the classical precursor to gauge-theory quantization. See *Chapter 8 (p. 89)*.

### VII. The bridge to quantum mechanics

Canonical quantization (Dirac): $\{\cdot,\cdot\} \to (i\hbar)^{-1}[\cdot,\cdot]$, so $\{q,p\}=1$ becomes $[\hat{q},\hat{p}]=i\hbar$. Deformation quantization (Moyal–Weyl; Bayen–Flato–Fronsdal–Lichnerowicz–Sternheimer; Kontsevich's formality theorem for general Poisson manifolds `[ESTABLISHED]`) deforms the commutative product into the noncommutative star product

$$f \star g = fg + \frac{i\hbar}{2}\{f,g\} + O(\hbar^2), \qquad [\hat{A},\hat{B}] = i\hbar\,\widehat{\{A,B\}} + O(\hbar^2).$$

The Feynman path integral $K = \int \mathcal{D}[q]\, e^{iS[q]/\hbar}$ localizes by stationary phase as $\hbar \to 0$ onto the classical least-action path. Thus least action is not a teleological mystery but the classical shadow of constructive/destructive interference of quantum amplitudes `[INFERENCE — the standard, widely held interpretation]`.

## Foundational assumptions

| ASSUMPTION | STATUS | JUSTIFICATION |
|---|---|---|
| Absolute, universal time $t$ as the single evolution parameter (Galilean spacetime). | historical-artifact | `[ESTABLISHED false fundamentally]` Special relativity demolishes absolute simultaneity; Newtonian time is the $c \to \infty$ limit of Lorentzian structure. Excellent for $v \ll c$, but a low-velocity artifact, not fundamental. |
| Configuration space is a smooth finite-dimensional manifold; states evolve by smooth ODEs and dynamics is deterministic. | likely-fundamental | `[INFERENCE/CONTESTED at the margins]` Picard–Lindelöf guarantees uniqueness for Lipschitz fields, but Lipschitz failure permits non-uniqueness (Norton's dome, $\ddot{x} = \sqrt{x}$) and finite-time blowup (Xia 1992) breaks completeness for $N \geq 5$. A robust working principle with genuine internal cracks. |

| | | |
|---|---|---|
| Forces are specified independently of the laws of motion and superpose; $L = T - V$ with $V = V(q)$ (possibly velocity-dependent). | conventional-choice | `[INFERENCE]` The kinetic/potential split and additivity is a modeling convention that works because gravity and electromagnetism derive from potentials/gauge fields. The form $L = T - V$ is itself a low-energy shadow (e.g. $-mc^2\sqrt{1 - v^2/c^2} \to \text{const} + \frac{1}{2}mv^2$). |
| Principle of stationary action: trajectories extremize $\int L\,dt$. | likely-fundamental | `[INFERENCE – strong consensus]` Derived, not assumed: it is the stationary-phase ( $\hbar \to 0$) limit of the path integral. Caveat: genuinely *stationary*, not *least* (saddles occur past conjugate points/caustics) — a common textbook imprecision. |
| Phase space carries a canonical symplectic structure $\omega$; observables form a Poisson algebra. | fundamental | `[ESTABLISHED within the framework]` The single most robust structural feature; it survives deformed into QM via $\{\cdot,\cdot\} \to [\cdot,\cdot]$ and is preserved by all canonical transformations and the flow (Liouville). Darboux's theorem makes it locally rigid and natural. |
| Space is Euclidean $\mathbb{E}^3$; global inertial frames exist (first law as existence axiom). | historical-artifact | `[ESTABLISHED false fundamentally]` GR replaces flat space and global inertial frames with dynamical curved spacetime; only *local* inertial frames exist. The existence axiom also smuggles in the unresolved origin of inertia (Mach's principle, `[OPEN]`). |
| Position and momentum have simultaneously sharp, arbitrarily precise real values independent of measurement. | historical-artifact | `[ESTABLISHED false fundamentally]` Forbidden by the uncertainty principle and noncommutativity; the phase-space point is the coarse-grained $S \gg \hbar$ idealization where $\hbar$-cells are negligible. |

| | | |
|---|---|---|
| The inertial mass tensor $g_{ij}$ is positive-definite and $L$ is regular (nondegenerate velocity Hessian). | conventional-choice | [INFERENCE] Regularity makes the Legendre transform invertible and is assumed for convenience; it holds for ordinary mechanical systems but is deliberately violated by gauge fields, reparametrization-invariant and relativistic point-particle theories (the "singular" Lagrangians of Dirac theory). |
| Time-reversal invariance: if $q(t)$ solves, so does $q(-t)$ (with $p \to -p$); no microscopic arrow of time. | likely-fundamental | [ESTABLISHED for standard $L$; CONTESTED as universal] Holds for $L$ quadratic in velocities with velocity-independent $V$; broken by magnetic ( $\mathbf{B} \to -\mathbf{B}$) and dissipative forces and, fundamentally, by CP-violation. Fundamental within idealized conservative mechanics; false as a universal law. |

See *Chapter 20 (p. 458)* for the cross-domain ledger and status taxonomy.

## Domain of validity

Classical mechanics is accurate in the joint regime: (1) $v \ll c$ (else special relativity); (2) characteristic action $S \gg \hbar$, equivalently de Broglie wavelength $\lambda = h/p$ small against system length scales (else quantum mechanics); (3) weak, slowly varying gravity, $GM/(rc^2) \ll 1$, with negligible spacetime curvature over the system (else general relativity); (4) fixed, not-too-large particle number — Hamiltonian mechanics *underlies* statistical mechanics, but the single-trajectory picture becomes useless for $\sim 10^{23}$ particles. Within its domain the framework is verified to extraordinary precision: celestial mechanics predicts eclipses and spacecraft trajectories to remarkable accuracy, and the anomalous perihelion precession of Mercury ($43''$/century) was the first clean empirical boundary where Newtonian gravity fails and GR is required [ESTABLISHED].

The classical limit is subtle and is *not* simply "$\hbar \to 0$" ($\hbar$ is a fixed constant): it is the regime of large action in units of $\hbar$, where decoherence destroys interference and Ehrenfest's theorem ( $d\langle x\rangle/dt = \langle p\rangle/m$, $d\langle p\rangle/dt = -\langle\partial_x V\rangle$) keeps wavepacket centroids on classical trajectories. The last condition fails for chaotic systems beyond the Ehrenfest time $t_E \sim \lambda^{-1}\ln(S/\hbar)$ — a known and unresolved tension `[OPEN]`. See *Chapter 16 (p. 308)*.

## Where it breaks down

- **High velocity,** $v \to c$ `[ESTABLISHED]`: Galilean kinematics and absolute time fail; $\mathbf{p} = m\mathbf{v}$, the energy–momentum relation, and velocity addition are wrong. Newtonian mechanics is the $c \to \infty$ contraction of the Poincaré-invariant theory.
- **Small action,** $S \sim \hbar$ `[ESTABLISHED]`: definite trajectories and simultaneously sharp $(q, p)$ cease to exist; uncertainty, interference, tunneling, discrete spectra, and entanglement appear. The phase-space point is replaced by a Hilbert-space state or a (possibly negative) Wigner quasi-probability.
- **Strong gravity / cosmological scales** `[ESTABLISHED]`: instantaneous $1/r^2$ action-at-a-distance violates causality and mispredicts light bending, perihelion precession, gravitational time dilation, and gravitational waves; replaced by general relativity. See *Chapter 10 (p. 121)*.
- **Non-Lipschitz force laws** `[OPEN/CONTESTED]`: Norton's dome and $\ddot{x} = x^{1/2}$ admit multiple solutions from identical initial data — determinism is a property of *sufficiently regular* force laws, not a theorem. Whether such models are physically admissible is debated.
- **Finite-time singularities** `[ESTABLISHED]`: in the Newtonian $N$ -body problem, solutions can blow up in finite time. Xia (1992) proved non-collision singularities exist for $N \geq 5$ (particles driven to infinity in finite time), so the flow is incomplete and global determinism fails for idealized point masses (the Painlevé-conjecture context).
- **Deterministic chaos** `[ESTABLISHED]`: not a failure of the equations but of predictability — positive Lyapunov exponents

preclude long-term forecasting from finite-precision data. The three-body problem has no general closed-form solution.

- **Nonholonomic and dissipative systems** `[ESTABLISHED structural limitation]`: rolling constraints obey Lagrange–d'Alembert, not the Euler–Lagrange equations of any action; dissipation breaks energy conservation, time-reversal, and Liouville volume, requiring contact geometry or non-Hamiltonian extensions.
- **Singular (gauge) Lagrangians** `[ESTABLISHED]`: a degenerate velocity Hessian makes the Legendre transform non-invertible; the naive Hamiltonian picture must be replaced by Dirac–Bergmann constraint analysis.
- **Continuum / infinite degrees of freedom** `[INFERENCE/OPEN]`: field theory extends mechanics to infinitely many DOF; finite-dimensional theorems (Liouville–Arnold, KAM) do not transfer cleanly, and global regularity of 3D incompressible Euler/Navier–Stokes is a Clay Millennium `[OPEN]` problem.

See *Chapter 14 (p. 191)* for the distinction between true inconsistencies and domain-of-validity mismatches.

## Open problems (internal)

- **Painlevé's conjecture / $N$-body singularities** `[OPEN]`: existence of non-collision singularities is proven (Xia, $N \geq 5$, 1992; Gerver), but a complete classification, the $N = 4$ case, and whether singular initial data has measure zero remain unresolved. Genuinely internal to classical mechanics.
- **Arnold diffusion** `[OPEN/partially resolved]`: for $n \geq 3$, KAM tori do not separate phase space, so trajectories can drift across resonance webs. Existence is established in examples and (via Mather, Cheng–Yan, and others) in some generic settings, but a complete theory of its prevalence and rate is missing.
- **Foundations of the classical limit and quantum chaos** `[OPEN/CONTESTED]`: how definite trajectories and the macroscopic absence of superpositions emerge — decoherence and the

measurement problem — straddles classical and quantum; the $\hbar \to 0$ limit is singular and nonuniform, and quantum chaos lacks a complete theory.

- **Regularity of classical field equations** `[OPEN]`: existence/uniqueness/regularity for 3D incompressible Euler and Navier–Stokes (a Clay Millennium problem) — inherited from classical dynamical principles.
- **Origin of inertia / Mach's principle** `[OPEN/CONTESTED]`: why do inertial frames coincide with the rest frame of distant matter? GR's frame dragging is only partially Machian. A question mechanics raises but cannot answer internally.
- **Reversibility vs the thermodynamic arrow** `[OPEN/CONTESTED]`: reconciling exact T-symmetry with observed irreversibility (Loschmidt). Coarse-graining, the $H$-theorem, and the past hypothesis are accepted ingredients, but a complete first-principles derivation remains debated.
- **Physical indeterminism in classical mechanics** `[CONTESTED]`: whether Norton's dome and supertasks reveal genuine indeterminism or merely unphysical idealizations.

These feed *Chapter 15 (p. 233)* and candidate resolutions in *Chapter 19 (p. 392)*.

## Connections to other frameworks

- ***Quantum mechanics* (Chapter 7 (p. 71))**: classical mechanics is the $\hbar \to 0$/large-action limit; QM is recovered by canonical or deformation quantization. The symplectic/Poisson structure is the part that survives. *Clash:* noncommutativity, indeterminacy, and the measurement problem have no classical counterpart, and the limit is singular for chaotic systems.
- **Special relativity** (within *Chapter 10 (p. 121)* and *Chapter 9 (p. 106)*): Galilean mechanics is the $c \to \infty$ limit of Poincaré-invariant dynamics; the free-particle $L = -mc^2\sqrt{1 - v^2/c^2}$ reduces to $\mathrm{const} + \frac{1}{2}mv^2$. *Clash:* absolute time and instantaneous forces are incompatible with a finite invariant speed.

- ***General relativity* (Chapter 10 (p. 121))**: Newtonian gravity is the weak-field, slow-motion limit (geodesic motion in $g_{00} \approx -(1 + 2\Phi/c^2)$); the least-action principle generalizes to spacetime geodesics. *Clash:* global inertial frames and action-at-a-distance gravity are replaced by dynamical curvature; mechanics cannot explain the equivalence of inertial and gravitational mass that GR postulates.
- ***Statistical mechanics* (Chapter 6 (p. 56)) and *Chapter 5 (p. 40)***: Hamiltonian flow plus Liouville's theorem is the microscopic foundation; the microcanonical measure is the Liouville measure on energy shells. *Tension:* deriving irreversibility from reversible, volume-preserving mechanics (Loschmidt, Poincaré recurrence) requires extra ingredients (mixing/ergodicity, special initial conditions).
- ***Mathematics* (Chapter 13 (p. 172)) — symplectic/differential geometry and dynamical systems**: Hamiltonian mechanics *is* dynamics on symplectic manifolds; Darboux's theorem, moment maps, Marsden–Weinstein reduction, KAM, Liouville–Arnold, Poincaré recurrence, and ergodic theory live here. Symplectic topology (the Arnold conjecture) grew partly from these mechanical questions.
- **Classical field theory and gauge theory** (see *Chapter 8 (p. 89)*): Lagrangian/Hamiltonian mechanics extends from finite to infinite DOF; Noether yields the stress–energy tensor and conserved currents; Dirac's constraint analysis is the classical foundation of gauge symmetry. Electromagnetism enters via minimal coupling.
- ***Quantum field theory* (Chapter 8 (p. 89))**: the path integral $e^{iS/\hbar}$ generalizes Hamilton's principle from particle paths to field histories, with the classical equations as the saddle point — the action principle being the organizing structure shared across all of physics. See *Chapter 18 (p. 349)*.

Confirmed boundary results and limits are logged in *Chapter 21 (p. 497)*; terminology in the Glossary (p. 747).

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## References

Full bibliographic entries and shared citation keys are maintained in the Bibliography (p. 776).

CHAPTER 5

# Thermodynamics

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Thermodynamics is the macroscopic theory of energy, heat, work, and entropy for systems at or near equilibrium, together with its statistical-mechanical foundations and its modern extensions into fluctuation theorems, information thermodynamics, and gravitational (black-hole) thermodynamics. It is the discipline that converts a small set of phenomenological laws into exact, substance-independent constraints on what physical processes are possible.

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## Scope

This page covers the macroscopic theory of energy, heat, work, and entropy for systems at or near equilibrium: the zeroth through third laws; state functions and the Legendre-transform structure relating internal energy to the free energies; the Carnot bound and reversibility/irreversibility; the second law as the thermodynamic arrow of time and its statistical reinterpretation; the far-from-equilibrium fluctuation theorems (Jarzynski, Crooks) that sharpen the second-law *inequality* into an *equality*; Landauer's principle tying logical irreversibility to a minimum dissipation; the Bekenstein–Hawking entropy and Hawking temperature with the generalized second law; and the status of negative and bounded-spectrum temperatures.

It does **not** derive the full microscopic ensemble theory — that is the job of *statistical mechanics* (Chapter 6 (p. 56)) — except where the micro–macro link is load-bearing. Kinetic theory and transport (the Boltzmann equation, Green–Kubo relations) are treated only at the interface. Detailed materials-science and chemical-engineering applications are out of scope.

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## Core formalism

### 1. State space and the fundamental relation

`[ESTABLISHED]` A simple one-component system is described by an *extensive* triple $(U, V, N)$ — internal energy, volume, particle number. Callen's postulatory formulation asserts the existence of an **entropy** $S = S(U, V, N)$, the *fundamental relation*, which is (i) single-valued, continuous, and differentiable; (ii) first-order homogeneous (extensive), $S(\lambda U, \lambda V, \lambda N) = \lambda S(U, V, N)$; (iii) monotonically increasing in $U$; and (iv) maximized over unconstrained internal variables at equilibrium. Everything thermodynamic about the system is encoded in this one function. Equivalently, one works in the energy representation $U = U(S, V, N)$, which is convex in its arguments.

The differential form is the **Gibbs relation**,

$$dU = T\,dS - p\,dV + \mu\,dN,$$

which simultaneously *defines* the conjugate intensive variables

$$T = \left(\frac{\partial U}{\partial S}\right)_{V,N}, \qquad p = -\left(\frac{\partial U}{\partial V}\right)_{S,N}, \qquad \mu = \left(\frac{\partial U}{\partial N}\right)_{S,V}.$$

In the entropy representation this reads $dS = \frac{1}{T}\,dU + \frac{p}{T}\,dV - \frac{\mu}{T}\,dN$, and the temperature appears in its most foundational guise:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{V,N}.$$

`[ESTABLISHED]` Because $S$ is first-order homogeneous, Euler's theorem gives the **Euler relation**$U = TS - pV + \mu N$; differentiating it against the Gibbs relation yields the **Gibbs–Duhem relation**

$$S\,dT - V\,dp + N\,d\mu = 0,$$

which expresses that the intensive variables $(T, p, \mu)$ are not independent — fixing two constrains the third.

## 2. The laws as axioms

- **Zeroth law** `[ESTABLISHED]`. Thermal equilibrium is an equivalence relation; its transitivity ("if $A \sim C$ and $B \sim C$ then $A \sim B$") guarantees a consistent empirical temperature labeling the equilibrium classes. This is logically *prior* to the definition of $T$ as a derivative.
- **First law** `[ESTABLISHED]`. $dU = \delta Q + \delta W$ (energy conservation including heat). $U$ is an exact differential (state function); $\delta Q$ and $\delta W$ are inexact (path-dependent). For quasi-static processes $\delta Q_{\rm rev} = T\,dS$ and $\delta W_{\rm rev} = -p\,dV$.
- **Second law** `[ESTABLISHED]`, with equivalent statements. *Clausius*: no cyclic process whose sole effect is heat transfer from a colder to a hotter body. *Kelvin–Planck*: no cyclic process whose sole effect is the complete conversion of heat into work. *Entropy form (Clausius–Carathéodory)*: there exists a state function $S$ with $dS \geq \delta Q/T$ (equality iff reversible), and for an isolated system $\Delta S \geq 0$. For a cycle, the **Clausius inequality** $\oint \delta Q/T \leq 0$.
- **Third law** `[ESTABLISHED, with caveats]`. Nernst's heat theorem: entropy *changes* of isothermal processes vanish as $T \to 0$. The stronger Planck statement fixes $S \to S_0$ (conventionally 0) as $T \to 0$ for a perfect crystal. The *unattainability* form: no finite sequence of operations reaches $T = 0$. The caveats (residual entropy, glasses) are domain mismatches discussed below.

## 3. Thermodynamic potentials and the Legendre structure

`[ESTABLISHED]` The free energies are **Legendre transforms** of $U(S, V, N)$ that trade an extensive variable for its conjugate intensive partner, making the natural control variables the ones an experimenter actually fixes:

$$F = U - TS \ \ (T,V,N), \qquad H = U + pV \ \ (S,p,N),$$

$$G = U - TS + pV \ \ (T,p,N), \qquad \Omega = U - TS - \mu N \ \ (T,V,\mu).$$

Their differentials,

$$dF = -S\,dT - p\,dV + \mu\,dN, \quad dG = -S\,dT + V\,dp + \mu\,dN, \quad d\Omega = -S\,dT - p\,dV - N\,d\mu,$$

yield the **Maxwell relations** by equality of mixed second partials — e.g. from $F$, $(\partial S/\partial V)_T = (\partial p/\partial T)_V$; from $G$, $(\partial S/\partial p)_T = -(\partial V/\partial T)_p$. These exactness constraints relate hard-to-measure entropy derivatives to mechanical equation-of-state data.

`[ESTABLISHED]` **Extremum principles** follow from $\Delta S_{\text{univ}} \geq 0$ applied to system-plus-reservoir: at fixed $(T,V,N)$, $F$ is minimized; at fixed $(T,p,N)$, $G$ is minimized. The Legendre structure also encodes **stability**: convexity of $U$ in extensive variables (equivalently concavity of $S$) forces $C_V \geq 0$ and isothermal compressibility $\kappa_T \geq 0$. Loss of convexity signals a phase transition, handled by the Maxwell equal-area construction.

The microscopic bridge to *statistical mechanics* (Chapter 6 (p. 56)): $F = -k_B T \ln Z$ with the canonical partition function $Z = \sum_i e^{-E_i/k_B T}$, $\Omega = -k_B T \ln \Xi$ (grand canonical), and the Gibbs entropy

$$S = -k_B \sum_i p_i \ln p_i,$$

which reduces to Boltzmann's $S = k_B \ln W$ for an equiprobable (microcanonical) ensemble.

## 4. The Carnot bound and engines

`[ESTABLISHED]` A reversible engine operating between reservoirs at $T_h > T_c$ attains the maximum possible efficiency,

$$\eta_{\text{Carnot}} = 1 - \frac{T_c}{T_h},$$

and *no* engine between the same two reservoirs can exceed it (Carnot's theorem, a direct corollary of the second law). This relation *defines* the **thermodynamic (absolute) temperature scale** via $T_c/T_h \equiv Q_c/Q_h$ for reversible operation, independent of working substance. Reversibility demands quasi-static operation and hence vanishing power. Finite-power optimization for endoreversible engines yields the Curzon–Ahlborn estimate $\eta_{CA} = 1 - \sqrt{T_c/T_h}$ [INFERENCE — a useful benchmark for a particular class of models, not a universal law].

## 5. Arrow of time and the statistical reframing

`[ESTABLISHED]` The microscopic laws — Hamiltonian mechanics, the Schrödinger/von Neumann equation — are time-reversal invariant (modulo weak CP-violation, irrelevant here), yet macroscopic entropy increase selects a direction of time. Boltzmann's $H$-theorem shows that under the *Stosszahlansatz* (molecular-chaos assumption) the kinetic-theory functional obeys $dH/dt \leq 0$, so $S_B = -k_B H$ increases. [INFERENCE; interpretation `[CONTESTED]`] Reconciling microscopic reversibility with macroscopic irreversibility rests on (a) a low-entropy initial condition (the "Past Hypothesis") and (b) the overwhelming phase-space volume of high-entropy macrostates. The arrow is thus *statistical*, not mechanical — a point developed further on *Chapter 11 (p. 139)*.

## 6. Fluctuation theorems — the second law as an equality

`[ESTABLISHED]` For a system driven by an external protocol $\lambda(t)$ from an initial canonical equilibrium, the **Jarzynski equality** holds *exactly*, arbitrarily far from equilibrium:

$$\left\langle e^{-\beta W} \right\rangle = e^{-\beta\, \Delta F}, \qquad \beta = 1/k_B T,$$

where $W$ is the stochastic work over a single realization and $\Delta F$ the equilibrium free-energy difference between endpoints. By Jensen's inequality $\langle e^{-\beta W} \rangle \geq e^{-\beta \langle W \rangle}$, this *implies* the second law as an average, $\langle W \rangle \geq \Delta F$, while the equality captures the rare trajectories that transiently "violate" it. The finer **Crooks fluctuation theorem**,

$$\frac{P_F(+W)}{P_R(-W)} = e^{\beta(W-\Delta F)},$$

relates the forward and time-reversed work distributions; Jarzynski is its integral consequence. The general **detailed fluctuation theorem** for total entropy production $\Sigma$ over time $\tau$ reads $P(+\Sigma)/P(-\Sigma) = e^{\Sigma/k_B}$, with integral form $\langle e^{-\Sigma/k_B}\rangle = 1$ and hence $\langle\Sigma\rangle \geq 0$. `[ESTABLISHED]` These relations have been verified experimentally (single-molecule pulling, colloidal beads in optical traps, electronic circuits). Stochastic thermodynamics extends $\delta Q$, $W$, and $S$ to individual trajectories via Langevin and master-equation dynamics with local detailed balance.

## 7. Landauer's principle (information thermodynamics)

[ESTABLISHED as a theorem within the stochastic/statistical framework; `[INFERENCE]` on the universality of its physical inevitability] Erasing one bit of information in contact with a bath at temperature $T$ dissipates at least

$$W_{\text{erase}} \geq k_B T \ln 2,$$

equivalently $\Delta S_{\text{env}} \geq k_B \ln 2$ per bit. It is *logical irreversibility* — a 2-to-1 logical map collapsing accessible phase-space volume by a factor of two — that carries the cost. This resolves the Maxwell's-demon paradox (Szilard, Landauer, Bennett): the demon's measurement record must eventually be reset, and that erasure pays the entropy bill. Reversible computation in principle costs nothing. The principle has been confirmed in colloidal and nanomagnetic single-bit experiments. The deep structural identity with Shannon information is developed on *information theory* (Chapter 12 (p. 155)).

## 8. Black-hole thermodynamics

[ESTABLISHED within semiclassical gravity (QFT on curved spacetime); the underlying microstate counting is `[OPEN]`] A stationary black hole obeys four laws isomorphic to the laws of thermodynamics (Bardeen–Carter–Hawking):

- **0th**: surface gravity $\kappa$ is constant over the horizon (analog of uniform $T$).
- **1st**: $dM = \dfrac{\kappa}{8\pi G}\, dA + \Omega_H\, dJ + \Phi\, dQ$.
- **2nd**: classically $dA \geq 0$ (Hawking's area theorem).

Bekenstein conjectured, and Hawking's derivation of thermal emission confirmed, the identifications

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar} = \frac{A}{4\ell_P^2}\, k_B, \qquad T_H = \frac{\hbar c^3}{8\pi G k_B M} = \frac{\hbar\kappa}{2\pi k_B c},$$

the latter Hawking temperature quoted for Schwarzschild. The **generalized second law (GSL)** posits that $S_{\text{gen}} = S_{\text{BH}} + S_{\text{out}}$ is non-decreasing, $dS_{\text{gen}} \geq 0$, rescuing the ordinary second law when matter falls through the horizon. That entropy scales with horizon *area* rather than enclosed *volume* is the seed of the **holographic principle** and the covariant (Bousso) entropy bound $S \leq A/4\ell_P^2$. See *general relativity* (Chapter 10 (p. 121)) and *quantum field theory* (Chapter 8 (p. 89)).

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# Foundational assumptions

| ASSUMPTION | STATUS | JUSTIFICATION |
|---|---|---|
| Equilibrium states exist and are fully specified by a small finite set of macroscopic state variables (a well-defined thermodynamic limit). | likely-fundamental | `[ESTABLISHED]` for short-range-interacting matter where extensivity and the thermodynamic limit provably hold; it is the precondition for the whole formalism. But it is a domain restriction, not a universal truth — it fails for long-range/gravitating systems, small systems where fluctuations dominate, and genuinely nonequilibrium matter. The *existence* of equilibrium is fundamental to this theory's applicability; the *universality* of that existence is a domain boundary, not a law of nature. |

| | | |
|---|---|---|
| Entropy is an extensive, additive state function (first-order homogeneous in extensive variables). | conventional-choice | `[ESTABLISHED]` for short-range systems; `[CONTESTED]`/limited in general. Additivity is what makes the Euler and Gibbs–Duhem relations work and is empirically excellent for normal matter, but it is a *consequence of short-range interactions*, not an axiom. Gravitating systems are non-extensive ($S_{\mathrm{BH}} \propto A$, not volume). Tsallis/Rényi entropies relax additivity; their physical *necessity* is contested. |
| Temperature is well-defined and positive (zeroth-law ordering; energy unbounded above). | conventional-choice | Zeroth-law transitivity is `[ESTABLISHED]` and genuinely structural. Positivity of $T$ is *not*: via $T = (\partial U/\partial S)^{-1}$... i.e. $1/T = (\partial S/\partial U)$, bounded-spectrum systems (nuclear spins, cold-atom lattices) can have $\partial S/\partial U < 0$, i.e. negative absolute temperature — experimentally realized and "hotter" than $+\infty$. "$T > 0$" is a feature of unbounded-spectrum systems, not an axiom. |
| The second law (entropy non-decrease for isolated systems) is an exact, exceptionless law. | historical-artifact | `[CONTESTED]`/refined. Phenomenology treats it as absolute, but statistical mechanics and the fluctuation theorems show it is *statistical*: $P(+\Sigma)/P(-\Sigma) = e^{\Sigma/k_B}$ makes transient decreases exponentially rare but nonzero, and Jarzynski/Crooks promote the inequality to an equality over trajectory ensembles. What is fundamental is the *probabilistic* statement; the deterministic inequality is its overwhelmingly-likely large-$N$ shadow. |
| A low-entropy past boundary condition (the "Past Hypothesis") underlies the thermodynamic arrow of time. | likely-fundamental | `[INFERENCE]`/`[OPEN]`. Time-symmetric microdynamics cannot yield entropy increase alone — a special initial condition is required. *That* a low-entropy past is needed is widely accepted `[INFERENCE]`; *why* the early universe had such low (especially gravitational) entropy is genuinely `[OPEN]` and exports to *Chapter 11 (p. 139)*. |

| | | |
|---|---|---|
| The third law: $S \to S_0$ (conventionally $0$) as $T \to 0$; absolute zero is unattainable. | fundamental | [ESTABLISHED] for systems with a non-degenerate (or sub-exponentially degenerate) ground state — a consequence of quantum statistics (discrete spectrum, unique low-degeneracy ground state), with unattainability following from vanishing heat capacities. The *zero* reference value is a conventional normalization (Planck); the vanishing of entropy *differences* and unattainability are genuine quantum facts. |
| Equal a priori probability of accessible microstates (microcanonical postulate) bridges micro and macro. | likely-fundamental | [INFERENCE]/[OPEN]. The load-bearing postulate linking mechanics to thermodynamics. Justified pragmatically and partially by ergodic theory, dynamical typicality, and the Eigenstate Thermalization Hypothesis (ETH), but a fully general first-principles derivation of thermalization for realistic Hamiltonians remains [OPEN] (many-body localization provides non-thermalizing counterexamples). More than convenience, less than a closed theorem. |
| Heat and work are cleanly separable, and quasi-static/reversible processes exist. | conventional-choice | [ESTABLISHED] as an idealization. Reversible processes are infinitely slow (zero-power) limits that never occur exactly; the heat/work split is unambiguous only for well-defined control protocols, and at the trajectory level (stochastic thermodynamics) it requires careful definition. Convenient and exact-in-limit, but not a physical state of affairs — it *bounds* real processes (Carnot, Clausius). |
| Boltzmann's constant $k_B$ is a fundamental energy–temperature conversion factor. | conventional-choice | [ESTABLISHED]. Since the 2019 SI redefinition, $k_B$ is a *fixed defined constant*, making temperature literally an energy unit ($k_BT$); entropy could be made dimensionless ($S/k_B$). A units convention — though the existence of a fundamental energy-per-degree-of-freedom scale is physical. See *constants and scales* (Chapter 16 (p. 308)). |

## Domain of validity

[ESTABLISHED] Classical equilibrium thermodynamics is quantitatively exact for systems satisfying all of:

1. **Macroscopic size** — $N \sim N_A$ degrees of freedom, so relative fluctuations $\sim N^{-1/2}$ are negligible.
2. **Short-range interactions** — so energy and entropy are extensive and the thermodynamic limit exists.
3. **Equilibrium or near-equilibrium** — state variables are well-defined (local equilibrium for irreversible thermodynamics à la Onsager and de Groot–Mazur).
4. **Separation of timescales** — observation times long compared to microscopic relaxation but short compared to any slow drift.

Within this regime, equations of state, Maxwell relations, the Carnot bound, and phase-coexistence conditions are exact constraints.

The framework *extends*, with modifications, into: small/mesoscopic systems and single molecules, where the laws become statistical and the fluctuation theorems carry the exact content; nonequilibrium steady states via stochastic thermodynamics and linear response (Onsager reciprocity, Green–Kubo) near equilibrium; and quantum regimes (quantum thermodynamics), where work becomes a two-point-measurement observable and coherence/entanglement modify the resource accounting. It is *reinterpreted* at the foundations by *statistical mechanics* (Chapter 6 (p. 56)): the deterministic laws are the $N \to \infty$, fluctuation-suppressed shadows of probabilistic ones. Boltzmann's constant sets the scale at which fluctuations matter; once $k_B T$ is comparable to characteristic energies and $N$ is small, thermodynamic certainties dissolve into distributions.

## Where it breaks down

- **Small systems and single molecules** `[ESTABLISHED — domain boundary, not inconsistency]`. With $N^{-1/2}$ fluctuations no longer negligible, the second-law inequality is routinely and transiently violated along individual trajectories. "Entropy never decreases" is false at this scale; the fluctuation theorems restore an exact statement only at the level of trajectory *ensembles*.
- **Long-range / self-gravitating systems** (stars, galaxies, black holes) `[ESTABLISHED]`. Energy and entropy are non-additive and non-extensive; the thermodynamic limit fails. Ensembles become inequivalent (microcanonical $\neq$ canonical), heat capacities can be *negative*, and black-hole entropy scales as area, not volume. Euler and Gibbs–Duhem do not apply; this forces holographic ideas and is only partially understood. See *general relativity* (Chapter 10 (p. 121)).
- **Black-hole evaporation and the information paradox** `[OPEN]`/`[CONTESTED]`. Hawking's semiclassical calculation yields *exactly thermal* radiation, suggesting a pure state evolves to a mixed one — apparent loss of unitarity and of any microstate count behind $S_{\mathrm{BH}}$. Whether information is preserved (now widely favored, supported by AdS/CFT, the Page curve, and island/replica-wormhole computations `[INFERENCE]`) or lost lacks an agreed bulk mechanism for our (asymptotically flat) universe. The clash is between thermodynamics + quantum unitarity and semiclassical gravity. Tracked in *gaps and contradictions* (Chapter 14 (p. 191)).
- **Microscopic origin of the arrow of time** `[OPEN]`. Time-reversal-invariant microdynamics cannot alone produce monotone entropy increase; the $H$-theorem smuggles in the time-asymmetric Stosszahlansatz, and the global arrow requires the cosmological Past Hypothesis. A genuine unsolved foundational gap, not an inconsistency — it exports to *Chapter 11 (p. 139)*.

- **Third law with residual entropy / glasses** `[ESTABLISHED — edge cases]`. For macroscopic ground-state degeneracy (ice "residual entropy", spin ices) or non-ergodic glasses frozen out of equilibrium, $S$ does not approach zero and equilibrium is operationally unreachable. The Nernst statement holds for true equilibrium with a unique ground state; apparent violations are degeneracy or broken ergodicity — domain mismatches, not failures of quantum statistics.
- **Negative absolute temperature** [ESTABLISHED; interpretation `[CONTESTED]`]. Bounded-spectrum systems reach population-inverted states with $\partial S/\partial U < 0$, hence $T < 0$, hotter than any positive temperature. Fully consistent thermodynamics, but it breaks the naive "$T > 0$"/unbounded-energy presumption. A live dispute (Dunkel–Hilbert vs. proponents of the Boltzmann surface entropy) concerns whether the Gibbs (volume) or Boltzmann (surface) entropy is the correct definition, and thus whether negative temperatures are admissible at all.
- **Maxwell's demon / measurement-feedback loops** `[ESTABLISHED resolution]`. A naive demon appears to violate the second law by sorting molecules; the resolution (Szilard, Landauer, Bennett) charges $k_B T \ln 2$ per erased bit. Not a breakdown but a demonstration that the accounting must include the information-bearing degrees of freedom — the law is sharpened, not refuted.
- **Far-from-equilibrium, turbulent, or strongly driven systems** `[OPEN]`. With steep gradients or strong driving, local equilibrium fails, temperature/entropy may not be definable, and there is no complete predictive macroscopic theory analogous to equilibrium thermodynamics. Extended irreversible thermodynamics, MaxEnt, and large-deviation theory are partial frameworks; a general entropy-production extremum principle (minimum or maximum) is *not* an established law.

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## Open problems (internal)

- **First-principles thermalization and the equal-a-priori-probability postulate.** [OPEN] Ergodic theory, dynamical typicality, and ETH are strong partial results [INFERENCE], but a general theorem for generic interacting Hamiltonians does not exist, and many-body-localized systems are explicit non-thermalizing counterexamples. See *open problems* (Chapter 15 (p. 233)).
- **Origin and magnitude of the cosmological low-entropy initial condition.** [OPEN] A low-entropy past is broadly accepted as logically necessary [INFERENCE], but *why* the early universe had such low gravitational entropy has no accepted explanation (Penrose's Weyl-curvature hypothesis is one [SPECULATIVE] proposal); it is entangled with quantum gravity.
- **Microstate identification behind $S_{\mathrm{BH}}$ for realistic black holes.** [OPEN] String theory reproduces $S = A/4$ by counting D-brane microstates for certain supersymmetric/extremal black holes [ESTABLISHED within that framework — Strominger-Vafa], and AdS/CFT gives a holographic account [INFERENCE], but a universally accepted derivation for generic astrophysical (e.g. Schwarzschild) black holes is lacking.
- **An explicit, agreed unitarity-preserving mechanism for black-hole evaporation in asymptotically flat spacetime.** [OPEN]/[CONTESTED] Island/replica-wormhole computations reproduce the unitary Page curve [INFERENCE, within semiclassical-gravity-plus-replica assumptions], strongly favoring information preservation, but the bulk physical mechanism (firewalls? complementarity? new physics?) is unsettled.
- **A predictive macroscopic theory for far-from-equilibrium systems.** [OPEN] Linear response and large-deviation theory work near equilibrium; far from it, proposed variational principles are not established as laws.

- **The correct entropy/temperature for small and bounded-spectrum systems** (Boltzmann surface vs. Gibbs volume entropy). [CONTESTED] The dispute bears on the precise meaning of $T$ and $S$ away from the thermodynamic limit.
- **Foundations of quantum thermodynamics.** [OPEN] Resource-theoretic and two-point-measurement frameworks are internally consistent, but which definition of "work" is physically privileged, and how coherence functions as a thermodynamic resource, are active research questions. See *quantum mechanics* (Chapter 7 (p. 71)).

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## Connections to other frameworks

- ***Statistical mechanics* (Chapter 6 (p. 56))** — the foundational substrate. It derives the state functions ($S = k_B \ln W$, $F = -k_B T \ln Z$) and reinterprets the second law as probabilistic. Thermodynamics is the $N \to \infty$, fluctuation-suppressed limit; the two agree wherever they overlap, with tension only over the *interpretation* of irreversibility and over small-$N$ regimes.
- ***Quantum mechanics* (Chapter 7 (p. 71)) / *quantum field theory* (Chapter 8 (p. 89))** — discrete spectra underwrite the third law and the microstate count; the von Neumann entropy $S = -k_B \operatorname{Tr}(\rho \ln \rho)$ generalizes the Gibbs entropy. Quantum thermodynamics adds coherence and entanglement as resources. Clash points: the definition of work, measurement back-action, and whether unitarity is compatible with thermal (Hawking) radiation.
- ***General relativity* (Chapter 10 (p. 121))** — black-hole mechanics maps exactly onto the four laws, with $S = A/4$ and Hawking temperature; this both validates thermodynamics gravitationally and exports it (GSL, holographic and Bousso bounds, Jacobson's [SPECULATIVE] derivation of the Einstein equations as an equation of state). The deepest clash — the

information paradox — sits at the GR/QM/thermodynamics triple point.

- ***Cosmology* (Chapter 11 (p. 139))** — supplies the low-entropy boundary condition (Past Hypothesis) that thermodynamics presupposes but cannot itself explain; the global entropy budget is dominated by gravitational/black-hole entropy, and the heat-death endpoint is a thermodynamic extrapolation `[INFERENCE]`.
- ***Information theory* (Chapter 12 (p. 155))** — the Shannon entropy $H = -\sum p \log p$ is formally identical to the Gibbs entropy up to $k_B$ and the logarithm base; Landauer's principle fixes the physical exchange rate ($k_B T \ln 2$ per bit). Maxwell's demon, Szilard engines, and Sagawa–Ueda feedback thermodynamics unify the two ledgers. A deep structural identity, not a mere analogy.
- ***Classical mechanics* (Chapter 4 (p. 25))** and **kinetic theory** — Hamiltonian dynamics plus the Boltzmann equation, Onsager reciprocal relations, and Green–Kubo formulas extend equilibrium thermodynamics to transport and the near-equilibrium frontier; stochastic thermodynamics carries the laws into the driven mesoscale.
- ***Mathematics* (Chapter 13 (p. 172))** — convex analysis and the Legendre transform supply the rigorous backbone of the potential structure, stability, and phase transitions.

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## Key references


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## References

See the Bibliography (p. 776) for the consolidated, cross-page reference list.

CHAPTER 6

# Statistical Mechanics

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Statistical mechanics is the bridge layer between microscopic, deterministic, time-reversible dynamics (Hamiltonian *Chapter 4 (p. 25)* flow or *Chapter 7 (p. 71)* unitary evolution) and the phenomenological, irreversible laws of *Chapter 5 (p. 40)*. Its central achievement is to derive the bulk behavior of matter built from $\sim 10^{23}$ constituents not by integrating their trajectories — which is intractable and, for chaotic systems, useless beyond microscopic times — but by replacing them with probability distributions over microstates and extracting the overwhelmingly typical macroscopic outcome.

## Scope

The domain partitions into five strata, all of which this page treats:

1. **Equilibrium ensemble theory** — the microcanonical, canonical, and grand-canonical ensembles; partition functions; free energies; and the systematic derivation of thermodynamic relations as derivatives of $\ln Z$.
2. **Fluctuations and linear response** — the fluctuation–dissipation theorem (FDT), Kubo/Green–Kubo formulas, and their modern far-from-equilibrium extensions (Jarzynski, Crooks).
3. **Collective phenomena** — phase transitions, critical exponents, scaling, universality, and Wilson's renormalization group (RG).
4. **Non-equilibrium foundations** — the Boltzmann equation, the $H$-theorem, the origin of macroscopic irreversibility from reversible microdynamics, ergodicity, mixing, and typicality.

5. **Quantum statistical mechanics** — Fermi–Dirac and Bose–Einstein statistics, and the eigenstate thermalization hypothesis (ETH) as the mechanism by which isolated quantum systems thermalize.

The recurring conceptual question that organizes all five is: **why does an ensemble average predict the behavior of a single individual macroscopic system?** `[OPEN]` — see *Open problems*.

# Core formalism

## Phase space, microstates, macrostates

A classical system of $N$ particles in $d$ dimensions has phase space $\Gamma = \mathbb{R}^{2dN}$ with point $x = (q_1, \ldots, q_{dN}, p_1, \ldots, p_{dN})$. The **microstate** is $x$; dynamics is the Hamiltonian flow $\Phi_t$ generated by $H(x)$ through $\dot{q}_i = \partial H/\partial p_i, \quad \dot{p}_i = -\partial H/\partial q_i.$ Liouville's theorem `[ESTABLISHED]` states the flow preserves the symplectic volume $d\Gamma = \prod_i dq_i\, dp_i$, so the phase-space density $\rho(x,t)$ obeys the Liouville equation

$$\partial_t \rho = \{H, \rho\} = -\sum_i \left( \frac{\partial H}{\partial p_i}\frac{\partial \rho}{\partial q_i} - \frac{\partial H}{\partial q_i}\frac{\partial \rho}{\partial p_i} \right), \qquad \frac{d\rho}{dt} =$$

0 along trajectories.

A **macrostate** $M$ is an equivalence class of microstates indistinguishable at the level of a few coarse observables (energy, volume, particle number, magnetization). The combinatorial heart of the subject is that the overwhelming majority of phase-space volume — quantum-mechanically, of Hilbert-space dimension — belongs to a single macrostate, the equilibrium one `[ESTABLISHED]`.

Quantum-mechanically the microstate is a ray $|\psi\rangle \in \mathcal{H}$; the statistical object is the density matrix $\hat{\rho}$, evolving by the von Neumann equation $i\hbar\, \partial_t \hat{\rho} = [\hat{H}, \hat{\rho}]$, with $S_{\mathrm{vN}} = -k_B \operatorname{Tr}(\hat{\rho} \ln \hat{\rho})$.

## The three equilibrium ensembles

**Microcanonical (NVE).** Isolated system, energy fixed in a shell $[E, E + \Delta]$. The postulate of equal *a priori* probabilities gives

$$\rho_{mc}(x) = \frac{1}{\Omega(E)}\,\delta\big(H(x)-E\big), \qquad \Omega(E) = \int_\Gamma d\Gamma\,\delta(H(x)-E),$$

with the Boltzmann entropy $S_B(E) = k_B \ln[\Omega(E)\,\Delta]$ (equivalently $k_B \ln \Sigma(E)$ with $\Sigma$ the phase volume below $E$; the two agree to $O(\ln N)$). Temperature, pressure, and chemical potential follow from $dS = \frac{1}{T}dE + \frac{P}{T}dV - \frac{\mu}{T}dN$ `[ESTABLISHED]`.

**Canonical (NVT).** System in weak thermal contact with a reservoir at temperature $T$, $\beta = 1/k_BT$. Maximizing Gibbs–Shannon entropy at fixed $\langle H\rangle$ — equivalently, tracing out a large short-range-coupled reservoir — yields the Boltzmann–Gibbs distribution

$$\rho_c(x) = \frac{1}{Z}e^{-\beta H(x)}, \qquad Z(\beta,V,N) = \frac{1}{N!\,h^{dN}}\int_\Gamma d\Gamma\,e^{-\beta H(x)} \quad (\text{quantum: } Z = \mathrm{Tr}\,e^{-\beta\hat{H}}).$$

The Helmholtz free energy is $F = -k_BT\ln Z$, with $dF = -S\,dT - P\,dV + \mu\,dN$. All equilibrium thermodynamics follows from derivatives of $\ln Z$ `[ESTABLISHED]`:

$$\langle E\rangle = -\partial_\beta \ln Z, \qquad C_V = \partial_T\langle E\rangle = k_B\beta^2\,\langle(\Delta E)^2\rangle, \qquad S = -\partial_T F.$$

The relation $C_V = k_B\beta^2\langle(\Delta E)^2\rangle$ is the first instance of a fluctuation–response identity: a static susceptibility ($C_V$) equals an equilibrium variance.

**Grand canonical ($\mu VT$).** Energy and particle exchange with a reservoir, fugacity $z = e^{\beta\mu}$:

$$\rho_{gc} = \frac{1}{\Xi}e^{-\beta(H-\mu N)}, \qquad \Xi(\mu,V,T) = \sum_{N=0}^{\infty} z^N Z(N,V,T) = \mathrm{Tr}\,e^{-\beta(\hat{H}-\mu\hat{N})},$$

grand potential $\Phi_G = -k_BT\ln\Xi = -PV$. For ideal quantum gases, $\ln\Xi = \mp\sum_k \ln(1\mp z\,e^{-\beta\epsilon_k})$, giving the Bose–Einstein ($-$) and Fermi–Dirac ($+$) occupations

$$\langle n_k\rangle = \frac{1}{e^{\beta(\epsilon_k-\mu)}\mp 1}.$$

These furnish blackbody radiation, Bose–Einstein condensation, degeneracy pressure, and the electronic theory of metals `[ESTABLISHED]`.

**Ensemble equivalence.** In the thermodynamic limit ($N, V \to \infty$ at fixed $N/V$) the three ensembles yield identical intensive predictions, because energy/number fluctuations are $O(1/\sqrt{N})$ relative to the mean; large-deviation theory makes this a statement about Legendre duality of rate functions `[ESTABLISHED]`. Equivalence *fails* at first-order transitions, for long-range interactions, and in finite systems `[ESTABLISHED]` — see *Where it breaks down*.

## Entropy: Boltzmann versus Gibbs/Shannon

- **Boltzmann entropy** $S_B(M) = k_B \ln |\Gamma_M|$ is a function of the *macrostate* — the log phase-space volume of microstates consistent with $M$. It is defined for an individual system and can increase along a single trajectory as the system migrates to larger macro-cells. This is the entropy of the second law `[INFERENCE – within the typicality program]`.
- **Gibbs/Shannon entropy** $S_G = -k_B \int \rho \ln \rho \, d\Gamma$ is a functional of the *distribution*. Under Liouville evolution $S_G$ is **exactly constant**, because $\rho$ is merely transported `[ESTABLISHED]`. It equals the thermodynamic entropy at equilibrium but cannot, by itself, describe approach to equilibrium without coarse-graining.

The reconciliation: the *fine-grained* Gibbs entropy is conserved, while a *coarse-grained* $\bar{\rho}$ has non-decreasing entropy, matching $S_B$. Jaynes recast equilibrium distributions as maximum-entropy (MaxEnt) distributions consistent with known macroscopic constraints, making statistical mechanics a special case of Bayesian inference `[CONTESTED – whether MaxEnt is physics or epistemology]`. See *Chapter 12 (p. 155)*.

## The Ising model and exact paradigms

The Ising Hamiltonian $H = -J\sum_{\langle ij\rangle} s_i s_j - h\sum_i s_i,\ s_i = \pm 1$, is the canonical interacting model. In $d = 1$ the transfer matrix

$$T = \begin{pmatrix} e^{\beta(J+h)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} \end{pmatrix}, \qquad Z = \mathrm{Tr}\, T^N,$$

has a unique largest eigenvalue for all $T > 0$, so there is no finite-temperature transition [ESTABLISHED]. Onsager's 1944 exact solution of the $d = 2$ zero-field model gives a continuous transition at $\sinh(2\beta_c J) = 1$; Yang's spontaneous magnetization $m = [1 - \sinh^{-4}(2\beta J)]^{1/8}$ yields the exponent $\beta = 1/8$, the first rigorous proof of non-mean-field critical behavior [ESTABLISHED].

## Phase transitions, scaling, universality, RG

Near a continuous transition the correlation length $\xi \sim |t|^{-\nu}$ diverges, $t = (T - T_c)/T_c$. Critical exponents $(\alpha, \beta, \gamma, \delta, \nu, \eta)$ satisfy scaling relations — Rushbrooke $\alpha + 2\beta + \gamma = 2$, Fisher $\gamma = \nu(2 - \eta)$ , and hyperscaling $2 - \alpha = d\nu$ [ESTABLISHED]. The singular free-energy density takes the homogeneous form

$$f_s(t, h) = b^{-d} f_s(b^{y_t} t,\ b^{y_h} h),$$

so exponents are fixed by the two RG eigenvalues $y_t, y_h$. **Universality** [ESTABLISHED]: exponents depend only on spatial dimensionality, order-parameter symmetry, and interaction range — not microscopic detail. **Wilson's renormalization group** explains why [ESTABLISHED]: iterated coarse-graining plus rescaling defines a flow in coupling space; critical points are unstable fixed points, and exponents are eigenvalues of the linearized flow. The $\epsilon = 4 - d$ expansion and momentum-shell RG (1971–72) put this on a calculational footing, with upper critical dimension $d_c = 4$ above which mean-field exponents hold. The RG is the deepest formal bridge to *Chapter 8 (p. 89)*.

## Fluctuation–dissipation and linear response

Equilibrium fluctuations encode the response to small perturbations. The classical FDT relates the response function $\chi(t)$ to the equilibrium autocorrelation $C(t) = \langle A(t)A(0)\rangle$ via $\chi(t) = -\beta\,\theta(t)\,\dot{C}(t)$. In frequency space, the quantum (Kubo) form reads

$$\chi''(\omega) = \frac{1}{2\hbar}\left(1 - e^{-\beta\hbar\omega}\right)\tilde{S}(\omega),$$

with $\tilde{S}(\omega)$ the power spectrum of fluctuations `[ESTABLISHED]`. Special cases: the Einstein relation $D = \mu k_B T$, Nyquist noise $S_V = 4k_B TR$, and Green–Kubo formulas for transport coefficients. Modern far-from-equilibrium extensions — the Jarzynski equality $\langle e^{-\beta W}\rangle = e^{-\beta\Delta F}$ and the Crooks fluctuation theorem — are exact for arbitrarily fast driving given a well-defined initial equilibrium state `[ESTABLISHED]`; by Jensen's inequality they recover $\langle W\rangle \geq \Delta F$, the second law as a corollary.

## Boltzmann equation, $H$-theorem, irreversibility

For a dilute gas the one-particle distribution $f(\mathbf{r}, \mathbf{v}, t)$ obeys

$$\partial_t f + \mathbf{v}\cdot\nabla_{\mathbf{r}} f + \tfrac{\mathbf{F}}{m}\cdot\nabla_{\mathbf{v}} f = \int d\mathbf{v}_1\, d\Omega\, |\mathbf{v} - \mathbf{v}_1|\, \sigma\, (f' f_1' - f f_1).$$

With $H(t) = \int f \ln f\, d\mathbf{v}\, d\mathbf{r}$, Boltzmann's $H$-theorem gives $dH/dt \leq 0$ (so $S = -k_B H$ increases), with equality only at the Maxwell–Boltzmann distribution. The decisive *non-mechanical* input is the **Stosszahlansatz** (molecular chaos): pre-collision velocities are uncorrelated, $f_2(\mathbf{v}, \mathbf{v}_1) = f(\mathbf{v})f(\mathbf{v}_1)$. This breaks time-reversal symmetry by hand `[ESTABLISHED that this is the locus of asymmetry]`. Loschmidt's reversibility paradox (momentum reversal yields $dH/dt > 0$) and Zermelo's recurrence paradox (Poincaré recurrence for bounded systems) prove the theorem cannot follow from mechanics alone. The resolution — Boltzmann's, sharpened by Lanford's 1975 rigorous short-time derivation — is that the $H$-theorem is *statistical/typical*: the overwhelming majority of microstates compatible with a low-entropy macrostate evolve toward higher entropy; recurrence times scale as $\sim e^N$ (astronomically long); and the arrow of time is inherited from a low-entropy initial condition (the "Past Hypothesis"), not from the dynamics [INFERENCE, broad consensus on the structure; the cosmological source is OPEN].

### Quantum thermalization (ETH)

For a closed quantum system, unitary evolution preserves $S_{\mathrm{vN}}$, yet local observables relax to thermal values. The eigenstate thermalization hypothesis (Deutsch 1991, Srednicki 1994) posits that matrix elements of few-body operators in energy eigenstates take the form

$$\langle E_n|\hat{O}|E_m\rangle = O(\bar{E})\,\delta_{nm} + e^{-S(E)/2}\, f_O(\bar{E},\omega)\, R_{nm},$$

with $\bar{E} = (E_n + E_m)/2$, $\omega = E_n - E_m$, smooth $O, f_O$, and $R_{nm}$ effectively random of unit variance. Each individual eigenstate then already encodes the microcanonical prediction, so the long-time average of a local observable is thermal [INFERENCE — strong numerical and random-matrix support, unproven for any realistic interacting Hamiltonian]. ETH provably *fails* for integrable systems (which relax to a Generalized Gibbs Ensemble) and for many-body-localized (MBL) systems. See *Chapter 7 (p. 71)*.

## Foundational assumptions

| ASSUMPTION | STATUS | JUSTIFICATION |
|---|---|---|
| Equal *a priori* probability on the energy shell | conventional-choice | [CONTESTED] It is the natural Liouville-invariant (uniform) measure on the energy surface, which is a strong dynamical justification — so it is not arbitrary. But no theorem derives it from mechanics for realistic systems; it is the unique measure consistent with the symmetries plus a MaxEnt/indifference principle. Any flow-invariant measure is formally admissible, so its privileged status is partly forced (Liouville), partly conventional. Justified pragmatically by predictive success. |

| | | |
|---|---|---|
| Ergodicity (time average = ensemble average) | historical-artifact | [ESTABLISHED that it is neither necessary nor sufficient as usually invoked.] Birkhoff/von Neumann made it rigorous, but KAM shows generic Hamiltonian systems are *not* ergodic (invariant tori survive) — yet statistical mechanics still works for them. Typicality, concentration of measure, and Khinchin's high-dimensional sum-function arguments show the equivalence holds for macroscopic observables at the vast majority of phase points regardless of strict ergodicity. Historically central, largely superseded. |
| Thermodynamic limit ($N, V \to \infty$, $N/V$ fixed) | conventional-choice | `[ESTABLISHED]` An idealizing modeling choice. Genuine non-analyticities (true phase transitions) exist only in the strict limit (Lee–Yang); finite systems show smooth crossovers. The physics is that the rounding scale is negligible for macroscopic $N$. It captures real behavior without asserting infinite systems exist. |
| Molecular chaos (Stosszahlansatz) | fundamental | `[ESTABLISHED as the locus of time-asymmetry]` The crucial physical input that injects irreversibility. It is provably false for *post*-collision velocities (collisions create correlations), and applying it symmetrically would be inconsistent. Lanford proved it holds for typical initial data for short times. Fundamental as the precise point where a typicality/initial-condition assumption enters; not derivable from reversible mechanics. |
| Coarse-graining (entropy relative to a macro-partition) | fundamental | `[CONTESTED in interpretation, structurally fundamental]` Jaynes regards the choice of macrovariables as observer-dependent (entropy subjective); Boltzmann/Lebowitz/Goldstein argue the relevant macrovariables are objectively singled out by scale separation. Either way, irreversibility cannot be *stated* without a description coarser than the microstate. The *need* for coarse-graining is fundamental; the specific partition has a conventional element. |
| Low-entropy initial condition ("Past Hypothesis") | fundamental | `[INFERENCE/OPEN]` Given time-reversible microdynamics, the observed arrow *must* be sourced from a boundary condition, not the laws — an essentially forced inference. *Why* the early universe had such low (gravitational) entropy is genuinely OPEN and belongs to *Chapter 11 (p. 139)*. As an assumption of statistical mechanics it cannot be eliminated. |

| | | |
|---|---|---|
| Gibbs $N!$ factor and cell size $h^{dN}$ | historical-artifact | `[ESTABLISHED]` In purely classical physics, $N!$ (indistinguishability, fixing the Gibbs paradox) and $h$ (phase-cell volume, fixing absolute extensive entropy) are inserted by hand. Quantum mechanics derives both: identical-particle statistics gives $N!$ automatically, and $h =$ Planck's constant sets the true cell size. Ad hoc classically; fundamental once QM is recognized as the correct microtheory. |
| Canonical form $e^{-\beta H}$ as the universal equilibrium law | likely-fundamental | `[ESTABLISHED within its domain; CONTESTED at the margins]` Follows rigorously from reservoir coupling with short-range interactions, from MaxEnt at fixed mean energy, and from ETH quantum-mechanically. Likely-fundamental for short-range, additive systems. Fails for long-range/non-additive systems (self-gravitating, some plasmas); proposed generalizations (Tsallis non-extensive statistics) are of disputed fundamental status. |
| Eigenstate Thermalization Hypothesis | likely-fundamental | `[INFERENCE, strong numerical support]` Unproven for any realistic interacting Hamiltonian, but supported by exact diagonalization and random-matrix arguments, and it explains thermalization without a bath. Likely-fundamental as the quantum mechanism of thermalization; provably fails for integrable (→ GGE) and MBL systems, so its domain is "generic non-integrable, non-localized." |

See *Chapter 20 (p. 458)* for the cross-domain register and *Chapter 2 (p. 8)* for marker definitions.

## Domain of validity

Equilibrium ensemble theory is rigorously controlled and quantitatively predictive for systems with many degrees of freedom ($N \gg 1$, ideally $N \to \infty$ for sharp statements); short-range, additive interactions (so energy and entropy are extensive and ensembles equivalent); systems that have equilibrated (times long compared to microscopic relaxation yet still subject to the relevant constraints); and weak reservoir coupling for the canonical/grand-canonical forms `[ESTABLISHED]`. The RG treatment of criticality is valid where $\xi$ greatly exceeds the lattice spacing. Linear-response/FDT holds for small perturbations about equilibrium; Jarzynski/Crooks extend exact statements arbitrarily far from equilibrium but still require a well-defined initial equilibrium state and Hamiltonian (or detailed-balance Markov) dynamics. The Boltzmann equation is valid in the

Boltzmann–Grad dilute limit (density $\to 0$, $N \to \infty$ at fixed $N\sigma^2$) for times short compared to recurrence. Quantum statistics (Fermi–Dirac, Bose–Einstein, the partition-function trace) apply wherever the microdescription is a many-body quantum Hamiltonian with a well-defined trace — essentially all condensed matter; ETH governs thermalization for generic non-integrable, non-localized closed systems. See *Chapter 16 (p. 308)* for $k_B$, $h$, and the relevant scale separations.

## Where it breaks down

- **Long-range / non-additive interactions** (self-gravitating clusters and galaxies, unscreened Coulomb systems, dipolar systems). Energy ceases to be extensive; the thermodynamic limit is ill-defined absent special (Kac) scaling; microcanonical and canonical ensembles become *inequivalent*. Self-gravitating systems exhibit negative specific heat (which the canonical ensemble cannot represent), the gravothermal catastrophe, and — for the idealized *point-mass* model — no global maximum-entropy equilibrium, the entropy being unbounded above until a short-distance cutoff is imposed (a property of the unregularized UV, distinct from the long-range non-additivity; cf. *GC-17* (Chapter 14 (p. 191))) `[ESTABLISHED]`. A genuine domain-of-validity breakdown, not an inconsistency — see *Chapter 11 (p. 139)* and *Chapter 10 (p. 121)*.
- **Small / nanoscale / single-molecule systems.** $O(1/\sqrt{N})$ fluctuations are no longer negligible; ensemble equivalence fails; the Boltzmann-vs-Gibbs (surface-vs-volume) entropy distinction and the work/heat split become operationally important. Quantities are non-self-averaging and must be replaced by full distributions (stochastic thermodynamics). Phase transitions are smooth crossovers `[ESTABLISHED]`.
- **The arrow of time.** Not a logical inconsistency but the deepest conceptual incompleteness: the $H$-theorem's monotonic decrease cannot follow from time-symmetric mechanics (Loschmidt) and conflicts with Poincaré recurrence (Zermelo). The framework delivers irreversibility only after a time-

asymmetric input (molecular chaos / a low-entropy past), pushing the explanation onto an unexplained boundary condition it cannot itself justify `[OPEN]`. See *Chapter 14 (p. 191)*.

- **Integrable and many-body-localized quantum systems.** ETH fails: extensively many (quasi-)conserved quantities prevent relaxation to the standard Gibbs ensemble. Integrable systems relax to a Generalized Gibbs Ensemble with one Lagrange multiplier per conserved charge; MBL systems retain initial-condition memory indefinitely. Whether MBL is a true stable phase in $d \geq 2$ in the thermodynamic limit is `[CONTESTED]`.
- **Glasses, granular media, active matter.** Ergodicity is broken or the system is driven; phase space is not explored; configurational entropy, aging, and history-dependence appear. Equilibrium ensembles do not apply; replica/cavity methods, effective temperatures, or fully non-equilibrium descriptions are required. No complete first-principles statistical mechanics of the glass transition exists; whether it is a true thermodynamic transition is `[CONTESTED]`.
- **First-order transitions and metastability.** Ensemble equivalence breaks at coexistence; the canonical free energy is the convex hull (Maxwell construction), losing the metastable branches, surface tension, and nucleation kinetics the system physically exhibits `[ESTABLISHED]`.
- **Negative absolute temperature / bounded spectra.** Population inversion in upper-bounded-spectrum systems (nuclear spins, certain optical lattices) gives formally negative $T$, exposing the Boltzmann-vs-Gibbs entropy dispute over whether negative temperatures and Carnot efficiencies $> 1$ are physical `[CONTESTED, unsettled post-2013]`.

# Open problems (internal)

- **Why ensemble averages predict an individual system.** `[OPEN/CONTESTED]` Competing programs — ergodic theory, typicality/measure-concentration (Boltzmann–Lebowitz–Goldstein), Jaynesian MaxEnt, and quantum canonical typicality

/ ETH — each work in some regime; none is a universally accepted single derivation. The pragmatic answer (it works; fluctuations are tiny) is undisputed; the principled one is unresolved.

- **Origin of the thermodynamic arrow and the Past Hypothesis.** `[OPEN]` That an asymmetric boundary condition is needed is INFERENCE bordering on ESTABLISHED; *why* the early universe had extraordinarily low (especially gravitational) entropy is open and bleeds into *Chapter 11 (p. 139)*. Penrose's Weyl-curvature hypothesis is one `[SPECULATIVE]` proposal.
- **First-principles theory of the glass transition.** `[OPEN/CONTESTED]` Whether an underlying ideal thermodynamic transition exists (Kauzmann temperature, random first-order transition theory from the replica/cavity solution of mean-field spin glasses) or the transition is purely kinetic is disputed. Mean-field/infinite-dimensional results are rigorous; their relevance to $d = 3$ is contested.
- **Rigorous ETH for a realistic Hamiltonian.** `[OPEN]` Strongly supported numerically and by random-matrix arguments but unproven for any realistic interacting model. The boundaries among thermalizing, integrable (GGE), prethermal, quantum-scarred, and MBL behavior are under active study and partly `[CONTESTED]`.
- **A predictive variational principle for non-equilibrium steady states.** `[OPEN]` Large-deviation theory, the Gallavotti–Cohen fluctuation theorem, and macroscopic fluctuation theory give powerful exact results, and stochastic thermodynamics is rigorous for Markov dynamics, but no universal organizing principle exists (min/max entropy-production proposals are of limited or `[CONTESTED]` validity).
- **Rigorous hydrodynamic limit.** `[OPEN]` Lanford derives the Boltzmann equation only for short times in the dilute limit; a full rigorous derivation of Navier–Stokes and finite transport coefficients from interacting Hamiltonian microdynamics remains a major problem (related to Hilbert's sixth problem).

- **Boltzmann vs Gibbs entropy.** `[CONTESTED]` A live post-2013 debate over which entropy obeys the second law and equipartition for small or bounded-spectrum systems; consensus is absent and it matters operationally for nanoscale thermodynamics.

See *Chapter 15 (p. 233)* and *Chapter 19 (p. 392)*.

## Connections to other frameworks

- ***Chapter 5 (p. 40)*** — Statistical mechanics is the microscopic foundation that derives the phenomenological laws: $F = -k_B T \ln Z$ reproduces the thermodynamic potentials, $dS = \delta Q/T$ emerges from coarse-grained $S_B$, and the third law connects to ground-state degeneracy. The reduction is subtle: the second law's strict irreversibility is statistical/typical, not absolute, and requires the extra Past-Hypothesis input thermodynamics simply posits.
- ***Chapter 4 (p. 25)* and *Chapter 7 (p. 71)*** — Supply the reversible, entropy-conserving microdynamics (Liouville/unitary). The central tension is that these conserve fine-grained entropy while statistical mechanics produces irreversible relaxation — resolved only via coarse-graining plus initial conditions.
- ***Chapter 8 (p. 89)* / *Chapter 9 (p. 106)*** — The Euclidean path integral at finite temperature is a partition function with imaginary-time period $\beta\hbar$ (Matsubara/thermal field theory). Wilsonian RG, born in critical phenomena, became the organizing principle of QFT; critical exponents are operator anomalous dimensions, and the $\epsilon$-expansion is shared technology.
- ***Chapter 12 (p. 155)*** — Shannon entropy is formally identical to Gibbs entropy; Jaynes recast the subject as MaxEnt inference. Landauer's principle ($k_B T \ln 2$ per bit erased) and Bennett's resolution of Maxwell's demon tie thermodynamic cost to computation. Whether entropy is fundamentally physical or epistemic is `[CONTESTED]`.

- ***Chapter 11 (p. 139) / Chapter 10 (p. 121)*** — Gravity's non-additivity breaks standard ensemble theory; black-hole thermodynamics ($S = A/4$ in Planck units) imports statistical-mechanical concepts into gravity and raises the still-open microstate-counting question (string theory, holography). The arrow of time connects to the cosmological initial condition.
- ***Chapter 13 (p. 172)*** — Ergodic theory, mixing, and Lyapunov/chaos theory underpin the ensemble postulates; large-deviation theory provides the rigorous language for fluctuations, free energies as rate functions, and ensemble equivalence. KAM's demonstration of generic non-ergodicity forced the foundational shift to typicality.

See *Chapter 3 (p. 15)*, *Chapter 18 (p. 349)*, and *Chapter 17 (p. 325)*.

## Key references

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See the Bibliography (p. 776) for full citation details.

CHAPTER 7

# Quantum Mechanics

## Scope

This page treats **non-relativistic quantum mechanics (NRQM)** of finitely many degrees of freedom in the standard Dirac–von Neumann Hilbert-space formulation. It covers: the state/observable/dynamics axioms; pure and mixed states; observables as self-adjoint operators and their spectral theory; the tension between unitary Schrödinger evolution and the Born/projection rule (the measurement problem); entanglement, tensor products, and density matrices; Bell/CHSH nonlocality and Kochen–Specker contextuality; decoherence and the classical limit; uncertainty relations; identical particles and quantum statistics; and the major interpretations together with their respective costs.

It deliberately excludes the full machinery of relativistic quantum field theory and quantum gravity, except where those frameworks **bound** the domain of validity of NRQM. Mathematics is given at the level of rigor of von Neumann, Reed–Simon, and modern quantum-information texts, with explicit flags where the physicist's working formalism (Dirac kets, $\delta$-normalization) diverges from the rigorous functional-analytic version. For the broader placement of this domain among physical frameworks see *Chapter 3 (p. 15)*; for the epistemic-marker conventions used throughout, see *Chapter 2 (p. 8)*.

## Core formalism

## The Dirac–von Neumann axioms

The standard mathematical skeleton (von Neumann 1932; Dirac 1930) is a small set of postulates. The **framework** is `[ESTABLISHED]` as the consensus working theory; the **interpretation** of several postulates is `[CONTESTED]`, which is a logically separate matter (see *Where it breaks down*).

**(A1) States.** To an isolated system is associated a complex separable Hilbert space $\mathcal{H}$, complete in the norm $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$ (physics convention: $\langle\cdot|\cdot\rangle$ linear in the ket). A *pure state* is a ray — a unit vector $|\psi\rangle$ modulo a global phase $e^{i\theta}$ — equivalently the rank-one projector $\rho = |\psi\rangle\langle\psi|$. The most general state is a *density operator* $\rho$: self-adjoint, positive ($\rho \geq 0$), trace-class with $\operatorname{Tr}\rho = 1$.

**(A2) Observables.** Measurable quantities correspond to densely-defined self-adjoint operators $A = A^\dagger$. Crucially, *self-adjoint* (equal domains, $A = A^\dagger$) is strictly stronger than *symmetric/Hermitian* ( $\langle A\phi|\psi\rangle = \langle\phi|A\psi\rangle$ on a domain). Only self-adjoint operators are guaranteed a real spectrum, a projection-valued spectral measure, and (via Stone's theorem) a unitary group $e^{-iAt}$. `[ESTABLISHED]`

**(A3) Spectral decomposition.** By the spectral theorem, every self-adjoint $A$ admits a unique projection-valued measure (PVM) $E_A(\cdot)$ on the Borel sets of $\mathbb{R}$ with

$$A = \int_{\sigma(A)} \lambda \, dE_A(\lambda).$$

The spectrum $\sigma(A)$ (the set of possible measured values) splits into discrete (point) and continuous parts. Position $\hat{x}$ and momentum $\hat{p} = -i\hbar\, d/dx$ have purely continuous spectrum and **no normalizable eigenvectors**: the "eigenstates" $|x\rangle, |p\rangle$ are distributions living in a rigged Hilbert space (Gel'fand triple) $\Phi \subset \mathcal{H} \subset \Phi'$, not elements of $\mathcal{H}$. `[ESTABLISHED]` This is the rigorous content behind $\delta$-normalization $\langle x|x'\rangle = \delta(x - x')$.

**(A4) Born rule.** For state $\rho$ and observable $A$ with PVM $E_A$, the probability that a measurement yields a value in Borel set $\Delta$ is

$$\Pr(A \in \Delta) = \mathrm{Tr}\big(\rho\, E_A(\Delta)\big), \qquad \langle A \rangle_\rho = \mathrm{Tr}(\rho A).$$

For a pure state and a discrete nondegenerate eigenvalue $a_n$ with eigenvector $|a_n\rangle$, this reduces to $\Pr(a_n) = |\langle a_n|\psi\rangle|^2$. **Gleason's theorem** (1957) shows that, for $\dim \mathcal{H} \geq 3$, the trace form is the *unique* noncontextual probability measure on projections — so given the Hilbert-space scaffolding the rule's *form* is forced, not free. `[ESTABLISHED]`

**(A5) Unitary dynamics.** A closed system evolves by the Schrödinger equation

$$i\hbar\, \partial_t |\psi(t)\rangle = H|\psi(t)\rangle, \qquad |\psi(t)\rangle = U(t)|\psi(0)\rangle, \quad U(t) = e^{-iHt/\hbar},$$

with $H$ the self-adjoint Hamiltonian; for $\rho$ this is the von Neumann equation $i\hbar\, \partial_t \rho = [H, \rho]$. **Stone's theorem** guarantees $U(t)$ is a strongly-continuous one-parameter unitary group **iff** $H$ is self-adjoint. `[ESTABLISHED]`

**(A6) Projection postulate (von Neumann–Lüders).** Upon a measurement yielding an outcome in $\Delta$, the state updates ("collapses") to

$$\rho \mapsto \frac{E_A(\Delta)\, \rho\, E_A(\Delta)}{\mathrm{Tr}\big(\rho\, E_A(\Delta)\big)}.$$

A6 is in logical tension with A5: the former is stochastic and non-unitary, the latter deterministic and unitary. Whether A6 is a genuine physical postulate or merely effective bookkeeping is the heart of the measurement problem. `[CONTESTED]`

**(A7) Composite systems.** The Hilbert space of a composite system is the tensor product $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$. Subsystem states are reduced density operators $\rho_A = \mathrm{Tr}_B\, \rho_{AB}$.

**Operational generalization.** A2/A4/A6 are special cases of a wider framework. The most general measurement is a **POVM** $\{E_i \geq 0,\ \sum_i E_i = 1\}$ with $\Pr(i) = \mathrm{Tr}(\rho E_i)$; the most general dynamics is a **CPTP map** (quantum channel) with Kraus form

$$\rho \mapsto \sum_k K_k \rho K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = 1.$$

By **Stinespring/Naimark dilation**, every POVM/channel is a projective measurement / unitary on a larger space — so A2/A5/A6 already generate the whole structure. `[ESTABLISHED]` See *Chapter 12 (p. 155)* for the resource-theoretic reframing.

## Canonical commutation and uncertainty

The canonical commutation relation (CCR) $[\hat{x},\hat{p}] = i\hbar$ has **no** finite-dimensional or bounded-operator realization (taking the trace would yield $0 = i\hbar \dim \mathcal{H}$). The **Stone–von Neumann theorem** `[ESTABLISHED]` states that, up to unitary equivalence, there is a *unique* irreducible representation of the Weyl (exponentiated, bounded) form of the CCR for finitely many degrees of freedom — which is precisely why Schrödinger's wave mechanics and Heisenberg's matrix mechanics are the same theory. The theorem **fails** for infinitely many degrees of freedom (QFT), where unitarily inequivalent representations proliferate (see *Connections*).

The **Robertson–Schrödinger uncertainty relation** follows from Cauchy–Schwarz:

$$\Delta A\,\Delta B \geq \tfrac{1}{2}\big|\langle[A,B]\rangle\big|, \qquad \text{hence} \quad \Delta x\,\Delta p \geq \tfrac{\hbar}{2}.$$

This is a theorem about the statistical spread of an ensemble prepared in a single state — **not**, in its standard form, a statement about measurement disturbance. The Heisenberg-microscope "disturbance" intuition requires separate, subtler relations (Ozawa; Busch–Lahti–Werner) that have been experimentally probed. The Robertson relation is `[ESTABLISHED]`; the measurement-disturbance refinements are an active and partly `[CONTESTED]` area.

## Entanglement and mixed states

A pure $|\psi\rangle_{AB}$ is *separable* iff $|\psi\rangle_{AB} = |\phi\rangle_A \otimes |\chi\rangle_B$; otherwise *entangled*. The **Schmidt decomposition**

$$|\psi\rangle_{AB} = \sum_i \sqrt{p_i}\,|i\rangle_A|i\rangle_B$$

makes the reduced states $\rho_A = \rho_B = \sum_i p_i|i\rangle\langle i|$ manifest, and the **entanglement entropy** $S(\rho_A) = -\operatorname{Tr}\rho_A \log \rho_A$ quantifies pure-state entanglement. A mixed $\rho_{AB}$ is separable iff $\rho_{AB} = \sum_k q_k\, \rho_A^k \otimes \rho_B^k$. Entanglement is the resource behind teleportation, dense coding, and the Bell violations below.

A subtle "gauge": many distinct ensembles $\{p_i, |\psi_i\rangle\}$ realize the same $\rho$, the freedom being exactly a unitary mixing of the "square roots" (the **Hughston–Jozsa–Wootters / GHJW theorem**). Consequently $\rho$, not the underlying ensemble, is physically operative for local statistics. `[ESTABLISHED]`

## Bell / CHSH and Kochen–Specker

**Bell/CHSH** `[ESTABLISHED – both as theorem and as experiment]`. Any *local hidden-variable* (local-causal) theory with a shared variable $\lambda$ and outcomes $A(a,\lambda), B(b,\lambda) \in \{\pm 1\}$ obeys

$$S = |E(a,b) - E(a,b') + E(a',b) + E(a',b')| \leq 2.$$

QM predicts, for the singlet with correlation $E(a,b) = -\cos\theta_{ab}$, a maximum $S = 2\sqrt{2}$ — **Tsirelson's bound**. Loophole-free experiments (2015 onward) close the locality and detection loopholes simultaneously and confirm the violation. The lesson: no theory reproducing QM statistics can be simultaneously *local* and have *predetermined outcomes*, subject to the measurement-independence ("free choice") assumption. That **local realism is excluded** is `[ESTABLISHED]`; *which* premise to drop is interpretation-dependent and `[CONTESTED]`.

**Kochen–Specker** `[ESTABLISHED theorem]`. For $\dim\mathcal{H} \geq 3$, there is no *noncontextual* assignment of definite values $v(A) \in \sigma(A)$ to all observables that respects functional relations ($v(f(A)) = f(v(A))$) and is consistent across all commuting sets. Concretely, one cannot color the projectors so that exactly one of each orthogonal triad is "1". Hence any hidden-variable completion must be *contextual*: the revealed value depends on which compatible set is co-measured. de Broglie–Bohm mechanics is an explicit contextual model demonstrating consistency.

## Decoherence and the classical limit

**Decoherence** [ESTABLISHED as dynamics; `[CONTESTED]` as a *solution* to measurement]. Coupling a system to an environment $\mathcal{E}$ and tracing it out drives $\rho_S$ toward approximate diagonality in a *pointer basis* selected by the interaction Hamiltonian (einselection; Zurek). Markovian open-system evolution is the **GKSL/Lindblad** master equation,

$$\dot{\rho} = -\tfrac{i}{\hbar}[H,\rho] + \sum_k \Big( L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \Big),$$

the most general generator of a CPTP semigroup. Decoherence explains why interference is unobservable for macroscopic superpositions, and *why* a quasi-classical preferred basis emerges, on extraordinarily short timescales. It does **not**, by itself, select a single outcome: the reduced $\rho_S$ is an *improper* mixture, while the global state remains a superposition. Conflating the two ("decoherence solves measurement") is a category error. The distinction is `[ESTABLISHED]`; its interpretive significance is `[CONTESTED]`.

**Classical limit.** Ehrenfest's theorem,

$$\tfrac{d}{dt}\langle \hat{x} \rangle = \tfrac{1}{m}\langle \hat{p} \rangle, \qquad \tfrac{d}{dt}\langle \hat{p} \rangle = -\langle \nabla V \rangle,$$

reproduces Newton's laws **only when** $\langle \nabla V(\hat{x}) \rangle \approx \nabla V(\langle \hat{x} \rangle)$, i.e. for narrow wavepackets in slowly varying potentials. The $\hbar \to 0$ limit is *singular*: WKB / stationary-phase analysis shows the phase $S/\hbar$ oscillates without bound and classical trajectories emerge as stationary-phase points, but the limit is non-uniform — caustics, tunneling $\sim e^{-S/\hbar}$, and chaotic systems (where the Ehrenfest time grows only logarithmically in $1/\hbar$) obstruct it. The Wigner–Weyl–Moyal phase-space formulation makes this precise: the Moyal star-product bracket reduces to the Poisson bracket as $\hbar \to 0$. That the limit is subtle/singular is `[ESTABLISHED]`; a fully general derivation of classicality is `[OPEN]`. See *Chapter 4 (p. 25)*.

### Identical particles and statistics

For indistinguishable particles the physical Hilbert space is a symmetry sector under particle exchange. In 3D the exchange is governed by the symmetric group: states are totally symmetric (**bosons**, integer spin) or antisymmetric (**fermions**, half-integer spin), the latter producing the Pauli exclusion principle and the Slater-determinant structure of atoms. The connection between spin and statistics is a **theorem of relativistic QFT** (the spin–statistics theorem; requires microcausality, positive energy, Lorentz invariance) — *not* derivable within NRQM, where it enters as an extra postulate. In 2D the exchange is governed by the braid group, permitting **anyons** with fractional statistics (relevant to the fractional quantum Hall effect). The spin–statistics connection is `[ESTABLISHED]` in QFT and an *imported input* to NRQM; 2D anyons are `[ESTABLISHED]` theoretically and via FQHE phenomenology. See *Chapter 9 (p. 106)* and *Chapter 6 (p. 56)*.

## Foundational assumptions

Status values follow the *Chapter 20 (p. 458)* convention: *fundamental* / *likely-fundamental* / *historical-artifact* / *conventional-choice* / *unclear*.

| Assumption | Status | Justification |
|---|---|---|
| States live in a complex Hilbert space; amplitudes superpose linearly (superposition principle). | fundamental | `[ESTABLISHED]` Linearity underlies all interference and is confirmed to extraordinary precision; searches for nonlinear corrections (Weinberg tests; Bialynicki-Birula–Mycielski) bound them severely. That the field is **complex** (not real or quaternionic) is subtler: real-QM was long thought an empirically equivalent convention, but recent multipartite Bell-type tests (≈2021–2022) argue standard complex QM cannot be reproduced by a real-Hilbert-space theory under standard tensor-product locality assumptions `[INFERENCE/CONTESTED]`, so complexity is plausibly fundamental modulo those framework assumptions. |

| | | |
|---|---|---|
| Observables are self-adjoint operators; measured values lie in the spectrum. | likely-fundamental | [ESTABLISHED as framework; partly conventional in scope] Self-adjointness (not mere symmetry) is forced if one wants real spectra, a spectral measure, and a unitary group (Stone). But identifying *all* self-adjoint operators with measurable quantities is an idealization: superselection rules forbid measuring some (e.g. coherent superpositions of distinct charge), and the POVM framework shows projective observables are a special case. The *form* is fundamental; the *exhaustiveness* is convenient idealization. |
| The Born rule $\Pr = \mathrm{Tr}(\rho E)$ gives outcome probabilities. | likely-fundamental | [ESTABLISHED empirically; derivation CONTESTED] Gleason fixes it as the unique noncontextual probability measure on projections for $\dim \geq 3$, so given the scaffolding it is essentially forced. Whether it can be *derived* from the unitary part alone (Deutsch–Wallace decision theory; Zurek envariance) rather than postulated is genuinely [CONTESTED]. As a postulate, fundamental; as a theorem, open. |
| The projection/collapse postulate (A6) is a real, distinct dynamical process. | unclear | [CONTESTED] The crux of the measurement problem. Collapse cannot be derived from unitary evolution, yet A5 and A6 are jointly applied with a vague "measurement" trigger (Bell's "shifty split"). Whether collapse is (i) real new dynamics (GRW/CSL), (ii) an artifact of not tracking environment/observer (Everett, decoherence), or (iii) an information-update rule (QBism) is exactly what interpretations dispute. The bare postulate is very likely *not* fundamental as stated, but its replacement is unknown. |
| Composite systems combine by tensor product (A7). | likely-fundamental | [ESTABLISHED operationally; INFERENCE as necessity] The tensor product encodes local tomography and the existence of entanglement, and is singled out among reasonable composition rules in GPT reconstructions. It could conceivably be otherwise (direct sum; theories without local tomography), so it is a strong physical principle rather than pure logical necessity. |
| Time is an external classical parameter, not an operator (no time observable). | historical-artifact | [INFERENCE] Pauli's theorem shows no self-adjoint time operator is conjugate to a Hamiltonian bounded below, so $t$ is a c-number while $x$ is an operator — an asymmetry inherited from non-relativistic, fixed-background thinking. Plausibly an artifact that becomes untenable in quantum gravity (problem of time) and is awkward even in relativistic QM. Relational/timeless formulations (Page–Wootters) recast time as correlation with a clock subsystem. |

| | | |
|---|---|---|
| Global phase and overall normalization are unphysical (states are rays). | conventional-choice | `[ESTABLISHED]` Pure representational convention with no observable content for closed systems — only *relative* phases matter. The projective structure (rays / rank-one projectors) is the invariant content; the unit-vector description is a chart on it. (Unobservability of *relative* phases between superselection sectors is a deeper, dynamical fact, distinct from this convention.) |
| Hilbert space is separable (countable basis). | conventional-choice | `[ESTABLISHED for NRQM]` Adequate for all standard single-system QM; non-separable spaces appear only in special constructions (some QFT representations, certain idealized limits). Not physically load-bearing for this domain. |
| Spin–statistics: integer↔symmetric/boson, half-integer↔antisymmetric/fermion. | fundamental | `[ESTABLISHED]` In NRQM an *added postulate*; a *theorem* in relativistic QFT (microcausality, positive energy, Lorentz invariance). Its truth is fundamental physics, but its status *within NRQM* is an imported input — a signal that NRQM is not closed. In 2D the exchange structure is richer (braid group → anyons), so the boson/fermion dichotomy is dimension-dependent, not absolute. |

| | | |
|---|---|---|
| Measurement-independence / "free choice" of settings (used in Bell theorems). | likely-fundamental | `[CONTESTED/OPEN]` Bell's no-go assumes the hidden variable $\lambda$ is statistically independent of the chosen settings. Denying it (superdeterminism or retrocausality) is a logically open escape preserving locality, but is widely regarded as conspiratorial and is not independently motivated. Best flagged as a load-bearing assumption rather than settled. |

## Domain of validity

`[ESTABLISHED]` NRQM is the empirically correct description for systems where:

1. Particle speeds and binding energies are small compared to rest-mass energy ($v \ll c$, $E \ll mc^2$), so particle number is conserved and pair creation is negligible;
2. Gravity is dynamically irrelevant (no need for a quantum spacetime);
3. The number of degrees of freedom is finite, so the Stone–von Neumann theorem applies and the CCR representation is essentially unique.

In this regime QM is among the most precisely tested theories in physics: atomic spectra, the quantum Hall effect, superconductivity and superfluidity, chemistry, semiconductors, NMR, and quantum-information experiments all confirm it. There is **no confirmed experimental deviation** from quantum predictions anywhere. The theory has been tested for superpositions of objects up to large molecules (matter-wave interferometry), continually pushing the boundary at which classicality must emerge. Relevant scales are catalogued in *Chapter 16 (p. 308)*.

## Where it breaks down

A disciplined accounting distinguishes (a) genuine internal *inconsistencies*, (b) *domain-of-validity* mismatches where a wider theory takes over, and (c) *unsolved-but-consistent* problems. See *Chapter 14 (p. 191)*.

- **The measurement problem (A5 vs A6).** `[ESTABLISHED as a structural tension; resolution OPEN/CONTESTED]` Unitary evolution (A5) is linear and never turns a superposition into a single outcome; the projection postulate (A6) does exactly that, stochastically. Applied to an apparatus + system + observer all treated quantum-mechanically, A5 predicts an entangled superposition of pointer states (Schrödinger's cat; Wigner's friend), while we only ever observe one outcome. The theory does not specify what physical process, system size, or interaction constitutes a "measurement" triggering A6. This is a genuine *incompleteness* — a type-(a) inconsistency between two purported universal dynamical laws, with no rule demarcating their jurisdictions.

- **Self-adjointness vs symmetry; domains and boundary conditions.** `[ESTABLISHED – a routinely glossed rigor gap, type (c)]` The physicist's "Hermitian operator" is often only symmetric. The distinction is physical: momentum on a half-line or interval has many inequivalent self-adjoint extensions (parameterized by deficiency indices / boundary conditions), each a different physics; some Hamiltonians are not even essentially self-adjoint, so dynamics is ambiguous until a boundary condition is fixed. Naive manipulation of $\hat{x}, \hat{p}$ "eigenstates" requires the rigged-Hilbert-space machinery to be rigorous. A breakdown of the *informal* formalism, repaired by functional analysis — not of the theory.

- **Born rule as postulate vs theorem.** `[CONTESTED, type (c)]` In no-collapse readings, probability is puzzling because the dynamics is deterministic and "all outcomes happen". Decision-theoretic (Deutsch–Wallace) and envariance (Zurek) derivations attempt to recover Born weights from symmetry/rationality, but critics charge circularity or that they fail to explain why branch weights should *matter* (the incoherence/quantitative problem).

Gleason constrains the *form* but presupposes the scaffolding. The rule's *origin* is unsettled though its *form* is forced.

- **Tension with relativistic locality.** [ESTABLISHED, but no operational inconsistency — a type-(b)/conceptual matter] Collapse (A6) and entanglement appear to act "instantaneously" across spacelike separation, and Bell violations show outcomes cannot be locally predetermined. Yet the **no-signaling theorem** guarantees no usable information travels faster than light: local marginals are independent of the distant setting. So there is a *conceptual* clash with relativistic intuition but *not* a contradiction with special relativity's operational content. Making collapse Lorentz-covariant is where objective-collapse models face their hardest challenge. See *Chapter 10 (p. 121)*.
- **The classical limit is singular.** `[ESTABLISHED, type (c)]` $\hbar \to 0$ is non-smooth: $e^{iS/\hbar}$ oscillates without bound, tunneling $\sim e^{-S/\hbar}$ vanishes non-analytically, and for chaotic systems the Ehrenfest time grows only logarithmically in $1/\hbar$. A robust general derivation of classicality remains incomplete — unsolved-but-consistent, not inconsistent.
- **Time and the energy–time uncertainty relation.** `[ESTABLISHED]` By Pauli's theorem there is no self-adjoint time operator conjugate to a lower-bounded $H$, so $\Delta E\, \Delta t \gtrsim \hbar$ is **not** an instance of the Robertson relation; $\Delta t$ must be reinterpreted (lifetime; characteristic evolution time, via the Mandelstam–Tamm bound). This exposes the asymmetric, external-parameter treatment of time as a structural limitation, acute in quantum gravity.
- **Infinite degrees of freedom and superselection.** `[ESTABLISHED, type (b)]` The single-Hilbert-space, Stone–von Neumann picture fails for infinitely many degrees of freedom: unitarily inequivalent representations, superselection sectors (charge; particle number in the relativistic vacuum; distinct thermodynamic phases), and the need for the algebraic ($C^*$ / von Neumann-algebra) formulation. Textbook QM silently

assumes the finite-dof setting; repaired by algebraic QFT and quantum statistical mechanics.

# Open problems (internal)

Tracked also in *Chapter 15 (p. 233)* and, where conjectural, *Chapter 19 (p. 392)*.

1. **The measurement problem.** `[OPEN/CONTESTED]` What physically (if anything) selects a single outcome, and where is the quantum–classical cut? The deepest internal problem: the axioms as stated are not a closed, consistent universal dynamics. No experiment yet distinguishes the no-collapse interpretations from one another; only collapse models make distinct predictions.
2. **Is collapse a real dynamical process?** `[OPEN, empirically testable — a genuine virtue]` Objective-collapse models (GRW; CSL; Diósi–Penrose gravitational collapse) add stochastic nonlinear terms that spontaneously localize macroscopic superpositions, predicting tiny deviations from unitary QM (anomalous heating; suppression of large-mass interference). Matter-wave interferometry, ultracold optomechanics, and underground spontaneous-radiation searches are progressively constraining or excluding parameter regions. The Diósi–Penrose proposal ties collapse to gravity and is being squeezed but not settled.
3. **Can the Born rule be derived rather than postulated?** `[OPEN/CONTESTED]` Gleason fixes its form given noncontextuality; Deutsch–Wallace and Zurek attempt full derivations within Everett. Whether any is non-circular is actively disputed; no universally accepted derivation exists.
4. **Probability in deterministic no-collapse interpretations (Everett).** `[OPEN/CONTESTED]` If all branches are equally real, the meaning of "the probability of an outcome" and the empirical confirmation of Born weights (the incoherence and quantitative problems) remain contested even among Everettians.

5. **Reconstructing QM from physical/operational axioms.** `[OPEN, active and fruitful]` The generalized-probabilistic-theories program (Hardy 2001; Chiribella–D'Ariano–Perinotti 2011; Masanes–Müller) derives the Hilbert-space formalism from information-theoretic postulates (purification; local tomography; continuous reversibility). Illuminating but not yet a single agreed-upon derivation; the axioms' physical necessity is debated. See *Chapter 17 (p. 325)*.

6. **Origin and necessity of complex amplitudes and the tensor-product rule.** `[OPEN/INFERENCE]` Recent (≈2021–2022) multipartite tests argue real-Hilbert-space QM is experimentally distinguishable from complex QM under standard locality assumptions, suggesting complexity is not mere convention — but the conclusion is conditional on those framework assumptions and is debated.

   **Status update 2026-07-02 (Track R2 — measurement problem operationalized; see Working note: *Track R2 — The Measurement Problem, Operationalized (2026-07 bound state)* (repo: `notes/2026-07-02-measurement-problem-operationalized.md`)).** *Item 2 (objective collapse):* the canonical **GRW parameter point ($r_C = 10^{-7}$ m, $\lambda = 10^{-16}$ s$^{-1}$) now deviates from the XENONnT best fit by 9.1σ — the first experimental exclusion of the originally proposed CSL values** for Markovian mass-proportional (white-noise) CSL, and the Diósi–Penrose smearing cutoff is forced to $R_0 > 4.9$ Å — both from the XENONnT spontaneous-radiation search (PRL **136**, 120201 (2026), arXiv:2506.05507; prior best: Majorana Demonstrator, PRL **129**, 080401 (2022)) `[ESTABLISHED]`. Parameter-free DP has been dead since 2021 (Donadi et al., *Nat. Phys.* **17**, 74 (2021)). The collapse class survives via colored-noise/dissipative extensions, which evade radiation bounds by construction — a qualitative parsimony cost, not merely a squeeze `[INFERENCE]`. *Item 1 (measurement problem):* extended-Wigner-friend results hardened 2023–2026 — Local Friendliness polytope = Bell polytope in a wide class of scenarios (*Quantum* **9**, 1819 (2025)); "thoughtful" LF (*Quantum* **7**, 1112 (2023)); first loophole-ridden LF-violation circuits on quantum hardware (*Quantum* **9**, 1851 (2025)) — but these constrain assumption *combinations* (absoluteness of observed events + local agency), never single interpretations `[ESTABLISHED theorems; CONTESTED significance]`. The relational-QM "relative facts" incompatibility exchange (Lawrence et al. *Quantum* **7**, 1015 (2023) → EJPS **15**, 16 (2025) vs Cavalcanti–Di Biagio–Rovelli EJPS **13**, 55 (2023)) remains live and unresolved `[CONTESTED]`.

## Connections to other frameworks

- ***Relativistic Quantum Field Theory* (Chapter 8 (p. 89)).** NRQM is the $c \to \infty$, fixed-particle-number effective limit of QFT. QFT supplies what NRQM must postulate — the spin–statistics connection, antiparticles, and the resolution of one-particle relativistic pathologies (Klein paradox, negative-energy states). It also breaks NRQM's uniqueness pillar: Stone–von Neumann fails for infinite dof, giving inequivalent representations, Haag's theorem, and the need for the algebraic formulation. *Clash:* NRQM's single unique CCR representation and global state do not survive.
- ***Classical mechanics* (Chapter 4 (p. 25)).** Recovered in the singular $\hbar \to 0$ limit via Ehrenfest, WKB/stationary phase, and the Wigner–Moyal phase-space formalism (Moyal bracket → Poisson bracket). The map is non-uniform (caustics, tunneling, chaos). Decoherence supplies the *effective* emergence of a classical pointer basis. *Clash:* definite trajectories and noncontextual values are excluded (Bell; Kochen–Specker).
- ***Quantum information / computation* (Chapter 12 (p. 155)).** QM is the operational substrate. The POVM/CPTP/Stinespring generalization, entanglement as a resource (teleportation; dense coding), the Tsirelson bound, and no-cloning are QM theorems reframed as information primitives. Device-independent cryptography rests directly on Bell violation. The GPT reconstruction program runs the arrow backward, deriving QM from information axioms.

- ***Quantum gravity* (Chapter 11 (p. 139)) and *general relativity* (Chapter 10 (p. 121)).** The sharpest clash. QM needs a fixed classical spacetime and external time; GR makes spacetime dynamical. The problem of time (the Wheeler–DeWitt constraint $H|\Psi\rangle = 0$ has no external $t$), black-hole information, and gravitationally induced collapse (Diósi–Penrose) all live at this interface. It is unresolved whether QM, GR, or both must be modified. See *Chapter 18 (p. 349)*.
- ***Statistical mechanics* (Chapter 6 (p. 56)) / *Chapter 5 (p. 40)*.** Density matrices and the von Neumann entropy $S = -\operatorname{Tr}\rho\ln\rho$ connect to thermal ensembles; the GKSL/Lindblad equation governs open-system thermalization and decoherence. The eigenstate thermalization hypothesis (ETH) addresses how isolated quantum systems appear to thermalize under unitary evolution. Quantum statistics (Bose–Einstein, Fermi–Dirac) follow from identical-particle symmetrization.
- **Foundations / interpretations (the explicit ledger of costs).** Each is empirically equivalent to standard QM *except* objective collapse, so the choice is currently philosophical/aesthetic rather than experimental — itself a striking feature of the domain. **Copenhagen/standard:** accepts a primitive classical–quantum cut, declines to describe its mechanism (instrumentalist; criticized as incomplete). **Many-worlds (Everett):** keeps only unitary A5, pays with branching multiplicity and an unsolved account of probability. **de Broglie–Bohm:** restores definite particle positions guided by $\psi$, pays with explicit nonlocality, contextuality, and relativistic/field awkwardness. **GRW/CSL objective collapse:** modifies the dynamics, pays with new fundamental stochasticity and Lorentz-covariance tension — but is empirically testable. **QBism:** reads $\psi$ and the Born rule as an agent's subjective degrees of belief, dissolving the measurement problem at the cost of a non-objective $\psi$. **Relational QM (Rovelli):** facts are relative to systems; no absolute state. **Consistent/decoherent histories:** assigns probabilities to coarse-grained history sets, criticized for a proliferation of

incompatible frameworks. Key findings cross-referenced in *Chapter 21 (p. 497). 2026-07 addendum:* the cost accounting is updated in Working note: *Track R2 — The Measurement Problem, Operationalized (2026-07 bound state)* (repo: `notes/2026-07-02-measurement-problem-operationalized.md`) — objective collapse (the one testable option) has had its canonical parameter targets excluded and now pays a moving-target cost; extended-Wigner-friend (Local Friendliness) theorems reprice all single-world objective readings without deciding among interpretations; the relational-QM "relative facts" consistency dispute is live `[CONTESTED]`.

## Key references

See the Bibliography (p. 776) for the consolidated, deduplicated citation list. Anchors for this page:


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- M. M. Wilde, *Quantum Information Theory* (Cambridge UP, 2nd ed. 2017) — channels, entropies, operational structure.


## References

Full citations consolidated in the Bibliography (p. 776). See also *Chapter 1 (p. 1)*, `AGENTS.md` (in the repository), *Chapter 40 (p. 739)*, and the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)) for wiki-level navigation and provenance.

CHAPTER 8

# Quantum Field Theory

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Quantum Field Theory (QFT) is the quantum theory of relativistic fields and the framework underlying the *Standard Model* (Chapter 9 (p. 106)), condensed-matter critical phenomena, and the modern language of effective theories across physics. It is simultaneously the most precisely tested theory of nature `[ESTABLISHED]` and one whose mathematical foundations remain incomplete for the cases we care about most `[OPEN]`. This tension — phenomenal empirical success atop a partly unproven scaffold — is the organizing theme of this page.

## Scope

This page treats QFT as a *foundational framework*, not a catalogue of computations. It covers:

1. The kinematic/dynamical core — fields as fundamental degrees of freedom, the action/path-integral, and canonical quantization.
2. The representation-theoretic definition of "particle" via Wigner's classification of Poincaré irreducible representations.
3. Renormalization, the renormalization group (RG), and the Wilsonian/effective-field-theory (EFT) reinterpretation.
4. The gauge principle, spontaneous symmetry breaking (SSB), the Brout–Englert–Higgs (BEH) mechanism, and anomalies.
5. Vacuum structure and zero-point energy.
6. The rigorous/axiomatic program (Wightman, Haag–Kastler, constructive QFT) and its structural theorems and obstructions.

7. The status of measurement in a relativistic field-theoretic setting.

Scope **excludes** quantum gravity and string theory except where they bound QFT's validity (see *Chapter 10 (p. 121)*); nonperturbative QCD, lattice gauge theory, and conformal field theory appear only insofar as they bear on foundations. For the broader inter-framework picture see *Chapter 3 (p. 15)* and *Chapter 18 (p. 349)*.

# Core formalism

## Fields, action, and the two quantization routes

The fundamental objects are operator-valued (more precisely, operator-valued-distribution-valued) **fields** $\phi_a(x)$ over Minkowski spacetime $\mathbb{R}^{1,3}$ with metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ (the signature is a conventional choice `[ESTABLISHED]`; physics is invariant under it, see *Chapter 20 (p. 458)*). Dynamics derive from a local action

$$S[\phi] = \int d^4x\, \mathcal{L}\big(\phi_a(x),\, \partial_\mu\phi_a(x)\big),$$

with Euler–Lagrange equations $\partial_\mu(\partial\mathcal{L}/\partial(\partial_\mu\phi_a)) - \partial\mathcal{L}/\partial\phi_a = 0$. The canonical example is the real scalar

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \tfrac{\lambda}{4!}\phi^4.$$

**Locality** — that $\mathcal{L}$ is a function of fields and finitely many derivatives at a single point — is the central structural assumption of standard QFT.

**Route A — Canonical quantization.** With conjugate momentum $\pi = \partial\mathcal{L}/\partial\dot{\phi}$, impose equal-time commutators (bosons) or anticommutators (fermions):

$$[\phi(\mathbf{x},t), \pi(\mathbf{y},t)] = i\,\delta^{(3)}(\mathbf{x}-\mathbf{y}).$$

Free fields expand in creation/annihilation operators, building a **Fock space** on a vacuum $|0\rangle$ with $a_{\mathbf{p}}|0\rangle = 0$.

**Route B — Path integral.** Time-ordered correlation (Green's) functions are

$$\langle 0|T\,\phi(x_1)\cdots\phi(x_n)|0\rangle = \frac{\int \mathcal{D}\phi\;\phi(x_1)\cdots\phi(x_n)\,e^{iS[\phi]/\hbar}}{\int \mathcal{D}\phi\;e^{iS[\phi]/\hbar}}.$$

The generating functional $Z[J] = \int \mathcal{D}\phi\, e^{i(S+\int J\phi)}$ yields connected correlators from $W[J] = -i\ln Z[J]$ and one-particle-irreducible (1PI) vertices from the effective action $\Gamma[\phi_{cl}]$ via Legendre transform. **Wick rotation** $t \to -i\tau$ converts $e^{iS}$ into a Boltzmann weight $e^{-S_E}$, mapping QFT onto Euclidean statistical mechanics and lending the path integral a (heuristically) convergent, measure-like form; the Osterwalder–Schrader (OS) axioms specify when a Euclidean theory reconstructs a Minkowski one. This is `[ESTABLISHED]` as a calculational framework, but the Lorentzian "measure" $\mathcal{D}\phi\, e^{iS}$ is *not* a rigorously defined measure in $d = 4$ `[ESTABLISHED limitation]` — see *Where it breaks down*.

## Particles = unitary irreps of the Poincaré group (Wigner)

A relativistic one-particle state space carries a unitary irreducible representation of the universal cover of the Poincaré group, $\mathrm{ISL}(2,\mathbb{C}) = \mathbb{R}^{1,3} \rtimes SL(2,\mathbb{C})$. Its Casimirs are

$$P^2 = P_\mu P^\mu\ (\text{mass}^2), \qquad W^2 = W_\mu W^\mu, \quad W_\mu = -\tfrac{1}{2}\epsilon_{\mu\nu\rho\sigma}P^\nu J^{\rho\sigma}\ (\text{Pauli–Lubanski}),$$

with $W^2 = -m^2 s(s+1)$ for massive states. **Wigner's classification** `[ESTABLISHED; Wigner 1939]` organizes irreps by the orbit of $P_\mu$ and its little group:

- $m > 0$: little group $SU(2) \to$ labels $(m, s)$, $s \in \{0, \frac{1}{2}, 1, \dots\}$, with $2s+1$ spin states.
- $m = 0$: little group $ISO(2)$; finite-helicity reps labelled by $h \in \frac{1}{2}\mathbb{Z}$ (massless particles have two helicity states for $h \neq 0$); plus exotic "continuous-spin" reps not realized in nature `[ESTABLISHED that they are unobserved]`.
- $P = 0$: the vacuum.

- $P^2 < 0$ (tachyonic) and $P^0 < 0$ orbits are excluded on physical grounds of stability and causality `[ESTABLISHED]`.

Fields are the *local, Lorentz-covariant carriers* that interpolate particle states. The connection between a **field** transforming in a finite-dimensional (non-unitary) representation of $SL(2, \mathbb{C})$ and a **particle** in a unitary representation, mediated by the mode expansion, is the structural origin of the **spin–statistics** and **CPT** theorems. See *Chapter 9 (p. 106)* and *Chapter 13 (p. 172)*.

## Renormalization and the renormalization group

Loop integrals produce ultraviolet (UV) divergences. One **regularizes** (dimensional regularization $d = 4 - \epsilon$, or a cutoff $\Lambda$), then **renormalizes** by absorbing divergences into a finite number of parameters (mass, couplings, field normalization). A theory is **perturbatively renormalizable** if finitely many counterterms suffice to all orders — equivalently, all couplings have mass dimension $\geq 0$ in $d = 4$. Bare and renormalized quantities relate by $Z$-factors, $\phi_0 = Z_\phi^{1/2} \phi$, $\lambda_0 = Z_\lambda \, \mu^\epsilon \lambda$.

The **renormalization group** expresses independence of physics from the arbitrary scale $\mu$:

$$\left[\mu \frac{\partial}{\partial \mu} + \beta(g) \frac{\partial}{\partial g} - \gamma_m \, m \frac{\partial}{\partial m} + n \, \gamma_\phi \right] \Gamma^{(n)} = 0, \qquad \beta(g) = \mu \frac{dg}{d\mu}.$$

The sign and zeros of $\beta$ govern UV/IR fate. For QCD ($SU(3)$ with $n_f$ flavors),

$$\beta(g) = -\frac{g^3}{16\pi^2}\left(11 - \tfrac{2}{3} n_f\right) + \mathcal{O}(g^5) \;\Rightarrow\; \text{asymptotic freedom for } n_f < \frac{33}{2}$$

`[ESTABLISHED; Gross-Wilczek and Politzer 1973]`. For QED $\beta > 0$, so the coupling grows in the UV toward a Landau pole. Fixed points $\beta(g_*) = 0$ define scale-invariant (typically conformal) theories — the endpoints of RG flows that anchor the space of QFTs.

## Wilsonian effective field theory

Wilson reframed renormalization physically `[ESTABLISHED as conceptual framework]`: define the theory with cutoff $\Lambda$, integrate out modes between $\Lambda$ and $b\Lambda$, and generate an effective action containing **all** operators allowed by symmetry,

$$\mathcal{L}_{\text{eff}} = \sum_i \frac{c_i}{\Lambda^{d_i-4}} \mathcal{O}_i, \qquad d_i = \text{mass dimension of } \mathcal{O}_i.$$

Operators are **relevant** ($d_i < 4$, grow in the IR), **marginal** ($d_i = 4$ ), or **irrelevant** ($d_i > 4$, suppressed by $(E/\Lambda)^{d_i-4}$). Renormalizability is thereby *derived*: low-energy physics is dominated by the finitely many relevant/marginal operators, while non-renormalizable operators are simply small. Every QFT becomes one rung in a tower of EFTs, and the Standard Model itself is an EFT valid up to some scale $\Lambda_{\text{new}}$ `[INFERENCE]`. This is the deep formal bridge to *Chapter 6 (p. 56)*: Euclidean QFT *is* classical statistical field theory, and universality classes/critical exponents are shared verbatim.

## Gauge principle, SSB, and the BEH mechanism

Promoting a global symmetry $\psi \to e^{i\alpha^a T^a}\psi$ to a local one forces a connection $A^a_\mu$ with covariant derivative $D_\mu = \partial_\mu - igA^a_\mu T^a$ and field strength

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu, \qquad \mathcal{L}_{YM} = -\tfrac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu}.$$

Geometrically, gauge fields are connections on a principal bundle and $F$ is curvature (see *Chapter 13 (p. 172)*). Quantization requires **gauge fixing** (Faddeev–Popov determinant → ghosts; BRST symmetry $s$ with $s^2 = 0$ controls unitarity).

**SSB** is a symmetric Lagrangian with a non-invariant vacuum. **Goldstone's theorem** `[ESTABLISHED]`: each spontaneously broken continuous *global* symmetry yields a massless scalar. The **BEH/Higgs mechanism** `[ESTABLISHED; confirmed 2012]`: when the broken symmetry is *gauged*, the would-be Goldstone bosons are "eaten," giving gauge bosons mass. For electroweak $SU(2)_L \times U(1)_Y \to U(1)_{\text{em}}$ with a doublet $\Phi$, $\langle\Phi\rangle = (0, v/\sqrt{2})$,

$$M_W = \tfrac{1}{2} g v, \quad M_Z = \tfrac{1}{2}\sqrt{g^2 + g'^2}\, v, \quad v \approx 246 \text{ GeV},$$

plus a physical Higgs scalar with measured $m_H \approx 125$ GeV. A subtlety often glossed in textbooks `[ESTABLISHED but underappreciated]`: in a gauge theory there is no gauge-invariant local order parameter — **Elitzur's theorem** forbids spontaneous breaking of a *local* gauge symmetry. The "Higgs phase" is gauge-fixing language; the gauge-invariant content lives in correlators, and the Fradkin–Shenker analysis shows the Higgs and confinement regimes are analytically connected. This bears directly on whether the gauge principle is "fundamental" — see the table below.

## Anomalies

An anomaly is a classical symmetry the quantized measure cannot preserve. The **chiral (Adler–Bell–Jackiw) anomaly** `[ESTABLISHED]`:

$$\partial_\mu j^{5\mu} = \frac{e^2}{16\pi^2}\, \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}.$$

Anomalies in *global* symmetries are physical and exact (e.g. the $\pi^0 \to 2\gamma$ rate). Anomalies in *gauged* symmetries are fatal and must cancel; in the Standard Model the hypercharge/$SU(2)$/$SU(3)$ anomalies cancel generation-by-generation, a strong structural constraint. 't Hooft anomaly matching constrains IR dynamics nonperturbatively — one of the few exact handles on strongly-coupled QFT.

## Vacuum and zero-point energy

The free Hamiltonian is $H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \big(a^\dagger_{\mathbf{p}} a_{\mathbf{p}} + \frac{1}{2}[a_{\mathbf{p}}, a^\dagger_{\mathbf{p}}]\big)$, the last term a formally infinite zero-point energy. *Differences* are physical (the Casimir effect `[ESTABLISHED]`); the *absolute* value couples to gravity and underlies the cosmological-constant problem (see *Where it breaks down* and *Chapter 11 (p. 139)*). The interacting vacuum is highly nontrivial, hosting condensates $\langle \bar{q}q \rangle$ and $\langle G^2 \rangle$ in QCD.

## The rigorous/axiomatic skeleton

**Wightman axioms** (operator/correlation formulation): a Hilbert space $\mathcal{H}$ carrying a positive-energy unitary Poincaré representation, a unique vacuum, fields as operator-valued tempered distributions on a dense domain, **microcausality** $[\phi(x), \phi(y)] = 0$ for spacelike $(x - y)$, and the spectrum condition $\mathrm{spec}(P) \subset \overline{V^+}$. The Wightman functions $\mathcal{W}_n = \langle 0|\phi(x_1) \cdots \phi(x_n)|0\rangle$ obey covariance, spectral support, locality, positivity, and cluster decomposition — and (reconstruction theorem) determine the theory.

**Haag–Kastler / algebraic QFT (AQFT)**: the primitive is a net of local von Neumann algebras $\mathcal{O} \mapsto \mathfrak{A}(\mathcal{O})$ over spacetime regions, satisfying isotony, locality (spacelike algebras commute), covariance, and the spectrum condition. Superselection structure (DHR), statistics, and charges emerge intrinsically; the local algebras are type $\mathrm{III}_1$, so there is **no tensor factorization** of $\mathcal{H}$ into local subsystems and no local number operator.

Theorems provable within these axioms `[ESTABLISHED within the framework]`: **spin–statistics** and **CPT** (Wightman axioms + analyticity); **Reeh–Schlieder** ($\overline{\mathfrak{A}(\mathcal{O})|0\rangle} = \mathcal{H}$ — the vacuum is cyclic and separating for any local algebra, implying ubiquitous vacuum entanglement); **Bisognano–Wichmann** (the modular flow of the Rindler wedge *is* the Lorentz boost, linking Tomita–Takesaki modular theory to the Unruh effect); and **Haag's theorem** (no unitary intertwines a free and an interacting representation in the same Hilbert space — the interaction picture, strictly, does not exist).

# Foundational assumptions

| Assumption | Status | Justification |
|---|---|---|
| Fields (not particles or paths) are the fundamental degrees of freedom | likely-fundamental | `[INFERENCE]` Locality + relativistic causality + cluster decomposition, with QM, essentially force a field description (Weinberg's "folk theorem"). Particles emerge as field excitations and are observer-ambiguous (Unruh; no global particle number in curved space). Whether fields are themselves emergent (strings, entanglement, pre-geometry) is `[OPEN]`. |
| Spacetime is a fixed, smooth, globally Lorentzian continuum (Minkowski) | likely-fundamental | `[INFERENCE/OPEN]` A fixed classical background is a spectacularly successful input but is expected to fail at the Planck scale where geometry is dynamical. The continuum is also idealized: whether a nonperturbative continuum limit exists is the constructive-QFT/triviality question. |
| Microcausality: spacelike-separated local observables commute | fundamental | `[ESTABLISHED as defining axiom]` The precise encoding of relativistic causality and no-signaling; drives spin–statistics, CPT, Reeh–Schlieder. The single most load-bearing, least negotiable assumption — relaxing it generically breaks Lorentz invariance or causality. |
| Poincaré invariance, unique stable vacuum, energy bounded below (spectrum condition) | fundamental | `[ESTABLISHED/INFERENCE]` Lorentz invariance is tested to extraordinary precision and grounds Wigner's classification; positive energy is required for stability. Globally exact only in flat space — an excellent *local* approximation cosmologically. Tiny Lorentz violation is `[OPEN]`, constrained but not excluded. |

| | | |
|---|---|---|
| Renormalizability as a selection criterion for fundamental Lagrangians | historical-artifact | `[ESTABLISHED reinterpretation]` Once a sacred requirement; Wilson's RG showed it is an *emergent* low-energy property (irrelevant operators are $E/\Lambda$-suppressed). Predictive non-renormalizable EFTs (chiral PT, Fermi theory, gravity-as-EFT) abound. A derived/organizational fact, not an axiom. |
| Existence of a well-defined functional measure $\mathcal{D}\phi$ and of the theory it defines | unclear | `[CONTESTED/OPEN]` The Lorentzian "measure" is not rigorous in $d = 4$; even the Euclidean measure is constructed only for super-renormalizable/low-$d$ models ($\phi^4_2$, $\phi^4_3$, 2D Yang–Mills, Gross–Neveu). For 4D QCD/electroweak no rigorous construction exists (Clay Millennium Problem). A phenomenally successful heuristic with incomplete foundations where it matters most. |
| The gauge "symmetry" is a fundamental interaction-generating principle | conventional-choice | `[CONTESTED interpretation]` Hugely productive, but gauge symmetry is a *redundancy*, not a physical symmetry (Elitzur forbids breaking it; observables are gauge-invariant). The physical core (massless spin-1 consistency, charge conservation) is fundamental; "derive interactions by gauging" is a heuristic — the bundle/connection geometry and Wilson loops are the invariant content. |
| Asymptotic in/out free states and a well-defined S-matrix exist (LSZ, asymptotic completeness) | likely-fundamental | `[INFERENCE]` Justified by LSZ and Haag–Ruelle scattering given isolated mass shells and a mass gap. Fails for confined theories (no free quarks), massless particles (IR divergences; coherent states), and CFTs (no particles). Asymptotic completeness is assumed, not generally proven. Fundamental where applicable, but restricted in domain. |

| | | |
|---|---|---|
| Wick rotation / OS reconstruction faithfully relates Euclidean and Lorentzian QFT | unclear | `[ESTABLISHED where conditions hold]` OS reconstruction is a theorem given reflection positivity; it underlies all rigorous and lattice work. But reflection positivity can fail (finite chemical potential — the sign problem) and real-time/non-equilibrium dynamics is not directly accessible. A powerful device of uncertai universal reach. |
| Unitarity (probability conservation) of evolution / the S-matrix | fundamental | `[ESTABLISHED]` Required for a consistent probabilistic quantum theory; enforced via BRST/Ward identities and the optical theore Apparent unitarity loss in effective/open descriptions reflects truncation, not failure of the closed theory. Non-negotiable withi QM. |
| A single fixed Hilbert (Fock) space with finite field content suffices | unclear | `[ESTABLISHED limitation — a convenience that is strictly fals` Haag's theorem: the interacting theory does *not* live in the free Fo space; inequivalent representations are generic (Stone–von Neumann uniqueness fails for infinitely many d.o.f.). AQFT repla the privileged representation with a net of algebras. |

See *Chapter 20 (p. 458)* for the cross-domain ledger and *Chapter 2 (p. 8)* for marker definitions.

## Domain of validity

QFT on a fixed (typically flat) spacetime is validated over an extraordinary range. `[ESTABLISHED]` QED predicts the electron anomalous magnetic moment to roughly twelve significant figures — among the most precise theory–experiment agreements in all of science. The electroweak Standard Model and perturbative QCD describe collider physics into the few-TeV regime probed at the LHC with no confirmed deviation. Lattice QCD computes the nonperturbative low-energy hadron spectrum from first principles. The same RG/EFT machinery quantitatively governs condensed-matter critical phenomena (see *Chapter 6 (p. 56)*) and is the working language across particle, nuclear, statistical, and condensed-matter physics. QFT is, in the current understanding, the *unique* known way to reconcile quantum mechanics (see *Chapter 7 (p. 71)*) with special relativity.

Boundaries of validity:

1. **Energy/UV.** Standard QFT presupposes a fixed classical spacetime and is expected to require replacement near $M_{\mathrm{Pl}} \sim 1.2 \times 10^{19}$ GeV, where gravitational backreaction makes the metric quantum. The SM is widely regarded as an EFT with a cutoff below $M_{\mathrm{Pl}}$, possibly far below `[INFERENCE]`.
2. **Gravity.** General relativity quantized as a QFT is a valid EFT at low energy but is perturbatively non-renormalizable, losing predictivity at $M_{\mathrm{Pl}}$ — the clearest internal signal of an upper validity bound. `[ESTABLISHED that GR-as-EFT works at low energy; OPEN what completes it` See *Chapter 10 (p. 121)*.
3. **Curved/dynamical spacetime.** QFT in curved backgrounds is well-developed (Hawking radiation, Unruh effect), but "particle" becomes observer-dependent and there is no preferred vacuum; full back-reacting quantum gravity lies outside the framework.
4. **Coupling/UV self-consistency.** QED and $\phi^4$ have Landau poles and are believed *trivial* (only the free theory survives the continuum limit) `[INFERENCE from lattice + perturbation theory]`, so they are EFTs with a finite UV cutoff, not complete theories.
5. **Confinement/IR.** For QCD the perturbative quark–gluon description fails near $\Lambda_{\mathrm{QCD}} \sim 200$ MeV; the correct degrees of freedom are hadrons, requiring nonperturbative (lattice) methods.
6. **Rigor.** Outside a list of super-renormalizable and low-dimensional models, no physically realistic 4D interacting QFT has been constructed to full mathematical rigor.

## Where it breaks down

Distinguishing the *kinds* of breakdown is essential (see *Chapter 14 (p. 191)*):

- **Quantum gravity / Planck scale** — `[ESTABLISHED breakdown, domain mismatch]`. Treating the metric as a quantum field gives a non-renormalizable EFT; graviton loops need infinitely many counterterms and predictivity fails at $M_{\mathrm{Pl}}$. This is a domain-of-validity mismatch (fixed background cannot host fluctuating geometry), *not* an internal inconsistency, and is the sharpest signpost of UV incompleteness.

- **Cosmological constant / vacuum energy** — `[OPEN, severe]`. Summed zero-point energies plus known condensates predict a vacuum energy vastly larger than the observed dark-energy scale (~120 orders of magnitude with a Planck cutoff; tens of orders even at the electroweak scale). Because it concerns how vacuum energy *gravitates*, it sits at the QFT/gravity interface and may signal that the naive QFT vacuum-energy estimate is the wrong object. See *Chapter 11 (p. 139)* and *Chapter 15 (p. 233)*.
- **Haag's theorem** — `[ESTABLISHED internal tension]`. The interaction picture does not exist (free and interacting representations are unitarily inequivalent), so the textbook perturbative starting point is mathematically inconsistent — yet renormalized perturbation theory gives correct answers. The resolution (an asymptotic recipe; rigorous objects are algebras/superselection sectors; a proper adiabatic/IR treatment evades the theorem) shows this is a foundational subtlety, not a falsification.
- **Triviality / Landau poles** — `[INFERENCE, strong]`. $\phi_4^4$ and QED couplings grow to a Landau pole; lattice and rigorous bounds indicate the only continuum limit is free. These are EFTs with an intrinsic cutoff — a consistent statement about domain of validity, refuting the idea that every renormalizable theory is UV-complete.
- **Constructive gap (Yang–Mills existence + mass gap)** — `[OPEN]`. No rigorous construction of 4D interacting Yang–Mills (or full QCD/electroweak) exists; existence + mass gap is a Clay Millennium Problem. Lattice evidence for the gap is strong; a proof satisfying the Wightman/OS axioms is missing. Unsolved-but-believed-consistent.
- **Measurement in relativistic QFT** — `[OPEN/CONTESTED]`. The measurement problem persists, sharpened by relativity: naive

projective state-update across spacelike slices is frame-dependent and can yield Sorkin's "impossible measurements" (superluminal signaling), while Reeh–Schlieder gives local operations global reach. No-signaling is preserved by microcausality, and consistent local-measurement frameworks exist (Fewster–Verch), but a fully agreed covariant account of state update is not settled. See *Chapter 7 (p. 71)*.

- **Reeh–Schlieder nonlocality vs. operational locality** — `[ESTABLISHED but counterintuitive]`. The vacuum is entangled across arbitrarily separated regions and cyclic/separating for every local algebra. This does not contradict no-signaling, but it forces the type III von Neumann algebra picture in which there is no local tensor factorization and no local number operator.
- **IR structure of gauge theories** — `[ESTABLISHED, controlled]`. Massless particles (photons, gluons) generate soft/collinear divergences; individual S-matrix elements are ill-defined and Fock asymptotic states are not the right notion. The cure (KLN theorem, inclusive cross sections, coherent/dressed states, soft-theorem/asymptotic-symmetry program) restores finite predictions while exposing the boundary of the naive particle idealization.
- **Nonperturbative/confining regime** — `[ESTABLISHED domain shift]`. Perturbation theory is at best asymptotic and fails outside asymptotic freedom; below ~1 GeV the quark–gluon variables are simply wrong, and one must use lattice or effective hadronic theories.

## Open problems (internal)

These are problems *internal* to QFT as a framework; cross-cutting items appear in *Chapter 15 (p. 233)*.

1. **Rigorous 4D construction: Yang–Mills existence and mass gap** `[OPEN]`. The flagship gap between empirical success and mathematical foundation (Clay Millennium Problem).
2. **UV completeness vs. EFT status** `[INFERENCE/OPEN]`. Triviality of $\phi^4$/QED suggests the SM Higgs and hypercharge

sectors need a UV completion. Whether asymptotic safety (a nontrivial UV fixed point) could rescue any sector — including gravity — is unresolved.

3. **Cosmological constant problem** `[OPEN; arguably the deepest in physics]`. Why is the observed vacuum energy so far below QFT estimates? Proposed resolutions (SUSY, anthropic/landscape, sequestering, modified gravity) are all `[SPECULATIVE]`.
4. **Covariant measurement theory** `[OPEN/CONTESTED]`. A complete, interpretation-neutral, manifestly covariant account of measurement/state-update.
5. **Nonperturbative dynamics** `[OPEN]`. An analytic confinement proof; real-time and finite-density QCD obstructed by the sign problem (limiting first-principles access to neutron-star interiors and the early-universe QCD transition).
6. **Inequivalent representations / type III algebras** `[OPEN]`. The right notion of local subsystems and finite, physical entanglement entropy in QFT (the naive region entropy is UV-divergent; modular Hamiltonians and relative entropy are clean but incomplete, especially with gravity/holography).
7. **Asymptotic safety in $d \geq 4$** `[SPECULATIVE/OPEN]`. Functional-RG evidence for an asymptotically safe gravity fixed point exists but is truncation-dependent and unproven.
8. **Continuum/locality vs. expected Planck-scale structure** `[OPEN]`. Whether spacetime, fields, and locality are fundamental or emergent (entanglement/holography/strings) bears directly on which axioms above are truly fundamental.

## Connections to other frameworks

- ***Quantum mechanics*** **(Chapter 7 (p. 71))** — QFT contains QM as a low-energy/single-particle limit and as the $d = 0{+}1$ case; conversely the Stone–von Neumann uniqueness of QM *fails* in QFT, the technical root of Haag's theorem, inequivalent vacua, and SSB. The measurement problem is inherited and sharpened.

- **Special relativity** (see *Chapter 4 (p. 25)*, *Chapter 10 (p. 121)*) — Poincaré invariance and microcausality are constitutive; relativity is what forces fields, antiparticles, particle creation, spin–statistics, and CPT.
- ***General relativity / quantum gravity* (Chapter 10 (p. 121))** — Clash at the deepest level: QFT presupposes fixed spacetime, GR makes it dynamical. GR is an excellent low-energy EFT, non-renormalizable at $M_{\mathrm{Pl}}$. The interface produces the hardest problems (cosmological constant, black-hole information, Hawking radiation) and motivates string theory, loop quantum gravity, and holography.
- ***Statistical mechanics* (Chapter 6 (p. 56))** and ***Chapter 5 (p. 40)*** — Deep formal equivalence via Wick rotation: Euclidean QFT = classical statistical field theory; the RG, universality, critical exponents, and relevant/irrelevant operators are literally shared. The path integral is a partition function; the Unruh/Hawking effects connect modular flow to temperature.
- **Condensed matter / many-body physics** — Two-way street: Green's functions, diagrams, RG, and gauge theory describe quasiparticles, superconductivity (Anderson/Nambu route to the Higgs mechanism via SSB of EM gauge symmetry), the quantum Hall effect, and topological phases. Emergent gauge symmetry and emergent Lorentz invariance suggest fundamental QFT structures may themselves be emergent (see *Chapter 17 (p. 325)*).
- ***Mathematics* (Chapter 13 (p. 172))** — QFT both consumes and generates mathematics: Tomita–Takesaki modular theory and von Neumann algebra classification (AQFT); index theorems and characteristic classes (anomalies); fiber-bundle geometry (gauge fields); representation theory of the Poincaré and conformal groups; and conjectural structures (mirror symmetry, Chern–Simons knot invariants, Langlands) that emerged from QFT reasoning.
- **Conformal field theory and the bootstrap** — CFTs are QFTs at RG fixed points; they anchor the space of QFTs (UV/IR endpoints of flows), are the most rigorously controlled

interacting QFTs (especially 2D), and via AdS/CFT relate to quantum gravity — a key node toward the gravity frontier.

- ***Particle physics* (Chapter 9 (p. 106))** and ***Chapter 11 (p. 139)*** — The Standard Model is QFT's empirical embodiment; its open phenomenology (neutrino masses, dark matter, baryon asymmetry, strong-CP, hierarchy) drives experiment, and early-universe cosmology (inflation, baryogenesis, the QCD/electroweak transitions) is QFT applied to the whole universe, looping back to vacuum-energy and measurement issues.
- ***Information theory* (Chapter 12 (p. 155))** — Entanglement entropy, modular Hamiltonians, and relative entropy of local algebras connect QFT to quantum information and, via holography, to the area law and the emergence of geometry.

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CHAPTER 9

# Particle Physics and the Standard Model

The Standard Model (SM) is the relativistic quantum field theory of the electromagnetic, weak, and strong interactions: a renormalizable, anomaly-free **chiral gauge theory** with gauge group $G_{SM} = SU(3)_c \times SU(2)_L \times U(1)_Y$, spontaneously broken to $SU(3)_c \times U(1)_{em}$ by the Higgs mechanism. It is the most precisely tested theory in the physical sciences and simultaneously, by broad consensus, an *effective* description awaiting a deeper completion.

## Scope

This page covers the gauge structure and fermion representations; the three-generation pattern; electroweak symmetry breaking (EWSB) and the Higgs boson; QCD with asymptotic freedom and confinement; flavor physics (CKM, PMNS) and CP violation; neutrino masses, oscillations, and the Dirac-vs-Majorana question; the free-parameter count; the hierarchy/naturalness and strong-CP problems; baryogenesis; beyond-SM (BSM) frameworks (SUSY, GUTs, extra dimensions, SMEFT); and the current anomaly landscape.

It **excludes** the detailed dynamics of quantum gravity (a neighboring domain — see *Chapter 10 (p. 121)*), nuclear structure, and condensed-matter applications of QFT, except where the latter illuminate SM concepts. For the underlying field-theoretic machinery (path integrals, renormalization, the renormalization group), see *Chapter 8 (p. 89)*; for foundational quantum structure, *Chapter 7 (p. 71)*.

# Core formalism

## 1. Gauge structure and field content

The SM is a Yang–Mills theory with local symmetry $G_{SM}$ and three gauge couplings $g_s, g, g'$. With non-abelian field strengths

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s f^{abc} G^b_\mu G^c_\nu \quad (a = 1, \ldots, 8),$$

$$W^i_{\mu\nu} = \partial_\mu W^i_\nu - \partial_\nu W^i_\mu + g\, \epsilon^{ijk} W^j_\mu W^k_\nu \quad (i = 1, 2, 3), \qquad B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu,$$

the gauge Lagrangian is $\mathcal{L}_{\text{gauge}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\,\mu\nu} - \frac{1}{4} W^i_{\mu\nu} W^{i\,\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu}$ `[ESTABLISHED]`.

The fermions come in three generations of identical quantum numbers. Per generation, the irreducible chiral content with charges $(SU(3)_c, SU(2)_L)_Y$ is `[ESTABLISHED]`:

- $Q_L = (u_L, d_L)^T \sim (3, 2)_{1/6}$
- $u_R \sim (3, 1)_{2/3}, \quad d_R \sim (3, 1)_{-1/3}$
- $L_L = (\nu_L, e_L)^T \sim (1, 2)_{-1/2}$
- $e_R \sim (1, 1)_{-1}$
- (no right-handed neutrino $\nu_R$ in the *minimal* SM).

Electric charge is $Q = T_3 + Y$ in this hypercharge convention; many texts instead write $Q = T_3 + Y/2$ with rescaled hypercharges (a [conventional-choice] bookkeeping difference, not physics). The decisive structural fact is **chirality**: left-handed fields are $SU(2)_L$ doublets while right-handed fields are singlets, so the weak interaction maximally violates parity (the V−A structure) `[ESTABLISHED]`. Gauge interactions enter only through the covariant derivative

$$D_\mu = \partial_\mu - ig_s\, G^a_\mu T^a - ig\, W^i_\mu T^i - ig' Y B_\mu,$$

with $T^a = \lambda^a/2$ (Gell-Mann) and $T^i = \sigma^i/2$ (Pauli) on the appropriate representations; the fermion kinetic term is $\mathcal{L}_{\text{ferm}} = \sum_\psi \bar{\psi}\, i\gamma^\mu D_\mu\, \psi$. Once the group and representations are fixed, the interactions are essentially determined by gauge invariance — a key economy of the construction `[ESTABLISHED]`.

**Anomaly cancellation.** Classical gauge symmetry must survive quantization: the triangle (Adler–Bell–Jackiw) anomalies must vanish. The conditions $[SU(3)]^2U(1)$, $[SU(2)]^2U(1)$, $[U(1)]^3$, and the mixed gravitational–$U(1)$ anomaly $\sum Y = 0$ are each satisfied *exactly, per generation*, by the hypercharges above `[ESTABLISHED]`. This ties quark and lepton charges together and effectively requires complete generations — a remarkably rigid consistency constraint.

## 2. Electroweak symmetry breaking (Higgs mechanism)

A complex scalar doublet $\phi \sim (1,2)_{1/2}$ has

$$\mathcal{L}_\phi = (D_\mu\phi)^\dagger(D^\mu\phi) - V(\phi), \qquad V(\phi) = -\mu^2\,\phi^\dagger\phi + \lambda\,(\phi^\dagger\phi)^2,$$

with $\mu^2 > 0$, $\lambda > 0$. The wrong-sign mass term drives a vacuum expectation value $\langle\phi\rangle = \frac{1}{\sqrt{2}}(0,\,v)^T$, with $v = \mu/\sqrt{\lambda} \approx 246$ GeV, breaking $SU(2)_L \times U(1)_Y \to U(1)_{em}$ `[ESTABLISHED]`. In unitary gauge $\phi = \frac{1}{\sqrt{2}}(0,\,v+h)^T$.

Three would-be Goldstone bosons are eaten by the gauge fields. The mass eigenstates are

$$W^\pm_\mu = \tfrac{1}{\sqrt{2}}(W^1_\mu \mp iW^2_\mu), \qquad \begin{pmatrix} Z_\mu \\ A_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & -\sin\theta_W \\ \sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} W^3_\mu \\ B_\mu \end{pmatrix},$$

with $\tan\theta_W = g'/g$, giving

$$m_W = \tfrac{1}{2}gv, \qquad m_Z = \tfrac{1}{2}v\sqrt{g^2+g'^2} = \frac{m_W}{\cos\theta_W}, \qquad m_\gamma = 0.$$

At tree level $\rho \equiv m_W^2/(m_Z^2\cos^2\theta_W) = 1$, a consequence of the **custodial** $SU(2)$ symmetry of the doublet potential; the measured $\rho$ agrees with unity at the per-mille level after radiative corrections — a genuine precision triumph `[ESTABLISHED]`. The physical scalar has $m_h = \sqrt{2\lambda}\,v = \sqrt{2}\,\mu \approx 125$ GeV, discovered by ATLAS and CMS in 2012 `[ESTABLISHED]`.

**Yukawa couplings** generate fermion masses (with $\tilde{\phi} = i\sigma_2\phi^*$):

$$-\mathcal{L}_Y = Y^d_{ij}\bar{Q}_{Li}\,\phi\,d_{Rj} + Y^u_{ij}\bar{Q}_{Li}\,\tilde{\phi}\,u_{Rj} + Y^e_{ij}\bar{L}_{Li}\,\phi\,e_{Rj} + \text{h.c.}$$

After EWSB, $m^f = Y^f v/\sqrt{2}$. Bi-unitary diagonalization of $Y^{u,d,e}$ rotates from flavor to mass eigenstates; the residual misalignment in the quark sector is the CKM matrix.

## 3. Flavor and CP violation

In the mass basis the charged current reads

$$\mathcal{L}_{cc} = \frac{g}{\sqrt{2}}\,W^+_\mu \bar{u}_{Li}\gamma^\mu (V_{CKM})_{ij} d_{Lj} + \text{h.c.},$$

with $V_{CKM} = U_L^{u\dagger} U_L^d$ a $3 \times 3$ unitary matrix carrying 3 mixing angles + 1 CP-violating phase (Kobayashi–Maskawa). CP violation in the SM requires $\geq 3$ generations and is captured by the rephasing-invariant Jarlskog quantity $J = \mathrm{Im}(V_{us}V_{cb}V^*_{ub}V^*_{cs}) \approx 3 \times 10^{-5}$ `[ESTABLISHED]`; the Wolfenstein parametrization expands in $\lambda \approx 0.225$. The leptonic analog (with massive neutrinos) is the PMNS matrix $U_{PMNS}$: 3 angles + 1 Dirac phase, plus 2 Majorana phases if neutrinos are Majorana.

Neutral currents couple to $J_Z^\mu \propto (T_3 - Q\sin^2\theta_W)$ and are flavor-diagonal at tree level; flavor-changing neutral currents (FCNC) are suppressed by the **GIM mechanism**, a structural prediction confirmed across kaon, charm, and $B$ physics `[ESTABLISHED]`.

## 4. QCD: asymptotic freedom and confinement

QCD is the $SU(3)_c$ Yang–Mills theory with quarks in the fundamental **3**. Its defining property is the running coupling. At one loop,

$$\mu\frac{d\alpha_s}{d\mu} = \beta(\alpha_s) = -\frac{\beta_0}{2\pi}\alpha_s^2 + \cdots, \qquad \beta_0 = 11 - \tfrac{2}{3}n_f.$$

For $n_f < 33/2$, $\beta_0 > 0$: the coupling **decreases** at high energy (asymptotic freedom; Gross–Wilczek–Politzer 1973 `[ESTABLISHED]`), making perturbative QCD predictive at colliders, and **grows** at low energy, generating a dynamical scale $\Lambda_{QCD} \sim 200$ MeV by dimensional transmutation:

$$\alpha_s(\mu) = \frac{2\pi}{\beta_0 \ln(\mu/\Lambda_{QCD})}.$$

The non-abelian gluon self-coupling (the "11") overwhelms quark screening (the "$n_f$") — the physical origin of the sign.

**Confinement** — that asymptotic states are color singlets and isolated colored quanta do not propagate — is an empirically certain fact, strongly supported by lattice QCD (area-law Wilson loops, a linear static potential $V(r) \sim \sigma r$) `[ESTABLISHED]`. A rigorous continuum analytic proof (the Yang–Mills mass-gap Clay Millennium Problem) remains `[OPEN]`. Spontaneous breaking of the light-quark chiral symmetry $SU(2)_L \times SU(2)_R \to SU(2)_V$ produces the pions as pseudo-Goldstone bosons and supplies most of the proton mass through QCD binding energy rather than Higgs Yukawa mass `[ESTABLISHED]`.

## 5. The theta term and strong CP

QCD admits a gauge-invariant, renormalizable term

$$\mathcal{L}_\theta = \theta \frac{g_s^2}{32\pi^2} G^a_{\mu\nu} \tilde{G}^{a\,\mu\nu},$$

a total derivative made physical by instantons. The observable parameter is $\bar{\theta} = \theta + \arg\det(Y^u Y^d)$, which would induce a neutron electric dipole moment; the experimental EDM bound forces $\bar{\theta} \lesssim 10^{-10}$`[ESTABLISHED bound]`. Nothing in the SM explains this smallness — the **strong-CP problem** (see §"Where it breaks down").

## 6. Quantization, renormalizability, and anomalies

The theory is quantized via the gauge-fixed path integral with Faddeev–Popov ghosts (or, covariantly, the BRST formalism). 't Hooft and Veltman proved that spontaneously broken non-abelian gauge theories are renormalizable, making the electroweak sector a predictive quantum theory `[ESTABLISHED]`. Observables are organized as a coupling expansion with renormalization-group running of all parameters. The accidental global symmetries $U(1)_B$ and $U(1)_L$ are individually anomalous under $SU(2)_L$ (only $B - L$ is anomaly-free); $B+L$ is violated nonperturbatively by **sphalerons**, a fact central to baryogenesis `[ESTABLISHED]`.

### Compact summary

$$\mathcal{L}_{SM} = -\tfrac{1}{4}\sum F_{\mu\nu}F^{\mu\nu} + \bar{\psi}\, i\gamma^{\mu} D_{\mu}\psi + |D_{\mu}\phi|^2 - V(\phi) - \left(Y_{ij}\,\bar{\psi}_{Li}\phi\,\psi_{Rj} + \text{h.c.}\right)\ (+\,\mathcal{L}_{\theta}).$$

Every confirmed non-gravitational phenomenon below the TeV scale follows from this Lagrangian. The leading window beyond it is the unique **dimension-5 Weinberg operator**

$$\mathcal{L}_5 = \frac{c_{ij}}{\Lambda}\,(\bar{L}^c_{Li}\tilde{\phi}^*)(\tilde{\phi}^{\dagger}L_{Lj}) + \text{h.c.},$$

which after EWSB gives Majorana neutrino masses $m_{\nu} \sim v^2/\Lambda$, naturally small for large $\Lambda$ (seesaw) `[INFERENCE]`.

## Foundational assumptions

| ASSUMPTION | STATUS | JUSTIFICATION |
|---|---|---|
| Nature is a local, Lorentz-invariant, unitary relativistic QFT | likely-fundamental | `[ESTABLISHED]` to extraordinary precision (electron $g-2$ to ~12 sig. figs.); locality + Lorentz + unitarity are tied together by spin-statistics and CPT. `[SPECULATIVE]` whether truly fundamental — QFT may be emergent (string theory, holography), and quantum gravity likely modifies locality near $M_{Pl}$. |
| Gauge group is exactly $SU(3)_c \times SU(2)_L \times U(1)_Y$ | likely-fundamental | `[ESTABLISHED]` as the low-energy description. The product form and hypercharge normalization look unifiable: $SU(5)$, $SO(10)$, Pati–Salam embed it, and coupling running hints at near-unification (imperfect in SM, better in MSSM) near $10^{16}$ GeV. Possibly a low-energy shadow of a simple group. `[INFERENCE]` |

| | | |
|---|---|---|
| The hypercharge/representation assignments (e.g. $Q_L \sim (3,2)_{1/6}$) | fundamental | `[ESTABLISHED]` fixed by anomaly cancellation plus measured charges. But *why* charge is quantized (proton/electron charge equality to ~$10^{-21}$) is not explained internally; GUT embedding explains it. The values are fixed; the reason points beyond the SM. |
| Exactly three generations with identical gauge numbers | unclear | `[ESTABLISHED]` empirically: the invisible $Z$ width gives 3 light active neutrinos with standard couplings, and a 4th chiral generation is excluded by Higgs data. *Why* three is `[OPEN]` — possibly flavor symmetry, extra-dimensional geometry, or contingency. |
| EWSB is driven by a single elementary scalar doublet | conventional-choice | `[ESTABLISHED]` that a ~125 GeV scalar with SM-like couplings exists and that EWSB occurs. Minimality is a choice: 2HDM, composite/pseudo-Goldstone Higgs, technicolor are alternatives. Elementarity is exactly what creates the hierarchy problem. `[CONTESTED]` whether elementary. |
| Neutrinos are massless (no $\nu_R$, exact lepton number) | historical-artifact | `[ESTABLISHED]` FALSE: oscillations prove nonzero mass. A minimality assumption of the original SM, now known wrong; minimally fixed by the Weinberg operator or added $\nu_R$. The clearest case of a falsified original assumption. |

| | | |
|---|---|---|
| Neutrino masses are Dirac (vs Majorana) | unclear | [OPEN/CONTESTED] genuinely unknown. Majorana (seesaw) elegantly explains lightness and is theoretically favored by many, but neutrinoless double-beta decay is unobserved. Dirac requires a tiny Yukawa or new structure to forbid the Majorana term. |
| The bare QCD $\theta$ is (or relaxes to) ~0 | unclear | [OPEN] strong-CP problem: nothing forces $\bar{\theta} \sim 0$; it is a free parameter $< 10^{-10}$. Possible resolutions: anthropic/contingent, a massless up quark (disfavored by lattice), or a Peccei–Quinn axion [SPECULATIVE but well-motivated]. |
| ~19 free parameters fixed only by experiment | likely-fundamental | [ESTABLISHED] counting fact for the minimal SM (3 gauge, 2 Higgs, 9 charged-fermion masses, 4 CKM, $\bar{\theta}$); rises to ~26–28 with neutrino masses. The vast hierarchies (e.g. top/electron Yukawa $\sim 10^5$) suggest an underlying flavor theory — the most "descriptive, not explanatory" aspect of the SM. |
| Spacetime is fixed 4D Minkowski; gravity neglected | conventional-choice | [ESTABLISHED] valid within the SM's domain (gravitational coupling $\sim (E/M_{Pl})^2$ is negligible at colliders). A deliberate restriction, not an error — but precisely the boundary where the SM must be completed, since gravity is non-renormalizable. |
| $B$ and $L$ are (accidental, approximate) global symmetries | historical-artifact | [ESTABLISHED] accidental symmetries of the renormalizable Lagrangian, not imposed. Anomalous (only $B-L$ survives), violated by sphalerons, and $B$ violation is *required* for baryogenesis. Conservation is an artifact of restricting to low-dimension operators. |

See *Chapter 20 (p. 458)* for the cross-domain ledger and *Chapter 2 (p. 8)* for marker definitions.

## Domain of validity

The SM is an effective field theory validated from sub-eV scales (atomic physics, neutrino oscillations) up to the ~TeV scale directly probed at the LHC, and indirectly — via precision electroweak fits and rare processes — well above it. Within this range it is the most precisely tested theory in science: QED predicts the electron anomalous magnetic moment to ~10–12 significant figures, and $Z$-pole observables, $m_W$, and Higgs couplings fit at the per-mille-to-percent level `[ESTABLISHED]`.

Its ultraviolet cutoff is at most the Planck scale $M_{Pl} \sim 1.2 \times 10^{19}$ GeV, where quantum gravity becomes strong and the QFT framework itself is expected to fail `[INFERENCE]`. Other plausible intermediate cutoffs include a GUT scale $\sim 10^{16}$ GeV (unification, proton decay), a seesaw scale $\sim 10^9$–$10^{15}$ GeV (neutrino mass), or — on naturalness grounds for the Higgs sector — new physics near the TeV scale `[SPECULATIVE]`. The SM also applies only to perturbatively accessible regimes for electroweak/QED processes; QCD is nonperturbative below ~1 GeV, where lattice methods or effective theories (chiral perturbation theory, HQET, SCET) are required `[ESTABLISHED]`. It says nothing about dark matter, dark energy, gravity, or cosmological initial conditions. See *Chapter 16 (p. 308)*.

## Where it breaks down

- **Neutrino masses and oscillations** — `[ESTABLISHED]`. The minimal SM predicts massless neutrinos; solar, atmospheric, reactor, and accelerator experiments decisively observe oscillations, with probability

  $$P(\nu_\alpha \to \nu_\beta) = \left| \sum_i U_{\alpha i} U^*_{\beta i} e^{-im_i^2 L/2E} \right|^2,$$

  nonzero only for massive, mixed neutrinos. This is the *only laboratory-confirmed* breakdown — an **incompleteness** (missing fields/operators), not an internal inconsistency, repaired by the Weinberg operator or $\nu_R$.
- **Gravity and the UV cutoff** — `[ESTABLISHED]`. The SM excludes gravity; perturbatively quantized GR is non-renormalizable. A **domain-of-validity mismatch**, not an internal contradiction, but it guarantees the SM is not fundamental. See *Chapter 10 (p. 121)*.

- **Hierarchy / naturalness** — `[OPEN, not an inconsistency]`. The Higgs mass-squared is quadratically sensitive to the cutoff, $\delta m_h^2 \sim \Lambda^2/16\pi^2$; with $\Lambda \sim M_{Pl}$, obtaining $m_h = 125$ GeV requires tuning to ~1 part in $10^{34}$. The theory is fully consistent (the sensitivity is absorbed into a counterterm); the "problem" is a naturalness *prior* that an elementary scalar's mass should not lie far below the cutoff without a protecting symmetry. `[CONTESTED]` whether this is a genuine problem.
- **Strong CP** — `[OPEN, not an inconsistency]`. $\bar{\theta} < 10^{-10}$ is allowed but unexplained; the SM is consistent for any $\bar{\theta}$.
- **Baryon asymmetry** — `[ESTABLISHED problem]`. The observed $\eta \sim 6 \times 10^{-10}$ cannot be generated by the SM: although the Sakharov conditions are in principle met (sphaleron $B$ violation, C/CP violation, departure from equilibrium), the SM CP violation is far too small and, for $m_h = 125$ GeV, the electroweak transition is a smooth crossover rather than first-order. An **incompleteness** requiring new CP violation and/or dynamics. See *Chapter 11 (p. 139)*.
- **Dark matter and dark energy** — `[ESTABLISHED]`. The SM offers no viable dark-matter candidate (neutrinos are too light/hot) and no account of the cosmological constant. Pure **incompleteness**, established by astrophysical/cosmological observation.
- **Landau pole and vacuum (meta)stability** — `[INFERENCE]`. The hypercharge $U(1)_Y$ has a Landau pole far above $M_{Pl}$ (academic in practice, but signaling the abelian sector is not UV-complete). Separately, RG running of $\lambda$ with the measured top and Higgs masses drives it slightly negative near $\sim 10^{10}$–$10^{11}$ GeV, implying a **metastable** (extremely long-lived) electroweak vacuum; the precise numbers are `[CONTESTED]`, sensitive to the top mass.
- **Flavor hierarchies** — `[OPEN, not an inconsistency]`. The SM accommodates but does not explain the Yukawa/mixing hierarchies, the generation count, or why CKM is nearly diagonal while PMNS has large angles. Descriptive, not explanatory.

For the inconsistency-vs-incompleteness taxonomy used above, see *Chapter 14 (p. 191)*.

## Open problems (internal)

- **The flavor puzzle** `[OPEN]`: no accepted theory explains the Yukawa hierarchies, $N_{\text{gen}} = 3$, or the CKM/PMNS patterns. Many `[SPECULATIVE]` frameworks (Froggatt–Nielsen, discrete flavor symmetries, extra-dimensional localization) exist; none is confirmed.
- **Hierarchy/naturalness** `[OPEN/CONTESTED]`: SUSY, composite Higgs, and extra dimensions were proposed as natural solutions; the LHC excludes the simplest TeV-scale versions, sharpening the debate over whether naturalness is the right guide versus anthropic/landscape reasoning.
- **Dirac vs Majorana neutrinos; absolute mass scale and ordering** `[OPEN]`: decided experimentally by neutrinoless double-beta decay (Majorana / lepton-number violation) and by cosmology/$\beta$-decay endpoints (absolute mass). The mass ordering and the leptonic phase $\delta_{CP}$ are being measured but not definitively resolved as of the knowledge cutoff.
- **Strong CP and the axion** `[OPEN]`: the Peccei–Quinn/axion mechanism is the most elegant `[SPECULATIVE]` proposal — and the axion is a dark-matter candidate — but searches (ADMX and others) have not detected it.
- **Baryogenesis** `[OPEN]`: requires beyond-SM CP violation and/or out-of-equilibrium dynamics; leptogenesis tied to heavy Majorana neutrinos is a leading `[INFERENCE/SPECULATIVE]` candidate but hard to test directly.
- **Confinement / Yang–Mills mass gap** `[OPEN]`: empirically certain and lattice-supported, but lacking a rigorous continuum proof (a Clay Millennium Problem). An internal hard problem, not an inconsistency.
- **Persistent anomalies** `[CONTESTED/OPEN, and shrinking]`: the muon $g-2$ discrepancy is now **largely resolved into SM-consistency** — the 2025 Muon $g-2$ Theory Initiative white

paper (arXiv:2505.21476) adopts a lattice-QCD HVP average, giving $a_\mu^{\rm SM} = 116\,592\,033(62) \times 10^{-11}$ and $\Delta a_\mu = 38(63) \times 10^{-11}$ (**no tension**), though a residual *internal* lattice-vs-$R$-ratio (CMD-3) HVP discrepancy persists as a theory/systematics question, not a BSM signal `[ESTABLISHED, 2025]`. The $B$-meson lepton-universality ratios $R_K, R_{K^*}$ have largely moved toward SM consistency with updated LHCb data, while some $b \to s\mu\mu$ angular/branching observables remain in mild tension. None has reached unambiguous discovery-level BSM significance.

- **Gauge unification and proton decay** `[OPEN/INFERENCE]`: couplings nearly unify near $\sim 10^{16}$ GeV (better with SUSY), motivating GUTs, which generically predict proton decay; the absence of observed decay (lifetime $>\sim 10^{34}$ yr) excludes minimal $SU(5)$ and constrains models.

A consolidated, ranked list lives in *Chapter 15 (p. 233)*; candidate resolutions in *Chapter 19 (p. 392)*.

## Connections to other frameworks

- **General relativity / quantum gravity** — clash and incompleteness. The SM uses a fixed background; GR makes spacetime dynamical, and naive quantization of GR is non-renormalizable, so neither is fundamental in present form. They coexist as an EFT below $M_{Pl}$; reconciliation requires quantum gravity (string theory, loop quantum gravity, asymptotic safety — all `[SPECULATIVE]`). See *Chapter 10 (p. 121)* and *Chapter 18 (p. 349)*.
- **Cosmology / early universe** — deep dependence and tension. The SM supplies the thermal bath, the electroweak and QCD transitions, and sphaleron processes relevant to baryogenesis and big-bang nucleosynthesis; cosmology in turn demands inflation, dark matter, dark energy, and a working baryogenesis mechanism the SM lacks. Neutrino properties feed $N_{\rm eff}$ and structure formation. See *Chapter 11 (p. 139)*.
- **Grand unified theories** — proposed UV completion. $SU(5)$, $SO(10)$, Pati–Salam embed $G_{SM}$ in a simple group, explaining

charge quantization and (in $SO(10)$) a full generation plus $\nu_R$ in one irrep, motivating unification and the seesaw. `[INFERENCE/SPECULATIVE]`; generically predicts (largely unobserved) proton decay.

- **Supersymmetry** — proposed naturalness fix. SUSY cancels the quadratic Higgs divergence, improves unification (MSSM), and supplies a dark-matter candidate (lightest neutralino). `[SPECULATIVE]`; no superpartners at the LHC weaken the original motivation.
- **Lattice gauge theory** — essential tool. Nonperturbative QCD (hadron spectrum, decay constants, the QCD contribution to $g-2$, confinement) is computed via Monte Carlo, currently central to the $g-2$ hadronic controversy. See *Chapter 8 (p. 89)*.
- **Neutrino / astroparticle physics** — the first crack and a bridge. Oscillations are the established BSM signal; the seesaw links neutrino mass to a high scale and to leptogenesis. See *Chapter 11 (p. 139)*.
- **Statistical and condensed-matter field theory** — shared formalism. The Higgs mechanism is the relativistic analog of Anderson–Higgs in superconductivity (Ginzburg–Landau); spontaneous symmetry breaking, the Wilsonian RG, effective field theory, and topological solitons are common language, with insight flowing both ways. See *Chapter 6 (p. 56)* and *Chapter 5 (p. 40)*.
- **Effective field theory (SMEFT)** — the modern organizing framework. Treating the SM as the renormalizable part of an EFT, BSM effects are parametrized by higher-dimension gauge-invariant operators (dim-5 for neutrino mass, dim-6 for collider/precision deviations) suppressed by powers of a new-physics scale $\Lambda$. This is the dominant model-independent strategy for confronting the SM with data. See *Chapter 3 (p. 15)* and *Chapter 17 (p. 325)*.

Connections to underlying mathematics (Lie groups, fiber bundles, index theorems behind anomalies) are catalogued in *Chapter 13 (p. 172)*; information-theoretic angles in *Chapter 12 (p. 155)*.

## Key references


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- S. Weinberg, *The Quantum Theory of Fields*, Vols. I–II (Cambridge Univ. Press, 1995–1996) — foundational treatment emphasizing why QFT takes its form; deep on EWSB.
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- M. D. Schwartz, *Quantum Field Theory and the Standard Model* (Cambridge Univ. Press, 2014) — formalism tied directly to SM phenomenology and EFT.
- Particle Data Group (R. L. Workman *et al.*), *Review of Particle Physics*, *Prog. Theor. Exp. Phys.* (updated regularly) — the standard reference for measured parameters, masses, mixing matrices, and reviews. Use for any numerical value.
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- A. D. Sakharov, *JETP Lett.* **5**, 24 (1967) — the baryogenesis conditions.

## References

See the Bibliography (p. 776) for the consolidated, cross-domain reference list.

CHAPTER 10

# General Relativity

General Relativity (GR) is the accepted classical relativistic theory of gravitation. It recasts gravity not as a force on a fixed stage but as the geometry of a four-dimensional spacetime: a smooth Lorentzian manifold whose curvature is sourced by stress-energy through the Einstein field equations. This page is the authoritative deep-dive for the GR domain in this wiki: its geometric formalism, variational origin, the equivalence and background-independence principles, exact solutions and causal structure, the singularity theorems and gravitational waves, the 3+1 initial-value formulation, the structural absence of local gravitational energy, and the precise senses in which the theory breaks down.

For epistemic-marker definitions see *Chapter 2 (p. 8)*; for the global theory layout see *Chapter 3 (p. 15)*.

## Scope

`[ESTABLISHED]` GR describes gravitation geometrically: spacetime is a pseudo-Riemannian manifold $(M, g)$, and matter-energy determines its curvature. The theory is empirically valid from laboratory and Solar-System scales through binary pulsars, black holes, gravitational-wave sources, and cosmology.

This page covers: the geometric formalism (metric, Levi-Civita connection, curvature); the field equations and their Einstein–Hilbert variational origin; the equivalence principle and diffeomorphism invariance / background independence; geodesics and causal structure; the canonical exact solutions (Schwarzschild, Kerr, FLRW) and their horizons; the Penrose–Hawking singularity theorems; gravitational waves; the ADM 3+1 initial-value formulation and its constraints; the problem of gravitational energy; and the structural reasons GR resists quantization.

Explicitly **excluded** (treated only as connections): detailed quantum-gravity programs — see *Chapter 8 (p. 89)* and *Chapter 18 (p. 349)* — detailed astrophysics, and numerical-relativity engineering.

# Core formalism

## 1. The arena: a Lorentzian manifold

[ESTABLISHED] Spacetime is modeled as a connected, 4-dimensional, smooth ($C^\infty$), Hausdorff, paracompact, time-orientable manifold $M$ equipped with a pseudo-Riemannian (Lorentzian) metric $g$ of signature $(-,+,+,+)$ (equivalently $(+,-,-,-)$; a pure convention). The metric is a smooth, symmetric, nondegenerate $(0,2)$-tensor field. In a chart $\{x^\mu\}$, $\mu = 0,1,2,3$,

$$ds^2 = g_{\mu\nu}\, dx^\mu dx^\nu, \qquad g_{\mu\nu} = g_{\nu\mu}, \qquad \det(g_{\mu\nu}) \neq 0,$$

with one negative and three positive eigenvalues at every point. Nondegeneracy yields an inverse $g^{\mu\nu}$ with $g^{\mu\alpha} g_{\alpha\nu} = \delta^\mu_\nu$. The metric raises and lowers indices and fixes lengths, angles, the causal cone, and the invariant volume element $\sqrt{-g}\, d^4x$.

Tangent vectors at $p \in M$ split into **timelike** ($g(v,v) < 0$), **null** ( $g(v,v) = 0$), and **spacelike** ($g(v,v) > 0$):

$$g(v,v) < 0 \text{ (timelike)}, \quad = 0 \text{ (null)}, \quad > 0 \text{ (spacelike)}.$$

The null cone at each point is the local light cone; time-orientability lets us globally label the two halves future and past. This Lorentzian causal structure is what the singularity theorems and Penrose diagrams exploit.

## 2. Connection and covariant derivative

[ESTABLISHED] GR uses the unique **Levi-Civita connection** $\nabla$: torsion-free and metric-compatible. Its coefficients are the

Christoffel symbols,

$$\Gamma^{\lambda}_{\mu\nu} = \tfrac{1}{2}\, g^{\lambda\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})\,.$$

Metric compatibility $\nabla_\lambda g_{\mu\nu} = 0$ together with zero torsion ($\Gamma^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\nu\mu}$) determines $\nabla$ uniquely — the fundamental theorem of (pseudo-)Riemannian geometry. On a vector,

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^{\nu}_{\mu\lambda} V^\lambda.$$

Crucially $\Gamma$ is **not** a tensor: it can be set to zero at any single point by choosing Riemann normal coordinates. This is the geometric expression of the equivalence principle — the local field is removable.

## 3. Curvature

`[ESTABLISHED]` The Riemann curvature tensor measures the path-dependence of parallel transport, i.e. the failure of covariant derivatives to commute:

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}, \qquad [\nabla_\mu, \nabla_\nu] V^\rho = R^{\rho}{}_{\sigma\mu\nu} V^\sigma.$$

Symmetries: $R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu} = -R_{\rho\sigma\nu\mu} = R_{\mu\nu\rho\sigma}$; the first (algebraic) Bianchi identity $R_{\rho[\sigma\mu\nu]} = 0$; and the second (differential) Bianchi identity

$$\nabla_{[\lambda} R_{\rho\sigma]\mu\nu} = 0 \;\Rightarrow\; \nabla^\mu G_{\mu\nu} = 0.$$

Contractions give the Ricci tensor $R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}$, the Ricci scalar $R = g^{\mu\nu} R_{\mu\nu}$, and the **Einstein tensor**

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu}, \qquad \nabla^\mu G_{\mu\nu} \equiv 0.$$

The trace-free part of Riemann is the **Weyl tensor** $C_{\rho\sigma\mu\nu}$ (conformally invariant), encoding the tidal and radiative degrees of freedom that survive in vacuum where $R_{\mu\nu} = 0$. In 4D, the 20 independent components of Riemann decompose as Ricci (10, fixed locally by the field equations from matter) plus Weyl (10, the free/radiative gravitational field).

## 4. The Einstein field equations

[ESTABLISHED]

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu},$$

where $T_{\mu\nu}$ is the symmetric stress-energy tensor of matter, $\Lambda$ the cosmological constant, $G$ Newton's constant, $c$ the speed of light. These are ten coupled, quasilinear, second-order PDEs for $g_{\mu\nu}$. The contracted Bianchi identity $\nabla^\mu G_{\mu\nu} \equiv 0$ makes only six independent; the remaining four are constraints, matching the four-fold coordinate gauge freedom. Local energy-momentum conservation $\nabla^\mu T_{\mu\nu} = 0$ is then a **consequence**, not an extra postulate. In geometric units $G = c = 1$: $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}$.

## 5. Variational origin: the Einstein–Hilbert action

[ESTABLISHED] The field equations follow from extremizing

$$S = \frac{c^4}{16\pi G} \int_M (R - 2\Lambda)\sqrt{-g}\, d^4x + S_{\text{matter}}, \qquad \frac{\delta S}{\delta g^{\mu\nu}} = 0,$$

with the matter stress tensor defined by $T_{\mu\nu} = -\,(2/\sqrt{-g})\,\delta S_{\text{matter}}/\delta g^{\mu\nu}$. Because $R$ contains second derivatives of $g$ , a well-posed variational principle on a region with boundary requires the **Gibbons–Hawking–York** boundary term $\propto \int_{\partial M} K\sqrt{h}\, d^3x$ ($K$ the extrinsic curvature, $h$ the induced metric); this term is central to black-hole thermodynamics and Euclidean quantum gravity. The Palatini (first-order) formulation treats $g$ and $\Gamma$ as independent and recovers the Levi-Civita connection as an equation of motion. See *Chapter 17 (p. 325)* on the action principle as a cross-domain organizing idea.

## 6. Matter coupling and the equivalence principle

[ESTABLISHED] Matter couples by **minimal coupling** (the comma-to-semicolon rule): take the special-relativistic Lagrangian and replace $\eta_{\mu\nu} \to g_{\mu\nu}$, $\partial \to \nabla$. Test bodies follow **geodesics** of $g$,

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0,$$

extremizing proper time for massive particles (timelike geodesics) and affine length for light (null geodesics). The operational, frame-invariant signature of gravity is **geodesic deviation**:

$$\frac{D^2\xi^\mu}{d\tau^2} = -R^\mu{}_{\alpha\nu\beta}\, u^\alpha\xi^\nu u^\beta,$$

the relative acceleration of neighboring free-fall worldlines, governed by Riemann. Nonzero Riemann is a genuine tidal field that cannot be transformed away — the invariant content of "gravity."

## 7. Diffeomorphism invariance / background independence

`[ESTABLISHED]` The action and field equations are invariant under arbitrary smooth diffeomorphisms $\varphi : M \to M$ acting by pullback (active diffeos), not merely passive coordinate relabelings. The metric is the only dynamical background; there is no fixed prior geometry. The **hole argument** (Einstein 1913–15; modern readings by Stachel and by Earman–Norton) shows that to preserve determinism one must regard diffeomorphic solutions $g$ and $\varphi^*g$ as the *same* physical state: bare manifold points have no physical individuation independent of the fields. This **background independence** is the deep structural feature distinguishing GR from field theories on a fixed background (Maxwell, Yang–Mills on Minkowski). See *Chapter 20 (p. 458)*.

## 8. The 3+1 / ADM Hamiltonian formulation

`[ESTABLISHED]` Foliate a globally hyperbolic spacetime by spacelike slices $\Sigma_t$ and write

$$ds^2 = -N^2dt^2 + h_{ij}\,(dx^i + N^i dt)(dx^j + N^j dt),$$

with lapse $N$, shift $N^i$, induced 3-metric $h_{ij}$, and extrinsic curvature $K_{ij}$. The Einstein–Hilbert action becomes a constrained

Hamiltonian system with canonical pair $(h_{ij}, \pi^{ij})$, $\pi^{ij} = \sqrt{h}\,(K^{ij} - K h^{ij})$, whose total Hamiltonian is a sum of constraints:

$$\mathcal{H} = \frac{1}{\sqrt{h}}\left(\pi^{ij}\pi_{ij} - \tfrac{1}{2}\pi^2\right) - \sqrt{h}\,{}^{(3)}R \approx 0, \qquad \mathcal{H}_i = -2\nabla_j \pi^j{}_i \approx 0,$$

with $N, N^i$ as Lagrange multipliers. The four constraints encode diffeomorphism invariance; "evolution" is partly gauge. `[ESTABLISHED]` The initial-value problem is well posed: Choquet-Bruhat (1952) and Choquet-Bruhat–Geroch (1969) proved local existence/uniqueness and the existence of a unique **maximal globally hyperbolic development** of constraint-satisfying Cauchy data $(h_{ij}, K_{ij})$. GR is deterministic in the PDE sense, given a Cauchy surface. Upon canonical quantization $\mathcal{H} \approx 0$ becomes the timeless Wheeler–DeWitt equation $\hat{\mathcal{H}}\Psi = 0$ — see "Where it breaks down."

## 9. Canonical exact solutions

`[ESTABLISHED]`

- **Minkowski**: $g = \eta$, flat — the special-relativistic vacuum.
- **Schwarzschild** (1916; unique static spherically symmetric vacuum by Birkhoff's theorem):

  $$ds^2 = -\left(1 - \tfrac{2GM}{r}\right)dt^2 + \left(1 - \tfrac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2.$$

  The event horizon at $r = 2GM$ is a *coordinate* singularity (removable via Kruskal–Szekeres); the genuine curvature singularity is at $r = 0$, where the Kretschmann scalar $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/r^6 \to \infty$. Basis of the classic Solar-System tests.
- **Kerr** (1963): stationary, axisymmetric, rotating vacuum black hole (mass $M$, angular momentum $J = Ma$), with an ergosphere and inner/outer horizons; **Kerr–Newman** adds charge. `[ESTABLISHED, under stated regularity/analyticity hypotheses]` The **no-hair / uniqueness theorems** (Israel, Carter, Hawking, Robinson) state that stationary electrovacuum black holes are Kerr–Newman, parameterized by $(M, J, Q)$.

- **FLRW cosmology**: $ds^2 = -dt^2 + a(t)^2\left[\,dr^2/(1-kr^2) + r^2 d\Omega^2\,\right]$, homogeneous and isotropic; with a perfect-fluid source it gives the Friedmann equation

  $$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3}.$$

  See *Chapter 11 (p. 139)* and *Chapter 16 (p. 308)*.

## 10. Causal structure and singularity theorems

`[ESTABLISHED]` Causal structure is fixed by the conformal class of $g$ (the light cones). Conformal compactifications (Penrose diagrams) display global causal structure, horizons, and the various infinities. The **Penrose (1965)** and **Hawking–Penrose (1970)** singularity theorems prove, under (i) an energy condition (e.g. the null/strong condition $R_{\mu\nu}k^\mu k^\nu \geq 0$), (ii) a causality/global condition, and (iii) a trapping condition (a closed trapped surface, or a cosmological Cauchy surface), that the spacetime is **geodesically incomplete**. Singularities are thus *generic*, not artifacts of symmetry. Crucially the theorems prove incompleteness, **not** curvature divergence and not "what happens" at the singularity. **Cosmic censorship** (weak: singularities hidden behind horizons; strong: the maximal Cauchy development is inextendible) is Penrose's `[OPEN]`/`[CONTESTED]` conjecture, with fine-tuned counterexamples known but believed generically true.

## 11. Gravitational waves

`[ESTABLISHED]` Linearize $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$; in Lorenz gauge the field equations give a wave equation for the trace-reversed perturbation,

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}, \qquad \bar{h}_{\mu\nu} = h_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu} h,$$

propagating at $c$ with two transverse-traceless polarizations $(+, \times)$, quadrupolar at leading order. The energy carried is described by the gauge-dependent Isaacson effective stress tensor after averaging over wavelengths. `[ESTABLISHED]` Indirect confirmation: the Hulse–Taylor binary pulsar PSR B1913+16 orbital decay matches the GR quadrupole prediction. Direct detection: LIGO/Virgo observed

GW150914 (a binary black-hole merger) and GW170817 (a binary neutron star with an electromagnetic counterpart, constraining the GW speed to equal $c$ to roughly one part in $10^{15}$).

## 12. Energy in GR

`[ESTABLISHED, structural]` Because covariant conservation $\nabla^\mu T_{\mu\nu} = 0$ yields an integral conservation law only in the presence of a Killing vector, and because the equivalence principle lets the local field be transformed away at a point, there is **no covariant local energy density for the gravitational field**. One uses coordinate-dependent **pseudotensors** (Einstein, Landau–Lifshitz, Møller), or quasi-local definitions (Hawking, Brown–York, Bartnik), or the well-defined *global* energies for asymptotically flat spacetimes: the **ADM mass** (at spatial infinity) and the **Bondi mass** (at null infinity, decreasing by radiated GW energy). `[ESTABLISHED]` The **positive energy theorem** (Schoen–Yau 1979; Witten 1981) proves the ADM mass is $\geq 0$, with equality iff Minkowski. The absence of a local gravitational energy density is an intrinsic feature, not a defect; see *Chapter 14 (p. 191)*.

## 13. Constants and conventions

`[ESTABLISHED]` $G \approx 6.674 \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$, $c \approx 2.998 \times 10^8\ \mathrm{m/s}$. Conventions vary — signature $(-+++)$ vs $(+---)$, the Riemann sign, and the $8\pi G$ vs $8\pi G/c^4$ prefactor — so always fix conventions before comparing formulae (the "MTW vs Weinberg" sign tables). See *Chapter 16 (p. 308)*.

# Foundational assumptions

| Assumption | Status | Justification |
|---|---|---|
| Spacetime is a smooth 4D Lorentzian manifold $(M, g)$, signature $(-, +, +, +)$ | likely-fundamental | `[INFERENCE]` The continuum/manifold structure and Lorentzian signature underwrite every confirmed prediction and the causal cone matching observation; signature is strongly constrained empirically. But smoothness is an idealization expected to fail near the Planck scale `[SPECULATIVE]`/`[OPEN]`, and the dimension "4," though empirically excellent, is not provable a priori (Kaluza–Klein/string scenarios add dimensions). The $(-+++)$ vs $(+---)$ choice is purely conventional. |
| Connection is the unique Levi-Civita one: torsion-free and metric-compatible | conventional-choice | `[ESTABLISHED as a choice]` Setting torsion $= 0$ and $\nabla g = 0$ uniquely fixes $\Gamma$ from $g$, but this is a choice among empirically equivalent reformulations. Einstein–Cartan allows spin-sourced torsion; teleparallel gravity uses a flat connection with torsion (the "geometric trinity"). These agree with GR for spinless matter and current data. Metric compatibility is more deeply tied to the equivalence principle (consistent rulers and clocks). |
| Field equations second-order; action linear in $R$ (plus $\Lambda$) | conventional-choice | `[ESTABLISHED structurally]` Lovelock's theorem: in 4D the only divergence-free symmetric 2-tensor built from $g$ and its first two derivatives, with second-order field equations, is $aG_{\mu\nu} + bg_{\mu\nu}$ — so the field equations $+\Lambda$ are essentially unique *given* those premises. But the premises are choices: higher-curvature terms ($R^2$, $R_{\mu\nu}R^{\mu\nu}$, Gauss–Bonnet) are generic EFT corrections expected near the Planck scale. "Linear in $R$" is the leading term of an effective expansion. |

| | | |
|---|---|---|
| Matter couples minimally and universally to one metric $g$ (equivalence principle) | likely-fundamental | [ESTABLISHED to extraordinary precision] The weak EP (universality of free fall) is tested to $\sim 10^{-15}$ (MICROSCOPE, torsion balances); the Einstein EP (local Lorentz + local position invariance) underpins geometrization. Caveat: a single universal metric for *all* matter is an assumption (multi-metric/modified theories posit otherwise, tightly constrained), and whether the EP survives for superposed/entangled masses is [OPEN]. |
| Diffeomorphism invariance / background independence | fundamental | [ESTABLISHED] The deepest structural principle. The hole argument shows determinism *forces* identifying $\varphi^* g$ with $g$. It dictates the constraint structure, the absence of a preferred time, and most obstructions to quantization — arguably the one feature any successful quantum gravity must reproduce. |
| Time-orientability; physically relevant spacetimes are globally hyperbolic | conventional-choice | [INFERENCE] Time-orientability is mild and near-universal. Global hyperbolicity is *imposed* to guarantee a well-posed initial-value problem (Choquet-Bruhat–Geroch). The bare field equations permit non-globally-hyperbolic solutions (closed timelike curves in Gödel, the Kerr interior, wormholes). Whether nature enforces it is the content of strong cosmic censorship and chronology protection — both [OPEN]/[CONTESTED]. |
| Matter satisfies (pointwise) energy conditions | historical-artifact | [CONTESTED] Energy conditions are hypotheses of the singularity, area, and positivity/censorship theorems, but they are not laws of GR — they are constraints on matter. Quantum fields violate the pointwise conditions (Casimir, squeezed states, Hawking radiation); the strong condition is violated by inflation and by dark energy / positive $\Lambda$. Modern work uses averaged or quantum versions (ANEC, QNEC). The pointwise form is a now-known-too-strong idealization. |

| | | |
|---|---|---|
| $\Lambda$ is a single constant of nature (not dynamical) | unclear | [OPEN]/[CONTESTED] $\Lambda$ is the unique extra term Lovelock allows, and $\Lambda$CDM fits data well, but its observed value ($\sim 10^{-52}$ m$^{-2}$) is $\sim$ 120 orders of magnitude below naive QFT vacuum-energy estimates — the cosmological-constant problem. Whether $\Lambda$ is truly constant, vacuum-energy artifact, anthropically selected, or dynamical (quintessence) is unresolved; recent surveys hint, inconclusively, at evolving dark energy. |
| Gravity is purely classical (the metric is a c-number field) | likely-fundamental | [INFERENCE]/[OPEN] GR as formulated is a classical field theory, hugely successful in its domain. But coupling a classical metric to quantum matter via $G_{\mu\nu} = 8\pi G\langle T_{\mu\nu}\rangle$ (semiclassical gravity) is at best approximate and internally problematic (measurement, mass superpositions). The near-universal expectation is that the metric must ultimately be quantum, so classicality is a feature of the *effective* theory. |

See *Chapter 20 (p. 458)* for the cross-domain register.

## Domain of validity

[ESTABLISHED] GR is confirmed across an enormous range:

- **Solar System**: perihelion precession of Mercury, light deflection ($\approx 1.75''$ at the limb), Shapiro time delay, and gravitational redshift, all confirmed at the $10^{-3}$–$10^{-5}$ level via the parametrized post-Newtonian (PPN) formalism; e.g. $\gamma - 1 \approx (2.1 \pm 2.3) \times 10^{-5}$ from Cassini.
- **Strong-field/dynamical**: the Hulse–Taylor and double pulsars (orbital decay matching GR to $< 0.1\%$), pericenter precession of stars near Sgr A*, Event Horizon Telescope black-hole shadows, and the tens of LIGO/Virgo/KAGRA mergers consistent with GR waveforms and the Kerr hypothesis.
- **Cosmology**: $\Lambda$CDM with GR fits the CMB (Planck), BBN light-element abundances, large-scale structure, BAO, and Type Ia supernovae (accelerating expansion).
- **Equivalence principle**: WEP to $\sim 10^{-15}$ (MICROSCOPE).

Quantitative window: from sub-millimeter laboratory inverse-square-law tests up to $\sim 10^{26}$ m, with curvature radii from cosmological down to near (but outside) astrophysical horizons.

`[ESTABLISHED]` Breakdown is *expected* — by dimensional analysis, not direct observation — at the **Planck scale**: $\ell_P = \sqrt{\hbar G/c^3} \approx 1.6 \times 10^{-35}$ m, $t_P \approx 5.4 \times 10^{-44}$ s, $E_P \approx 1.2 \times 10^{19}$ GeV, where curvature reaches $\ell_P^{-2}$ and geometry fluctuations become $O(1)$. `[INFERENCE]` GR is a low-energy effective field theory valid for curvatures $\ll \ell_P^{-2}$; its perturbative quantization is non-renormalizable, signaling new physics at or below $E_P$. No experiment yet probes this regime. See *Chapter 16 (p. 308)*.

## Where it breaks down

- **Curvature singularities** ($r \to 0$ in black-hole interiors; the Big Bang $t \to 0$). `[ESTABLISHED]` The singularity theorems prove generic geodesic incompleteness, and in the symmetric cases invariants diverge. GR predicts its *own* breakdown: the equations cease to define evolution. This is a domain-of-validity limit, not a logical inconsistency. What "replaces" the singularity (bounce, resolution, information fate) is `[OPEN]`.
- **Perturbative quantization of the metric.** `[ESTABLISHED]` Quantizing $h$ in $g = \eta + h$ gives a non-renormalizable theory: one-loop pure gravity is finite on-shell ('t Hooft–Veltman) but two-loop pure gravity diverges (Goroff–Sagnotti 1985), and gravity+matter diverges at one loop. Infinitely many counterterms with undetermined coefficients are needed, so the naive graviton QFT loses predictivity above $E_P$ — a true incompleteness of GR-as-a-fundamental-QFT, though consistent as an EFT below $E_P$.
- **Semiclassical gravity and black-hole evaporation.** `[INFERENCE]`/`[OPEN]` Hawking's calculation predicts thermal radiation at $T_H = \hbar c^3/(8\pi G M k_B)$. Iterated to complete evaporation it appears to map a pure state to a mixed state — the **information paradox**, a clash between GR's local causal structure and quantum unitarity. Recent island/replica-

wormhole computations `[SPECULATIVE-but-influential]` reproduce a unitary Page curve, but a complete, agreed mechanism in realistic (non-AdS) gravity is `[OPEN]`/`[CONTESTED]`. See *Chapter 12 (p. 155)*.

- **The cosmological constant / vacuum energy.** `[OPEN]` QFT vacuum energy gravitates via $\Lambda_{\text{eff}}$; naive estimates exceed observation by $\sim 120$ orders of magnitude. GR offers no mechanism for why $\Lambda$ is so tiny yet nonzero — a quantitative crisis at the GR/QFT interface, not an inconsistency within classical GR (where $\Lambda$ is a free parameter).
- **Local gravitational energy density.** `[ESTABLISHED structural]` By the equivalence principle the field is locally removable, so no covariant local stress-energy tensor for gravity exists — only pseudotensors and global/quasi-local notions.
- **The problem of time.** `[OPEN]`/structural In the ADM Hamiltonian, $\mathcal{H}$ is a sum of constraints vanishing on shell, generating what is partly gauge; observables must be diffeomorphism-invariant (hence nonlocal and hard to construct), and on quantization $\hat{\mathcal{H}}\Psi = 0$ (Wheeler–DeWitt) is timeless. A feature of background independence classically, a conceptual obstruction for quantum gravity `[CONTESTED]`.
- **Closed timelike curves / chronology violation.** `[CONTESTED]` Exact solutions (Gödel, the interior Kerr, van Stockum, Tipler/Gott constructions, traversable wormholes with exotic matter) contain CTCs, undermining global determinism. Whether physics forbids them is Hawking's chronology-protection conjecture — heuristically supported but `[OPEN]`.

These are catalogued cross-domain in *Chapter 14 (p. 191)* and *Chapter 15 (p. 233)*.

## Open problems (internal)

- **Quantum gravity** `[OPEN]`: a consistent theory reducing to GR below $E_P$ and to QFT in flat space. Perturbative quantization fails; candidate frameworks (string theory, loop quantum gravity, asymptotic safety, causal sets, causal dynamical triangulations) exist but none is experimentally confirmed, and the very form of the answer (is spacetime fundamental? is the metric emergent?) is unsettled. See *Chapter 19 (p. 392)*, *Chapter 18 (p. 349)*.

- **Black-hole information paradox** `[OPEN]`/`[CONTESTED]`: whether evaporation is unitary, where/how information escapes, and whether the horizon is smooth (no firewall). The islands/quantum-extremal-surface program is influential but its scope and interpretation remain contested — two named objections as of 2026-07: the ensemble-vs-single-theory reading of replica wormholes, and the massive-graviton/gravitating-bath critique (all controlled $d > 2$ island constructions have massive gravitons; islands in long-range massless gravity strain the gravitational Gauss law). A replica-independent algebraic leg (crossed-product Type II algebras; von Neumann entropy $= S_{\text{gen}}$ ; perturbative GSL/QFC) supports unitarity at the perturbative level. See *Chapter 14 (p. 191)* GC-4/GC-13.
- **Nature and fate of singularities** `[OPEN]`: the theorems guarantee incompleteness but say nothing about resolution (bounce vs replacement vs other), currently untestable.
- **Cosmic censorship (weak and strong)** `[OPEN]`/`[CONTESTED]`: no general proof; fine-tuned counterexamples (critical collapse, certain charged/rotating interiors, $\Lambda < 0$ and near-extremal results) exist, but censorship is widely believed generically true.
- **Cosmological constant / dark energy** `[OPEN]`: why $\Lambda \sim 10^{-120}$ in Planck units, and whether dark energy is constant or dynamical; recent hints of evolving dark energy are `[CONTESTED]` and not decisive.
- **Energy and quasi-local quantities** `[OPEN]`/active: ADM and Bondi masses are well defined, but a fully satisfactory quasi-local energy and angular momentum (e.g. resolving supertranslation ambiguities in Bondi angular momentum, recently linked to the BMS group and memory/soft theorems) remains active.

- **Global existence and nonlinear stability** `[OPEN]`/active: stability of Minkowski is proven (Christodoulou–Klainerman 1993) and there is strong recent progress on (sub-extremal) Kerr stability, but full nonlinear Kerr stability and the structure of generic collapse are still being established.
- **The problem of time and Dirac observables** `[OPEN]`/`[CONTESTED]`: defining local, physical, diffeomorphism-invariant observables and the meaning of "evolution" absent a preferred time; relational/partial-observable proposals are promising but not decisive.

## Connections to other frameworks

- **Special relativity** `[ESTABLISHED]`: GR contains SR as the tangent-space / local-inertial-frame limit — at any point one can set $g_{\mu\nu} = \eta_{\mu\nu}$, $\partial g = 0$ (Einstein EP). GR generalizes the *global* Lorentz symmetry of SR to *local* diffeomorphism invariance.
- **Newtonian gravity** `[ESTABLISHED]`: recovered in the weak-field, slow-motion, quasi-static limit, $g_{00} \approx -(1 + 2\Phi/c^2)$, with the geodesic equation reducing to $\ddot{\mathbf{x}} = -\nabla\Phi$ and the 00-equation to $\nabla^2\Phi = 4\pi G\rho$. This fixes the $8\pi G/c^4$ coupling. See *Chapter 4 (p. 25)*.
- **QFT / the Standard Model** `[INFERENCE]`/clash: QFT lives on a fixed background; GR makes the background dynamical. They coexist as QFT-in-curved-spacetime and semiclassical gravity (Hawking radiation, the Unruh effect, cosmological particle production) but clash fundamentally — GR is non-renormalizable as a QFT, and reconciling background independence with superposed geometries is unsolved. See *Chapter 8 (p. 89), Chapter 9 (p. 106)*.
- **Thermodynamics / statistical mechanics** `[ESTABLISHED-theory, INFERENCE-physics]`: black-hole mechanics mirrors thermodynamics — the four laws, $S_{\text{BH}} = k_B c^3 A/(4G\hbar)$ (entropy $\propto$ horizon area), and $T_H \propto$ surface gravity. This bridge, with the holographic principle and AdS/CFT, suggests gravity/spacetime may be

emergent/thermodynamic, though microscopic state-counting and general validity remain partly `[SPECULATIVE]`. See *Chapter 5 (p. 40)*, *Chapter 6 (p. 56)*, *Chapter 12 (p. 155)*.

- **Cosmology and astrophysics** `[ESTABLISHED]`: GR is the dynamical backbone of cosmology (FLRW, Friedmann, ΛCDM, structure growth, inflation as GR + scalar field) and of compact-object astrophysics (neutron stars via the TOV equation, black holes, lensing, GW astronomy). See *Chapter 11 (p. 139)*.
- **Differential geometry / global analysis** `[ESTABLISHED]`: GR is essentially Lorentzian geometry and both uses and drives mathematics — hyperbolic PDE theory and the Cauchy problem, global causal geometry (Penrose, Hawking), and geometric analysis (positive mass via minimal surfaces/spinors). See *Chapter 13 (p. 172)*.
- **Gauge theory** `[INFERENCE]`/analogy: GR can be cast in connection variables (tetrad/spin-connection, Palatini, Ashtekar), resembling a gauge theory of the local Lorentz (or de Sitter/Poincaré) group, with diffeomorphisms as the gauge symmetry — underlying loop quantum gravity, though gravity's gauge group differs qualitatively from internal Yang–Mills groups. See *Chapter 17 (p. 325)*.
- **Modified / alternative gravity** `[CONTESTED]`: $f(R)$, scalar-tensor (Brans–Dicke, Horndeski), Einstein–Cartan, teleparallel/ $f(T)$, massive gravity, and MOND-inspired theories relax one GR assumption. The "geometric trinity" shows curvature-, torsion-, and nonmetricity-based formulations can be empirically equivalent to GR, highlighting which assumptions are conventions. All current data remain consistent with standard GR.

## Key references

The following are real, standard sources; see the Bibliography (p. 776) for the consolidated list and full details.


- C. W. Misner, K. S. Thorne, J. A. Wheeler, *Gravitation* (Freeman 1973; Princeton 2017) — the canonical comprehensive reference

("MTW"); geometry, EP, ADM, gravitational waves, sign conventions.

- R. M. Wald, *General Relativity* (Univ. Chicago Press 1984) — standard rigorous graduate text; causal structure, singularity theorems, energy, asymptotic flatness, QFT in curved spacetime.
- S. W. Hawking and G. F. R. Ellis, *The Large Scale Structure of Space-Time* (Cambridge 1973) — definitive on global causal structure, energy conditions, and the Hawking–Penrose theorems.
- S. M. Carroll, *Spacetime and Geometry* (Addison-Wesley 2004) — modern graduate text; variational derivation, diffeomorphisms, observational connections.
- R. Penrose, "Gravitational Collapse and Space-Time Singularities," *Phys. Rev. Lett.* **14**, 57 (1965) — original trapped-surface singularity theorem (2020 Nobel Prize).
- R. Arnowitt, S. Deser, C. W. Misner, "The Dynamics of General Relativity" (1962; arXiv:gr-qc/0405109 reprint) — the ADM 3+1 formulation and ADM energy.
- Y. Choquet-Bruhat and R. Geroch, "Global aspects of the Cauchy problem in general relativity," *Commun. Math. Phys.* **14**, 329 (1969) — maximal globally hyperbolic development; rigorous determinism.
- B. P. Abbott et al. (LIGO/Virgo), "Observation of Gravitational Waves from a Binary Black Hole Merger," *Phys. Rev. Lett.* **116**, 061102 (2016) — GW150914, first direct detection.
- R. Schoen and S.-T. Yau, *Commun. Math. Phys.* **65**, 45 (1979); E. Witten, *Commun. Math. Phys.* **80**, 381 (1981) — positive energy theorem.
- D. Christodoulou and S. Klainerman, *The Global Nonlinear Stability of the Minkowski Space* (Princeton 1993) — nonlinear stability of Minkowski.
- S. W. Hawking, "Particle Creation by Black Holes," *Commun. Math. Phys.* **43**, 199 (1975) — Hawking radiation; origin of the information paradox.

- C. M. Will, "The Confrontation between General Relativity and Experiment," *Living Reviews in Relativity* **17**, 4 (2014) — authoritative PPN/experimental-test review.

## References

See the Bibliography (p. 776) for the consolidated, formatted reference list.

CHAPTER 11

# Cosmology

Relativistic / physical cosmology studies the dynamics, thermal history, structure, and global geometry of the observable universe, treating it as a single physical system governed by general relativity (GR) coupled to a quantum-field-theoretic matter sector. It is the discipline in which gravity, particle physics, statistical mechanics, and the foundations of time meet a single dataset of unrepeatable scope.

## Scope

This page covers the homogeneous–isotropic background (the FLRW geometry plus Friedmann dynamics), the ΛCDM concordance model and its six base parameters, the three classical observational pillars — the cosmic microwave background (CMB), primordial nucleosynthesis (BBN), and large-scale structure / baryon acoustic oscillations (BAO) — cosmological perturbation theory and structure formation, cosmic inflation (the problems it solves and the ones it raises), the constituents of the cosmic energy budget (baryons, photons, neutrinos, cold dark matter, dark energy / Λ), the cosmological-constant problem, the thermodynamic arrow of time and the low-entropy "past hypothesis," eternal inflation and the measure problem, and the current empirical tensions ($H_0$, $S_8/\sigma_8$ ).

It excludes the detailed machinery of quantum gravity (see *Chapter 10 (p. 121)* and the quantum-gravity discussion in *Chapter 3 (p. 15)*), string-landscape constructions per se, and the astrophysics of individual objects except where they serve as cosmological probes.

# Core formalism

## 1. Geometric kinematics: the FLRW metric

[ESTABLISHED] The **cosmological principle** posits that on large scales space is spatially homogeneous and isotropic. The most general metric consistent with a foliation by maximally symmetric spatial slices is the Friedmann–Lemaître–Robertson–Walker (FLRW) line element. In reduced-circumference polar coordinates with signature $(-,+,+,+)$ and $c=1$:

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2\left(d\theta^2 + \sin^2\theta\, d\phi^2\right)\right].$$

Equivalently, with comoving radial coordinate $\chi$,

$$ds^2 = -dt^2 + a^2(t)\left[d\chi^2 + S_k^2(\chi)\, d\Omega^2\right], \qquad S_k(\chi) = \begin{cases} \sin\chi & k=+1 \\ \chi & k=0 \\ \sinh\chi & k=-1 \end{cases}$$

Here $a(t)$ is the scale factor (conventionally $a_0 \equiv a(t_0) = 1$ today) and $k \in \{-1, 0, +1\}$ (after rescaling) labels spatial curvature — hyperbolic, flat, spherical. The isometry group is six-dimensional (three translations + three rotations of each spatial slice). The coordinate $t$ is the cosmic (proper) time of **comoving observers**, who are everywhere at rest in these coordinates and whose worldlines are geodesics orthogonal to the constant-$t$ slices.

**Redshift.** A photon emitted at scale factor $a_e$ and observed at $a_0$ is redshifted by

$$1 + z = \frac{a_0}{a_e} = \frac{\lambda_{\text{obs}}}{\lambda_{\text{emit}}},$$

the operational link between observed spectra and cosmic epoch. The Hubble parameter $H \equiv \dot{a}/a$ measures the fractional expansion rate; its present value is $H_0 = 100\, h\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ with $h \approx 0.67$–$0.73$ (the spread is the substance of the $H_0$ tension below).

## 2. Dynamics: the Friedmann equations

[ESTABLISHED] Feeding the FLRW metric into the Einstein field equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ with a perfect-fluid stress tensor $T^{\mu}{}_{\nu} = \mathrm{diag}(-\rho, p, p, p)$ yields two independent equations. The Friedmann (00) equation,

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3},$$

and the acceleration / Raychaudhuri (ii) equation,

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3}.$$

Local energy conservation $\nabla_\mu T^{\mu\nu} = 0$ — a consequence of the contracted Bianchi identity, hence not independent — gives the continuity equation

$$\dot{\rho} + 3H(\rho + p) = 0.$$

With an equation of state $p = w\rho$ this integrates to $\rho \propto a^{-3(1+w)}$: matter ($w = 0$) scales as $a^{-3}$, radiation ($w = 1/3$) as $a^{-4}$, vacuum/$\Lambda$ ($w = -1$) stays constant. Absorbing $\Lambda$ into $\rho$ as a fluid with $\rho_\Lambda = \Lambda/(8\pi G)$, defining the critical density $\rho_c = 3H^2/(8\pi G)$ and density parameters $\Omega_i = \rho_i/\rho_c$, the Friedmann equation becomes the expansion-history master relation:

$$\frac{H^2(a)}{H_0^2} = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda, \qquad \sum_i \Omega_i + \Omega_k = 1,$$

with $\Omega_k \equiv -k/(a_0^2 H_0^2)$. This single equation encodes which component dominates at each epoch (radiation → matter → $\Lambda$). Accelerated expansion ($\ddot{a} > 0$) requires $w < -1/3$ for the dominant component — a key structural fact: ordinary matter and radiation decelerate, so observed acceleration demands a negative-pressure component. See *Chapter 16 (p. 308)* for the numerical density budget.

## 3. Distances and horizons

is $D_M = S_k(D_C)$. The angular-diameter distance is $D_A = D_M/(1+z)$ and the luminosity distance is $D_L = (1+z)^2 D_A$. The relation $D_L = (1+z)^2 D_A$ (Etherington reciprocity) holds in any metric theory with photon-number conservation and is itself an observational test. The **particle horizon** (causal-patch radius) is $d_H(t) = a(t)\int_0^t dt'/a(t')$; in pure radiation- or matter-dominated eras it is finite — the seed of the **horizon problem**. The comoving Hubble radius $1/(aH)$ shrinks during acceleration and grows during deceleration; this single quantity organizes both the horizon problem and inflation's resolution of it.

## 4. Thermal history and the hot Big Bang

`[ESTABLISHED]` In thermal equilibrium the radiation energy density is $\rho_r = (\pi^2/30)\, g_*(T)\, T^4$ with $g_*(T)$ the effective relativistic degrees of freedom. Reaction rates $\Gamma = n\langle\sigma v\rangle$ compete with $H$; a species **freezes out** (decouples) when $\Gamma \lesssim H$. Key epochs, in order of decreasing temperature: the electroweak transition ($\sim 100$ GeV), QCD confinement ($\sim 150$ MeV), neutrino decoupling ($\sim 1$ MeV), $e^+e^-$ annihilation, Big Bang Nucleosynthesis ($\sim 0.1$ MeV, $t \sim$ minutes), matter–radiation equality ($1+z_{\rm eq} \approx 3400$), recombination and photon decoupling ($z_* \approx 1090$, $t \approx 3.8\times10^5$ yr) producing the CMB last-scattering surface, and reionization ($z \sim 6$–$10$).

**BBN** `[ESTABLISHED]`: With a single free parameter (the baryon-to-photon ratio $\eta = n_b/n_\gamma$, equivalently $\Omega_b h^2$), the coupled Boltzmann nuclear-reaction network predicts the primordial abundances of D, $^3$He, $^4$He ($Y_p \approx 0.247$), and $^7$Li. Deuterium and helium agree with observation across orders of magnitude. The **lithium problem** `[OPEN/CONTESTED]` is a persistent factor-$\sim 3$ overprediction of $^7$Li relative to metal-poor halo-star measurements.

**CMB** `[ESTABLISHED]`: A near-perfect blackbody at $T_0 = 2.7255$ K with relative temperature anisotropies $\Delta T/T \sim 10^{-5}$. The statistics live in the angular power spectrum $C_\ell$, defined by $\langle a_{\ell m} a^*_{\ell' m'}\rangle = C_\ell\, \delta_{\ell\ell'}\delta_{mm'}$ where $\Delta T(\hat n) = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\hat n)$. Acoustic peaks arise from baryon–photon plasma oscillations frozen at recombination;

the first peak at $\ell \approx 220$ pins spatial flatness; peak ratios measure $\Omega_b h^2$ and $\Omega_c h^2$; polarization (E-modes from scalar perturbations, B-modes from tensors and lensing) provides consistency checks and the cleanest window onto primordial gravitational waves.

## 5. Cosmological perturbation theory and structure formation

[ESTABLISHED] Write $g_{\mu\nu} = \bar{g}_{\mu\nu} + \delta g_{\mu\nu}$ and decompose into scalar/vector/tensor (SVT) modes, which decouple at linear order. In conformal Newtonian (longitudinal) gauge for scalar modes,

$$ds^2 = a^2(\tau)\left[-(1+2\Psi)\,d\tau^2 + (1-2\Phi)\,\delta_{ij}\,dx^i dx^j\right].$$

The density contrast $\delta = \delta\rho/\bar{\rho}$ obeys, in the sub-horizon Newtonian limit for cold matter,

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G\bar{\rho}_m\,\delta = 0,$$

a competition between Hubble friction and gravitational instability that yields a growing mode $\delta \propto D(a)$ (the linear growth function). The linear matter power spectrum $P(k) = \langle|\delta_k|^2\rangle$ has the primordial form $P \propto k^{n_s}$ shaped by the transfer function $T(k)$, which encodes the turnover at the equality scale and the BAO wiggles. Its amplitude is parameterized by $\sigma_8$ (rms linear fluctuation in $8\,h^{-1}$ Mpc spheres) and increasingly by $S_8 \equiv \sigma_8(\Omega_m/0.3)^{1/2}$, the combination weak-lensing surveys constrain best. BAO imprints a **standard ruler** — the sound horizon at the drag epoch, $r_d \approx 147$ Mpc — used as a geometric distance probe. Full predictions come from numerically integrating the Einstein–Boltzmann hierarchy (CAMB, CLASS) for the coupled photon/baryon/CDM/neutrino/metric system.

## 6. ΛCDM: the concordance model

[ESTABLISHED] The base model is specified by six parameters,

$$\{\Omega_b h^2,\ \Omega_c h^2,\ \theta_*\ (\text{or } H_0),\ \tau,\ A_s,\ n_s\},$$

assuming flatness, a cosmological constant ($w = -1$), three neutrino species, and adiabatic, nearly scale-invariant, Gaussian primordial fluctuations. Derived quantities include $\Omega_m \approx 0.31$, $\Omega_\Lambda \approx 0.69$, $\Omega_b \approx 0.049$, an age $\approx 13.8$ Gyr, and $n_s \approx 0.965$ (a slight red tilt, mildly favoring inflation). This handful of numbers simultaneously fits the CMB acoustic spectrum, the BBN light elements, BAO, the late-time supernova acceleration, and the growth of structure — the empirical core of modern cosmology and the strongest single entry in *Chapter 21 (p. 497)*.

## 7. Inflation

[ESTABLISHED that the model class exists and fits data; INFERENCE/SPECULATIVE that it actually occurred] Postulate an early epoch of quasi-exponential accelerated expansion driven by a scalar inflaton $\phi$ with action

$$S = \int d^4x \sqrt{-g}\left[\tfrac{1}{16\pi G}R - \tfrac{1}{2}(\partial\phi)^2 - V(\phi)\right].$$

The homogeneous field obeys $\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0$ with $H^2 = \frac{8\pi G}{3}(\frac{1}{2}\dot{\phi}^2 + V)$. In slow-roll ($\dot{\phi}^2 \ll V$, $\ddot{\phi}$ negligible) define the potential slow-roll parameters

$$\epsilon = \frac{1}{16\pi G}\left(\frac{V'}{V}\right)^2, \qquad \eta = \frac{1}{8\pi G}\frac{V''}{V},$$

with inflation requiring $\epsilon \ll 1$ and lasting $N = \int H\,dt \gtrsim 50$–$60$ e-folds. Quantum fluctuations of $\phi$ and the metric are stretched super-horizon and freeze, seeding a near-scale-invariant curvature power spectrum

$$\mathcal{P}_{\mathcal{R}}(k) = \frac{1}{8\pi^2}\frac{H^2}{\epsilon\, M_{\rm Pl}^2}\bigg|_{k=aH}, \qquad n_s - 1 = 2\eta - 6\epsilon, \qquad r = 16\epsilon,$$

where $r$ is the tensor-to-scalar ratio. Confirmed predictions: spatial flatness, near scale-invariance with a slight red tilt, adiabaticity, Gaussianity, and super-horizon correlations (the low-$\ell$ CMB TE anticorrelation). A primordial B-mode detection ($r \neq 0$) would be the cleanest remaining signature and is not yet in hand.

# Foundational assumptions

| Assumption | Status | Justification |
|---|---|---|
| Cosmological principle (large-scale homogeneity + isotropy) | likely-fundamental | [ESTABLISHED on $\gtrsim 100$ Mpc] Isotropy is directly observed $\sim 10^{-5}$ after dipole subtraction); homogeneity is inferred via t principle plus isotropy and confirmed statistically by galaxy su idealization holding statistically above a scale, not an exact ax truly fundamental or an emergent product of inflation is `[OPE` anomalies/bulk-flow claims remain `[CONTESTED]`. |
| GR is the correct gravity theory on cosmological scales | likely-fundamental | `[ESTABLISHED in tested regimes; INFERENCE on horizon scal` exquisitely tested in the solar system, binary pulsars, and GW those probe scales $\sim 10^{30}\times$ smaller than the Hubble radius. C application is an extrapolation; modified-gravity alternatives Horndeski) are constrained but not all excluded. See *Chapter* |
| Perfect-fluid matter, $T^{\mu}{}_{\nu} = \mathrm{diag}(-\rho, p, p, p)$, simple EoS | conventional-choice | `[INFERENCE]` At the background level the fluid form is *forced* l symmetry (isotropy forbids heat flux and anisotropic stress). "dust" dark matter and exactly $w = -1$ dark energy are replac choices, not forced truths. |
| Spatial flatness ($\Omega_k = 0$) | conventional-choice | `[INFERENCE]` A generic inflationary prediction, consistent with |
| Cold, collisionless, non-baryonic dark matter dominates the matter sector | likely-fundamental | `[ESTABLISHED that something beyond baryons+GR is needed;` Multi-pillar concordance: rotation curves, cluster lensing (Bu mass–gas offset), the CMB third-peak height, the $\Omega_b \ll \Omega_m$ bu very growth of structure. "Cold" is favored because hot/warm small-scale power. The particle identity is `[OPEN]`. |
| Dark energy is a cosmological constant ($w = -1$, constant) | conventional-choice | [ESTABLISHED that acceleration occurs; OPEN/CONTESTE exactly $\Lambda$] Acceleration is established (SNe Ia, CMB+BAO, ISV minimal description; $w = -1$ exactly is a choice. Some recent have been read as hinting at dynamical $w(z)$ `[CONTESTED, not` |
| Primordial fluctuations adiabatic, Gaussian, nearly scale-invariant | likely-fundamental | `[ESTABLISHED as data description; INFERENCE for inflation` statistics ($n_s \approx 0.965$, super-horizon correlations) are measur that *inflation* produced them is compelling but not proven (so bounces/ekpyrotic models reproduce the spectrum). |
| Inflation occurred | likely-fundamental | `[INFERENCE bordering SPECULATIVE]` Solves horizon, flatness, problems and predicts the fluctuation spectrum, but (i) no pri detection, (ii) inflaton unidentified with any SM field, (iii) ete measure problem, (iv) Penrose's low-entropy-initial-conditio Leading paradigm, not established fact. See *Chapter 19 (p. 39* |

| | | |
|---|---|---|
| The universe began in an extraordinarily low-entropy macrostate (past hypothesis) | fundamental | `[INFERENCE/OPEN]` The thermodynamic arrow demands a special low-*gravitational*-entropy boundary condition (near-zero initial Weyl curvature, Penrose), since microphysics is CPT-symmetric. *Why* the initial state was special is genuinely `[OPEN]`; it functions as a law-like posit, not a derivation. |
| Equilibrium-thermodynamic / single-arrow reasoning applies to the cosmos as a whole | conventional-choice | `[INFERENCE/CONTESTED]` Self-gravitating systems have negative specific heat and no global max-entropy equilibrium, so a global cosmic temperature/entropy is a useful idealization that breaks down conceptually for gravity. See *Chapter 6 (p. 56)*. |
| Exactly three light neutrino species ($N_{\rm eff} \approx 3.044$), standard radiation | conventional-choice | `[ESTABLISHED to good precision]` Measured from BBN and the CMB damping tail; the slight excess over 3 is a real QFT prediction (non-instantaneous decoupling). Held fixed as a well-motivated choice, easily promoted to a free parameter (dark radiation as an $H_0$-tension lever). |
| Cosmic time $t$ is a globally well-defined, preferred time coordinate | historical-artifact | `[INFERENCE]` FLRW symmetry singles out a preferred slicing and a cosmic rest frame (the CMB frame). Not a contradiction with relativity — it is spontaneous breaking of Lorentz invariance by the matter distribution — but treating cosmic time as fundamental is partly an artifact of the high symmetry of the idealized model; no global time exists in a genuinely inhomogeneous universe. |

See *Chapter 20 (p. 458)* for the cross-domain ledger and *Chapter 2 (p. 8)* for the marker definitions.

## Domain of validity

`[ESTABLISHED]` The FLRW + Friedmann background is valid where the cosmological principle holds: averaged over scales $\gtrsim 100$ Mpc, from shortly after the inflationary/Planck era down to today. The hot-Big-Bang thermal history is quantitatively trustworthy from BBN ($T \sim 1$ MeV, $t \sim 1$ s) — the earliest epoch with direct empirical corroboration — through recombination ($z \approx 1090$, the CMB) to the present. Linear perturbation theory is valid while $|\delta| \ll 1$, i.e. on large scales and at early times; smaller scales and later times require higher-order perturbation theory, the halo model, or $N$-body/hydrodynamic simulations (the mildly-to-fully nonlinear regime, $k \gtrsim 0.1\text{–}1\, h\,\mathrm{Mpc}^{-1}$). Slow-roll inflationary predictions hold while $\epsilon, |\eta| \ll 1$.

Boundaries: (i) Below $\sim 100$ Mpc the universe is manifestly inhomogeneous; FLRW is a statistical average, and the "backreaction" of inhomogeneities on the averaged expansion is `[CONTESTED]` but generally believed small. (ii) Before BBN ($t < 1$ s, $T > 1$ MeV) the thermal history is theoretically modeled (electroweak, QCD epochs) but lacks direct cosmological data; reheating after inflation is well before any direct probe. (iii) At the classical Big Bang singularity ($a \to 0$) curvature diverges and GR predicts its own breakdown — outside the domain entirely. (iv) The framework presumes GR on scales/curvatures never independently tested. Dark matter and dark energy mark the edge of "known physics": ΛCDM works phenomenologically while $\sim 95\%$ of the energy budget is described by placeholders whose microphysics lies outside the tested Standard Model (see *Chapter 9 (p. 106)*).

## Where it breaks down

- **The initial singularity ($a \to 0$).** `[ESTABLISHED breakdown]` The Hawking–Penrose singularity theorems show that, under reasonable energy conditions, FLRW expansion implies a past geodesically incomplete singularity where curvature invariants diverge and GR ceases to be predictive. This is a *domain-of-validity failure*, not an internal inconsistency: it signals the need for quantum gravity. The Borde–Guth–Vilenkin theorem shows that even eternally inflating spacetimes are past-incomplete, so inflation relocates but does not remove the boundary.
- **The cosmological-constant problem.** `[OPEN, severe]` Naive QFT vacuum-energy estimates, cut off at the Planck (or even electroweak) scale, exceed the observed $\rho_\Lambda$ by $\sim 10^{60}\text{–}10^{120}$:

$$\rho_\Lambda^{\text{obs}} \sim 10^{-47}\,\text{GeV}^4 \;\ll\; \rho_\Lambda^{\text{QFT}} \sim M_{\text{cut}}^4.$$

A quantitative clash between GR and QFT — a fine-tuning / domain-mismatch, not a strict logical contradiction. No accepted mechanism cancels the bulk while leaving the tiny observed remainder. Catalogued in *Chapter 14 (p. 191)*.

- **Dark matter and dark energy as unidentified components ($\sim 95\%$ of the budget).** `[OPEN]` No laboratory detection of either: decades of null WIMP direct/indirect/collider searches, and an unexplained dark-energy scale. This is *incompleteness*, not inconsistency; conceivably part of the effect signals a GR breakdown rather than new substance (disfavored by the multi-pillar concordance).
- **The Hubble ($H_0$) tension.** `[CONTESTED, possibly a breakdown]` The early-universe value from CMB+ΛCDM ($H_0 \approx 67$–$68$) disagrees at the $\sim 4$–$6\sigma$ level with the late-universe distance-ladder value (Cepheid-calibrated SNe Ia, $H_0 \approx 73$). If not a systematic, it is the most likely current crack in ΛCDM, possibly requiring new early-universe physics (early dark energy, extra $N_{\text{eff}}$) that reduces $r_d$. Unresolved as of this writing.
- **The $S_8/\sigma_8$ tension.** `[CONTESTED, milder]` Some weak-lensing and cluster-count surveys infer a lower $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ (a less-clustered late universe) than Planck-ΛCDM extrapolation predicts, at $\sim 2$–$3\sigma$. More sensitive to baryonic-feedback systematics than $H_0$, but a watched anomaly; could signal suppressed growth (warm/interacting DM, neutrino mass) or under-modeled astrophysics.
- **The measure problem in eternal inflation.** `[OPEN, foundational]` If inflation is generically eternal, computing observation probabilities requires regulating ratios of infinities; different measures (proper-time, scale-factor, causal-patch) give different and sometimes pathological predictions (Boltzmann brains, the youngness paradox). This undermines the predictivity that motivated inflation — a conceptual

incompleteness in the multiverse extension, not a flaw in the minimal fit to data.

- **The low-entropy past / arrow-of-time problem.** `[OPEN]` Time-symmetric microphysics plus a thermodynamic arrow forces a special low-entropy boundary condition; the early universe's extreme gravitational smoothness is wildly improbable on any uniform measure (Penrose). ΛCDM/inflation does not explain *why*. An explanatory gap consistent with, but not derivable within, the framework. See *Chapter 5 (p. 40)*.
- **The lithium problem.** `[OPEN/CONTESTED, narrow]` BBN with the CMB baryon density overpredicts $^7$Li by $\sim 3\times$. A localized failure (stellar depletion, nuclear rates, or speculatively new physics) that does not threaten BBN broadly.
- **CMB large-scale "anomalies."** `[CONTESTED/SPECULATIVE]` Low quadrupole, hemispherical asymmetry, the "cold spot" — mild ($\sim 2$–$3\sigma$) and possibly a-posteriori flukes (look-elsewhere effect). Watched, not established. Relatedly, the trans-Planckian problem (inflation stretches sub-Planckian wavelengths to cosmic scales) makes predictions formally depend on unknown UHE physics, though robustly only at unobservably small corrections.

## Open problems (internal)

- **The cosmological-constant problem and the coincidence problem.** `[OPEN]` Why is $\rho_\Lambda$ nonzero yet tens of orders of magnitude below every known-physics vacuum-energy contribution (the electron loop alone overshoots by ~30 orders; the "120 orders" is a cutoff-regularization artifact — see *Chapter 16 (p. 308)* §8.1), and why is it comparable to $\rho_m$ *today*? The deepest quantitative puzzle in fundamental physics. Weinberg's anthropic bound predicted a small nonzero Λ before its detection, but relies on the contested multiverse+measure.
- **The microphysical identity of dark matter.** `[OPEN]` Gravitational evidence is overwhelming and concordant; null particle searches (direct detection approaching the neutrino

floor, indirect, collider, ongoing axion) leave WIMP/axion/sterile-neutrino/primordial-black-hole/hidden-sector options open.

- **Is dark energy a true constant or dynamical?** `[OPEN/CONTESTED]` $w = -1$ fits most data; DESI DR2 BAO + CMB prefer evolving $w(z)$ at 3.1σ (2.8–4.2σ with SNe, sample-dependent; arXiv:2503.14738), but the preference is SN-systematics-fragile (< 2σ under low-z recalibration, arXiv:2502.04212; Bayesian evidence from DESI+CMB alone modestly favors ΛCDM, arXiv:2511.10631). Near-decisive with the 2027 five-year DESI $w(z)$; see *Chapter 14 (p. 191)* GC-3.
- **What set the low-entropy initial state and the arrow of time?** `[OPEN]` No dynamical derivation; the past hypothesis is a posited boundary condition; inflation does not resolve it. Connects to quantum gravity and the nature of the Big Bang.
- **Did inflation occur, and what is the inflaton? Will primordial B-modes ($r \neq 0$) be detected?** `[OPEN]` A B-mode detection at the predicted level would be near-decisive; current bounds ($r \lesssim 0.03$–$0.06$) already exclude the simplest large-field models. Alternatives (bounce, ekpyrosis, string gas) remain logically possible.
- **Is the multiverse / eternal inflation real, and can the measure problem be solved?** `[OPEN/SPECULATIVE]` Borders the limits of empirical science; without a principled measure there are no well-defined probabilities. Whether it is even a scientific question is itself contested.
- **The Hubble tension: new physics or systematics?** `[CONTESTED/OPEN]` The leading candidate symptom of ΛCDM breakdown. Early-time fixes can worsen $S_8$ or need fine-tuning; late-time-only fixes are constrained by BAO. See *Chapter 15 (p. 233)*.
- **Averaging / backreaction (the "fitting problem").** `[CONTESTED]` Most experts judge backreaction too small to mimic dark energy, but a rigorous derivation of the averaged dynamics (Buchert equations) is incomplete and its magnitude is debated.

- **Reconciling the cosmic preferred frame and global cosmic time with frame-independence, and the nature of time at/before the Big Bang.** `[OPEN, partly philosophical]` Not a contradiction (symmetry breaking by matter), but the status of cosmic time in a future singularity-free quantum-gravitational description ties into the "problem of time."

## Connections to other frameworks

- ***General relativity* (Chapter 10 (p. 121))** — Parent framework. Cosmology *is* GR applied to the whole universe; the Friedmann equations are Einstein's equations under maximal spatial symmetry. The classical singularity and the cosmological constant inherit their status from GR, and any cosmological test of gravity is a test of GR on the largest scales.
- ***Quantum field theory* (Chapter 8 (p. 89)) / *particle physics* (Chapter 9 (p. 106))** — Supplies the matter sector (thermal history, the BBN nuclear network, neutrino decoupling, baryogenesis) and clashes sharply via the cosmological-constant problem (QFT vacuum energy vs observed $\Lambda$). The inflaton and dark matter are sought as SM extensions; $N_{\text{eff}}$, neutrino masses, and the baryon asymmetry are shared observables.
- **Quantum gravity** — Needed where cosmology breaks down: the initial singularity, the Planck-era pre-inflationary state, and possibly $\Lambda$ itself (the string "landscape" as the multiverse substrate for an anthropic $\Lambda$). The quantum-gravity "problem of time" bears on cosmic time. No empirically confirmed input yet. See *Chapter 18 (p. 349)*.
- ***Statistical mechanics* (Chapter 6 (p. 56)) & *Chapter 5 (p. 40)*** — Source of the arrow-of-time problem: the second law's cosmological grounding is the low-entropy past hypothesis. Self-gravitating thermodynamics (negative specific heat) and horizon entropy via Bekenstein–Hawking $S = A/4G$ link gravitational entropy to areas and to holography.

- ***Information theory* (Chapter 12 (p. 155)) / holography** — [SPECULATIVE] Covariant entropy bounds, de Sitter entropy, and the cosmological horizon as a holographic screen aim to constrain cosmology's degrees of freedom and possibly reframe $\Lambda$ and the measure problem. An active but unestablished interface; see *Chapter 17 (p. 325)*.
- **Astrophysics & observational astronomy** — Provides the data: SNe Ia (acceleration), the Cepheid distance ladder ($H_0$), galaxy redshift surveys and BAO (geometry + growth), weak lensing ($S_8$), 21-cm and Lyman-$\alpha$ forest (high-$z$ matter power), cluster counts. Cosmology in turn sets the boundary conditions for galaxy and star formation. Probe systematics are central to the current tensions.
- **Philosophy of science / foundations** — Cosmology stresses the limits of empirical method: the multiverse and measure problem raise testability questions; the Copernican principle and the past hypothesis are partly metaphysical posits; the anthropic principle is invoked (controversially) for $\Lambda$; the single-universe sample size challenges standard inference (cosmic variance). See *Chapter 2 (p. 8)*.

For the cross-domain unification picture, see *Chapter 3 (p. 15)* and *Chapter 18 (p. 349)*.

## Key references


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- R. Penrose, "Singularities and Time-Asymmetry," in *General Relativity: An Einstein Centenary Survey*, eds. Hawking & Israel (CUP, 1979); and *The Road to Reality* (Jonathan Cape, 2004) — the Weyl curvature hypothesis and the low-entropy past.
- Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020) — the definitive CMB determination of the six ΛCDM parameters; baseline for the $H_0$ and curvature discussions. (Numerical values above are stated approximately by design.)

## References

Full bibliographic entries are collected in the Bibliography (p. 776); the Key references section above lists the works directly load-bearing for this page.

CHAPTER 12

# Information Theory and Physics

This page is the authoritative deep-dive on the mathematical structures linking **information** and **physics**, spanning four strata of decreasing certainty: classical (Shannon/Kolmogorov) information, quantum information, the thermodynamics of computation, and the information-theoretic structure of gravity and spacetime. The organizing discipline throughout is to mark exactly where each layer is a theorem, where it is well-motivated inference, and where it is principled speculation. See *Chapter 2 (p. 8)* for the marker definitions and *Chapter 3 (p. 15)* for this domain's place in the larger landscape.

## Scope

The domain comprises four nested layers:

1. **Classical information theory** — Shannon entropy, mutual information, channel capacity, the data-processing inequality, and Kolmogorov (algorithmic) complexity. This layer is `[ESTABLISHED]` mathematics with no physical domain restriction.
2. **Quantum information** — von Neumann entropy, qubits, no-cloning, teleportation, entanglement measures, and entanglement entropy in QFT. `[ESTABLISHED]` theory; the operational primitives are experimentally confirmed.
3. **Thermodynamics of computation** — Landauer's principle, reversible computation, the Maxwell-demon resolution, and physical limits on computation (Margolus–Levitin,

Bremermann, Bekenstein). A mix of [ESTABLISHED] bounds and [CONTESTED] interpretive bookkeeping.

4. **Information and gravity** — the Bekenstein and Bousso bounds, holography, the Ryu–Takayanagi formula, AdS/CFT as quantum error correction, and the "it-from-bit" / "spacetime-from-entanglement" programs. This layer ranges from [INFERENCE] (RT *within* AdS/CFT) to [SPECULATIVE]/[OPEN] (entanglement as the fundamental substrate of spacetime).

# Core formalism

## 1. Classical (Shannon) information theory — [ESTABLISHED]

A discrete random variable $X \sim p(x)$ over alphabet $\mathcal{X}$ has **Shannon entropy**

$$H(X) = -\sum_{x\in\mathcal{X}} p(x)\log p(x),$$

in bits if $\log = \log_2$, nats if natural. $H$ is the *unique* function (up to scale) satisfying continuity, monotonicity in $n$ for the uniform distribution, and the grouping/recursion axiom (Shannon 1948; Khinchin's axiomatization) [ESTABLISHED]. Operationally, $H(X)$ is the asymptotic optimal lossless compression rate: by the **Source Coding Theorem**, $N$ i.i.d. copies compress to $\approx NH(X)$ bits with vanishing error and no lower rate is achievable. The engine is the **Asymptotic Equipartition Property** (AEP): $-\frac{1}{N}\log p(X_1,\ldots,X_N) \to H(X)$ in probability, concentrating measure on $\approx 2^{NH}$ "typical" sequences each of probability $\approx 2^{-NH}$.

Joint, conditional, relative entropy and mutual information:

$$H(X,Y) = -\sum p(x,y)\log p(x,y), \qquad H(X\mid Y) = H(X,Y) - H(Y),$$

$$D(p\|q) = \sum_x p(x)\log\frac{p(x)}{q(x)} \;\geq\; 0 \quad \text{(Gibbs' inequality)},$$

$I(X;Y) = H(X) - H(X \mid Y) = H(X) + H(Y) - H(X,Y) = D\big(p(x,y)\|p(x)p(y)\big) \geq 0.$

Mutual information is symmetric, vanishes iff $X \perp Y$, and equals the relative entropy from the joint to the product distribution — hence its nonnegativity.

A discrete memoryless channel $p(y \mid x)$ has **capacity** $C = \max_{p(x)} I(X;Y)$. The **Noisy Channel Coding Theorem** makes $C$ a sharp threshold: reliable communication is possible for any rate $R < C$ and impossible for $R > C$ `[ESTABLISHED]`. For the additive white Gaussian noise channel of bandwidth $B$, signal power $P$, noise power $N$, the **Shannon–Hartley** capacity is $C = B\log_2(1 + P/N)$.

The **data-processing inequality** (DPI): if $X \to Y \to Z$ is a Markov chain, then $I(X;Z) \leq I(X;Y)$ `[ESTABLISHED]`. Post-processing cannot increase information about the source. Its quantum generalization (monotonicity of relative entropy under CPTP maps) is the structural backbone of the entropy bounds in gravity discussed below.

## 2. Kolmogorov (algorithmic) complexity — `[ESTABLISHED]` math, `[OPEN]` computability caveats

For a universal prefix Turing machine $U$, the **Kolmogorov complexity** of a finite string $x$ is

$K(x) = \min\{|p| : U(p) = x\},$

the length of the shortest program outputting $x$. The **invariance theorem** gives $|K_U(x) - K_V(x)| \leq c_{U,V}$ for any two universal machines, so $K$ is well-defined up to an additive constant `[ESTABLISHED]`. $K$ is **uncomputable** — no algorithm computes it for all $x$ (equivalent to undecidability of halting; the root of Chaitin's incompleteness results) `[ESTABLISHED]`. For a computable source, $\mathbb{E}[K(X)] = H(X) + O(1)$`[ESTABLISHED]`. The distinction is conceptual: $H$ measures average uncertainty of an *ensemble*; $K$ measures the intrinsic descriptive content of a *single object*. For a single string there is no canonical "amount of information" — see *Where it breaks down*.

## 3. Quantum information — `[ESTABLISHED]`

A quantum system is a Hilbert space $\mathcal{H}$; a state is a density operator $\rho \succeq 0$, $\mathrm{Tr}\,\rho = 1$. A **qubit** is $\mathcal{H} = \mathbb{C}^2$, with pure states $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ parameterized on the Bloch sphere. The **von Neumann entropy** is

$$S(\rho) = -\mathrm{Tr}(\rho \log \rho) = -\sum_i \lambda_i \log \lambda_i,$$

$\lambda_i$ the eigenvalues of $\rho$. $S = 0$ iff $\rho$ is pure; $S = \log d$ for the maximally mixed state. Key inequalities `[ESTABLISHED]`:

- **Subadditivity:** $S(\rho_{AB}) \le S(\rho_A) + S(\rho_B)$.
- **Araki–Lieb:** $S(\rho_{AB}) \ge |S(\rho_A) - S(\rho_B)|$.
- **Strong subadditivity (SSA)** (Lieb–Ruskai 1973): $S(\rho_{ABC}) + S(\rho_B) \le S(\rho_{AB}) + S(\rho_{BC})$, equivalently the conditional mutual information $I(A;C \mid B) \ge 0$. SSA is the deep theorem of the subject; nearly every entropy inequality in quantum gravity descends from it.

Unlike the classical case, $S(\rho_A)$ can exceed $S(\rho_{AB})$ when $AB$ is pure and entangled — the diagnostic of entanglement. The **quantum mutual information** is $I(A;B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \ge 0$, and the **quantum relative entropy** $S(\rho\|\sigma) = \mathrm{Tr}\,\rho(\log \rho - \log \sigma)$ is **monotone under any CPTP map** $\mathcal{N}$: $S(\mathcal{N}\rho\|\mathcal{N}\sigma) \le S(\rho\|\sigma)$ — the quantum DPI, equivalent in strength to SSA.

The **no-cloning theorem** (Wootters–Zurek, Dieks 1982): no unitary $U$ satisfies $U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$ `[ESTABLISHED]`. Linearity plus inner-product preservation would force $\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$, impossible for non-orthogonal, non-identical states. No-cloning underwrites quantum cryptography and the sharpness of the black-hole information puzzle.

**Teleportation** (Bennett et al. 1993): one shared Bell pair plus two classical bits transmits one unknown qubit exactly `[ESTABLISHED]`. The qubit is destroyed at the sender (consistent with no-cloning) and no superluminal signalling occurs (classical bits are required). **Superdense coding** is its dual: one qubit plus one Bell pair sends two classical bits.

For a **pure bipartite state**, the canonical entanglement measure is the **entanglement entropy** $E(|\psi\rangle_{AB}) = S(\rho_A) = S(\rho_B)$ (equal by the Schmidt decomposition $|\psi\rangle = \sum_i \sqrt{\lambda_i}\,|i\rangle_A|i\rangle_B$). For **mixed states**, inequivalent measures appear — entanglement of formation, distillable entanglement $E_D$, entanglement cost $E_C$, and the relative entropy of entanglement $E_R(\rho) = \min_{\sigma\in\mathrm{SEP}} S(\rho\|\sigma)$ — all reducing to $S(\rho_A)$ on pure states and obeying $E_D \le E_C$ with strict gaps (bound entanglement) [ESTABLISHED]. Asymptotically reversible LOCC manipulation holds for pure states ($E_D = E_C = S(\rho_A)$) but generally fails for mixed states.

**Entanglement entropy in QFT.** For a spatial region $A$, $S(\rho_A)$ of the reduced state is UV-divergent and obeys an **area law**: the leading divergence in $d$ spatial dimensions is $S \sim c\,\mathrm{Area}(\partial A)/\epsilon^{d-1}$ with cutoff $\epsilon$ [ESTABLISHED]. In a $1+1$ CFT, for an interval of length $\ell$, $S = \frac{c}{3}\log(\ell/\epsilon)$ with $c$ the central charge (Holzhey–Larsen–Wilczek; Calabrese–Cardy) [ESTABLISHED]. The area-law scaling is the bridge to the holographic entropy bounds. See *Chapter 8 (p. 89)*.

## 4. Thermodynamics of computation — [ESTABLISHED] core, [CONTESTED] interpretation

**Landauer's principle** (1961): erasing one bit in contact with a bath at temperature $T$ dissipates at least

$$\Delta Q \ge k_B T \ln 2$$

of heat, tying *logical* irreversibility (a many-to-one map on logical states) to *thermodynamic* irreversibility. The bound is derivable from phase-space contraction and from the Jarzynski/Crooks fluctuation theorems, and has been experimentally confirmed near saturation (colloidal-particle and single-spin experiments, c. 2012 onward) [ESTABLISHED]. Its logical *status* as an independent principle is [CONTESTED] (see breakdowns).

**Reversible computation** (Bennett 1973): any computation embeds in a logically reversible one (e.g. via the universal Toffoli gate) that, run quasi-statically, costs arbitrarily little energy per step — only erasure costs `[ESTABLISHED]`. The **Maxwell-demon / Szilard-engine resolution** (Szilard 1929; Landauer; Bennett 1982): the demon's apparent extraction of $k_B T \ln 2$ per bit of which-side information is repaid when its memory is reset, restoring the second law. The modern **Sagawa–Ueda generalized second law** reads $\langle W_{\text{ext}} \rangle \leq k_B T \, \Delta I$ — information is a thermodynamic resource convertible to work, demonstrated experimentally `[ESTABLISHED]`. See *Chapter 5 (p. 40)* and *Chapter 6 (p. 56)*.

Physical limits on computation:

- **Margolus–Levitin theorem** (1998) `[ESTABLISHED]`: a system with mean energy $E$ above its ground state needs time $t_\perp \geq \frac{\pi\hbar}{2E}$ to reach an orthogonal state — a maximum rate of distinct operations $\leq 2E/(\pi\hbar)$. Combined with the Mandelstam–Tamm bound $t_\perp \geq \frac{\pi\hbar}{2\Delta E}$, the operative speed limit is the min of the two.
- **Bremermann's limit** (1962) `[INFERENCE/heuristic]`: $\sim c^2/h \approx 1.36 \times 10^{50}$ bit s$^{-1}$ kg$^{-1}$, a back-of-envelope ceiling from $E = mc^2$ plus an energy–time argument, not a theorem.
- **Bekenstein bound** (1981) `[INFERENCE]`: $S \leq \dfrac{2\pi k_B R E}{\hbar c}$ for a system of radius $R$ and energy $E$.
- **Holographic ('t Hooft–Susskind) bound** `[INFERENCE]`: maximal entropy is the boundary area in Planck units, $S_{\max} = \frac{k_B A}{4\ell_P^2}$, matching the **Bekenstein–Hawking** entropy $S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar} = \frac{k_B A}{4\ell_P^2}$. Lloyd's "ultimate laptop" combines Margolus–Levitin with Bekenstein into operations- and memory-bounds.

## 5. Holography, RT, and emergent spacetime

**Bousso covariant entropy bound** (1999) `[INFERENCE]`: entropy flux through a light-sheet (a null hypersurface of non-positive expansion) from a surface of area $A$ obeys $S \leq A/4G\hbar$ (Planck units) — the covariant refinement of the holographic bound, reducing to Bekenstein in the appropriate limit.

**Ryu–Takayanagi (RT) formula** (2006) `[INFERENCE, within AdS/CFT]`: in a holographic CFT with AdS dual, the entanglement entropy of a boundary region $A$ is

$$S(A) = \frac{\mathrm{Area}(\gamma_A)}{4G_N},$$

$\gamma_A$ the minimal bulk surface homologous to $A$. The covariant generalization is **HRT** (extremal surface), and the all-orders form is the **quantum extremal surface (QES)** prescription (Engelhardt–Wall): extremize $\mathrm{Area}/4G_N + S_{\mathrm{bulk}}$. RT geometrizes entropy — boundary SSA becomes a statement about minimal surfaces, and bulk geometry is reconstructible from boundary entanglement (entanglement-wedge reconstruction).

**Quantum error correction (QEC) and bulk emergence** `[INFERENCE]`: Almheiri–Dong–Harlow (2015) reframed AdS/CFT as a quantum error-correcting code — bulk interior operators are redundantly (nonlocally) encoded in the boundary, like logical qubits protected against boundary erasure. The **HaPPY code** (Pastawski–Yoshida–Harlow–Preskill) and random tensor networks realize RT-type entropies exactly in toy models. Subregion duality is sharpened by the **JLMS relation** (bulk/boundary relative entropies coincide) and by modular-flow / Petz-map reconstruction (Faulkner–Lewkowycz). See *Chapter 10 (p. 121)*.

**Entanglement as the fabric of spacetime** `[SPECULATIVE]`: Van Raamsdonk argued that reducing the mutual entanglement of two CFTs (the thermofield double) disconnects the dual spacetime, suggesting spatial connectivity is built from entanglement; the **ER=EPR** conjecture (Maldacena–Susskind) identifies entanglement with an Einstein–Rosen bridge. The linearized Einstein equations have been *derived* from the first law of entanglement, $\delta S_A = \delta\langle K_A\rangle$ (modular Hamiltonian $K_A$), in holographic setups (Faulkner–Guica–Hartman–Myers–Van Raamsdonk 2014; Jacobson's 1995 "Einstein equation of state" is the precursor) [SPECULATIVE as a route to full GR — only the linearized equations, under specific assumptions]. These are striking but framework-dependent results, not a complete theory.

# Foundational assumptions

| ASSUMPTION | STATUS | JUSTIFICATION |
|---|---|---|
| Information has an observer-independent measure $H$, fixed by the continuity/monotonicity/recursion axioms | fundamental | Shannon's uniqueness theorem and Khinchin's axiomatization single out $H$; its operational meaning (compression/communication limits) is proven and validated daily `[ESTABLISHED]`. |
| The logarithm and base-2 "bit" are the natural units of information | conventional-choice | The log form is *forced* by additivity over independent sources (fundamental); base-2 vs nats vs base-$d$ is pure convention. Physics uses $k_B \ln 2$ to bridge to thermodynamics. |
| States are density operators on a Hilbert space; information measures are spectral functions ( $S(\rho)$) | fundamental | Follows from the Hilbert-space + Born-rule postulates; von Neumann entropy is singled out by Schumacher compression / the quantum AEP. Fundamental *conditional* on QM — whose finality is `[OPEN]`. |
| Gravitational entropy bounds scale with AREA, not volume (holographic principle) | likely-fundamental | $S_{\mathrm{BH}}$ is area-extensive on multiple independent derivations (Hawking, Euclidean action, string microstate counting, loop area spectra); area-law entanglement and the Bousso bound reinforce it. Near-universal `[INFERENCE]`, but no general background-independent proof. |

| | | |
|---|---|---|
| Landauer: logically irreversible ops cost $\geq k_B T \ln 2$ per erased bit | fundamental | Derivable from statistical mechanics and fluctuation theorems; measured near the bound. The *inequality* is fundamental `[ESTABLISHED]`; its status as an *independent* principle is `[CONTESTED]`. |
| Spacetime geometry is emergent from entanglement (RT / "it-from-qubit") | unclear | Rigorous only *within* AdS/CFT (negatively curved, asymptotically-AdS, large-$N$). RT/JLMS/QEC/first-law results are real there `[INFERENCE]`; generalization to our de Sitter-like universe is unknown `[SPECULATIVE]`. |
| AdS/CFT holds exactly: a bulk quantum-gravity theory equals a boundary CFT | likely-fundamental | Overwhelming consistent checks (symmetries, spectra, correlators, integrability, BH thermodynamics); proven in some low-$d$/BPS cases. No general nonperturbative proof — exactness/genericity `[OPEN]`. |
| ER=EPR: entanglement is a geometric (wormhole) connection | unclear | Bold unifying conjecture supported by traversable-wormhole and thermofield-double constructions, but `[SPECULATIVE]` — no first-principles derivation, status for generic entanglement unknown. |
| A physical Church–Turing thesis holds (no device computes a Turing-uncomputable function) | unclear | Suggested by all known physics but not a theorem; Malament–Hogarth spacetimes, real-valued analog computation, and hypercomputation proposals leave it `[OPEN/CONTESTED]`. The *efficient* thesis is separately challenged by quantum computing (BQP vs BPP). |
| Information is conserved (unitarity): pure $\to$ pure | fundamental | A pillar of quantum theory; the apparent conflict with BH evaporation is widely believed resolved in favor of unitarity (Page curve reproduced via QES/islands `[INFERENCE]`; AdS/CFT manifestly unitary). The mechanism in our universe is still being worked out. |

| | | |
|---|---|---|
| The continuum (smooth fields/manifolds) is the fundamental substrate vs. a discrete "it-from-bit" description | historical-artifact | Continuum QFT/GR are spectacularly successful, but UV divergences, finite BH entropy, and holographic finiteness all hint the continuum is *effective*. Wheeler's "it from bit" suggests otherwise; arguably "unclear" rather than settled. |

See *Chapter 20 (p. 458)* for the cross-domain ledger and *Chapter 17 (p. 325)*.

## Domain of validity

**Layer 1 (Shannon/Kolmogorov)** is mathematically exact with no physical domain restriction — it applies to any probabilistic source or string. Two caveats: (a) the operational theorems are *asymptotic* (i.i.d., large $N$), so finite-blocklength corrections (Polyanskiy–Poor–Verdú) matter for real codes; (b) $K(x)$ is uncomputable and thus a theoretical, not algorithmic, quantity.

**Layer 2 (quantum information)** is valid wherever non-relativistic or relativistic quantum theory holds; the operational primitives (no-cloning, teleportation, superdense coding, distillation) are experimentally confirmed to high precision. In continuum QFT, $S(\rho_A)$ is UV-divergent and requires a regulator, so only universal/finite pieces (mutual information, differences, central-charge coefficients) are cutoff-independent and physically meaningful.

**Layer 3 (thermodynamics of computation):** Landauer's bound holds in the quasi-static, weak-coupling, thermal-bath regime; it is a *lower* bound saturated only in the reversible limit at fixed $T$. Margolus–Levitin and Mandelstam–Tamm are exact theorems of unitary QM. The Bekenstein bound holds for weakly gravitating systems ($R$ larger than the Schwarzschild radius) and is saturated by black holes; where "radius" and "energy" become ambiguous (strong gravity, cosmology) the Bousso covariant bound is the correct statement.

**Layer 4 (RT/holography/emergent spacetime)** is rigorously formulated and tested essentially only within AdS/CFT: asymptotically anti-de Sitter spacetimes, large central charge / large $N$, strong 't Hooft coupling (Einstein-gravity regime), with a static (or HRT for time-dependent), homology-respecting setting; quantum corrections need QES/islands. Validity for de Sitter space (our accelerating universe), cosmological horizons, and non-holographic theories is **unestablished** — this is the central domain-of-validity frontier. See *Chapter 11 (p. 139)*.

## Where it breaks down

- **Continuum QFT entanglement entropy.** $S(\rho_A)$ is UV-divergent and the leading area-law term is regulator-dependent; only differences, mutual information, and universal coefficients are physical. More deeply, the local algebra of a QFT region is a **Type III$_1$ von Neumann factor**, on which no trace — and hence no standard density matrix or finite entropy — exists. The naive "entanglement entropy of a region" is ill-defined without specifying the operator algebra and a split/cutoff. This is structural, not merely calculational. `[ESTABLISHED]` (the obstruction); `[OPEN]` (the resolution).
- **Black-hole information paradox.** Hawking's semiclassical calculation predicts thermal (mixed) radiation from a pure collapse, implying non-unitary evolution — a genuine tension between EFT on a curved background and unitarity. QES/island computations now reproduce the unitary **Page curve** `[INFERENCE]`, strongly suggesting EFT was applied outside its domain (missing replica-wormhole contributions), but a complete first-principles mechanism for a realistic, asymptotically-flat black hole is `[OPEN]`. See *Chapter 10 (p. 121)*.
- **Bekenstein bound in strong gravity / cosmology.** The bound presupposes a well-defined region radius and energy in a weakly-curved background; near horizons and in cosmological spacetimes these become observer-dependent and the bound loses sharp meaning. Bousso's light-sheet construction repairs this covariantly, but even it requires the expansion/light-sheet

conditions and a suitable entropy-flux definition, and lacks a fully general first-principles proof. `[INFERENCE]`.

- **RT/holography outside AdS.** De Sitter space, flat-space holography, and finite-$N$/stringy regimes have no established RT analog; applying "spacetime = entanglement" to cosmology is extrapolation, not derived physics. This is a domain-of-validity boundary, **not** an internal inconsistency. `[OPEN]`.
- **Landauer — interpretive status.** Not a failure of the inequality (which is measured) but a `[CONTESTED]` claim about its logical standing: Earman–Norton and Hemmo–Shenker argue it is either a consequence of, or insufficient to independently rescue, the second law, and that the Maxwell-demon resolution admits inequivalent framings. The physics is agreed; the bookkeeping of where the irreducible cost "lives" is disputed.
- **Kolmogorov uncomputability / Shannon mismatch.** $K(x)$ is uncomputable and defined only up to a constant, so it cannot be operationalized for finite strings; the identity $\mathbb{E}[K] = H + O(1)$ holds only for computable sources, in expectation. There is no canonical "amount of information" for a single object.
- **Type III algebras and "a qubit of the field."** Because local subsystems are not tensor factors, much "entanglement as fabric of spacetime" discourse implicitly assumes a Type-I tensor-product structure the QFT lacks. Crossed-product / large-$N$ constructions (yielding Type II algebras with well-defined traces) partially repair this, but the foundational gap is real and recent. `[OPEN]`.
- **ER=EPR and "spacetime from entanglement" as general claims.** No demonstration exists that *generic* entanglement produces a smooth geometric connection, nor that our universe is so constituted. The first-law derivation recovers only the *linearized* Einstein equations; the full nonlinear, background-independent reconstruction is not achieved. `[SPECULATIVE]`.

## Open problems (internal)

- **BH information recovery in a realistic black hole.** `[OPEN → partially INFERENCE]`. The Page curve is reproduced via QES/islands and replica wormholes, indicating unitarity, but *how* information reaches an external observer, the role of the interior, and the firewall/complementarity debate remain unresolved. Islands are computed in controlled models, not yet shown to resolve the paradox in nature. *(2026-07-02 refinement)* Two named objections remain live `[CONTESTED]`: the **ensemble-vs-single-theory** reading of replica wormholes (Marolf–Maxfield; PSSY; contra Schlenker–Witten's sub-threshold theorem (3d)), and the **massive-graviton/gravitating-bath** critique (Geng–Karch et al.: all controlled $d > 2$ island constructions have massive gravitons; islands strain the gravitational Gauss law in long-range massless gravity). A replica-independent second leg — crossed-product Type II algebras with von Neumann entropy $= S_{\text{gen}}$ and a perturbative GSL/QFC — now supports unitarity-compatible entropy bookkeeping at the perturbative level (see the Type III bullet under "Where it breaks down" and *Chapter 14 (p. 191)* GC-13).
- **Does holography/RT extend beyond AdS — to de Sitter and cosmology?** `[OPEN]`. No established dS holographic dual or RT analog; dS/CFT proposals are far less constrained than AdS/CFT. Arguably the central open question of the program.
- **The correct definition of entanglement entropy in quantum gravity** given Type III algebras and no fundamental tensor factorization. `[OPEN]`. Crossed-product / "algebra of observables in gravity" constructions are promising but incomplete.
- **A physical realization of hypercomputation vs. a Church–Turing law of nature.** `[OPEN/CONTESTED]`. All proposed loopholes face severe physical-plausibility objections

(infinite energy/precision, instability); no consensus, no decisive theorem.

- **Reconcile area-extensive (finite, holographic) gravitational degrees of freedom with the volume-extensive, infinite-dimensional Hilbert space of local QFT.** `[OPEN]`. The cosmological-constant problem, the species bound, and holographic finiteness all point to over-counting in continuum QFT; the precise reorganization (and whether spacetime is discrete) is unknown.
- **Derive the FULL nonlinear Einstein equations from entanglement/information principles, background-independently.** `[OPEN/SPECULATIVE]`. Linearized equations follow from entanglement first laws and Jacobson's thermodynamic derivation gives Einstein's equations as an equation of state, but a complete nonperturbative generic derivation is not in hand.
- **Is quantum mechanics itself derivable from information-theoretic axioms?** `[OPEN]`. Reconstructions (Hardy 2001; Chiribella–D'Ariano–Perinotti; Clifton–Bub–Halvorson "no signalling / no broadcasting / no bit commitment") derive QM from operational postulates — suggestive — but multiple inequivalent axiom sets succeed, so which (if any) is fundamental is open.

See *Chapter 15 (p. 233)*, *Chapter 14 (p. 191)*, and *Chapter 19 (p. 392)*.

## Connections to other frameworks

- ***Chapter 6 (p. 56)* & *Chapter 5 (p. 40)*:** a deep structural identity — Gibbs/Boltzmann and Shannon entropy share the $-\sum p \log p$ form; Jaynes recast statistical mechanics as MaxEnt inference. Landauer and the fluctuation theorems (Jarzynski, Crooks, Sagawa–Ueda) make information a thermodynamic resource at $k_B T \ln 2$ per bit. The second law and relative-entropy monotonicity are reflections of the same arrow `[INFERENCE on the depth of the identity]`.

- ***Chapter 7 (p. 71)*****:** no-cloning, no-signalling, and monogamy of entanglement constrain what is operationally possible; the field both *uses* QM and seeks to *reconstruct* it. Decoherence and the Born rule are reinterpreted informationally (quantum Darwinism, Zurek).
- ***Chapter 10 (p. 121)*****:** the strongest and most contested link — Bekenstein–Hawking entropy ties horizon area to information; holography, RT, and "spacetime from entanglement" propose geometry *is* organized entanglement; Jacobson derived Einstein's equations thermodynamically. Here the domain is most ambitious and least empirically grounded.
- ***Chapter 8 (p. 89)* & string theory / AdS/CFT:** the concrete arena where RT, JLMS, entanglement-wedge reconstruction, QEC/HaPPY tensor networks, and the island formula are computable. Strominger–Vafa string microstate counting gave the first statistical-mechanical derivation of $S_{\mathrm{BH}}$, anchoring "it-from-bit" in a UV-complete theory. See also *Chapter 9 (p. 106)*.
- ***Chapter 13 (p. 172)* & computer science:** Kolmogorov complexity, the Church–Turing thesis, and the extended (efficient) thesis bridge to CS; quantum computing challenges the efficient thesis. The "complexity = volume / complexity = action" conjectures (Susskind) speculatively link computational complexity to black-hole interiors `[SPECULATIVE]`.
- **Condensed matter / many-body physics:** entanglement-entropy area laws, topological entanglement entropy (Kitaev–Preskill, Levin–Wen), and tensor networks (MERA, PEPS) connect the same structures to phases of matter; MERA's geometry inspired discrete holography.
- ***Chapter 11 (p. 139)*****:** a clashing frontier — holographic and Bekenstein/Bousso bounds constrain the observable universe's information content; the de Sitter horizon has an entropy by analogy with black holes, but dS holography is unestablished. Whether "entanglement = geometry" survives in an accelerating, non-AdS universe is a live tension, not a settled connection.

See *Chapter 18 (p. 349)*, *Chapter 16 (p. 308)*, and the Glossary (p. 747).

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## References

See the Bibliography (p. 776) for the consolidated, canonical reference list across the wiki.

CHAPTER 13

# Mathematics for Physics

This page treats mathematics not as a neutral notation but as the *substrate* of physical theory, examined layer by layer for where it is a theorem versus where it is a heuristic. The organizing question throughout is epistemic: which mathematical structures underlying physics are rigorously established, which are productive abuses of notation, and which mark genuine open frontiers. See *Chapter 2 (p. 8)* for the marker conventions and *Chapter 3 (p. 15)* for how this domain sits beneath the others.

## Scope

Covered: linear algebra and infinite-dimensional Hilbert spaces; functional analysis (unbounded self-adjoint operators, the spectral theorem, rigged Hilbert spaces); operator algebras (C*- and von Neumann algebras, Tomita–Takesaki modular theory); differential geometry (manifolds, fiber bundles, connections, curvature) as the geometry of gauge theory and gravity; Lie groups, Lie algebras, and representation theory; measure theory and probability; topology and its physical role (anomalies, instantons, topological phases); the ill-defined path-integral measure and the rigor gap of interacting QFT; constructive and algebraic QFT; category theory and higher structures (TQFT, factorization algebras); and the "unreasonable effectiveness" meta-question.

Excluded: detailed physical predictions, numerical methods, and the internal physics of particular models except as illustrations of mathematical structure. Those live in the *domains/* (Chapter 3 (p. 15)) pages proper.

# Core formalism

## 1. Linear algebra and Hilbert space

The kinematic arena of quantum mechanics is a separable complex Hilbert space $\mathcal{H}$: a complex vector space with inner product $\langle\cdot,\cdot\rangle$ (antilinear in the first slot by physics convention), complete in the induced norm $\|\psi\| = \sqrt{\langle\psi,\psi\rangle}$. Separability (a countable orthonormal basis exists) is a standing background assumption. The canonical model is $L^2(\mathbb{R}^n, d^n x)$. `[ESTABLISHED]`

States are not vectors but rays — unit vectors up to phase, i.e. points of projective space $\mathbb{P}(\mathcal{H})$, equivalently rank-one projections. **Wigner's theorem** `[ESTABLISHED]` states that any bijection of $\mathbb{P}(\mathcal{H})$ preserving transition probabilities $|\langle\psi,\phi\rangle|^2$ is implemented on $\mathcal{H}$ by an operator that is *either* unitary *or* antiunitary. Combined with continuity (one-parameter groups) this forces unitarity, and via Stone's theorem a self-adjoint generator.

## 2. Unbounded operators and the spectral theorem

Observables are self-adjoint operators. The subtlety separating physics from naïve linear algebra is that the fundamental observables — position $\hat{x}$, momentum $\hat{p} = -i\hbar\, d/dx$, Hamiltonians — are **unbounded** and defined only on a dense domain $D(A) \subsetneq \mathcal{H}$. One must distinguish:

- **symmetric**: $\langle A\psi,\phi\rangle = \langle\psi, A\phi\rangle$ for $\psi,\phi \in D(A)$;
- **self-adjoint**: symmetric *and* $D(A) = D(A^*)$.

Only self-adjointness — not mere symmetry ("Hermiticity") — guarantees a real spectrum and unitary evolution $e^{-itA}$. The gap is measured by the **deficiency indices** $n_\pm = \dim\ker(A^* \mp i)$: a symmetric operator is essentially self-adjoint iff $n_+ = n_- = 0$, and admits self-adjoint extensions iff $n_+ = n_-$. Different extensions are genuinely different physics (boundary conditions; the particle on a half-line; singular potentials). `[ESTABLISHED]`

**Spectral theorem (multiplication form).** For any self-adjoint $A$ there is a measure space $(M, \mu)$, a unitary $U : \mathcal{H} \to L^2(M, \mu)$, and a real measurable $f$ with $UAU^{-1}$ = multiplication by $f$. Equivalently there is a projection-valued measure $E(\cdot)$ on the Borel sets of $\mathbb{R}$ with

$$A = \int_{\sigma(A)} \lambda \, dE(\lambda), \qquad f(A) = \int f(\lambda) \, dE(\lambda).$$

This is what makes "probability that a measurement of $A$ lies in $\Delta$" equal $\|E(\Delta)\psi\|^2$ — the Born rule on the spectral side. `[ESTABLISHED]`

**Stone's theorem.** Strongly continuous one-parameter unitary groups $U(t)$ are in bijection with self-adjoint generators: $U(t) = e^{-itA}$, with $A\psi = i \lim_{t\to 0}(U(t)\psi - \psi)/t$ on the domain where the limit exists. This underwrites time evolution (generator: the Hamiltonian) and translations (generator: momentum). `[ESTABLISHED]`

**Rigged Hilbert spaces.** The Dirac eigenkets $|x\rangle, |p\rangle$ are not elements of $\mathcal{H}$ (continuous spectrum has no eigenvectors). Their rigorous home is a **Gelfand triple** $\Phi \subset \mathcal{H} \subset \Phi'$, with $\Phi$ a nuclear space (e.g. Schwartz functions) and $\Phi'$ its dual (tempered distributions); "eigenkets" are distributional eigenvectors in $\Phi'$ via the nuclear spectral theorem. It is `[ESTABLISHED]` that this makes Dirac's bra-ket bookkeeping rigorous; the physicist's manipulations are heuristic shorthand for it. See *Chapter 7 (p. 71)*.

## 3. Operator algebras

The algebraic approach takes the observables themselves as primary, rather than fixing $\mathcal{H}$.

A **C*-algebra** is a complex Banach algebra $\mathcal{A}$ with involution $*$ satisfying the C*-identity $\|a^*a\| = \|a\|^2$. The **Gelfand–Naimark theorem** `[ESTABLISHED]` shows every abstract C*-algebra is isometrically $*$-isomorphic to a norm-closed $*$-subalgebra of $B(\mathcal{H})$; the commutative case is $C_0(X)$ for locally compact Hausdorff $X$ — the seed of noncommutative geometry ("spaces" $\leftrightarrow$ commutative

C*-algebras, so general C*-algebras are "noncommutative spaces"). **States** are positive normalized linear functionals $\omega$, and the **GNS construction** reconstructs from $(\mathcal{A}, \omega)$ a representation $\pi_\omega$ on a Hilbert space with cyclic vector $\Omega$ such that $\omega(a) = \langle \Omega, \pi_\omega(a)\Omega \rangle$. `[ESTABLISHED]`

A **von Neumann algebra** is a $*$-subalgebra $M \subset B(\mathcal{H})$ equal to its own double commutant, $M = M''$ (equivalently weakly closed, by the bicommutant theorem). Murray–von Neumann classified factors (trivial center) into types $\mathrm{I}_n, \mathrm{I}_\infty$ (ordinary QM, $B(\mathcal{H})$), $\mathrm{II}_1, \mathrm{II}_\infty$ (continuous trace), and III (no trace). `[ESTABLISHED]` A structurally crucial fact for QFT: the **local algebras of relativistic quantum field theory are generically type $\mathrm{III}_1$ factors**. This is *why* there is no global number operator on a single local algebra, no normalizable vacuum localized there, and why the **Reeh–Schlieder theorem** holds (the vacuum is cyclic and separating for any local algebra). `[ESTABLISHED]` See *Chapter 8 (p. 89)*.

**Tomita–Takesaki modular theory.** Given a von Neumann algebra $M$ with a cyclic and separating vector $\Omega$, the closure of $S_0 : a\Omega \mapsto a^*\Omega$ has polar decomposition $S = J\Delta^{1/2}$, yielding the modular conjugation $J$ and the positive modular operator $\Delta$. The deep theorems: $JMJ = M'$ (the commutant), and the **modular automorphism group** $\sigma_t(a) = \Delta^{it} a \Delta^{-it}$ maps $M$ to itself. The state $\omega = \langle \Omega, \cdot\, \Omega \rangle$ satisfies the **KMS condition** at inverse temperature $\beta = 1$ for $\sigma_t$ — a structural identity between equilibrium statistical mechanics (KMS states) and the intrinsic dynamics a state induces on its algebra. `[ESTABLISHED]` This underlies the **Bisognano–Wichmann theorem** (the Rindler/Unruh modular flow is the Lorentz boost) and modern work on entropy and energy conditions. See *Chapter 6 (p. 56)*.

## 4. Differential geometry and gauge theory

A smooth manifold $M$ models spacetime/configuration space. The objects: tangent/cotangent bundles, tensor fields, the exterior algebra $\Omega^k(M)$ with exterior derivative $d$ ($d^2 = 0$), Stokes' theorem

$\int_M d\omega = \int_{\partial M} \omega$, and de Rham cohomology $H^k_{dR}(M) = \ker d/\operatorname{im} d$. For gravity one adds a (pseudo-)Riemannian metric $g$, the unique torsion-free metric (Levi-Civita) connection $\nabla$, curvature $R^\rho{}_{\sigma\mu\nu}$, and Einstein's equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}$. See *Chapter 10 (p. 121)*.

**Principal bundles and connections** are the geometry of gauge theory. A principal $G$-bundle $P \xrightarrow{\pi} M$ carries a free fiberwise $G$-action. A **connection** is a Lie-algebra-valued 1-form $A \in \Omega^1(P, \mathfrak{g})$ (the gauge potential); its **curvature** is

$$F = dA + \tfrac{1}{2}[A, A] = dA + A \wedge A, \qquad dF + [A, F] = 0 \ \ (\text{Bianchi}).$$

Matter fields are sections of associated vector bundles; the gauge-covariant derivative is $D = d + \rho(A)$ in representation $\rho$. Gauge transformations act as $A \mapsto gAg^{-1} - (dg)g^{-1}$, $F \mapsto gFg^{-1}$, and the Yang–Mills action is

$$S_{YM} = \frac{1}{2g^2} \int_M \operatorname{tr}(F \wedge \star F).$$

The dictionary "gauge field = connection on a principal bundle; field strength = curvature; Wilson line = parallel transport / holonomy" is exact. `[ESTABLISHED]` Characteristic classes (Chern, Pontryagin, Euler) built from $F$ are topological invariants; e.g. the instanton number / second Chern number $\frac{1}{8\pi^2} \int \operatorname{tr}(F \wedge F) \in \mathbb{Z}$.

**Index theory.** The Atiyah–Singer index theorem equates the analytic index of an elliptic operator (e.g. a Dirac operator \slashed$D$ coupled to a gauge field) with a topological integral of characteristic classes, controlling fermion zero modes, chiral anomalies, and instanton contributions:

$$\operatorname{ind}(\text{\slashed}D) = \dim\ker \text{\slashed}D - \dim\operatorname{coker} \text{\slashed}D = \int_M \hat{} A(M) \wedge \operatorname{ch}(F).$$

## 5. Lie groups and representation theory

Continuous symmetries are Lie groups $G$; their infinitesimal version are Lie algebras $\mathfrak{g}$ with bracket $[\cdot, \cdot]$, related by $\exp : \mathfrak{g} \to G$. Physical states transform in **unitary representations**. Structural results: Peter–Weyl (compact groups), highest-weight classification of finite-dimensional irreps of semisimple Lie algebras (Cartan–Weyl, root/weight lattices), and induced representations for noncompact groups. `[ESTABLISHED]`

The relativistic kinematic backbone is **Wigner's classification**: relativistic particles are unitary irreducible representations of the Poincaré group $\mathrm{ISO}(1,3)^{\uparrow}_{+} = \mathbb{R}^{1,3} \rtimes \mathrm{SL}(2,\mathbb{C})$, labeled by mass $m \geq 0$ (Casimir $P^2$) and spin/helicity (little group: $\mathrm{SU}(2)$ for $m > 0$, $\mathrm{ISO}(2)$ /helicity for $m = 0$). The need for *projective* representations and the universal cover $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}(1,3)$ is *why* half-integer spin (spinors) exist. `[ESTABLISHED]` See *Chapter 9 (p. 106).*

## 6. Measure, probability, and the path integral

Classical probability is Kolmogorov's: a measure space $(\Omega, \mathcal{F}, P)$ with $P(\Omega) = 1$, random variables as measurable functions, expectation as Lebesgue integral. Quantum theory generalizes this to a **noncommutative probability** over a von Neumann algebra with a state, and to POVM-valued observables. `[ESTABLISHED]` See *Chapter 12 (p. 155).*

The **path integral** $\langle q_f | e^{-iHT/\hbar} | q_i \rangle = \int \mathcal{D}q \, e^{iS[q]/\hbar}$ is rigorous in two regimes and heuristic in general:

- QM (finitely many degrees of freedom), **Euclidean** ($t \to -i\tau$): the Feynman–Kac formula expresses $e^{-\tau H}$ via the genuine **Wiener measure** on continuous paths — fully rigorous. `[ESTABLISHED]`
- The **Lorentzian** functional measure $\mathcal{D}q \, e^{iS}$ is *not* a measure: there is no translation-invariant $\sigma$-additive measure on infinite-dimensional spaces, and $e^{iS}$ oscillates without absolute convergence. `[ESTABLISHED that no such measure exists; the object is heuristic.]`
- In QFT the would-be measure $\mathcal{D}\phi$ lives on distributions, requires renormalization, and exists rigorously only for super-renormalizable models in low dimensions (constructive QFT: $P(\phi)_2$, $\phi^4_3$; Glimm–Jaffe, Nelson, Osterwalder–Schrader). The **Osterwalder–Schrader axioms** give precise conditions

(reflection positivity above all) under which a Euclidean measure reconstructs a relativistic QFT. `[ESTABLISHED]`

## 7. Topology in physics

Topology enters through *global* features invisible to local field equations: homotopy groups classify defects and solitons ($\pi_1$ vortices, $\pi_2$ monopoles, $\pi_3$ Skyrmions/instantons via $\pi_3(G) = \mathbb{Z}$ for simple $G$); cohomology and characteristic classes classify anomalies and flux quantization (Dirac quantization $\leftrightarrow c_1 \in H^2(M, \mathbb{Z})$); $K$-theory and twisted/equivariant variants classify topological phases of matter and D-brane charges. The integer quantum Hall conductance is a first Chern number (TKNN invariant). Anomalies are computed by index theory and descend via the Stora–Zumino descent equations. `[ESTABLISHED]`

## 8. Category theory and higher structures

Functorial QFT (Atiyah–Segal axioms) recasts a TQFT as a symmetric monoidal functor $Z : \mathrm{Bord}_n \to \mathcal{C}^{\otimes}$: manifolds $\mapsto$ vector spaces, bordisms $\mapsto$ linear maps, gluing $\mapsto$ composition. The **cobordism hypothesis** (Baez–Dolan; proof program by Lurie) characterizes fully extended TQFTs by their value on a point — a fully dualizable object in a symmetric monoidal $(\infty, n)$-category. The framework is rigorous and Lurie's proof is widely accepted `[INFERENCE]`, but the full written details remain less universally vetted than a textbook theorem. **Factorization algebras** (Costello–Gwilliam) rigorously encode the operator-product structure of *perturbative* QFT; this is `[ESTABLISHED as mathematics]`, though its full equivalence to physicists' nonperturbative QFT is only partial.

# Foundational assumptions

| Assumption | Status | Justification |
|---|---|---|
| Pure states are rays in a complex, separable Hilbert space (field $= \mathbb{C}$, not $\mathbb{R}$ or $\mathbb{H}$). | likely-fundamental | `[ESTABLISHED]` complex Hilbert space reproduces all data. `[CONTESTED]` whether $\mathbb{C}$ is *forced*: Stueckelberg's real-QM-with-superselection reproduces standard QM, and device-independent "ruling out real QM" results (Renou et al. 2021) reject real QM only under network/independence assumptions. Separability is a near-universal convenience; nonseparable spaces appear in some continuum settings. Quaternionic QM (Finkelstein–Adler) is logically consistent but empirically unsupported. |
| Observables are self-adjoint (not merely symmetric/Hermitian). | fundamental | `[ESTABLISHED]` self-adjointness is exactly what the spectral and Stone theorems require for real spectra and unitary dynamics; deficiency-index theory shows symmetric-but-not-self-adjoint operators give different or no physics. Choosing a self-adjoint extension is a physical choice (boundary conditions). PT-symmetric/pseudo-Hermitian QM weakens this to similarity-to-self-adjoint — a reformulation, not a refutation. |
| Dynamical symmetries act by unitary (or antiunitary) reps; time evolution is unitary. | fundamental | `[ESTABLISHED]` via Wigner's theorem plus Stone's theorem; antiunitarity is forced for time reversal. Strict unitarity is in tension with measurement "collapse," a `[CONTESTED]` interpretational issue, not a mathematical defect. |

| | | |
|---|---|---|
| Spacetime is a smooth ($C^\infty$) finite-dimensional manifold; fields are smooth sections of bundles over it. | conventional-choice | [INFERENCE] Smoothness is a powerful idealization, plausibly a low-energy effective description. Causal sets, spin foams, noncommutative geometry, and string-theoretic minimal length all posit that manifold structure dissolves near the Planck scale. Enormously effective, with no established reason it is fundamental. Even classically, weak/distributional solutions relax smoothness. |
| Gauge fields *are* connections on principal $G$-bundles; gauge transformations are bundle automorphisms (gauge = redundancy). | fundamental | [ESTABLISHED] as a mathematical identity (Wu–Yang/Trautman dictionary; Aharonov–Bohm and Dirac quantization confirm the bundle topology is physical). The "redundancy" reading is the standard [INFERENCE]; large-gauge/boundary "soft" charges refine but do not refute it. The structure group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ is fixed by experiment, hence physical, not conventional. |
| Probabilities follow the Born rule $ | \langle\phi | \psi\rangle |
| Relativistic QFT can be axiomatized (Wightman / Haag–Kastler / Osterwalder–Schrader) and a 4D interacting example satisfying these axioms exists. | likely-fundamental | [ESTABLISHED] the axiom systems are consistent with nontrivial low-dimensional models. [OPEN] whether central 4D theories (Yang–Mills, the Standard Model) satisfy them — the Yang–Mills mass-gap problem. A minority view holds perturbatively-defined theories like QED/$\phi^4$ are "trivial" ($\phi^4_4$ triviality is a theorem of Aizenman–Duminil-Copin), existing only as cutoff effective theories — a domain-of-validity matter, not an inconsistency. |

| | | |
|---|---|---|
| The Lorentzian path-integral "measure" $\mathcal{D}\phi\, e^{iS}$ is a legitimate integration measure. | historical-artifact | [ESTABLISHED] no $\sigma$-additive translation-invariant measure exists on infinite-dimensional space and $e^{iS}$ is non-absolutely-convergent. The rigorous content is via Euclidean reflection-positive measures (Wiener/OS) and analytic continuation, oscillatory-integral surrogates, or perturbation theory. Productive abuse of notation, astonishingly reliable as bookkeeping. |
| Renormalization is physically meaningful; the Wilsonian effective-field-theory / RG picture organizes cutoff dependence. | fundamental | [ESTABLISHED] empirically (QED, electroweak precision) and conceptually (Wilson's RG; perturbative renormalizability is BPHZ-rigorous order by order). The shift from "infinity subtraction" to "EFT with a physical cutoff" is a genuine conceptual correction; the older framing was partly a historical artifact. Nonperturbative existence of the continuum limit for non-asymptotically-free theories is [OPEN/CONTESTED]. |
| Local QFT algebras are type $\mathrm{III}_1$ von Neumann factors (no normal trace, no local number operator, vacuum cyclic & separating). | likely-fundamental | [ESTABLISHED] in algebraic QFT for standard models (Reeh–Schlieder, Bisognano–Wichmann, type-III results). It explains entanglement-area divergences and the need for *relative* (not absolute) entropy. The algebra type can shift in special limits (e.g. type II in some gravitational/large-$N$ crossover analyses). |
| Symmetry groups are Lie groups with continuous unitary reps; superselection sectors organize inequivalent representations. | fundamental | [ESTABLISHED] Wigner's particle classification, the role of universal covers (spin), and Doplicher–Haag–Roberts superselection theory are rigorous and predictive. Generalized/categorical symmetries and quantum groups (in low dimensions) extend but do not overturn this. |

See *Chapter 20 (p. 458)* for the cross-domain rollup.

## Domain of validity

Each layer is a theorem in a sharp regime and a heuristic beyond it.

1. **Hilbert-space QM** with the spectral/Stone machinery is fully rigorous for finitely many degrees of freedom with well-posed self-adjoint Hamiltonians; the subtleties (domains, self-adjoint extensions) are controllable. `[ESTABLISHED]`
2. **Stone–von Neumann uniqueness** holds *only* for finitely many canonical pairs. With infinitely many degrees of freedom (QFT, the thermodynamic limit) there are uncountably many unitarily *inequivalent* representations — not a pathology but the correct setting for distinct phases, superselection, and spontaneous symmetry breaking. This is precisely where the "one Hilbert space, one Fock vacuum" intuition breaks. `[ESTABLISHED]`
3. **Differential geometry / gauge theory as connections** is exact classically and as a quantum *kinematic* framework; its *quantization* (the Yang–Mills measure) is rigorous only perturbatively in 4D.
4. **Constructive QFT** rigorously builds interacting models in $d = 2$ ($P(\phi)_2$, Yukawa$_2$, Thirring) and $d = 3$ ($\phi^4_3$), exploiting super-renormalizability; $d = 4$ interacting theories are *not* constructed. `[ESTABLISHED]`
5. **Algebraic QFT** (Haag–Kastler nets) is a rigorous *structural* theory wherever local nets exist, including QFT on a fixed curved background; it does not by itself construct interacting 4D dynamics.
6. **Euclidean methods / OS reconstruction** are valid when reflection positivity holds and the continuation back to Lorentzian signature is justified — generically true for static/stationary spacetimes, problematic for generic Lorentzian/cosmological geometries.
7. The whole edifice presupposes a **fixed (possibly curved) background spacetime**. It has no established validity once

the metric itself is quantized: quantum gravity lies outside the domain of all these frameworks.

## Where it breaks down

- **Interacting QFT in 4D (Yang–Mills, the Standard Model).** No rigorous nonperturbative construction exists. Perturbation series are asymptotic (Dyson) and generically divergent; QED has a Landau pole and $\phi^4_4$ is *provably trivial* (Aizenman 1981; Aizenman–Duminil-Copin 2021), i.e. its continuum limit is free — strongly suggesting survival only as a cutoff EFT. A true incompleteness of the rigorous program (the Clay mass-gap problem), not a logical inconsistency. `[OPEN]`
- **The Lorentzian functional measure.** Literal nonexistence; meaning only via Euclidean continuation, oscillatory-integral surrogates, or perturbation theory. Domain-of-validity / abuse-of-notation issue. `[ESTABLISHED]`
- **Quantization ambiguities.** The classical $\to$ quantum map is not a homomorphism: the **Groenewold–van Hove no-go theorem** proves there is no quantization map respecting $\{\cdot,\cdot\} \mapsto \frac{1}{i\hbar}[\cdot,\cdot]$ on the *whole* Poisson algebra. Operator ordering, choice of polarization (geometric quantization), and self-adjoint extensions are genuine ambiguities fixed only by extra physical input. A fundamental obstruction, not a gap to be closed. `[ESTABLISHED]`
- **Wick rotation in generic curved/Lorentzian spacetimes.** The Euclidean$\leftrightarrow$Lorentzian dictionary requires a real Euclidean section and reflection positivity, generically absent in time-dependent or cosmological spacetimes (e.g. de Sitter). Without it, neither constructive techniques nor OS reconstruction apply. `[OPEN]` in those settings.
- **Gauge-fixing and the global configuration space.** The **Gribov ambiguity**: in non-abelian gauge theory no global continuous gauge-fixing section exists (**Singer's theorem** — the gauge orbit space is topologically obstructed), so Faddeev–Popov gauge-fixing is only locally valid and the BRST/path-

integral construction is incomplete nonperturbatively. A genuine topological obstruction. [ESTABLISHED]

- **Measurement vs. unitary evolution.** Unitary Schrödinger evolution and projective Born-rule update are two different dynamical laws; reconciling them (the measurement problem) is unresolved. Interpretational, not a mathematical inconsistency — each interpretation is internally consistent but they disagree on ontology. [CONTESTED]
- **QFT + dynamical gravity.** Perturbatively quantized GR is non-renormalizable ('t Hooft–Veltman one-loop matter result; Goroff–Sagnotti two-loop pure-gravity divergence). The smooth-manifold + quantized-metric framework breaks at the Planck scale — the outer boundary of this entire domain. [OPEN]

See *Chapter 14 (p. 191)* for how these are classified (inconsistency vs. domain mismatch vs. unsolved-but-consistent).

## Open problems (internal)

- **Existence of 4D quantum Yang–Mills with a mass gap** (Clay Millennium Problem). Construct a nontrivial 4D YM theory satisfying the Wightman/OS axioms with $\sigma(H) \subset \{0\} \cup [\Delta, \infty)$, $\Delta > 0$. Expected true on lattice/numerical and confinement grounds; no rigorous proof. [OPEN]
- **Nonperturbative construction of the Standard Model** (or any realistic 4D non-asymptotically-free theory). $\phi^4_4$ triviality [ESTABLISHED] and the QED Landau pole [INFERENCE] suggest some sectors are only EFTs; lattice chiral fermions and the sign problem add obstructions. [OPEN/CONTESTED]
- **Rigorous nonperturbative definition of the (Lorentzian) 4D path integral / functional measure**, beyond known low-dimensional and Euclidean reflection-positive cases. [OPEN]
- **Derivation vs. postulation of the Born rule** and resolution of the measurement problem. Gleason and envariance constrain it; Everettian derivations are disputed. [OPEN/CONTESTED]

- **Why complex Hilbert space?** Necessity of $\mathbb{C}$ over $\mathbb{R}$ or $\mathbb{H}$, and of the Hilbert-space framework itself. Operational reconstructions (Hardy 2001; Chiribella–D'Ariano–Perinotti) derive QM from axioms, but the complex-field choice and its experimental tests rest on auxiliary independence assumptions. `[OPEN]`
- **Complete proof/dissemination of the cobordism hypothesis** and a nonperturbative higher-categorical formulation of realistic 4D QFT. `[OPEN to partially settled]`
- **Reconciling QFT with dynamical/quantum gravity** at and below the Planck scale; the smooth-manifold substrate may not survive. `[OPEN]`
- **The "unreasonable effectiveness of mathematics"** (Wigner). Not a conjecture with an ordinary truth value but a genuine, unresolved meta-question. Partial deflationary accounts (selection of effective math, symmetry/representation-theoretic constraint, cognitive/anthropic) exist; none is decisive. Flag as a *question*, never a settled doctrine. `[OPEN/philosophical]`

See *Chapter 15 (p. 233)* and *Chapter 19 (p. 392)*.

## Connections to other frameworks

- ***Quantum mechanics* (Chapter 7 (p. 71))** — Mathematics supplies the entire kinematic substrate (Hilbert space, self-adjoint observables, spectral theorem, Stone, Gleason, Wigner). The clash is at measurement: the linear/unitary structure is silent on collapse, so the foundational debate is *downstream* of, not contained in, the mathematics.
- ***Quantum field theory* (Chapter 8 (p. 89))** — Where the mathematics is most powerful *and* most incomplete: operator algebras (type III), representation theory (Wigner classification), and geometry (gauge connections) are exact, but the dynamics/measure are rigorous only in low dimensions. The Yang–Mills mass gap sits exactly on this seam.

- ***General relativity* (Chapter 10 (p. 121))** and quantum gravity — Differential geometry is shared bedrock. The clash: GR makes the metric dynamical, but every rigorous QFT framework presupposes a fixed background; the smooth-manifold assumption is the suspected casualty.
- ***Statistical mechanics* (Chapter 6 (p. 56))** / ***Chapter 5 (p. 40)*** — Deep unification via KMS states and Tomita–Takesaki: equilibrium states *are* modular states of the observable algebra, so "temperature" and "time" share an algebraic origin. Reflection positivity links Euclidean QFT to classical lattice systems; phase transitions correspond to inequivalent representations.
- ***Classical mechanics* (Chapter 4 (p. 25))** — The symplectic geometry of phase space, Poisson brackets, and the Groenewold–van Hove obstruction define the classical → quantum boundary discussed above.
- ***Information theory* (Chapter 12 (p. 155))** and probability — Quantum theory is a noncommutative generalization of Kolmogorov probability (states on a von Neumann algebra, POVMs); information-theoretic reconstructions try to derive the Hilbert-space formalism, and modular relative entropy feeds back into QFT energy conditions and holography.
- ***Cosmology* (Chapter 11 (p. 139))** — Inherits the Wick-rotation breakdown: defining path integrals and QFT in de Sitter/time-dependent geometries is open.
- **Pure mathematics** (see the in-text references and *Chapter 13 (p. 172)* cross-links) — Bidirectional and historically transformative: physics (QFT/string theory) conjectured mirror symmetry, Seiberg–Witten invariants, the Jones polynomial via Chern–Simons, and monstrous moonshine, later proved by mathematicians; mathematics supplies index theory, operator algebras, and representation theory. The recurring clash is rigor: physical derivations are heuristic until mathematized, and some (path integrals) resist full rigor.

See *Chapter 18 (p. 349)* and *Chapter 17 (p. 325)* for the cross-cutting roles of symmetry, locality, and the algebra/state duality;

*16 (p. 308)* for where these structures meet physical scales.

## Key references

Full entries in the Bibliography (p. 776). Standard, real sources only:


- M. Reed and B. Simon, *Methods of Modern Mathematical Physics*, Vols. I–IV (Academic Press, 1972–1979) — the canonical rigorous reference for functional analysis: self-adjointness and extensions, the spectral theorem, Stone's theorem, scattering.
- R. Haag, *Local Quantum Physics: Fields, Particles, Algebras*, 2nd ed. (Springer, 1996) — algebraic QFT, type-III local algebras, modular theory, DHR superselection, Reeh–Schlieder, Bisognano–Wichmann.
- J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed. (Springer, 1987) — constructive QFT, the rigorous Euclidean path integral, OS axioms, $P(\phi)_2$ and $\phi^4_3$.
- M. Takesaki, *Theory of Operator Algebras*, Vols. I–III (Springer, 1979/2003) — definitive treatise on C*- and von Neumann algebras, the Murray–von Neumann/Connes type classification, and Tomita–Takesaki modular theory with the KMS condition.
- M. Nakahara, *Geometry, Topology and Physics*, 2nd ed. (IOP, 2003) — manifolds, fiber bundles, connections, curvature, characteristic classes, Atiyah–Singer, anomalies, instantons.
- A. O. Barut and R. Rączka, *Theory of Group Representations and Applications* (PWN, 1977) — Lie groups, Lie algebras, unitary representations, induced representations, Wigner's classification.
- A. Jaffe and E. Witten, "Quantum Yang–Mills Theory" (Clay Mathematics Institute Millennium Problem description, 2000) — the official statement of the existence and mass-gap problem.
- M. Aizenman and H. Duminil-Copin, "Marginal triviality of the scaling limits of critical 4D Ising and $\phi^4_4$ models," *Annals of*

*Mathematics* **194** (2021), 163–235 (with M. Aizenman, *Phys. Rev. Lett.*, 1981) — rigorous triviality of $\phi^4$ in four dimensions.

- J. Lurie, "On the Classification of Topological Field Theories," *Current Developments in Mathematics 2008* (2009), 129–280 — the cobordism-hypothesis program.
- K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory*, Vols. 1–2 (Cambridge Univ. Press, 2017/2021) — rigorous perturbative QFT via factorization algebras and the BV formalism.
- E. P. Wigner, "The Unreasonable Effectiveness of Mathematics in the Natural Sciences," *Comm. Pure Appl. Math.* **13** (1960), 1–14 — the original framing of the meta-question.
- Landmark structural theorems: V. Bargmann, "On unitary ray representations of continuous groups," *Ann. Math.* **59** (1954); A. M. Gleason, "Measures on the closed subspaces of a Hilbert space," *J. Math. Mech.* **6** (1957); and the Gelfand–Naimark / Segal foundational papers.

## References

See the Bibliography (p. 776) for full citation entries and additional standard sources.

PART III

# WHERE PHYSICS BREAKS

A map is only useful if it marks the cliffs. This part is the book's catalogue of exactly where, and exactly how, the established frameworks of Part II fail to fit together — compiled with one discipline that distinguishes it from most accounts of the "crisis in physics": every clash is traced to its technical root, and none is exaggerated for drama.

Chapter 14 catalogues the seventeen inter-framework clashes, each with a stable identifier (GC-1 through GC-17) used throughout the rest of the book. The headline finding deserves emphasis precisely because it is undramatic: not one of the seventeen is a proven logical inconsistency. They are domain mismatches, extrapolation failures, and unsolved-but-consistent problems — the black-hole information paradox, the cosmological-constant problem, the measurement problem among them — concentrated where both pillars are pushed past their tested regimes.

Chapter 15 is the open-problems registry (the OP-n series), the running ledger of genuinely unresolved questions that Part V's investigation will attack one by one. Chapter 16 turns to the quietest scandal in fundamental physics: the constants and scales

— the free parameters no current theory predicts, the hierarchies no mechanism explains.

What this part establishes is the demand side of unification: the precise list of debts any universal theory must pay. Part IV surveys the candidates that promise to pay them.

CHAPTER 14

# Gaps and Contradictions: The Seventeen Clashes

This page registers the principal *gaps, tensions, and apparent contradictions* among the accepted frameworks of physics. Its central methodological commitment — see *Chapter 2 (p. 8)* — is to **distinguish four very different things** that are routinely conflated in informal discourse:

| KIND | MEANING | STATUS OF THE UNDERLYING THEORIES |
|---|---|---|
| **logical-inconsistency** | A contradiction derivable *within a single, fixed, consistent axiom system* (rare; almost never what is actually meant). | At least one framework, as stated, is internally broken. |
| **domain-of-validity mismatch** | Two frameworks are each internally consistent but rest on incompatible idealizations; the clash appears only where their domains are forced to overlap. | Both frameworks fine in their regimes. |
| **unsolved-but-consistent problem** | A well-posed question with no accepted answer, but no contradiction. | Frameworks consistent; we lack a derivation/mechanism. |
| **conceptual tension** | A clash of interpretation or principle, with no operational/empirical contradiction. | Empirically fine; the dispute is about *meaning* or *mechanism*. |

Every entry carries a **REFEREE VERDICT** distilling an adversarial review of the original claim: a *refined statement*, the *corrections* applied, a *recommended epistemic tag*, and a *severity*.

Where the referee judged the framing materially wrong (e.g., a `kind` mislabel), the **refined statement governs** and supersedes the headline. No entry below was rejected outright (`keep=false`); the single entry whose *internal logic* was found unsound (GC-17) is retained but corrected, and its erroneous sub-claim is quarantined in *Corrected / not actually contradictions*.

Cross-references: *Chapter 15 (p. 233)* (the forward-looking research agenda), *Chapter 3 (p. 15)*, *Chapter 18 (p. 349)*, *Chapter 20 (p. 458)*, *Chapter 16 (p. 308)*.

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# Index

| ID | Title | kind | Tag | Severity |
|---|---|---|---|---|
| *GC-1* | Background dependence vs. background independence (QFT/GR) | domain-mismatch + OPEN foundational | `[ESTABLISHED]`/`[OPEN]` | major |
| *GC-2* | Perturbative non-renormalizability of quantized gravity | established theorem / open UV | `[ESTABLISHED]`/`[OPEN]` | minor |
| *GC-3* | Cosmological-constant problem | domain-mismatch / fine-tuning | `[OPEN]` | minor |
| *GC-4* | Black-hole information paradox | domain-of-validity tension | `[ESTABLISHED]`/`[OPEN]` | minor |
| *GC-5* | AMPS firewall puzzle | conceptual tension (rigorous no-go) | `[INFERENCE]`/`[CONTESTED]` | minor |
| *GC-6* | Measurement problem | structural incompleteness | `[ESTABLISHED]`/`[OPEN]` | minor |

| | | | | |
|---|---|---|---|---|
| *GC-7* | Thermodynamic arrow vs. time-symmetric microlaws | established structure / open origin | [ESTABLISHED]/[OPEN] | minor |
| *GC-8* | Problem of time (Wheeler–DeWitt) | conceptual tension | [OPEN]/[CONTESTED] | minor |
| *GC-9* | Haag's theorem vs. the interaction picture | rigor gap / domain-mismatch | [ESTABLISHED] | minor |
| *GC-10* | Triviality / Landau poles vs. "renormalizable = fundamental" | established refutation / open SM-UV | [ESTABLISHED]/[INFERENCE] | minor |
| *GC-11* | GR singularities (geodesic incompleteness) | domain-mismatch | [ESTABLISHED] | minor |
| *GC-12* | Quantum nonlocality vs. relativistic locality | conceptual tension (no operational clash) | [ESTABLISHED]/[CONTESTED] | minor |
| *GC-13* | Holographic area-law vs. volume-extensive QFT Hilbert space | conceptual tension on [ESTABLISHED] parts | [OPEN] | minor |
| *GC-14* | Stone–von Neumann uniqueness vs. inequivalent representations | domain-mismatch | [ESTABLISHED] | minor |
| *GC-15* | Semiclassical gravity vs. quantum superposition of mass | domain-mismatch / breakdown | [INFERENCE]/[CONTESTED] | minor |
| *GC-16* | ER=EPR / "spacetime from entanglement" generality | conceptual tension / evidential gap | [SPECULATIVE] on [INFERENCE] base | minor |

| | | | | |
|---|---|---|---|---|
| *GC-17* | Ensemble statistical mechanics vs. long-range gravitating systems | domain-mismatch | [ESTABLISHED] | minor (one excised sub-claim) |

## GC-1 — Background dependence vs. background independence: QFT and GR

**Frameworks:** *QFT* (Chapter 8 (p. 89)), *GR* (Chapter 10 (p. 121)), *QM* (Chapter 7 (p. 71)), *Particle Physics / Standard Model* (Chapter 9 (p. 106)). **Kind:** domain-of-validity boundary + OPEN foundational problem (explicitly **not** a logical inconsistency).

**Precise statement.** Standard *Minkowski* QFT presupposes a fixed, non-dynamical metric $\eta_{\mu\nu}$. That metric anchors (i) the global Poincaré group, whose unitary irreducible representations classify particles via the Casimirs $P^2 = m^2$ and $W^2 = -m^2 s(s+1)$ (mostly-minus signature, $c = 1$; the sign is convention-dependent) [ESTABLISHED]; (ii) global microcausality $[\phi(x), \phi(y)] = 0$ for spacelike separation [ESTABLISHED]; and (iii) a Poincaré-invariant vacuum satisfying the spectrum condition [ESTABLISHED]. GR's defining feature is the opposite: diffeomorphism invariance / **background independence**, where the metric is dynamical, there is no prior geometry, and physical states are diffeomorphism-equivalence classes (Leibniz equivalence). [ESTABLISHED]

**Technical root.** In the ADM formulation the GR Hamiltonian is a sum of constraints, $H \approx 0$, generating diffeomorphisms; there is no fixed causal structure to anchor microcausality, and "particle" loses observer-independent meaning even semiclassically (Unruh effect; no preferred vacuum in curved spacetime). The two frameworks treat the metric as *fixed input* vs. *dynamical output*. [ESTABLISHED]

**Why it matters.** This contrast is the deepest structural reason the naive union of the two frameworks fails; it underlies the problem of time (*GC-8*), the non-renormalizability of quantized gravity (*GC-2*), and the difficulty of even formulating "QFT on a superposition of spacetimes." See *Chapter 18 (p. 349)*, *Chapter 15 (p. 233)*.

**Current status.** Both frameworks are confirmed to extreme precision in their domains (electron $g-2$; binary-pulsar decay; gravitational-wave detections). They coexist successfully as (a) QFT on fixed curved backgrounds (Hawking, Unruh; algebraic/Haag–Kastler nets on curved spacetime) and (b) perturbative graviton EFT, $g = \eta + h$, valid below the Planck scale. The clash bites only when the metric must be *both quantum and dynamical and non-perturbative*.

> **REFEREE VERDICT — isSound: false (overstated as written); severity: major; keep. Corrections.** The original "mutually exclusive kinematic foundations / cannot combine in principle" is **false as an in-principle claim** and is downgraded. QFT does *not* in general require Poincaré invariance, a preferred vacuum, or a flat metric: QFT on fixed *curved* backgrounds is rigorous and already makes "particle" observer-dependent, replacing the global spectrum condition with the microlocal/Hadamard condition. Microcausality needs a *definite Lorentzian causal structure*, not a flat or non-dynamical one. The hole-argument paraphrase ("determinism FORCES identifying $\phi^*g$ with $g$") overstates a philosophically contested inference: Leibniz equivalence is the conventional *resolution* chosen to retain determinism, not a logical compulsion. Wigner's classification is a flat-space crutch, not a universal QFT requirement. **Refined statement.** Standard Minkowski QFT and GR rest on opposite treatments of geometry. This opposition is the deepest structural source of the difficulty in quantizing gravity, but it is a **non-perturbative/UV and conceptual incompatibility, not an in-principle kinematic mutual-exclusion**. Perturbative graviton EFT consistently places a dynamical, superposable metric perturbation inside QFT below $M_{\mathrm{Pl}}$, failing by *non-renormalizability* (Goroff–Sagnotti), not by inconsistency. The genuine, unsolved obstruction: a *superposed/fluctuating causal structure* deprives microcausality, the global $S$-matrix, and the global particle/vacuum concepts of their usual definitions, and no background-independent, UV-complete quantum theory of the

> metric with well-defined local diffeomorphism-invariant observables (the problem of time) has been constructed. **RECOMMENDED TAG.** [ESTABLISHED] for the structural background-(in)dependence contrast and the non-renormalizability of perturbative gravity; [OPEN] for a background-independent UV-complete quantum theory of geometry and the resolution of superposed causal structure.

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## GC-2 — Perturbative non-renormalizability of quantized gravity

**Frameworks:** *GR* (Chapter 10 (p. 121)), *QFT* (Chapter 8 (p. 89)), *Particle Physics / SM* (Chapter 9 (p. 106)). **Kind:** ESTABLISHED theorem (perturbative non-renormalizability) + OPEN UV completion.

**Precise statement.** Treating the metric as a quantum field, $g = \eta + h$, and quantizing $h$ yields a perturbatively non-renormalizable theory: graviton-loop divergences require infinitely many independent counterterms, so the perturbative expansion loses predictivity in the deep UV. [ESTABLISHED]

**Technical root.** Newton's constant $G$ has mass dimension $-2$ in 4D natural units, so the loop expansion is organized in $E^2/M_{\text{Pl}}^2$ (higher-derivative graviton operators are irrelevant about the Gaussian fixed point). Concretely: pure gravity is **one-loop finite on-shell** ('t Hooft–Veltman 1974, the counterterm vanishing on the vacuum equations / removable by field redefinition); gravity + matter diverges already at one loop; pure gravity **diverges at two loops** with a nonvanishing Weyl-cubed ($C^3$) counterterm (Goroff–Sagnotti 1985, confirmed by van de Ven). [ESTABLISHED] Contrast the renormalizable Standard Model (dimensionless or positive-dimension couplings).

**Why it matters.** By QFT's own internal criteria this sets an upper bound on the validity of the perturbative GR+QFT description at or below $E_{\text{Pl}} \sim 1.2 \times 10^{19}$ GeV (see *Chapter 16 (p. 308)*), and reframes

GR as a predictive low-energy EFT (Donoghue): the leading Einstein–Hilbert term plus $M_{\rm Pl}$-suppressed $R^2$, $R_{\mu\nu}R^{\mu\nu}$ corrections, with unambiguous leading quantum corrections (e.g., to the Newtonian potential).

**Current status.** The perturbative non-renormalizability and the EFT status are **proven mathematics**. The UV completion is `[OPEN]` (string theory, asymptotic safety, LQG; see *Chapter 18 (p. 349)*). This is **not** a logical inconsistency: an EFT with a cutoff is fully consistent below it.

> **REFEREE VERDICT — isSound: true; SEVERITY: MINOR; KEEP. Corrections.** The original `kind: unsolved-problem` is split: the perturbative non-renormalizability is `[ESTABLISHED]`, the *resolution* is `[OPEN]`. The inference "GR cannot be a fundamental QFT" is weakened to "**cannot be a *perturbatively renormalizable* QFT in the Dyson sense**": a nontrivial UV fixed point (Weinberg asymptotic safety `[SPECULATIVE/OPEN]`) would render gravity non-perturbatively renormalizable and predictive at all scales; higher-derivative gravity (Stelle 1977) is perturbatively renormalizable but contains a unitarity-violating ghost. Predictivity is *not* lost everywhere — only the perturbative expansion / finite parameter set fails in the deep UV; low-energy quantum-gravity predictions are unambiguous. **Recommended tag.** `[ESTABLISHED]` (theorem + EFT status); `[OPEN]` (UV completion / "fundamental QFT" question).

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## GC-3 — The cosmological-constant problem: QFT vacuum energy vs. observed Λ

**Frameworks:** *QFT* (Chapter 8 (p. 89)), *GR* (Chapter 10 (p. 121)), *Cosmology* (Chapter 11 (p. 139)), *Particle Physics / SM* (Chapter 9 (p. 106)). **Kind:** domain-of-validity mismatch / extreme fine-tuning (**not** a logical inconsistency, **not** a falsified prediction).

**Precise statement.** A Lorentz-invariant QFT vacuum carries energy density $\rho_{\rm vac}$ with $T_{\mu\nu} = -\rho_{\rm vac}\, g_{\mu\nu}$ (equation of state $w = -1$); by **Lovelock's theorem** (1971–72) the term $\Lambda g_{\mu\nu}$ is the unique additional divergence-free metric piece beyond $G_{\mu\nu}$ in 4D, so a constant vacuum energy is indistinguishable from $\Lambda$ and gravitates. Estimates of $\rho_{\rm vac}$ vastly exceed the observed dark-energy density $\rho_\Lambda^{\rm obs} \sim 10^{-47}\ {\rm GeV}^4 \approx (2.3\ {\rm meV})^4$. `[ESTABLISHED tension]`

**Technical root (robust vs. heuristic).**

- *Heuristic (regulator-dependent):* a zero-point sum cut at the Planck scale gives $\rho_{\rm vac} \sim M_{\rm cut}^4 \sim 10^{76}\ {\rm GeV}^4$, exceeding observation by $\sim 10^{123}$ ($\sim$120 orders); an electroweak cutoff still leaves $\sim50–60 orders. The quartic estimate is regulator-dependent (dimensional regularization yields an $m^4\ln structure, not $M_{\rm cut}^4$).
- *Robust (regulator-independent):* **finite physical condensates** already overshoot — the QCD chiral/gluon condensate $\sim \Lambda_{\rm QCD}^4 \sim 10^{-3}\ {\rm GeV}^4$ (~44 orders too large) and the electroweak Higgs potential $\sim (100\ {\rm GeV})^4$ (~59 orders). `[ESTABLISHED]`
- Unbroken SUSY gives exactly zero vacuum energy, but SUSY breaking at $\gtrsim 1$ TeV reinstates $\rho \sim M_{\rm SUSY}^4 \sim 10^{12}\ {\rm GeV}^4$ (~59 orders too large) — SUSY does not rescue it.

**Why it matters.** It may signal the naive QFT vacuum-energy calculation is the wrong object, that gravity "degravitates" vacuum energy, or that selection (anthropic/landscape) or sequestering is required. Weinberg's anthropic bound predicted a small nonzero $\Lambda$ before detection, but relies on the contested multiverse + measure. See *Chapter 15 (p. 233)*, *Chapter 20 (p. 458)*.

**Current status (updated 2026-07-02).** Cosmic acceleration is `[ESTABLISHED]` (SNe Ia, CMB+BAO). That dark energy is *exactly* a constant $\Lambda$ ($w = -1$) is the simplest fit but a `[CONTESTED]` assumption. DESI DR2 BAO + CMB prefer an evolving $w(z)$ ($w_0 > -1$, $w_a < 0$) over $\Lambda$CDM at 3.1σ, and 2.8–4.2σ when SNe are added, depending on sample (arXiv:2503.14738, *PRD* **112**, 083515 (2025)); but the preference is SN-sample-driven, drops below 2σ under low-z SN-calibration systematics (arXiv:2502.04212), and inverts to a modest *Bayesian* preference for ΛCDM from DESI+CMB alone, the $w_0 w_a$ signal tracing to a DESI↔DES-SN5YR inter-dataset tension that largely vanishes with recalibrated DES-Dovekie SNe (arXiv:2511.10631) — `[CONTESTED]`; five-year DESI $w(z)$ verdict expected 2027. All proposed solutions are `[SPECULATIVE]`. Regularization-aware restatement + 2024–2026 approach status: *Chapter 16 (p. 308)* §8.1.

> **REFEREE VERDICT — isSound: true; severity: minor; keep. Corrections.** No technical errors; the canonical numbers check out. Sharpen the driver: vacuum energy gravitates as Λ because a Lorentz-invariant vacuum has $w = -1$ (the equivalence-principle phrasing is a heuristic). In strict logic there is **no failed prediction and no inconsistency**: in EFT the cosmological constant is a renormalized free parameter (a bare-Λ counterterm absorbs the divergence, exactly as for the electron mass). What "fails" is naturalness — the renormalized value must cancel radiative/condensate contributions to ~100+ decimal places with no known dynamical mechanism. "Largest mismatch in physics" is a defensible standard characterization (Weinberg 1989), but a rhetorical superlative whose magnitude depends on the unjustified cutoff choice. **Recommended tag.** `[OPEN]` — severe fine-tuning / domain-mismatch at the QFT–GR interface; the original `domain-mismatch` kind is correct.

**Status block (2026-07-02 — R3 "without folklore" sweep).** Theorem-level inventory for this entry: Lovelock uniqueness `[ESTABLISHED]`; $w = -1$ for a Lorentz-invariant vacuum `[ESTABLISHED]`; **Weinberg's no-go** under its stated assumptions (local 4D Lagrangian theory; finitely many fields; translation-invariant constant-field vacuum; no external constraints) `[ESTABLISHED theorem]`; **radiative instability** of the renormalized Λ in dimensional regularization ($m^4 \ln \mu$ per massive field; retuning at each loop order and each mass threshold; known particles alone overshoot by $10^{30}$–$10^{56}$) `[ESTABLISHED]`; physical phase-transition shifts (EW, QCD —

epoch-dependent tuning) `[ESTABLISHED]`; the "$10^{120}$" [folklore — regulator artifact]; whether $\langle T_{\mu\nu}\rangle_{\text{vac}}$ is the object that gravitates `[OPEN]`. Approach verdicts 2024–2026 (citations in *Chapter 16 (p. 308)* §8.1): **unimodular gravity** reallocates the tuning into an untuned integration constant — does not solve (arXiv:2207.08499; arXiv:2406.12958); **sequestering** cancels matter-loop vacuum energy (global + manifestly-local versions; graviton loops beyond the originals) `[SPECULATIVE]`; **running vacuum** fits DESI DR2 comparably to $w_0 w_a$CDM `[CONTESTED]`; **swampland/dS-conjecture** reroutes to rolling quintessence carrying its own instability `[SPECULATIVE]`. No 2024–2026 result changes this entry's kind (domain-of-validity mismatch / fine-tuning, **not** inconsistency).

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## GC-4 — Black-hole information paradox: unitarity vs. semiclassical thermal evaporation

**Frameworks:** *QM* (Chapter 7 (p. 71)), *QFT* (Chapter 8 (p. 89)), *GR* (Chapter 10 (p. 121)), *Thermodynamics* (Chapter 5 (p. 40)), *Statistical Mechanics* (Chapter 6 (p. 56)), *Information Theory* (Chapter 12 (p. 155)). **Kind:** domain-of-validity / inter-framework tension (**not** a logical inconsistency).

**Precise statement.** Hawking's **leading-order** semiclassical calculation (Bogoliubov mode-mixing between in/out vacua across the horizon) yields an outgoing radiation state that is *mixed* — a thermal/graybody density matrix at Hawking temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}.$$

Naively iterating to complete evaporation maps a pure collapsing state to a mixed final state, in apparent conflict with the unitarity of global QM/QFT evolution. `[ESTABLISHED tension]`

**Technical root.** Unitarity requires the radiation's *fine-grained* von Neumann entropy to follow the **Page curve** (rise, then return to $\sim 0$), whereas the naive semiclassical entropy rises monotonically. The Bekenstein–Hawking entropy

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar}$$

should count microstates that the semiclassical computation does not exhibit. [Microstate counting `[ESTABLISHED]` for specific supersymmetric/BPS black holes via Strominger–Vafa; general case `[OPEN]`.]

**Why it matters.** It is the cleanest place where QM, GR, QFT, and thermodynamics collide simultaneously; resolving it constrains quantum gravity and forces the question of whether semiclassical QFT is being used outside its domain near/inside the horizon. See *GC-5, GC-13, Chapter 15 (p. 233).*

**Current status.** `[OPEN]`/`[CONTESTED]` as to mechanism. Unitarity is now widely favored: AdS/CFT is manifestly unitary, and post-2019 quantum-extremal-surface (QES) / island and replica-wormhole computations reproduce a unitary Page curve in controlled (largely AdS, lower-dimensional, large-$c$) settings `[INFERENCE]`. No agreed first-principles bulk mechanism exists for a realistic, asymptotically-flat, dynamically-formed 4D black hole.

> **REFEREE VERDICT — isSound: true; severity: minor; keep. Corrections.** The original `kind: logical-inconsistency` is **wrong** and corrected to a domain-of-validity / inter-framework tension: each framework is internally consistent; the conflict arises only when semiclassical gravity is pushed past its validity (precisely what the island/QES results indicate by recovering the Page curve). "Exactly thermal to all orders" is overstated — the spectrum is a *graybody* (Planckian × frequency/spin-dependent transmission factors), and the mixedness is a leading-order, finite-order statement, not exact/robust. Unitarity *violation* is not demonstrated; the leading calculation, naively iterated, is what is incompatible with unitarity. "Where information is" (interior, horizon, entanglement wedge, soft hair) is interpretation-dependent. **Recommended tag.** `[ESTABLISHED]` as a structural tension

between semiclassical gravity and unitarity; `[OPEN]`/`[CONTESTED]` as to resolution. **Not** a logical inconsistency. **Status update (2026-07-02, track R1 — dated append; verdict-neutral; all coordinates web-verified this sweep).**

- **What islands/replica wormholes establish `[INFERENCE, within their setting]`:** the semiclassical gravitational path integral, *including replica-wormhole saddles*, computes the fine-grained radiation entropy via the quantum-extremal-surface/island formula and returns a unitary Page curve in controlled settings (AdS/JT + matter; end-of-world-brane models): Penington, *JHEP* **09** (2020) 002; Almheiri–Engelhardt–Marolf–Maxfield, *JHEP* **12** (2019) 063; Penington–Shenker–Stanford–Yang, *JHEP* **03** (2022) 205; Almheiri–Hartman–Maldacena–Shaghoulian–Tajdini, *JHEP* **05** (2020) 013. Consensus reading: Hawking's monotone entropy omitted path-integral saddles, not physics beyond the semiclassical regime. Unestablished: the *mechanism* of escape; the interior encoding (leading proposal: non-isometric codes protected by complexity, Akers–Engelhardt–Harlow–Penington–Vardhan, arXiv:2207.06536) `[SPECULATIVE]` (leading proposal); any controlled derivation for a realistic asymptotically-flat 4D collapse black hole `[OPEN]`.
- **Standing objection 1 — ensemble vs. single theory `[CONTESTED]`:** replica wormholes are most naturally read as computing *ensemble-averaged* entropies (explicit in PSSY; Marolf–Maxfield baby-universe/α-sector superselection with null states, *JHEP* **08** (2020) 044); against genericity: Schlenker–Witten prove — in three-dimensional pure gravity, with arguments against ensemble existence in higher-dimensional examples — that connected geometries never contribute to sub-black-hole-threshold observables (*JHEP* **07** (2022) 143). The 2025–2026 continuation runs through OP-50 (closed-universe Hilbert space) and the Kudler-Flam–Witten fixed-$N$ non-convergence / averaging-over-$N$ line already registered at OP-44 — no

theorem either way for a single unitary theory underlying the island entropies.

- **Standing objection 2 — massive graviton / gravitating bath** `[CONTESTED]`**:** every controlled island construction in $d > 2$ (Karch–Randall brane + *non-gravitating* bath) has a massive graviton, with the island contribution vanishing in the massless limit (Geng–Karch, *JHEP* **09** (2020) 121); islands as disconnected entanglement wedges strain the gravitational Gauss law in long-range massless gravity (Geng et al., *JHEP* **01** (2022) 182); with a gravitating bath the radiation entropy becomes time-independent (*SciPost Phys.* **10**, 103 (2021)). Post-2023: string-embedded islands with parametrically light (nonzero-mass) gravitons within massive-gravity EFT validity (Demulder–Gnecchi–Lavdas–Lüst, *JHEP* **02** (2023) 016); the Gauss-law argument extended to low dimensions incl. JT (Geng, *SciPost Phys.* **19**, 146 (2025)); the island formula derived via replica wormholes on the KR brane (Geng, *JHEP* **01** (2025) 063); position consolidated in Geng, arXiv:2509.22775. The opposing reading (Raju et al. holography of information: in massless gravity boundary data is complete, islands unnecessary — arXiv:2012.05770; the wiki's A2 cluster) and its Dec-2025 dressing refinement (arXiv:2512.18912, iteration 11) now likewise locate the key structure in the gravitational dressing (converging vocabulary; interpretive reading) while disagreeing on what it implies.
- **The algebraic second leg** `[ESTABLISHED as mathematics / INFERENCE as physics]`**:** independent of replicas, crossed-product constructions give Type II horizon/subregion algebras whose von Neumann entropy is $S_{\text{gen}}$ (Witten, *JHEP* **10** (2022) 008; CLPW, *JHEP* **02** (2023) 082; CPW, *JHEP* **04** (2023) 009 — a replica-free special case of the QES prescription and a version of the GSL; KLS, *PRD* **111**, 025013 — arbitrary Killing horizons incl. Kerr/collapse; arXiv:2601.07910 — complete constraint group + edge modes; arXiv:2601.07915 — GSL for non-stationary linearized perturbations and a

> perturbative-regime QFC proof). Scope limits (unchanged): perturbative/large-$N$; the observer/clock/corner symmetry is *installed*, not derived; the entropy carries an unfixed additive constant (no $A/4$ microstate count). See *GC-13* and OP-44.
>
> - **Net for GC-4:** unitarity is now favored on **two independent legs** (replica path integral; Type II algebraic entropies), each with named scope limits. Kind and tags unchanged.

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## GC-5 — The AMPS firewall puzzle

**Frameworks:** *GR* (Chapter 10 (p. 121)), *QM* (Chapter 7 (p. 71)), *Information Theory* (Chapter 12 (p. 155)), *QFT* (Chapter 8 (p. 89)). **Kind:** conceptual tension built on a **rigorous no-go argument**.

**Precise statement.** AMPS (2012) showed that for an *old* (post-Page-time) evaporating black hole, four individually well-motivated assumptions cannot all hold: (1) unitarity of evaporation/$S$-matrix; (2) validity of low-energy EFT outside the stretched horizon; (3) "no drama" at the horizon for an infalling observer (the near-horizon state is the local Hadamard vacuum, from the equivalence principle **plus** the short-distance vacuum entanglement structure of QFT); and (4) the black hole is a conventional quantum system with $e^{A/4}$ states (so the Page argument applies). `[INFERENCE — rigorous given premises]`

**Technical root.** The obstruction is entanglement monogamy, made rigorous via **strong subadditivity** of von Neumann entropy applied to {early radiation $E$, exterior late mode $b$, interior partner $a$}:

$$S(b,a) + S(b,E) \;\geq\; S(a) + S(E).$$

A smooth horizon requires $b$ nearly maximally entangled with $a$ (small $S(b,a)$); unitarity past the Page time requires $b$ nearly maximally entangled with $E$ (small $S(b,E)$). SSA forbids both. Severing the $a$–$b$ entanglement to restore unitarity excites the would-be vacuum, depositing high-energy quanta at the horizon — a **firewall** — violating (3).

**Why it matters.** It shows the information paradox threatens the equivalence principle itself at the horizon of an old black hole, where curvature is low and GR "should" be reliable. Proposed escapes — black-hole complementarity, ER=EPR (*GC-16*), state-dependence — each pay a conceptual price.

**Current status.** `[OPEN]`/`[CONTESTED]`: no consensus on whether old black holes have firewalls. ER=EPR (Maldacena–Susskind) is a `[SPECULATIVE]` proposed resolution with no first-principles derivation for generic entanglement. The island/QES program is read by many to suggest no firewall, but the interpretation is disputed.

> **REFEREE VERDICT — isSound: true; severity: minor; keep. Corrections.** The rigorous engine is strong subadditivity; "monogamy of maximal entanglement" is the heuristic special case (the physical modes are only *nearly* maximally entangled, so the contradiction is quantitative). Postulate (3) is precisely "no drama," following from EP **plus** the local-vacuum (Hadamard) condition — calling it "the equivalence principle" alone is loose. The canonical framing includes the distinct fourth postulate (dimension $e^{A/4}$), which the original folds into "unitarity." AMPS proves only *joint incompatibility*; the firewall is **one** resolution, and which postulate fails is genuinely contested (complementarity, ER=EPR, state-dependence, islands/replica wormholes all argue the smooth horizon can survive). **Recommended tag.** `[INFERENCE]` for the AMPS no-go itself (a rigorous, widely accepted derivation of joint incompatibility given its premises); `[CONTESTED]`/`[OPEN]` for whether a firewall actually forms. **Status update (2026-07-02, track R1 — dated append).** Post-island readings increasingly locate the failing AMPS premise in the *tensor-factorization* of {early radiation, late mode, interior partner}: in the QES/island account the interior partner lies inside the radiation's entanglement wedge after the Page time, so monogamy is not violated

> [INFERENCE, within island models]. The algebraic line sharpens this structurally: Type $\mathrm{III}_1$ local algebras admit no tensor factorization, and gravitational dressing further obstructs subsystem independence (cf. *GC-13*, OP-44). Tags unchanged.

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## GC-6 — The measurement problem: unitary linear evolution vs. definite single outcomes

**Frameworks:** *QM* (Chapter 7 (p. 71)), *QFT* (Chapter 8 (p. 89)), *Classical Mechanics* (Chapter 4 (p. 25)). **Kind:** structural **incompleteness** / domain-demarcation gap (**not** a bare logical inconsistency).

**Precise statement.** Unitary Schrödinger evolution (axiom A5; linear) maps a von Neumann premeasurement $\sum_i c_i \, |s_i\rangle|\text{ready}\rangle \to \sum_i c_i \, |s_i\rangle|\text{pointer}_i\rangle$ — an entangled macroscopic superposition, never a single branch. The Born/projection rule (A6) instead replaces the state by one eigenprojection with probability $\text{Tr}(\rho E)$ — non-unitary, non-linear, stochastic. If A5 is taken as a *universal* dynamical law covering apparatus + observer, it conflicts with the observed single definite outcome (Schrödinger cat, Wigner's friend). [ESTABLISHED tension]

**Technical root.** Linearity of $U(t) = e^{-iHt/\hbar}$ forbids a single branch. Standard QM avoids formal contradiction only by leaving "measurement" a primitive, undefined notion, with no rule fixing when each law applies (Bell's "shifty split"). **Decoherence** (GKSL/Lindblad dynamics) explains rapid, robust suppression of interference in a pointer basis and the emergence of an effective classical regime, but yields an *improper mixture* — the global state stays pure-entangled — so it does not by itself select an outcome. **Gleason's theorem** (dim $\geq 3$) fixes the *form* $\text{Tr}(\rho E)$ of the probability rule but not why one outcome occurs.

**Why it matters.** It is the central foundational problem of QM, bounds the QM→classical correspondence, and is where the only currently-distinguishable alternatives (objective-collapse models) make testable deviations. See *Chapter 15 (p. 233)*, *Chapter 19 (p. 392)*.

**Current status.** `[OPEN]`/`[CONTESTED]` as to resolution; `[ESTABLISHED]` as a structural tension. Interpretations pay different prices: Everett (keep unitarity; pay with branching + an unsolved account of Born weights), de Broglie–Bohm (definite positions; pay with explicit nonlocality/contextuality and, relativistically, a preferred foliation), GRW/CSL objective collapse (modify dynamics; pay with new stochasticity and Lorentz-covariance tension — but empirically testable and being constrained by matter-wave/optomechanics/spontaneous-radiation experiments). All except objective collapse are empirically equivalent to standard QM so far.

> **REFEREE VERDICT — isSound: true; SEVERITY: MINOR; KEEP. Corrections.** The original `kind: logical-inconsistency` is corrected to **structural incompleteness / domain-demarcation gap**: A5 and A6 have disjoint stipulated domains and are not formally contradictory; the contradiction appears *only* under the added premise that A5 is universal — precisely the contested premise. The "single experienced outcome" imports an extra determinacy premise that Everettian readings deny. The concrete technical claims (premeasurement entanglement; non-unitarity of projection; decoherence → improper mixture; Gleason fixing form not occurrence) are all correct. Optional sharpeners (Frauchiger–Renner 2018; extended-Wigner's-friend no-go theorems) tighten the tension but are themselves interpretation-dependent — do not cite as settled. **Recommended tag.** `[ESTABLISHED]` (the structural tension/incompleteness is real); `[OPEN]`/`[CONTESTED]` (its resolution). Not a settled logical inconsistency.

## GC-7 — Thermodynamic arrow of time vs. time-symmetric microscopic laws

**Frameworks:** *Thermodynamics* (Chapter 5 (p. 40)), *Statistical Mechanics* (Chapter 6 (p. 56)), *Classical Mechanics* (Chapter 4 (p. 25)), *QM* (Chapter 7 (p. 71)), *Cosmology* (Chapter 11 (p. 139)). **Kind:** ESTABLISHED structural result + OPEN cosmological origin (a consistent-but-incomplete problem, not a contradiction).

**Precise statement.** The microscopic laws (Hamiltonian/Schrödinger) are time-reversal invariant up to the tiny CP/T-violation of the weak interaction, which is *irrelevant* to the thermodynamic arrow: a perfectly T-invariant world would still exhibit it given a low-entropy past. Yet isolated macroscopic systems show entropy non-decrease. **Monotonic entropy increase cannot follow from time-symmetric dynamics alone** — Loschmidt's reversibility objection and Zermelo's Poincaré-recurrence objection both rule out any purely dynamical derivation. `[ESTABLISHED]`

**Technical root.** Liouville's theorem (classical) and unitarity (quantum) conserve the fine-grained Gibbs/von Neumann entropy $S = -k_B \operatorname{Tr}(\rho \ln \rho)$ *exactly* for a closed system. Irreversibility appears only for a *coarse-grained* entropy (or, for open subsystems, via environmental entanglement/decoherence — a second route the naive "coarse-graining only" story omits), and in Boltzmann's $H$-theorem only after inserting the molecular-chaos assumption (Stosszahlansatz), applied asymmetrically to pre- vs. post-collision correlations — the precise point where time-asymmetry enters (rigorous only in the Boltzmann–Grad limit for short times, per Lanford 1975). The resolution requires a time-asymmetric **input**: a special low-entropy boundary condition (the *Past Hypothesis*) plus a coarse-graining/typicality choice.

The **fluctuation theorems** refine the second law into exact statistical statements — Crooks: $P_F(+W)/P_R(-W) = e^{\beta(W-\Delta F)}$; Jarzynski: $\langle e^{-\beta W}\rangle = e^{-\beta \Delta F}$ — making transient entropy *decreases* exponentially rare but nonzero. They presuppose an equilibrium initial state (itself a time-asymmetric input) and so illustrate, rather than independently derive, the arrow.

**Why it matters.** The second law is not a fundamental microlaw but a consequence of dynamics + a cosmological boundary condition; this exports the deepest part of the puzzle — *why* the early universe had such low, especially gravitational, entropy — to cosmology and quantum gravity. See *GC-17* (gravitational non-extensivity), *Cosmology* (Chapter 11 (p. 139)), *Chapter 15 (p. 233)*.

**Current status.** That a time-asymmetric input is *necessary* is `[ESTABLISHED]`. *Why* the early universe had extraordinarily low gravitational entropy (near-zero Weyl curvature; Penrose's Weyl-curvature hypothesis is one `[SPECULATIVE]` proposal) is genuinely `[OPEN]`. A complete first-principles derivation of thermalization for realistic Hamiltonians is also `[OPEN]` (the eigenstate thermalization hypothesis is strong but unproven and fails for integrable / many-body-localized systems).

> **REFEREE VERDICT — isSound: true; severity: minor; keep. Corrections.** Sharpen the CP/T remark: weak-interaction T-violation is essentially *irrelevant*, not merely insufficient. Fine-grained entropy conservation is exact for the *total isolated* system; open-subsystem entanglement (decoherence) is a distinct second route to observed irreversibility. The Stosszahlansatz is inserted as molecular chaos and applied asymmetrically; Lanford's theorem justifies the Boltzmann equation only for short times in the Boltzmann–Grad limit. The original `unsolved-problem` label applies **only** to the cosmological residue (origin of the Past Hypothesis), not to the established structural result. **Recommended tag.** Mixed: `[ESTABLISHED]` (necessity of a time-asymmetric input; Loschmidt/Zermelo; fine-grained-entropy conservation; Stosszahlansatz as the locus of asymmetry; exactness of fluctuation theorems); `[OPEN]` (origin/justification of the low-entropy Past Hypothesis).

## GC-8 — The problem of time in canonical quantum gravity vs. QM's external time

**Frameworks:** *GR* (Chapter 10 (p. 121)), *QM* (Chapter 7 (p. 71)), *QFT* (Chapter 8 (p. 89)). **Kind:** conceptual / structural tension (**not** a logical inconsistency, **not** an in-principle impossibility).

**Precise statement.** QM presupposes an external classical time parameter $t$: $i\hbar\,\partial_t|\psi\rangle = H|\psi\rangle$, and by **Pauli's theorem** a lower-bounded $H$ admits no self-adjoint operator canonically conjugate to it ($[T, H] = i\hbar$), so $t$ is a fixed background c-number. GR, being background-independent, has no preferred external time; its canonical quantization *formally* yields the **Wheeler–DeWitt equation** $\hat{H}_\perp|\Psi\rangle = 0$ — a constraint with no $\partial_t$ term and hence no Schrödinger-type evolution (the "frozen formalism"). `[ESTABLISHED facts]`

**Technical root.** The total ADM Hamiltonian is purely a sum of constraints,

$$H = N\,\mathcal{H}_\perp + N^i\,\mathcal{H}_i, \qquad \mathcal{H}_\perp \approx 0, \quad \mathcal{H}_i \approx 0\;(\text{weakly}),$$

with lapse $N$ and shift $N^i$ as Lagrange multipliers. On the standard reading, coordinate-time reparametrization is part of the diffeomorphism gauge group, so "evolution" in coordinate time is pure gauge — **though whether $\mathcal{H}_\perp$ generates genuine gauge or genuine many-fingered-time dynamics is itself contested (the Kuchař critique)**. Dirac observables must commute with all constraints, making them diffeomorphism-invariant, generically nonlocal, and hard to construct.

**Why it matters.** It is a direct obstruction to writing a Schrödinger equation for the universe and to defining unitary evolution and probabilities in quantum gravity — one of the sharpest conceptual barriers any canonical program must overcome. Connected to *GC-1* (background independence). See *Chapter 15 (p. 233)*.

**Current status.** `[OPEN]`/`[CONTESTED]`. Proposed (unconfirmed) resolutions: relational / Page–Wootters conditional-probability time (time as correlation with an internal clock subsystem), Rovelli's partial observables / evolving constants, and an emergent semiclassical WKB time from a Born–Oppenheimer expansion. Within *classical* GR the absence of preferred time is a feature; it becomes a problem only upon quantization. Cosmology's preferred cosmic time relies on FLRW symmetry and does not exist in an inhomogeneous universe.

> **REFEREE VERDICT — isSound: true; severity: minor; keep. Corrections.** "Canonical quantization *yields* WDW" → "**formally** yields": the WDW operator is heuristic (operator-ordering ambiguities, regularization/UV ill-definedness; no rigorous self-adjoint definition). Flag that the pure-gauge reading of $\mathcal{H}_\perp$ is contested (Kuchař) and that this contest is constitutive of the difficulty. Pauli's theorem is used correctly. The tension is a deep unsolved-but-consistent structural problem, **not** a demonstrated impossibility nor a logical contradiction. Canonical references: Dirac's constrained-Hamiltonian analysis, DeWitt (1967), and the Isham and Kuchař reviews of the problem of time. **Recommended tag.** `[OPEN]`/`[CONTESTED]` structural problem (conceptual tension). Underlying technical facts (ADM constraint structure, Pauli's theorem, the formal WDW equation) are `[ESTABLISHED]`; the gauge-vs-dynamics status of $\mathcal{H}_\perp$ and the resolution are `[CONTESTED]`.

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## GC-9 — Haag's theorem vs. the working interaction picture of perturbative QFT

**Frameworks:** *QFT* (Chapter 8 (p. 89)), *QM* (Chapter 7 (p. 71)), *Mathematics for Physics* (Chapter 13 (p. 172)). **Kind: rigor gap / domain-of-validity mismatch** — a no-go for a formal construction (**not** a logical inconsistency of QFT or of renormalized perturbation theory).

**Precise statement.** The textbook interaction picture presupposes a unitary $U(t)$, on one Poincaré-covariant Hilbert space with a unique vacuum, intertwining free and interacting fields at fixed time. **Haag's theorem** (Hall–Wightman 1957, after Haag 1955; see Streater–Wightman, *PCT, Spin and Statistics, and All That*) proves no such $U$ exists for a genuinely interacting field: were it to exist, the interacting Wightman functions would equal the free ones (so the "interacting" theory would be free). `[ESTABLISHED]`

**Technical root. Stone–von Neumann uniqueness** holds only for finitely many canonical pairs (in the Weyl form, with regularity); QFT has infinitely many degrees of freedom, so it fails, yielding uncountably many unitarily inequivalent representations of the CCR. The free and interacting theories generically inhabit different ones — the interacting theory does not live in the free Fock space. (See *GC-14*.) The textbook *derivation* of the Dyson series via a global interaction-picture unitary is therefore ill-defined as literally stated.

**Resolution / repair.** Rigorous content is recovered with **no** global $U$: (a) IR/UV regularization (finite volume breaks the load-bearing translation-invariance hypothesis); (b) LSZ / adiabatic switching defines asymptotic in/out fields in their own representations; (c) **Epstein–Glaser causal perturbation theory** builds the perturbative $S$-matrix directly from microcausality with no interaction picture and no Haag obstruction; (d) the invariant content lives in Haag–Kastler nets (generically type $\mathrm{III}_1$ factors) and Doplicher–Haag–Roberts superselection theory. Dyson's observation that the QED series is asymptotic (not convergent) is a logically *separate* point.

**Why it matters.** Textbook QFT, despite its empirical triumphs (electron $g-2$), is not the rigorous theory it appears to be; this motivates algebraic QFT and the constructive program, and clarifies that spontaneous symmetry breaking and distinct phases live precisely in inequivalent representations.

**Current status.** `[ESTABLISHED]` as a theorem and as a foundational subtlety — **not** a falsification: it does not impugn the empirical correctness of renormalized perturbation theory, recovered as an asymptotic expansion controlled order-by-order (BPHZ). 4D constructive existence remains `[OPEN]`.

> **REFEREE VERDICT — isSound: true; SEVERITY: MINOR; KEEP. Corrections.** The original `kind: logical-inconsistency` is **downgraded** to a rigor-gap / domain-mismatch: Haag's theorem is a no-go for one formal scaffolding, not a contradiction within a fixed consistent axiom system. "Mathematically inconsistent" → "the textbook *derivation* is not well-defined / rests on a nonexistent object"; the *output* is an asymptotic expansion of quantities definable by other means (Epstein–Glaser, Wightman/OS). Stone–von Neumann uniqueness is conditional (requires the bounded Weyl form **and** regularity); it can fail even in finite dof for non-regular representations or on non-simply-connected configuration spaces (Aharonov–Bohm $\theta$-vacua). The sharp dividing line is "Weyl form + regularity + finiteness," not merely "finite vs. infinite dof." **Recommended tag.** `[ESTABLISHED]` (as a rigor gap / structural no-go and its resolution). The "logical-inconsistency" label is itself an overclaim, replaced by "rigor-gap / domain-of-validity mismatch."

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## GC-10 — Triviality / Landau poles refute "renormalizable = fundamental"

**Frameworks:** *QFT* (Chapter 8 (p. 89)), *Particle Physics / SM* (Chapter 9 (p. 106)), *Mathematics for Physics* (Chapter 13 (p. 172)). **Kind:** ESTABLISHED refutation + OPEN (Planck/gravity-dominated) SM UV completion.

**Precise statement.** Perturbative renormalizability does **not** guarantee a nontrivial UV-complete continuum theory. $\phi^4$ in 4D is **proven trivial** (Aizenman 1981 for $d > 4$; Aizenman–Duminil-Copin, *Annals of Math.* 2021, for the marginal $d = 4$ case, for reflection-positive lattice $\phi^4$ / Ising-class models): the continuum limit is the free (Gaussian) theory. QED is strongly **believed** trivial (perturbative Landau pole + lattice evidence), though its pole lies far above $M_{\rm Pl}$ and is academic. [ESTABLISHED for the refutation and $\phi^4$; INFERENCE for QED]

**Technical root.** The $\beta$-function $\beta(g) = \mu\, dg/d\mu$ is positive for QED and $\phi^4$, driving the coupling toward a Landau pole. Contrast **asymptotic freedom** in QCD:

$$\beta_{\rm QCD} = -\frac{g^3}{16\pi^2}\left(11 - \tfrac{2}{3}n_f\right) < 0 \quad (n_f < 33/2),$$

so the coupling vanishes logarithmically in the UV — QCD is UV-safe where QED/$\phi^4$ are not. **Caveat:** the Landau pole is a one-loop *perturbative diagnostic*, distinct from the rigorous triviality theorem (the ADC proof does **not** proceed via the pole).

**Why it matters.** It overturns renormalizability-as-fundamental-axiom (Wilson's RG already reinterpreted renormalizability as an emergent low-energy property) and *contributes* to the view that the SM is an EFT. It connects to the still-`[OPEN]` constructive question of whether *any* realistic interacting 4D QFT exists in the continuum. See *GC-9*, *Chapter 15 (p. 233)*.

**Current status.** $\phi^4_4$ triviality is `[ESTABLISHED]` (theorem). QED triviality is `[INFERENCE]`. Asymptotic safety as a possible nontrivial UV fixed point is `[SPECULATIVE]`/`[OPEN]`. This is a consistent statement about domain of validity, **not** a logical contradiction.

> **REFEREE VERDICT — ISSOUND: TRUE; SEVERITY: MINOR; KEEP. Corrections.** Re-categorize: the headline content is `[ESTABLISHED]`, not `unsolved-problem`; only the SM-specific UV completion is open, and that is *gravity-dominated* at $M_{\rm Pl}$, not driven by triviality. Two overclaims fixed: (1) the SM-as-EFT conclusion rests primarily on **empirical** incompleteness (gravity, neutrino masses, dark matter, baryon asymmetry), with triviality a contributing motivation; (2) the SM Higgs sector's dominant UV concern is **quartic metastability** — for the measured top mass the Higgs quartic $\lambda$ runs slightly

> *negative* near $\sim 10^{10}$–$10^{11}$ GeV (electroweak-vacuum metastability) — **not** a quartic Landau pole; the $U(1)_Y$ Landau pole is academic, far above $M_{\text{Pl}}$. State the ADC hypotheses (lattice, reflection positivity, Ising universality class) rather than presenting triviality as unconditional. **Recommended tag.** `[ESTABLISHED]` (the refutation "renormalizable $\neq$ UV-complete" and $\phi^4_4$ triviality); `[INFERENCE]` (QED triviality; SM-as-EFT inheritance); the SM's own UV completion is `[OPEN]` but Planck/gravity-dominated. **Not**`unsolved-problem`.

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## GC-11 — Singularities: GR predicts its own breakdown (geodesic incompleteness)

**Frameworks:** *GR* (Chapter 10 (p. 121)), *Cosmology* (Chapter 11 (p. 139)), *QFT* (Chapter 8 (p. 89)), *QM* (Chapter 7 (p. 71)). **Kind:** domain-of-validity limit / breakdown (**not** an internal logical inconsistency).

**Precise statement.** The Penrose–Hawking singularity theorems prove, via the Raychaudhuri focusing equation plus an energy condition and an initial focusing condition, that classical solutions are **geodesically incomplete**:

- *Penrose (1965):* null energy condition (NEC) + a **trapped surface** + global hyperbolicity (non-compact Cauchy surface) $\Rightarrow$ null geodesic incompleteness;
- *Hawking / Hawking–Penrose (1970):* strong energy condition (SEC) (or the generic condition) + cosmological expansion or a trapped set $\Rightarrow$ timelike/null incompleteness. `[ESTABLISHED]`

Incompleteness means the equations cease to define evolution. Curvature invariants (e.g., the Kretschmann scalar $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$) diverge in the **symmetric** exact cases (Schwarzschild $r \to 0$; FLRW $a \to 0$), though geodesic incompleteness does not by itself entail curvature blow-up in general (cf. conical/quasiregular singularities).

**Technical root & the energy-condition caveat.** The energy conditions are hypotheses on matter that quantum fields *violate pointwise* (Casimir effect, squeezed states, near-horizon Hawking fluxes for the NEC; positive $\Lambda$ and slow-roll inflation, with $\rho + 3p = -2\rho < 0$, decisively for the SEC). Crucially:

- **SEC violation** by $\Lambda > 0$ and inflation is a genuine, decisive loophole that *evades* the SEC-based cosmological theorems — this is why inflationary cosmology does not need an initial singularity from those theorems. The eternal-inflation case is instead closed by the **Borde–Guth–Vilenkin theorem** (a kinematic average-Hubble-rate condition, not an energy condition): even eternal inflation is past-incomplete.
- **Pointwise NEC violation** does **not** generally defeat the collapse theorem: quantum fields obey *averaged / quantum energy inequalities* (Ford–Roman QEIs; ANEC), and the singularity theorems have been re-derived from these weaker averaged conditions (Tipler–Borde–Roman; Fewster–Galloway; Fewster–Kontou) and largely survive.

**Why it matters.** Singularities mark precisely where GR must hand off to quantum gravity (Big Bang, black-hole interiors) at curvatures $\sim \ell_P^{-2}$; whether they are resolved (bounces, replacement) is `[OPEN]` and currently untestable. See *GC-1*, *GC-7*, *Chapter 16 (p. 308)*.

**Current status.** `[ESTABLISHED]` that GR is geodesically incomplete (the theorems are rigorous). The pointwise energy conditions are known too strong but are robustified by averaged/quantum conditions. What replaces the singularity is `[OPEN]`. This is a domain-mismatch/breakdown, not a contradiction within classical GR.

> **REFEREE VERDICT — ISSOUND: TRUE; SEVERITY: MINOR; KEEP. Corrections.** Penrose's theorem is conditional on the existence of a trapped surface plus global hyperbolicity; that trapped surfaces *form* in realistic collapse is a separate result (Christodoulou). Hawking radiation's NEC violation is a late-time near-horizon evaporation effect, not relevant to the

> *collapse-stage* premises — do not list it alongside Casimir/squeezed states as a threat to the theorem's hypotheses. Distinguish: SEC violation is a decisive loophole for the cosmological theorems; pointwise NEC violation is largely *repaired* by ANEC/QEIs for the collapse theorem. Overstated "premises in tension with quantum matter" → "pointwise violated, but the theorems are substantially robust under the physically appropriate averaged/quantum energy conditions." **Recommended tag.** `[ESTABLISHED]` (theorem content + domain-of-validity reading); the energy-condition sub-claim is `[ESTABLISHED]` as to pointwise violation but `[INFERENCE]`/partially-resolved given the averaged-energy / QEI robustification program.

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## GC-12 — Quantum nonlocality (Bell violation) vs. relativistic locality

**Frameworks:** *QM* (Chapter 7 (p. 71)), Special Relativity (see *GR* (Chapter 10 (p. 121)), *Classical Mechanics* (Chapter 4 (p. 25))), *QFT* (Chapter 8 (p. 89)). **Kind:** conceptual tension with **no operational/empirical contradiction**.

**Precise statement.** Loophole-free Bell/CHSH experiments (Hensen, Giustina, Shalm et al., 2015) decisively violate the local-hidden-variable bound

CHSH: $S = |E(a,b) - E(a,b') + E(a',b) + E(a',b')| \leq 2,$

with QM correlations bounded above by Tsirelson's bound $S \leq 2\sqrt{2}$ (Cirel'son 1980); real experiments record values strictly between 2 and $2\sqrt{2} \approx 2.828$ (the ideal QM ceiling is not attained). `[ESTABLISHED]` Entanglement correlations and (on collapse-type readings) the projection postulate *appear* to act instantaneously across spacelike separation.

**Technical root.** Bell's theorem (local causality, Bell 1976) shows the empirical correlations cannot arise from any theory simultaneously satisfying **local causality**, **measurement-**

**independence** (free choice / no superdeterminism), and **no retrocausality**, holding the QM predictions fixed; experiment forces abandoning at least one, and which is interpretation-dependent. *("Realism" is not a separate premise — it is part of local causality;* ***noncontextuality*** *is the distinct Kochen–Specker assumption and should not be conflated with the Bell premises.)* The clash is **not** operational: QM respects **parameter independence** — the no-signaling theorem, $\rho_A = \mathrm{Tr}_B(\rho_{AB})$ is independent of $B$'s setting choice — so no usable information propagates superluminally. In the Jarrett/Shimony decomposition of local causality, QM violates only **outcome independence**, which carries no signal. At the QFT level, microcausality $[\phi(x), \phi(y)] = 0$ for spacelike separation enforces no-signaling. `[ESTABLISHED]`

**Why it matters.** It pins down exactly which classical intuitions must be abandoned and shows the clash with relativity is *conceptual* (about hidden mechanism), not operational. Making collapse manifestly Lorentz-covariant is genuinely hard and is where objective-collapse models (GRW/CSL) face their stiffest challenge; relativistic QFT measurement also confronts frame-dependent state-update, Sorkin's "impossible measurements," and Reeh–Schlieder vacuum nonlocality. See *GC-6*, *GC-15*.

**Current status.** `[ESTABLISHED]` that Bell violation is real and that no-signaling holds — so **no contradiction with special relativity's operational content**. The tension is interpretational. Measurement-independence/free choice is load-bearing; denying it (superdeterminism/retrocausality) is a logically `[OPEN]` but widely-regarded-as-conspiratorial escape, actively `[CONTESTED]` by a minority. A fully covariant, agreed account of relativistic state-update is `[OPEN]`.

> **REFEREE VERDICT — ISSOUND: TRUE; SEVERITY: MINOR; KEEP. Corrections.** State the Bell premises precisely: {local causality, measurement-independence, no-retrocausality} with predictions held fixed. "Realism" is subsumed in local causality (CHSH = "local realism"); noncontextuality is the distinct

> Kochen–Specker premise and must not be listed inside the Bell trichotomy. Tsirelson's $2\sqrt{2}$ is the ideal QM ceiling, not an experimental value. The Jarrett/Shimony split (parameter independence respected = no-signaling; outcome independence violated = no signal) sharpens the SR reconciliation. Collapse is interpretation-dependent (absent in Everett and de Broglie–Bohm; stochastic/Lorentz-challenged in GRW/CSL). **Recommended tag.** `[ESTABLISHED]` (a conceptual tension with no operational contradiction); `[CONTESTED]`/interpretation-dependent (which premise of local causality fails; the physical status of collapse).

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## GC-13 — Holographic area-law degrees of freedom vs. volume-extensive QFT Hilbert space

**Frameworks:** *Information Theory* (Chapter 12 (p. 155)), *QFT* (Chapter 8 (p. 89)), *GR* (Chapter 10 (p. 121)), *Statistical Mechanics* (Chapter 6 (p. 56)), *Thermodynamics* (Chapter 5 (p. 40)). **Kind:** conceptual tension built on `[ESTABLISHED]` components; resolution `[OPEN]`. **Domain-of-validity mismatch**, not a logical inconsistency.

**Precise statement.** Black-hole and covariant entropy bounds — $S_{\mathrm{BH}} = k_B A/(4\ell_P^2)$; the Bousso bound $S \leq A/4$ in Planck units — cap the entropy of a *gravitating* region by its boundary **area**, implying a finite, area-extensive count of physical gravitational degrees of freedom. Local QFT, by contrast, assigns independent oscillators to every point, giving a volume-extensive Hilbert-space dimension. The conceptual tension is over "how many degrees of freedom live in a region." `[ESTABLISHED components]`

**Technical root.** Ground-state entanglement entropy of a region obeys an **area law** but with a UV-*divergent*, cutoff-dependent coefficient,

$$S \sim \frac{\text{Area}}{\epsilon^{D-2}} \quad (D = \text{spacetime dimension; e.g. } S \sim \text{Area}/\epsilon^2 \text{ in 4D}),$$

reflecting infinitely many short-distance modes; only mutual information, entropy *differences*, and universal subleading coefficients (central charge in 2D; log / $a$-anomaly terms) are physical. Rigorously, local algebras in QFT are **type III$_1$ von Neumann factors** (Reeh–Schlieder; cyclic-separating vacuum): they admit **no trace and no tensor factorization** of the Hilbert space into local subsystems, so there is no well-defined local density matrix or number operator. The resolution of the apparent over-counting is gravitational: a region whose field-theoretic entropy would exceed $A/4$ has enough energy to have *already collapsed* to a black hole (the Susskind spherical-entropy-bound argument) — the QFT count is being applied *outside its gravitational domain of validity*.

**Why it matters.** It suggests continuum QFT badly over-counts degrees of freedom and is effective, not fundamental (connecting to *GC-3* and the species/Bekenstein bounds), and underlies the holographic principle and "spacetime from entanglement" programs (*GC-16*).

**Current status.** Area-law BH entropy is `[ESTABLISHED]`-theory, near-universally accepted across independent derivations (Hawking radiation, Euclidean action, string microstate counting for extremal/BPS black holes — Strominger–Vafa; loop area spectra), but lacks a *general background-independent* proof `[INFERENCE]`. The type-III structure is `[ESTABLISHED]` (algebraic QFT). Recent crossed-product / large-$N$ constructions (Witten; Chandrasekaran–Longo–Penington–Witten) yield type II algebras with a well-defined trace and finite *renormalized* (generalized) entropy — a `[INFERENCE]` **partial bridge** that does not yet derive the $A/4$ microstate count. How to define entanglement entropy in full quantum gravity, and whether spacetime is fundamentally discrete, is `[OPEN]`.

**REFEREE VERDICT — isSound: true; severity: minor; keep. Corrections.** Fix the dimensional slip: the leading divergence is $S \sim \mathrm{Area}/\epsilon^{D-2}$ in $D$ spacetime dimensions (not $\epsilon^{d-1}$). The *Hilbert space* is volume-extensive, but the *ground-state* entanglement entropy obeys an area law (a volume law holds only for generic excited/random states) — the original verbally conflated these. The relationship is a domain-of-validity mismatch (QFT without gravity vs. with gravity), not a contradiction: once gravitational backreaction is included, the over-counting is removed. The crossed-product type-II bridge addresses factorization/trace and matches generalized entropy but does not derive the finite count — "partial bridge" is fair; more would overclaim. **Recommended tag.** `[OPEN]` conceptual tension built on `[ESTABLISHED]` components ($S_{BH}$; the area-law UV divergence and which pieces are universal; type $III_1$ algebras + Reeh–Schlieder; the Susskind resolution; the Bousso bound); the crossed-product bridge is `[INFERENCE]`; the full reconciliation is `[OPEN]`. **Status update (2026-07-02, track R1).** The crossed-product "partial bridge" is now theorem-grade at linearized order for the *complete* perturbative constraint group, with boost-supertranslation edge modes and entropy $S_{\mathrm{gen}}$ + edge-mode + constant (Klinger–Kudler-Flam–Satishchandran, arXiv:2601.07910), and yields a generalized second law for non-stationary linearized perturbations of Killing horizons plus a proof of the quantum focusing conjecture within the perturbative regime (Chandrasekaran–Flanagan, arXiv:2601.07915). Founding coordinates now verified: Witten, *JHEP* **10** (2022) 008; Chandrasekaran–Longo–Penington–Witten, *JHEP* **02** (2023) 082; Chandrasekaran–Penington–Witten, *JHEP* **04** (2023) 009. Unchanged: the bridge does not derive the $A/4$ microstate count; the entropy carries an unfixed additive constant; the observer/boost/corner symmetry is installed, not derived (OP-44; A-CP-KILLING). "Partial bridge" `[INFERENCE]` stands.

## GC-14 — Stone–von Neumann uniqueness vs. inequivalent representations (infinite dof)

**Frameworks:** *QM* (Chapter 7 (p. 71)), *QFT* (Chapter 8 (p. 89)), *Statistical Mechanics* (Chapter 6 (p. 56)), *Mathematics for Physics* (Chapter 13 (p. 172)). **Kind:** domain-of-validity boundary (**not** an inconsistency).

**Precise statement.** `[ESTABLISHED]` The **Stone–von Neumann theorem** (von Neumann 1931): for a *finite* number of canonical pairs, every irreducible representation of the **Weyl (exponentiated) form** of the CCR that is **weakly/strongly continuous (regular)** is unitarily equivalent to the Schrödinger representation. *Both* the Weyl-form and the continuity hypotheses are essential. `[ESTABLISHED]` For *infinitely* many degrees of freedom (QFT, the thermodynamic limit) the theorem **fails**: there are uncountably many unitarily inequivalent irreducible representations of the CCR.

**Technical root.** The deep reason is **not** "compactness" but the nonexistence of an infinite-dimensional translation-invariant (Lebesgue) measure: in finite dimensions the canonical Gaussian construction is essentially unique up to unitary equivalence, while in infinite dimensions Gaussian measures with different covariances are *mutually singular* (disjoint), yielding inequivalent representations.

**Why it matters.** This is the precise mathematical seam between the "one Hilbert space" intuition of textbook QM and the algebraic reality of QFT and many-body physics. It underlies — as **related but logically distinct** phenomena, each needing extra hypotheses — inequivalent vacua and spontaneous symmetry breaking; superselection sectors (charge, particle number, distinct thermodynamic phases); Haag's theorem (*GC-9*, which additionally requires Poincaré covariance and vacuum uniqueness); and the necessity of the algebraic C*/von Neumann formulation. In statistical mechanics, distinct phases live in inequivalent GNS representations and genuine non-analyticities (phase transitions) appear only in the thermodynamic limit (illustrated for ferromagnetic lattice models by the Lee–Yang circle theorem on partition-function zeros).

**Current status.** `[ESTABLISHED]` on both sides: Stone–von Neumann is a theorem (finite dof); inequivalent representations are a rigorous, physically essential feature of infinite-dof systems. A clean domain mismatch, fully understood mathematically (algebraic QFT / quantum statistical mechanics), not an open problem.

> **REFEREE VERDICT — isSound: true; severity: minor; keep.** **Corrections.** State both load-bearing hypotheses (Weyl form **and** regularity/continuity) — without continuity, inequivalent representations exist even in finite dof. The mechanism is the missing infinite-dimensional Lebesgue measure / mutual singularity of Gaussian measures, **not** "compactness" (a garbled attribution). SSB, superselection, and Haag's theorem are logically distinct, not uniform direct corollaries (superselection can arise without infinite dof; Haag needs Poincaré covariance + vacuum uniqueness). SvN gives uniqueness of the *irreducible* rep up to unitary equivalence; "Fock vacuum" is anachronistic for the bare SvN (Schrödinger-representation) setting. Lee–Yang specifically concerns partition-function zeros of ferromagnetic Ising/lattice-gas models. SvN is a *theorem* about the chosen representation, not a foundational axiom QM "rests on." References: von Neumann 1931; Reed–Simon; Haag, *Local Quantum Physics*; Bratteli–Robinson. **Recommended tag.** `[ESTABLISHED]`.

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## GC-15 — Semiclassical gravity $G_{\mu\nu} = 8\pi G \langle T_{\mu\nu} \rangle$ vs. quantum superposition of mass

**Frameworks:** *GR* (Chapter 10 (p. 121)), *QM* (Chapter 7 (p. 71)), *QFT* (Chapter 8 (p. 89)). **Kind:** domain-of-validity breakdown of an approximation that becomes inconsistent if promoted to fundamental.

**Precise statement.** Semiclassical gravity (Møller 1962, Rosenfeld 1963), $G_{\mu\nu} = 8\pi G \langle \psi | T_{\mu\nu} | \psi \rangle$, couples a classical metric to quantum matter via the *expectation value* of the stress-energy — a source quadratic (hence nonlinear) in the state. For a macroscopic superposition $(|\text{mass at } A\rangle + |\text{mass at } B\rangle)/\sqrt{2}$, the equation produces a *single* mean-field geometry sourced by both locations at once — **not** a superposition of two geometries. `[INFERENCE that this fails for superpositions]`

**Technical root.** The mean-field source is nonlinear in $|\psi\rangle$ (the map $|\psi\rangle \mapsto |\psi\rangle\langle\psi|$ is nonlinear). The mean-field prediction is empirically disfavored (Page–Geilker 1981), and — combined with stochastic state reduction — the nonlinearity is pathological: the general theorem that **deterministic nonlinear modifications of quantum dynamics permit superluminal signaling** (Gisin 1990; Polchinski 1991) applies, and the non-relativistic Schrödinger–Newton limit conflicts with Born-rule statistics under collapse (Bahrami–Großardt–Donadi–Bassi 2014).

**Why it matters.** It is a concrete internal reason semiclassical gravity is at best an approximation, motivating either quantizing the metric or modifying QM via gravitational collapse. It drives gravitationally-induced-collapse proposals (Diósi–Penrose) and tabletop gravity-mediated-entanglement experiments aiming to test whether gravity must be quantized. See *GC-1*, *GC-12*, *Chapter 15 (p. 233)*.

**Current status.** `[INFERENCE]`/`[OPEN]`. The near-universal expectation is that the metric must be quantized (or QM modified by gravitational collapse), but this is not experimentally settled.

Diósi–Penrose collapse is being *constrained* (not excluded) by underground spontaneous-radiation and interferometry experiments. Proposed gravity-mediated-entanglement tests would, if entanglement were observed, argue gravity is non-classical — but they have not been performed at the required sensitivity. Semiclassical gravity remains an excellent approximation in its domain (Hawking, Unruh; the leading term of a $1/N$ expansion).

> **REFEREE VERDICT — isSound: true; SEVERITY: MINOR; KEEP. Corrections.** The signaling/Born-rule pathology does **not** follow from the nonlinearity of $\langle T \rangle$ alone: unitary-only semiclassical gravity is nonlinear but does not by itself signal; the inconsistency arises specifically when the mean-field source is *combined* with stochastic state reduction (then Gisin/Polchinski applies). The "single mean-field geometry" is not self-evidently inconsistent in isolation — it is empirically disfavored (Page–Geilker) and pathological only with collapse. "A classical metric cannot consistently couple to quantum sources" is **too strong as a universal claim**: it is the *mean-field (expectation-value)* coupling that fails; fundamentally *stochastic* classical–quantum models (Diósi–Penrose; Oppenheim-type CQ dynamics with intrinsic decoherence/diffusion) evade the deterministic-nonlinearity no-go and remain `[CONTESTED]`-but-open. Do not lean on Eppley–Hannay (1977), whose gedanken argument is criticized as relying on unphysical idealizations. **Recommended tag.** `[INFERENCE]` for the breakdown (mean-field failure for superpositions + nonlinearity-implies-signaling are well-established); the universal no-go ("no classical metric can couple to quantum matter") is downgraded to `[CONTESTED]`/`[SPECULATIVE]`.

## GC-16 — ER=EPR / "spacetime from entanglement" vs. its unestablished generality beyond AdS

**Frameworks:** *Information Theory* (Chapter 12 (p. 155)), *GR* (Chapter 10 (p. 121)), *QFT* (Chapter 8 (p. 89)), *Cosmology* (Chapter 11 (p. 139)). **Kind:** conceptual tension / **evidential gap** (domain-of-validity / generality-of-evidence limitation, **not** an internal logical inconsistency).

**Precise statement.** The Ryu–Takayanagi formula $S(A) = \mathrm{Area}(\gamma_A)/4G_N$, entanglement-wedge reconstruction, the "first law of entanglement ⇒ linearized Einstein equations" result, and the ER=EPR conjecture jointly motivate the program that spacetime geometry emerges from the entanglement structure of an underlying quantum theory. The tension: all of this is rigorously formulated and *derived* essentially **only** within AdS/CFT (asymptotically anti-de Sitter, large central charge, strong 't Hooft coupling — the bulk Einstein-gravity regime). `[INFERENCE within AdS]`

**Technical root.** RT/HRT and entanglement-wedge reconstruction are derived (Lewkowycz–Maldacena 2013 gravitational replica trick; FLM/JLMS modular-Hamiltonian relation; QES via Engelhardt–Wall; islands via Penington / Almheiri–Engelhardt–Marolf–Maxfield / Penington–Shenker–Stanford–Yang 2019–20) — but every derivation presupposes the holographic dictionary plus the Einstein-gravity (large-$c$, large-$\lambda$) regime, with a homology constraint and extremal-surface prescriptions. Crucially: (i) the gravity-from-entanglement derivations recover only the **linearized** Einstein equations (Lashkari–Van Raamsdonk; Faulkner–Guica–Hartman–Myers–Van Raamsdonk 2014; second order only under restrictive assumptions); the full nonlinear, background-independent reconstruction is **not** achieved; (ii) ER=EPR has no first-principles derivation for generic, non-holographic entanglement; (iii) de Sitter holography (dS/CFT, relevant to our positive-$\Lambda$ universe) is far less constrained and speculative, and flat-space / finite-$N$ holography lack established RT analogs; (iv) an algebraic-QFT obstruction (local algebras are type $III_1$ factors lacking factorization and a finite entanglement entropy — see *GC-13*) sits beneath the heuristic "entanglement-as-fabric" discourse.

**Why it matters.** It is the most ambitious proposed reconciliation of GR and quantum information — potentially reframing the GR/QFT clash (*GC-1*) by making geometry emergent — but its applicability to real cosmology (where the deepest puzzles, including Λ and the arrow of time, live) is exactly where it is least grounded. See *Chapter 18 (p. 349)*, *Chapter 15 (p. 233)*.

**Current status.** RT within AdS/CFT is `[INFERENCE]` (well-supported, extensively checked, proven in some lower-dim/BPS cases; AdS/CFT itself lacks a general nonperturbative proof). "Spacetime from entanglement" as a *universal* principle, ER=EPR for generic entanglement, and de Sitter holography are `[SPECULATIVE]`/`[OPEN]`. Whether this is profound universal insight or an AdS-specific artifact is arguably the central open question of the program.

> **REFEREE VERDICT — ISSOUND: TRUE; SEVERITY: MINOR; KEEP. Corrections.** No genuine technical errors; the claim is deliberately anti-overclaiming. Minor: within AdS/CFT these results are *derived*, not merely "tested" (the stated domain restriction is unaffected). The status hierarchy is endorsed: RT/entanglement-wedge in AdS `[INFERENCE]`; linearized-Einstein-from-first-law `[INFERENCE, linearized only]`; full nonlinear background-independent gravity-from-entanglement `[SPECULATIVE/OPEN]`; ER=EPR for generic entanglement `[SPECULATIVE]`; dS/CFT and flat/finite-$N$ analogs `[SPECULATIVE/OPEN]`. The label is more precisely a *generality-of-evidence* limitation than a logical tension — there is no contradiction, only an unwarranted extrapolation outside AdS. **Recommended tag.** `[SPECULATIVE]` for the program's generality (AdS-restricted RT and entanglement-wedge results at `[INFERENCE]`); the meta-claim that this

> generality is currently *unestablished* beyond AdS is `[ESTABLISHED]` (contested only by optimists).

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## GC-17 — Standard ensemble statistical mechanics vs. long-range gravitating systems

**Frameworks:** *Statistical Mechanics* (Chapter 6 (p. 56)), *Thermodynamics* (Chapter 5 (p. 40)), *GR* (Chapter 10 (p. 121)), *Cosmology* (Chapter 11 (p. 139)). **Kind:** domain-of-validity mismatch (**not** a logical inconsistency).

> **Note.** This entry's original framing was the **only** one the referee found internally *unsound* (`isSound: false`). It is retained and corrected; one erroneous sub-claim (black-hole area-entropy as a *consequence* of Newtonian non-additivity) is **excised to** *Corrected / not actually contradictions*.

**Precise statement.** Equilibrium statistical mechanics and thermodynamics presuppose interactions decaying faster than $r^{-d}$ in $d$ dimensions (short-range), so that energy and entropy are extensive/additive, the standard fixed-density thermodynamic limit exists, and the microcanonical and canonical ensembles are equivalent. Self-gravitating systems (star clusters, galaxies, the cosmological matter distribution) violate this: the potential $\sim -Gm^2/r$ is long-range and unscreened, making energy non-additive. The standard ensemble machinery breaks down — as a **domain mismatch**, not a logical inconsistency. `[ESTABLISHED]`

**Technical root & genuine consequences (classical self-gravitating systems).**

- (a) **Ensemble inequivalence:** the microcanonical and canonical ensembles are *inequivalent*.
- (b) **Negative microcanonical specific heat** — impossible in the canonical ensemble, where $C_V = \mathrm{Var}(E)/(k_B T^2) \geq 0$

identically; the negative microcanonical $C_V$ is precisely the *diagnostic signature* of (a), not a logically independent fact.

- (c) The **gravothermal catastrophe / Antonov instability** (Antonov 1962; Lynden-Bell–Wood 1968), with long-lived metastable *local* equilibria below threshold (the Antonov radius) but **no global entropy maximum** for the idealized confined point-mass model.
- (d) Failure of the **Euler and Gibbs–Duhem relations**, which rest on first-order homogeneity (extensivity).

The "entropy unbounded above" feature is a property of the *unregularized* point-mass model and stems from the short-distance ($r \to 0$, UV) singularity of $-Gm^2/r$ — distinct from the long-range (IR) non-additivity that is the headline subject. Short-distance cutoffs (quantum degeneracy pressure, hard cores; Padmanabhan) restore boundedness and metastable equilibria. A nontrivial mean-field thermodynamic limit *does* exist under **Kac rescaling** (coupling $\sim 1/N$), though non-additivity and ensemble inequivalence persist.

**Why it matters.** The foundations of thermodynamics/statistical mechanics do not straightforwardly apply to the dominant long-range force at large scales, complicating any global entropy budget for the universe (relevant to the arrow-of-time / Past-Hypothesis problem, *GC-7*).

**Current status.** `[ESTABLISHED]` as a domain breakdown: non-additivity, negative specific heat, and ensemble inequivalence of gravitating systems are well understood and not in dispute (Antonov 1962; Lynden-Bell–Wood 1968; Thirring 1970; Padmanabhan 1990; Ruelle/Fisher stability theory). Treating the universe as a global thermodynamic system with well-defined temperature/entropy for the gravitational sector is `[INFERENCE]`/`[CONTESTED]` and not rigorously founded.

> **REFEREE VERDICT — isSound: false (one sub-claim miscategorized); severity: minor; keep (with excision). Corrections.** The headline domain-mismatch

> claim is **SOUND** and well-established. Excised category error: black-hole area-entropy $S_{\text{BH}} \sim A$ was listed as a *consequence* of Newtonian self-gravity non-additivity *on par* with (a)–(d). It is not — Bekenstein–Hawking area scaling is a relativistic/semiclassical horizon/holographic result (Bekenstein 1973; Hawking 1975), a *different* domain that does not follow from the Euler-relation failure; ordinary self-gravitating gas entropy does not scale as area (see quarantine below). "Entropy unbounded / no equilibrium" is a property of the idealized unregularized point-mass model (a UV effect), not of the long-range non-additivity. The standard fixed-density thermodynamic limit fails, but a Kac-rescaled mean-field limit exists. Negative $C_V$ is the *signature* of ensemble inequivalence, not an independent consequence. **Recommended tag.** `[ESTABLISHED]` (the core domain-mismatch claim and consequences (a)–(d) for classical self-gravitating systems).

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## Corrected / not actually contradictions

This section quarantines specific sub-claims the referee judged miscategorized. The *parent entries are retained*; only the flagged sub-claim is moved here with an explanation.

### CN-1 — Black-hole area-entropy is not a consequence of Newtonian self-gravity non-additivity (from *GC-17*)

**Excised claim.** "Self-gravitating systems exhibit … area- rather than volume-scaling entropy for black holes ($S_{\text{BH}} \sim A$)" — listed in the original *GC-17* as a *consequence* of the non-additivity of classical Newtonian self-gravitating systems, alongside negative specific heat and ensemble inequivalence.

**Why it is not a valid consequence.** Bekenstein–Hawking entropy,

$$S_{\mathrm{BH}} = \frac{k_B c^3 A}{4G\hbar},$$

is a result of **relativistic / semiclassical gravity** — event horizons, the holographic bound, Hawking 1975 / Bekenstein 1973 — a *different domain* from the classical statistical mechanics of self-gravitating point masses. It does **not** follow from the failure of the Euler relation that drives the gravothermal catastrophe (Antonov 1962; Lynden-Bell–Wood 1968). Moreover, ordinary self-gravitating gas / star-cluster entropy does **not** scale as area — it has its own non-extensive (and, for unsoftened point masses, unbounded) behavior. The two phenomena share only the *heuristic* slogan "gravity defeats naive volume-extensivity."

**Disposition.** Black-hole area scaling is genuine, established physics and a real instance of gravity defeating volume-extensivity — but as a *separate, relativistic* phenomenon. It belongs with *GC-13* (holographic area-law vs. volume-extensive QFT) and *GC-4*, not as a corollary of the Newtonian argument in *GC-17*. No contradiction is lost by removing it from GC-17; one was merely *misattributed*.

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# References

See the Bibliography (p. 776) for full citations. Landmark sources invoked above (real, standard) include: 't Hooft–Veltman (1974) and Goroff–Sagnotti (1985) on graviton-loop divergences; Stelle (1977) on higher-derivative gravity; Donoghue on GR as an EFT; Weinberg (1989) review of the cosmological-constant problem; Lovelock (1971–72); Hawking (1975) and Bekenstein (1973) on black-hole radiation/entropy; Page on the Page curve; Strominger–Vafa (1996) microstate counting; Almheiri–Marolf–Polchinski–Sully (AMPS, 2012); Bell (1964, 1976), CHSH (1969), Cirel'son/Tsirelson (1980), Hensen/Giustina/Shalm et al. (2015); von Neumann (1932) measurement; Gleason (1957); GRW/CSL objective-collapse literature; Boltzmann's $H$-theorem, Loschmidt, Zermelo, Lanford (1975), Crooks and Jarzynski fluctuation theorems; Dirac constrained-Hamiltonian theory, DeWitt (1967), Isham and Kuchař problem-of-time reviews; Page–Wootters, Rovelli; Haag (1955), Hall–Wightman (1957), Streater–Wightman, Epstein–Glaser, Doplicher–Haag–Roberts; Aizenman (1981) and Aizenman–Duminil-Copin (2021) on $\phi^4$ triviality; Penrose (1965), Hawking–Penrose (1970), Borde–Guth–Vilenkin, Ford–Roman quantum energy inequalities, Fewster–Galloway, Fewster–Kontou, Christodoulou; Møller (1962), Rosenfeld (1963), Page–Geilker (1981), Gisin (1990), Polchinski (1991), Bahrami et al. (2014), Diósi–Penrose; Ryu–Takayanagi, Lewkowycz–Maldacena (2013), Faulkner et al. (2014), Maldacena–Susskind (ER=EPR), Penington / Almheiri–Engelhardt–Marolf–Maxfield / Penington–Shenker–Stanford–Yang (2019–20), Witten and Chandrasekaran–Longo–Penington–Witten on crossed-product algebras; Antonov (1962), Lynden-Bell–Wood (1968), Thirring (1970), Padmanabhan (1990), Ruelle/Fisher stability theory; Haag, *Local Quantum Physics*; Bratteli–Robinson; Streater–Wightman, *PCT, Spin and Statistics, and All That*.

CHAPTER 15

# The Open-Problems Registry

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This page is the canonical, stably-numbered list of genuinely unresolved problems that a program toward a universal theory must confront. Each entry carries a **precise statement**, a **why-it-is-hard** analysis, the **current leading approaches**, an **epistemic tag** (per *Chapter 2 (p. 8)*), and cross-links.

A central discipline (from *Chapter 2 (p. 8)* §5, applied in *Chapter 14 (p. 191)*) is to classify *what kind* of problem each entry is:

- **Logical inconsistency** — two accepted statements cannot both hold. Rare.
- **Domain-of-validity mismatch** — frameworks clash only where neither is trusted (a *frontier*, not a paradox).
- **Fine-tuning / naturalness puzzle** — the theory is consistent for any value, but a value looks "unreasonably" special.
- **Unsolved-but-consistent problem** — no inconsistency, just no derivation/solution yet.

> **Convention.** IDs (`OP-N`) are permanent. When a problem is resolved, it is *not* deleted — its status is updated and the resolution logged in the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)). New problems append at the next free ID. Tags reflect the state of knowledge at the *Last updated* date; nothing here asserts a speculative resolution as fact.

**Index by theme.** A. *Quantum gravity & unification* (OP-1, OP-15, OP-23, OP-24, OP-25) · B. *Foundations of QM* (OP-2, OP-26, OP-27, OP-28, OP-29) · C. *Cosmology & the dark sector* (OP-3, OP-5,

OP-6, OP-17, OP-18, OP-19, OP-30, OP-31) · D. *Particle physics & naturalness* (OP-4, OP-7, OP-8, OP-9, OP-14, OP-20, OP-21, OP-22, OP-32) · E. *Arrow of time & statistical foundations* (OP-10, OP-16, OP-33, OP-34, OP-35) · F. *Mathematical foundations of QFT & dynamics* (OP-11, OP-12, OP-36, OP-37) · G. *Other* (OP-13, OP-38, OP-39).

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# A. Quantum gravity & unification

### OP-1 — Quantum gravity: a consistent theory reducing to GR and to QFT

**Area:** Quantum gravity / unification (GR + QFT). **Tag:** `[OPEN]`. **Kind:** structural domain-mismatch (frontier), not a logical inconsistency.

**Statement.** There is no experimentally confirmed, mathematically consistent theory that reduces to general relativity in the low-curvature limit and to quantum field theory on flat spacetime while treating the spacetime metric itself as a dynamical quantum variable. Perturbatively quantized GR (write $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and quantize $h_{\mu\nu}$) is non-renormalizable: pure gravity first diverges at two loops (Goroff–Sagnotti 1985) and gravity-plus-matter already at one loop, requiring infinitely many counterterms with undetermined coefficients, so predictivity is lost at the Planck scale $E_P \sim 1.22 \times 10^{19}$ GeV ($\ell_P \sim 1.6 \times 10^{-35}$ m; see *Chapter 16 (p. 308)*).

**Why it is hard.** QFT presupposes a fixed smooth classical background and an external time parameter; GR makes geometry dynamical and background-independent (diffeomorphism invariance), and the canonical theory has *no* preferred time — the Wheeler–DeWitt constraint $\hat{H}\,|\Psi\rangle = 0$ is timeless (the "problem of time", cf. OP-25). Reconciling background independence with superposition of geometries, plus the non-renormalizability of the graviton, makes the conflict structural rather than merely technical. `[ESTABLISHED]` that perturbative GR is non-renormalizable; `[OPEN]`

**Leading approaches.** String theory / M-theory (with AdS/CFT holography as its best-understood corner); loop quantum gravity and spin foams; asymptotic safety (a nontrivial UV fixed point for gravity via functional RG, OP-22); causal dynamical triangulations and causal sets; treating GR as a low-energy effective field theory valid below $E_P$ [INFERENCE, robust]; emergent-spacetime / "spacetime from entanglement" programs (OP-23, OP-24).

**Cross-links.** *Chapter 10 (p. 121), Chapter 8 (p. 89), Chapter 18 (p. 349), Chapter 3 (p. 15), Chapter 14 (p. 191).*

## OP-15 — Black-hole information paradox: unitarity vs. the equivalence principle

**Area:** Quantum gravity / GR–QFT–thermodynamics interface. **Tag:** [OPEN] (mechanism [CONTESTED]). **Kind:** triple-point domain clash.

**Statement.** Hawking's semiclassical calculation predicts thermal (mixed) radiation at $T_H = \hbar c^3/(8\pi G k_B M)$; iterated to complete evaporation it appears to map an initial pure state to a mixed state, violating unitarity and severing any link between the Bekenstein–Hawking entropy $S_{BH} = k_B c^3 A/(4G\hbar)$ and a microscopic state count. Whether, where, and how information escapes — and whether the horizon stays smooth (no "firewall") — is unresolved.

**Why it is hard.** It is a clash between GR (smooth horizon, local causality), QM (unitarity), and thermodynamics (entropy as state-counting), in a regime mixing strong gravity with quantum fields where the correct degrees of freedom are unknown. Recent island / quantum-extremal-surface and replica-wormhole computations reproduce a unitary Page curve within controlled (largely AdS, semiclassical-plus-replica) settings [INFERENCE], strongly favoring information preservation, but the bulk mechanism in a realistic asymptotically-flat, dynamically-formed black hole — and how information reaches an external observer — is not settled [OPEN]. Do not overstate: islands are computed in toy models, not yet shown to resolve the paradox in nature.

**Leading approaches.** Quantum extremal surfaces / islands and replica wormholes (Page curve); AdS/CFT (manifestly unitary boundary description); black-hole complementarity and the firewall (AMPS) debate [CONTESTED]; string microstate counting (Strominger–Vafa for extremal/BPS black holes) and fuzzballs; holographic / quantum-error-correction models of bulk reconstruction.

**Cross-links.** *Chapter 10 (p. 121)*, *Chapter 5 (p. 40)*, *Chapter 12 (p. 155)*, OP-33 (BH microstate counting), OP-23.

**Status (2026-07-02, R1 sweep — dated append).** As of mid-2026: (i) island/replica-wormhole computations of a unitary Page curve are consensus-grade *within* controlled settings (verified coordinates in the Bibliography (p. 776)); the interior mechanism and any realistic asymptotically-flat 4D derivation remain [OPEN]. (ii) Two named objections stand [CONTESTED]: **ensemble-vs-single-theory** (PSSY's ensemble reading; Marolf–Maxfield α-sectors; contra Schlenker–Witten's sub-threshold no-averaging theorem (proven in 3d), *JHEP* **07** (2022) 143) and **massive-graviton/gravitating-bath** (Geng–Karch et al.: all controlled $d > 2$ islands have massive gravitons, *JHEP* **09** (2020) 121; islands vs the gravitational Gauss law, *JHEP* **01** (2022) 182; entropy time-independence with a gravitating bath, *SciPost Phys.* **10**, 103 (2021)). (iii) A replica-independent second leg now exists: crossed-product Type II algebras with von Neumann entropy $= S_{\text{gen}}$, a GSL for non-stationary linearized perturbations, and a perturbative-regime QFC proof (Witten *JHEP* **10** (2022) 008; CLPW *JHEP* **02** (2023) 082; CPW *JHEP* **04** (2023) 009; KLS *PRD* **111**, 025013; arXiv:2601.07910; arXiv:2601.07915) — perturbative, observer/corner-symmetry-installed, additive constant unfixed. See the GC-4/GC-13 status appends of this date and OP-44 for the nonperturbative fate.

## OP-23 — Does holography / "spacetime from entanglement" extend beyond AdS?

**Area:** Quantum gravity / holography / information. **Tag:** [SPECULATIVE] (AdS results [ESTABLISHED] within their setting). **Kind:** unsolved-but-consistent; possible over-generalization.

**Statement.** Within AdS/CFT, geometry appears encoded in boundary entanglement: the Ryu–Takayanagi formula $S(A) = \mathrm{Area}(\gamma_A)/(4G_N)$ ties entanglement entropy to minimal bulk surfaces, entanglement-wedge reconstruction realizes a quantum-error-correcting code, and the entanglement "first law" yields the *linearized* Einstein equations. Whether these results — and the broader "it-from-qubit" / ER=EPR program — are a universal principle or an artifact of the special negatively-curved large-$N$ AdS corner is unknown; there is no established de Sitter (positive-$\Lambda$, like our universe) or flat-space analog.

**Why it is hard.** All rigorous results live in asymptotically-AdS, large-central-charge, strong-coupling (Einstein-gravity) regimes; dS/CFT is far less constrained and speculative. Local QFT algebras are type $\mathrm{III}_1$ von Neumann factors with no tensor factorization and no finite entanglement entropy, so "entanglement = fabric of spacetime" lacks a clean foundational footing in full quantum gravity (OP-36). Only the *linearized* Einstein equations have been recovered; a background-independent nonlinear derivation does not exist, and our accelerating universe has no working holographic dual.

**Leading approaches.** AdS/CFT with RT/HRT, quantum extremal surfaces, entanglement-wedge reconstruction; tensor-network models (MERA, HaPPY codes); crossed-product / algebra-of-observables constructions giving well-defined gravitational entropies (OP-37); dS/CFT and flat-space holography [SPECULATIVE]; Jacobson-style thermodynamic derivations of Einstein's equations.

**Cross-links.** *Chapter 12 (p. 155)*, *Chapter 17 (p. 325)*, *Chapter 19 (p. 392)*, OP-15, OP-24, OP-36, OP-37.

## OP-24 — Derive the full nonlinear Einstein equations from entanglement/information

**Area:** Quantum gravity / holography / information. **Tag:** [OPEN]/[SPECULATIVE]. **Kind:** unsolved-but-consistent.

**Statement.** Linearized Einstein equations have been obtained from entanglement first laws, and Jacobson (1995) derives

Einstein's equations as a local equation of state from the Clausius relation $\delta Q = T\,\delta S$ on local Rindler horizons. A complete, non-perturbative, background-independent derivation of the *full* nonlinear (and quantum-corrected) gravitational dynamics from information-theoretic principles is not in hand.

**Why it is hard.** The linear results rely on perturbing around a known background; promoting them to a generic, nonlinear, observer-independent statement requires a notion of entanglement entropy in dynamical gravity that does not yet exist rigorously (OP-36, OP-37). It is unknown whether "gravity from entanglement" is fundamental or an emergent re-description valid only in special regimes (OP-23).

**Leading approaches.** Entanglement-first-law expansions to higher order; Jacobson's thermodynamic / entanglement-equilibrium derivations; tensor-network and holographic-complexity programs; crossed-product algebra methods.

**Cross-links.** *Chapter 17 (p. 325)*, *Chapter 12 (p. 155)*, OP-23, OP-1.

## OP-25 — The problem of time and diffeomorphism-invariant observables

**Area:** General relativity / quantum gravity. **Tag:** [OPEN]/[CONTESTED]. **Kind:** conceptual tension bordering on structural.

**Statement.** In a fully background-independent theory there is no external time; canonical quantization yields the timeless constraint $\hat{H}\,|\Psi\rangle = 0$. How to define local, physical (Dirac) observables that commute with the constraints, and what "time evolution" means absent a preferred time, remains conceptually unresolved.

**Why it is hard.** Diffeomorphism invariance means coordinate time is pure gauge; genuine observables must be relational (correlations between physical quantities), but constructing a complete set of local relational observables for full GR is technically open, and the issue compounds in the quantum theory.

space and deparametrization methods. Promising but not decisive.

**Cross-links.** *Chapter 10 (p. 121)*, *Chapter 7 (p. 71)*, OP-1, OP-10, OP-31.

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## B. Foundations of quantum mechanics

### OP-2 — The measurement problem and the quantum–classical cut

**Area:** Foundations of quantum mechanics. **Tag:** `[OPEN]`/`[CONTESTED]`. **Kind:** genuine incompleteness (axioms not a closed universal dynamics), *not* a calculational gap.

**Statement.** Linear unitary Schrödinger evolution never turns a superposition into a single outcome, yet the Born/projection rule does exactly that, stochastically. Treating apparatus + system + observer quantum-mechanically, unitary dynamics predicts an entangled superposition of macroscopically distinct pointer states (Schrödinger's cat; Wigner's friend), while experiment always records one definite outcome. The theory supplies no rule for what physical process, system size, or interaction constitutes a "measurement" (Bell's "shifty split").

**Why it is hard.** Unitary evolution and the projection postulate are mutually inconsistent as *universal* dynamical laws, so the standard axioms are not a closed, consistent universal dynamics. Decoherence sharply suppresses interference and selects a pointer basis (einselection) but does not by itself pick a single outcome or supply a fundamental boundary. Every interpretation that keeps unitary QM unmodified is empirically equivalent to standard QM, so the choice is currently philosophical/aesthetic; only objective-collapse models make distinct, testable predictions (OP-26).

**Leading approaches.** Everett / many-worlds (keep only unitary evolution; pay with branching and the probability problem, OP-28); de Broglie–Bohm pilot wave (definite positions, explicit nonlocality); objective collapse (GRW, CSL, Diósi–Penrose, OP-26); QBism / Copenhagen-instrumentalist / relational QM;

consistent/decoherent histories; decoherence + einselection as shared dynamical input.

**Cross-links.** *Chapter 7 (p. 71), Chapter 14 (p. 191), Chapter 20 (p. 458)*, OP-26, OP-27, OP-28.

**Status 2026-07-02 (extended-Wigner-friend developments).** The Local Friendliness (LF) no-go line hardened 2023–2026: derivation from strictly-weaker-than-Bell "thoughtful" assumptions (Wiseman–Cavalcanti–Rieffel, *Quantum* **7**, 1112 (2023); critique: Kent arXiv:2302.12707); **LF polytope = Bell polytope** in a wide class of multipartite/sequential scenarios, with a systematic Kochen–Specker→LF translation (*Quantum* **9**, 1819 (2025)); a "Noncontextual Friendliness" generalization — absoluteness of observed events + noncontextual agency jointly inconsistent with QT (arXiv:2502.02461 [preprint]); and first proof-of-principle, loophole-ridden LF-violation circuits on quantum hardware (*Quantum* **9**, 1851 (2025)) `[ESTABLISHED theorems]`. Counterweight: the six-argument review arXiv:2308.16220 [preprint] argues EWF arguments hinge on assumptions about in-principle-inaccessible correlations `[CONTESTED]`. Yield: these theorems constrain assumption *combinations* (absoluteness + local/noncontextual agency), never single interpretations — they reprice the interpretation ledger without creating a distinguishing test. See Working note: *Track R2 — The Measurement Problem, Operationalized (2026-07 bound state)* (repo: `notes/2026-07-02-measurement-problem-operationalized.md`) §3.

## OP-26 — Is collapse a real dynamical process? (Objective-collapse models)

**Area:** Foundations of QM. **Tag:** `[OPEN]`, empirically testable — a genuine virtue. **Kind:** unsolved-but-consistent (a distinct *physical* theory).

**Statement.** Objective-collapse models (GRW, Continuous Spontaneous Localization, Diósi–Penrose gravitational collapse) add stochastic nonlinear terms that spontaneously localize macroscopic superpositions, predicting tiny deviations from unitary QM: anomalous energy heating, spontaneous radiation, and suppression of large-mass matter-wave interference. Whether such collapse occurs is unknown.

**Why it is hard.** The predicted effects scale with mass/size and are extraordinarily small; isolating them requires exquisite control of decoherence backgrounds. Matter-wave interferometry, ultracold optomechanics, and underground spontaneous-radiation searches progressively constrain (and in regions exclude) the CSL and Diósi–Penrose parameter spaces, but the models are not falsified as a class, and the bare Diósi–Penrose proposal is being squeezed rather than settled.

**Leading approaches.** Large-mass matter-wave interferometry; cryogenic optomechanical force/heating searches; X-ray spontaneous-emission searches; refinement of CSL collapse-rate / correlation-length parameters.

**Cross-links.** *Chapter 7 (p. 71)*, OP-2, OP-1 (gravitationally-induced collapse links to quantum gravity).

**Status 2026-07-02.** `[ESTABLISHED]` XENONnT (PRL **136**, 120201 (2026), arXiv:2506.05507): $\lambda/r_C^2 < 3.0 \times 10^{-3}$ s$^{-1}$m$^{-2}$ (90% CL) for white-noise CSL — **canonical GRW point deviates from best fit by 9.1σ — the paper's stated first experimental exclusion of the originally proposed CSL values**; ×135 over the Majorana Demonstrator (PRL **129**, 080401 (2022)); DP $R_0 >$ 4.9 Å (90% CL). LISA-Pathfinder updates: rotational-noise bound ~2× better for $r_C \sim 10^{-5.5}$–$10^{-3.5}$ m (PRA **111**, L020203 (2025)); reanalysis of final acceleration data reports $\lambda \lesssim 8.3 \times 10^{-11}$ s$^{-1}$ at $r_C = 10^{-7}$ m (arXiv:2411.17588 [preprint]). Within the Markovian mass-proportional family *both* historically motivated CSL points (GRW, Adler) are now excluded; survival requires colored-noise/dissipative extensions or unnatural corners `[INFERENCE]`. The model class is not falsified — but the *original parameter targets* are. Full bound table and window analysis: Working note: *Track R2 — The Measurement Problem, Operationalized (2026-07 bound state)* (repo: `notes/2026-07-02-measurement-problem-operationalized.md`).

## OP-27 — Can the Born rule be derived rather than postulated?

**Area:** Foundations of QM / mathematical foundations. **Tag:** [OPEN]/[CONTESTED]. **Kind:** unsolved-but-consistent / interpretation-dependent.

**Statement.** Whether the probability rule $P = |\langle\phi|\psi\rangle|^2$ can be *derived* from the other axioms (especially unitary dynamics) or must stand as an independent postulate is unresolved.

**Why it is hard.** Gleason's theorem fixes the rule's *form* given noncontextuality and $\dim \mathcal{H} \geq 3$, but assumes the measure structure rather than deriving probability from dynamics. Deutsch–Wallace decision-theoretic derivations and Zurek's envariance attempt full derivations within Everett; whether any is non-circular (does it covertly assume a probability measure or a preferred decomposition?) is actively disputed. No clean, universally accepted derivation exists.

**Leading approaches.** Gleason's theorem; Deutsch–Wallace decision theory; Zurek envariance; reconstructions from operational axioms (OP-29).

**Cross-links.** *Chapter 7 (p. 71)*, *Chapter 13 (p. 172)*, OP-2, OP-28, OP-29.

## OP-28 — Probability in deterministic no-collapse (Everett) interpretations

**Area:** Foundations of QM. **Tag:** [OPEN]/[CONTESTED]. **Kind:** conceptual / interpretational.

**Statement.** If all branches are equally real, the meaning of "the probability of an outcome" is unclear (the *incoherence problem*: why assign probabilities to events that all occur?) as is the empirical confirmation of Born weights (the *quantitative problem*). These remain contested even among Everett's proponents.

**Why it is hard.** Deterministic universal unitarity provides branch *amplitudes* but no obvious notion of self-locating uncertainty or a non-question-begging rule connecting amplitudes to credences and to observed relative frequencies.

**Leading approaches.** Decision-theoretic (Deutsch–Wallace) and envariance arguments (OP-27); self-locating-uncertainty accounts; explicit denial that probability is fundamental.

**Cross-links.** *Chapter 7 (p. 71)*, OP-2, OP-27.

## OP-29 — Reconstructing QM from physical/operational axioms (and why complex Hilbert space?)

**Area:** Foundations of QM / mathematical foundations. **Tag:** `[OPEN]`, active and fruitful. **Kind:** unsolved-but-consistent (a "why this formalism" question).

**Statement.** Can the Hilbert-space formalism — including the use of the complex field $\mathbb{C}$ (over $\mathbb{R}$ or quaternions $\mathbb{H}$) and the tensor-product rule for composite systems — be *derived* from operational/information-theoretic postulates, and if so which axioms are physically necessary versus conventional?

**Why it is hard.** Multiple inequivalent axiom sets succeed in reconstructing QM (Hardy 2001; Chiribella–D'Ariano–Perinotti 2011; Masanes–Müller; Clifton–Bub–Halvorson "no broadcasting / no signaling / no bit commitment"), which is suggestive but leaves open which (if any) is fundamental. Recent (2021–2022) multipartite experiments argue real-Hilbert-space QM is distinguishable from complex QM under standard locality/tomographic-independence assumptions, indicating complexity is not mere convention — but the conclusion is conditional on those framework assumptions and is debated.

**Leading approaches.** Generalized-probabilistic-theories program (purification, local tomography, continuous reversibility); information-theoretic reconstructions; experimental tests of real vs complex QM.

**Cross-links.** *Chapter 7 (p. 71), Chapter 12 (p. 155), Chapter 13 (p. 172), Chapter 17 (p. 325)*, OP-27.

# C. Cosmology & the dark sector

## OP-3 — The cosmological constant problem

**Area:** QFT/gravity interface; cosmology. **Tag:** [OPEN] (arguably the deepest quantitative puzzle). **Kind:** fine-tuning / domain-clash, *not* internal inconsistency.

**Statement.** The observed dark-energy / vacuum-energy density ( $\rho_\Lambda \sim 10^{-47}\ \mathrm{GeV}^4$, $\Lambda \sim 10^{-52}\ \mathrm{m}^{-2}$) is smaller than naive QFT zero-point-energy estimates by roughly 120 orders of magnitude with a Planck cutoff (still $\sim$ 50–60 orders with an electroweak/SUSY cutoff). No mechanism is known for why the *gravitating* vacuum energy is so tiny yet nonzero. The *coincidence problem* adds that $\rho_\Lambda$ happens to be comparable to the matter density today.

**Why it is hard.** It sits exactly at the unresolved QFT/gravity boundary: it concerns how vacuum energy gravitates, and the naive estimate may be the wrong object once gravity is properly quantum (OP-1). In classical GR, $\Lambda$ is a free parameter, so this is fine-tuning, not inconsistency. No known symmetry forces cancellation to $\sim$ 120 digits while leaving the tiny observed remainder, and supersymmetry — which could cancel it — is broken at accessible scales. The "$10^{120}$" figure is regularization-dependent; see *Chapter 16 (p. 308)*.

**Leading approaches.** Anthropic/landscape selection (Weinberg's 1987 bound, which anticipated a small nonzero $\Lambda$; relies on a multiverse + a measure, OP-19); supersymmetric cancellations (constrained by no TeV superpartners); sequestering and degravitation; modified gravity / dynamical dark energy (OP-6); reformulating which vacuum-energy object sources curvature.

**Status (2026-07-02).** Weinberg's no-go (Rev. Mod. Phys. **61**, 1 (1989)) organizes the approach space: each viable mechanism must violate one of its stated assumptions — constancy of the vacuum configuration (→ relaxation), dynamical determination without external constraints (→ unimodular gravity, sequestering), finitely many fields in 4D (→ extra dimensions / towers) — or exit to anthropics. 2024–2026 verdicts: **unimodular gravity does not solve it** — $\Lambda$ becomes a radiatively stable integration constant, relocating the unexplained value into an initial condition (status report: arXiv:2207.08499, *CQG* **39**, 243001; explicit statement: arXiv:2406.12958, *PLB* **856** (2024) 138920). **Sequestering** remains self-consistent for matter loops (*PRL* **112**, 091304; *PRL* **116**, 051302) `[SPECULATIVE]`. **Running-vacuum models** fit DESI DR2 with $w_0 w_a$CDM-comparable evidence, inheriting its SN-systematics fragility (arXiv:2512.20616) `[CONTESTED]`. The regularization-aware restatement (radiative instability vs cutoff folklore; which parts are theorem-level) is at *Chapter 16 (p. 308)* §8.1.

**Cross-links.** *Chapter 11 (p. 139)*, *Chapter 8 (p. 89)*, *Chapter 10 (p. 121)*, *Chapter 16 (p. 308)*, OP-1, OP-6, OP-19.

## OP-5 — The microphysical identity of dark matter

**Area:** Cosmology / astroparticle / particle physics. **Tag:** [OPEN] (existence of missing mass [ESTABLISHED]; particle identity [OPEN]). **Kind:** established incompleteness; unknown microphysics.

**Statement.** Independent concordant evidence — galaxy rotation curves; cluster dynamics and lensing (the mass–gas offset in the Bullet Cluster); the CMB acoustic-peak structure; BBN+CMB showing $\Omega_b \ll \Omega_m$; the amplitude and timing of structure formation — requires a dominant, cold, non-baryonic, gravitating component absent from the Standard Model. Its particle/physical identity is unknown after decades of null direct, indirect, and collider searches.

**Why it is hard.** The gravitational evidence establishes that *something* beyond baryons + GR is needed but constrains the microphysics weakly: viable candidates span $\sim 90$ orders of magnitude in mass (ultralight axions to primordial black holes). WIMP direct-detection has reached the neutrino-floor regime with no signal; axion searches are ongoing. A minority view that part of the effect signals a breakdown of GR (MOND-like dynamics) is disfavored by the multi-probe concordance — especially the Bullet-

Cluster offset and the CMB peaks — but not fully excluded for all phenomena [CONTESTED].

**Leading approaches.** WIMPs (thermal relics); QCD axions / axion-like particles (also addressing strong-CP, OP-8); sterile neutrinos; primordial black holes; hidden/dark sectors with self-interactions; warm dark matter; modified gravity (MOND/TeVeS) as a contested alternative.

**Cross-links.** *Chapter 11 (p. 139)*, *Chapter 9 (p. 106)*, OP-8, OP-31.

## OP-6 — The nature of dark energy / cosmic acceleration

**Area:** Cosmology. **Tag:** [OPEN]/[CONTESTED] (acceleration itself [ESTABLISHED]). **Kind:** unsolved-but-consistent, entangled with OP-3.

**Statement.** Late-time accelerated expansion is established (Type Ia supernovae; CMB+BAO; integrated Sachs–Wolfe), implying a component with equation of state $w < -1/3$ comprising $\sim 70\%$ of the energy budget. Whether this is exactly a cosmological constant ( $w = -1$, e.g. vacuum energy), a slowly evolving dynamical field (quintessence, $w(z) \neq -1$), or an artifact of misapplying GR on the largest scales is unresolved.

**Why it is hard.** $\Lambda$ is the simplest description and fits almost all data; distinguishing $w = -1$ from a slowly varying $w(z)$ demands percent-level control of geometry and growth across cosmic time, where systematics (SN Ia calibration, photometric redshifts, baryonic feedback) limit. DESI DR2 BAO + CMB prefer evolving dark energy ($w_0 > -1$, $w_a < 0$) over ΛCDM at 3.1σ, and at 2.8–4.2σ when SNe are added, depending on sample (arXiv:2503.14738); but the preference is SN-sample-driven, drops below 2σ under low-z SN-calibration systematics (arXiv:2502.04212), and inverts to a modest Bayesian preference *for* ΛCDM from DESI+CMB alone once evidence rather than best-fit $\Delta\chi^2$ is compared (arXiv:2511.10631) — [CONTESTED]; near-decisive with the 2027 five-year DESI $w(z)$. The problem is entangled with the cosmological-constant problem (OP-3) and possible GR modification.

**Leading approaches.** $\Lambda$ / vacuum energy ($\Lambda$CDM baseline); quintessence and dynamical scalar fields; modified gravity ($f(R)$, Horndeski); inhomogeneity backreaction (contested, generally believed small, OP-30); next-generation surveys (DESI, Euclid, LSST-class) to pin down $w(z)$.

**Cross-links.** *Chapter 11 (p. 139)*, OP-3, OP-18, OP-30.

## OP-17 — Did inflation occur, and what is the inflaton? (Primordial gravitational waves)

**Area:** Cosmology / early universe. **Tag:** [OPEN]. **Kind:** unsolved-but-consistent; under-determined by data.

**Statement.** Cosmic inflation — a hypothesized early phase of slow-roll accelerated expansion — solves the horizon, flatness, and monopole problems and predicts the observed adiabatic, nearly-Gaussian, nearly-scale-invariant fluctuation spectrum (spectral index $n_s \approx 0.965$). But its occurrence is not confirmed: the inflaton is not identified with any known field, the cleanest signature (primordial tensor $B$-mode polarization, tensor-to-scalar ratio $r \neq 0$) has not been detected, and current bounds ($r \lesssim 0.03$–$0.06$) already exclude the simplest large-field models.

**Why it is hard.** The energy scales are far above any laboratory probe, so inference is indirect and degenerate: alternatives (some bouncing/ekpyrotic models) can mimic the observed scalar spectrum, so data underdetermine the mechanism. Generic inflation tends to become eternal, producing a multiverse and a measure problem that undermines predictivity (OP-19). Inflation also appears to require its own special low-entropy initial patch, so it may not fully solve the puzzle it was invoked to address (OP-10, OP-16).

**Leading approaches.** CMB $B$-mode searches for primordial gravitational waves ($r$); single-field slow-roll model building constrained by $n_s$, running, non-Gaussianity; alternatives (ekpyrotic/bouncing, string gas); connecting the inflaton to BSM physics (Higgs inflation, GUT-scale fields).

**Cross-links.** *Chapter 11 (p. 139)*, OP-10, OP-16, OP-19.

## OP-18 — The Hubble tension (and the $S_8$ tension)

**Area:** Observational cosmology. **Tag:** [CONTESTED]/[OPEN]. **Kind:** possible systematics vs. possible new physics — undecided.

**Statement.** The present expansion rate inferred from the early universe (CMB + BAO within ΛCDM, $H_0 \approx 67\text{–}68\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$) disagrees at roughly the 4–6$\sigma$ level with the late-universe distance-ladder value (Cepheid-calibrated Type Ia supernovae, $H_0 \approx 73\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$). A milder ($\sim 2\text{–}3\sigma$) tension exists in the structure-growth amplitude $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$ between early-universe extrapolations and some weak-lensing/cluster measurements. If not systematics, these are the leading empirical cracks in the ΛCDM concordance model.

**Why it is hard.** The discrepancy persists across multiple independent methods on each side, making a single overlooked systematic less likely but not excluded. Early-time solutions that reduce the sound horizon $r_d$ (early dark energy, extra relativistic species $N_{\text{eff}}$, modified recombination) tend to worsen other fits (e.g. $S_8$) or require tuning, while late-time-only solutions are tightly constrained by BAO. As of the knowledge horizon it is genuinely unresolved.

**Leading approaches.** Distance-ladder cross-checks (Cepheids, TRGB, surface-brightness fluctuations) and independent $H_0$ probes (lensing time delays, gravitational-wave standard sirens); early dark energy and extra dark radiation; modified recombination; interacting/evolving dark sectors; systematics audits of both pipelines.

**Cross-links.** *Chapter 11 (p. 139)*, *Chapter 14 (p. 191)*, OP-6.

## OP-19 — Eternal inflation, the multiverse, and the measure problem

**Area:** Cosmology / foundations. **Tag:** [SPECULATIVE]. **Kind:** borders the limits of empirical science.

**Statement.** If inflation is generically eternal — quantum fluctuations keep some regions inflating, nucleating infinitely many pocket universes — then computing the probability of any observation requires regulating ratios of infinite quantities. Different regularization schemes ("measures": proper-time, scale-factor, causal-patch, etc.) yield different and sometimes pathological predictions (Boltzmann brains, the youngness paradox), so there is no agreed way to extract well-defined probabilities.

**Why it is hard.** The infinities are physical to the scenario, not a calculational artifact, and no principle selects a unique pathology-free measure. If observers are spread across an unobservable multiverse, standard inference — and even the testability of the proposal — is challenged. It is entangled with anthropic explanations of fine-tuning (e.g. Λ, OP-3) and with the string landscape, neither independently confirmed.

**Leading approaches.** Proposed measures (causal-patch / census-taker, scale-factor cutoff, stationary measure) tested against Boltzmann-brain domination; string-landscape vacuum ensembles; anthropic reasoning constrained by predictivity; debate over whether the multiverse is falsifiable.

**Cross-links.** *Chapter 11 (p. 139)*, *Chapter 2 (p. 8)*, OP-3, OP-17, OP-35 (Boltzmann brains relate to the arrow of time).

## OP-30 — Averaging / backreaction: emergence of the smooth FLRW background

**Area:** Cosmology / GR. **Tag:** [CONTESTED]. **Kind:** unsolved-but-consistent (magnitude debated).

**Statement.** How does the smooth FLRW background emerge from, and how is it affected by, the inhomogeneous real universe? Because GR is nonlinear, the average of the inhomogeneous metric need not satisfy the Einstein equations of the averaged matter (the "fitting problem"; Buchert's averaged equations).

**Why it is hard.** A rigorous, gauge-invariant averaging scheme for tensorial quantities on a Lorentzian manifold is itself nontrivial, and estimating the effective backreaction term requires controlling structure across scales. Most experts judge backreaction far too small to mimic dark energy [INFERENCE], but a rigorous derivation of the averaged dynamics is incomplete and the magnitude is debated.

**Leading approaches.** Buchert averaging formalism; relativistic $N$-body / numerical-relativity cosmology; perturbative estimates of the backreaction parameter.

**Cross-links.** *Chapter 11 (p. 139)*, *Chapter 10 (p. 121)*, OP-6.

## OP-31 — Singularities, cosmic censorship, and the fate of the Big Bang / black-hole interiors

**Area:** General relativity. **Tag:** [OPEN]/[CONTESTED]. **Kind:** unsolved-but-consistent (some fine-tuned counterexamples known).

**Statement.** The Penrose–Hawking singularity theorems guarantee geodesic incompleteness under reasonable energy/causality conditions but say nothing about *resolution*. Whether quantum gravity replaces the Big Bang and black-hole singularities with a bounce or something else is unknown. Separately, the weak and strong cosmic-censorship conjectures (no generic naked singularities; deterministic evolution) lack a general proof, and fine-tuned counterexamples (critical collapse; certain charged/rotating interiors; recent $\Lambda < 0$ and near-extremal results) exist, though censorship is widely believed to hold "generically".

**Why it is hard.** Singularity resolution is a quantum-gravity question with no testable handle (OP-1). Cosmic censorship is a hard global problem in nonlinear PDE / Lorentzian geometry; a precise, generically-true formulation is still being sought, and counterexamples show the naive statements are false.

**Leading approaches.** Quantum-gravity models of bounces (LQC, string-theoretic); rigorous study of strong censorship in Kerr–Newman–(A)dS interiors; critical-collapse analysis; the nonlinear-stability program (OP-13 family).

**Cross-links.** *Chapter 10 (p. 121)*, *Chapter 11 (p. 139)*, OP-1, OP-10.

# D. Particle physics & naturalness

## OP-4 — The hierarchy / naturalness problem

**Area:** Particle physics / Standard Model. **Tag:** `[CONTESTED]`. **Kind:** aesthetic/naturalness expectation, *not* a logical inconsistency or domain breakdown.

**Statement.** The Higgs mass-squared receives radiative corrections quadratically sensitive to the cutoff, $\delta m_h^2 \sim \Lambda^2/(16\pi^2)$. With $\Lambda$ near $M_{\mathrm{Pl}}$, obtaining $m_h = 125$ GeV requires a cancellation between bare term and corrections fine-tuned to roughly one part in $10^{34}$. Equivalently: why is the electroweak scale ($\sim 246$ GeV) so far below any high fundamental cutoff, when no symmetry protects an elementary scalar's mass?

**Why it is hard.** The Standard Model is consistent and predictive for any value (the cutoff sensitivity is absorbed into a counterterm), so the "problem" is a naturalness expectation — which is itself why it is contested. The simplest natural solutions (low-scale SUSY, technicolor, lightest composite-Higgs scenarios) predicted new particles near the TeV scale that the LHC has not found, intensifying debate over whether naturalness is the right guiding prior at all.

**Leading approaches.** Supersymmetry (cancels the quadratic divergence; simplest natural versions disfavored by LHC); composite / pseudo-Goldstone Higgs and technicolor; extra dimensions (large or warped) lowering the fundamental scale; anthropic/landscape and relaxion (cosmological relaxation); rejecting naturalness as a valid prior.

**Cross-links.** *Chapter 9 (p. 106)*, *Chapter 20 (p. 458)*, *Chapter 14 (p. 191)*, OP-3, OP-14.

## OP-7 — Baryon asymmetry of the universe (baryogenesis)

**Area:** Particle physics / early-universe cosmology. **Tag:** `[OPEN]` (the asymmetry and SM's inability to produce it are `[ESTABLISHED]`).

**Kind:** established incompleteness; mechanism unknown.

**Statement.** The observed baryon-to-photon ratio ($\eta \sim 6 \times 10^{-10}$) cannot be generated by the Standard Model. Although the SM in principle satisfies Sakharov's three conditions — baryon-number violation (nonperturbative electroweak sphalerons), C and CP violation, and departure from equilibrium — the CKM CP violation (Jarlskog invariant $J \sim 3 \times 10^{-5}$) is far too small, and for $m_h =$ 125 GeV the electroweak transition is a smooth crossover, not first-order, so no net asymmetry survives.

**Why it is hard.** It requires beyond-SM physics: a new CP-violation source and/or new out-of-equilibrium dynamics, operating in the early universe (pre-BBN, $t < 1$ s) with no direct empirical probes. Leading mechanisms tie the asymmetry to very high mass scales (heavy Majorana neutrinos) hard to test directly.

**Leading approaches.** Leptogenesis via heavy Majorana neutrino decays (linked to the seesaw, OP-9); electroweak baryogenesis with an extended Higgs sector (first-order transition + new CP violation); Affleck–Dine baryogenesis; GUT baryogenesis; searches for new CP violation (electric dipole moments; leptonic phase $\delta_{CP}$).

**Cross-links.** *Chapter 9 (p. 106)*, *Chapter 11 (p. 139)*, OP-9, OP-20.

## OP-8 — The strong-CP problem

**Area:** Particle physics / QCD. **Tag:** [OPEN]. **Kind:** fine-tuning puzzle pointing to missing physics, not an inconsistency.

**Statement.** The QCD Lagrangian admits a CP-violating term $\bar{\theta}\,\frac{g_s^2}{32\pi^2} G^a_{\mu\nu}\tilde{G}^{a\,\mu\nu}$. Naturalness expects $\bar{\theta} \sim \mathcal{O}(1)$, but the bound on the neutron electric dipole moment forces $\bar{\theta} \lesssim 10^{-10}$. Nothing within the SM explains why this physical, CP-violating angle is so extraordinarily small.

**Why it is hard.** $\bar{\theta}$ is the sum of the bare QCD angle and the phase of the quark mass matrix, mixing strong and electroweak sectors; no SM symmetry sets it to zero (a massless up quark would, but is disfavored by lattice QCD). The theory is consistent for any $\bar{\theta}$. The most elegant fix predicts a new light particle (the axion) not yet detected.

**Leading approaches.** Peccei–Quinn symmetry with a dynamical axion (also a dark-matter candidate, OP-5; searched by ADMX and others); massless-up-quark solution (disfavored by lattice); Nelson–Barr / spontaneous-CP-violation models; anthropic or contingent-smallness arguments.

**Cross-links.** *Chapter 9 (p. 106)*, OP-5, OP-12.

## OP-9 — Neutrino mass mechanism: Dirac vs Majorana

**Area:** Particle physics / neutrino physics. **Tag:** [OPEN] (nonzero mass [ESTABLISHED]). **Kind:** unsolved-but-consistent; the unique confirmed BSM signal.

**Statement.** Neutrino oscillations prove nonzero mass and mixing (PMNS matrix) — the only laboratory-confirmed breakdown of the minimal SM. It is unknown whether masses are Dirac (requiring right-handed neutrinos and conserved lepton number, with tiny Yukawas) or Majorana (the neutrino is its own antiparticle, lepton number violated, lightness explained by the seesaw). The absolute scale, the ordering (normal vs inverted), and the leptonic CP phase are not fully determined.

**Why it is hard.** The decisive Majorana signature — neutrinoless double-beta decay — has not been observed, and its rate, if Majorana, may lie below current sensitivity. Majorana masses are favored theoretically (seesaw naturalness; link to leptogenesis, OP-7) but not established. The absolute scale requires combining beta-decay endpoint, cosmology, and oscillation data, each with its own systematics.

**Leading approaches.** Neutrinoless double-beta-decay experiments (the key Majorana test); seesaw mechanisms (types I/II/III); the Weinberg dimension-5 operator as leading effective description; beta-decay endpoint (KATRIN-class) and cosmological $\sum m_\nu$ bounds; long-baseline oscillation experiments for ordering and $\delta_{CP}$.

**Cross-links.** *Chapter 9 (p. 106)*, OP-7, OP-14.

## OP-14 — Origin of the Standard Model parameters (the flavor puzzle)

**Area:** Particle physics / Standard Model. **Tag:** [OPEN]. **Kind:** unsolved-but-consistent; the SM's most "descriptive, not explanatory" feature.

**Statement.** The SM has $\sim 19$ free parameters ($\sim 26$–$28$ with neutrino masses/mixings) fixed only by experiment: three gauge couplings, the Higgs mass and vacuum expectation value, nine charged-fermion masses, the CKM (and PMNS) mixing angles and CP phases, and the QCD $\theta$ angle. The framework offers no explanation for the enormous hierarchies (top/electron Yukawa $\sim 10^5$), why there are exactly three generations, the specific gauge group $SU(3) \times SU(2) \times U(1)$, or charge quantization.

**Why it is hard.** The SM accommodates these values but is purely descriptive about them. The patterns (mass hierarchies; near-diagonal CKM vs large PMNS angles) strongly suggest an underlying flavor theory, but no proposed mechanism is confirmed and the relevant scales may be far above reach. Some structure is fixed by quantum consistency (anomaly cancellation fixes hypercharges), and gauge unification hints at a GUT embedding (OP-20), but the deep origin is unknown and may be partly contingent/anthropic.

**Leading approaches.** Grand unified theories (SU(5), SO(10), Pati–Salam) explaining charge quantization and near-unification; flavor symmetries and the Froggatt–Nielsen mechanism; extra-dimensional wavefunction localization (split fermions); anthropic/landscape selection; SMEFT parametrization to probe deviations.

**Cross-links.** *Chapter 9 (p. 106)*, *Chapter 16 (p. 308)*, *Chapter 20 (p. 458)*, OP-9, OP-20.

## OP-20 — Gauge-coupling unification, proton decay, and grand unification

**Area:** Particle physics / BSM. **Tag:** [OPEN]/[INFERENCE]. **Kind:** unsolved-but-consistent; an inference, not established fact.

**Statement.** The three SM gauge couplings, run to high energies, nearly meet near $\sim 10^{16}$ GeV (more precisely in supersymmetric extensions), suggesting an underlying grand unified gauge group (SU(5), SO(10), Pati–Salam) explaining charge quantization and the fermion-representation pattern. Such theories generically predict proton decay, unobserved to date (partial-lifetime bounds exceed $\sim 10^{34}$ yr), excluding minimal $SU(5)$ and constraining the rest.

**Why it is hard.** The unification scale is $\sim 13$ orders of magnitude above collider energies, so the primary handle is the rare-decay rate of the proton, requiring enormous detectors and exposure. Precise unification depends on the unknown particle content between the electroweak and GUT scales (e.g. whether SUSY exists), so the plain-SM near-miss is suggestive but inconclusive. Whether nature is grand-unified remains an inference.

**Leading approaches.** Large-volume proton-decay searches (water-Cherenkov, liquid-argon); supersymmetric GUTs improving unification; SO(10) accommodating right-handed neutrinos and the seesaw (OP-9); precision running and threshold-correction analyses.

**Cross-links.** *Chapter 9 (p. 106)*, *Chapter 18 (p. 349)*, OP-9, OP-14.

## OP-21 — Persistent BSM experimental anomalies (muon $g-2$, flavor)

**Area:** Particle physics (experiment–theory). **Tag:** [CONTESTED]/[OPEN]. **Kind:** watched anomalies, not established breakdowns.

**Statement.** Several measured quantities sit in tension with SM predictions without reaching unambiguous discovery significance. **Update (2025):** the muon anomalous magnetic moment $a_\mu =$

$(g-2)/2$ — long the flagship anomaly — is now **largely SM-consistent**: the Theory Initiative white paper WP2025 (arXiv:2505.21476) adopts a lattice-QCD hadronic-vacuum-polarization average, giving $a_\mu^{\mathrm{SM}} = 116\,592\,033(62) \times 10^{-11}$ and $\Delta a_\mu = 38(63) \times 10^{-11}$ (no tension); the residual issue is an *internal* lattice-vs-$e^+e^-$($R$-ratio, CMD-3) HVP discrepancy, a theory/systematics question rather than a BSM signal `[ESTABLISHED, 2025]`. Certain $B$-meson decay observables (some $b \to s\mu\mu$ angular and branching measurements) remain in mild tension, while the lepton-universality ratios $R_K$, $R_{K^*}$ largely moved back toward SM consistency with updated data.

**Why it is hard.** For $g-2$, the dominant theory uncertainty is the hadronic vacuum polarization, where the long-standing data-driven ($e^+e^-$) evaluation and recent lattice-QCD results (e.g. BMW) *disagree* — so the very size of any SM deviation is contested and may be a theory/systematics issue rather than new physics. For flavor observables, hadronic form-factor uncertainties and the statistical evolution of measurements complicate interpretation. None has crossed the threshold of a clean, corroborated BSM discovery.

**Leading approaches.** Improved $g-2$ measurement (Fermilab) and reconciliation of lattice vs data-driven HVP; updated LHCb and Belle II flavor measurements and global fits; BSM model-building (leptoquarks, $Z'$) tailored to surviving anomalies; SMEFT global analyses.

**Cross-links.** *Chapter 9 (p. 106)*, *Chapter 14 (p. 191)*, *Chapter 21 (p. 497)*.

## OP-22 — (Meta)stability of the electroweak vacuum

**Area:** Particle physics / Standard Model. **Tag:** `[CONTESTED]`. **Kind:** unresolved structural question, *not* an inconsistency.

**Statement.** RG running of the Higgs quartic coupling $\lambda$, using the measured top-quark and Higgs masses, drives $\lambda$ slightly negative at high energies (around $\sim 10^{10}$–$10^{11}$ GeV), implying the electroweak vacuum is metastable (extremely long-lived) rather than absolutely stable. Whether it is stable, metastable, or unstable depends sensitively on inputs and possible new physics.

**Why it is hard.** The verdict is exquisitely sensitive to the precise top-quark mass and strong coupling $\alpha_s$, whose central values place the SM near the stability/metastability boundary, so the numbers are contested. The high-scale extrapolation assumes no new physics over many decades in energy; any BSM field coupling to the Higgs can change the result. A metastable vacuum with lifetime vastly exceeding the age of the universe is empirically fine — so this is a structural question, not a paradox.

**Leading approaches.** Precision top / Higgs mass and $\alpha_s$ feeding higher-order RG; two- and three-loop effective-potential and tunneling-rate computations; BSM scalars that stabilize the potential; cosmological consistency during inflation/reheating.

**Cross-links.** *Chapter 9 (p. 106)*, *Chapter 16 (p. 308)*, OP-4, OP-17.

## OP-32 — Confinement in QCD (analytic proof) & the mass gap

**Area:** Particle physics / QCD. **Tag:** `[OPEN]` (empirically certain; analytic proof missing). **Kind:** internal hard problem, not an inconsistency. *Closely tied to OP-11/OP-36.*

**Statement.** QCD is believed to confine color: quarks and gluons do not appear as free asymptotic states, only color-singlet hadrons do, and the static quark–antiquark potential rises linearly at large separation, $V(r) \sim \sigma r$. This is overwhelmingly supported by experiment and lattice QCD, but there is no analytic, first-principles continuum proof of confinement (tied to the Yang–Mills mass-gap problem, OP-11).

**Why it is hard.** Confinement is intrinsically nonperturbative and strong-coupling: perturbation theory in $\alpha_s$ is at best asymptotic and breaks down below $\sim 1$ GeV, where the quark–gluon variables are the wrong description. The mechanism (dual superconductivity / monopole condensation, center vortices) is not rigorously established, the continuum limit has not been proven to exist with a gap, and there is no analytic order parameter universally valid for full QCD with dynamical quarks.

**Leading approaches.** Lattice QCD (linear potential, string tension, hadron spectrum); dual-superconductor / monopole-condensation and center-vortex pictures; Schwinger–Dyson / functional-RG studies of gluon and ghost propagators; connection to the Clay Yang–Mills program (OP-11).

**Cross-links.** *Chapter 9 (p. 106)*, *Chapter 8 (p. 89)*, OP-11, OP-36.

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## E. Arrow of time & statistical foundations

### OP-10 — The arrow of time and the low-entropy past (the Past Hypothesis)

**Area:** Statistical mechanics / thermodynamics / cosmology. **Tag:** `[OPEN]`. **Kind:** unsolved-but-consistent; requires a time-asymmetric boundary condition.

**Statement.** Microscopic dynamics are (CPT) time-symmetric, so the thermodynamic arrow — entropy increase, the second law — cannot follow from the dynamical laws alone; it requires a special, extraordinarily low-entropy past boundary condition (the *Past Hypothesis*). Crucially, the early universe's near-perfect thermal uniformity is *high* thermal entropy but *very low gravitational* entropy (smoothness is special when gravity dominates). Why the initial state was so special is unexplained.

**Why it is hard.** Reversible mechanics plus monotone entropy increase is impossible without a time-asymmetric input (the H-theorem smuggles in the Stosszahlansatz; Loschmidt's reversibility and Zermelo's recurrence objections). The needed low-entropy past is a *posited* boundary condition, not a derivation, and its origin pushes into cosmology and quantum gravity (e.g. quantifying gravitational entropy via the Weyl curvature). Inflation arguably relocates rather than removes the puzzle (OP-17).

**Leading approaches.** Boltzmann coarse-graining + typicality + the Past Hypothesis as an explicit posit; Penrose's Weyl-curvature hypothesis (near-zero initial Weyl tensor) [SPECULATIVE]; inflationary smoothing [CONTESTED] as a complete resolution; proposals tying the arrow to cosmological initial conditions / quantum gravity.

**Cross-links.** *Chapter 6 (p. 56)*, *Chapter 5 (p. 40)*, *Chapter 11 (p. 139)*, OP-16, OP-17, OP-33.

## OP-16 — Foundations of statistical mechanics: why ensemble averages predict individual systems, and the origin of thermalization

**Area:** Statistical mechanics / foundations. **Tag:** [OPEN]/[CONTESTED]. **Kind:** unsolved-but-consistent (the practice is undisputed; the principled justification is not).

**Statement.** There is no single, universally accepted first-principles derivation of why equilibrium ensemble averages correctly predict the behavior of an individual macroscopic system, nor a general proof that realistic isolated systems thermalize. The microcanonical equal-a-priori-probability postulate and the canonical $e^{-\beta H}$ form work spectacularly but rest on assumptions (ergodicity, typicality, or — quantum-mechanically — the Eigenstate Thermalization Hypothesis, ETH) that are not theorems for generic interacting Hamiltonians.

**Why it is hard.** Strict ergodicity is neither necessary nor generic (KAM tori survive in typical Hamiltonian systems, OP-39), so justification shifts to high-dimensional typicality / measure concentration, which lacks a fully general proof. Quantum-mechanically, ETH (Deutsch, Srednicki) is strongly supported numerically and by random-matrix arguments but is unproven for any realistic interacting model and provably fails for integrable systems (which relax to a Generalized Gibbs Ensemble) and many-body-localized systems (which never thermalize, OP-34). Deriving hydrodynamics/transport rigorously from reversible microdynamics (Hilbert's sixth problem) is also open (OP-35).

**Leading approaches.** Ergodic theory, mixing, dynamical-systems methods; typicality / canonical-typicality and concentration-of-measure; ETH for closed quantum systems; GGE for integrable systems; study of MBL and prethermalization; large-deviation theory and rigorous hydrodynamic-limit programs.

**Cross-links.** *Chapter 6 (p. 56)*, *Chapter 5 (p. 40)*, OP-10, OP-34, OP-35, OP-39.

## OP-33 — Statistical-mechanical microstates of the Bekenstein–Hawking entropy

**Area:** Thermodynamics / quantum gravity. **Tag:** [OPEN] (counted [ESTABLISHED] for special cases). **Kind:** unsolved-but-consistent.

**Statement.** What microstates does the Bekenstein–Hawking entropy $S_{BH} = k_B c^3 A/(4G\hbar)$ count for a realistic (e.g. Schwarzschild) black hole? String theory reproduces $S = A/4$ by counting D-brane microstates for certain supersymmetric/extremal black holes (Strominger–Vafa 1996) [ESTABLISHED within that framework], and AdS/CFT gives a holographic account [INFERENCE], but a universally accepted microscopic derivation for generic astrophysical black holes is lacking.

**Why it is hard.** The controlled microstate counts exploit supersymmetry/extremality and BPS protection that real astrophysical black holes lack; extending to non-extremal, dynamically-formed, asymptotically-flat black holes leaves the strong-gravity regime where the correct degrees of freedom are unknown (OP-1, OP-15).

**Leading approaches.** D-brane microstate counting (extremal/BPS); fuzzballs; AdS/CFT and holographic entropy; loop-quantum-gravity horizon-state counting; the crossed-product / generalized-entropy program (OP-37).

**Cross-links.** *Chapter 5 (p. 40)*, *Chapter 12 (p. 155)*, OP-15, OP-23, OP-37.

## OP-34 — Rigorous ETH and the boundary between thermalizing and non-thermalizing systems

**Area:** Statistical mechanics. **Tag:** [OPEN] (MBL stability [CONTESTED]). **Kind:** unsolved-but-consistent.

**Statement.** Prove ETH for a physically realistic non-integrable Hamiltonian, and sharply characterize which systems thermalize. ETH is strongly supported numerically and by random-matrix arguments but unproven for any realistic interacting model. The boundary between thermalizing, integrable (Generalized Gibbs Ensemble), prethermal, quantum-many-body-scarred, and many-body-localized (MBL) behavior — and even whether MBL is a true stable phase in the thermodynamic limit in $d \geq 2$ — is under active investigation.

**Why it is hard.** ETH is a statement about *all* eigenstates of a generic many-body Hamiltonian, which is analytically intractable; numerics reach only small sizes. MBL stability hinges on rare-region (avalanche) effects that are exponentially hard to probe numerically, leaving the asymptotic phase structure genuinely contested.

**Leading approaches.** Numerical diagonalization and tensor-network studies; random-matrix and ETH ansatz analyses; avalanche-instability theory for MBL; rigorous results in special models.

**Cross-links.** *Chapter 6 (p. 56)*, *Chapter 7 (p. 71)*, OP-16.

## OP-35 — Nonequilibrium foundations: thermalization derivations, transport, and a far-from-equilibrium variational principle

**Area:** Statistical mechanics / thermodynamics. **Tag:** [OPEN]. **Kind:** unsolved-but-consistent; major mathematical-physics gap.

**Statement.** Two linked open problems: (i) there is no general, predictive variational/ensemble principle for nonequilibrium *steady states* (an analogue of MaxEnt far from equilibrium); proposed minimum/maximum entropy-production principles are of limited or contested validity. (ii) Deriving hydrodynamic (Navier–Stokes / Boltzmann) and transport equations *rigorously* from microscopic reversible dynamics for interacting systems remains open: Lanford's theorem derives the Boltzmann equation only for short times in the dilute limit, and a full rigorous hydrodynamic limit with finite transport coefficients is unproven (related to Hilbert's sixth problem).

**Why it is hard.** Far-from-equilibrium systems lack a universal organizing principle comparable to equilibrium MaxEnt; the rigorous derivation of irreversible macroscopic equations from reversible microdynamics confronts the same time-asymmetry issue as OP-10, plus formidable control of long-time, many-body correlations.

**Leading approaches.** Large-deviation theory; Gallavotti–Cohen fluctuation theorem; macroscopic fluctuation theory; stochastic thermodynamics (rigorous for Markov dynamics); Lanford-type and hydrodynamic-limit programs.

**Cross-links.** *Chapter 6 (p. 56)*, *Chapter 5 (p. 40)*, OP-10, OP-16, OP-12.

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# F. Mathematical foundations of QFT & dynamics

## OP-11 — Rigorous non-perturbative foundation of interacting QFT; Yang–Mills mass gap (Clay Millennium Problem)

**Area:** Mathematical physics / QFT foundations. **Tag:** [OPEN]. **Kind:** unsolved-but-consistent; flagship rigour gap.

**Statement.** No physically realistic interacting QFT in 4 spacetime dimensions (Yang–Mills, full QCD, the electroweak theory) has been constructed to full mathematical rigour satisfying the Wightman / Osterwalder–Schrader axioms. The flagship instance is the Clay problem: prove that 4D quantum Yang–Mills theory exists as a well-defined QFT and has a mass gap $\Delta > 0$ (spectrum in $\{0\} \cup [\Delta, \infty)$).

**Why it is hard.** The Lorentzian path-integral "measure" $\mathcal{D}\phi\, e^{iS}$ is not a genuine measure (no $\sigma$-additive translation-invariant measure exists in infinite dimensions; $e^{iS}$ is not absolutely convergent); rigorous control requires Euclidean reflection-positivity and analytic continuation. Constructive QFT succeeds only in $d = 2, 3$ (super-renormalizable models). Perturbation series are at best asymptotic; $\phi^4_4$ is provably trivial and QED has a Landau pole, showing some renormalizable theories survive only as cutoff effective theories. Confinement and the mass gap are seen on the lattice and physically certain (OP-32), but a continuum existence-and-gap proof is missing. Local algebras are type $\mathrm{III}_1$ factors, so naive subsystem/tensor-factor intuition fails (OP-36).

**Leading approaches.** Constructive QFT (rigorous $d = 2, 3$); algebraic / axiomatic QFT (Wightman, Haag–Kastler nets, Osterwalder–Schrader reconstruction); lattice gauge theory (strong numerical evidence for gap and confinement); conformal bootstrap for fixed-point theories; asymptotic-safety / functional-RG methods.

**Cross-links.** *Chapter 8 (p. 89)*, *Chapter 13 (p. 172)*, OP-32, OP-36, OP-37.

## OP-12 — Turbulence and global regularity of Navier–Stokes / Euler

**Area:** Classical mechanics / fluid dynamics / mathematical physics. **Tag:** `[OPEN]`. **Kind:** unsolved-but-consistent; Clay Millennium Problem.

**Statement.** There is no complete predictive theory of fully developed turbulence — the statistical and dynamical behavior of high-Reynolds-number flow, including the energy cascade, intermittency, and anomalous scaling of structure functions (corrections to Kolmogorov 1941 scaling). Relatedly, it is unproven whether smooth solutions of the 3D incompressible Navier–Stokes equations exist globally and stay regular for all time, or can develop finite-time singularities (a Clay problem); 3D Euler blow-up is similarly open.

**Why it is hard.** Turbulence is strongly nonlinear, multi-scale, far-from-equilibrium, with no small parameter and no separation into tractable modes; the nonlinear advection term couples all scales (the *closure problem*: every moment equation depends on higher moments). Finite-dimensional theorems (KAM, Liouville–Arnold) do not transfer to the infinite-dimensional PDE setting, and the regularity question is a genuine open problem in nonlinear PDE analysis. No first-principles derivation of the observed intermittency exponents exists.

**Leading approaches.** Kolmogorov 1941 phenomenology and refined-similarity / multifractal models; direct numerical simulation and large-eddy simulation; renormalization-group / field-theoretic methods; rigorous PDE analysis (partial regularity à la Caffarelli–Kohn–Nirenberg; conditional Beale–Kato–Majda criteria); Onsager's conjecture and convex-integration results for Euler.

**Cross-links.** *Chapter 4 (p. 25)*, *Chapter 13 (p. 172)*, OP-35.

## OP-36 — Local algebras, type III factors, and entanglement entropy in QFT/gravity

**Area:** Mathematical physics / QFT foundations / information. **Tag:** `[OPEN]`. **Kind:** unsolved-but-consistent (conceptual + technical).

**Statement.** What is the right notion of local subsystems and entanglement entropy in QFT, and in gravity? The entanglement entropy of a region is UV-divergent, and the local von Neumann algebras of relativistic QFT are type $\mathrm{III}_1$ — there are no density matrices and no clean tensor factorization across a boundary, so the naive "entanglement entropy of a region" is ill-defined. With gravity the question is sharper still (it underlies "spacetime from entanglement", OP-23/OP-24).

**Why it is hard.** Type III algebras lack a trace and minimal projections; defining finite physical quantities (mutual information, relative entropy, modular Hamiltonians via Tomita–Takesaki theory) is mathematically clean but a complete physical picture — especially the area law and holographic finiteness with dynamical gravity — is still developing.

**Leading approaches.** Tomita–Takesaki modular theory; relative-entropy and mutual-information regularizations; the Bekenstein/area-law and holographic-entanglement programs; crossed-product constructions (OP-37).

**Cross-links.** *Chapter 8 (p. 89)*, *Chapter 12 (p. 155)*, *Chapter 13 (p. 172)*, OP-11, OP-23, OP-37.

## OP-37 — A well-defined entanglement entropy for (quantum) gravity

**Area:** Quantum gravity / information / mathematical physics. **Tag:** `[OPEN]`. **Kind:** unsolved-but-consistent.

**Statement.** Given the type III algebras of QFT (OP-36) and the absence of a fundamental tensor factorization, what is the correct definition of entanglement entropy *in gravity*? Recent crossed-product and "algebra of observables in gravity" constructions (large-$N$, with gravitational dressing yielding type II algebras and well-defined traces) are promising but new and incomplete; the conceptual status of "the entanglement entropy of a region" in full quantum gravity is unsettled.

**Why it is hard.** Promoting the boundary/observer to a dynamical (gravitating) part of the system changes the algebra type (III $\to$ II) and renders certain entropies finite, but a background-independent, non-perturbative formulation valid beyond the semiclassical large-$N$ regime does not exist.

**Leading approaches.** Crossed-product / modular-flow constructions; generalized entropy $S_{\text{gen}}$ and quantum extremal surfaces; relating area-extensive (holographic, finite) to volume-extensive (local-QFT) degree-of-freedom counts (species bound, holographic finiteness).

**Cross-links.** *Chapter 12 (p. 155)*, *Chapter 8 (p. 89)*, OP-15, OP-23, OP-33, OP-36.

## G. Other

### OP-13 — Classical-mechanics internal open problems (N-body singularities, Arnold diffusion, Norton's dome, inertia/Mach)

**Area:** Classical mechanics. **Tag:** mixed (see below). **Kind:** mostly unsolved-but-consistent; some [CONTESTED].

**Statement & status (a bundle of genuinely internal classical problems).**

- **Painlevé's conjecture / N-body singularities** [OPEN]: non-collision singularities exist (Xia 1992 for $N \geq 5$; Gerver), but the complete classification, the $N = 4$ case, and whether singular initial data has measure zero are unresolved.
- **Arnold diffusion** [OPEN]/partially resolved: for $n \geq 3$ degrees of freedom KAM tori do not separate phase space, so trajectories can drift across resonance webs. Existence is established in examples and some generic settings (Mather; Cheng–Yan), but a complete theory of its prevalence and speed is missing.
- **Nonlinear stability of physical GR solutions** [OPEN]/partial: stability of Minkowski space is [ESTABLISHED] (Christodoulou–Klainerman 1993), with strong recent progress on (sub-extremal) Kerr black-hole stability; full nonlinear Kerr stability and generic dynamical collapse are still being established. *(Listed here for the dynamical-systems content; physically it belongs with OP-31.)*
- **Norton's dome / indeterminism in classical mechanics** [CONTESTED]: whether non-Lipschitz forces and idealized boundary conditions model physical reality, or are unphysical idealizations, is a live philosophy-of-physics debate.
- **Origin of inertia / Mach's principle** [OPEN]/[CONTESTED]: why inertial frames coincide with the rest frame of distant matter,

and whether inertia is relational, is a foundational question mechanics raises but cannot answer internally (GR's frame-dragging is only partially Machian).

**Why they are hard.** The N-body and Arnold-diffusion problems are infinite-codimension / high-dimensional dynamical-systems questions with no general technique; the PDE-stability problems are nonlinear and global; the indeterminism and Mach questions are partly conceptual and turn on what counts as an admissible physical model.

**Cross-links.** *Chapter 4 (p. 25), Chapter 10 (p. 121), Chapter 13 (p. 172)*, OP-12, OP-31, OP-10.

## OP-38 — High-temperature (unconventional) superconductivity

**Area:** Condensed matter / many-body physics. **Tag:** [OPEN]. **Kind:** unsolved-but-consistent; an emergent-physics hard problem.

**Statement.** There is no accepted, predictive microscopic theory of the mechanism behind high-$T_c$ superconductivity in the cuprates (and related unconventional superconductors), where pairing occurs far above what conventional phonon-mediated BCS theory plausibly explains, with $d$-wave order parameters and an anomalous "strange-metal" / pseudogap normal state.

**Why it is hard.** These are strongly correlated electron systems with Coulomb interaction comparable to the kinetic energy — no small parameter for a controlled perturbative (BCS-style) treatment. The relevant low-energy models (e.g. the 2D Hubbard model) are not exactly solvable and are hard numerically (the fermion sign problem obstructs quantum Monte Carlo). The intertwining of superconductivity with antiferromagnetism, charge order, and the pseudogap, plus the non-Fermi-liquid strange-metal phase, means the *normal* state itself is not understood.

**Leading approaches.** Hubbard / $t$–$J$ studies via DMFT, DMRG, tensor networks; spin-fluctuation / resonating-valence-bond pairing theories; quantum-critical-point and strange-metal scenarios (including holographic / AdS-CMT models); cluster and embedding methods to partially bypass the sign problem; materials discovery (high-pressure hydrides as a separate, more BCS-like high-$T_c$ route).

**Cross-links.** *Chapter 6 (p. 56)*, *Chapter 7 (p. 71)*, OP-16.

## OP-39 — Foundational meta-problems: classical limit / quantum chaos, the glass transition, entropy definitions, hypercomputation, and "unreasonable effectiveness"

**Area:** Cross-domain foundations. **Tag:** mixed (see below). **Kind:** mostly unsolved-but-consistent; some [CONTESTED]; one explicitly meta/philosophical.

**Statement & status (a bundle of distinct foundational meta-problems, kept together to avoid ID inflation).**

- **Rigorous classical limit & quantum chaos** [OPEN]/[CONTESTED]: how definite classical trajectories and the macroscopic absence of superpositions emerge (decoherence + the measurement problem, OP-2) straddles classical and quantum; the $\hbar \to 0$ limit is singular and nonuniform (Ehrenfest-time breakdown), and quantum chaos lacks a complete theory.
- **First-principles theory of the glass transition** [OPEN]/[CONTESTED]: whether there is an underlying ideal thermodynamic transition (Kauzmann temperature; random first-order transition theory from the mean-field replica/cavity solution) or the glass transition is purely kinetic is disputed; mean-field (infinite-$d$) results are rigorous but their $d = 3$ relevance is contested.
- **Boltzmann (surface) vs Gibbs (volume) entropy and negative temperatures** [CONTESTED]: which entropy is the consistent thermodynamic entropy for small/bounded-spectrum systems (the Dunkel–Hilbert debate, post-2013) is unresolved and matters operationally at the nanoscale.
- **Physical Church–Turing thesis / hypercomputation** [OPEN]/[CONTESTED]: whether Malament–Hogarth spacetimes, real-

number/analog computation, or exotic GR setups permit hypercomputation, or a physical Church–Turing thesis holds as a law of nature, has no consensus and no decisive theorem either way; all proposed loopholes face severe physical-plausibility objections.

- **Foundations of quantum thermodynamics** `[OPEN]`: which operational definition of "work" is physically privileged at the quantum scale, and how coherence/entanglement function as thermodynamic resources, is unresolved (resource-theoretic and two-point-measurement frameworks are internally consistent but not uniquely privileged).
- **The "unreasonable effectiveness of mathematics" (Wigner)** `[OPEN]`/philosophical: a genuine unresolved meta-question, not a conjecture with a truth value; partial deflationary accounts (selection of effective math; symmetry / representation-theoretic constraints; anthropic/cognitive explanations) exist but none is decisive. **Flag as a question, never as settled doctrine.**

**Why they are hard.** Each sits at a boundary where a domain's tools run out: the classical limit needs the unresolved measurement problem; the glass and entropy debates concern finite-size / non-thermodynamic-limit regimes where standard ensembles are ambiguous; hypercomputation and Wigner's puzzle are partly meta-physical and resist formalization as testable claims.

**Cross-links.** *Chapter 4 (p. 25)*, *Chapter 6 (p. 56)*, *Chapter 5 (p. 40)*, *Chapter 12 (p. 155)*, *Chapter 13 (p. 172)*, OP-2, OP-16.

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## Iteration 2 additions (2026-06-08)

*New open problems surfaced by the iteration-2 research tracks (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`)); each passed the adversarial referee. Tags are shown inline (not yet folded into the status table below).*

- **OP-40 — Does gravitationally-induced entanglement (BMV/QGEM) decisively establish that gravity is non-**

**classical?** [OPEN] The LOCC no-go is a theorem, but the inference "entanglement via gravity ⇒ non-classical mediator" needs an additional *local-mediation* premise (no direct $A$–$B$ Hamiltonian term; a mediator carrying its own classical observables) that is logically **stronger** than relativistic no-signaling. Unresolved: (i) whether the static Newtonian potential is a mediated local exchange or a direct nonlocal-but-causal (Coulomb-gauge-like) interaction; (ii) whether fundamentally semiclassical/CQ theories (Hall–Reginatto; Oppenheim stochastic-CQ) are genuine loopholes or violate independent locality axioms. *Why hard:* the candidate theories disagree on the definition of "classical," and the experiment ($m \sim 10^{-14}$ kg, ~μm superpositions, ~1 s coherence) is far beyond current capability. *Approaches:* QGEM/BMV experiments + complementary gravitational-decoherence bounds; the $m^2$-vs-$m^3$ scaling discriminator (HYP-GIE-DISCRIM). Bears on *GC-15* (Chapter 14 (p. 191)). → *[notes* (Working note: *Would gravitationally-induced entanglement prove the gravitational field is quantum?* (repo: `notes/2026-06-08-tabletop-quantum-gravity.md`))]

- **OP-41 — A de Sitter entanglement first law.** [OPEN] Derive, in a controlled $\Lambda > 0$ setting, that $\delta S = \delta\langle K \rangle$ (modular Hamiltonian $K$) implies the *linearized* Einstein equations — the genuine dS analog of the AdS FGHMVR result. None exists; CLPW supplies the Type II₁ entropy structure but no equation of motion. **The single highest-leverage missing step for the emergence program.** → *[notes* (Working note: *Can emergent-spacetime / holography graduate to our Λ>0 universe?* (repo: `notes/2026-06-08-de-sitter-and-emergent-spacetime.md`))]
- **OP-42 — Is there a genuine unitary holographic dual of de Sitter?** [CONTESTED] Whether double-scaled SYK / static-patch / $T\bar{T} + \Lambda$ proposals realize unitary dS or only an effective subsector is unsettled; Rahman–Susskind vs Narovlansky–Verlinde entropy↔area scalings disagree by factors diverging as $N \to \infty$. Distinct from OP-41 (dynamics): this concerns the *existence/correctness of a dual.* → *[notes* (Working note: *Can*

*emergent-spacetime / holography graduate to our Λ>0 universe?* (repo: `notes/2026-06-08-de-sitter-and-emergent-spacetime.md`))]

- **OP-43 — The quantitative Born-rule problem.** `[OPEN]` In a strictly unitary (no-collapse) theory, derive that branch weights $|c|^2$ function as chances over single outcomes using premises provably *not* equivalent to Born. Every extant program (frequency operators, envariance, Deutsch–Wallace, Carroll–Sebens) re-imports a Born-equivalent symmetry/rationality/typicality input at the unequal-amplitude step. The *form* is forced (Gleason+Busch); the *meaning* is not. Sharpens the "probability in Everett" facet of *GC-6* (Chapter 14 (p. 191)). → *[notes* (Working note: *Is the Born Rule Fundamental or Derivable, and Is Quantum Mechanics Exactly Linear?* (repo: `notes/2026-06-08-born-rule-and-quantum-linearity.md`))]
- **OP-44 — Nonperturbative fate of the crossed-product Type II structure.** `[OPEN]` Does the III→II crossed-product / generalized-entropy structure (established in the $G_N \to 0$ / large-$N$ expansion) admit a nonperturbative completion in which a von Neumann-algebraic description of subregions survives, or does gravitational dressing ("algebraic spacetime disruption," Giddings 2505.22708) force a non-algebraic replacement (e.g. a network of Hilbert-space inclusions)? No no-go currently constrains either answer. → *[notes* (Working note: *Does the modular / crossed-product (von Neumann algebra) program deliver background independence?* (repo: `notes/2026-06-08-algebraic-background-independence.md`))]
- **OP-45 — Microscopic origin of horizon area-entropy for a generic horizon.** `[OPEN]` Identify degrees of freedom whose count is $A/4\ell_P^2$ for a *generic* (non-extremal, non-supersymmetric, asymptotically-flat) horizon — and that predict an independently measurable deviation from GR, rather than re-deriving the area law the entropic-gravity programs assume. (Strominger–Vafa and Ryu–Takayanagi cover only special cases; Sakharov instead *outputs* the area law.) → *[notes* (Working note: *Entropic / emergent gravity: genuine derivation, or suggestive*

*analogy?* (repo: `notes/2026-06-08-entropic-and-emergent-gravity.md`))]

- **OP-46 — Encode vs generate: emergent geometry from non-symmetric modular flow.** [OPEN] Is there an (algebra, state) pair with *non-geometric* modular flow from which a semiclassical metric/horizon can nonetheless be reconstructed — breaking the Bisognano–Wichmann / CHM circularity in which geometry is recognized only because a symmetric vacuum on a fixed spacetime is assumed? Is there even an agreed operational criterion distinguishing "encode" from "generate"? Until resolved, the algebraic program is *reorganization*, not *derivation*. **The problem is posed for external solvers, in its sharpest current form, in *the carrier-problem dossier* (Chapter 35 (p. 674)) (iteration 15).** → *[notes* (Working note: *Does the modular / crossed-product (von Neumann algebra) program deliver background independence?* (repo: `notes/2026-06-08-algebraic-background-independence.md`))]
- **OP-47 — Derive the entropic-gravity decoherence rate from a specified screen model.** [CONTESTED] A phenomenological open-system model already bounds entropic-gravity centre-of-mass decoherence with ultracold-neutron data ($\sigma \gtrsim 250$; Schimmoller et al. 2021) and admits a decoherence-free limit. The open residue: can such a master equation be *derived from Verlinde's holographic-screen / equipartition postulates* (not assumed), is the decoherence-free regime natural rather than fine-tuned, and does it genuinely answer Kobakhidze's reversibility objection? → *[notes* (Working note: *Entropic / emergent gravity: genuine derivation, or suggestive analogy?* (repo: `notes/2026-06-08-entropic-and-emergent-gravity.md`))]

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## Iteration 3 additions and progress (2026-06-08)

*From the iteration-3 attempts (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter3-*). One new problem;*

*progress notes on existing ones.*

- **OP-48 — Can the GPT framework-fixing axioms be formulated on a Type III1 algebra?** `[OPEN]` (candidate-no-go "OP-GPT-TYPE3"). The generalized-probabilistic-theory axioms that single out the Hilbert-space framework — tensor-product composition / local tomography, purification, continuous reversibility — presuppose a tensor factorization $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$ and finite-dimensional structure. Local relativistic QFT algebras are Type III1 (no factorization, no density matrices), and the load-bearing axioms provably fail in infinite dimensions. Can *any* formulation fix **composition** (not merely the Born-rule *form*, which the 2026 causal-rigidity theorem already lifts) on a Type III1 factor — or is this a genuine no-go? Decisive for whether Program A (geometry-from-algebra) and Program B (quantum-framework-final) can fully merge. → *[notes* (Working note: *Attempt: Merge Program A (geometry-from-algebra) and Program B (quantum-framework-final)* (repo: `notes/2026-06-08-iter3-merge-attempt-operational-substrate.md`))]

**Progress on existing problems:**

- **OP-41 (dS entanglement first law)** — obstruction now *located*: one Killing vector / one region (no continuum to invert to $\delta G_{\mu\nu}$) + a sign-reversed *maximization* principle ($dE = -T\,dS$, empty dS maximizes $S_{\text{gen}}$). One un-foreclosed escape: an $SO(d+1,1)$ family of static-patch modular Hamiltonians (Chen–Xu 2511.00622) or sub-patch CHM diamonds (Fröb 2308.14797). A conjecture **HYP-dS-NOMIN** (no holographic dS first law in the strict AdS sense) is recorded `[OPEN]`, *not* a no-go. → *[notes* (Working note: *Attempt: a de Sitter entanglement first law (OP-41)* (repo: `notes/2026-06-08-iter3-de-sitter-first-law-attempt.md`))]
- **OP-46 (encode vs generate)** — now equipped with a decision-usable criterion (axes $\Delta_{\text{geo}}$, $\kappa_\Phi$ + a 5-item smuggling checklist) and a verdict table over six constructions; finding: none generates Lorentzian spacetime from factorization-/region-/symmetric-state-free data. → *[notes*

— now equipped with a decision-usable criterion (axes $\Delta_{\text{geo}}$, $\kappa_\Phi$ + a 5-item smuggling checklist) and a verdict table over six constructions; finding: none generates Lorentzian spacetime from factorization-/region-/symmetric-state-free data. → *[notes* (Working note: *Formalize: an encode-vs-generate criterion for emergent geometry (OP-46)* (repo: `notes/2026-06-08-iter3-encode-vs-generate-criterion.md`))]

- **OP-44 (nonperturbative fate of crossed-product Type II)** — sharpened by the verified Giddings results (2505.22708, 2510.24833): dressed operators generically fail to commute spacelike at leading order in $G_N$; the III→II story may be a perturbative truncation. → *[notes* (Working note: *Attempt: Merge Program A (geometry-from-algebra) and Program B (quantum-framework-final)* (repo: `notes/2026-06-08-iter3-merge-attempt-operational-substrate.md`))]

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# Iteration 4 progress (2026-06-08)

*From the iteration-4 attacks + convergence audit (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter4-*).*

- **OP-41 (dS entanglement first law) — promoted toward near-no-go.** The $SO(d+1,1)$-family escape was executed and fails on a dimension count (finite orbit → ≤10 global charges, no Radon inversion to local $\delta G_{\mu\nu}$) + geometric-only-at-Bunch–Davies + concave-GSL second-order character (HYP-dS-NOMIN-R3). Surviving routes: non-unitary dS/CFT pseudo-entropy; Jacobson small-ball continuum inside a CLPW II₁ algebra. → *[notes* (Working note: *Execute the dS first-law escape: an SO(d+1,1) family of modular Hamiltonians (OP-41)* (repo: `notes/2026-06-08-iter4-ds-first-law-escape.md`))]
- **OP-48 (GPT-on-Type-III) — settled as a disjunction.** No-go for exact composition on a single III₁ factor; split-property qualified *yes* (intermediate Type I) that relocates the BW/CHM circularity (HYP-SPLIT-GEOM); full merge needs a state-/collar-free composition surrogate. `[OPEN, narrowed]` →

*[notes* (Working note: *Resolve OP-48: can GPT framework-axioms live on a Type III1 algebra? (split-property test)* (repo: `notes/2026-06-08-iter4-gpt-type3-resolution.md`))]

- **OP-46 (encode vs generate) — near-vacuous under the standard readout.** HYP-CKV-VACUITY (Borchers/Wiesbrock + Caminiti et al. 2502.02633); the live target is a non-CKV, Lorentzian readout (Connes reconstruction gives only Riemannian — GAP-NCG-LORENTZIAN-SIGNATURE). → *[notes* (Working note: *Break the Bisognano–Wichmann/CHM circularity: geometry from non-symmetric modular flow? (OP-46)* (repo: `notes/2026-06-08-iter4-modular-circularity.md`))]
- **OP-21 (muon $g-2$) — largely resolved.** WP2025 (arXiv:2505.21476) lattice-HVP prediction gives $\Delta a_\mu = 38(63) \times 10^{-11}$ (SM-consistent); see the OP-21 entry above and *Chapter 9 (p. 106).*
- **Convergence:** trend diminishing; analytical saturation ≈1–3 iterations away (remaining: OP-48 point (c), OP-44/OP-CP1, OP-46 non-CKV readout); object-level non-forecastable. Recommendation HYBRID. See *Chapter 21 (p. 497)* § Iteration 4 update and *the audit note* (Working note: *Convergence / completeness audit — how far from a conclusion?* (repo: `notes/2026-06-08-iter4-convergence-audit.md`)).

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## Iteration 5 progress (2026-06-08) — closing push, saturation

*From the closing analytical push (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter5-*). Consolidated in Chapter 38 (p. 705); object-level channels tracked in Chapter 39 (p. 725).*

- **OP-48(c) — SETTLED (conditional no-go).** "Collar-free + state-independent" composition and "singles out QM" are mutually exclusive on a single Type III1 factor (the Haagerup/Connes-fusion surrogate is a universal embezzler $\kappa_{\max} = 0$; QM-selection forces a Type I split factor re-importing the collar). → *[notes* (Working note: *OP-48 point (c): a state-/collar-free composition surrogate on Type III₁?* (repo: `notes/2026-06-08-iter5-op48c-collarfree-composition.md`))]

- **OP-46 — FORECLOSED on all known readouts.** The non-CKV Lorentzian readout produced the first emergent Lorentzian metric (Huang–Ma $dS_2$) but with signature *inherited from symmetry*; the obstruction is now readout-independent (HYP-CKV-VACUITY). Formally open only for a hypothetical no-symmetry readout. → *[notes* (Working note: *OP-46: emergent LORENTZIAN geometry from a non-CKV modular flow? (the verdict-flipping move)* (repo: `notes/2026-06-08-iter5-op46-lorentzian-nonckv-readout.md`))]
- **OP-44/OP-CP1 — regime boundary located; sole residual open lever.** Crossed-product Type II survives as a leading-order / QRF-relational truncation below $N_* \sim R/2GE_f$; degenerates above it (Giddings algebraic spacetime disruption). No theorem on either side. → *[notes* (Working note: *OP-44/OP-CP1: nonperturbative fate of the crossed-product Type II structure* (repo: `notes/2026-06-08-iter5-op44-nonperturbative-crossed-product.md`))]
- **OP-49 (≡ OP-dS-PSEUDO-REALITY) — new.** `[OPEN]` Can the imaginary pseudo-entropy of the dS/CFT first law (arXiv:2511.07915) be physically nulled (genuine post-selection / real reduced density matrix) while retaining nontrivial $\delta G$? A sharp decidable test isolating whether the dS/CFT pseudo-entropy route could move Program A toward *generates*. → *[notes* (Working note: *The two dS routes surviving the cardinality no-go: pseudo-entropy and Jacobson-in-CLPW* (repo: `notes/2026-06-08-iter5-surviving-dS-routes.md`))]
- **Saturation:** the named decidable list (OP-48c, OP-46, OP-44) is settled/bounded and the verdict-flipping lever did not flip; the analytical program is **saturated** (4th consecutive iteration at the same verdict). See *Chapter 38 (p. 705)* and *the iteration-5 synthesis* (Chapter 25 (p. 612)).

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## Iteration 6 additions and progress (2026-06-10)

*From the new-input iteration (6 research + 4 attack tracks, all refereed; see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter6-*). One new problem (OP-50, `[CONTESTED]`); OP-49 closes to near-no-go; OP-44 and OP-48(c) restructured; two experiment-side updates (OP-21, OP-18). Each action below names the residual lever it moves or is filed as watchlist/landscape maintenance.*

### OP-50 — The closed-universe Hilbert space: is it one-dimensional, and is QM only observer-relative there?

**Area:** Quantum gravity / quantum cosmology / holography. **Tag:** `[CONTESTED]`. **Kind:** structural frontier with an active two-camp dispute; no theorem on either side.

**Statement.** Nonperturbative topology-sum (spacetime-wormhole) arguments claim that the Hilbert space of a spatially closed universe (no asymptotic boundary, no external observer) is **one-dimensional (and real) per α-sector**: including wormholes in the inner product degenerates the Gram matrix of perturbatively-distinct closed-universe states to rank 1 — proven in JT gravity + matter and a topological model (Usatyuk–Wang–Zhao, SciPost Phys. 17, 051 (2024)), claimed general via QES/path-integral arguments, with lineage to McNamara–Vafa (arXiv:2004.06738). On this view standard quantum mechanics is recovered **only relative to an internal observer** with $S_{Ob}$ degrees of freedom, described by an effective Hilbert space of dimension $\sim e^{S_{Ob}}$ with errors exponentially small in $S_{Ob}$ (Harlow–Usatyuk–Zhao, arXiv:2501.02359 [JHEP ref unverified]; Nomura–Ugajin via partial observability, arXiv:2505.20390, 2602.13387). Is the one-dimensionality claim true — and if so, is observer-relative recovery of QM forced?

**Why it is hard.** The premise is itself contested by three distinct counter-positions, none refuted: (i) Antonini–Rath–Sasieta–

Swingle–Vilar López (arXiv:2507.10649) — one-dimensionality is an *external-access artifact* of the CFT description, with closed-universe EFT recoverable via a final-state-projection reading; (ii) Hong Liu (arXiv:2509.14327) — subregion-subalgebra duality plus an averaged large-$N$ dictionary implies a semiclassical description of a macroscopic closed universe exists at small finite $G_N$, an *observer-free* framing that contests whether algebra selection counts as observer-anchored input; (iii) Engelhardt–Gesteau (arXiv:2504.14586) — the actual CFT dual is the spacetime *without* a semiclassical baby universe. Higginbotham's "observer complementarity" (arXiv:2512.17993 [JHEP ref unverified — arXiv page says only "Submitted to JHEP"]) argues the descriptions are observer-relationally reconcilable. The dispute turns on what the gravitational path integral's wormhole sector means — exactly the regime with no independent definition of the theory.

**Why it matters here.** `[INFERENCE, medium-high]` The "observer rescue" is structurally the same move as the wiki's OP-48(c) conditional no-go: the fundamental description (Type $III_1$ factor there; one-dimensional Hilbert space here) fails to support QM's composition/probability structure, and standard QM is recovered only relative to an observer-anchored factorization — with the observer-selecting move coinciding with the geometry-importing input. This is a structural ANALOGY, not a derivation in either direction: cross-framework convergence reinforces, but cannot harden, horn (C), because the premise is `[CONTESTED]`. On the observer-rescue and external-access branches the OP-48(c) conditional structure is enforced; on Liu's branch it holds only if subalgebra selection is granted observer status — *consistent with* every branch, *enforced by* only some. **Residual lever moved:** this problem is the path-integral face of the OP-48(c) horn-C residue.

**Leading approaches.** Topology-sum/Gram-matrix computations in low-dimensional models; QRF/crossed-product methods (DEHK PW=CLPW, arXiv:2405.00114; follow-up 2412.15502, verified); Harlow's α-parameter-free third-quantized holography (arXiv:2602.03835, argued not shown); subregion-subalgebra duality (Liu); boundary-operator reconstructions (Engelhardt–Gesteau).

**Cross-links.** OP-48 (the algebraic face), OP-44 (observer-relational crossed products), OP-25 (problem of time), HYP-dS-CARDINALITY-R1 (*Chapter 19 (p. 392)*), *Chapter 38 (p. 705)* §5. → *[notes* (Working note: *Iteration 6 — de Sitter and closed-universe quantum gravity: the 2023–2026 literature delta* (repo: `notes/2026-06-10-iter6-closed-universe-and-dS-holography.md`))]

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## Progress notes (iteration 6)

- **OP-44 / OP-CP1 — interior restructured; bulk $N_*$ boundary still theorem-free on both sides.** *(Moves the OP-44 residual lever.)* (i) Jensen–Raju–Speranza (arXiv:2412.21185; JHEP 06 (2025) 242) prove the AdS boundary time-band algebra **strictly has no commutant** nonperturbatively (a theorem-grade *boundary-anchored* instance of the failure pattern — it brackets, not settles, the bulk $N_*$ question), plus **crossed-product rigidity**: the modular crossed product is the *only* crossed product of a Type III1 algebra yielding a trace — any tracial truncation surviving at higher orders that is a crossed product must be the modular one. [ESTABLISHED] (ii) De Vuyst–Eccles–Höhn–Kirklin (arXiv:2411.19931; JHEP 2025, 211) locate the **order-$G_N^2$ proof obligation**: the quadratic (boost) linearization-instability constraints — "essentially unambiguous" to impose for spatially closed spacetimes, contingent for bounded ones; transferring the closed-slice cleanliness to the dS static-patch II1 requires an additional step `[INFERENCE, medium]`. [ESTABLISHED] (iii) Survival side theorem-grade at linearized order for the **complete** constraint family with edge modes (KLS II, arXiv:2601.07910 — supersedes iter-4's "incremental, action=none" gloss; entropy is $S_{\text{gen}}$ + edge-mode + constant term), including a **GSL for non-stationary linearized perturbations and a proof of the quantum focusing conjecture in the perturbative regime** (Chandrasekaran–Flanagan, arXiv:2601.07915 — a theorem *within the linearized/perturbative regime around a Killing-horizon background*; interacting QFT and genuinely dynamical non-Killing horizons remain fully open). `[ESTABLISHED]` (iv) **Fourth verdict option**: Kudler-Flam–Witten (arXiv:2510.06376) — baby-universe state sequences fail to converge at large $N$ unless one **averages over** $N$ (conditional on analytic continuation in $N$ + superpolynomially small fluctuations, arXiv:2605.15180); the trichotomy (survives / fails / regime-dependent) becomes four options with **survives-after-averaging**; the transfer to subregion crossed products is `[INFERENCE]`. (v) The iter-5 "split-property downside" is realized in print — by the stringy Hagedorn mechanism at sub-string-length separations (Herderschee–Kudler-Flam, arXiv:2510.01556), not by $O(\kappa)$ dressing (still open); every published post-breakdown candidate remains *algebraic*, weakening (publication-bias-prone evidence) Giddings' "beyond algebras" horn. No Giddings follow-up or rebuttal post-2510.24833 `[INFERENCE, high — negative-after-search, as of 2026-06]`. **Net: OP-44 stays `[OPEN]`, "no theorem on either side of $N_*$ " intact; statement updated with the four-option verdict space.**→ *[notes* (Working note: *OP-44 & OP-48(c) iteration-6 attack: theorem-shaped brackets on the crossed-product fate, and a hardened horn C* (repo: `notes/2026-06-10-iter6-op44-op48c-levers.md`)), *notes* (Working note: *OP-44 literature delta (2024–2026): the survival side hardens perturbatively; the failure side turns out algebraic; no theorem on either side of $N_*$* (repo: `notes/2026-06-10-iter6-algebraic-frontier.md`))]

- **OP-48(c) horn (C) — PARTIALLY DISCHARGED; the hedge shrinks.** *(Moves the OP-48c horn-C residual lever.)* Under the GPT-faithful **full-pair** reading ("recovers QM" = existence of a normal product state across the commuting pair, the precondition of local tomography's dimension count): **exact** — a normal product state across $(\mathcal{M}_A, \mathcal{M}_B)$ forces a Type I interpolating factor (GNS factorization + normality; split-inclusion equivalences anchored to Doplicher–Longo, Invent. Math. 75 (1984), D'Antoni–Longo 1983, Summers, Rev. Math. Phys. 2 (1990) [unverified this session]; Jäkel hep-th/9811227 (solo author) only as thermal-context corroboration — the original "et al." anchoring is corrected); for the Haag-dual pair $(\mathcal{M}, \mathcal{M}')$ a normal product state exists **iff $\mathcal{M}$ is Type I**, so on III1 any composition rule with normal product states must retreat to a proper sub-pair — *which is the collar*. `[INFERENCE, high – theorem-shaped under the full-pair reading]` **Approximate** — every normal state of a III1 pair is at norm distance $\geq 1 - 1/\sqrt{2}$ from every product/separable state (Summers–Werner maximal Bell violation + Tsirelson; own derivation, refereed). **The irreducible residue (corrected by referee):** operationally there is a proven *spectrum* of composition regimes in one-to-one correspondence with the von Neumann types (van Luijk, arXiv:2510.07563) — not a binary; the two physically motivated endpoints — product-preparations/Type I (textbook QM) and universal-embezzlement/III1 (local QFT; now with an explicit consistent *multipartite* III1 composition theory, van Luijk–Stottmeister–Wilming, Quantum 9, 1818 (2025), arXiv:2409.07646) — are each internally consistent and Bell-gap-separated; the algebra proves the classification but cannot choose the endpoint. The choice is a physical/axiomatic input — encodes-not-generates at the composition layer. Plus the GPT formalization choice (full-pair vs subsystem reading). **Net: hedge 2 shrinks but does not vanish; *Chapter 38 (p. 705)* §5 item 4 re-worded.→** *[notes* (Working note: *OP-44 & OP-48(c) iteration-6 attack: theorem-shaped brackets on the crossed-product fate, and a hardened horn C* (repo: `notes/2026-06-10-iter6-op44-op48c-levers.md`))]

- **OP-49 (≡ OP-dS-PSEUDO-REALITY) — CLOSED-NEGATIVE at NEAR-NO-GO grade (conditional exclusion; iter-5 "sharpened expectation: no" upgraded to an exclusion argument).** *(Closes the OP-49 residual lever, negative direction; mirrors OP-41 house language.)* In dS3/CFT2 the pseudo-entropy of a boundary cap is $S_A = -i(\tilde{c}/3)\log[(2/\epsilon)\sin(\theta_0/2)] + \pi\tilde{c}/6, \tilde{c} = 3R/2G \in \mathbb{R}$: the **real** part is the finite universal *region-independent* constant $S_{\mathrm{dS}}/2$; the **entire** region- and state-dependence (divergence + finite shape-dependent log) is **imaginary** (Doi–Harper–Mollabashi–Takayanagi–Taki, PRL 130, 031601 (2023)) — *corrects the inverted iter-5 parenthetical*. In the arXiv:2511.07915 first law the linearized-Einstein content is carried exclusively by Im δS (coefficient = the imaginary central charge). **Horn (i), fixed dictionary:** enforcing reality forces $\Delta S_A \equiv 0$, hence (assuming CHM-kernel invertibility over caps on $S^2$ `[INFERENCE, FGHMVR-pattern, not re-derived]`) $\delta\langle T_{tt}\rangle \equiv 0$ — the probe vanishes; conversely $\delta G \neq 0$*forces* non-reality via the Guo–He–Zhang necessity direction (pseudo-Hermiticity is necessary and sufficient for real trace moments, diagonalizable case; arXiv:2209.07308, JHEP 05 (2023) 021). **Horn (ii), deformed states/dictionary:** every reality-restoring deformation *examined* (Θ-invariant states; modular-conjugate post-selection; $c \to$ real) deletes the relative Lorentzian phase and deforms AdS-ward — an encodes-type Euclidean reconstruction [INFERENCE over an open class; supported by the unitary-AdS mirror arXiv:2511.17098 — where Im S is an inert constant cancelling in variations and only Re δS carries the Einstein equation, the exact image under $c \to ic$ — and by the rigidity of Im S against renormalization, shape deformation, and contour choice]. **Three named residuals keep it short of a theorem:** (a) exceptional-point (non-diagonalizable) transition matrices; (b) the SVD-entanglement-entropy protocol (Parzygnat–Takayanagi–Taki–Wei, JHEP 12 (2023) 123) — no first law exists as of 2026-06; a holomorphy-breaking argument predicts none survives [conjecture, decidable by the stated hemisphere computation]; (c) horn-(ii) protocol-class openness. **Retag `[OPEN]` → near-no-go / resolved-negative (conditional); paired CHANGELOG line in the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)).**→ *[notes* (Working note: *OP-49 resolved-negative (conditional): reality of the dS/CFT pseudo-entropy excludes nontrivial first-law content* (repo: `notes/2026-06-10-iter6-op49-pseudo-entropy-reality.md`))]

- **OP-21 — flavor anomaly sharpened (watchlist maintenance).** LHCb's comprehensive amplitude-level analysis of $B^0 \to K^{*0}\mu^+\mu^-$ on the full Run 1+2 dataset (arXiv:2512.18053; PRL 2026) reports ~4σ tension with the SM [reported], with supporting CMS results (2025). Interpretation `[CONTESTED]` (new physics vs charm-loop/form-factor hadronics). The entry's "remain in mild tension" wording is stale as of 2026; paired CHANGELOG line. Muon $g-2$ remains closed (final FNAL + WP2025 re-verified, iteration 6).
- **OP-18 — split verdict (watchlist maintenance).** $H_0$: persists at roughly ~5σ and sharpened under JWST cross-checks (ladder near 73; CCHP TRGB/JAGB ≈ 70.4; SBF ≈ 73.8 — method-internally contested) [reported]. $S_8$: substantially eased **on the KiDS cosmic-shear axis** — KiDS-Legacy $S_8 = 0.815^{+0.016}_{-0.021}$, 0.73σ from Planck (arXiv:2503.19441); joint KiDS-Legacy+DES Y3+Pantheon++DESI Y1 Planck-consistent (arXiv:2503.19442) — but survey-dependent: DES Y6 remains in significant tension per a 2026 review (arXiv:2602.12238) `[CONTESTED]`. The two tensions decouple; paired CHANGELOG line.

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## Iteration 7 progress (2026-06-10)

*From the lever-execution iteration (4 attack tracks on the four named decidable levers, all adversarially refereed; referee corrections binding; see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter7-*). No new OP is opened (next free ID remains OP-51); two new named hypothesis items (HYP-MODULAR-TRICHOTOMY, HYP-FACTORIZATION-IS-GEOMETRY) enter Chapter 19 (p. 392). Each action below names the lever it moves; every retag has a paired line in the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)). Headline verdict untouched — sixth consecutive iteration.*

- **OP-44 / OP-CP1 — the order-$G_N^2$ obligation restructured P1 → P1a/P1b/P1c; first failure-side perturbative foothold; $N_*$ untouched.** *(Moves the OP-44 residual lever.)* **(P1a — descent: DISCHARGED at inference grade.)** Quadratic-order descent of the *complete* global-dS constraint set to the CLPW static-patch crossed product, via the stabilizer decomposition $\mathfrak{so}(d+1,1) = (\mathbb{R}_B \oplus \mathfrak{so}(d)) \oplus \mathfrak{m}$ relative to the pode observers: the boost constraint is the CLPW constraint itself; the compact $\mathfrak{so}(d)$ twirl preserves Type II (trace restricts faithfully; AKL centrality criterion); the $2d$ coset constraints are solved by the observer worldline as a complete dynamical frame (group averaging factors through the coset part, type decided by the stabilizer alone). `[INFERENCE, high – conditional on the inherited quantum-imposition hedge (P1b context, below); fixed-point factoriality not separately proven (FJLRW give II₁-factor criteria under their hypotheses); generic frame states assumed]` Independently confirmed at the literature level: DEHK §2.5.1 verbatim (additional isometry constraints "turn out not to play a role in the type transition", citing FJLRW arXiv:2403.11973 and AKL arXiv:2407.01695). Classical input: the FMM/AMM cone theorem (quadratic Taub conditions necessary AND sufficient on closed slices; all higher $C^{(n>2)}$ solvable) is `[ESTABLISHED]` at $\Lambda = 0$; its dS consequence carries `[INFERENCE, medium-high]` (no published $\Lambda > 0$ verbatim statement; Higuchi 1991 and Altas–Tekin support). Note: the discharge consumes pre-installed $\mathfrak{so}(d+1,1)$ structure — itself an encoding step; HYP-ENCODING-SCREEN mildly reinforced, OP-46 hedge untouched by this track. **(P1b = R1 — located classical residue.)** DEHK footnote 12 (verbatim, primary PDF): sufficiency of the Taub conditions *with clock insertions* "has not been checked" — and the check is non-routine: worldline clocks have distributional stress-energy, exiting the $H^s$ Sobolev spaces of the FMM/AMM proofs. Decidable forks: smeared-observer extension of AMM, or a distributional extension of linearization-stability theory (absent from the literature as of 2026-06). `[OPEN – located, decidable]` The global hedge stands: DEHK's "not possible to rigorously prove" that the quantum Taub constraints must be imposed (argued-not-proven; inherited by P1a). **(P1c = R2 — located quantum residue; the foothold.)** The one-sided boost-weighted cubic charge ($\mathcal{O}_3 \sim h\,\partial\varphi\partial\varphi$, $\Delta_3 = (3d+1)/2$) has vacuum fluctuations $\sim A/\delta^{2(d-1)}$ vs $A/\delta^{d-1}$ for the quadratic charge (the latter reproducing the known EE/modular-fluctuation area law — consistency check): the cubic charge is globally well-defined on the closed slice but **does not split across the horizon**, the relative cocycle is *not manifestly affiliated* to the CLPW algebra (strictly weaker than provably outer), and subdominance requires $\delta \gtrsim \ell_P$ in every dimension — a **Planckian dressing-collar floor**, the gravitational analog of OP-48's stringy $\ell_s$ floor (the iter-4/iter-6 collar-tax leg sharpened from a second independent direction). Both horns theorem-shaped: discharge = cubic edge modes (KLS-II-style) rendering the cocycle inner in a finite extension; obstruction = no affiliated renormalization ⟹ the cubic derivation is outer and Type II stability fails to be automatic at first subleading order. `[INFERENCE, medium — scaling estimate; caveats binding: constrained-graviton tensor structure or cross-cell correlations could soften the exponent]`**The obstruction horn is the first identified perturbative foothold for the failure-side reading of OP-44 — at order $\kappa^3$, not at $N_*$; it is a candidate, not an instance.** Literature delta 2024–2026 `[INFERENCE, high — negative-after-search, dated]`: nearest machinery exists (Kirklin all-orders GSL, arXiv:2412.01903, JHEP 07 (2025) 192 — builds on, does not re-derive, the crossed product; Ali Ahmad–Jefferson arXiv:2501.01487, order-by-order) but no 2025–2026 paper performs the cubic edge-mode construction or the footnote-12 check. **Net: OP-44 stays `[OPEN]`; "no theorem on either side of $N_*$" intact; the obligation is no longer vague — it is P1a (done, conditional) + R1 + R2.**→ *[notes* (Working note: *OP-44 iteration-7 attack: the order-$G_N^2$ obligation for the dS static patch — quadratic descent discharged, residue located at footnote 12 and the cubic charge* (repo: `notes/2026-06-10-iter7-op44-orderG2-descent.md`))]

- **OP-46 — residual note: the J-route is CLOSED (executed, geometry-void); hedge MEDIUM → MEDIUM-HIGH (conditional); target re-compressed.** *(Moves — and hardens — the OP-46 residual lever, the designated verdict-flipper.)* The compressed iteration-6 target ("derive the indefinite pairing from modular data") was *achieved literally and voided*: $\eta_0 = \mathrm{sgn}(\ln\Delta)$ is a canonical, smuggle-free Krein fundamental symmetry from pure $(\mathcal{M}, \Omega)$ modular data — one of an infinite $J$-odd family $\varepsilon(\Delta)$ (referee correction; uniqueness needs an added modular-rescaling-invariance axiom) — and is **provably geometry-void**: unitarily universal in the Borchers/HSMI class ($\Delta_{\mathrm{geo}} = 0$ proven), generically algebra-incompatible, non-localizable without a carrier; the modular-form trichotomy lemma (HYP-MODULAR-TRICHOTOMY, ansatz-scoped) closes the route generally; $J$ structurally carries reflection positivity (cone self-duality), upgrading the $g_3$ trap from symmetric-state inheritance to a structural fact. **Retag: the OP-46 residual hedge `MEDIUM` → `MEDIUM-HIGH (conditional on the modular-constructibility ansatz)`; paired CHANGELOG line. The faithful residual target: derive an algebra-compatible, localized (fiberwise) $(n, n)$ pairing from modular data — the carrier problem (HYP-FACTORIZATION-IS-GEOMETRY), the fifth independent route to converge there.** Direction-map (negative search, referee-corroborated): all prior art runs Krein → modular (Gottschalk JMP 43 (2002) 4753; Jakóbczyk 1985, OSTI 5135323; Devastato–Farnsworth–Lizzi–Martinetti JHEP 03 (2018) 089), never modular → Krein. → *[notes* (Working note: *OP-46 attack, iteration 7: η-from-modular-data — the modular sign η0 exists, is geometry-void, and a trichotomy lemma closes the J-route* (repo: `notes/2026-06-10-iter7-indefinite-pairing-attempt.md`))]
- **OP-48(c) horn (C) — definitional residue RESOLVED (full-pair forced); horn C now theorem-conditional-on-framework; residue relocated to the product-preparation primitive.** *(Moves the OP-48c horn-C residual lever.)* All three GPT reconstructions (Hardy quant-ph/0101012;

CDP arXiv:1011.6451; Masanes–Müller arXiv:1004.1483 — axioms verbatim-checked against the primary PDFs) quantify composition universally over THE designated pair and install product states at *framework* level, prior to local tomography; the subsystem reading is inexpressible as a weakening of any reconstruction axiom and vacuous where expressible — adopting it means rejecting OP-48's question (the referee's "forced-modulo-the-question" gloss is binding). New mathematics (referee re-derived, correct): **the separation half of local tomography HOLDS on the Type III1 Haag-dual pair** — so the failure locus is exactly the **normal-product-preparation primitive (PS)**, violated with the uniform Summers–Werner gap $1 - 1/\sqrt{2}$; conversely (PS) on the pair forces Type I. **Horn C final form: theorem-conditional-on-framework** (the framework condition is shared by all three reconstructions and weaker than any single axiom set), conditional exactly on (i) the carried Summers–Werner primary spot-check, (ii) the carried split-equivalence technicalities, (iii) the normal-state postulate. **New residue statement: the normal-state postulate plus the composition-primitive choice — (X1) singular product states (existence via the referee-corrected Schlieder/Roos + Hahn–Banach route; admitting them abandons the density-matrix/σ-additive state notion) or (X2) the non-OPT III1-embezzlement composition endpoint; both exits leave the framework rather than reinterpret it.** Retag paired in CHANGELOG; CONCLUSION §5 item 4 re-worded. → *[notes* (Working note: *OP-48(c) iteration-7 attack: the full-pair reading is forced — horn C loses its definitional residue; plus the OP-50 watch* (repo: `notes/2026-06-10-iter7-op48c-fullpair-forcing.md`))]

- **OP-49 — the SVD residual CLOSED-NEGATIVE (conditional); stays NEAR-NO-GO; residual list rewritten (a)/(b)/(c′).** *(Closes the OP-49 SVD residual lever, negative direction, on referee-narrowed grounds.)* The iteration-6 stated decidable computation (δS_SVD for the dS3 hemisphere under the arXiv:2511.07915 perturbation family) was performed. **Unconditional pillar:** the literal first law

$\delta S_{\mathrm{SVD}} = \delta\langle K_A\rangle$ is false — $S_{\mathrm{SVD}}$ is real *by construction* while the verified carrier ($\delta\langle K_A\rangle$, imaginary $T_{tt}$) is purely imaginary and nonzero unless the probe vanishes; the iter-6 non-holomorphy conjecture is confirmed as an exact operator identity ( $\delta\,\mathrm{Tr}(\mathcal{T}^\dagger\mathcal{T})^m$ = Re-projection of a $\mathbb{C}$-linear functional). **Conditional mechanism:** the SVD replica object is an alternating $c/c^*$ branched cover; the sheetwise-additive twist dimension carries $c + c^* = 0$ (cap-size-blind spectrum) — but exact blindness requires unproven heterogeneous-cover **junction additivity** at the branch points, where interface-CFT precedent (Sakai–Satoh-type transmission-dependent $c_{\mathrm{eff}}$) shows naive additivity generically fails (referee: four majors, all kept with corrections). Either way, no local Einstein constraint survives in the SVD channel (a θ0-independent response fails $(\Box_{\mathrm{dS}_2} + 2)\delta S = 0$ and CHM inversion; a cap-dependent real response still cannot equal the imaginary carrier). **Residual list now: (a) exceptional-point (non-diagonalizable) transition matrices; (b) beyond-2511.07915-family generalization; (c′) heterogeneous-cover junction additivity** — three items (the old (c)-SVD residual is closed-negative-conditional and replaced by the narrower (c′)). Byproduct $S_{\mathrm{SVD}} = S_{\mathrm{dS}}/2$ (region-independent) **downgraded to** `[SPECULATIVE]` (the $q = 1$ self-normalization is in tension with the trace-norm inequality given the verified complex pseudo-spectrum); region-independence `[INFERENCE, conditional on (c′)]`. Absence re-confirmed `[INFERENCE – timestamped search-bounded, 2026-06-10]`: no SVD-in-dS computation and no SVD first law in the literature; the computation is new. Retag paired in CHANGELOG. → *[notes* (Working note: *OP-49 SVD residual closed: the SVD first law fails on the dS*$_3$ *hemisphere because* $c + c^* = 0$ (repo: `notes/2026-06-10-iter7-op49-svd-computation.md`))]

- **OP-50 — now three-cornered; stays** `[CONTESTED]` **(watch maintenance).** Three verified post-2026-01 developments: (1) Harlow–Usatyuk–Zhao published — JHEP 02 (2026) 108 (upgrades the iter-6 `[unverified]` flag); (2) Nomura–Ugajin, arXiv:2602.13387 — accepts per-sector one-dimensionality, derives statistically stable semiclassical predictions from observers' limited access; (3) Higginbotham, arXiv:2512.17993, JHEP 03 (2026) 183 (sole author, building on EGH — attribution corrected; journal ref verified this session) — refined SWAP-test operators softening the observer-complementarity tension. Plus a **third position**: Miyata–Horikoshi–Tajima–Ugajin, arXiv:2606.04575 — symmetric-orbifold closed-universe sectors whose dominant large-N sector, *before* the $S_N$ gauge constraint, is exponentially large with hyperfinite **Type II₁** structure, and whose *post-constraint* physical Hilbert space grows only **polynomially** in $N$ (matching wormhole path-integral counting); Hartle–Hawking alone fails to reproduce the CFT results and agreement is restored by coupling to an external observer system. Neither strictly 1-dim nor $e^{S_{Ob}}$; cross-feeds OP-44's survival side (model-specific construction, not a theorem). **OP-50 framing: strict 1-dim / prediction-framework-on-1-dim / polynomial-dim II₁ sector.** Negative sweep (dated process note, 2026-06-10): no EGH/ARSSV/Liu direct responses post-2026-01; on next rewrite the dossier should also note arXiv:2509.14338 ("A no-go theorem for large N closed universes", Sep 2025, pre-window). → *[notes* (Working note: *OP-48(c) iteration-7 attack: the full-pair reading is forced — horn C loses its definitional residue; plus the OP-50 watch* (repo: `notes/2026-06-10-iter7-op48c-fullpair-forcing.md`))]

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## Iteration 8 progress (2026-06-10)

*From the net/family lever-execution iteration (4 refereed attack tracks; see notes/ (repo:* `notes/`*; see the notes index, repo:* `notes/README.md`*) iter8-*). Verdict unchanged (7th consecutive); principal hedge hardened toward HIGH on a narrowed condition. No new numeric IDs consumed.*

- **OP-44 — R1/R2 sharpened to theorem-shape; $N_*$ and the $\ell_P$ collar floor untouched.** `[SHARPENED-OPEN]` **R1:** the bare distributional worldline-clock quadratic Taub charge

**diverges** ($\varepsilon^{2-d}$; $\ln\varepsilon$ at $d=2$) — the FMM/AMM cone theorems and Hintz's solvability theorem (arXiv:2306.07715) are proven only for smooth, spatially-compact sources, so DEHK footnote-12 clock-sufficiency cannot be discharged in the naive distributional form (Pound 1506.06245: $\delta^2 G[h_1, h_1] \sim 1/r^4$ non-integrable). The escape is **Gralla–Wald / Pound puncture (effective-source) regularization** (referee correction: *not* an EM-style mass renormalization) — sufficiency holds iff the regularized charge meets the FMM cone condition (conditional; the regularization supplies the clock structure as INPUT — HYP-ENCODING-SCREEN reinforced). **R2:** all three cubic-$\delta^{-2(d-1)}$ cancellations fail at leading order (smearing relocates the cutoff; KLS-II edge modes are provably quadratic-order only, arXiv:2601.07910; normal-ordering leaves the connected channel invariant); the obstruction horn — the first perturbative failure-side foothold, order $\kappa^3$ — is reinforced as INFERENCE, with the constrained-graviton transverse contraction the sole surviving escape. → *[notes* (Working note: *OP-44 iteration-8: the two located residues executed — R1 (worldline-clock Taub finiteness) resolves two-horn with a renormalization condition; R2 (cubic non-splittability) reinforced against smearing, edge modes, and normal-ordering* (repo: `notes/2026-06-10-iter8-op44-R1-R2.md`))]

- **OP-46 residual / the carrier problem — net/family no-go; hedge hardens MEDIUM-HIGH → toward HIGH on a narrowed condition.** `[INFERENCE — HIGH classification; closure conditional on the net-constructibility ansatz]` The net upgrade — relative modular operators $\Delta_{\sigma|\rho}$, families $\{J_O\}$, half-sided modular inclusions, modular intersection, the modular-Berry connection over a net $O \mapsto \mathcal{M}(O)$ — adds strictly more *operators* but no new localized indefinite *form*: by the covariance–locality obstruction every modular-covariant pairing is positive-and-local, signature-free, or indefinite-but-non-local (numerically verified). Localization lives only in the net's index set, which **is** the geometry (BGL positivity⇔isotony; Morinelli–Neeb–Ólafsson Euler-element-as-input; Lashkari–Leung–

Moosa–Ouseph, JHEP 10 (2025) 153). **Count correction (binding):** this is the net-level extension of the fifth route's boost-covariance engine — the carrier-convergence count **stays at five**, not six. Two named obligations remain before HIGH is unconditional (replace the ansatz by a naturality theorem; rule out a modular-Berry-holonomy-natural localized indefinite form). → *[notes* (Working note: *OP-46 / carrier-problem attack, iteration 8: the net upgrade — net-modular data adds operators, not a localized indefinite form; the parametrization is the geometry* (repo: `notes/2026-06-10-iter8-net-carrier-no-go.md`))]

- **OP-48(c) — X1 promoted to a theorem-with-a-price-tag; horn C unchanged.** `[INFERENCE, high]` **X1-EXISTENCE (theorem):** a singular (non-normal) product state across the Type III1 Haag-dual pair $(\mathcal{M}, \mathcal{M}')$ exists — factor ⟹ Schlieder (the **central-carrier fact**, Kadison–Ringrose: every nonzero projection in a factor has central carrier $I$, so $ef \neq 0$; *not* Halvorson Def 3.4, which is only the *definition*) ⟹ C*-independence (Halvorson implication box) ⟹ a product-state extension on **plain C*-independence** (Roos 1970, via the Rédei–Summers post-Def-1 sentence — no product-sense isomorphism needed; Rédei–Summers Prop. 1: plain ⇏ product-sense in general) ⟹ Hahn–Banach; non-normal by **Takesaki 1958** (Tôhoku Math. J. **10**, 116–119, via Rédei–Summers Prop. 2: $\mathcal{M} \vee \mathcal{M}' = B(\mathcal{H})$ is Type I while $\mathcal{M} \overline{\otimes} \mathcal{M}'$ is Type III). *(Florig–Summers 1997 is the separate C*-side faithful-product-state characterization, Rédei–Summers Prop. 4 — not the non-normality obstruction; the joint attribution is dropped.)* **X1-DISQUALIFICATION (sharper than posed):** *no* reconstruction axiom *fails* on a singular state — Hardy/CDP/Masanes–Müller quantify only over finite-dim, $\sigma$-additive (normal) states, so a singular state is upstream of every axiom. Horn C therefore **stays theorem-conditional-on-framework**; X1 is a costed framework *exit* (**X1-COST**, theorem: admitting singular product states relinquishes density-matrix states + $\sigma$-additivity + finite tomographic dimension *jointly*). The normal-state postulate is thereby proven a load-bearing, non-derivable installation choice. → *[notes* (Working note: *OP-48(c) EXIT X1: singular product states on the III1 Haag-dual pair exist (theorem) and constitute a framework exit, not a closable residue* (repo: `notes/2026-06-10-iter8-op48c-X1-singular-states.md`))]

- **OP-49 — (c′) reframed-and-narrowed, NOT closed; stays NEAR-NO-GO (the submitted RESOLVED-POSITIVE was overturned by the referee).** `[INFERENCE, medium]` The transmissive-orientation-reversal / dimension-zero-junction mechanism (which would make the $c + c^* = 0$ cap-blindness exact) is valid at real-$c$ orientation reversal but **extrapolated to the non-unitary imaginary-$c = \tilde{\imath}c$ regime** where transmission coefficients violate the unitary bounds and the Karch–Kusuki–Ooguri–Sun–Wang (SSA / $c$-theorem) machinery does not apply — "not rigorously excluded." Residual list stands: **(a) exceptional points; (b) beyond-2511.07915-family; (c′) heterogeneous-cover junction additivity (narrowed to the complex-$c$ extension)**. The unconditional pillar — $S_{\mathrm{SVD}} \in \mathbb{R}_{\geq 0}$ by construction vs the verified carrier $\delta\langle K_A\rangle \in i\mathbb{R} \setminus \{0\}$ — is re-verified and untouched; no CONCLUSION hedge moves. (Citation fix: arXiv:0705.3129 is Fuchs–Gaberdiel–Runkel–Schweigert; the transmissive-defect criterion is Bachas–Brunner–Roggenkamp.) → *[notes* (Working note: *OP-49 residual (c′) closed: the dS SVD bra/ket junction is the transmissive orientation-reversal defect, so cap-blindness is exact* (repo: `notes/2026-06-10-iter8-op49-junction-additivity.md`))]

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## Iteration 9 progress (2026-06-10)

*From the carrier-naturality lever iteration (4 refereed attack tracks; see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter9-*). Verdict unchanged (eighth consecutive confirmation); principal hedge grade unchanged, condition sharpened; one named obligation discharged; one foothold de-hardened. No new numeric IDs consumed.*

- **OP-46 / the carrier problem — the net-constructibility ansatz replaced by a naturality formulation; one of the two iteration-8 obligations discharged.** `[INFERENCE — HIGH classification; conditional on the net-naturality theorem + a located commutant-to-generated gap]` (A1) Under the BFV/GMA naturality formulation the Gram operator of an $\mathrm{Aut}(\mathcal{N}, \Omega)$-natural modular-covariant carrier is *forced* into the commutant of $\{\Delta_O^{it}\}$ (Lemma N1), and the boost-covariance engine is promoted to a naturality obstruction (Lemma N2): a localized indefinite natural form has a flow-carried sign locus with no fixed point in the open index set, so localized + natural + indefinite is impossible. The **irreducible residual**: "localized" is defined relative to the index set's region order — a base functor that is part of the source datum — so it presupposes $n_1$, *which is the geometry* (BGL; Neeb–Ólafsson Euler element; Lashkari–Leung–Moosa–Ouseph). **Referee correction (binding):** Lemma N1 forces $\eta$ into the *commutant* of $\Delta$, not the *abelian algebra generated by* $\Delta$ (unequal in the generic Type-III degenerate-spectrum case) — so the constructibility ansatz is *weakened, not eliminated*; a commutant-to-generated gap survives. (A2) The **modular-Berry-holonomy escape closes RESOLVED-NEGATIVE**: the modular-Berry curvature is antisymmetric = the Kirillov–Kostant symplectic (Crofton) form on a coadjoint orbit of $SO(d, 2)$ — *signature-free*, not a localized indefinite pairing; the only indefinite object (the kinematic-space Lorentzian $dS_2$ metric) is *vacuum/symmetry-inherited* from the conformal vacuum (Huang–Ma 2110.08703), the iteration-5 closure. **Hedge: HYP-CKV-VACUITY-R5 → -R6; grade unchanged (MEDIUM-HIGH→HIGH conditional); condition sharpened**; the net-naturality theorem (plus the commutant-to-generated gap) is the sole residual between HIGH and a theorem. Carrier-convergence count **stays at five**. → *[notes* (Working note: *OP-46 / carrier obligation (i), iteration 9: the naturality promotion — constructibility becomes a theorem, localization is the irreducible residual (and is the geometry)* (repo: `notes/2026-06-10-iter9-`

carrier-naturality.md)), *notes* (Working note: *Carrier obligation (ii): the modular-Berry-holonomy escape closes — the holonomy-natural form is symplectic (signature-free) or symmetry-inherited, never a localized indefinite carrier* (repo: notes/2026-06-10-iter9-modular-berry-holonomy.md))]

- **Reverse-Weinberg–Witten = the empirical-side shadow of the carrier problem (the central iteration-9 identification).** [INFERENCE, high — structural identification, conditional on the carrier ansatz] (A3) A1 ("every observable is an invariant of the GNS pair, net + vacuum"), in its load-bearing concrete form, is a **conditional theorem**: by **Weiner's algebraic Haag theorem** (arXiv:1006.4726, verified) split + geometric modular action ⟹ the vacuum-sector net is a complete invariant; Doplicher–Roberts closes the charged sector and global/topological data (DHR / Haag-duality-violation invariants, Casini–Magán 2511.21810, the H2 closure re-used). But Weiner's hypotheses **are the geometry** (GMA demands the modular data act geometrically on a pre-given causal index set $= n_1$; split is the OP-48 collar). **Identification:** A1 unconditional ⟺ the carrier problem has no positive solution ⟺ no reverse-WW witness — proving any one proves all three; the reverse-WW *witness* is strictly harder than a positive carrier solution. This **pins** the standing reverse-WW target ([OPEN, as of 2026-06]) to the §8 reopening condition (HYP-FACTORIZATION-IS-GEOMETRY) rather than leaving it free-floating; it remains un-registered (NOT OP-50). No headline flip, no route added (count stays five), no hedge moved. → *[notes* (Working note: *The Reverse-Weinberg–Witten Target Is the Carrier Problem Seen From the Empirical Side (2026-06-10)* (repo: notes/2026-06-10-iter9-reverse-weinberg-witten.md))]
- **OP-44 R2's graviton escape stays OPEN; the failure-side foothold de-hardened after a fatal computational error.** [SHARPENED-OPEN] (A4, overturned by the referee from RESOLVED-NEGATIVE) The submitted claim that the constrained-graviton (transverse-traceless) tensor structure

leaves the order-$\kappa^3$ cubic horizon divergence robust is **wrong**: the decisive contraction $\Pi_{ij,kl}\hat{n}^i\hat{n}^j\hat{n}^k\hat{n}^l = 0$ in every dimension (the TT projector annihilates its own axis, $P_{ij}\hat{n}^j = 0$), so the TT structure **annihilates** the naive leading $\delta^{-2(d-1)}$ all-radial term rather than leaving it robust. The surviving same-power coefficient (full-Hessian contraction, not performed) is nonzero for transverse dim $\geq 3$ but **zero for transverse dim 2** (the central $D = 4/d = 3$ codim-2 case). **R2's escape is NOT closed on the failure side; it stays OPEN**, residual coefficient uncomputed and dimension-dependent. The $\ell_P$ collar floor / first perturbative failure-side foothold **reverts** `[INFERENCE, medium-high]` → `[INFERENCE, medium]`; $N_*$ and the rest of OP-44 untouched; the $\ell_P$ floor's *in-principle* status untouched. → *[notes* (Working note: *OP-44 R2 closed on the failure side: the constrained-graviton tensor structure does not cancel the cubic boost-charge horizon divergence* (repo: `notes/2026-06-10-iter9-op44-graviton-escape.md`))]

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## Iteration 10 progress (2026-06-10)

*From the carrier-gap lever iteration (3 refereed attack tracks; see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter10-*). Verdict unchanged (ninth consecutive confirmation); the principal hedge's grade AND condition unchanged — the designated verdict-bearing lever (close the commutant-to-generated gap) was executed and did not close; two submitted outcome labels corrected by the referee. No new numeric IDs consumed (next free remains: OP-51, A-24, GC-18, H9).*

- **OP-46 / the carrier problem — the designated verdict-bearing lever (close the commutant-to-generated gap → unconditional HIGH) was executed and did NOT fire; the hedge sheds no condition; no carrier lives in the gap.** *(Targets the OP-46 residual lever, the verdict-flipper; verdict-neutral outcome.)* `[INFERENCE – verdict-neutral null result; closure unchanged, conditional on the net-naturality theorem + the located commutant-to-generated` `gap]` (A1) The claim that ergodicity of the Bisognano–Wichmann vacuum empties the iteration-9 commutant-to-generated gap (repairing the Lemma-N1 deficiency, so the hedge sheds its commutant-gap condition) is **STRUCK as a binding major error**: ergodicity empties the *centralizer*$\mathcal{M}_\omega = \mathcal{M} \cap \{\Delta^{it}\}'$ (Borchers Lebesgue boost spectrum; Longo 1978), but the iteration-9 gap is the *representation-space* multiplicity gap $\langle\Delta\rangle'' \subsetneq \{\Delta^{it}\}'$ driven by spectral degeneracy of Δ, and since $\Delta \notin \mathcal{M}$ the gap is non-empty even in the BW case and contains the canonical indefinite $\eta_0 = \mathrm{sgn}(\ln \Delta)$. **HYP-CKV-VACUITY stays at -R6** (conditional HIGH on $n_1$ AND the located commutant-to-generated gap); it does NOT advance to "$n_1$ alone." (A2) No carrier lives in the gap (**RESOLVED-NEGATIVE**, referee-upheld with two corrections): where geometry lives the centralizer is trivial (Lebesgue spectrum, no modular eigenvectors — Shlyakhtenko free-quasi-free structure theory; Houdayer 0809.3827 / Boutonnet–Houdayer 1602.01741 corroborate), so no algebra-internal carrier; where the gap is non-empty (almost-periodic sector) the strongest-ever candidate $\mathrm{sgn}(h) \in \mathcal{M}$ clears bars (i)/(iv) by an inner automorphism but is geometry-void ($\Delta_{\text{geo}} = 0$). **Corrections (binding):** the complete $\mathrm{Aut}(\mathcal{M}, \omega)$-invariant is the finer per-modular-eigenspace signature, not the global $(p, q)$ pair (4 conjugacy classes in the toy, not 1; Skolem–Noether); the "boost-orbit-constant ⟹ wedge-filling" mechanism is struck (sector-inconsistent), localization-failure surviving via the inherited $n_1$ presupposition. **Combined:** the commutant residual is substantially closed for algebra-internal (Ad-inner) carriers, but a residual sliver — algebra-compatible carriers in $\{\Delta^{it}\}' \setminus \mathcal{M}$ — is NOT discharged, so the hedge narrows substantially toward but **not provably to $n_1$ alone**; grade and condition unchanged. Two kept sub-results re-establish (not advance) the picture: the III1 centralizer is genuinely large for degenerate spectrum (integrable weight ⟹ Type $\mathrm{II}_\infty$ centralizer, Connes–Takesaki flow of weights; Popa realization, arXiv:math/0011084) and the state restricts to a faithful normal trace on it (Takesaki, semifinite). Carrier-convergence count **stays at five** (centralizer-completion of route (5)). → *[notes* (Working note: *OP-46 / carrier obligation (i), iteration 10: the net-naturality theorem (negative horn) — the closure attempt was STRUCK (the gap does NOT close); hedge stays at R6* (repo: `notes/2026-06-10-iter10-naturality-commutant-gap.md`)), *notes* (Working note: *OP-46 / carrier obligation, iteration 10: the positive dual executed — the commutant-to-generated gap is geometry-void too (centralizer carrier sgn(h) is empty where geometric, universal-and-non-local where it exists)* (repo: `notes/2026-06-10-iter10-carrier-in-the-gap.md`))]

- **OP-44 R2 — the decidable D=4 cubic-coefficient sub-computation does NOT resolve negative; R2 stays SHARPENED-OPEN, the iteration-9 posture upheld.** *(Moves the OP-44 residual lever; outcome neutral on the standing verdict.)* `[SHARPENED-OPEN]` (A3, submitted RESOLVED-NEGATIVE → **overturned to SHARPENED-OPEN** by the referee, REFUTE SUSTAINED) The attempt to re-harden the R2 obstruction at the physical $D = 4$ ($m = 2$) by substituting a symmetric-traceless-on-full-cut graviton projector $\Pi^{ST}$ for iteration-9's radial-subtracting $\Pi^{\text{rad}}$ fails on the physics (the coincidence limit has $\hat{k} \parallel \hat{n}$, so the physically-correct TT projector IS radial-subtracting; the all-radial term is annihilated and the full Hessian vanishes at $m = 2$ in every $D$) and, independently, on the arithmetic (even granting $\Pi^{ST}$, the genuine leading $s^2$ coefficient is $(m-1)(m-2)/2$, which vanishes at $m = 2$; only an $s^4/2$ remnant survives). The submission's "$\Pi^{\text{rad}} \Rightarrow 0$ graviton polarizations in 4D" reductio is a category error. **Net:** the $\ell_P$ collar floor in 4D is NOT re-established by this track; the R2 obstruction horn does NOT re-harden; OP-44 R2's failure-side foothold stays `[INFERENCE, medium]`. $N_*$, the rest of OP-44, and the OP-46 hedge untouched; carrier-convergence count stays FIVE. → *[notes* (Working note: *OP-44 R2 D4 coefficient: the surviving cubic-divergence coefficient is nonzero at the physical D=4 — the iteration-9 transverse-dim-2 vanishing used the wrong graviton* *projector; the ell_P collar floor stands* (repo: `notes/2026-06-10-iter10-op44-R2-D4-coefficient.md`))]

## Iteration 11 progress (2026-06-10)

*From the carrier-reduction + net-completeness + terminal-saturation-audit iteration (3 refereed attack tracks; see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter11-*). Verdict unchanged (tenth consecutive confirmation); the principal hedge's grade AND registry-condition unchanged — the located commutant-gap residual re-grounds its vocabulary on a NAMED external open problem (Connes' bicentralizer conjecture, OPEN in general as of 2026) without closing to a theorem; A1's submitted logical-equivalence reduction was MAJOR-CORRECTED by the referee. Terminal analytical saturation CONFIRMED. No new numeric IDs consumed (next free remains: OP-51, A-24, GC-18, H9).*

- **OP-46 / the carrier problem — the inner (algebra-internal, J∈M) horn re-grounds on Connes' bicentralizer problem; it does NOT close to a proved theorem; the hedge sheds no condition.** *(Targets the OP-46 residual lever, the verdict-flipper; verdict-neutral outcome.)* `[INFERENCE — verdict-neutral; closure unchanged, conditional on the net-naturality theorem + the located commutant-to-generated gap]` (A1) The algebra-internal carrier horn lives in the modular centralizer $M_\psi = \mathcal{M} \cap \{\Delta_\psi^{it}\}'$, so centralizer/bicentralizer structure theory is the correct vocabulary for the iteration-9 commutant-gap residual. **Binding referee correction (centerpiece MAJOR-CORRECTED):** the submitted "the carrier no-go REDUCES (logical equivalence) to Connes' bicentralizer problem" is **downgraded to "is RELATED TO centralizer/bicentralizer structure theory"** — the spine conflated trivial centralizer ($M_\psi = \mathbb{C}1$, killing a carrier, **unconditionally achievable on every $\text{III}_1$ factor** — Marrakchi–Vaes, Crelle 809 (2024) 247, Thm A) with

Haagerup's bicentralizer-triviality (an irreducible/large-centralizer state in which a carrier **exists**) — opposite conditions, so no equivalence. **Second binding correction:** the mission's "Connes' bicentralizer problem RESOLVED by Houdayer–Marrakchi–Tomatsu" premise is **FALSE**; the conjecture is **OPEN in general as of 2026** (only large classes and equivalent reformulations settled; Marrakchi 2308.15163 solves Kadison's problem only for $\mathrm{III}_1$ factors *satisfying* it). **Net:** the located commutant-gap residual now connects to a named, externally-tracked open problem with its own resolution clock (cleaner and more honest), but **HYP-CKV-VACUITY stays at -R6** — grade MEDIUM-HIGH→HIGH and registry condition both UNCHANGED — and the no-go does NOT close to a theorem; no R-advance. The full no-go stays **two-bar**: the bicentralizer route touches only the inner horn (and there as a loose structural connection, not a reduction); the $\{\Delta^{it}\}' \setminus \mathcal{M}$ sliver ($\eta_0 = \mathrm{sgn}(\ln \Delta)$ and its $J$-mirror) is disposed of separately by non-localizability + algebra-incompatibility (iters 7/9/10), unchanged. Carrier-convergence count **stays at five** (reduction-to-known-problem framing of route (5), not a sixth route). → *[notes* (Working note: *OP-46 / carrier obligation, iteration 11 (track A1): the carrier no-go's inner horn is RELATED TO Connes' bicentralizer problem (structural connect…* (repo: `notes/2026-06-10-iter11-carrier-bicentralizer.md`))]

- **Reverse-Weinberg–Witten — re-grounded from a third (holography / T-duality / Tsirelson) direction; no witness; stays `[OPEN, as of 2026-06]`, un-registered, NOT OP-50.** *(Targets the §8 reopening condition / HYP-ENCODING-SCREEN; verdict-neutral.)* `[INFERENCE, high]` (A2) Three independent handles on "is net+vacuum a complete invariant" all land on the existing carrier gate without flipping it: holography of information is a STATE-completeness theorem conditional on the gravitational Gauss law (= the geometry), and its Dec-2025 failure mode (Geng–Jafferis–Shrivastava–Tata, arXiv:2512.18912) reproduces the carrier gap via an emergent localized observer installed by background dressing; T-duality/mirror symmetry shows net+vacuum does not determine the target GEOMETRY but class-symmetrically (no class-separating observable), supporting the emergent-geometry ontology; MIP*=RE (arXiv:2001.04383) underdetermines the split/tensor (Type-I) structure — the OP-48 collar — not net-completeness. The iteration-9 RWW≡carrier identity is independently corroborated; reverse-WW stays a literature-absence `[OPEN, as of 2026-06]`, NOT a structural impossibility, NOT OP-50; count stays FIVE; no hedge moves. New sub-question NQ-1 (does the holographic-completeness *failure* regime coincide with carrier existence?) recorded un-IDed inside HYP-ENCODING-SCREEN's portability corollary. → *[notes* (Working note: *Net + Vacuum as a Complete Invariant: Three Independent Handles, One Carrier Gate (2026-06-10)* (repo: `notes/2026-06-10-iter11-net-completeness-handles.md`))]

- **Terminal analytical saturation CONFIRMED — every verdict-bearing residual is decidable only by external input; no reasoning lever capable of moving the verdict or the hedge remains.** *(Meta-audit; moves no lever by design.)* `[INFERENCE, high]` (A3) Of the seven OPEN/decidable residuals, two (OP-44 R2 higher-D coefficient; OP-49 a/b/c′) are reasoning-decidable but headline-inert by their own scoping; the other five are gated on external input — four on the same carrier/net-naturality math breakthrough, OP-50 on a community resolution that cannot harden the headline, and the 2027 DESI $w(z)$ verdict bears on the target not the wager. Marginal-yield trend monotone to zero (grade-derivative zero at iter 9, condition-derivative zero at iter 10; iter 11 added precision-only re-grounding, no R-advance). A targeted 2026-06 live search found no decidable external lever (nearest constructions arXiv:2504.06698, arXiv:2602.11806 evade rather than invert the encode pattern). **Recommendation: consolidate + watch-mode on the carrier problem and the 2027 DESI verdict; no genuine remaining decidable verdict-moving iteration.** $N_*$, OP-44 R2, the OP-46 hedge, and the watchlist untouched. → *[notes* (Working note: *Terminal-*

*saturation audit — is the analytical program done?* (repo: `notes/2026-06-10-iter11-terminal-saturation-audit.md`))]

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## Iteration 12 progress (2026-06-19)

*From the physics-net-bicentralizer-lever + external-sweep iteration (3 refereed tracks, all UPHELD-WITH-CORRECTION; see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter12-*), plus a read-only frontier scout that closed iteration-11 A1 subquestion #3. Verdict unchanged (ELEVENTH consecutive confirmation); the principal hedge's grade AND condition unchanged; terminal analytical saturation re-confirmed as of 2026-06. No new numeric IDs consumed (next free remains: OP-51, A-24, GC-18, H9).*

- **OP-46 — iteration-11's "sharp next lever" (physics-net relative-bicentralizer triviality) RETIRED as a NON-LOAD-BEARING lever; the no-go for physics nets stands on the inherited bars.** *(Targets the OP-46 residual; verdict-neutral.)* `[INFERENCE — verdict-neutral; closure unchanged, conditional on the net-naturality theorem + the located commutant-to-generated gap]` (B1, referee UPHELD-WITH-CORRECTION) The physics-relevant local algebras $\mathcal{M}(O)$ are injective $\mathrm{III}_1$, so the bicentralizer $B(\mathcal{M}(O), \varphi) = \mathbb{C}1$ unconditionally (Haagerup affirmative-for-injective) — but that is the wrong object. The actual gating object is the **relative** bicentralizer of the irreducible, infinite-index, **non-discrete** inclusion $\mathcal{M}(O) \subset \mathcal{M}(W)$, which is **NOT settled** by Ando–Haagerup–Houdayer–Marrakchi (Math. Ann. 376 (2020) 1145), whose relative-bicentralizer theorem is restricted to **discrete** inclusions with $N$ amenable. So the triviality question is **GENUINELY OPEN as pure operator algebra** (record never as "trivial" nor "settled"). It is **non-load-bearing** because, per the binding iteration-11 corrections, closing the inner carrier horn needs a **trivial centralizer** $M_\psi = \mathbb{C}1$ — unconditional on every $\mathrm{III}_1$ factor (Marrakchi–Vaes Thm A, no bicentralizer hypothesis) — while **trivial (relative)**

**bicentralizer is the carrier-FRIENDLY** condition (Haagerup's large-centralizer biconditional yields a carrier-admitting state). So resolving it either way cannot close the no-go, which holds for physics nets on the inherited geometry-void / non-localizability bars (the iteration-9 boost-no-fixed-locus engine). HYP-CKV-VACUITY stays **-R6**; count stays **FIVE**. *(Citation-precision repair of the standing record: the iter-11 A1 claim that Marrakchi–Vaes Thm A is "conditional on the bicentralizer conjecture" is corrected — Thm A is* ***unconditional****.)→ [notes* (Working note: *Track B1 (iteration 12): the relative-bicentralizer lever for physics nets is NON-load-bearing — subquestion #1 closes NEGATIVE* (repo: `notes/2026-06-19-iter12-physics-net-bicentralizer-lever.md`))]

- **OP-46 — the indefinite-metric / PT-symmetric / Krein-space carrier route (iteration-11 A1 subquestion #3) CLOSED-NEGATIVE.** *(Targets the OP-46 residual / HYP-FACTORIZATION-IS-GEOMETRY; verdict-neutral.)* `[INFERENCE, high]` (read-only frontier scout) Every verified indefinite-metric / PT-symmetric / Krein modular structure collapses to encode-not-generate by one of **three mutually-exhaustive mechanisms**: **(A) gauge artifact** (Strocchi's theorem forces the indefinite metric for locality+covariance, but Gupta–Bleuler/BRST mod it out — $\mathcal{H}_{\mathrm{phys}} = \ker Q_B / \mathrm{im}\, Q_B$ is positive-definite; no content reaches the observable net); **(B) installed, not derived** (Gottschalk's BW-on-Krein arXiv:math-ph/0408048; the rank-1-only $\Pi_1$ J-symmetric Tomita theorem, which fails the $(\infty, \infty)$ signature bar; Sablevice–Millington pseudo-Hermitian QFT arXiv:2307.16805, global/non-localized; Kawamura — all run Krein→modular with $J$ as input); **(C) $\eta_0$-equivalent / positivity-recovering / complex-pseudo-entropy** (Takesaki KMS-uniqueness forces a Hermitian $\Delta$ where the metric is genuinely modular; the metric is non-unique à la Mostafazadeh, mirroring the iter-7 $\eta_0$ non-uniqueness; genuinely-indefinite instances give complex pseudo-entropy, already closed at OP-49). The structural closer: indefiniteness enters only by being **built from** positive modular data (→ iter-7 trichotomy, $\eta_0$-class, void) or **installed by hand** (→ gauge or geometry-smuggling); non-self-adjointness is governed by the Connes cocycle (a unitary *in*$\mathcal{M}$), not a new indefinite localized involution. The indefinite-metric route is the encode-not-generate screen in Krein vocabulary; **count stays FIVE** (same route-5 carrier problem, not a sixth route). → *[notes* (Working note: *Scoping memo — iter-11 A1 subquestion #3: does a non-self-adjoint / PT-symmetric / Krein-space relative modular structure furnish a CARRIER outside...* (repo: `notes/2026-06-19-iter12-indefinite-metric-carrier-closed.md`))]

- **Net-naturality residual + reverse-WW — no 2023–2026 external math; terminal analytical saturation re-confirmed.** *(Meta; moves no lever.)* `[INFERENCE, high]` (B2/B3, both CONFIRMS-SATURATION) A web-verified 2023–2026 sweep found no result closing/refuting/addressing the net-naturality theorem or the commutant-to-generated gap; the one genuinely-new 2026 carrier-adjacent mathematics (Morinelli–Neeb–Olafsson Euler-pair geometric-BW / Spin–Statistics theorems, arXiv:2603.26390 / 2508.10960) is a cleaner statement of HYP-ENCODING-SCREEN (symmetry = input), NOT a carrier, NOT a sixth route. Connes' bicentralizer re-confirmed OPEN in general (Ando–Goldbring, arXiv:2605.12776). Every verdict-bearing residual stays external-math- or external-experiment-gated; the marginal-yield curve is flat to second order. Reverse-WW stays `[OPEN, as of 2026-06]`, un-registered, NOT OP-50. → *[notes* (Working note: *Track B2 (iteration 12): the 2023-2026 external-math sweep finds no lever on the net-naturality residual - CONFIRMS-SATURATION* (repo: `notes/2026-06-19-iter12-external-math-sweep.md`)), *notes* (Chapter 32 (p. 648))]

## Iteration 16 progress (2026-07-03) — the multi-wedge closure; the OP-46 residual pinned to (E_O)

*From the clock-(1)-triggered iteration (tracks G1–G4, referees R1/R2/R3 binding; see notes/ (repo:* `notes/`*; see the notes index, repo:* `notes/README.md`*) iter16-* and the synthesis (Chapter 36 (p. 684))). Verdict unchanged (FIFTEENTH consecutive confirmation) — but the principal hedge advances for the first time since iteration 9. No new numeric IDs consumed (next free remains: OP-51, A-24, GC-18, H9); three named items enter (LEM-NET-NATURALITY-G, (E_O), DCNG).*

- **OP-46 — the commutant-to-generated gap CLASSIFIED (not closed); the residual is now the single named open hypothesis (E_O).** *(The verdict-bearing move; referee R1 UPHELD-WITH-CORRECTIONS.)*

  `[INFERENCE, high – referee-verified proof package]` The multi-wedge naturality lever (intersecting the constraints of ALL wedge modular flows — genuinely unused in iterations 9–10, which were single-Δ) yields **LEM-NET-NATURALITY-G**: **(G-loc)** every localized natural carrier is ±1 (proof-grade: Borchers scaling ⟹ $\mathrm{Fix}(\Delta_W^{it}) = \mathbb{C}\Omega$ ⟹ vacuum pinning ⟹ Reeh–Schlieder kill); **(G-glob)** the wedge boosts of all translated wedges generate $\mathcal{P}_+^\uparrow$, so every fully-natural non-localized algebra-compatible carrier is a gauge-symmetry implementer with Poincaré-invariant eigenprojections, $\Delta_{\mathrm{geo}} = 0$ (proof-grade; the sliver is NON-empty — Fock parity $(-1)^N$ — and exhaustively geometry-void). The sole surviving branch is massive double-cone fibers, pinned to **(E_O)**: vacuum ergodicity on massive double-cone algebras, $M(O)_\omega = \mathbb{C}1$, `[OPEN]`. Direction: (E_O) ⟹ the carrier no-go is a theorem on the stated hypotheses; ¬(E_O) necessary-but-NOT-sufficient for a counterexample. **HYP-CKV-VACUITY advances R6 → R7** (condition {n1, (E_O)}; grade unchanged, conditional HIGH). → *[notes* (Working note: *Track G1, iteration 16: direct assault on* *the net-naturality theorem — Lemma G formalized; the multi-wedge lever closes the non-localized sliver; the localized horn is a three-line theorem; the residual shrinks to double-cone vacuum ergodicity (E_O)* (repo: `notes/2026-07-03-iter16-G1-multiwedge-naturality.md`))]

- **OP-46 constructive side — three untried constructions BLOCKED; the state-family extension closed by a new trichotomy.** `[INFERENCE, high]` (Referee R2 UPHELD-WITH-CORRECTIONS.) State-family naturality (the co-moving evasion of Lemma N2 — real off-vacuum) closes via Lemma S2 (sketch-grade; an *extension* of the posed conjecture, N2 and the dossier untouched); the Doplicher–Longo flip fails algebra-compatibility without an installed leg-identification η; LEM-CORE-DESCENT (proof-grade transport corollary) exhausts core-built carriers. → *[notes* (Working note: *Iteration 16 (track G2): the constructive direction — three untried carrier constructions (state-family naturality, split-inclusion canonical data, core-descent), executed and graded* (repo: `notes/2026-07-03-iter16-G2-constructive-carrier.md`))]
- **The Marrakchi lever (watch-sweep #1's sharpened target) — RESOLVED-NEGATIVE at the level of the object; target re-sharpened.** `[INFERENCE, high]` (Referee R3 SUSTAINED.) The "with expectation" hypothesis of arXiv:2606.23636 is pervasive AND definitional (the flow is undefined without it); for the physics inclusions no ambient state's modular flow preserves M(O), so the sweep-#1 target was **ill-posed** — superseded by the three-prong + (E_O) + DCNG watch state (*synthesis §6* (Chapter 36 (p. 684))). Standing-record repairs: "irreducible" for M(O) ⊂ M(W) is FALSE (collar algebra in the relative commutant; conclusions survive via Takesaki's criterion); Doplicher–Longo is Invent. Math. **75** (1984). → *[notes* (Working note: *Track G3 (iteration 16): full technical audit of Marrakchi arXiv:2606.23636 — the expectation hypothesis is pervasive-and-definitional, the physics-net transfer fails at the level of the object, and the AHHM hyperfinite-subfactor output is*

*VACUOUS for local inclusions* (repo: `notes/2026-07-03-iter16-G3-marrakchi-transfer.md`))]

- **Untried-readout sweep — 7 of 8 frameworks REPRODUCE-SCREEN; one conditional survivor (DCNG).** `[INFERENCE, high]` (Referee R3 SUSTAINED-WITH-CORRECTIONS.) NC Dirichlet forms, free Araki–Woods, infinite-index subfactor invariants, entropic order parameters, QRF crossed products, net cohomology, and the 2024–26 web sweep all close at a located smuggle step. Survivor: **DCNG** (definable-carrier no-go; continuous model theory makes the gap a *definability* question; no-go direction only; gated on an $\eta_0$-definability lemma) `[OPEN — new named target]`. Morinelli–Neeb (Adv. Math. **458** (2024) 109960) prove the two-directional, BW-conditional interchangeability of the two smuggle items — theorem-level, deriving neither from (algebra, state). → *[notes* (Working note: *Iter-16 Track G4: the untried-readouts sweep — eight candidate frameworks graded against the carrier screen, and a definable-carrier no-go as the surviving lever* (repo: `notes/2026-07-03-iter16-G4-untried-readouts.md`))]

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## Status summary (epistemic tags at the *Last updated* date)

| TAG | IDs |
|---|---|
| `[OPEN]` | OP-1, OP-5, OP-6*, OP-7, OP-8, OP-9, OP-10, OP-11, OP-12, OP-14, OP-15, OP-16, OP-17, OP-20*, OP-24, OP-25*, OP-26, OP-27*, OP-28*, OP-29, OP-31*, OP-32, OP-33, OP-34, OP-35, OP-36, OP-37, OP-38 |
| `[CONTESTED]` | OP-4, OP-18*, OP-21*, OP-22, OP-30 |
| `[SPECULATIVE]` | OP-19, OP-23 |
| Bundles (mixed) | OP-13, OP-39 |

* carries a secondary tag noted in its entry (e.g. `[OPEN]/[CONTESTED]`). The single quantities/processes that are themselves `[ESTABLISHED]` — cosmic acceleration (OP-6), the missing-mass evidence (OP-5),

nonzero neutrino mass (OP-9), the baryon asymmetry (OP-7), non-renormalizability of perturbative GR (OP-1) — are flagged inside each entry; what is [OPEN] is the *mechanism/identity*, not the phenomenon.

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## References

Specific landmark results referenced inline (Goroff–Sagnotti two-loop divergence; Weinberg's anthropic bound; Strominger–Vafa microstate counting; Christodoulou–Klainerman stability of Minkowski; Xia's non-collision singularities; the Clay Millennium Problems for Yang–Mills and Navier–Stokes; Gleason's theorem; Hardy and Chiribella–D'Ariano–Perinotti reconstructions; Jacobson's thermodynamic derivation; Ryu–Takayanagi) are collected with full bibliographic detail in the Bibliography (p. 776). Per *Chapter 2 (p. 8)* §4, no citation, number, or theorem on this page is invented; quantitative figures carry their context, and any bibliographic detail not independently verified should be marked `[unverified]` in BIBLIOGRAPHY.md.

CHAPTER 16

# Constants, Scales, and Free Parameters

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This page catalogs the dimensionful and dimensionless parameters of fundamental physics, the unit conventions that distinguish *physical* content from *bookkeeping*, the characteristic scales (especially the Planck scale), the free-parameter count of the Standard Model + ΛCDM concordance cosmology, and the deepest fine-tuning puzzles (hierarchy, cosmological constant). It is the quantitative companion to *Chapter 3 (p. 15)*, *Chapter 18 (p. 349)*, and the *Chapter 20 (p. 458)*. For the meaning of epistemic tags see *Chapter 2 (p. 8)*.

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## 1. Dimensionful constants: physics versus convention

The "fundamental constants" most often quoted are:

| SYMBOL | NAME | ROLE |
|---|---|---|
| $c$ | speed of light | limiting speed; spacetime metric signature scale |
| $\hbar$ | reduced Planck constant | quantum of action |
| $G$ | Newton's gravitational constant | strength of gravity |
| $k_B$ | Boltzmann constant | energy-per-temperature conversion |
| $e$ | elementary charge | quantum of electric charge |

A crucial and frequently muddled point: **only dimensionless ratios carry frame- and unit-independent physical content**

`[ESTABLISHED]`. The numerical *value* of a dimensionful constant depends on the chosen unit system and can be altered (even set to 1) by a change of units without changing any predicted experimental outcome. The cleanest articulation is due to Duff, Okun, and Veneziano, who debated how many dimensionful constants are "truly fundamental" and concluded that the number of *operationally meaningful* constants is a matter of convention; the physically invariant content lives entirely in dimensionless combinations `[CONTESTED in emphasis, but the operational core is ESTABLISHED]`. See the Glossary (p. 747) for "dimensionless constant."

Concretely:

- $c$ and $\hbar$ and $k_B$ are best understood as **unit-conversion factors**: $c$ converts seconds to meters (relating the time and space components of one four-vector), $\hbar$ converts energy to frequency, and $k_B$ converts temperature to energy `[INFERENCE — widely accepted; this is the logic behind natural units]`. Setting $c = \hbar = k_B = 1$ removes them entirely without loss of physics.
- $G$ sets the scale at which gravity becomes strong relative to quantum effects; its dimensionful value defines the Planck scale (§3). Whether $G$ is "more physical" than $c$ or $\hbar$ is itself partly conventional, but it cannot be removed *simultaneously* with $c$ and $\hbar$ — it instead fixes the unit of mass/length/time once those are chosen.
- $e$ appears physically only through the dimensionless fine-structure constant $\alpha$ (§6).

This is why one can build a system (Planck units, §3) in which $c = \hbar = G = k_B = 1$: dimensionful constants are partly the residue of historically chosen units, and partly markers of where one regime of physics crosses into another.

> Caveat to flag `[ESTABLISHED]`: a *variation* of a single dimensionful constant (e.g. "did $c$ change?") is not, by itself, an operationally well-posed question without specifying which dimensionless ratio is held fixed. Only variation of

dimensionless constants is observable (§7). This is the "constants of nature are truly constant" assumption examined in the *Chapter 20 (p. 458)*.

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## 2. The 2019 SI redefinition

Since 20 May 2019, the SI base units are **defined by fixing exact numerical values of seven defining constants** rather than by physical artifacts `[ESTABLISHED]`. The relevant fixed values are:

$$\begin{aligned}
\Delta\nu_{\mathrm{Cs}} &= 9\,192\,631\,770\ \mathrm{Hz} \quad (\text{Cs-133 hyperfine transition}),\\
c &= 299\,792\,458\ \mathrm{m/s},\\
h &= 6.626\,070\,15 \times 10^{-34}\ \mathrm{J{\cdot}s},\\
e &= 1.602\,176\,634 \times 10^{-19}\ \mathrm{C},\\
k_B &= 1.380\,649 \times 10^{-23}\ \mathrm{J/K},\\
N_A &= 6.022\,140\,76 \times 10^{23}\ \mathrm{mol}^{-1},\\
K_{\mathrm{cd}} &= 683\ \mathrm{lm/W}.
\end{aligned}$$

These are now **exact by definition** (zero uncertainty), which inverts the historical relationship: the kilogram is defined *via* the fixed $h$ (and the Kibble balance / Avogadro sphere realize it), not the other way around `[ESTABLISHED]`.

Interpretive note `[INFERENCE]`: this codifies the §1 lesson institutionally. Fixing $c$, $h$, $e$, $k_B$ turns them into pure conventions of the unit system; what remains experimentally determined are the *dimensionless* constants and the values of constants — such as $G$ and the fine-structure constant $\alpha$ — that the SI does **not** fix. In particular $G$ remains the least precisely measured fundamental constant (relative uncertainty $\sim 10^{-5}$, with persistent inter-experiment discrepancies) `[ESTABLISHED]`, and $\alpha$ is measured, not fixed.

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## 3. Natural units and the Planck scale

**Natural (particle-physics) units** set $c = \hbar = 1$, so masses, momenta, energies, and inverse lengths/times are all measured in energy units (typically GeV). Useful conversions `[ESTABLISHED]`:

$$\hbar c \approx 0.1973\ \text{GeV}\cdot\text{fm}, \qquad (\hbar c)^2 \approx 0.3894\ \text{GeV}^2\,\text{mb}.$$

**Planck units** further set $G = k_B = 1$, defining scales from $c, \hbar, G$ alone:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \approx 1.616 \times 10^{-35}\ \text{m}, \qquad t_P = \sqrt{\frac{\hbar G}{c^5}} \approx 5.39 \times 10^{-44}\ \text{s},$$

$$M_P = \sqrt{\frac{\hbar c}{G}} \approx 2.176 \times 10^{-8}\ \text{kg} \approx 1.22 \times 10^{19}\ \text{GeV}/c^2,$$

$$T_P = \frac{M_P c^2}{k_B} \approx 1.42 \times 10^{32}\ \text{K}.$$

(Conventions vary: the *reduced* Planck mass $\overline{M}_P = M_P/\sqrt{8\pi} \approx 2.435 \times 10^{18}$ GeV is standard in cosmology and quantum gravity, because $8\pi G$ appears in the Einstein equations. Always check which is meant — see *Chapter 16 (p. 308)* §3 here as the canonical reference.)

**What the Planck scale means.** `[INFERENCE – strong consensus]` The Planck energy $M_P c^2 \sim 1.22 \times 10^{19}$ GeV is the scale at which the dimensionless gravitational coupling of a process, $\sim (E/M_P c^2)^2$, becomes $O(1)$: graviton exchange is no longer perturbatively small, and the effective-field-theory treatment of general relativity as a quantum field theory loses predictivity (an infinite tower of counterterms with uncontrolled coefficients). This is the clearest *internal* signal that quantum field theory on a fixed background must be replaced; see *Chapter 8 (p. 89)* and *Chapter 10 (p. 121)*.

**Critical caveat — what the Planck scale does NOT establish** `[ESTABLISHED that this is the correct caution]`:

- It is **not** a proven minimum length. The frequent claim "$\ell_P$ is the shortest measurable distance" is a *heuristic/conjecture*, motivated by gedanken arguments (a probe localizing to $\ell_P$ has energy $\sim M_P$, forming a black hole) and realized in *some* quantum-gravity frameworks (generalized uncertainty principles, string theory's minimal length, loop quantum gravity's discrete area/volume spectra). But no framework with this feature is experimentally confirmed, and a fundamental minimum length is in tension with exact Lorentz invariance unless implemented carefully (e.g. via deformed/doubly-special relativity, which is `[SPECULATIVE]`). Tag: `[SPECULATIVE / OPEN]`.
- The Planck scale marks where *naive perturbative* gravity-as-EFT fails, **not** necessarily where new physics first appears. New physics may intervene far below $M_P$ (a GUT scale $\sim 10^{16}$ GeV, a seesaw scale $\sim 10^{9\text{–}15}$ GeV, TeV-scale states), or — in scenarios like large extra dimensions or asymptotic safety — the *effective* scale of strong gravity could differ from $M_P$ `[SPECULATIVE]`. See *Chapter 15 (p. 233)*.

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# 4. Free parameters of the Standard Model

The Standard Model is a renormalizable QFT (see *Chapter 9 (p. 106)*) whose Lagrangian, once the gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$ and the fermion representations are fixed, contains a finite set of numbers determined only by experiment. The commonly cited count is **~19 free parameters for the minimal SM** (massless neutrinos), grouped as follows `[ESTABLISHED as a counting fact, modulo convention]`:

| Group | Count | Parameters |
|---|---|---|
| Gauge couplings | 3 | $g_s, g, g'$ (equivalently $\alpha_s,\ \alpha_{\text{em}},\ \sin^2\theta_W$) |
| Charged-fermion masses / Yukawas | 9 | 6 quark + 3 charged-lepton masses |
| Quark mixing (CKM) | 4 | 3 angles + 1 CP-violating phase |
| Higgs sector | 2 | vev $v \approx 246$ GeV (or $\mu^2$) and quartic $\lambda$ (equivalently $m_h, v$) |

| | | |
|---|---|---|
| QCD vacuum angle | 1 | $\bar{\theta}_{\mathrm{QCD}}$ |
| **Total (minimal SM)** | **19** | |

> Convention warning [ESTABLISHED]: the exact integer depends on bookkeeping. Some authors quote 18 (omitting $\bar{\theta}$, since it is observationally consistent with zero — see strong-CP, §8 / *Chapter 15 (p. 233)*); some quote $G$ as a 20th if gravity is appended. Treat "19" as a convention, not a theorem.

**Neutrino sector.** Oscillations prove neutrinos are massive (the one laboratory-confirmed breakdown of the *minimal* SM — see *Chapter 14 (p. 191)*), adding parameters:

- **Dirac neutrinos:** $+3$ masses $+3$ PMNS angles $+1$ Dirac phase $= +7 \rightarrow$ **~26**.
- **Majorana neutrinos:** as above $+2$ additional Majorana phases $= +9 \rightarrow$ **~28**.

Whether neutrinos are Dirac or Majorana is [OPEN], decided in principle by neutrinoless double-beta decay (see *Chapter 9 (p. 106)*). So "the SM has ~26–28 parameters including neutrinos" is the honest statement [ESTABLISHED as counting; the *physics* of the neutrino sector is partly OPEN].

**Cosmology.** The base ΛCDM concordance model adds **6 parameters** (§5), giving the often-quoted grand total of **~26–31** for "SM + ΛCDM" depending on neutrino assumptions and on whether one counts $G$ and overlapping inputs. [ESTABLISHED as a rough convention; the precise number is not canonical.]

The brute existence of these ~19–31 *inputs* — and especially the wild hierarchies among them (the top/electron Yukawa ratio is $\sim 10^5$; see §6) — is the strongest sense in which the SM is *descriptive, not explanatory*. See *Chapter 15 (p. 233)* (the flavor puzzle) and *Chapter 19 (p. 392)*.

# 5. ΛCDM cosmological parameters

The base **six-parameter ΛCDM** model (see *Chapter 11 (p. 139)*) is conventionally specified as `[ESTABLISHED as the standard parametrization]`:

| PARAMETER | MEANING |
|---|---|
| $\Omega_b h^2$ | physical baryon density |
| $\Omega_c h^2$ | physical cold dark matter density |
| $100\,\theta_{MC}$ (or $H_0$) | acoustic angular scale / expansion rate |
| $\tau$ | reionization optical depth |
| $A_s$ (or $\ln(10^{10} A_s)$) | primordial scalar amplitude |
| $n_s$ | scalar spectral index |

Representative values (round numbers; consult primary sources for precision and error bars — see the Bibliography (p. 776)): the scalar tilt is measured to be slightly red, $n_s \approx 0.965$ `[ESTABLISHED]`; the universe is spatially flat to sub-percent level `[ESTABLISHED]`; and $H_0 \approx 67$–$73$ km/s/Mpc depending on probe — the unresolved **Hubble tension** at the $\sim 4$–$6\sigma$ level `[CONTESTED]`. Spatial curvature $\Omega_k$, the dark-energy equation of state $w$, neutrino mass $\sum m_\nu$, and extra relativistic species $N_{\text{eff}}$ are held fixed in the base model but are standard *extensions* (each a probe of physics beyond base ΛCDM). $N_{\text{eff}} \approx 3.044$ is the SM prediction for three neutrino species including non-instantaneous decoupling `[ESTABLISHED]`. See *Chapter 14 (p. 191)* for the $H_0$ and $S_8$ tensions.

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# 6. Important dimensionless numbers

These are the genuinely physical, unit-independent constants `[ESTABLISHED values; explanations as noted]`:

- **Fine-structure constant:**

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c} \approx \frac{1}{137.036} \approx 7.297 \times 10^{-3}.$$

It is the coupling of QED at low (zero-momentum) energy. It **runs**: at the $Z$ pole $\alpha(M_Z) \approx 1/128$ `[ESTABLISHED]`. The integer "137" is famously suggestive but has no known fundamental significance — attempts to derive it (Eddington and others) are regarded as numerology [CONTESTED → effectively OPEN, with strong consensus that no accepted derivation exists]. The value of $\alpha$ is *not* explained by the SM; it is an input (§4).

- **Proton-to-electron mass ratio:**

  $$\mu \equiv \frac{m_p}{m_e} \approx 1836.15.$$

  Most of the proton mass is QCD binding/gluon-field energy, *not* the sum of quark rest masses `[ESTABLISHED]`; thus $\mu$ is largely a statement about $\Lambda_{\text{QCD}}$ relative to the electron Yukawa, intertwining the strong sector with electroweak symmetry breaking.
- **Strong coupling:** $\alpha_s(M_Z) \approx 0.118$ `[ESTABLISHED]`, running to small values at high energy (asymptotic freedom) and to strong coupling near $\Lambda_{\text{QCD}} \sim 200$ MeV.
- **Weak mixing angle:** $\sin^2\theta_W \approx 0.231$ (scheme- and scale-dependent) `[ESTABLISHED]`.
- **Gravitational coupling of two protons:** $\alpha_G = Gm_p^2/(\hbar c) \approx 5.9 \times 10^{-39}$ `[ESTABLISHED]` — quantifying how absurdly weak gravity is at particle masses, and the numerical core of the hierarchy problem (§7).

These dimensionless numbers are the *real* content; everything in §1–§3 is, by contrast, partly convention. See *Chapter 17 (p. 325)*.

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# 7. The hierarchy / naturalness problem

### Statement

`[ESTABLISHED as a technical fact; CONTESTED as a "problem"]`: in the SM, the Higgs mass-squared receives radiative corrections that, in a cutoff regularization, scale quadratically with the cutoff,

$$m_h^2 = (m_h^2)_{\text{bare}} + \delta m_h^2, \qquad \delta m_h^2 \sim \frac{\Lambda^2}{16\pi^2}.$$

If $\Lambda \sim M_P$, reproducing the observed $m_h \approx 125$ GeV requires the bare term and the correction to cancel to $\sim 1$ part in $10^{34}$. Equivalently, the electroweak scale $v \approx 246$ GeV sits $\sim 10^{16}$ below $M_P$ with no symmetry protecting a scalar mass (unlike fermion masses, protected by chiral symmetry, or gauge-boson masses, protected by gauge symmetry).

**Epistemic status** `[CONTESTED]`: this is **not** a logical inconsistency and **not** a domain-of-validity failure — the SM is perfectly predictive (the cutoff sensitivity is absorbed into a counterterm). It is a *naturalness* expectation: that dimensionless ratios should be $O(1)$ absent a protecting mechanism. The LHC's exclusion of the simplest TeV-scale solutions (low-energy SUSY, technicolor, simplest composite Higgs) has intensified debate over whether naturalness is the correct guide at all, versus environmental/anthropic selection `[CONTESTED/OPEN]`. See *Chapter 19 (p. 392)*, *Chapter 15 (p. 233)*, and the EFT-decoupling entry of the *Chapter 20 (p. 458)*.

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# 8. The cosmological-constant problem

The observed dark-energy density is $\rho_\Lambda^{\text{obs}} \sim 10^{-47}\ \text{GeV}^4$ (equivalently a mass scale $\sim 2 \times 10^{-3}$ eV) `[ESTABLISHED]`. A naive QFT estimate of the vacuum (zero-point) energy density, cut off at a scale $M_{\text{cut}}$, gives $\rho_\Lambda^{\text{QFT}} \sim M_{\text{cut}}^4$. The disagreement is the **cosmological-constant problem** — widely called the worst quantitative mismatch in physics `[ESTABLISHED that it is a severe puzzle]`.

**The famous "$10^{120}$" — read carefully** `[ESTABLISHED, and the nuance is the important part]`:

- With $M_{\text{cut}} \sim M_P$: $\rho_\Lambda^{\text{QFT}} \sim M_P^4 \sim 10^{74}\ \text{GeV}^4$, so the ratio $\rho^{\text{QFT}}/\rho^{\text{obs}} \sim 10^{120}$. Hence the headline number.

- With $M_{\rm cut}$ at the electroweak/TeV scale: $\rho_\Lambda^{\rm QFT} \sim (10^3\ {\rm GeV})^4 = 10^{12}\ {\rm GeV}^4$, giving a ratio $\sim 10^{59\text{–}60}$.
- Even the QCD chiral condensate alone contributes $\sim (\Lambda_{\rm QCD})^4 \sim 10^{-3}\ {\rm GeV}^4$, already $\sim 10^{44}$ times too large.

**The key conceptual point** `[INFERENCE – broad consensus]`: the " $10^{120}$" is the ratio of a **cutoff-dependent, regularization-dependent** estimate to the observed value. It is *not* a clean prediction. The naive sum-of-zero-point-energies is not Lorentz-invariant as written (a momentum cutoff breaks Lorentz invariance), and a proper treatment (dimensional regularization) makes the vacuum energy proportional to the masses of the fields, not to a quartic cutoff — changing the numerology though not the qualitative disaster. The robust statement is therefore: *no symmetry or mechanism is known that cancels the large vacuum-energy contributions while leaving the tiny observed remainder*, and the contributions that survive any reasonable accounting (e.g. the electroweak and QCD phase-transition condensates) already overshoot by dozens of orders of magnitude `[ESTABLISHED]`. The problem genuinely lives at the **QFT/gravity interface**: it is about how vacuum energy *gravitates*, and may signal that the naive QFT vacuum-energy object is simply the wrong thing to feed into Einstein's equations `[INFERENCE/OPEN]`. See *Chapter 11 (p. 139), Chapter 8 (p. 89)*, and *Chapter 14 (p. 191)*.

Proposed resolutions — supersymmetry (would cancel boson/fermion zero-point energies if unbroken; broken SUSY leaves a residue still far too large), the string landscape + anthropic selection (Weinberg's anthropic *upper bound* on $\Lambda$ famously preceded the discovery of acceleration), sequestering, degravitation, modified gravity — are all `[SPECULATIVE]`; none is established. See *Chapter 18 (p. 349)*.

## 8.1 The problem without folklore — regularization-aware statement and 2024–2026 status *(added 2026-07-02)*

*Sharpens §8 into theorem-level / robust / folklore layers and records the current status of the serious approaches. Companion* *entry: Chapter 14 (p. 191) GC-3.*

**(a) Theorem-level** `[ESTABLISHED]`:

1. *Lovelock uniqueness:* in 4D the term $\Lambda g_{\mu\nu}$ is the unique divergence-free local addition to $G_{\mu\nu}$; a constant vacuum energy, if it gravitates, gravitates exactly as $\Lambda$.
2. *Lorentz invariance:* a Lorentz-invariant vacuum has $\langle T_{\mu\nu}\rangle = -\rho_{\text{vac}}\, g_{\mu\nu}$ $(w = -1)$ — vacuum energy is observationally indistinguishable from $\Lambda$.
3. *Weinberg's no-go* (Rev. Mod. Phys. **61**, 1 (1989); pedagogical: Padilla, arXiv:1502.05296), with its exact assumptions: in a **local 4D Lagrangian field theory** with (i) **finitely many fields**, (ii) a **translation-invariant vacuum** in which all fields and the metric are spacetime-constant, and (iii) field values fixed **dynamically by the field equations** (no external constraints), no adjustment mechanism yields zero vacuum energy at the stationary point without fine-tuning: the required trace condition forces a scale symmetry under which the vacuum energy transforms homogeneously, and its vanishing re-imports the original tuning. Every serious approach exits through a named assumption: dynamical relaxation violates (ii); unimodular gravity and sequestering violate (iii) via global/external constraints; extra dimensions and infinite towers violate (i)/4D-locality; anthropic selection steps outside the theorem's scope.

**(b) The genuine puzzle: radiative instability, not a failed prediction** `[ESTABLISHED]`. QFT does not predict a value of $\Lambda$; it is a renormalized input, like the electron mass. What fails is *stability*. In dimensional regularization — Lorentz-preserving, no quartic cutoff — each massive field contributes $\sim m^4 \ln \mu$ (Martin, arXiv:1205.3365; Burgess, arXiv:1309.4133): the electron loop alone overshoots $\rho_\Lambda^{\text{obs}}$ by roughly **30 orders of magnitude**, the top quark by roughly **55**. The renormalized $\Lambda$ must be retuned order-by-order in perturbation theory and again at every mass threshold — using only experimentally confirmed particles and no assumption about Planck-scale physics. Separately, the electroweak

($\sim (100\,\mathrm{GeV})^4$) and QCD ($\sim \Lambda_{\mathrm{QCD}}^4$) phase transitions shift the vacuum energy by regulator-independent amounts: a $\Lambda$ tuned small today was enormous before the transitions — the tuning is epoch-dependent, not a one-time convention. *Folklore layer:* the "$10^{120}$" is the ratio of a Lorentz-violating quartic-cutoff estimate to observation (§8); the honest known-physics mismatch is $10^{30}$–$10^{56}$.

**(c) Genuinely open** [OPEN]: whether $\langle T_{\mu\nu}\rangle_{\mathrm{vac}}$ is the object that sources gravity at all — semiclassical gravity is untested at vacuum-energy level; every approach below is an attempt to modify that coupling.

**(d) Status of the serious approaches (2024–2026):**

- **Unimodular gravity — verdict: reallocates, does not solve.** UG is equivalent to GR except $\Lambda$ enters as an integration constant/initial condition rather than a coupling; the status report finds *no further differences* at classical, semiclassical, or quantum level (Carballo-Rubio–Garay–García-Moreno, *Class. Quantum Grav.* **39**, 243001 (2022), arXiv:2207.08499; misconception-clearing: Bengochea et al., *JCAP* 11 (2023) 011, arXiv:2308.07360). Salvio (*Phys. Lett. B* **856** (2024) 138920, arXiv:2406.12958) states explicitly that UG *does not solve the new CC problem* — why the constant has its observed value — and resorts to anthropics for the value. The radiative instability of §(b) is moved into an undetermined constant, not removed. [ESTABLISHED as the literature verdict; CONTESTED whether "$\Lambda$ is technically natural in UG" counts as progress.]
- **Vacuum-energy sequestering** (Kaloper–Padilla, *PRL* **112**, 091304 (2014), arXiv:1309.6562; manifestly local version: Kaloper–Padilla–Stefanyszyn–Zahariade, *PRL* **116**, 051302 (2016), arXiv:1505.01492): cancels *matter-loop* vacuum energy including loop corrections and renders phase-transition contributions cosmologically innocuous; the original global version requires a finite spacetime 4-volume and predicts transient dark energy plus future collapse. Graviton loops are not covered by these foundational versions. Alive but minority; no established observational discriminant. [SPECULATIVE]

- **Running-vacuum models (RVM) vs DESI:** flipped/threshold RVM variants fit DESI DR2 BAO + Planck PR4 + SNe with dynamical-dark-energy evidence comparable to $w_0w_a$CDM, strongest with the DES-Y5 SN sample (de Cruz Pérez–Gómez-Valent–Solà Peracaula, arXiv:2512.20616, accepted 2026) — hence inheriting the SN-systematics fragility of the underlying preference (see (e)). `[CONTESTED]`
- **Swampland / dS-conjecture angle:** if metastable de Sitter is excluded `[SPECULATIVE]`, $\Lambda$ cannot be the endpoint and dark energy must roll; DESI-compatible constructions satisfying the distance, dS, and TCC conjectures exist (S-dual quintessence: Anchordoqui–Antoniadis–Lüst, arXiv:2503.19428). This *re-labels* the problem rather than solving it: quintessence presupposes the old-$\Lambda$ cancellation and has its own radiative-instability problem (field mass $\sim H_0 \sim 10^{-33}$ eV). `[SPECULATIVE]`
- **Anthropic/landscape:** unchanged from §10; note its one predictive success (Weinberg's bound) targets a *constant* $\Lambda$ and would be weakened if evolving dark energy is confirmed `[INFERENCE]`.

**(e) DESI evolving-dark-energy hints — what they would and would not change.** DESI DR2 BAO + CMB prefer $w_0 > -1$, $w_a < 0$ over $\Lambda$CDM at 3.1σ, and at 2.8–4.2σ when SNe are added, depending on sample (2.8σ Pantheon+, 3.8σ Union3, 4.2σ DES-Y5) (*Phys. Rev. D* **112**, 083515 (2025), arXiv:2503.14738); the signal is robust to parameterization choice and prefers a phantom crossing (arXiv:2503.14743). Counter-analyses: the preference is SN-sample-driven (Efstathiou, arXiv:2408.07175: a ~0.04 mag low/high-z offset in DES-Y5 suffices); it drops below 2σ under low-z SN recalibration (Huang–Cai–Wang, *Sci. China Phys. Mech. Astron.* **68** (2025) 100413, arXiv:2502.04212); and Bayesian model comparison finds DESI DR2+CMB *alone* modestly favors $\Lambda$CDM (log-Bayes $-0.57 \pm 0.26$), the $w_0w_a$ preference tracing to a 2.95σ DESI↔DES-SN5YR inter-dataset tension that largely vanishes with recalibrated DES-Dovekie SNe (ln B drops to −0.30±0.19) (Ong–Yallup–Handley, arXiv:2511.10631, revised 2026-05). `[CONTESTED]`;

). If evolving dark energy survives: (i) the **old problem** — §(b) — is untouched: a rolling component is *additive to*, not a substitute for, the cancellation `[ESTABLISHED logical point]`; (ii) the anthropic success story weakens `[INFERENCE]`; (iii) the preferred phantom crossing excludes any single minimally coupled quintessence field, pushing model space toward multi-field / interacting / modified-gravity / running-vacuum descriptions `[INFERENCE, conditioned on a CONTESTED observation]`; (iv) the coincidence ("why now") problem is re-posed, not solved.

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## 9. Do the constants vary?

Only variation of **dimensionless** constants is physically meaningful (§1). The most-constrained probes are $\alpha$ and $\mu = m_p/m_e$:

- The **Oklo natural fission reactor** ($\sim 2$ Gyr ago) bounds $|\Delta\alpha/\alpha| \lesssim 10^{-7}$ over that interval `[ESTABLISHED — order of magnitude; exact bound is analysis-dependent]`.
- **Atomic-clock comparisons** bound present-day drift $|\dot{\alpha}/\alpha|$ at roughly $\lesssim 10^{-17}\,\mathrm{yr}^{-1}$ *ESTABLISHED — order of magnitude; consult primary metrology literature in [BIBLIOGRAPHY.md* (the Bibliography (p. 776)) for current best values].
- **Quasar absorption spectra** probe $\alpha$ and $\mu$ at cosmological lookback. Claims of a nonzero spatial/temporal variation of $\alpha$ (the "Webb dipole") have appeared but are **not confirmed** and are widely regarded as not yet established, with systematics a serious concern `[CONTESTED]`.
- **BBN and CMB** consistency constrain combinations of constants in the early universe `[ESTABLISHED as constraints; INFERENCE in interpretation]`.

Bottom line: there is **no confirmed evidence** that any fundamental dimensionless constant varies `[ESTABLISHED — i.e. the null result is robust]`, but small variation is **not excluded** and is a generic prediction of theories with light dynamical scalars (quintessence, dilatons, the string landscape) `[INFERENCE/OPEN]`. This is the empirical face of the "constants are constant" entry in the *Chapter 20 (p. 458)*. See also *Chapter 15 (p. 233)*.

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## 10. Anthropic reasoning (clearly flagged)

Several constants appear "fine-tuned" for complexity/life: the smallness of $\Lambda$ (§8), the proximity of the up/down quark mass difference and electromagnetic effects that make the proton lighter than the neutron, the strength of the strong force relative to electromagnetism (Hoyle-state carbon production), etc.

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## References

See the Bibliography (p. 776) for canonical sources: CODATA recommended values and the BIPM SI Brochure (9th ed., 2019 redefinition); Particle Data Group *Review of Particle Physics* (parameter values, $\alpha$, $\alpha_s$, masses); Planck collaboration cosmological-parameter papers ($\Lambda$CDM); Weinberg's review of the cosmological constant problem; and the Duff–Okun–Veneziano "trialogue" on the number of fundamental constants. Specific numerical precision and current best bounds should always be taken from these primary sources rather than from this page.

PART IV

# THE CANDIDATE ROADS

Given the map and the cracks in it, what are the serious proposals for going deeper? This part surveys them — critically, comparatively, and without a favorite to protect.

Chapter 17 collects the candidate deep principles: the recurring structural ideas — symmetry, holography, entanglement-first ontologies, thermodynamic and informational reformulations — that individual programs draw on. Chapter 18 then surveys the unification landscape itself: string theory, loop quantum gravity, asymptotic safety, causal sets, emergent-gravity programs, and the rest, each assessed on the same questions — what does it assume, what has it actually delivered, what would falsify it, and what does it quietly import?

Chapter 19 sharpens the survey into refereed hypotheses (the H-series): specific candidate directions, each subjected to the adversarial verdict process described in the Preface, with the referee's findings preserved on the page. Chapter 20 is the assumptions ledger, perhaps the most distinctive instrument in the book: the field's foundational assumptions sorted along a separate axis — not how true, but how load-bearing and how revisable — from genuine bedrock down to historical artifact and mathematical convenience. Moving items down that ladder is, in a real sense, what unification research is.

What this part establishes is the supply side: the live candidate roads, honestly graded. One wager — that causal, algebraic, and entanglement structure precedes geometry — emerges as the most developed. Part V puts it on trial.

CHAPTER 17

# Candidate Unifying Principles

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This page catalogues the principal candidates for *truly foundational* organizing principles of physics, distinguished sharply from principles that are best read as *emergent* or *effective*. For each, we give a precise statement, the evidence in its favor, its scope and limits, the arguments for and against genuine foundational status, and where it surfaces across the domain pages. The page concludes with a ranked, epistemically tagged view.

A recurring methodological theme — see *Chapter 2 (p. 8)* — is to separate three things that are easily conflated:

1. A **logical inconsistency** (two claims that cannot both be true).
2. A **domain-of-validity mismatch** (each claim true in its regime, with no overlap).
3. An **unsolved-but-consistent problem** (a genuine open question within a coherent framework).

Most of the "tensions" between the principles below are of types 2 and 3, not type 1. The deepest live tension — whether **geometry** or **information/entanglement** is the deeper layer — is type 3 (an open question), not a contradiction. See *Chapter 14 (p. 191)* and *Chapter 18 (p. 349)*.

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# 1. Symmetry, representation theory, and the gauge principle

**Statement.** Physical laws are constrained by invariance under transformation groups. Three logically distinct sub-claims must be held apart:

- **(a) Noether's theorem.** `[ESTABLISHED]` In any variational theory, every continuous global symmetry of the action yields a conserved current $\partial_\mu j^\mu = 0$ and charge. This is a *mathematical theorem*, not a contingent empirical fact. Energy–momentum conservation follows from spacetime-translation invariance, angular momentum from rotational invariance, and so on. In Hamiltonian form, a quantity $J$ is conserved iff it Poisson-commutes with the Hamiltonian, $\{J, H\} = 0$ (see *Chapter 4 (p. 25)*).
- **(b) Wigner's classification.** `[ESTABLISHED as a classification result]` A relativistic "particle" *is* a unitary irreducible representation of the Poincaré group, labelled by the two Casimir invariants

  $$P^2 = m^2, \qquad W^2 = -m^2\, s(s+1),$$

  where $W^\mu$ is the Pauli–Lubanski vector. Spacetime symmetry thereby dictates the kinematic catalogue of matter: mass and spin *are* the Poincaré Casimirs. See *Chapter 8 (p. 89)* and *Chapter 9 (p. 106)*.
- **(c) The gauge principle.** `[CONTESTED interpretation]` The heuristic that *demanding local invariance generates the interactions* (Yang–Mills) is enormously productive but interpretively delicate: gauge symmetry is a **redundancy of description**, not a physical symmetry. Elitzur's theorem forbids the spontaneous breaking of a local gauge symmetry; physical observables are gauge-invariant (e.g. Wilson loops $W[C] = \mathrm{tr}\, \mathcal{P} \exp \oint_C A$); and the same physics can be expressed without manifest gauge redundancy.

**Evidence for.**

- Noether's theorem is proven and pervades every variational theory, from point particles to the QFT stress tensor.
- Wigner's irrep classification underpins the entire particle content of QFT and the spin–statistics structure. `[ESTABLISHED]`
- Lorentz invariance is tested to extreme precision; the Standard Model is built as an **anomaly-free chiral gauge theory** $SU(3)_c \times SU(2)_L \times U(1)_Y$, where anomaly cancellation — a quantum-consistency requirement — fixes hypercharge assignments. `[ESTABLISHED]`
- Bundle/connection geometry, with curvature $F = dA + A \wedge A$, is confirmed to be physical via the Aharonov–Bohm effect and Dirac charge quantization $eg = 2\pi n\hbar$. `[ESTABLISHED]`

**Scope and limits.** Global Poincaré invariance is *exact only in flat spacetime*; in our expanding universe it is a local approximation, and the cosmic matter distribution + cosmic time furnish a preferred frame at large scales (see *Chapter 11 (p. 139)*). Diffeomorphism invariance in general relativity is *qualitatively different* from internal Yang–Mills gauge symmetry: it is the gauge group of GR, tied to background independence and the hole argument (see §9 and *Chapter 10 (p. 121)*). The gauge framing is a redundancy, so "derive the interactions by gauging" is heuristic; large-gauge and asymptotic-symmetry charges complicate the pure-redundancy reading. Possible tiny Lorentz violation is `[OPEN]` but tightly constrained.

**Foundational vs emergent.**

- *For:* the **representation-theoretic content** (what can exist, given a symmetry) and **Noether's theorem** (symmetry ⇒ conservation) are about as close to foundational as physics offers, recurring identically across classical mechanics, QFT, and the SM.
- *Against:* (i) gauge "symmetry" being a redundancy means at least that part is descriptive scaffolding, not ontology; (ii) the *specific* gauge group and the three generations are empirical inputs the SM cannot explain, suggesting a low-energy shadow of deeper (e.g. GUT) structure — see *Chapter 15 (p. 233)*; (iii)

symmetries can themselves be emergent — condensed-matter systems exhibit emergent gauge fields and emergent Lorentz invariance in the infrared, which would make even Lorentz invariance an IR accident rather than a UV principle. `[SPECULATIVE]`

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## 2. The action principle and the path integral

**Statement.** `[INFERENCE, strong consensus]` Dynamics across classical mechanics, field theory, and gravity follows from stationarity of an action functional, $\delta S = 0$. The modern reading *inverts* the classical hierarchy: least action is not a teleological postulate but the stationary-phase ($\hbar \to 0$) limit of the Feynman path integral, which weights every history by $e^{iS/\hbar}$:

$$K(b,a) = \int \mathcal{D}[q]\; e^{iS[q]/\hbar}.$$

**Evidence for.**

- The Euler–Lagrange and Hamilton equations, Maxwell, Yang–Mills, and the Einstein field equations all derive from action stationarity. The latter follow from the Einstein–Hilbert action

  $$S_{\text{EH}} = \frac{c^4}{16\pi G} \int (R - 2\Lambda)\,\sqrt{-g}\;d^4x.$$

- Stationary phase recovers the classical trajectory as $\hbar \to 0$, *explaining why* nature appears to extremize. See *Chapter 7 (p. 71)* and *Chapter 8 (p. 89)*.
- **Lovelock's theorem**: in four dimensions, the most general metric action giving second-order field equations yields essentially uniquely the Einstein tensor plus a cosmological constant — a rigidity result showing how much variational + symmetry structure fixes. `[ESTABLISHED]`
- Noether's theorem operates *within* the variational framework, binding symmetry to conservation.

**Scope and limits.** The principle is *stationary*, not *least* (saddle points past conjugate points/caustics). Dissipative and nonholonomic systems are **not** derivable from a standard action: rolling constraints obey the Lagrange–d'Alembert principle, not the Euler–Lagrange equations of any action, and dissipation is naturally framed in contact geometry. Singular (gauge) Lagrangians require Dirac constraint analysis. Most severely, the Lorentzian "measure" $\mathcal{D}\phi\, e^{iS}$ is **not** a rigorous measure in $d = 4$: there is no $\sigma$-additive translation-invariant measure on an infinite-dimensional space, and $e^{iS}$ is not absolutely integrable. It is a phenomenally successful heuristic, rigorous only via Euclidean, reflection-positive (Osterwalder–Schrader) constructions in low dimensions (see *Chapter 13 (p. 172)*).

**Foundational vs emergent.**

- *For:* extraordinary universality — the one formal skeleton common to every dynamical theory in this wiki — and the path-integral derivation gives the classical action principle a genuine mechanistic ground rather than leaving it teleological.
- *Against:* (i) the fundamental object is arguably the path integral / quantum amplitude, with the action principle *derivative*; (ii) that very object lacks rigorous existence in 4D, so we do not actually possess the thing we invoke; (iii) it is a *framework/language* for encoding dynamics, not a substantive law about the world's content. Best read as a near-universal **structural** principle whose own foundation (the quantum measure) is mathematically incomplete. `[INFERENCE]`

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## 3. Renormalization group, universality, and effective field theory

**Statement.** `[ESTABLISHED]` Physics organizes into a tower of effective descriptions, each valid below some energy scale, connected by renormalization-group (RG) flow of couplings. Two profound consequences:

- **Universality.** Near continuous phase transitions, microscopically distinct systems share identical critical exponents because irrelevant operators die under RG flow; the diverging correlation length $\xi \sim |t|^{-\nu}$ washes out microscopic detail. See *Chapter 6 (p. 56)*.
- **Reinterpretation of renormalizability.** Once treated as a sacred constraint, renormalizability is, after Wilson, an *emergent* low-energy property: non-renormalizable (irrelevant) operators are suppressed by powers of $E/\Lambda$. Non-renormalizable EFTs (Fermi theory, chiral perturbation theory, GR-as-EFT, SMEFT) are perfectly predictive at low energy.

**Evidence for.**

- Critical exponents and universality classes are confirmed quantitatively across condensed-matter and statistical systems; scaling relations such as hyperscaling $2 - \alpha = d\nu$ hold.
- Euclidean QFT = classical statistical field theory under Wick rotation; Wilsonian RG originated in critical phenomena and became the organizing language of QFT (running couplings, beta functions). `[ESTABLISHED]`
- Asymptotic freedom of QCD ($\beta < 0$ for $n_f < 33/2$) and the Landau pole / triviality of $\phi^4$ and QED are RG facts that recast these as cutoff EFTs. See *Chapter 9 (p. 106)*.
- The SM is now understood as the renormalizable part of an EFT (SMEFT): the dimension-5 Weinberg operator for neutrino mass, dimension-6 operators for BSM physics — the dominant model-independent analysis framework.

**Scope and limits.** RG control is sharpest near fixed points (CFTs anchor the space of theories) and for weak-to-moderate coupling; strongly coupled nonperturbative flows (QCD confinement below $\Lambda_{\text{QCD}} \sim 200$ MeV) require lattice methods. Universality is a near-critical, large-$\xi$ statement. The existence of a genuine nonperturbative continuum limit (vs triviality) is `[OPEN]` for the physically central 4D theories. RG presupposes notions of scale and locality that may not survive quantum gravity.

**Foundational vs emergent.** This is arguably the strongest candidate for a *truly foundational meta-principle*, precisely because it **reframes the question**.

- *For:* established, recurs identically across stat-mech, QFT, and the SM, and explains *why* effective theories work.
- *Against:* it is a principle *about theories* rather than a law of nature's content, and it points toward emergence as primary, in tension with any single foundational substrate. Its deepest implication may be **anti-foundationalist**: that "emergent" and "fundamental" are scale-relative. `[ESTABLISHED]` (with a `[CONTESTED]` philosophical reading). See *Chapter 17 (p. 325)* cross-links in *Chapter 3 (p. 15)*.

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# 4. The holographic principle and entropy/area bounds

**Statement.** `[INFERENCE, widely accepted]` The maximum entropy in a region of gravitating spacetime scales with the boundary **area**, not the volume: $S \leq A/4$ in Planck units. Anchors:

$$S_{\mathrm{BH}} = \frac{k_B c^3 A}{4G\hbar} \quad \text{(Bekenstein–Hawking)}, \qquad S \leq \frac{2\pi k_B R E}{\hbar c} \quad \text{(Bekenstein bound)},$$

with Bousso's covariant light-sheet bound as the strong-gravity generalization. The **strong form** (gauge/gravity duality) asserts that a $(d{+}1)$-dimensional gravitational theory is *exactly equivalent* to a $d$-dimensional non-gravitational QFT on its boundary (AdS/CFT).

**Evidence for.**

- $S_{\mathrm{BH}}$ is area-extensive on *multiple independent* derivations: Hawking radiation, the Euclidean gravitational action, string-theoretic D-brane microstate counting for extremal/BPS black holes (Strominger–Vafa —

`[ESTABLISHED within that framework]`), and loop-quantum-gravity area spectra.

- **Ryu–Takayanagi**: boundary entanglement entropy equals a bulk minimal-surface area,

  $$S(A) = \frac{\mathrm{Area}(\gamma_A)}{4G_N},$$

  geometrizing entanglement — rigorous within AdS/CFT. See *Chapter 12 (p. 155)*.
- AdS/CFT passes an enormous body of consistent checks (symmetries, spectra, correlators, integrability, BH thermodynamics).
- Area-law scaling of ground-state entanglement entropy in QFT reinforces the area-counting of degrees of freedom.

**Scope and limits.** The strong (RT / AdS-CFT) form is rigorously formulated and tested essentially **only** in asymptotically anti-de Sitter spacetimes at large $N$ / strong coupling (Einstein-gravity regime). There is **no** established de Sitter holographic dual or RT analog — yet our universe is de Sitter-like (accelerating; see *Chapter 16 (p. 308)*). The Bekenstein bound is sharp only for weak gravity ($R$ larger than the Schwarzschild radius); in strong gravity Bousso's covariant bound is the correct statement but lacks a fully general first-principles proof. AdS/CFT has no general nonperturbative proof — it is a conjecture with overwhelming evidence. See *Chapter 13 (p. 172)*.

**Foundational vs emergent.**

- *For:* the area-law for entropy is robust and multiply-derived; any quantum gravity must reproduce it, and it strongly suggests gravitational degrees of freedom are *area-extensive* (radically fewer than local QFT's volume-extensive count), pointing to a holographic reorganization of the fundamental Hilbert space.
- *Against:* the only setting where holography is a *derived equivalence* rather than a heuristic bound is AdS/CFT — a special, negatively-curved corner with the wrong cosmological-constant sign for our universe. Whether holography is universal

or an AdS artifact is arguably **the** central open question of the program. Net: the area **bound** is likely-fundamental `[INFERENCE]`; the strong holographic **duality** as a *universal* fact is `[SPECULATIVE]` outside AdS.

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## 5. Information-theoretic reconstructions of quantum mechanics

**Statement.** `[OPEN, active and fruitful]` The Hilbert-space formalism of QM can be *derived* from a small set of operationally/information-theoretically motivated axioms rather than postulated. Representative programs: Hardy (2001), from continuity/reversibility plus simplicity; Chiribella–D'Ariano–Perinotti (2011), from purification and local tomography; Clifton–Bub–Halvorson, from no-signaling, no-broadcasting, no-bit-commitment; Masanes–Müller, from operational postulates. The thrust: QM's structure (complex amplitudes, tensor-product composition, the Born-rule form) may be the *unique* theory satisfying reasonable constraints on information processing.

**Evidence for.**

- Multiple distinct axiom sets recover the Hilbert-space formalism, showing the structure is highly constrained by operational requirements.
- **Gleason's theorem** (dim $\geq 3$): $\rho \mapsto \mathrm{Tr}(\rho E)$ is the *unique* noncontextual probability measure on projections — the Born rule's *form* is forced given the Hilbert-space scaffolding. `[ESTABLISHED]`
- Tensor-product composition is singled out among operationally reasonable composition rules by local tomography in generalized-probabilistic-theory (GPT) reconstructions.
- Recent (2021–2022) multipartite Bell-type tests argue that *complex* (vs real) Hilbert space is experimentally distinguishable under standard locality assumptions, suggesting complexity is not mere convention. `[CONTESTED]` (conditional on the tests' framework assumptions).

**Scope and limits.** No *single* agreed axiom set exists; the multiplicity of successful, inequivalent reconstructions itself undercuts any claim to be *the* foundation. The axioms' physical *necessity* (vs naturalness) is debated. Reconstructions are cleanest in finite dimensions; the infinite-dimensional / field-theoretic / relativistic case is harder. The programs reconstruct the *formalism* but do not, by themselves, resolve the measurement problem or fix an interpretation. See *Chapter 7 (p. 71)* and *Chapter 19 (p. 392)*.

**Foundational vs emergent.**

- *For:* if QM is the unique information theory consistent with a few operational principles, then "physics is information" acquires real content and QM stops being an arbitrary postulate set — a genuinely foundational reframing.
- *Against:* the multiplicity of successful axiom sets means we do not know *which* principles are fundamental vs conventional; it presupposes the probabilistic/operational (agent, preparation, measurement) framing, itself contested as primitive. Best classified as a promising but unresolved route to foundations. `[OPEN]`

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## 6. Thermodynamic / entropic origin of gravity (Jacobson)

**Statement.** `[SPECULATIVE, principled]` The Einstein field equations may be not a fundamental dynamical law but an **equation of state**. Jacobson (1995) derived them from the Clausius relation $\delta Q = T\,dS$ applied to local Rindler (acceleration) horizons, with entropy $\propto$ horizon area and $T$ the Unruh temperature, *demanded to hold for all local observers*. On this view spacetime geometry and its dynamics are emergent, macroscopic, thermodynamic phenomena of unknown microscopic degrees of freedom — analogous to hydrodynamics emerging from molecular statistics.

**Evidence for.**

- Jacobson's derivation is internally consistent: imposing $\delta Q = T\,dS$ on all local causal horizons, with $S \propto A$ and $T$ the Unruh temperature, yields the full nonlinear Einstein equations. See *Chapter 10 (p. 121)* and *Chapter 5 (p. 40)*.
- Black-hole mechanics maps exactly onto the four laws of thermodynamics ($S_{\mathrm{BH}} \propto A$, $T_H \propto \kappa$, the surface gravity) — a deep `[ESTABLISHED]` structural correspondence.
- The holographic first law $\delta S_A = \delta\langle K_A \rangle$ (with $K_A$ the modular Hamiltonian) yields the *linearized* Einstein equations from entanglement in AdS/CFT — a parallel "gravity from entropy/entanglement" derivation.
- The non-renormalizability of quantized GR is naturally explained if the metric is an emergent collective variable (like a phonon field), not a fundamental field to be quantized.

**Scope and limits.** The derivation *assumes* the area-entropy law and Unruh temperature as inputs — it explains the *form* of the field equations given thermodynamic premises, not why those premises hold microscopically. The entropy functional can be generalized (higher-curvature corrections $\leftrightarrow$ different entropy functionals). The entanglement-route derivations recover only the *linearized* equations under specific assumptions; a full nonlinear, background-independent derivation is **not** achieved. The microscopic degrees of freedom remain unidentified.

**Foundational vs emergent.** This is the *inverse* of "geometry as primary" (§9): it makes geometry emergent and thermodynamics/entropy fundamental.

- *For:* elegantly explains why GR resists quantization (one should not quantize an equation of state any more than the Navier–Stokes equations), and unifies the otherwise suspicious coincidence of black-hole thermodynamics.
- *Against:* it derives the equations *given* thermodynamic inputs that themselves require explanation; it is `[SPECULATIVE]` (no experimental discrimination from ordinary GR); and "entropy" presupposes a coarse-graining whose substrate is unknown. It *relocates* the mystery rather than closing it. `[SPECULATIVE]`

*(Iteration-2 disaggregation: Jacobson 1995 is a genuine theorem **given** the area law + Unruh T — the AdS first-law route yields only the linearized equations; Padmanabhan is an `[INFERENCE]`-level structural rewriting; Sakharov a mechanism that **outputs** the area law; Verlinde-2011 a heuristic; Verlinde-2016 dark matter testable but disfavored at cluster/CMB scales. See Working note: Entropic / emergent gravity: genuine derivation, or suggestive analogy? (repo: `notes/2026-06-08-entropic-and-emergent-gravity.md`) and Chapter 19 (p. 392) HYP-ENTROPIC-GRAVITY-SPLIT.)*

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# 7. Entanglement as the structural glue of spacetime

**Statement.** `[SPECULATIVE → INFERENCE, setting-dependent]` The connectivity and geometry of spacetime emerge from the entanglement structure of an underlying quantum theory. Within AdS/CFT: Ryu–Takayanagi ties boundary entanglement entropy to bulk minimal-surface area; **Van Raamsdonk's** argument shows that *disentangling* boundary degrees of freedom pinches off and disconnects bulk regions, suggesting entanglement literally sews spacetime together; **ER = EPR** (Maldacena–Susskind) conjectures that entanglement between systems *is* a geometric (Einstein–Rosen-bridge) connection. Slogan: "spacetime is built from entanglement."

**Evidence for.**

- RT / HRT and **entanglement-wedge reconstruction** are rigorous within AdS/CFT, with quantum error correction modeling how bulk operators are encoded in boundary entanglement. See *Chapter 12 (p. 155)*.
- Van Raamsdonk's thought experiment: smoothly reducing entanglement between two CFTs corresponds to the two bulk regions separating.
- ER = EPR is supported by the thermofield-double / eternal-black-hole correspondence and traversable-wormhole

constructions.

- The entanglement first law $\delta S_A = \delta\langle K_A \rangle$ reproduces the linearized Einstein equations, deriving (linearized) geometric dynamics from entanglement.

**Scope and limits.** Demonstrated rigorously only in asymptotically-AdS, large-$N$, strong-coupling holographic settings. ER = EPR is a bold conjecture with **no** first-principles derivation and unknown status for generic (non-holographic) entanglement. There is no demonstration that *generic* entanglement produces a smooth geometric connection, nor that our universe's spacetime is so constituted. Continuum QFT entanglement entropy is UV-divergent, and local algebras are **Type III** von Neumann factors with *no* tensor factorization — so the naive "qubits sewing space" picture lacks a rigorous subsystem structure (see *Chapter 13 (p. 172)*). De Sitter / cosmological applicability is unestablished.

**Foundational vs emergent.**

- *For:* the most concrete realization of "spacetime is emergent," converting a slogan into computable statements (RT) and a derivation of linearized gravity. If it generalizes, entanglement is deeper than geometry.
- *Against:* heavily AdS-dependent (wrong $\Lambda$ sign for us), conjectural beyond linear order, and resting on a subsystem/tensor-factor picture that algebraic QFT (Type III) says does not strictly exist. Net: `[SPECULATIVE]` overall, with the RT-within-AdS sub-claim rising to `[INFERENCE]`. Like §6, it points to information/entanglement — not geometry — as the deeper layer, without claiming universality. See *Chapter 14 (p. 191)*.

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## 8. Locality and causality (microcausality, no-signaling)

**Statement.** `[ESTABLISHED as defining axioms]` Physical influence is local and causally ordered. Precise encodings:

- **(a) Microcausality.** Spacelike-separated local field observables commute, $[\phi(x), \phi(y)] = 0$ for $(x - y)^2 < 0$ — the QFT axiom enforcing relativistic causality and driving the spin–statistics and CPT theorems.
- **(b) No-signaling.** Entanglement and "collapse" never transmit usable information faster than light; local marginals are independent of distant settings (Bell violation notwithstanding). The Tsirelson bound caps quantum correlations at $2\sqrt{2}$.
- **(c) Cluster decomposition.** Distant experiments have independent statistics, essentially forcing (with QM + Poincaré) a *local field* description (Weinberg's "folk theorem").

**Evidence for.**

- Microcausality is the load-bearing axiom of QFT, generating spin–statistics, CPT, and the Reeh–Schlieder structure. See *Chapter 8 (p. 89)*.
- The no-signaling theorem guarantees no usable superluminal information despite Bell-inequality violation (confirmed by loophole-free Bell tests) and despite apparent instantaneous collapse. See *Chapter 7 (p. 71)*.
- Cluster decomposition + Poincaré invariance + QM essentially force the field description.
- Causal structure is built into GR's Lorentzian metric (light cones sort timelike/null/spacelike), and global hyperbolicity secures determinism. See *Chapter 10 (p. 121)*.

**Scope and limits.** The **Reeh–Schlieder theorem** shows the QFT vacuum is entangled across arbitrarily separated regions (cyclic and separating for every local algebra), violating *naive* locality intuitions while preserving no-signaling — so "locality" is subtle. GR permits closed timelike curves (Gödel, the Kerr interior, wormholes) unless censored by chronology protection `[OPEN/CONTESTED]`. Quantum gravity is widely expected to modify locality at the Planck scale (holography is intrinsically non-local). In algebraic QFT there is no tensor factorization into strictly local subsystems (Type III).

**Foundational vs emergent.**

- *For:* microcausality and no-signaling are the *least negotiable* principles within any relativistic quantum theory; relaxing them generically destroys Lorentz invariance or causality. They have unmatched theorem-generating power (spin–statistics, CPT).
- *Against:* (i) holography and the Reeh–Schlieder / Type-III structure show locality is not fundamental in the naive sense and is expected to be emergent in quantum gravity; (ii) entanglement-as-geometry treats non-local entanglement as *more* fundamental than local spacetime. Net: foundational *for* relativistic QFT on a fixed background, but a prime suspect for **emergence** at the quantum-gravity level. **Causal order** may be more robust than **spatial separability**. `[ESTABLISHED]` (in-domain).

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## 9. Geometry as primary (background independence)

**Statement.** `[CONTESTED]` Gravity and physics are fundamentally geometric: matter is the curvature of a dynamical spacetime (GR), and the deepest structural principle is **diffeomorphism invariance / background independence** — there is no prior fixed geometry, and physical states are diffeomorphism-equivalence classes of fields. Geometry is the dynamical actor, not a stage, and any fundamental theory must be background-independent.

**Evidence for.**

- GR's geometric account is confirmed across an enormous range: perihelion precession, light deflection, Shapiro delay, binary-pulsar decay, gravitational-wave detection, EHT shadows, and cosmology. See *Chapter 10 (p. 121)*.
- The **hole argument** shows that determinism *forces* treating diffeomorphic metrics as physically identical: spacetime points have no identity apart from the fields. Background independence is not optional in GR.

- Geodesic motion geometrizes gravity (free fall = inertial motion in curved spacetime); the equivalence principle (WEP tested to $\sim 10^{-15}$, MICROSCOPE) makes gravity universal and hence geometrizable. `[ESTABLISHED]`
- Lovelock rigidity: the geometric Einstein–Hilbert action is essentially the unique 4D second-order metric theory.

**Scope and limits.** GR is a *classical* effective theory; its perturbative quantization is non-renormalizable, signaling breakdown at the Planck scale where the smooth-manifold continuum is expected to fail. The "geometric trinity" (curvature-, torsion-, and nonmetricity-based formulations are empirically equivalent) shows the specific geometric framing is partly conventional. GR has **no** local gravitational energy density (only pseudotensors / quasi-local notions), and the *problem of time* afflicts its canonical quantization. Competing programs (§§6–7, holography) treat geometry as **emergent**, directly contesting its primacy.

**Foundational vs emergent.** This principle is in direct tension with §§3, 4, 6, 7.

- *For primacy:* background independence is GR's deepest, best-confirmed structural lesson, widely held as a non-negotiable requirement for quantum gravity; geometry's explanatory reach (unifying gravity, inertia, causal structure) is vast.
- *Against primacy:* the strongest current research currents (Jacobson, Van Raamsdonk, holography, area-law entropy) point the *other* way — geometry emerging from entropy/entanglement/information, with the metric a collective/thermodynamic variable; non-renormalizability is naturally read as evidence geometry is emergent. Net: a serious but actively **contested** candidate. **Background independence** (no prior structure) may survive even if geometry itself is emergent. `[CONTESTED]`

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## 10. Decoherence and the observer (the quantum–classical transition)

**Statement.** `[ESTABLISHED dynamics; CONTESTED significance]` **Decoherence** is the established dynamical process by which a system entangling with its environment loses local phase coherence, suppressing interference and selecting a robust *pointer basis* (einselection), explaining the *effective* emergence of classicality. The *stronger* claim — that decoherence (plus the observer, or branching, or information acquisition) *resolves* the measurement problem and explains the appearance of single outcomes — is **contested** and interpretation-dependent.

**Evidence for.**

- Decoherence dynamics are rigorous and confirmed: the GKSL/Lindblad master equation governs open-system evolution, and interference vanishes controllably as environmental coupling increases (matter-wave interferometry with large molecules). See *Chapter 7 (p. 71)* and *Chapter 6 (p. 56)*.
- Decoherence supplies the effective pointer basis and the practical quantum–classical boundary, underlying quantum Darwinism / Zurek's einselection.
- The classical limit $\hbar \to 0$ is singular and non-uniform (WKB oscillation, logarithmic Ehrenfest time for chaotic systems), so decoherence — not the bare limit — robustly yields classical behavior. See *Chapter 4 (p. 25)*.

**Scope and limits.** Decoherence by itself does **not** solve the measurement problem: it explains why interference is unobservable and which basis is robust, but does not select a single *actual* outcome from the improper mixture — unitary evolution never reduces a superposition to one branch. Its significance is interpretation-dependent (Everett: branching, no collapse; objective collapse: real reduction; QBism: agent's information

**Foundational vs emergent.**

- *Against foundational status (as usually hoped):* decoherence is a *derived* consequence of unitary QM plus many environmental degrees of freedom — emergent, not a new law — and it explicitly does not close the measurement problem. The "observer" is almost certainly **not** fundamental; treating it so is the *shifty split* of the standard axioms.
- *For a weaker role:* the **emergence of classicality** via environmental entanglement is a robust, established mechanism, and its information-theoretic reading (information leaking irreversibly to the environment, redundantly encoded) connects to the entropic/information themes above. Net: an important **emergent bridge**, not a foundational principle. `[ESTABLISHED]` (as dynamics).

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## 11. The low-entropy past (the Past Hypothesis) and the arrow of time

**Statement.** `[INFERENCE that it is required; OPEN why]` Because the microscopic dynamics are essentially time-symmetric (Hamiltonian, unitary, CPT), the observed thermodynamic arrow — the monotone increase of entropy, the second law — *cannot* follow from the dynamical laws alone. It requires a time-asymmetric **boundary condition**: an extraordinarily low-entropy initial macrostate of the universe (the **Past Hypothesis**). This single posit grounds the arrow across thermodynamics, statistical mechanics, and cosmology.

**Evidence for.**

- **Loschmidt's** reversibility paradox and **Zermelo's** recurrence objection show the $H$-theorem's monotonicity cannot come from reversible dynamics alone — a time-asymmetric input (molecular chaos / low-entropy past) is logically necessary. See *Chapter 6 (p. 56)* and *Chapter 5 (p. 40)*.

- The early universe was thermally near-uniform (high thermal entropy) but gravitationally extremely smooth (very *low* gravitational entropy, since clumping increases entropy when gravity dominates) — Penrose quantifies this via near-zero initial Weyl curvature. See *Chapter 11 (p. 139)*.
- **Fluctuation theorems** (Jarzynski, Crooks) show the second law is statistical, with transient violations exponentially suppressed but nonzero — consistent with a typicality + special-initial-condition account.
- The same low-entropy boundary condition is required across thermodynamics, statistical mechanics, and cosmology, unifying the arrow.

**Scope and limits.** That an asymmetric arrow *requires* an asymmetric boundary condition is `[INFERENCE]` bordering on established. *Why* the early universe had such low (especially gravitational) entropy is genuinely `[OPEN]` and exports to cosmology and quantum gravity. Inflation does **not** resolve it — inflation arguably requires its own special low-entropy initial patch to begin (Penrose's critique), relocating rather than removing the puzzle. The very notion of *gravitational* entropy is not rigorously defined (self-gravitating systems are non-extensive, have negative specific heat, and lack a global maximum-entropy equilibrium), so the bookkeeping is conceptually incomplete.

**Foundational vs emergent.**

- *For:* a genuine candidate for a fundamental, law-like posit that is logically *independent* of the dynamical laws — not derivable within any current framework, functioning as an irreducible boundary condition, and the deepest known source of time's directionality.
- *Against:* it is a boundary *condition*, not a law, and many hope it will eventually be *explained* dynamically (quantum cosmology / the initial quantum state of the universe), making it derived. Moreover, unlike symmetry or RG, it is a statement about a *particular contingent initial condition of our universe*, which sits uneasily with a *universal* foundational principle. Net:

explanatory necessity robust; ultimate status (brute fact vs derivable) `[OPEN]`. See *Chapter 15 (p. 233)*.

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## 12. Unitarity / information conservation

**Statement.** `[ESTABLISHED within quantum theory]` Closed-system quantum evolution is unitary: pure states evolve to pure states, total probability is conserved, and information is never truly destroyed. This is enforced in QFT order-by-order via BRST/Ward identities and the optical theorem, and is a non-negotiable requirement of a consistent probabilistic quantum theory. Its sharpest stress test — black-hole evaporation appearing to map pure to mixed states (the **information paradox**) — is now widely believed to resolve *in favor* of unitarity.

**Evidence for.**

- Unitarity is structurally required for conservation of total probability and is built into the $S$-matrix ($S^\dagger S = 1$); apparent unitarity loss (open systems, non-Hermitian effective descriptions) reflects truncation, not failure of the closed theory. See *Chapter 8 (p. 89)*.
- AdS/CFT is manifestly unitary on the boundary, providing a concrete framework in which black-hole evaporation *must* be unitary.
- Recent quantum-extremal-surface / island / replica-wormhole computations reproduce the unitary **Page curve** for black-hole evaporation entropy, strongly suggesting EFT was being used outside its domain when it predicted information loss. See *Chapter 10 (p. 121)* and *Chapter 12 (p. 155)*.

**Scope and limits.** The *complete*, first-principles mechanism by which information escapes a realistic (asymptotically-flat, dynamically-formed) evaporating black hole — and reaches an external observer — remains `[OPEN]`; islands are computed in controlled AdS/toy models, not yet shown to fully resolve the paradox in nature. Unitarity is in apparent tension with the projective collapse postulate (the measurement problem), resolved differently by each interpretation. Objective-collapse models (GRW/CSL) deliberately *violate* strict unitarity with new stochastic dynamics and are empirically testable — so universal unitarity is not absolutely beyond question.

**Foundational vs emergent.**

- *For:* among the most robust pillars of quantum theory, with theorem-level enforcement and a strong (if incomplete) resolution of its hardest challenge; any fundamental theory is widely expected to be unitary. Connects to information-conservation as a candidate deep principle.
- *Against:* it is a feature *of* quantum theory rather than an independent meta-principle, and whether quantum theory itself is final is open (objective-collapse models; the measurement problem). Best seen as foundational *conditional on* quantum mechanics — itself the subject of the reconstruction program (§5). `[ESTABLISHED]` (conditionally).

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## Synthesis: a ranked view of fundamental vs emergent

The principles split into two families with a shared, striking feature: **several of the strongest "emergence" arguments converge on information/entanglement/entropy as the deeper layer, beneath geometry and locality.** This convergence is the wiki's central live hypothesis — see *Chapter 19 (p. 392)* and *Chapter 18 (p. 349)* — but it is not established and is largely confined to the AdS corner.

The ranking below is *provisional* and intended as a working map, not a verdict. "Likely fundamental" means *plausibly part of the deepest layer of any future theory*; "likely emergent" means *plausibly a derived, scale- or regime-dependent feature*. Tags reflect confidence in the *fundamental/emergent classification*, not in the principle's empirical validity.

**Most likely genuinely fundamental (in roughly descending confidence):**

1. **Noether + representation theory** (§1a,b) — `[INFERENCE, high]`. Theorem-level and regime-universal; the part of "symmetry" not tainted by gauge redundancy.
2. **Unitarity / information conservation** (§12) — `[INFERENCE, high]`, *conditional on QM being final*.
3. **Microcausality / causal order** (§8) — `[INFERENCE]`. Least negotiable for relativistic QFT, but a suspect for emergence in quantum gravity; *causal order* likely outlives *spatial locality*.
4. **The Past Hypothesis** (§11) — `[OPEN, but fundamental-if-irreducible]`. Uniquely a *boundary condition* rather than a law; foundational status hinges on whether it is ever derived.

**Foundational as meta-principles (about theories, not contents):**

5. **Renormalization group / EFT** (§3) — `[ESTABLISHED]`. Arguably the deepest *meta*-principle; its lesson may be anti-foundationalist (scale-relativity of "fundamental").
6. **The action principle / path integral** (§2) — `[INFERENCE]`. Near-universal structural skeleton, but plausibly derivative of the (mathematically incomplete) quantum measure.

**Likely fundamental as a constraint, but with a speculative strong form:**

7. **Holographic area bound** (§4) — bound `[INFERENCE]`; universal *duality* `[SPECULATIVE outside AdS]`.

**Most likely emergent / derived:**

8. **Geometry as primary** (§9) — `[CONTESTED]`; multiple currents suggest geometry is emergent, though *background independence* may survive.
9. **Entanglement as spacetime glue** (§7) — `[SPECULATIVE]`; concrete only within AdS; if correct, would *demote* geometry and *promote* information.

10. **Entropic gravity** (§6) — `[SPECULATIVE]`; relocates rather than closes the mystery.
11. **Decoherence / the observer** (§10) — dynamics `[ESTABLISHED]` but *emergent*; the "observer" almost certainly not fundamental.

**Genuinely open at the foundational level:**

12. **Information-theoretic reconstructions of QM** (§5) — `[OPEN]`; if successful, would make "physics is information" substantive and reposition §§5, 12, and 7 as facets of one principle.

A consistency note (cf. *Chapter 14 (p. 191)*): the apparent conflict between §9 (geometry primary) and §§6–7 (geometry emergent) is **not** a logical contradiction. It is an *unsolved-but-consistent* question about which description is more fundamental, currently undecidable because the only regime with a derived answer (AdS/CFT) has the wrong cosmological-constant sign for our universe. Treat claims of "spacetime from entanglement" as profound conjectures, not results, outside that corner.

See *Chapter 20 (p. 458)* for the load-bearing assumptions each principle rests on, *Chapter 21 (p. 497)* for cross-cutting conclusions, and *Chapter 40 (p. 739)* / the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)) for revision history.

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## References

See the Bibliography (p. 776) for full citations. Standard sources underpinning this page include: Noether (1918); Wigner (1939); Weinberg, *The Quantum Theory of Fields* (microcausality, cluster decomposition, the folk theorem); Peskin & Schroeder and Srednicki (QFT); Wald, *General Relativity* and *Quantum Field Theory in Curved Spacetime* (BH thermodynamics, Reeh–Schlieder); Misner–Thorne–Wheeler (GR); Wilson (RG / critical phenomena); Bekenstein (1973), Hawking (1975), Bousso (covariant entropy bound); Maldacena (1998) and Witten (AdS/CFT); Ryu–Takayanagi (2006); Van Raamsdonk (2010); Maldacena–Susskind (ER=EPR, 2013); Jacobson (1995); Strominger–Vafa (1996); Hardy (2001) and Chiribella–D'Ariano–Perinotti (2011) (QM reconstructions); Gleason (1957); Penrose, *The Road to Reality* (Weyl curvature hypothesis, Past Hypothesis); Zurek (decoherence, einselection). Only sources the author is confident exist are cited; any claim whose attribution is uncertain is flagged inline.

CHAPTER 18

# The Unification Landscape: A Critical Survey

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This page surveys the major programs that aim to unify the known interactions and/or to quantize gravity. The organizing thesis is uncomfortable but important: **the programs that are most empirically testable are the least ambitious, and the programs that are most ambitious are the least testable** `[INFERENCE]`. We assess each program on six axes — core idea, what it claims to unify, mathematical status, empirical status, successes, problems, testability, verdict — and tag each nontrivial claim with an epistemic marker per *Chapter 2 (p. 8)*.

A recurring structural fact frames everything below: the natural scale of quantum gravity is the Planck scale,

$$M_{\mathrm{Pl}} = \sqrt{\frac{\hbar c}{G}} \approx 1.22 \times 10^{19}\ \mathrm{GeV}, \qquad \ell_{\mathrm{Pl}} = \sqrt{\frac{\hbar G}{c^3}} \approx 1.6 \times 10^{-35}\ \mathrm{m},$$

roughly $10^{15}$ times beyond the LHC's energy reach `[ESTABLISHED]`. Direct experimental access to this scale is not available with foreseeable technology `[ESTABLISHED]`, which is why "no distinguishing empirical confirmation" recurs as a verdict and is **not** a moral failing of any single program. See *Chapter 16 (p. 308)* for scale definitions and *Chapter 10 (p. 121)*, *Chapter 8 (p. 89)* for the frameworks being reconciled.

For the deeper conceptual currents shared across programs (background independence, holography, emergence, "spacetime conjectures extracted from these programs, see *Chapter 19 (p. 392)*. For the catalog of what remains unsolved, see *Chapter 15 (p. 233)*.

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# 0. Two kinds of "unification" — a necessary distinction

Two logically distinct ambitions are routinely conflated and must be kept apart `[INFERENCE]`:

1. **Unification of forces / matter** within quantum field theory — embedding the Standard Model gauge group into a larger structure (GUTs), or unifying bosons and fermions (supersymmetry). This is conventional QFT and is, in principle, testable at sub-Planckian scales.
2. **Quantization of gravity** — reconciling general relativity with quantum mechanics. The obstruction here is that the Einstein–Hilbert action quantized as a point-particle QFT is perturbatively non-renormalizable `[ESTABLISHED]`: Newton's constant $G$ has mass dimension $-2$ (in $\hbar = c = 1$ units, $[G] = M^{-2}$), so loop divergences require infinitely many counterterms. Goroff and Sagnotti (1985) showed pure gravity diverges at two loops `[ESTABLISHED]`. Different programs respond to this obstruction differently.

A "theory of everything" attempts both at once. Most programs below attempt only one. We flag which.

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# 1. String / M-theory and Supergravity

**Core idea.** Replace point particles with one-dimensional extended objects (strings) whose vibrational modes constitute the particle spectrum. The closed-string spectrum *necessarily* contains a massless spin-2 state identified with the graviton, so gravity is mandatory rather than added `[ESTABLISHED, within the framework]`. Consistent superstring theories require $D = 10$ spacetime dimensions; their strong-coupling and low-energy limits are conjecturally unified by an 11-dimensional **M-theory** whose low-energy limit is 11D supergravity `[INFERENCE]`. Supergravity is the locally supersymmetric extension of GR and arises as the strings' low-energy effective field theory.

**What it claims to unify.** All four interactions plus all matter in one quantum-consistent framework; bosons and fermions via supersymmetry; and the five 10D superstring theories plus 11D SUGRA via a web of dualities (T-, S-, U-duality).

**Mathematical status.**

- Perturbative string theory is well-developed and internally consistent order by order `[ESTABLISHED]`. The critical dimension is fixed by Weyl-anomaly cancellation on the worldsheet: for the bosonic string the conformal anomaly vanishes when the central charge $c = 26$; for the superstring the corresponding condition gives $D = 10$ `[ESTABLISHED]`. Modular invariance of the one-loop amplitude and the GSO projection (ensuring spacetime supersymmetry and a tachyon-free spectrum) are rigorous structural results `[ESTABLISHED]`.
- The low-energy effective action reproduces (super)gravity plus an infinite tower of higher-curvature $\alpha'$ corrections `[INFERENCE]`.
- **There is no non-perturbative, background-independent definition of M-theory** `[OPEN]`. It is characterized by its limits and dualities, most of which are conjectural; S-duality has strong evidence but is largely unproven, and AdS/CFT (§6) is the one non-perturbative handle, valid only in special backgrounds. The perturbative genus expansion is asymptotic, not convergent `[ESTABLISHED]`.

**Empirical status (blunt).** There is **no distinguishing experimental confirmation** of string/M-theory `[ESTABLISHED — as a statement about the empirical record]`. No string-specific signature — extra dimensions, superpartners, Regge excitations, modifications to Newtonian/GR gravity — has been observed. Supersymmetry, the most testable necessary ingredient, has **not** been detected: the LHC excludes the simplest naturalness-motivated low-scale SUSY up to the multi-TeV range

`[ESTABLISHED]`. The generic string scale sits near $M_{\rm Pl}$, so the framework makes essentially no falsifiable prediction at accessible energies. It is consistent with all data largely because it predicts almost nothing testable `[INFERENCE]`.

**Main successes.**

- A perturbatively finite, consistent quantum theory of gravity: graviton scattering amplitudes are UV-finite order by order, evading point-particle non-renormalizability `[INFERENCE]`.
- **Strominger–Vafa (1996):** microscopic D-brane state counting reproduces the Bekenstein–Hawking entropy

  $$S_{\rm BH} = \frac{k_B c^3 A}{4G\hbar} = \frac{A}{4\ell_{\rm Pl}^2}$$

  *exactly* for certain supersymmetric/extremal black holes — a genuine quantitative achievement `[ESTABLISHED, within the framework]`.
- Gave rise to **AdS/CFT** (Maldacena 1997), the most concrete realization of holography (§6) `[ESTABLISHED]`.
- Naturally incorporates gauge fields, chiral fermions, and Yang–Mills structure; deep two-way exchange with mathematics (mirror symmetry, knot invariants, monstrous/umbral moonshine) `[INFERENCE]`.

**Main problems / criticisms.**

- No non-perturbative or background-independent formulation `[OPEN]`.
- **The landscape problem.** Flux compactifications produce an enormous number of metastable vacua (the figure "$\sim 10^{500}$" is a heuristic estimate, not a rigorous count), with no known selection principle, undermining predictivity `[CONTESTED]`. The **swampland** program attempts to constrain which low-energy EFTs admit UV completion, but its conjectures are themselves `[SPECULATIVE]`.
- Constructing a fully realistic, stable de Sitter (positive-$\Lambda$) vacuum reproducing the Standard Model and observed dark

energy remains controversial — the KKLT construction versus the de Sitter swampland conjecture debate `[CONTESTED]`.
- Requires SUSY (unobserved) and extra dimensions (unobserved) `[ESTABLISHED]`.
- Most defining dualities are conjectural `[OPEN]`.

**Testability.** Generically untestable at accessible energies `[CONTESTED]`. Model-dependent (not generic) signatures — superpartners, large/warped extra dimensions, cosmic superstrings, specific inflationary non-Gaussianity — have not appeared and are not guaranteed. The "not even wrong" critique holds that the landscape renders it unfalsifiable in practice; defenders cite consistency constraints and AdS/CFT results.

**Verdict.** The most mathematically developed and influential quantum-gravity program, with real internal successes (perturbatively finite gravity, black-hole microstate counting, AdS/CFT). But after decades it has produced no distinguishing empirical confirmation, requires unobserved ingredients, and the landscape erodes predictivity. **Best regarded as a rich, possibly-correct mathematical framework whose connection to our universe is unestablished — not confirmed physics.** Epistemic tag: `[SPECULATIVE]` (as a description of nature; many internal *results* are `[ESTABLISHED]` within the framework).

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## 2. Loop Quantum Gravity (LQG) and Spin Foams

**Core idea.** A background-independent, non-perturbative canonical (and covariant) quantization of GR itself, without extra dimensions or added matter. In Ashtekar connection variables, the kinematical Hilbert space is spanned by **spin-network** states — graphs with edges labeled by $SU(2)$ irreps and vertices by intertwiners. Geometric operators acquire discrete spectra; e.g. the area operator eigenvalues take the form

$$A = 8\pi\gamma\ell_{\text{Pl}}^2 \sum_i \sqrt{j_i(j_i+1)},$$

with $j_i$ the spins piercing the surface and $\gamma$ the Barbero–Immirzi parameter, so space is fundamentally quantized at the Planck scale `[ESTABLISHED, within the chosen quantization]`. **Spin foams** (e.g. the EPRL model) provide the covariant histories/path-integral formulation, summing over 2-complexes interpolating between spin networks.

**What it claims to unify.** It does **not** aim to unify the forces. It unifies quantum mechanics with general relativity by quantizing geometry directly. Its central ambition is background independence and a quantum theory of spacetime.

**Mathematical status.**

- The kinematical level is rigorous: the spin-network basis, the Ashtekar–Lewandowski measure (and the LOST/Fleischhack uniqueness theorem for the diffeomorphism-invariant representation), and the discreteness of area/volume spectra are well-defined results `[ESTABLISHED, within the quantization]`.
- **The dynamics is the central unsolved problem** `[OPEN]`. The Hamiltonian (scalar/Wheeler–DeWitt) constraint suffers quantization ambiguities; demonstrating correct closure of the constraint algebra (anomaly freedom) and recovery of the right semiclassical limit is not established.
- Spin-foam models are better-defined covariantly, but recovering smooth GR in the continuum/large-spin limit and controlling the sum over 2-complexes remain unresolved `[OPEN]`.

**Empirical status (blunt).** No experimental confirmation `[ESTABLISHED — empirical record]`. Predicted discreteness is at $\sim \ell_{\text{Pl}}$, beyond direct probes. Searches for Lorentz-invariance violation (LIV) via gamma-ray-burst photon dispersion (e.g. Fermi-LAT) have found **no effect** and tightly constrain LIV, disfavoring naive discreteness-induced Lorentz breaking `[ESTABLISHED]`. Loop quantum cosmology (LQC) predicts a "big bounce" replacing the initial singularity, but this is unconfirmed `[OPEN]`.

**Main successes.**

- A concrete, background-independent kinematical quantization of geometry with discrete area/volume spectra — conceptually distinct from string theory's background-dependent starting point `[ESTABLISHED, within the framework]`.
- Reproduces $S_{\mathrm{BH}} \propto A$ via counting horizon spin-network punctures, after fixing $\gamma$ to one specific value `[INFERENCE]`. (That $\gamma$ must be fixed by this matching is also a weakness — see below.)
- LQC resolves the classical cosmological singularity into a bounce in symmetry-reduced models `[INFERENCE/SPECULATIVE]`.

**Main problems / criticisms.**

- Ambiguous, non-unique dynamics; recovery of the classical Einstein equations / smooth spacetime in the semiclassical limit is unproven `[OPEN]`.
- The Barbero–Immirzi parameter $\gamma$ is a free input fixed (somewhat circularly) by black-hole entropy matching `[CONTESTED]`.
- No clear derivation of low-energy QFT or the Standard Model; matter coupling is underdeveloped `[OPEN]`.
- Whether manifest discreteness is compatible with the absence of observed Lorentz violation is debated; defenders argue the discreteness is Lorentz-invariant in a subtle spectral (not lattice-like) sense `[CONTESTED]`.

**Testability.** Some LQC and LIV phenomenology is in principle testable (CMB signatures of a bounce, photon dispersion), but no positive signal exists and LIV bounds are already very tight `[CONTESTED]`. Core Planck-scale predictions are practically inaccessible.

**Verdict.** A serious, mathematically careful program whose kinematic results (geometric discreteness, background independence) are real and distinctive, but which is stuck on dynamics and the semiclassical limit, lacks contact with the matter sector, and has no empirical confirmation. **Genuinely open rather than refuted.** Epistemic tag: `[OPEN]`.

*Iteration-6 status refresh (2026-06-10):* numerically Lorentzian era — complex-critical-point machinery for 4d Lorentzian EPRL/Conrady–Hnybida amplitudes (Han–Liu–Qu, arXiv:2404.10563; PRD 111, 024021 (2025)); and a quantum Oppenheimer–Snyder phenomenology industry (Lewandowski–Ma–Yang–Zhang, PRL 130, 101501 (2023): quantum-corrected exterior, bounce, minimal mass — model-dependent, no positive signal; this section's first concrete effective-metric astrophysics contact point). → *[notes* (Working note: *Iteration 6 — R4: The Other Background-Independent Programs, 2023–2026 (Landscape Refresh)* (repo: `notes/2026-06-10-iter6-background-independent-programs.md`))]

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## 3. Asymptotic Safety (UV completion via a nontrivial fixed point)

**Core idea.** Gravity may be **non-perturbatively renormalizable**: although perturbatively non-renormalizable, its renormalization-group flow could approach a nontrivial ultraviolet fixed point with a *finite-dimensional* critical surface (UV-attractive directions). If so, the theory is predictive to arbitrarily high energy with only finitely many free parameters, and no new degrees of freedom (strings, loops) are needed — ordinary QFT, properly continued, suffices. This generalizes Weinberg's 1979 notion of asymptotic safety.

**What it claims to unify.** Quantum mechanics with gravity, inside ordinary QFT. Some variants attempt to constrain the matter sector (asymptotically safe gravity–matter systems, predictions for SM couplings).

**Mathematical status.**

- Substantial functional-renormalization-group (FRG) evidence — via the Wetterich equation for the effective average action $\Gamma_k$,

$$\partial_t \Gamma_k = \tfrac{1}{2}\,\mathrm{STr}\left[\left(\Gamma_k^{(2)} + R_k\right)^{-1} \partial_t R_k\right], \qquad t = \ln k,$$

indicates a suitable fixed point (the "Reuter fixed point"), robustly across many truncations `[INFERENCE/CONTESTED]`.

- **All evidence relies on truncating the infinite-dimensional theory space** `[OPEN/CONTESTED]`. There is no proof the fixed point survives untruncated; the standard FRG works in Euclidean signature, with subtleties continuing to Lorentzian gravity. Scheme/truncation dependence is the central caveat.

**Empirical status (blunt).** No distinguishing experimental confirmation `[ESTABLISHED — empirical record]`. The fixed point governs Planck-scale physics. Some claimed retrodictions — e.g. the Shaposhnikov–Wetterich argument (2009) for a Higgs mass near the later-measured value, or constraints on SM couplings — are intriguing but model- and truncation-dependent and not unambiguous tests `[CONTESTED]`.

**Main successes.**

- A candidate UV completion of gravity entirely within standard QFT — the most economical hypothesis (no new dimensions or objects) `[INFERENCE]`.
- Robust (across truncations) FRG evidence for the fixed point's existence `[INFERENCE]`.
- Some phenomenological predictions for SM parameters and improved high-energy behavior `[CONTESTED]`.

**Main problems / criticisms.**

- No proof the fixed point exists beyond truncations; truncation/scheme dependence unresolved `[OPEN]`.
- Lorentzian formulation and unitarity are not fully controlled; higher-derivative truncations risk ghosts / negative-norm states `[OPEN]`.
- Relationship to other approaches and to a positive $\Lambda$ is unsettled `[CONTESTED]`.

**Testability.** Largely Planck-scale; some indirect leverage via SM-coupling constraints, but no clean experimental test exists `[CONTESTED]`.

**Verdict.** The most conservative quantum-gravity proposal — it asks only that QFT keep working. The FRG evidence is real and suggestive but rests entirely on truncations, so the foundational claim is unproven; no empirical confirmation. **A live, respectable, but unestablished possibility.** Epistemic tag: **`[OPEN]`**.

*Iteration-6 status refresh (2026-06-10):* Lorentzian consolidation — dS-background Einstein–Hilbert flows (D'Angelo–Ferrero–Fröb, CQG 42 (2025)) and foliated Wick-rotated fluctuation studies; the "first Lorentzian evidence" attribution needs a caveat (multiple routes share inputs); the Higgs-mass retrodiction is framed by proponents as a top-mass-sensitive ratio post/prediction, not a clean test `[CONTESTED]`. Signature is still installed by hand (Wick rotation, foliation, or a chosen dS background) — corroborating HYP-CKV-VACUITY's moral. → *[notes* (Working note: *Iteration 6 — R4: The Other Background-Independent Programs, 2023–2026 (Landscape Refresh)* (repo: `notes/2026-06-10-iter6-background-independent-programs.md`))]

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# 4. Causal Dynamical Triangulations (CDT)

**Core idea.** Define the gravitational path integral non-perturbatively as a sum over discrete geometries built from simplices, with a crucial **causality** constraint: each configuration admits a global time foliation (only causally well-behaved Lorentzian geometries, Wick-rotated for computation). The continuum limit is studied by Monte Carlo, asking whether a sensible macroscopic spacetime emerges. Schematically the partition function regularizes

$$Z = \int \mathcal{D}g\, e^{iS_{\mathrm{EH}}[g]} \;\longrightarrow\; \sum_T \frac{1}{C_T}\, e^{iS_{\mathrm{Regge}}[T]},$$

summing over triangulations $T$ via the Regge action.

**What it claims to unify.** Quantum mechanics and GR, via a regularized, computable gravitational path integral. Not a unification of forces.

**Mathematical status.** Well-defined as a lattice regularization `[INFERENCE]`. The causality (foliation) condition distinguishes it from earlier *Euclidean* dynamical triangulations, which produced degenerate "crumpled" or "branched-polymer" phases `[ESTABLISHED, computationally]`. Existence and universality of a genuine continuum limit (a second-order transition removing the lattice spacing) is not rigorously established; results are numerical `[OPEN]`.

**Empirical status (blunt).** No experimental confirmation; CDT is a theoretical/computational program at the Planck scale `[ESTABLISHED – empirical record]`.

**Main successes.**

- Generates a phase with a macroscopic, extended, approximately 4-dimensional de Sitter-like universe emerging dynamically — nontrivial, since Euclidean DT failed at exactly this `[ESTABLISHED, computationally]`.
- **Dimensional reduction:** the spectral dimension flows from $\approx 4$ at large scales to $\approx 2$ at short (Planckian) scales — a striking, reproducible feature also seen in asymptotic safety and Hořava–Lifshitz gravity `[ESTABLISHED, computationally]`. The cross-program recurrence of UV dimensional reduction is itself a notable hint (see *Chapter 17 (p. 325)*).

**Main problems / criticisms.**

- No proof of a true continuum limit; the relevant transition's order/universality is still under investigation `[OPEN]`.
- Whether the imposed global time foliation is physical or an artifact that breaks the spirit of full diffeomorphism invariance / Lorentz symmetry is debated `[CONTESTED]`.
- Matter coupling and recovery of particle physics are underdeveloped; results largely qualitative `[OPEN]`.

- Heavy reliance on numerics, limited analytic control `[OPEN]`.

**Testability.** In principle short-scale dimensional reduction could have indirect consequences, but there is no current experimental test `[CONTESTED]`.

**Verdict.** A concrete, computationally honest attempt to define and evaluate the gravitational path integral, notable for dynamically producing a 4D universe and dimensional reduction. Remains a numerical program without a proven continuum limit, without matter, without empirical contact. **Genuinely open.** Epistemic tag: **`[OPEN]`**.

*Iteration-6 status refresh (2026-06-10):* first quantitative CDT–FRG contact — Ambjørn et al. (arXiv:2408.07808; PRD 110, 126006 (2024)) identify CDT's de Sitter-phase infinite-volume limit with the FRG Gaussian/IR fixed point; data *permit* but do not establish a UV lattice fixed point; the 2026 Ambjørn–Loll review's own status: "UV fixed point indicated, not established". → *[notes* (Working note: *Iteration 6 — R4: The Other Background-Independent Programs, 2023–2026 (Landscape Refresh)* (repo: `notes/2026-06-10-iter6-background-independent-programs.md`))]

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## 5. Causal Set Theory (causets)

**Core idea.** Spacetime is fundamentally a discrete **partially-ordered set**: discrete elements with a causal (order) relation, plus a counting rule — *"order plus number equals geometry"*: Lorentzian geometry is recovered from the causal order (which fixes conformal/light-cone structure) and the element count (which fixes volume). Continuum spacetime is emergent and coarse-grained. Discreteness is implemented to preserve Lorentz invariance *statistically*, via random Poisson **sprinkling** (no preferred frame, unlike a lattice).

**What it claims to unify.** Quantum mechanics and gravity via a discrete substrate; uniquely, it combines discreteness with Lorentz invariance. Not a unification of forces.

**Mathematical status.** The kinematic framework (causal sets, sprinkling, the Hauptvermutung that two non-isometric continuum spacetimes cannot both approximate the same causet) is clearly defined; Lorentz invariance of sprinkling is a real strength `[ESTABLISHED, conceptually]`. The quantum dynamics — a sum over causets / classical sequential growth models — is far from complete; deriving a unique, physically correct dynamics and the continuum limit is unresolved `[OPEN]`.

*Iteration-6 status refresh (2026-06-10):* the bare "unproven Hauptvermutung" is now too coarse — Müller (arXiv:2503.01719, preprint, v2 Dec 2025) splits it into precise variants with claimed truth values (one formulation false, one true; the countable-set version holds); whether the proven variant is the physically faithful (finite-Poisson-sprinkling) one is itself undecided `[OPEN]`. → *[notes* (Working note: *Iteration 6 — R4: The Other Background-Independent Programs, 2023–2026 (Landscape Refresh)* (repo: `notes/2026-06-10-iter6-background-independent-programs.md`))]

**Empirical status (notable).** Causal set theory's most-cited claim to fame is a heuristic, order-of-magnitude **prediction** (Sorkin, ca. 1990, *before* the 1998 discovery of cosmic acceleration) that Poisson fluctuations in the number of spacetime elements yield a small, fluctuating cosmological constant of roughly the observed magnitude:

$$\Lambda \sim \frac{1}{\sqrt{V}} \quad \text{(in Planck units)},$$

where $V$ is the spacetime volume to the present. This is genuinely striking and predates the data, but it is a heuristic argument, not a derivation from a complete theory, and yields no sharp falsifiable number `[CONTESTED/INFERENCE]`. Beyond this, no experimental confirmation.

**Main successes.**

- Reconciles fundamental discreteness with Lorentz invariance via Poisson sprinkling — solving a problem that defeats naive lattice discretization `[ESTABLISHED, conceptually]`.

- The Sorkin heuristic for a small fluctuating Λ of the right order, made before acceleration was observed, is arguably the most interesting near-prediction of any discrete approach `[CONTESTED/INFERENCE]`.
- Defines non-local but Lorentz-invariant d'Alembertian / curvature operators on causets `[INFERENCE]`.

**Main problems / criticisms.**

- No complete quantum dynamics; the sum-over-causets and its continuum limit are unsolved `[OPEN]`.
- The **entropy problem**: generic causets are non-manifoldlike (Kleitman–Rothschild "three-layer" posets dominate the counting), so recovering manifold-like spacetime is a known difficulty `[OPEN]`.
- Matter coupling and the Standard Model are largely undeveloped `[OPEN]`.
- The status of the Λ result as a genuine test versus a suggestive heuristic `[CONTESTED]`.

**Testability.** The fluctuating-Λ scenario has potential cosmological signatures (dynamically varying dark energy), of current interest given hints of evolving dark energy in DESI-era data; nothing is confirmed and most predictions are inaccessible `[CONTESTED]`.

*Iteration-6 status refresh (2026-06-10):* current concrete everpresent-Λ implementations are CMB-disfavored (Das–Nasiri–Yazdi, arXiv:2307.13743, preprint — program-internal stress test), and as of 2026-06-10 **no published confrontation with the DESI** $w(z)$ **posterior exists** — an identified gap `[OPEN]`. → *[notes* (Working note: *Iteration 6 — R4: The Other Background-Independent Programs, 2023–2026 (Landscape Refresh)* (repo: `notes/2026-06-10-iter6-background-independent-programs.md`))]

**Verdict.** A philosophically clean, minimal proposal whose strengths (Lorentz-invariant discreteness; an early order-of-magnitude Λ heuristic) are real, but lacking complete dynamics, a continuum-limit proof, and matter. **Open and underdeveloped**

**relative to string/LQG, but conceptually distinctive.** Epistemic tag: `[OPEN]`.

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## 6. AdS/CFT Correspondence and Holography

**Core idea.** A quantum gravity theory in a $(d+1)$-dimensional asymptotically anti-de Sitter (negatively curved) bulk is *exactly equivalent* (dual) to a non-gravitational conformal field theory on its $d$-dimensional boundary. The bulk geometry, including its radial dimension, is encoded in boundary data; bulk gravity emerges from boundary entanglement structure. The **Ryu–Takayanagi (RT)** formula identifies boundary entanglement entropy with the area of a bulk minimal surface,

$$S_A = \frac{\text{Area}(\gamma_A)}{4G_N},$$

geometrizing entanglement. The canonical example is type IIB string theory on $\text{AdS}_5 \times S^5$ dual to $\mathcal{N} = 4$ super-Yang–Mills.

**What it claims to unify.** Quantum gravity with ordinary gauge/QFT — two descriptions of one system. Concretely realizes the holographic principle ('t Hooft, Susskind): gravitational degrees of freedom scale with boundary area, and spacetime geometry is linked to quantum information.

**Mathematical status.**

- An enormous, consistent body of checks (matching symmetries, operator/state spectra, correlators, integrability, thermodynamics) supports the conjecture; it is proven in some lower-dimensional and highly supersymmetric cases `[INFERENCE – very strong]`.
- RT (and its covariant HRT and quantum-extremal-surface refinements) is a derived, internally rigorous result here `[ESTABLISHED, within the framework]`.
- No general non-perturbative proof exists; it remains a conjecture (with overwhelming evidence) in the general case

`[OPEN]`.

**Empirical status (blunt).** No direct experimental confirmation as a theory of *our* universe, which is approximately de Sitter (positive $\Lambda$), **not** anti-de Sitter `[ESTABLISHED — empirical record]`. There is no established dS/CFT analog. AdS/CFT-inspired methods (holographic QCD; the shear-viscosity-to-entropy bound $\eta/s \approx \hbar/4\pi k_B$) give qualitatively useful, sometimes quantitatively suggestive results for strongly-coupled systems (quark–gluon plasma, some condensed-matter systems), but these are applications/analogies, not confirmations of holography as fundamental physics `[INFERENCE]`.

**Main successes.**

- The most concrete, computationally productive realization of holography and "spacetime from entanglement"; manifestly unitary, giving a controlled setting where black-hole information is preserved `[INFERENCE]`.
- Reproduces a unitary **Page curve** via quantum-extremal-surface / island computations, strongly supporting information conservation in evaporation within these models `[INFERENCE]`.
- RT and entanglement-wedge reconstruction give a precise geometry–entanglement dictionary; linearized Einstein equations have been *derived* from the entanglement "first law" $\delta S = \delta\langle K\rangle$ (Faulkner et al.) `[ESTABLISHED, within the framework]`.
- A practical tool generating real, if model-dependent, results for strongly-coupled QFT and transport `[INFERENCE]`.

**Main problems / criticisms.**

- Validity is established essentially only for asymptotically-AdS, large-$N$, strong-coupling (Einstein-gravity) regimes; extension to de Sitter, flat space, finite $N$, and cosmology is unestablished — the central limitation `[OPEN]`.
- No general non-perturbative proof `[OPEN]`.
- Bulk locality, the black-hole interior, and observer-level information recovery remain incompletely understood `[OPEN]`.

- Whether holographic lessons transfer to realistic, positively-curved cosmology `[CONTESTED]`.

**Testability.** Not directly testable as fundamental physics of our universe (wrong sign of $\Lambda$); indirect analog tests via strongly-coupled lab/collider systems are suggestive but not decisive `[CONTESTED]`.

**Verdict.** A profound, well-supported theoretical equivalence — the crown jewel of holography and the most rigorous arena for "spacetime from entanglement" and unitary black-hole evaporation. But it is a conjecture (overwhelmingly evidenced) restricted to anti-de Sitter space, and our universe is not anti-de Sitter. **`[INFERENCE]` for the duality within AdS; `[SPECULATIVE]` as a statement about our cosmos.** Epistemic tag: **`[INFERENCE]`** (within its domain of validity).

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## 7. Entropic / Thermodynamic / Emergent Gravity

(Sakharov, Jacobson, Padmanabhan, Verlinde.)

**Core idea.** Gravity may not be fundamental but **emergent** — a thermodynamic/statistical effect of underlying microscopic degrees of freedom, analogous to how hydrodynamics or elasticity emerges from molecules.

- **Sakharov (1967):** "induced gravity" — the Einstein–Hilbert action arises from quantum vacuum fluctuations of matter fields.
- **Jacobson (1995):** the Einstein equations are *derived* as a local equation of state, applying the Clausius relation $\delta Q = T\,\delta S$ to all local Rindler horizons with entropy $\propto$ horizon area and the Unruh temperature $T = \hbar a/2\pi c k_B$.
- **Padmanabhan:** gravitational field equations carry a thermodynamic structure (equipartition, surface "degrees of freedom").

- **Verlinde (2011):** gravity as an "entropic force" from information on holographic screens; later extended to galactic dynamics aiming to mimic dark matter.

**What it claims to unify.** Reframes gravity as emergent thermodynamics, unifying gravitational dynamics with thermodynamics/information theory and suggesting spacetime is a coarse-grained statistical entity. Verlinde's later work additionally aims to explain galactic "dark matter" phenomenology without a particle. See *Chapter 5 (p. 40)*, *Chapter 6 (p. 56)*, *Chapter 12 (p. 155)*.

**Mathematical status.**

- **Jacobson's 1995 derivation is a genuine, theorem-level result** `[ESTABLISHED]`: given Bekenstein–Hawking entropy proportionality and the Clausius relation on local causal horizons, the Einstein equations follow as an equation of state.
- Sakharov's induced-gravity mechanism is well-defined `[ESTABLISHED]`.
- Verlinde's entropic-force derivation is heuristic and was criticized as ill-defined/circular; his emergent-dark-matter proposal is qualitative, not derived from a complete microscopic theory `[CONTESTED/SPECULATIVE]`. Padmanabhan's reformulations are suggestive structural rewritings, not a microscopic theory `[INFERENCE]`.

**Empirical status (blunt).** The core claim (gravity is emergent/thermodynamic) has **no distinguishing experimental confirmation**: by construction the derivation versions reproduce GR and make no new predictions `[ESTABLISHED]`. Verlinde's emergent-gravity formula gave some early successful galaxy rotation-curve and lensing fits (comparable to MOND), but fares poorly against the full data — clusters, the CMB acoustic peaks, the Bullet Cluster mass–gas offset, and structure formation — which strongly favor particle dark matter `[CONTESTED / refuted at cosmological scale]`. See *Chapter 11 (p. 139)*.

- Jacobson's derivation of Einstein's equations as a horizon equation of state — a deep, rigorous structural result connecting gravity, thermodynamics, and horizon entropy `[ESTABLISHED]`.
- Sakharov induced gravity: a concrete mechanism for an emergent Einstein–Hilbert action `[ESTABLISHED]`.
- Sharpens the well-supported view that the horizon-entropy area law and the holographic principle point to emergent geometry, connecting to AdS/CFT-era "spacetime from entanglement" `[INFERENCE]`.
- Verlinde's framework offered some economical, MOND-like galaxy-scale fits `[CONTESTED]`.

**Main problems / criticisms.**

- None of these specifies the microscopic degrees of freedom whose thermodynamics *is* gravity; "emergent from what?" is unanswered `[OPEN]`.
- Jacobson's result, though elegant, is a consistency/structural statement that by itself yields neither a quantum theory nor new predictions; whether "gravity is thermodynamics" is more than a suggestive analogy is debated `[CONTESTED]`.
- Verlinde's emergent-dark-matter proposal fails against clusters, the CMB, and the Bullet Cluster `[CONTESTED / refuted at cosmological scale]`.
- Recovering full nonlinear GR, quantum corrections, and the matter sector from a complete emergent picture is not achieved `[OPEN]`.

**Testability.** The pure-derivation versions make no new testable predictions by design `[CONTESTED]`. Verlinde's dark-matter variant *is* testable — and is largely disfavored at **cluster and cosmological scales** (Bullet Cluster, CMB acoustic peaks, structure formation) `[CONTESTED → disfavored]`. *Correction (iteration 2):* galaxy–galaxy weak-lensing radial-acceleration tests (Brouwer et al. 2017, 2021) are fit *comparably well* by Verlinde EG and MOND and do **not** themselves falsify it — the decisive evidence against is at cluster/CMB scales, while a ≥6σ early/late-type RAR dependence challenges *all* baryon-only modifications; see Working note: *Entropic / emergent gravity: genuine derivation, or suggestive analogy?* (repo: `notes/2026-06-08-entropic-and-emergent-gravity.md`).

**Verdict.** Two distinct things bundled together. (1) The thermodynamic-derivation insight (Sakharov, Jacobson, Padmanabhan) is genuinely deep and partly rigorous (Jacobson's is a real result), strongly suggesting gravity has thermodynamic/holographic character — but it identifies no microscopic substrate and makes no new predictions. (2) Verlinde's entropic-force and emergent-dark-matter extensions are heuristic and, at cosmological scales, empirically disfavored. **Treat the structural insight as established-but-non-predictive, and the emergent-dark-matter claim as contested-to-refuted.** Epistemic tag: `[CONTESTED]`.

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## 8. ER=EPR and It-from-Qubit / Tensor-Network Spacetime

**Core idea.** Spacetime geometry and connectivity emerge from quantum entanglement.

- **ER=EPR (Maldacena–Susskind 2013):** the Einstein–Rosen bridge ("ER", a non-traversable wormhole) connecting two black holes is identical to the entanglement ("EPR") between them — every entangled pair is connected by a (possibly Planckian) geometric bridge.
- **It-from-qubit:** the fundamental ontology is quantum information; spacetime is built from the entanglement structure of an underlying quantum system.
- **Tensor networks** (MERA, HaPPY codes, random tensor networks) provide discrete toy models in which bulk geometry and a quantum error-correcting code emerge from a network of entangled degrees of freedom, concretely realizing RT.

**What it claims to unify.** Geometry/gravity with quantum information: entanglement *is* connectivity; emergent spacetime is the organization of an entangled state. Connects black-hole physics, quantum error correction, and holography. See *Chapter 17 (p. 325)* ("spacetime from entanglement") and *Chapter 12 (p. 155)*.

**Mathematical status.**

- Tensor-network models rigorously reproduce key holographic features (RT entropy, bulk reconstruction as quantum error correction) in toy settings `[INFERENCE, within AdS/CFT]`. The "first-law → linearized Einstein equations from entanglement" result is concrete.
- **ER=EPR itself is a bold conjecture** supported by suggestive constructions (traversable wormholes via double-trace deformations; the thermofield-double / eternal-black-hole correspondence) but has no first-principles derivation and no proof it holds for generic entanglement `[SPECULATIVE]`. The general claim "spacetime = entanglement" is not a theorem.

**Empirical status (blunt).** No experimental confirmation `[ESTABLISHED – empirical record]`. There is no laboratory or astrophysical test of ER=EPR or of entanglement literally building spacetime. (Claims of "wormhole" simulation on quantum processors are simulations of toy holographic models, **not** observations of physical wormholes, and were widely criticized as overstated `[CONTESTED]`.) These ideas live almost entirely within AdS/CFT-style theory.

**Main successes.**

- Tensor networks give a concrete, calculable mechanism for emergent bulk geometry and bulk reconstruction as quantum error correction — a genuine conceptual advance `[INFERENCE]`.
- ER=EPR offers an elegant, unifying handle on several black-hole puzzles (reconciling entanglement structure with smooth horizons; aspects of the firewall paradox) `[INFERENCE]`.
- Sharpens the RT "geometry = entanglement" dictionary and the derivation of linearized gravity from entanglement first laws `[ESTABLISHED, structurally]`.

**Main problems / criticisms.**

- ER=EPR has no first-principles derivation; whether *generic* entanglement yields a smooth geometric connection is unknown `[SPECULATIVE]`.
- Demonstrated only within AdS / large-$N$ holography; relevance to our de Sitter universe is unestablished `[OPEN]`.
- The type-III von Neumann algebra structure of QFT means there is **no fundamental tensor factorization** of local regions, so "qubits of the field" and clean entanglement entropy are subtle/ill-defined without a cutoff — a real foundational gap for the literal it-from-qubit picture `[OPEN]`.
- Full nonlinear, background-independent reconstruction of Einstein gravity from entanglement is not achieved (only linearized) `[OPEN]`.

**Testability.** Essentially untestable with current means; proposals concern Planck/quantum-gravity-scale structure and special holographic settings `[CONTESTED]`.

**Verdict.** An exciting, influential direction with real results inside AdS/CFT (tensor-network holography, bulk reconstruction as error correction, gravity-from-entanglement first laws). But ER=EPR and "spacetime is built from entanglement" as universal claims are bold conjectures, demonstrated only in special holographic models, with no empirical confirmation and unresolved foundational issues (type-III algebras, de Sitter). **Principled and promising, but speculative as fundamental physics.** Epistemic tag: **`[SPECULATIVE]`**.

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## 9. Twistor Theory and the Scattering-Amplitudes Program (Amplituhedron)

**Core idea.**

- **Twistor theory** (Penrose, late 1960s): reformulate physics with **twistor space** — a complex projective space encoding null structure/light rays — as more fundamental than spacetime points; spacetime events become derived objects, with the hope that quantum and gravitational structure are more natural in twistor variables.

- **Modern amplitudes program:** scattering amplitudes in gauge theory (especially planar $\mathcal{N} = 4$ super-Yang–Mills) have striking hidden simplicity invisible in Feynman diagrams. Twistor-inspired methods (BCFW recursion, twistor-string theory, Grassmannian formulations) and the **amplituhedron** (Arkani-Hamed–Trnka) recast amplitudes as the "volume" of a positive-geometry object, with locality and unitarity *emergent* rather than assumed.

**What it claims to unify.** Twistor theory aims to unify quantum mechanics and spacetime/gravity by changing the geometric arena. The amplitudes/amplituhedron program unifies scattering data with positive geometry and combinatorics, seeking to derive spacetime locality and quantum unitarity as emergent consequences.

**Mathematical status.**

- Twistor methods are rigorous, productive mathematical physics (the Penrose transform, twistor descriptions of massless fields and self-dual sectors, links to integrable systems and algebraic geometry) `[ESTABLISHED]`.
- The amplitudes program has produced concrete, proven machinery: BCFW recursion, the Grassmannian formulation of $\mathcal{N} = 4$ SYM amplitudes, and the amplituhedron reproducing planar $\mathcal{N} = 4$ SYM integrands `[ESTABLISHED, within its domain]`.
- Generalizing twistors to full nonlinear gravity (the "googly" / nonlinear-graviton problem) is a longstanding obstruction; the amplituhedron is established chiefly for planar $\mathcal{N} = 4$ SYM — a maximally supersymmetric, conformal toy theory, not realistic theories with gravity `[OPEN]`.

**Empirical status (blunt).** No new physical predictions / no distinguishing confirmation as fundamental physics `[ESTABLISHED — empirical record]`. Crucially, the machinery is enormously useful *practically*: modern on-shell/unitarity methods dramatically streamline real QCD and gravitational (post-Minkowskian) amplitude computations used for the LHC and for gravitational-wave waveform modeling. So it has indirect empirical relevance as a **calculational technology** — but the deeper claim that twistor/positive geometry is the fundamental substrate of spacetime is untested `[INFERENCE]`.

**Main successes.**

- BCFW recursion, on-shell/unitarity methods, and the amplituhedron reveal genuine, proven hidden structure in gauge amplitudes (planar $\mathcal{N} = 4$ SYM especially) `[ESTABLISHED]`.
- These methods are practically transformative for amplitude computation in QCD (LHC) and gravitational two-body dynamics (gravitational-wave templates) — a real empirical payoff at the level of technique `[ESTABLISHED]`.
- Suggestive evidence that locality and unitarity may be emergent from a deeper geometric/combinatorial object — a genuinely novel reframing `[INFERENCE]`.
- Deep, ongoing two-way exchange with pure mathematics (algebraic geometry, combinatorics, positive geometry) `[ESTABLISHED]`.

**Main problems / criticisms.**

- No extension of twistor theory to full nonlinear GR (the nonlinear-graviton problem) `[OPEN]`.
- The amplituhedron is established for planar $\mathcal{N} = 4$ SYM, not the real Standard Model or gravity; whether positive geometry reaches realistic theories is unresolved `[OPEN]`.
- "Spacetime/locality is emergent from positive geometry" is a research aspiration, not an established result `[SPECULATIVE]`.
- As a *fundamental* theory it supplies no dynamics for our universe and no contact with quantum gravity beyond toy models `[INFERENCE]`.

**Testability.** No direct test of twistors-as-fundamental `[CONTESTED, as fundamental physics]`. Indirectly, the computational methods are validated daily against collider and gravitational-wave data — but that validates the *techniques*, not the foundational claim.

**Verdict.** Mathematically rich and genuinely successful — but in two senses that must not be conflated. As **computational technology** (BCFW, unitarity methods, amplituhedron for $\mathcal{N} = 4$ SYM) it is `[ESTABLISHED]` and even empirically useful for LHC/LIGO calculations. As a candidate **foundational theory** (twistors/positive geometry as the substrate from which spacetime, locality, and quantum mechanics emerge) it is `[SPECULATIVE]`, restricted to toy models, and untested. **The tag reflects the deeper foundational ambition.** Epistemic tag: **`[SPECULATIVE]`**.

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## 10. Group Field Theory (GFT)

**Core idea.** A quantum field theory whose fields live on group manifolds (not spacetime), constructed so that its Feynman diagrams generate spin-foam amplitudes and its quanta are the "atoms of space" of LQG-style discrete geometry. GFT is a second-quantized reformulation of LQG/spin foams in which spacetime and geometry are meant to emerge from collective dynamics (e.g. a condensate phase) of geometric quanta.

**What it claims to unify.** A common field-theoretic framework unifying LQG, spin foams, and (via the perturbative expansion) discrete random-geometry / tensor-model approaches; aims to derive emergent continuum spacetime and cosmology from many-body dynamics of geometric quanta. Not a unification of forces.

**Mathematical status.** Well-defined as a class of QFTs on group manifolds; the GFT–spin-foam amplitude correspondence is concrete `[INFERENCE]`. Renormalization of GFT models (fixed-point/asymptotic-safety structure of certain GFTs) is an active, partially-developed area. Establishing the right model, proving the existence of a geometric (condensate) phase and a controlled continuum limit, and recovering smooth GR are unresolved `[OPEN]`.

**Empirical status (blunt).** No experimental confirmation `[ESTABLISHED – empirical record]`. Connects to phenomenology mainly through **GFT cosmology**, where a condensate of GFT quanta is argued to yield effective Friedmann dynamics with a bounce — a theoretical derivation, not an observation.

**Main successes.**

- A principled second-quantized framework tying together spin foams, LQG, and tensor/random-geometry models, supplying a notion of summing over spin-foam complexes `[INFERENCE]`.
- GFT condensate cosmology derives effective cosmological (bounce) dynamics from the microscopic theory — a concrete attempt at the micro-to-macro link that LQG lacks `[INFERENCE]`.
- RG analyses give some control over continuum behavior `[INFERENCE]`.

**Main problems / criticisms.**

- Which GFT model is correct, and whether any admits a geometric phase with the right continuum limit, is unresolved `[OPEN]`.

*Iteration-6 status refresh (2026-06-10):* mean-field/Landau–Ginzburg evidence for a condensate transition now exists in realistic Lorentzian models (Marchetti–Oriti–Pithis–Thürigen, PRL 130, 141501 (2023); Lorentzian Barrett–Crane follow-up Dekhil–Jercher–Pithis, PRD 111, 026014 (2025), DOI 10.1103/PhysRevD.111.026014 [author list web-verified 2026-06-19]); whether that phase is geometric/continuum-GR remains `[OPEN]`. → *[notes* (Working note: *Iteration 6 — R4: The Other Background-Independent Programs, 2023–2026 (Landscape Refresh)* (repo: `notes/2026-06-10-iter6-background-independent-programs.md`))]

- Recovery of full classical GR and standard QFT/matter is not established `[OPEN]`.
- Inherits the unsolved dynamics / semiclassical-limit problems of spin foams and LQG `[INFERENCE]`.

- Largely formal; limited contact with observation beyond model-dependent cosmology `[OPEN]`.

**Testability.** GFT cosmology could in principle leave imprints (bounce signatures in the CMB), but nothing is confirmed and predictions are model-dependent and currently inaccessible `[CONTESTED]`.

**Verdict.** A technically sophisticated unifying framework for the discrete/loop family, valuable for a field-theoretic handle on spin-foam sums and a route to emergent cosmology. But highly formal, model-dependent, sharing the unsolved dynamics/continuum-limit problems of its parents, with no empirical confirmation. **Open and developmental.** Epistemic tag: **`[OPEN]`**.

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## 11. Noncommutative Geometry (Connes program)

**Core idea.** Generalize geometry by replacing the commutative algebra of functions on a space with a **noncommutative algebra**, encoded in a *spectral triple* $(\mathcal{A}, \mathcal{H}, D)$ — an algebra, a Hilbert space, and a Dirac operator. Geometry is read off spectrally, allowing "spaces" with no ordinary points (Connes' reconstruction theorem recovers a Riemannian manifold from commutative spectral data). In physics: spacetime is taken as the product of ordinary 4D spacetime with a tiny *finite* noncommutative internal space; the Standard Model — its gauge group, fermion content, and Higgs sector — is then *derived* as the geometry of that internal space, with the Higgs appearing as part of the connection. Separately, noncommutativity of the coordinates themselves, $[x^\mu, x^\nu] = i\theta^{\mu\nu} \neq 0$ , is a candidate Planck-scale structure providing a minimal length.

**What it claims to unify.** Gravity and the Standard Model geometrically: gauge fields and the Higgs arise on the same footing as gravity, as aspects of one noncommutative geometry. The **spectral action** yields the Einstein–Hilbert action plus the full SM action from a single geometric principle. See *Chapter 9 (p. 106)*, *Chapter 13 (p. 172)*.

**Mathematical status.** Noncommutative geometry is rigorous, deep mathematics (Connes; index theory, operator algebras, spectral reconstruction) `[ESTABLISHED]`. The spectral-triple formulation of the SM is mathematically precise. The "almost-commutative" construction reproduces the SM gauge group and representation content remarkably economically and relates couplings at a unification scale `[INFERENCE/CONTESTED]`. Specific predictions from the spectral action depend on assumptions about boundary conditions and the precise internal algebra `[CONTESTED]`.

**Empirical status (blunt).** No confirmed distinguishing prediction, and a notable setback: an early, much-cited version predicted a Higgs mass around $\sim 170$ GeV, **excluded** by the 2012 discovery of the $\approx 125$ GeV Higgs `[ESTABLISHED]`. The framework was subsequently modified (adding a scalar "$\sigma$" field) to accommodate 125 GeV — a post-hoc adjustment, not a successful prediction `[ESTABLISHED]`. Coordinate noncommutativity could induce Lorentz violation / modified dispersion, tightly constrained by high-energy astrophysics with no positive signal `[ESTABLISHED]`.

**Main successes.**

- A genuinely deep mathematical framework (operator-algebraic geometry, spectral reconstruction) of independent value `[ESTABLISHED]`.
- Derives the SM's intricate gauge group, fermion representations, and Higgs sector from a compact geometric input — an elegant organizing principle that "explains" some SM structure as geometry `[INFERENCE]`.
- Unifies gravitational and SM actions in one spectral action; offers a natural geometric origin for the Higgs as a connection `[INFERENCE]`.

**Main problems / criticisms.**

- The original $\sim 170$ GeV Higgs-mass prediction was experimentally falsified; the model was adjusted afterward `[ESTABLISHED]`.

- The construction is largely classical/effective-action; a full quantum theory (quantizing the spectral triple, controlling spectral-action quantum corrections) is not established `[OPEN]`.
- The choice of internal algebra and the inputs yielding the SM are not uniquely forced; some appear engineered to match known physics `[CONTESTED]`.
- Coordinate-noncommutativity models face strong LIV constraints and unresolved unitarity/causality issues (UV/IR mixing) `[OPEN]`.

**Testability.** The SM-from-geometry version makes relations among couplings/masses that *are* in principle testable — and one such prediction was falsified, forcing modification. Coordinate noncommutativity is constrained by LIV searches. **Some genuine, if limited, falsifiability** `[CONTESTED]`.

**Verdict.** Mathematically profound and conceptually attractive — it comes closer than most programs to "deriving" the SM's structure from one geometric principle, and it is honestly falsifiable (its original Higgs-mass prediction was killed by data, then patched). But it remains an effective/classical construction without a complete quantum theory, with non-unique inputs and no surviving confirmed prediction. **A serious, partly-falsified, still-open program.** Epistemic tag: **`[OPEN]`**.

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## 12. Grand Unified Theories (GUTs)

**Core idea.** Embed the SM gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$ into a single larger *simple* gauge group — $SU(5)$, $SO(10)$, or Pati–Salam $SU(4) \times SU(2)_L \times SU(2)_R$ — spontaneously broken at $\sim 10^{16}$ GeV down to the SM. A single unified coupling splits into the three observed couplings; quarks and leptons share multiplets, explaining charge quantization. $SO(10)$ places a full SM generation plus a right-handed neutrino in one **16**-dimensional irreducible representation.

**What it claims to unify.** The strong, weak, and electromagnetic gauge interactions into one group with one gauge coupling; quarks and leptons into shared multiplets. **Does not include gravity.** Explains electric charge quantization (commensurate quark/lepton charges) and, in $SO(10)$, motivates the seesaw mechanism for small neutrino masses. See *Chapter 9 (p. 106)*.

**Mathematical status.** GUTs are well-defined, renormalizable QFTs built from standard, rigorous machinery — far more mature and conventional than any quantum-gravity program `[ESTABLISHED]`. Their group theory, symmetry breaking, and RG running are fully under control.

**Empirical status (and partly negative).** Three pieces of indirect support, one sharp negative:

1. The three SM gauge couplings, run up via the RG, *nearly* meet near $\sim 10^{16}$ GeV — imperfectly in the plain SM, more accurately with low-scale supersymmetry `[INFERENCE]`.
2. Charge quantization is observed (proton and electron charges equal to $\sim 1$ part in $10^{21}$) `[ESTABLISHED]`.
3. $SO(10)$ naturally accommodates confirmed nonzero neutrino masses via right-handed neutrinos and the seesaw `[INFERENCE]`.
4. **But the signature prediction — proton decay — has not been observed.** Super-Kamiokande and predecessors set lifetime bounds exceeding $\sim 10^{34}$ years (channel-dependent), which **exclude minimal non-supersymmetric** $SU(5)$ and tightly constrain other models `[ESTABLISHED]`. GUTs are thus partially falsified (the simplest version is dead) and otherwise unconfirmed.

**Main successes.**

- Approximate gauge-coupling unification near $10^{16}$ GeV (notably improved with SUSY) is real, suggestive support `[INFERENCE]`.
- Explains charge quantization elegantly — a fact the SM merely accommodates `[ESTABLISHED]`.
- $SO(10)$ organizes a full fermion generation (with right-handed neutrino) into one multiplet and naturally yields the seesaw,

fitting observed neutrino masses `[INFERENCE]`.
- Provides a framework for baryogenesis via leptogenesis (tied to heavy Majorana neutrinos) `[INFERENCE]`.

**Main problems / criticisms.**

- No proton decay observed; minimal $SU(5)$ excluded, other models squeezed — the central prediction is so far negative `[ESTABLISHED]`.
- The unification scale ($\sim 10^{16}$ GeV) is far beyond direct reach, so most predictions are inaccessible `[OPEN]`.
- Introduces a large hierarchy (the doublet–triplet splitting problem) and depends on unobserved heavy fields; SUSY GUTs require unobserved superpartners `[OPEN]`.
- Does not include gravity; not a theory of everything `[OPEN]`.

**Testability.** GUTs make the sharp, falsifiable prediction of **proton decay** with specific branching ratios and mass/mixing relations; ongoing and next-generation detectors (Hyper-Kamiokande, DUNE) probe the relevant lifetimes `[ESTABLISHED — genuinely testable, a virtue]`. Far more testable than any quantum-gravity program — and the simplest model has already been falsified.

**Verdict.** The most empirically grounded and falsifiable program here — conventional, mature QFT with concrete, testable predictions (proton decay, coupling unification, neutrino-mass relations). Its motivations are real and attractive. But the flagship prediction has not been seen, killing minimal $SU(5)$ and constraining the rest, while the unification scale keeps most physics out of reach. **A live, partly-falsified, genuinely scientific program — distinct in kind from the speculative quantum-gravity proposals.** Epistemic tag: **`[OPEN]`**.

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## 13. Discrete / Computational Approaches (Wolfram hypergraph model; digital physics)

**Core idea.** The universe is fundamentally a discrete computational/combinatorial structure. In Wolfram's model, reality is a hypergraph evolving by simple local rewriting rules; space, time, particles, and laws are meant to emerge as large-scale features, with relativity emerging from causal-graph structure and quantum mechanics from multiway (branching/merging) evolution. More broadly, "digital physics" (Zuse, Fredkin, 't Hooft's cellular-automaton interpretation of QM) posits that physics is at bottom computation, and that continuum spacetime/QFT are emergent approximations.

**What it claims to unify.** Everything — spacetime, gravity, quantum mechanics, particle physics — as emergent features of a single underlying computational rule; a common discrete-computational substrate beneath relativity and quantum theory.

**Mathematical status.** Rich in suggestive constructions and terminology (causal graphs, multiway systems, "branchial space", claimed analogies to GR and QM) but, as of the knowledge cutoff, the Wolfram Physics Project has **not** produced a rigorous derivation of the Einstein equations, the Standard Model, or quantitative quantum-mechanical predictions from a specific rule `[CONTESTED/SPECULATIVE]`. Arguments are largely qualitative/heuristic and have not been validated by the broader physics community. 't Hooft's cellular-automaton interpretation is more carefully argued but still `[SPECULATIVE/CONTESTED]` and runs into the standard obstacle that Bell-inequality violations require either superdeterminism or abandoning measurement independence.

**Empirical status (blunt).** No distinguishing experimental confirmation and no novel quantitative prediction that has been tested `[ESTABLISHED – empirical record]`. No specific rule has been shown to reproduce known physics, let alone predict something new. Fundamentally discrete, Lorentz-violating substrates are constrained (and disfavored) by the same high-energy LIV tests that constrain other discrete approaches, though proponents argue Lorentz invariance emerges.

**Main successes.**

- A vivid conceptual picture of emergent space/time from causal structure, reproducing some qualitative features (causal-graph analogs of light cones) — at the level of analogy `[CONTESTED]`.
- Connects to legitimate ideas (causal sets, holographic/computational-limit literature, the physical Church–Turing question) and to the broadly respectable view that the continuum may be effective `[INFERENCE]`.

**Main problems / criticisms.**

- No derivation of quantitative physics: no specific Einstein equations, no SM spectrum, no tested numbers — the gap between analogy and physics is enormous and unbridged `[CONTESTED/SPECULATIVE]`.
- Generic discrete substrates violate Lorentz invariance; exact emergent Lorentz invariance (as causal sets achieve via sprinkling) is not established here `[OPEN]`.
- Reproducing QM (Bell violations, contextuality) from underlying determinism requires superdeterminism or giving up measurement independence — widely regarded as conspiratorial/unmotivated `[CONTESTED]`.
- Methodological criticism: heavy reliance on suggestive terminology and rule-search without falsifiable, quantitative output; not validated by mainstream peer review at the level of a physical theory `[CONTESTED]`.
- **Rule selection** — *which* rule is the universe? — has no principle; the framework can a posteriori accommodate almost anything, undermining predictivity `[OPEN]`.

**Testability.** Proponents suggest possible signatures (discreteness effects, dimension fluctuations, deviations in extreme-energy or black-hole regimes), but no concrete, falsifiable, quantitative prediction has been put forward and tested `[CONTESTED]`. In practice currently untestable.

**Verdict.** The most speculative and least empirically/mathematically grounded program in this survey. The *general* idea that physics might have a discrete or computational substrate is legitimate and overlaps with more disciplined approaches (causal sets, holographic computational bounds). But the *specific* computational/hypergraph proposals have not derived known physics quantitatively, have made no tested predictions, face the standard discreteness (Lorentz) and quantum (Bell) obstacles, and are not validated by the community. **Suggestive metaphysics, not yet physics.** Epistemic tag: **`[SPECULATIVE]`**.

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## 13a. Postquantum Classical–Quantum (CQ) Hybrid Gravity (Oppenheim program) — *added iteration 6 (2026-06-10)*

**Core idea.** Gravity is **fundamentally classical**; the conflict with quantum matter is resolved not by quantizing the metric but by coupling a classical metric to quantum matter through linear, completely positive, trace-preserving (CPTP) classical–quantum dynamics (Oppenheim, *A postquantum theory of classical gravity?*, arXiv:1811.03116; PRX 13, 041040 (2023)). Consistency forces the evolution of **both** metric and matter to be fundamentally stochastic, with built-in decoherence and no measurement postulate; the dynamics reduces to GR in the classical limit.

**What it claims to unify.** GR and quantum matter — by *denying* that gravity must be quantized. The direct negation of the gate assumption (gravity is non-classical) shared by Programs A and B.

**Mathematical status.** The CQ master-equation framework is explicit and the central structural result is a theorem within it: the **decoherence–diffusion trade-off** (Oppenheim–Sparaciari–Šoda–Weller-Davies, arXiv:2203.01982; Nat. Commun. 14, 7910 (2023)) — maintaining quantum coherence forces a minimum stochastic diffusion of the classical sector, observable as gravitational force noise `[ESTABLISHED — theorem within the CQ framework]`. Published consistency critiques exist (diffeomorphism invariance of the CQ path integral; pathologies of ultralocal/continuous realizations) `[CONTESTED]`.

**Empirical status (the rare program with a distinguishing near-term test).** The trade-off is squeezed from both sides by existing data: Janse–Uitenbroek–van Everdingen–Plugge–Hensen–Oosterkamp (arXiv:2403.08912; Phys. Rev. Research 6, 033076 (2024)) compile force-noise bounds from single atoms to kg masses, lowering the diffusion upper bound by **at least 15 orders of magnitude** against the initial benchmarks for explicit models. Status of the three explicit kernels `[ESTABLISHED]`: ultralocal-continuous — dead (ruled out in the original paper); ultralocal-discrete — the atom-interferometry figure of merit now sits *below* the model's trade-off lower bound (excluded modulo the authors' stated caveats); non-local continuous — survives by ~1 order of magnitude. The three kernels are representative, not an exhaustive parametrization of CQ dynamics `[INFERENCE]`.

**Main successes.** A mathematically explicit, internally consistent "gravity stays classical" alternative; a falsifiable, theorem-backed prediction being tested *now* with existing data; defines the **asymmetric gate-closure route** — complete falsification of the CQ kernels would close Program A/B's gate (is gravity non-classical?) from the classical side, possibly before BMV ever runs (see *Chapter 39 (p. 725)*).

**Main problems / criticisms.** Fundamental stochasticity and energy/momentum non-conservation pressure `[CONTESTED]`; **Hotta–Murk–Terno "QG in disguise" counter-pressure** **`[CONTESTED]`**: any CP classical–quantum dynamics admits an embedding into unitary quantum theory on an enlarged Hilbert space (plus an exhibited closed CP system violating angular-momentum conservation) — if it stands, the hybrid description is an effective sector of a quantum theory and the quantum-vs-classical gate dichotomy loses foundational sharpness, even while the diffusion *phenomenology* stays falsifiable; no published concession found as of 2026-06. Nonlocal classical mediators (Hall–Reginatto-type) sit outside the trade-off's reach.

**Testability.** Yes — the program's defining virtue: decoherence–diffusion force noise, currently running on archival data; complementary to (and faster than) BMV.

**Verdict.** A genuine, falsifiable competitor at the gate. Its existence sharpens, by contrast, the wiki's NOT-YET-PHYSICS demotion of the central wager: a surveyed program *can* carry a distinguishing test, so the demotion cannot be excused as "no quantum-gravity-adjacent program is testable." Simplest realizations close to excluded; class not excluded; fundamentality contested. Epistemic tag: `[CONTESTED]` (framework established as mathematics; physics contested on both experimental and embedding fronts). → *[notes* (Working note: *Iteration 6 — R4: The Other Background-Independent Programs, 2023–2026 (Landscape Refresh)* (repo: `notes/2026-06-10-iter6-background-independent-programs.md`)), *notes* (Working note: *Quantum-gravity phenomenology 2022–2026: the distinguishing-prediction hunt, audited* (repo: `notes/2026-06-10-iter6-qg-phenomenology.md`))]

## 13b. The Swampland Dark-Dimension Scenario (string-phenomenology corner) — *added iteration 6 (2026-06-10)*

**Core idea.** Combine the swampland distance/duality conjectures with the observed value of Λ to argue for **one mesoscopic extra dimension** at $\ell \sim \lambda\,\Lambda^{-1/4}$ — central value ~1 μm, with the 0.1–10 μm band reflecting the O(1) spread in λ — carrying a Kaluza–Klein graviton tower, neutrino-mass relations tied to $\Lambda^{1/4}$, and a species scale ~$10^9$–$10^{10}$ GeV (Montero–Vafa–Valenzuela, arXiv:2205.12293; JHEP 02 (2023) 022). KK-graviton dark matter is the separate follow-up (Gonzalo–Montero–Obied–Vafa, arXiv:2209.09249 [unverified]).

**What it claims to unify.** Not a new framework — a falsifiable *corner* of string theory connecting Λ, neutrino masses, dark matter, and short-distance gravity.

**Mathematical status.** Conjecture-stack: rests on swampland conjectures that are themselves `[SPECULATIVE]`; internally consistent dimensional analysis with O(1) ambiguities acknowledged by proponents.

**Empirical status / Testability.** The defining virtue: **micron-scale deviations from Newtonian gravity** — torsion-balance inverse-square-law tests need roughly an order-of-magnitude improvement to reach the predicted band; a null through the band would *strongly disfavor* (not logically exclude, given the O(1) ambiguities) the scenario `[INFERENCE]`. Extension to an evolving dark sector: Bedroya–Obied–Vafa–Wu (arXiv:2507.03090, v3 May 2026) tie dark-energy evolution to a varying dark-matter mass via a rolling scalar and claim DESI and SN data independently favor the predicted O(1) couplings, with apparent phantom ($w < -1$) behavior naturally mimicked `[CONTESTED — proponent fit, prior-dependent]`. This adds independent weight to the wiki's eternal-dS wrong-idealization hedge (A-20) without resolving it.

**Related string-side status note.** The dS-vacua debate moved from "no construction" to "candidates, control unestablished": McAllister–Moritz–Nally–Schachner, *Candidate de Sitter Vacua* (arXiv:2406.13751; PRD 111, 086015 (2025)) assemble for the first time all KKLT components in explicit Calabi–Yau orientifolds, exhibiting ~30 candidate metastable dS vacua (per the arXiv listing) at leading order in $\alpha'$ and $g_s$; stability beyond leading order is `[OPEN]`.

**Main problems / criticisms.** Conjecture-dependence; O(1) ambiguities make the falsifier a band, not a line; the DESI fit is a proponent analysis `[CONTESTED]`; tension between the dark-dimension corner and the KKLT-candidate corner of the same theory is unaddressed.

**Verdict.** A falsifiable swampland corner with a dated experimental hook (micron-scale ISL) and a contested claim on live cosmology data. Like the CQ entry, its testability sharpens by contrast the not-yet-physics status of the central wager. Epistemic tag: `[SPECULATIVE]` (consequences falsifiable; foundations conjectural). → *[notes* (Working note: *Iteration 6 — R4: The Other Background-Independent Programs, 2023–2026 (Landscape Refresh)* (repo: `notes/2026-06-10-iter6-background-independent-programs.md`))]

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## 14. Comparative table

Maturity is a coarse judgment of mathematical/internal development, not of truth.

| PROGRAM | MATURITY | DISTINGUISHING EMPIRICAL TEST? | MAIN STRENGTH | MAIN WEAKNESS |
|---|---|---|---|---|
| String / M-theory + SUGRA | Very high (perturbative); low (non-perturbative) | None at accessible energies; model-dependent SUSY/extra-dim signatures absent | Perturbatively finite gravity; black-hole microstate counting; AdS/CFT | Landscape destroys predictivity; SUSY & extra dimensions unobserved; no non-perturbative definition |
| Loop Quantum Gravity / Spin Foams | High (kinematics); low (dynamics) | LIV / LQC bounce signatures — no positive signal; LIV tightly bounded | Background-independent quantization; discrete area/volume spectra | Dynamics ambiguous; semiclassical limit unproven; no matter sector |
| Asymptotic Safety | Medium–high (FRG) | None clean; indirect SM-coupling hints (contested) | Most economical: QFT all the way up; robust FRG fixed-point evidence | Truncation-dependent; Lorentzian/unitarity issues; no proof beyond truncations |

| | | | | |
|---|---|---|---|---|
| Causal Dynamical Triangulations | Medium (numerical) | None; possible UV dimensional-reduction echoes | Dynamically yields 4D de Sitter-like universe; dimensional reduction | No proven continuum limit; foliation may break Lorentz spirit; no matter |
| Causal Set Theory | Medium (kinematics); low (dynamics) | Fluctuating-$\Lambda$ heuristic (predates 1998!); varying dark energy | Lorentz-invariant discreteness via sprinkling; early $\Lambda$ heuristic | No complete dynamics; entropy/manifoldness problem; no matter |
| AdS/CFT & Holography | Very high (within AdS) | None for our universe (wrong sign of $\Lambda$); analog tests only | Exact gauge/gravity duality; unitary Page curve; RT entanglement geometry | Restricted to AdS, large-$N$; our universe is de Sitter; no general proof |
| Entropic / Emergent Gravity | High (Jacobson); low (Verlinde DM) | Derivation versions: none by design. Verlinde DM: testable & disfavored | Jacobson derives Einstein eqs as equation of state | No microscopic substrate; Verlinde DM fails clusters/CMB/Bullet |
| ER=EPR / It-from-Qubit / Tensor Networks | Medium (within AdS) | None; "wormhole" sims are toy-model, not physical | Tensor-network holography; bulk reconstruction = error correction | ER=EPR unproven for generic entanglement; type-III algebra obstruction; AdS-only |
| Twistors / Amplituhedron | High (as technique); low (as foundation) | None as foundation; techniques validated daily (LHC/LIGO) | Proven hidden amplitude structure; transformative computation | Planar $\mathcal{N} = 4$ SYM toy model; no nonlinear gravity; emergence is aspiration |

| | | | | |
|---|---|---|---|---|
| Group Field Theory | Medium (formal) | None; GFT-cosmology bounce signatures (inaccessible) | Field-theoretic handle on spin-foam sums; emergent cosmology route | Model-dependent; inherits LQG dynamics problem; highly formal |
| Noncommutative Geometry (Connes) | High (math); medium (physics) | Yes — and a Higgs-mass prediction was falsified, then patched | "Derives" SM gauge group/Higgs from one geometric principle | Classical/effective only; non-unique inputs; falsified original prediction |
| Grand Unified Theories | Very high (conventional QFT) | **Yes — proton decay; minimal $SU(5)$ already excluded** | Charge quantization; coupling unification; seesaw in $SO(10)$ | Proton decay unseen; no gravity; unification scale inaccessible |
| Discrete / Computational (Wolfram, digital physics) | Low | None concrete; in practice untestable | Vivid emergence picture; overlaps legitimate discrete ideas | No quantitative derivation of known physics; Lorentz & Bell obstacles; rule selection arbitrary |
| Postquantum CQ hybrid gravity (Oppenheim) | Medium (explicit framework + trade-off theorem) | **Yes — decoherence–diffusion trade-off; simplest explicit models squeezed ≥15 orders (PRResearch 2024); class not excluded** | Falsifiable "gravity stays classical" alternative; gate-closure from the classical side | Stochasticity/conservation pressure; "QG in disguise" embedding claim `[CONTESTED]` |

| | | | | |
|---|---|---|---|---|
| Swampland dark dimension | Low–medium (conjecture stack) | **Yes — micron-scale inverse-square-law band; DESI evolving-dark-energy fit claimed** [CONTESTED] | Dated experimental hook tying Λ, neutrinos, dark matter | Rests on [SPECULATIVE] conjectures; O(1) band, not a line; proponent-fit cosmology |

## 15. Honest assessment of the field

A few conclusions that the survey supports, stated plainly:

1. **No program has distinguishing experimental confirmation as fundamental physics** [ESTABLISHED — empirical record]. The single program that makes a sharp, falsifiable, near-term prediction (GUTs / proton decay) has had its simplest version *falsified*, and the noncommutative-geometry program had a genuine prediction (Higgs mass) *falsified* and then patched. Honest falsifiability and being correct are different things; only the former has clearly occurred, and only negatively.
2. **The empirical–ambition trade-off is structural, not accidental** [INFERENCE]. The Planck scale is $\sim 10^{15}$ times beyond direct reach, so any program reaching it is forced into Planck-scale invisibility. GUTs are testable precisely because they stop short of gravity. This is a feature of where the relevant physics lives, not (mainly) a failure of imagination.
3. **Some internal results are genuinely established and will survive** regardless of which program (if any) is ultimately correct [INFERENCE]: black-hole entropy $S = A/4\ell_{\mathrm{Pl}}^2$ and its microscopic countings (Strominger–Vafa; LQG punctures); AdS/CFT as a calculational duality; Jacobson's equation-of-state derivation; RT and the entanglement first-law derivation of linearized gravity; BCFW/unitarity amplitude methods;

CDT/asymptotic-safety UV dimensional reduction. These are the durable deposits.

4. **Convergent hints recur across independent programs** `[INFERENCE]`, suggesting they track something real even though no single program is confirmed: (i) the holographic area-law of entropy and "spacetime from entanglement"; (ii) UV dimensional reduction to $\approx 2$; (iii) some form of minimal length / discreteness near $\ell_{\mathrm{Pl}}$; (iv) the deep role of horizon thermodynamics. See *Chapter 17 (p. 325)*.

5. **The de Sitter problem is the field's quiet scandal** `[INFERENCE/OPEN]`. Our universe has a small *positive* cosmological constant, yet the most rigorous tool (AdS/CFT) requires *negative* curvature, string theory struggles to construct stable de Sitter vacua (KKLT vs. swampland), and most programs say little about realistic cosmology. The approach with the best (if heuristic) track record on $\Lambda$ — causal sets — is among the least developed dynamically.

6. **"Untestable" should be parsed carefully** `[INFERENCE]`. It can mean *currently inaccessible* (most quantum-gravity programs — a contingent limitation) or *unfalsifiable in principle* (the strongest critique, leveled at the string landscape and at rule-selection in computational models — a methodological objection). These are not the same, and conflating them is sloppy. A domain-of-validity limitation (AdS-only holography) is also distinct from a logical inconsistency: none of these programs is known to be *internally inconsistent*; they are unconfirmed and, in places, incomplete. See *Chapter 14 (p. 191)* and *Chapter 2 (p. 8)*.

7. **Recommended posture.** Treat the most-developed mathematical frameworks (string theory, AdS/CFT) as *possibly-correct mathematics whose relation to nature is unestablished*; treat the conservative QFT programs (asymptotic safety, GUTs) as the most scientifically orthodox bets; treat the emergent/information-theoretic ideas as *deep structural clues* rather than finished theories; and treat the

computational/hypergraph proposals as metaphysics awaiting a quantitative result. Update sharply on any of: a proton-decay detection, a SUSY discovery, a controlled de Sitter holography, or a Planck-scale LIV signal — none of which has occurred `[ESTABLISHED — empirical record]`.

The field is not in crisis so much as in a long evidential drought imposed by the Planck scale. Intellectual honesty requires holding the best ideas as *promising and unconfirmed*, neither overselling decades of beautiful mathematics as established physics nor dismissing it for failing a test no foreseeable experiment can administer.

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## References

See the Bibliography (p. 776) for full citations. Landmark sources referenced above include (real, standard works): Polchinski, *String Theory* (vols. I–II); Maldacena (1997) on AdS/CFT; Strominger & Vafa (1996) on black-hole microstate counting; Rovelli, *Quantum Gravity*, and Rovelli & Vidotto, *Covariant Loop Quantum Gravity*; Weinberg (1979) and Reuter on asymptotic safety; Ambjørn, Jurkiewicz & Loll on CDT; Sorkin and Bombelli–Lee–Meyer–Sorkin on causal sets; Ryu & Takayanagi (2006); Jacobson (1995), *"Thermodynamics of Spacetime"*; Sakharov (1967) induced gravity; Verlinde (2011, 2016); Maldacena & Susskind (2013) on ER=EPR; Penrose & Rindler, *Spinors and Space-Time*, and Arkani-Hamed & Trnka on the amplituhedron; Connes, *Noncommutative Geometry*, and Chamseddine–Connes–Marcolli on the spectral Standard Model; Goroff & Sagnotti (1985) on two-loop gravity divergences. Flag: specific numerical bounds (e.g. proton-lifetime limits, LIV bounds) are order-of-magnitude as stated and should be checked against the current Particle Data Group review before citation.

CHAPTER 19

# Hypotheses, Refereed

> **DISCLAIMER — READ FIRST.** Nothing on this page is a claimed result. These are **research directions**: structured conjectures, each built by *synthesizing* established mathematics and landmark results, then stress-tested by an internal referee. **No new physics, no new theorem, and no new experimental number is asserted as established here.** Where a sub-result *is* established, it is cited to its real source and tagged accordingly; the *unifying readings* that turn those sub-results into a "direction" are explicitly SPECULATIVE or, at most, INFERENCE. Several directions converge on the same wager — that **causal/algebraic/entanglement structure is more fundamental than metric geometry** — and several share the same fatal-if-unmet obstacles (Weinberg–Witten, the Type $\mathrm{III}_1$ obstruction, AdS-confinement of the rigorous results). Each entry carries a **REFEREE VERDICT** recording errors, overclaims, the refined statement, and a severity grade. One sub-thesis is recorded as a *negative/limiting* result. See *Chapter 2 (p. 8)* for the marker definitions and *Chapter 14 (p. 191)* for the contradictions these directions target.

These directions are ordered by the **lens that produced them**: the algebraic/modular lens (H0, H4), the causal-order lens (H1, H3, H6), the holographic/information lens (H2, H7), the amplitudes/positive-geometry lens (H5), and the operational/post-quantum lens (H8).

# Index

| ID | Lens | One-line | Tag | Confidence | Severity |
|---|---|---|---|---|---|
| *H0* | Modular / algebraic | Linearized Einstein eq = stationarity of relative entropy | SPECULATIVE (AdS sub-results INFERENCE) | medium | minor |
| *H1* | Causal order | Entanglement fixes conformal metric; relational time fixes the scale | SPECULATIVE | low | minor |
| *H2* | Holographic (negative) | Entanglement-geometry is **not yet** a universal theory — graduation criteria | INFERENCE (negative meta-claim) | medium | minor |
| *H3* | Causal order / amplitudes | Causal structure primary; celestial symmetry as the boundary observable | SPECULATIVE | low | minor |
| *H4* | Modular / algebraic | Type $III_1$ + modular flow as primitive; crossed product gives time & entropy | SPECULATIVE (inputs ESTABLISHED) | medium | minor |
| *H5* | Amplitudes | Positive geometry as primitive; locality & unitarity derived | SPECULATIVE (N=4 core ESTABLISHED) | low | minor |

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|---|---|---|---|---|---|
| *H6* | Causal order | Causal-set substrate: order + counting = geometry | SPECULATIVE | low | minor |
| *H7* | Holographic / information | QECC substrate; Type III QFT & bulk geometry as code limit | SPECULATIVE | low | minor |
| *H8* | Operational | QM as the rigid limit of an operational info theory; post-quantum room mapped | SPECULATIVE (scaffolding ESTABLISHED) | very low | minor |

All eight passed the referee (`keep=true`) at **minor** severity. None is relegated to the rejected section, but **H2 is itself a negative/limiting result** and is flagged as such.

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## H0 — Modular / Type-III algebraic emergence: linearized Einstein from relative-entropy stationarity

**Statement.** Take the underlying ontology to be a Haag–Kastler-type net of von Neumann algebras (operationally: *which operations are jointly performable*) with states, and with **no prior length metric beyond an assumed causal order**, no manifold, and no tensor-factorized Hilbert space. The conjecture: an emergent semiclassical metric exists precisely for states whose modular (Tomita–Takesaki) flow is, in a coarse-grained large-$N$ limit, *geometrically localizable*; and the **linearized** Einstein equation is the statement that relative entropy is stationary at first order around such states (the entanglement "first law"). Geometry becomes a derived bookkeeping of modular Hamiltonians; the metric is not quantized because it was never a fundamental field. `[SPECULATIVE for the universal claim; the AdS sub-results are INFERENCE.]`

**Motivation.** Attacks the *background-dependence vs background-independence* and *holographic area-law vs volume-extensive Hilbert space* contradictions at their shared technical root: local QFT algebras are **Type III$_1$ factors** with no trace and no tensor factorization, so there is *literally no local density matrix* to "sew space from." The honest move is to make the Type III modular structure primary rather than treat it as an obstruction. See *Chapter 14 (p. 191)* and *Chapter 17 (p. 325)* (entanglement-as-glue).

**Mathematical sketch.** For $(\mathcal{A}, \Omega)$ cyclic-separating, Tomita–Takesaki gives $S = J\Delta^{1/2}$, modular Hamiltonian $K = -\log \Delta$, modular flow $\sigma_t(a) = \Delta^{it} a \, \Delta^{-it}$. The Araki relative entropy

$$S(\omega \,\|\, \psi) = -\langle \Omega | \log \Delta_{\psi|\Omega} | \Omega \rangle$$

(built from the **relative** modular operator $\Delta_{\psi|\Omega}$) is finite and monotone even where individual entropies diverge. For a ball-shaped region with vacuum modular Hamiltonian $K_A = \int_A f(x)\, T_{00} + \text{const}$, the entanglement first law $\delta S_A = \delta\langle K_A \rangle$ for all small regions is equivalent (Faulkner–Guica–Hartman–Myers–Van Raamsdonk) to

$$\delta G_{\mu\nu} = 8\pi G \, \delta T_{\mu\nu},$$

the **linearized** Einstein equation. The crossed product (Witten; Chandrasekaran–Longo–Penington–Witten) adjoins the modular-flow generator together with an observer/clock, promoting Type III $_1$ to **Type II**, which admits a (observer-relative) trace and a renormalized generalized entropy $S_{\text{gen}} = \langle K \rangle + \text{Area}/4G + \text{const}$. Monotonicity of relative entropy yields the QNEC $2\pi \langle T_{vv} \rangle \geq S''_A$ (Ceyhan–Faulkner), playing the role of an averaged null energy condition.

**What it would explain.** Why perturbative metric quantization fails (the metric is a *collective modular response*, not a fundamental field); why holographic entropy is area-extensive (the finite quantity is relative entropy / renormalized $S_{\text{gen}}$, not the bare divergent von Neumann entropy); the thermodynamic flavor of the linearized Einstein equation (Jacobson); the operational survival of no-signaling and a coarse causal order even when naive Reeh–Schlieder locality fails.

**Predictions / tests.** (1) Whether the crossed-product Type II construction extends *beyond AdS/large-N* to a state-independent generalized-entropy functional in de Sitter and in dynamically formed asymptotically-flat black holes — failure marks the program AdS-specific. (2) Whether QNEC can be derived purely from modular monotonicity with **no** assumed background metric. (3) Whether the **nonlinear** Einstein equation follows from second-order relative-entropy (Fisher-information) data — persistent failure is evidence the lens is incomplete.

**Strongest objections.** (i) **Weinberg–Witten** is sharpest: the algebraic theory must not contain a Lorentz-covariant conserved stress tensor in the *same* spacetime where a massless spin-2 emerges; the only escape is the emergent-extra-dimension/holographic route. (ii) Everything rigorous is AdS/large-$N$; de Sitter (our $\Lambda > 0$ universe) has no established dual. (iii) Only *linearized* equations are obtained. (iv) Modular Hamiltonians are geometric only for special regions/states (Rindler, CFT balls); generically the flow is non-local, so "geometry = modular localizability" risks being stipulative. (v) It explains the *form* of dynamics given thermodynamic inputs but does not identify the microscopic degrees of freedom.

**No-go theorems to confront.** Weinberg–Witten (survived only via emergent extra dimension); Reeh–Schlieder (used constructively, but forbids strict local factorization); Haag's theorem / Stone–von Neumann failure (the framework is natively algebraic, so this is structural, not a bug); singularity theorems (the emergent metric must degrade gracefully where modular localizability fails); Coleman–Mandula / Haag–Łopuszański–Sohnius (emergent spacetime symmetry must factorize from internal symmetry in the IR).

**Relation to existing programs.** A *synthesis*, claiming no novelty for: Tomita–Takesaki modular theory and Haag–Kastler AQFT; the FGHMVR linearized-Einstein-from-first-law derivation; Jacobson's thermodynamic and entanglement-equilibrium derivations; the Casini–Huerta relative-entropy/QNEC line; and the Witten / CLPW crossed-product Type II reformulation. See *Chapter 18 (p. 349)* and *Chapter 8 (p. 89)*.

**Epistemic tag:** SPECULATIVE (universal claim) / INFERENCE (AdS sub-results). **Confidence:** medium.

> **REFEREE VERDICT (H0) — KEEP, SEVERITY: MINOR.** *Sound as a live research program; sub-results correctly attributed.* Key corrections: **(1)** Geometricity of modular flow is established only for Rindler wedges (Bisognano–Wichmann) and CFT-vacuum balls (Casini–Huerta–Myers); for generic states it is non-geometric, so the emergence criterion is partly *stipulative and potentially circular*. **(2)** The ansatz $K_A = \int_A f(x)T_{00} + \text{const}$ already imports geometry — $T_{\mu\nu}$ is defined by variation w.r.t. $g_{\mu\nu}$ and $f(x)$ is the CHM conformal weight; FGHMVR relates *two geometric descriptions*, not geometry-from-a-metric-free substrate. **(3)** "No prior metric" is too strong: an abstract causal poset already fixes the conformal class of the metric (Malament / Hawking–King–McCarthy). Read it as "no prior *length* metric beyond the assumed causal order." **(4)** Notation: relative entropy uses the *relative* modular operator $\Delta_{\psi|\Omega}$ (Araki). **(5)** The crossed-product trace/entropy is *observer-relative* (requires adjoining a clock). **Overclaims flagged:** universal scope (every concrete result is asymptotically-AdS, large-$N$, strong-coupling); "linearized Einstein *is* relative-entropy stationarity" holds only at first order in holographic CFTs; "$\Lambda$-energy conditions $\Leftrightarrow$ monotonicity" is established only as monotonicity $\Rightarrow$ QNEC, not the biconditional. **Recommended tag:** SPECULATIVE (universal) + INFERENCE (AdS), equivalent to the author's "principled-speculation."

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## H1 — Causal order before geometry: entanglement fixes the conformal metric, a relational clock fixes the scale

**Statement.** Separate the two pieces of a Lorentzian metric that the entanglement program conflates. **Causal/conformal structure** (light cones, $g_{\mu\nu}$ up to a local scale) is more fundamental and robust than the **conformal factor** (local volumes/proper times). Conjecture: causal order is primitive (a discrete or algebraic partial order on events); entanglement/mutual-information structure reconstructs the conformal metric on the coarse-grained order; and the missing scalar conformal factor — equivalently *local clock rate* — is supplied by a relational/thermodynamic time (Page–Wootters correlation with a clock subsystem, or modular flow as an intrinsic clock). This makes time emergent and relational from the start. `[SPECULATIVE.]`

**Motivation.** Targets three contradictions jointly: the *problem of time* (Wheeler–DeWitt $H|\Psi\rangle = 0$, no external $t$); *background dependence vs independence* (a partial order is background-free); *nonlocality vs relativistic locality* (causal order may survive where spatial separability fails). The conformal/scale split is principled: a Lorentzian metric *is* (conformal class) + (volume element), and a causal order already fixes the conformal metric in the continuum (Malament / Hawking–King–McCarthy). So entanglement need supply only *one scalar*.

**Mathematical sketch.** Primitive: a locally finite partial order $(C, \prec)$ (causal set), or the inclusion order of local algebras. Continuum recovery: a bijective causality-preserving map between distinguishing spacetimes is a conformal isometry, so causal order $\Rightarrow g_{\mu\nu} = \Omega^2(x)\bar{g}_{\mu\nu}$ with $\Omega$ undetermined. Discreteness fixes the scale via $V \sim N$ (number of elements) — "order + number = geometry." Relational time: Page–Wootters, $H_{\text{tot}}|\Psi\rangle\rangle = 0$ with $H_{\text{tot}} = H_{\text{clock}} \otimes 1 + 1 \otimes H_{\text{sys}}$; the conditional state $|\psi(t)\rangle = \langle t|_{\text{clock}}|\Psi\rangle\rangle$ obeys $i\, d_t|\psi\rangle = H_{\text{sys}}|\psi\rangle$ with no external time. Modular-clock version: the Connes–Rovelli thermal time hypothesis takes the modular flow $\sigma_t$ of a KMS state as physical time. Entanglement enters via $I(A{:}B)$ as a regulator-independent proxy and a candidate distance $d(A, B) \sim -\log(\text{correlations})$.

**What it would explain.** Why there is no external time (time is correlation, reducing to Schrödinger evolution via Page–Wootters); how background independence coexists with QFT's fixed causal structure (microcausality as IR shadow of the partial order); a heuristic for small positive $\Lambda$ (Sorkin's causet $\Lambda \sim 1/\sqrt{V}$, a genuine if crude pre-observation order-of-magnitude estimate); Lorentz-invariant discreteness via Poisson sprinkling.

**Predictions / tests.** Causal-set discreteness predicts **no** generic frame-dependent dispersion — a confirmed energy-dependent photon time-of-flight from GRBs attributable to vacuum dispersion would *falsify* the Poisson-sprinkling version (Fermi-LAT bounds already strongly constrain linear-in-$E$ Lorentz violation). The everpresent-$\Lambda$ scenario predicts a fluctuating dark energy $\sim 1/\sqrt{V_{\text{past}}}$; evolving-$w(z)$ analyses are a live discriminator. Causet propagation predicts tiny momentum diffusion ("swerves"), bounded by null results. Self-consistency: the relational clock must reproduce unitary QFT in the appropriate limit.

**Strongest objections.** Causal-set *dynamics* is unsolved (no demonstrated 4D Lorentzian continuum limit); "entanglement fixes the conformal metric" is asserted by analogy to AdS, not derived, and $I(A{:}B)$ is UV-divergent in continuum QFT; thermal time is state-dependent, hard to reconcile with universal laboratory time; Page–Wootters faces the Kuchař criticism (two-time correlators, Pauli's no-time-operator theorem); no matter sector.

**No-go theorems to confront.** Pauli's theorem (evaded by making time *relational*, not an operator); singularity theorems (discreteness must be *shown* to resolve incompleteness); Bell / no-signaling (must reproduce Tsirelson correlations without conspiratorial superdeterminism); Borde–Guth–Vilenkin (a genuine beginning or well-defined replacement); empirical Lorentz-invariance bounds (near-theorem-strength).

**Relation to existing programs.** Malament/HKM theorems; causal set theory (Bombelli–Lee–Meyer–Sorkin), including "order + number = geometry" and Sorkin's everpresent-$\Lambda$; Page–Wootters relational time and Rovelli's partial observables; Connes–Rovelli thermal time. No novelty claimed; the candidate content is the *division of labor* (entanglement $\to$ conformal metric; relational/modular time $\to$ conformal factor). See *Chapter 18 (p. 349)*, *Chapter 11 (p. 139)*.

**Epistemic tag:** SPECULATIVE. **Confidence:** low.

> **REFEREE VERDICT (H1) — KEEP, SEVERITY: MINOR.** *Internally consistent; conflicts with no result*. Central defect: the load-bearing identification **"conformal factor = local clock rate" is asserted, not derived**, and equivocates between (i) a *local* geometric volume/proper-time scale $\Omega(x)$ on a 4-manifold and (ii) a *single global* PW time parameter or a *single* modular flow. **Routes A (causets, where $V \sim N$ already fixes $\Omega$ combinatorially) and B (relational/modular time) are ALTERNATIVES, not complementary** — route A makes route B redundant unless an explicit map from a global/modular time to a local $\Omega(x)$ is supplied. The construction tacitly assumes the unproven causal-set **Hauptvermutung**, and treats the *continuum* Malament theorem as if it bootstraps geometry from a bare order. The Connes–Rovelli flow is dimensionless and state-dependent, so using it to *supply* the scale risks circularity. **Overclaim:** "addresses the problem of time" should read "addresses one (kinematic) facet" — Dirac observables, the multiple-choice and global-time problems, and Kuchař's objections remain. Mutual information does not in general define a true metric. **Recommended tag:** SPECULATIVE.

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## H2 — A negative/limiting result: entanglement-geometry is *not yet* a supported route to a universal theory

> **This entry is itself a negative result.** Recording where a lens *fails to yield* a supported direction is a legitimate output (see *Chapter 2 (p. 8)*).

**Statement.** Judged as a candidate for a **universal** theory (not as a tool *within* AdS/CFT), "spacetime and gravity emergent from entanglement" currently fails to clear two identifiable bars and should be pursued as a *constrained program*, not asserted as a framework. **Bar 1:** all rigorous RT/HRT/QES and entanglement-wedge results, and the first-law derivation of gravity, hold only in asymptotically-AdS / large-$N$ / strong-coupling Einstein gravity, recover only the *linearized* Einstein equation, and use the *wrong sign of* $\Lambda$ for our accelerating universe, with no established de Sitter or asymptotically-flat analog. **Bar 2:** local QFT algebras are Type $\mathrm{III}_1$ — no normal trace, no factorization $\mathcal{H} = \mathcal{H}_O \otimes \mathcal{H}_{O'}$ — so the slogan's "entanglement entropy of region $A$" and "qubits in $A$" are *not literally well-defined* without a cutoff; the naive subsystem picture is mathematically **ill-defined** (not realizable), though **not logically inconsistent**, since modular theory provides the rigorous reformulation. [The negative meta-conclusion is an INFERENCE from established items below; CONTESTED by proponents.]

**Motivation.** Several supplied contradictions point the same way: ER=EPR is tagged SPECULATIVE precisely because generality beyond AdS is unestablished; the holographic-vs-volume contradiction is OPEN with the Type III caveat; the entanglement-glue principle is SPECULATIVE with RT-in-AdS only at INFERENCE. Asserting a universal framework from these would violate epistemic discipline. The honest deliverable is a precise set of *graduation criteria*.

**Mathematical sketch (what IS vs is NOT established).** (a) RT/HRT $S(A) = \mathrm{Area}(\gamma_A)/4G_N$ and linearized-Einstein-from-first-law: derived only in the large-$c$, large-$\lambda$ asymptotically-AdS limit; linear order only. (b) Local algebras are Type $\mathrm{III}_1$ (Buchholz–D'Antoni–Fredenhagen; Fredenhagen): no trace, $\mathcal{H} \neq \mathcal{H}_O \otimes \mathcal{H}_{O'}$; the crossed-product fix yields Type II *after* adjoining modular data, again mainly in AdS/large-$N$ and de Sitter static-patch settings. (c) ER=EPR has *no* first-principles derivation for **generic** (non-thermofield-double) entanglement — no theorem that a generic entangled state is a smooth geometric connection; support is limited to the eternal black hole (TFD) and engineered traversable wormholes. The gap is structural, not a missing computation.

**What it would explain.** Why decades of AdS results yielded no de Sitter / real-cosmology counterpart (tractability may be intrinsic to the negatively-curved corner); why the deepest puzzles ($\Lambda$ sign/value, the arrow of time, dynamically-formed black holes) live exactly where the lens is least grounded; why the "qubits sewing space" intuition keeps hitting the Type III wall.

**Falsifiable graduation criteria.** (1) A de Sitter ($\Lambda > 0$) RT/QES analog with a controlled dual and derived entropy. (2) A state-independent generalized-entropy functional and entanglement-wedge reconstruction in a *dynamically formed, asymptotically-flat* black hole (not a toy model). (3) Derivation of the **full nonlinear** Einstein equation from an entanglement principle without a background metric. (4) A concrete operational test of ER=EPR for *generic* entanglement — none currently exists, which is itself diagnostic.

**Strongest objections (to this negative framing).** "Programs are judged by fertility, not current universality" — and by that standard it thrives (Page curve in controlled models, crossed-product reorganization of QFT algebras, unification of black-hole thermodynamics with entanglement). AdS may be a legitimate solvable laboratory whose lessons transfer. Linearized Einstein from entanglement is already nontrivial; partial nonlinear extensions exist. Risk: a negative framing may discourage a promising direction; the calibrated stance is "support the program, withhold the universal claim."

**No-go theorems to confront.** Weinberg–Witten (any emergent graviton must route through an emergent extra dimension; same-spacetime versions are dead); Reeh–Schlieder + Type $III_1$ (make the local-qubit picture ill-posed); singularity theorems and Borde–Guth–Vilenkin (singularities unreplaced outside toy models); the persistent *absence* of a unitary, normalizable de Sitter dual (not a theorem, but the mathematical fact the universal claim must overcome).

**Relation to existing programs.** A faithful restatement of established status: Maldacena's AdS/CFT (a conjecture restricted to anti-de Sitter); Ryu–Takayanagi/HRT; Van Raamsdonk's disentangling argument; Buchholz–Fredenhagen Type III results;

the recognized open status of de Sitter holography. The contribution is purely meta-level. See *Chapter 18 (p. 349)*, *Chapter 15 (p. 233)*, *Chapter 14 (p. 191)*.

**Epistemic tag:** INFERENCE (the negative meta-conclusion), resting on ESTABLISHED structural facts plus a SPECULATIVE element (ER=EPR). **Confidence:** medium.

> **REFEREE VERDICT (H2) — KEEP, SEVERITY: MINOR.** *Substantively correct and well-sourced.* Two fixes. **(1)** Bar 2 said "mathematically inconsistent"; downgrade to **"the naive tensor-factor/qubit picture is ill-defined (not realizable) for Type III$_1$ algebras, expressed instead via modular theory and relative entropy."** Nonexistence of factorization is a domain-of-validity issue, **not** a logical inconsistency — no contradiction is derivable, and the technical RT/first-law results never relied on Type-I factorization (the looseness lives in popularizations). **(2)** "All rigorous derivations are AdS-confined" — true for the RT/HRT/QES/entanglement-wedge machinery, but Jacobson-type horizon-thermodynamic derivations (1995 Clausius "equation of state"; 2015 entanglement equilibrium) are non-AdS, though heuristic and equally short of a universal theory; restrict the word "all" accordingly. Citations verified: RT (2006), HRT (2007), QES (Engelhardt–Wall 2014), RT derivation (Lewkowycz–Maldacena 2013), FGHMVR (2013/14, linear order only), ER=EPR (Maldacena–Susskind 2013), wrong-sign-$\Lambda$ obstruction. **Recommended tag:** INFERENCE (negative meta-claim) on ESTABLISHED scaffolding + one SPECULATIVE element; better classed as a well-supported inference than as speculation.

# H3 — Causal structure primary; conformal + volume reconstruct the metric; celestial symmetry as the boundary observable

**Statement.** Take the partial causal order, not the metric, as the irreducible primitive. A Lorentzian metric factorizes as $g_{\mu\nu} = \Omega^2 \tilde{g}_{\mu\nu}$ ($\tilde{g}$ = light cones / causal order; $\Omega$ = scale/volume). Proposal: a fundamental theory specifies only the causal order plus a counting measure; the smooth metric, locality, and the Poincaré "particle" notion are emergent IR shadows. The organizing symmetry is then the asymptotic (BMS / celestial / Carrollian) symmetry on the boundary of causal diamonds, with the S-matrix (or its celestial-CFT avatar) as a *candidate* primary observable in the asymptotically-flat sector. `[SPECULATIVE program built on one ESTABLISHED theorem.]`

**Motivation.** Targets the background-dependence contradiction: QFT needs a fixed metric for Poincaré irreps, microcausality, and a vacuum; GR forbids prior geometry. Causal order is the one piece GR keeps and that needs no prior length scale. Microcausality is reframed as an emergent IR property. The causal-diamond boundary is exactly where Bousso's covariant bound and celestial data live.

**Mathematical sketch.** Metric split $g_{\mu\nu} = \Omega^m \tilde{g}_{\mu\nu}$ ($\sqrt{-g} = \Omega^n \sqrt{-\tilde{g}}$ in $n$ dimensions). `[ESTABLISHED, HKM 1976 / Malament 1977]` For *distinguishing* (or strongly causal) spacetimes, the causal order determines topology, differentiable structure, and conformal metric; a volume element then fixes $g_{\mu\nu}$. Discrete sharpening (causal set, hedged, not endorsed): a locally finite poset with $N \sim V/\ell^4$. Boundary observable: at null infinity the residual symmetry is BMS = supertranslations $\rtimes$ global Lorentz $SL(2,\mathbb{C})$, possibly extended to superrotations/Virasoro `[CONTESTED]`; celestial amplitudes transform as 2D-conformal correlators with weights $h = \frac{1}{2}(1 + s + i\lambda)$.
`[ESTABLISHED at proven orders, Strominger et al.]`
Leading/subleading soft theorems equal the

supertranslation/superrotation Ward identities, e.g. $\lim_{\omega\to 0}\omega A_{n+1} = (\sum_i \cdots)A_n$.

**What it would explain.** Microcausality and a fixed light-cone in IR QFT as the emergent conformal causal order; the problem of time (primary observable is the boundary S-matrix, not bulk Schrödinger evolution); natural area-counting on diamond boundaries; Wigner's particle classification as the flat-space limit where the conformal causal structure degenerates to exact Poincaré.

**Predictions / tests.** Celestial-CFT consistency (associativity/crossing, celestial OPE) are sharp constraints flat-space gravity amplitudes must satisfy. A discreteness-scale imprint in the volume sector (causet $\Lambda \sim 1/\sqrt{V}$, order-of-magnitude only). Lorentz invariance must remain exact: a robust energy-dependent photon dispersion would disfavor the Lorentz-invariant-discreteness form.

**Strongest objections.** Weinberg–Witten (the boundary theory must lack a local covariant $T_{\mu\nu}$ — exactly where least proven); the metric = conformal + volume reconstruction is rigorous only for smooth distinguishing/globally-hyperbolic spacetimes, not off the smooth locus; celestial holography is perturbative and the 2D "CFT" is non-unitary (continuous principal-series weights, distributional correlators); specifying consistent quantum dynamics on a poset is unsolved.

**No-go theorems to confront.** Weinberg–Witten; Coleman–Mandula / Haag–Łopuszański–Sohnius (BMS evades by being asymptotic, not a symmetry of a nontrivial massive S-matrix); Reeh–Schlieder (vacuum entanglement across the diamond boundary); Penrose–Hawking singularity theorems; Haag's theorem / Stone–von Neumann failure (the "one celestial CFT" picture must confront inequivalent representations).

**Relation to existing programs.** The metric/causal split is classic (Weyl; Ehlers–Pirani–Schild; Malament 1977; HKM). Causal-set theory supplies "order + number = geometry." Celestial holography, BMS/asymptotic-symmetry/soft-theorem equivalences (Strominger and collaborators), and Carrollian field theory supply the boundary side. The only novelty is the *framing*. See *Chapter 18 (p. 349)*, *Chapter 9 (p. 106)*, *Chapter 10 (p. 121)*.

**Epistemic tag:** SPECULATIVE. **Confidence:** low.

> **REFEREE VERDICT (H3) — KEEP, SEVERITY: MINOR.** *Internally consistent; conflicts with no result or no-go.* Corrections. **(1)** Reconstruction theorem: attribute precisely — **HKM (1976)** (topology + differentiable + conformal structure under a causal-isomorphism hypothesis) and **Malament (1977)** (weakened to a pure order bijection); the **distinguishing** hypothesis is essential and must be stated. This is the one genuinely ESTABLISHED piece. **(2)** "Order + number = geometry" is the causal-set **Hauptvermutung** — UNPROVEN, complicated by Kleitman–Rothschild dominance of non-manifoldlike posets; label SPECULATIVE, *not* on Malament's footing. **(3)** Tensorial care: $\det \tilde{g}$ fixed is coordinate-dependent; the geometric object is the volume density $\Omega^n$, not $\Omega^2$. **(4)** Distinguish **BMS** (global Lorentz) from **extended/generalized BMS** (superrotations/Virasoro — CONTESTED). **(5)** Celestial CFT is non-unitary (Mellin-transform continuous/complex weights, no standard OPE convergence); a complete celestial CFT for 4D gravity is **not** constructed. **(6)** "Crossing + Ward identities fix correlators with no bulk time" overstates: asymptotic symmetries fix only the *soft/universal* sector, not the hard dynamics. The S-matrix is privileged only under asymptotically-flat boundary conditions and is IR-subtle (Faddeev–Kulish/KLN); it is **not** defined in de Sitter/cosmology. **Recommended tag:** SPECULATIVE.

## H4 — Operator-algebraic background independence: Type III$_1$ + modular flow as primitive geometry; crossed product gives time, entropy, area law

**Statement.** Make the net of observable algebras and their Tomita–Takesaki modular structure the primary object, prior to any metric. A "region" is a von Neumann algebra $\mathcal{A}$ (generically Type III$_1$ in QFT), not a set of points; the modular automorphism group $\sigma_t^\omega$ of a state $\omega$ supplies an intrinsic, state-dependent flow that can serve as emergent time. Coupling to gravity (an observer/constraint) implements the crossed product $\mathcal{A} \rtimes_\sigma \mathbb{R}$, converting Type III$_1$ into Type II, restoring a trace and a renormalized entropy whose leading term is the generalized/area entropy. `[SPECULATIVE for the emergent-geometry thesis; inputs ESTABLISHED.]`

**Motivation.** Meets three contradictions with established mathematics: the *problem of time* (Tomita–Takesaki gives a canonical flow with no external clock — Connes–Rovelli thermal time); the *holographic/area-law vs volume* tension and the Type III obstruction (the crossed-product yields Type II *with* a trace and a finite, area-leading entropy); Haag's theorem / Stone–von Neumann failure are the central structural facts, not pathologies. Locality survives only as the net structure; subsystem/qubit pictures are explicitly denied.

**Mathematical sketch.** `[ESTABLISHED]` Tomita–Takesaki: $Sa|\Omega\rangle = a^\dagger|\Omega\rangle$, $S = J\Delta^{1/2}$, $\Delta^{it}\mathcal{A}\Delta^{-it} = \mathcal{A}$, $J\mathcal{A}J = \mathcal{A}'$. Connes: local QFT algebras are Type III$_1$ (no trace, no minimal projections, no density matrix). KMS: $\omega(a\,\sigma_t(b)) = \omega(b\,\sigma_{t+i\beta}(a))$ — the modular flow is KMS *with respect to its own generator* (convention $\beta = 1$); the *physical* Unruh temperature of the Rindler/boost state is $\beta = 2\pi$ (Bisognano–Wichmann). `[INFERENCE, recent]` Crossed product: adjoin a *bounded-below observer Hamiltonian* (clock) and impose the modular constraint; the result is Type II$_\infty$ (black hole) or II$_1$ (de Sitter static patch), admitting a semifinite trace and a renormalized

$S(\rho) = -\mathrm{Tr}(\rho \ln \rho) = \langle \mathrm{Area} \rangle / 4G + S_{\mathrm{out}} + \cdots$. For a Rindler wedge, $K = 2\pi \int_{\mathrm{wedge}} \xi^{\mu} T_{\mu\nu} \, d\Sigma^{\nu}$ (boost generator), and $\delta \langle K \rangle = \delta S_{\mathrm{ent}}$ gives the *linearized* Einstein equations.

**What it would explain.** The problem of time (modular/thermal time intrinsic to (algebra, state)); why naive QFT over-counts degrees of freedom (the true objects are Type III, only the Type II renormalized entropy is physical and area-leading); Unruh/Hawking thermality as a theorem about $\Delta^{it}$; black-hole $S = A/4G + S_{\mathrm{out}}$ as a Type II trace obeying a second law.

**Predictions / tests.** Primarily structural: the crossed-product entropy must reproduce known generalized-entropy results across de Sitter, black-hole, and Rindler horizons — failure to extend bounds its scope. Predicts that *any* consistent quantum-gravity entropy is the trace of a Type II algebra (entropy *differences* UV-finite, absolute entropy not — matching the physical fact that only differences are observable). Modular/thermal time should agree with proper time along stationary observers in KMS states.

**Strongest objections.** Crossed-product results are recent and shown mainly in semiclassical / large-$N$ / AdS / de Sitter static-patch settings; full background-independent dynamical asymptotically-flat cosmology is OPEN. Thermal time is state-dependent — recovering a single shared classical time risks circularity (must pick the equilibrium state). Modular Hamiltonians are geometric only for special regions. **This lens makes geometry emergent from algebra+entanglement, partly *self-undermining* within a "geometry-primary" framing —** it is honestly an "algebra/information-primary" direction.

**No-go theorems to confront.** Reeh–Schlieder (the *enabling* cyclic-separating hypothesis, but it kills strict locality); Haag's theorem / Stone–von Neumann failure (the whole point); Weinberg–Witten (the emergent linearized Einstein eq must not be misread as a composite graviton with a covariant boundary $T_{\mu\nu}$); Bekenstein/Bousso bounds (the Type II trace-entropy must respect area bounds); Coleman–Mandula (modular flow is not an S-matrix spacetime symmetry).

**Relation to existing programs.** Built on ESTABLISHED Tomita–Takesaki and Connes classification, Haag–Kastler AQFT, Bisognano–Wichmann; plus Connes–Rovelli thermal time (SPECULATIVE but principled) and the recent (2021–2024) Witten / Chandrasekaran–Longo–Penington–Witten crossed-product Type II generalized entropy (INFERENCE, young), and the FGHMVR / Jacobson first-law → linearized Einstein. No novelty for the ingredients. See *Chapter 8 (p. 89)*, *Chapter 12 (p. 155)*, *Chapter 18 (p. 349)*. Closely allied with *H0*.

**Epistemic tag:** SPECULATIVE (emergent-geometry thesis) / ESTABLISHED (modular-theory inputs) / INFERENCE (crossed-product entropy). **Confidence:** medium.

> **REFEREE VERDICT (H4) — KEEP, SEVERITY: MINOR.** *The author's original sketch was marked* `isSound=false` *for mislabeling the inferential bridge as nearly established; the refined statement above repairs this and is kept.* Load-bearing corrections, now incorporated: **(1)** "$\beta = 1$ KMS" is a *convention* of the abstract modular flow, **not** the physical Unruh temperature ($\beta = 2\pi$ w.r.t. boost energy) — the original conflated them. **(2)** Modular flow is geometric only for special (algebra, state) pairs (Bisognano–Wichmann wedge; Hislop–Longo CFT double-cone); generically non-geometric, so geometry is recognized *a posteriori* in symmetric cases, **not derived in general**. **(3)** The crossed product adjoins a bounded-below *observer Hamiltonian*, not "the modular Hamiltonian" (which is unbounded both ways and not affiliated to the Type $III_1$ algebra); the trace/entropy are *observer-relative*, perturbative in $G$, and presuppose a **fixed AdS or dS background** — the program *encodes/reconstructs* geometry, it does not yet *generate* spacetime. **(4)** First-law ⇒ Einstein is *linearized* and holographic. **Headline overclaim curbed:** "geometry and a preferred time DERIVED from algebra + state" is not delivered background-independently. **Recommended tag:** SPECULATIVE (synthesis) / ESTABLISHED (inputs) / INFERENCE (recent crossed-product result).

## H5 — Positive geometry as primitive: spacetime, locality, unitarity as derived boundary properties

**Statement.** Posit that the fundamental object generating observables is a **positive geometry** — a region with boundaries equipped with a canonical differential form having logarithmic singularities exactly on those boundaries — and that physical amplitudes *are* the canonical form. Spacetime locality and unitarity are then emergent: factorization/positivity properties of the geometry's boundaries, not input axioms. The amplituhedron for planar $\mathcal{N} = 4$ super-Yang–Mills is the proof-of-concept; the speculative extension is that gravity and the realistic Standard-Model S-matrix likewise arise from (associahedron-/cosmological-polytope-like) positive geometries. [SPECULATIVE, with an ESTABLISHED-within-planar-$\mathcal{N} = 4$ core.]

**Motivation.** Engages the "scattering amplitudes / positive geometry" clause and the principle that *the continuum and locality may be derived, not assumed*. In the amplituhedron, BCFW recursion and even unitarity emerge from combinatorics, not from a Lagrangian with built-in microcausality. If the S-matrix data come from a geometry with no reference to a bulk metric, a fixed background is not a prerequisite for defining observables.

**Mathematical sketch.** [ESTABLISHED, within $\mathcal{N} = 4$ SYM] A positive geometry $(X, X_{\geq 0})$ has a unique canonical form $\Omega$ with $d\log$ singularities on all boundary components: for $[a, b]$, $\Omega = dx\left(\frac{1}{x-a} - \frac{1}{x-b}\right)$; for the ABHY associahedron $A_{n-3}$, $\Omega$ yields the tree-level bi-adjoint $\phi^3$ amplitude $m(\alpha|\alpha)$ for the corresponding ordering. The amplituhedron $A_{n,k,L}$ (image of $G_{\geq 0}(k, n)$ under a $Z$-map into $G(k, k+4)$ at tree level, with an extended loop-level geometry) has canonical form equal to the planar $\mathcal{N} = 4$ SYM **loop integrand**. Locality: physical poles = codimension-1 boundaries; factorization $A \to A_L \times A_R$ = a boundary being a product of smaller positive geometries. Unitarity is encoded *separately*, via the $d\log$

/logarithmic-singularity structure (no double poles; leading singularities $\pm 1$) and, at loop level, forward-limit/positivity properties. Cosmological polytope (Arkani-Hamed–Benincasa–Postnikov): wavefunction coefficients for conformally-coupled scalars with polynomial couplings in FRW/de-Sitter-limit backgrounds.

**What it would explain.** Why amplitudes are local (poles = boundaries) and unitarity-respecting ($d\log$/forward-limit structure) without those being axioms; the hidden simplicity (no spurious poles, all-loop integrands from one object) invisible in Feynman diagrams; a possible origin for cosmological correlators bypassing bulk time evolution; positivity-constrained EFT coefficients linking to the bootstrap.

**Predictions / tests.** Sharp integrand predictions (sign/positivity, singularity structure) checkable against independent unitarity computations — agreement is ESTABLISHED for many $\mathcal{N} = 4$ cases; a discrepancy would falsify. Cosmological-polytope predictions for total-energy/partial-energy singularities of inflationary correlators are testable against in-in QFT. If extendable to gravity, it must reproduce graviton amplitudes via KLT/double-copy (ESTABLISHED); failure to embed the double copy would bound the program.

**Strongest objections.** Established only for *planar, maximally supersymmetric* $\mathcal{N} = 4$ SYM — not gravity (whose amplitudes lack the simple $d\log$/pole-positivity structure), not non-planar, not massive non-SUSY matter, not the Higgs/confinement sectors. It computes the *integrand*; loop *integration* (IR/UV, regularization, transcendental functions) is outside the construction. Even in the proof-of-concept, momentum-twistor kinematics presuppose 4D conformal spacetime and planarity — so "spacetime emergence" is partial; the background is smuggled into the kinematics.

**No-go theorems to confront.** Coleman–Mandula / HLS ($\mathcal{N} = 4$ saturates maximal SUSY; a real-world theory must respect the limits); Weinberg–Witten (a positive-geometry gravity theory must evade the massless-spin-2 stress-tensor obstruction — the double-copy origin is the likely route); Froissart-type unitarity/analyticity bounds (must hold after integration); Haag's theorem (sidestepped by working with on-shell integrand data — but the nonperturbative object the geometry computes must then be defined).

**Relation to existing programs.** The amplituhedron / positive-geometry program (Arkani-Hamed–Trnka 2013; associahedron/ABHY; cosmological polytope; curve-integral/surfacehedron work) is ESTABLISHED as computational technology and a reformulation of planar $\mathcal{N} = 4$ SYM. Twistor theory (Penrose), BCFW recursion, generalized unitarity, and KLT/double copy are the backdrop. No novelty claimed beyond framing against the contradictions. See *Chapter 9 (p. 106)*, *Chapter 13 (p. 172)*, *Chapter 18 (p. 349)*.

**Epistemic tag:** SPECULATIVE (gravity/real-SM extension OPEN; integrand-level $\mathcal{N} = 4$ core ESTABLISHED). **Confidence:** low.

> **REFEREE VERDICT (H5) — KEEP, SEVERITY: MINOR.** *Internally consistent; conflicts with no no-go; the ESTABLISHED-within-$\mathcal{N} = 4$ core is genuine.* Scope/attribution fixes (incorporated above). **(1)** The associahedron canonical form computes $m(\alpha|\alpha)$ for a *single* ordering; the full bi-adjoint $m(\alpha|\beta)$ matrix needs associahedron intersections — do not imply one polytope gives all. **(2)** The amplituhedron geometrizes the **loop integrand**, not the integrated amplitude. **(3)** "Linear map into $G(k, k + 4)$ " is exact at tree level only; the loop geometry is more elaborate. **(4) Conceptual:** locality (boundary/factorization) and unitarity ($d\log$/forward-limit positivity) are encoded by *different* geometric features — do not collapse unitarity into "factorization of boundaries." **Overclaims curbed:** "spacetime emergent" is partial even in $\mathcal{N} = 4$ (momentum twistors presuppose 4D conformal spacetime + planarity); the gravity/real-SM extension is *highly* speculative (no positive geometry for gravity, non-planar, or massive non-SUSY matter; associahedron handles only massless colored $\phi^3$); cosmological polytope rigorously covers conformally-coupled scalars with polynomial couplings in FRW, with de Sitter in

> specific limits. **RECOMMENDED TAG:** SPECULATIVE (ESTABLISHED planar-$\mathcal{N} = 4$ integrand core; OPEN gravity/SM extension).

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## H6 — Causal-set substrate: order + counting as primitive, continuum geometry as coarse-grained limit

**Statement.** `[SPECULATIVE]` The fundamental description of spacetime is a locally finite partially ordered set (causal set) $C = (S, \preceq)$: elements are discrete spacetime atoms, $\preceq$ is the microscopic causal relation, and the only other datum is counting (cardinality). A smooth globally hyperbolic $(M, g)$ emerges only as a coarse-grained approximation of a causet well-approximated by it via Poisson sprinkling — "**order + number** $\approx$ **geometry**" (causal order $\to$ conformal/light-cone structure, atom-counting $\to$ scale). Dynamics is a quantum sum-over-causets (a discrete path integral / quantum sequential growth), with the gravitational action replaced by a discrete functional (Benincasa–Dowker).

**Motivation.** Most directly motivated by four contradictions jointly: (i) *background dependence vs independence* (no prior metric — causal order is primitive); (ii) holographic/area-law vs volume and the discreteness signaled by $S_{\mathrm{BH}} = A/4\ell_P^2$ (a discrete substrate gives a finite, countable d.o.f. budget); (iii) singularity theorems (curvature singularities as artifacts of pushing the continuum past $\sim \ell_P$); (iv) the cosmological-constant problem ( $\Delta N \sim \sqrt{N} \Rightarrow \Delta\Lambda \sim 1/\sqrt{V}$ in Planck units, a parametrically small fluctuating $\Lambda$ of roughly the right order — the program's one quantitative success *and* its most contested claim). The deep-principle anchor: *causality may be more robust than locality* — causal sets keep causal order and drop the continuum.

**Mathematical sketch.** $C = (S, \preceq)$: reflexive, antisymmetric, transitive, locally finite ($|I(x, y)| = |\{z : x \preceq z \preceq y\}| < \infty$). Continuum recovery: Poisson sprinkling, $P(n \text{ in } V) = (\rho V)^n e^{-\rho V}/n!$

: reflexive, antisymmetric, transitive, locally finite ($|I(x,y)| = |\{z : x \preceq z \preceq y\}| < \infty$). Continuum recovery: Poisson sprinkling, $P(n \,\text{in}\, V) = (\rho V)^n e^{-\rho V}/n!, \quad \rho = \ell_P^{-d}$. **KEY Lorentz-invariance theorem** `[ESTABLISHED, Bombelli-Henson-Sorkin]`: no Lorentz-equivariant measurable rule selects a preferred direction from a Minkowski sprinkling — statistically Lorentz invariant, unlike a regular lattice. Metric recovery: timelike distance $\approx$ longest-chain length; Myrheim–Meyer dimension from chain/antichain counts. Discrete curvature: the Benincasa–Dowker operator (a sum over small inclusive-past *layers* with *d*-dependent alternating coefficients) $\to (\Box - R/2)f$ in the continuum mean; the action reduces to $\sim \ell^{-(d-2)} \int \sqrt{-g}\, R$. Dynamics: Classical Sequential Growth (Rideout–Sorkin), with growth probabilities fixed by discrete general covariance + Bell causality; the quantum sum $Z = \sum_C e^{iS_{\rm BD}/\hbar}$ is the central OPEN problem. Matter: free scalar via the discrete retarded Green function; the Sorkin–Johnston construction yields a distinguished state from the causal Green operator with no preferred time (but can fail to be Hadamard).

**What it would explain.** Why GR resists fundamental quantization (the metric is emergent/coarse-grained); how Lorentz invariance survives discreteness without a preferred frame (the sprinkling theorem — the obstacle that defeats lattice/digital-physics accounts); finite area-scaling d.o.f. behind black-hole entropy; singularities as continuum breakdown; an everpresent, fluctuating, small $\Lambda$ of roughly the observed order (heuristic, contested).

**Predictions / tests.** Lorentz invariance preserved on average $\Rightarrow$ **absence** of the linear-in-$E/E_{\rm Pl}$ photon dispersion many discrete approaches predict; tight Fermi-LAT GRB-timing and polarization bounds thus *confirm* rather than refute this approach, distinguishing it from naive lattices. An everpresent $\Lambda$ fluctuating $\sim \pm 1/\sqrt{V_4}$, with $w \neq -1$ — constrainable by $w(z)$ reconstructions. Non-local "swerves" / momentum diffusion, bounded by precision tests.

**Strongest objections.** No complete tractable quantum dynamics ($e^{iS}$ over causets is even harder to define than the continuum path integral); the **entropy/inverse problem** — generic random causets are non-manifoldlike (Kleitman–Rothschild orders dominate), and dynamics must suppress them (unproven); no contact with the Standard Model; the everpresent-$\Lambda$ heuristic is not a controlled calculation; Reeh–Schlieder vacuum entanglement and Type III structure must be *recovered*, not obviously delivered by Sorkin–Johnston.

**No-go theorems to confront.** Lorentz-invariance no-go for discreteness (EVADED by Bombelli–Henson–Sorkin — the signature strength); Penrose–Hawking singularity theorems (reinterpreted as the signal that the emergent description fails); Weinberg–Witten (no fundamental flat-space graviton, so premises do not apply — must be checked in any continuum limit); Bell / Tsirelson (quantum dynamics must reproduce $2\sqrt{2}$, not a local-hidden-variable cap of 2); Reeh–Schlieder (the emergent vacuum must be cyclic/separating).

**Relation to existing programs.** This *is* Causal Set Theory (Bombelli, Lee, Meyer, Sorkin 1987; Sorkin; Dowker; Henson; Surya). The sprinkling result, Benincasa–Dowker action, Classical Sequential Growth, Bell causality, Sorkin–Johnston vacuum, and everpresent-$\Lambda$ (Ahmed–Dodelson–Greene–Sorkin) are all real, published. No novelty for the framework; the candidate direction is methodological emphasis on causal order as the load-bearing primitive. See *Chapter 11 (p. 139)*, *Chapter 10 (p. 121)*, *Chapter 16 (p. 308)*.

**Epistemic tag:** SPECULATIVE. **Confidence:** low.

> **REFEREE VERDICT (H6) — KEEP, SEVERITY: MINOR.** *Sound as a research program; conflicts with no result or no-go.* Fixed-background no-go theorems (Stone–von Neumann, Groenewold–van Hove, Haag) do not bear against a background-independent substrate. Corrections (incorporated). **(1)** The Benincasa–Dowker action is built from counts of small order-interval cardinalities with $d$-dependent

> alternating coefficients, tending to Einstein–Hilbert *in the mean* — **not** a naive continuum-Ricci sum. **(2)** "Order + number = geometry" silently assumes the **Hauptvermutung** (no causet well-approximated by two macroscopically distinct manifolds), which is OPEN; the *continuum* theorem (Malament/HKM) is established, its discrete analogue is not. **(3)**$Z = \sum_C e^{iS/\hbar}$ is the central OPEN problem; only *Classical* Sequential Growth is rigorously defined. **(4)** The BD d'Alembertian is intrinsically *non-local* (a consequence of Lorentz invariance), with large fluctuations needing an intermediate mesoscale — omitted in the sketch. **(5)** The Sorkin–Johnston state can fail to be Hadamard — omitted. **(6)** Kleitman–Rothschild non-manifoldlike dominance is a serious obstacle. **For balance:** Sorkin's $\Lambda \sim V^{-1/2}$ estimate, predating the 1998 acceleration discovery, is a noted (CONTESTED, non-decisive) point in favor. **Recommended tag:** SPECULATIVE.

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## H7 — Quantum error-correcting code as algebraic substrate; bulk geometry and Type III QFT as the code's continuum limit

**Statement.** `[SPECULATIVE]` The fundamental description (at least of asymptotically-AdS gravitating systems) is a controlled sequence of finite-dimensional quantum systems carrying an (approximate) **quantum error-correcting code** (QECC): a small "logical" (bulk/low-energy/geometric) algebra is redundantly encoded into the entanglement of many "physical" (boundary/microscopic) degrees of freedom. Smooth geometry, the Type III von Neumann structure of continuum QFT, and the (approximate) emergence of a local bulk are properties of the *code* in a large-system limit — not fundamental. "Distance/area in the bulk" = entanglement/min-cut in the code (Ryu–Takayanagi as a code property); bulk locality = subregion/entanglement-wedge reconstruction = code recoverability.

**Motivation.** Anchored in the holographic area-law vs volume-law contradiction, the Type III obstruction, the black-hole information paradox / Page curve, and firewalls. The Type III contradiction is sharpest: continuum local algebras have no trace and no factorization, so "qubits sewing spacetime" literally does not exist at the continuum level. A QECC substrate turns this from paradox into a *limit statement*: the **fundamental** algebra is Type I (finite-dim, has a trace, factorizes — genuine subsystems exist), and the Type III, trace-less structure **emerges** only in the strict large-$N$ /continuum limit. Unitarity (vindicated by the Page curve) is automatic because the substrate is a closed finite quantum system. See *Chapter 17 (p. 325)* (entanglement as glue).

**Mathematical sketch.** Code subspace $\mathcal{H}_{\text{code}} \subset \mathcal{H}_{\text{phys}}$, isometric encoding $V : \mathcal{H}_{\text{code}} \to \mathcal{H}_{\text{phys}}$, $V^\dagger V = 1$. Bulk locality = recoverability: $O$ in the entanglement wedge of boundary $A$ is reconstructable as $O = V^\dagger(O_A \otimes 1_{A^c})V$. RT as a code theorem [ESTABLISHED for finite-dim codes with *exact* complementary recovery, Harlow 2017]: $S(\rho_A) = \text{Area}(\gamma_A)/4G_N + S_{\text{bulk}}(a)$, the area term being the code's center/edge-mode contribution — derived, not assumed. Toy substrate: HaPPY perfect-tensor code / random tensor networks, where network min-cut tracks the RT surface (large bond dimension, on average). Type flow: at finite $N$ the algebra is Type $\text{I}_n$ ; the GNS limit $N \to \infty$ in the TFD/KMS state yields Type $\text{III}_1$; the crossed product by the modular flow (with an observer clock) yields Type $\text{II}_\infty$ (black hole) or $\text{II}_1$ (de Sitter static patch), with a renormalized trace making $S_{\text{gen}} = \langle A \rangle/4G + S_{\text{out}}$ an honest von Neumann entropy. Dynamics: $U(t) = e^{-iH_{\text{bdy}}t}$; the first law $\delta S_A = \delta\langle K_A \rangle \Rightarrow$ *linearized* bulk Einstein equations.

**What it would explain.** The Type III structure of continuum QFT as an artifact of the continuum limit (the "no local subsystems" contradiction becomes emergent, not fundamental); the area-law d.o.f. budget and $S_{\text{BH}} = A/4$ as a code property; the unitary Page curve and information escape (islands = entanglement-wedge recovery); why bulk EFT is robust yet breaks down (code distance sets when low-energy operators become non-reconstructable — a sharp firewall/AMPS diagnostic); linearized Einstein from the first law.

**Predictions / tests.** Largely structural: it must reproduce the **full nonlinear** Einstein equations from a code condition — currently NOT achieved (a concrete success/failure criterion). Quantum-simulation analog tests of RT entropy relations and subregion recovery on small holographic codes (validating the mathematics, not nature). A genuine physical prediction requires a de Sitter/cosmological version with observable consequences — none currently exists (the central empirical weakness). Predicts no firewall for young black holes but possible bulk-EFT breakdown at the code-distance/Page-time threshold.

**Strongest objections.** Almost everything rigorous is AdS-specific (negative $\Lambda$, large $N$, strong coupling); de Sitter holography lacks an established RT analog (arguably *the* central open question). AdS/CFT itself is a conjecture without general nonperturbative proof. Only *linearized* Einstein is recovered. ER=EPR for *generic* entanglement has no first-principles derivation. It presupposes QM exactly, so it does **not** address the measurement problem and is not itself post-quantum.

**No-go theorems to confront.** Weinberg–Witten (EVADED cleanly: the graviton lives in a higher-dimensional bulk; the boundary theory has no graviton and no covariant bulk $T^{\mu\nu}$ — emergence is across dimensions); Type III no-factorization (confronted: substrate is Type I, Type III emerges, crossed-product is the bridge); Reeh–Schlieder (vacuum entanglement from redundant encoding); Haag's theorem / Stone–von Neumann (emerge from the $N \to \infty$ limit; the finite substrate sidesteps Haag by construction); singularity theorems / Bousso bound (reproduced via RT; singularities as code breakdown).

**Relation to existing programs.** The "It-from-Qubit" / holographic-QEC program: Almheiri–Dong–Harlow 2014; Pastawski–Yoshida–Harlow–Preskill (HaPPY); Hayden et al. (random tensor networks); Harlow's RT-from-QEC theorem; Van Raamsdonk and Maldacena–Susskind (ER=EPR); FGHMVR; the crossed-product/Type II work (Chandrasekaran–Penington–Witten and collaborators); the Page curve via quantum extremal surfaces / islands (Penington; Almheiri et al.). No novelty for these. See *Chapter 12 (p. 155)*, *Chapter 8 (p. 89)*, *Chapter 18 (p. 349)*. Allied with *H0*/*H4*; the negative counterpart is *H2*.

**Epistemic tag:** SPECULATIVE (with an ESTABLISHED finite-dim-code integrand core). **Confidence:** low.

> **REFEREE VERDICT (H7) — KEEP, SEVERITY: MINOR.** *Coherent, principled, active; conflicts with no result or no-go.* Corrections (incorporated). **(1)** Type flow: which type ($II_\infty$ vs $II_1$) is setting-dependent and physically loaded; the crossed product adjoins a *bounded-below observer Hamiltonian*, not "the modular Hamiltonian." **(2)** Harlow's RT-as-code theorem is rigorous for finite-dim codes with **exact** complementary recovery; realistic holography has only *approximate* recovery, so transfer to actual gravity is INFERENCE. **(3)** HaPPY/random-tensor-network min-cut = RT holds exactly only for restricted regions / in the large-bond on-average sense — the flat "=" overclaims slightly. **(4)** First-law ⇒ *linearized* Einstein only (correctly stated). **Overclaims curbed:** the headline "fundamental description IS a QECC" extrapolates from AdS/CFT toy/holographic examples to a de Sitter universe with no RT analog and no generic-entanglement-to-geometry theorem; Type $III_1$ is an *independently* established theorem of any local QFT (Buchholz–D'Antoni–Fredenhagen), so the code **relocates** rather than **derives** it ab initio; there is a genuine (self-acknowledged) tension between a finite-dim tensor-factorized substrate and the Type III continuum, only partially repaired by the crossed product. **Recommended tag:** SPECULATIVE.

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## H8 — Is quantum theory fundamental, or the rigid limit of an operational information theory? A constrained — and

## largely walled-off — post-quantum direction

**Statement.** [SPECULATIVE framing on an ESTABLISHED scaffold; partly a negative result.] The disciplined version of "the fundamental theory may be post-quantum" is **not** to posit an arbitrary post-quantum theory, but to ask: within generalized-probabilistic theories (GPTs), is standard complex-Hilbert-space QM the unique fixed point selected by a few information-theoretic / discreteness axioms? The honest finding is that the reconstruction program makes QM look **rigid** (nearly forced by operational axioms), and a battery of no-go and experimental results (Bell/Tsirelson, PBR, real-vs-complex tests, no-signaling, higher-order interference) **largely (not totally) walls off** viable post-quantum deformations. So the defensible output is *cartographic*: a precise map of where post-quantum room remains (essentially only untested regimes — quantum gravity / cosmological scales, or settings violating the axioms) and where it is closed.

**Motivation.** Motivated by (i) the measurement problem (unitary linearity vs definite outcomes); (ii) the Stone–von Neumann uniqueness vs inequivalent-representations *domain mismatch*; (iii) the information-theoretic reconstructions of QM (Hardy, CDP, CBH), tagged OPEN. If QM is reconstructible from operational axioms, then "the exact quantum framework is approximate" becomes a precise question: which axiom, relaxed, yields a post-quantum theory — and is that relaxation empirically open? The discipline demanded is to *not* assert a post-quantum TOE; the value is delineating the boundary.

**Mathematical sketch.** GPT scaffold: a system is a convex state space $\Omega$ in an ordered vector space; effects $e : \Omega \to [0, 1]$; composition via local tomography $\dim V_{AB} = \dim V_A \cdot \dim V_B$. Classical = simplex; quantum = density matrices (qubit Bloch ball). **Hardy (2001):** continuity of reversible transformations rules out classical $r = 1$; together with the simplicity axiom, subspace axiom, and composite-system consistency this excludes $r \geq 3$ and selects $r = 2$ (complex QM) via $K = N^r$. **CDP (2011):** purification + local tomography + causality $\Rightarrow$ finite-dim QM. **Post-quantum candidates killed:**$r = 3$ higher-order (Sorkin) interference — bounded *small but nonzero* experimentally, consistent with $r = 2$; real-Hilbert-space QM — excluded by Renou et al. (2021) network-Bell experiments *under the assumption that independent sources combine as local real tensor factors* (global-conjugation real QM, à la Stueckelberg, is not excluded); Popescu–Rohrlich boxes (CHSH $= 4 > 2\sqrt{2}$) — no-signaling but not quantum, with information causality recovering Tsirelson *for CHSH* (but not provably the full quantum set — OPEN; cf. Almost-Quantum correlations). **Born-rule rigidity:** Gleason ($\dim \geq 3$, assuming noncontextual frame functions) fixes $p = \mathrm{Tr}(\rho E)$. **PBR:** under preparation independence, within the ontological-models framework, the quantum state is $\psi$-ontic.

**What it would explain.** Why QM has its specific structure (complex amplitudes, tensor composition, Born rule) as the unique operational info theory — demoting it from arbitrary postulate set to derived framework; *where* post-quantum modification could still hide (untested QG/cosmological/strong-curvature regimes, connecting to the semiclassical-gravity-vs-superposition contradiction); a reframing of the measurement problem as the question of which axiom (e.g. universal reversibility across all scales) might fail — the same locus where testable objective-collapse models (GRW/CSL, Diósi–Penrose) live.

**Predictions / tests.** Tightened Sorkin triple-slit bounds on the third-order interference parameter $\kappa \to 0$ (any nonzero $\kappa$ is a genuine post-quantum signal). Precision CHSH never exceeding $2\sqrt{2}$. Objective-collapse constraints from matter-wave interferometry, optomechanics, and underground spontaneous-X-ray-emission experiments — progressively excluding parameter space. Gravity-mediated-entanglement (Bose–Marletto–Vedral): if two masses entangle via gravity, the gravitational field cannot be classical (bearing on whether the quantum framework extends to gravity).

**Strongest objections.** Multiple inequivalent successful axiom sets (Hardy, CDP, CBH, Masanes–Müller) show QM is rigid but *not which* principle is fundamental vs conventional. Reconstructions are cleanest in finite dimensions; the infinite-dim/relativistic-field-theoretic case (where Type III and inequivalent representations live) is harder, so the bridge to QFT is incomplete. The no-go gauntlet leaves so little room that the *expected outcome of the strictly post-quantum sub-claim is largely negative*. The program reconstructs the formalism but does not resolve the measurement problem or fix an interpretation. The real-vs-complex exclusion is conditional on framework/locality assumptions (CONTESTED).

**No-go theorems to confront.** Bell + Tsirelson (must reproduce CHSH up to $2\sqrt{2}$, not exceed it; PR boxes excluded; loophole-free Bell violations kill local-hidden-variable substrates absent conspiratorial superdeterminism); PBR (excludes broad $\psi$-epistemic models *within the ontological-models framework, assuming preparation independence*); Gleason (fixes $\mathrm{Tr}(\rho E)$); no-signaling (any signaling deformation excluded); Kochen–Specker (a value-assignment substrate must be contextual); Stone–von Neumann and its failure (must reproduce both finite-dof uniqueness and infinite-dof inequivalent representations in the appropriate limits).

**Relation to existing programs.** Synthesizes QM reconstruction (Hardy 2001; Chiribella–D'Ariano–Perinotti 2011; Clifton–Bub–Halvorson; Masanes–Müller; Barrett's GPTs), device-independent/correlation bounds (Tsirelson; Popescu–Rohrlich; Pawłowski et al. information causality), Gleason and PBR, the Sorkin higher-order-interference program and its null results, the Renou et al. real-vs-complex test, and objective-collapse phenomenology. No novelty. The deliverable is the explicit *verdict*: reconstruction supports "QM is rigid," but post-quantum room is near-closed in any tested regime. See *Chapter 7 (p. 71)*, *Chapter 12 (p. 155)*, *Chapter 15 (p. 233)*, *Chapter 20 (p. 458)*.

**Epistemic tag:** SPECULATIVE (framing thesis) on ESTABLISHED scaffolding; "which single principle carves out the quantum correlation set" is OPEN. **Confidence:** very low.

> **REFEREE VERDICT (H8) — KEEP, SEVERITY: MINOR.** *Sound as a research-direction statement; its "mostly a negative result" framing is its main virtue. The cited results are real and point the claimed way, but several need tightening (incorporated above).* **(1)** Hardy's uniqueness needs the *full* axiom set — continuity rules out classical $r = 1$; *simplicity* + composite-system consistency exclude $r \geq 3$. "Continuous reversibility alone forces QM" is false. **(2)** Higher-order (Sorkin) interference is experimentally *bounded-small*, not "null to high precision," and is consistent with QM rather than a proof of $r = 2$. **(3)** Renou et al. rules out real QM only under the *local-tensor-product rule for independent sources* in a network; global-conjugation real QM (Stueckelberg) is not excluded. **(4)** Information causality recovers Tsirelson for CHSH but does **not** provably carve out the full quantum set (OPEN; Almost-Quantum). **(5)** PBR's $\psi$-ontology is internal to the ontological-models framework and assumes preparation independence. **(6)** The "$q \to 1$ / $N \to \infty$ limit of a finite computational information theory" framing is a loose metaphor, **not** an output of the reconstruction program — flag SPECULATIVE and decouple from the established results. **Note:** the self-tag "highly-speculative" *under-credits* the rigorous core; the reconstruction-rigidity and no-go results are ESTABLISHED. **Recommended tag:** SPECULATIVE-with-established-scaffolding.

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## Rejected / not currently supported

*No candidate direction was judged unsound or vacuous by the referee.* All eight passed at `keep=true`, **minor** severity. Two caveats are recorded here so future iterations do not re-litigate settled points:

- **H2 is retained as a negative/limiting result, not as a positive framework.** Its claim is precisely that the entanglement-geometry lens *is not yet* a supported route to a *universal* theory. It belongs above (it is a refereed, kept direction) but should never be cited as endorsement of emergent-spacetime universality.

- **H4's original mathematical sketch was marked** `isSound=false` for presenting the inferential bridge (algebra+state $\Rightarrow$ background-independent geometry) as nearly established and for the $\beta = 1/\beta = 2\pi$ conflation. The **refined statement** repairs both; the *refined* H4 is what is kept. The original, unrefined phrasing is *not* supported.

When a future direction is judged genuinely unsound or vacuous, record it here with: its ID, the statement, and the specific reason for rejection (logical inconsistency vs no-go violation vs vacuity vs domain-mismatch masquerading as a result — see *Chapter 2 (p. 8)* for the distinction).

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## Cross-cutting observations

- **Convergence.** H0, H1, H3, H4, H6 all wager that *causal or algebraic structure precedes metric geometry*. H1/H3/H6 take **causal order** as primitive; H0/H4 take the **von Neumann algebra / modular flow** as primitive; H7 takes the **code / entanglement** as primitive. These are not independent — H1 and H4 both invoke relational/modular time; H0, H4, H7 all lean on the crossed-product Type II construction. `[INFERENCE]`
- **Shared load-bearing obstacles.** Three recur across nearly every direction and gate all of them: **Weinberg–Witten** (forces any emergent graviton through an emergent extra dimension), the **Type III$_1$ obstruction** (forbids the naive local-qubit picture, forcing algebraic reformulation), and **AdS-confinement** of the rigorous results (the wrong sign of $\Lambda$ for our universe). *ESTABLISHED that these are the binding constraints; see [GAPS_AND_CONTRADICTIONS.md* (Chapter 14 (p. 191)).]
- **The recurring honest limit.** Every direction that derives gravity derives only the *linearized* Einstein equation; no full

nonlinear, background-independent reconstruction from an entanglement/algebraic/causal principle exists. `[ESTABLISHED as the current frontier.]`

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## Iteration 2 refinements (2026-06-08)

*Refinements from the iteration-2 deep-research tracks (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`)), each refereed (0 flagged unsound). These update H0/H2/H4 in place without renumbering.*

- **H2-R1 (de Sitter graduation, partial update).** `[INFERENCE]` Among H2's graduation criteria, criterion (1) (a dS RT/QES analog with a controlled dual and derived entropy) is now **partially met in a weak sense** — CLPW's Type II1 static-patch algebra gives a *derived finite Λ>0 entropy* with a maximal-entropy state (= empty dS) — **but with no RT surface and no boundary dual.** Criteria (3) (nonlinear Einstein from entanglement) and (4) (operational generic ER=EPR) remain entirely unmet in dS, and the dS entanglement first law (*OP-41* (Chapter 15 (p. 233))) does not exist. *Referee correction:* it is **not** that "algebra type cleanly encodes the Λ sign" — local QFT algebras are Type III1 for *all* Λ signs; it is the II1-vs-II∞ distinction (existence of a tracial maximal-entropy state) that carries the dS-specific information. H2's negative meta-conclusion stands. → *[notes* (Working note: *Can emergent-spacetime / holography graduate to our Λ>0 universe?* (repo: `notes/2026-06-08-de-sitter-and-emergent-spacetime.md`))]
- **H4-R2 (graded status of the crossed-product program).** Replace any blanket "ESTABLISHED" reading of H4 with graded conjuncts: **`[ESTABLISHED, given a background]`** the crossed product turns Type III1 into Type II with a finite $S_{\text{gen}} = \langle A \rangle/4G + S_{\text{out}}$ (CLPW/CPW 2022; Witten 2308.03663), and **KLS (2024–25)** extends this to any bifurcate Killing horizon (Kerr, collapse, Schwarzschild–dS); **`[INFERENCE, semiclassical]`** a crossed-product generalized

second law (Faulkner–Speranza 2024), with the beyond-semiclassical fate open; **[scope-limit, ESTABLISHED]** none of this derives geometry or time — a fixed metric with a symmetry and a stationary state is input throughout, and Witten 2308.03663's "background independence" means state-independence on a *fixed* worldline algebra. Net: a **rigorous reorganization that encodes, not generates, geometry**; the nonperturbative fate is OPEN (*OP-44* (Chapter 15 (p. 233)), Giddings 2505.22708). → *[notes* (Working note: *Does the modular / crossed-product (von Neumann algebra) program deliver background independence?* (repo: `notes/2026-06-08-algebraic-background-independence.md`))]

- **H4-R3 (observer-relativity of gravitational entropy).** `[INFERENCE]` The Type II crossed-product (generalized) entropy is observer / quantum-reference-frame dependent and fixed only up to a state-independent additive constant; the **leading area term remains a robust per-observer geometric invariant**, but the *full* von Neumann entropy of a gravitational subregion is frame-relative, not an absolute scalar attached to "the geometry." Reinforces the encode-not-generate verdict. → *[notes* (Working note: *Does the modular / crossed-product (von Neumann algebra) program deliver background independence?* (repo: `notes/2026-06-08-algebraic-background-independence.md`))]
- **HYP-GIE-DISCRIM (BMV scaling discriminator).** `[INFERENCE]` In an achievable parameter window, genuine quantum-gravity gravitationally-induced entanglement scales as $m^2$, whereas the classical-gravity-plus-quantum-matter loophole (Aziz–Howl) scales as $m^3$; a mass/distance scaling test could separate them (and from EM/Casimir backgrounds), tying *OP-40* (Chapter 15 (p. 233)) to a concrete discriminator. → *[notes* (Working note: *Would gravitationally-induced entanglement prove the gravitational field is quantum?* (repo: `notes/2026-06-08-tabletop-quantum-gravity.md`))]
- **HYP-ENTROPIC-GRAVITY-SPLIT (disaggregation of "entropic/emergent gravity").** Split the bundled principle (*Chapter 17 (p. 325)* §6, *Chapter 18 (p. 349)* §7) into: **(a)**

`[ESTABLISHED]` GR carries genuine thermodynamic structure — Jacobson 1995, *given* the area law + Unruh *T* (the AdS first-law route yields only the *linearized* equations; Padmanabhan is an `[INFERENCE]`-level rewriting, not a co-equal theorem); **(b)** `[SPECULATIVE]` gravity is foundationally emergent from *identified* microscopic DOF (none identified — *OP-45* (Chapter 15 (p. 233))); **(c)** `[CONTESTED → disfavored]` Verlinde-type entropic gravity as a dark-matter replacement (RAR fit comparable to MOND; disfavored at cluster/CMB scales; a ≥6σ early/late-type RAR dependence defeats all baryon-only modifications). → *[notes* (Working note: *Entropic / emergent gravity: genuine derivation, or suggestive analogy?* (repo: `notes/2026-06-08-entropic-and-emergent-gravity.md`))]

**Unified-program note.** The iteration-2 synthesis (*notes* (Chapter 22 (p. 571))) judges these directions to cohere **partially**: a single *geometry-from-algebra* program (H0/H1/H3/H4/H6/H7 + entropic gravity) and a distinct *is-the-quantum-framework-final* program (Born/linearity + BMV), joined only by no-signaling/causal-order and by BMV as a logical gate. The defensible overall posture is a **program of demotion-and-derivation**, not a theory-of-everything; see *Chapter 21 (p. 497)* § "Iteration 2 update."

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## Iteration 3 refinements (2026-06-08)

*From the iteration-3 attempts + red-team (see notes/ (repo:* `notes/`*; see the notes index, repo:* `notes/README.md`*) iter3-*).*

- **H2-R1 further correction — $II_1$ is a bounded-area feature, not a Λ-diagnostic.** `[ESTABLISHED]` The iteration-2 H2-R1 correctly removed "algebra type encodes the Λ sign," but even the $II_1$-vs-$II_\infty$ distinction does *not* track Λ: Schwarzschild–de Sitter (Λ>0) is $II_\infty \times II_\infty$ (Kudler-Flam–Leutheusser–Satishchandran, PRD 111 025013). The static patch is Type $II_1$ because its area is *bounded* (finite maximal entropy), set by the

- **HYP-dS-NOMIN (conjecture).** [OPEN] There is no holographic de Sitter entanglement first law in the strict AdS sense (boundary CFT + anchored RT minimization ⇒ pointwise $\delta G_{\mu\nu}$), because dS lacks both a timelike conformal boundary (no continuum of anchoring regions) and a minimization principle (the static-patch variational principle is a concave *maximization*, $dE = -T\,dS$). **Status: OPEN, not a no-go —** one un-foreclosed escape exists (an $SO(d+1,1)$ modular-Hamiltonian family); whether it yields a pointwise field equation is untested. → *[notes* (Working note: *Attempt: a de Sitter entanglement first law (OP-41)* (repo: `notes/2026-06-08-iter3-de-sitter-first-law-attempt.md`)), *OP-41* (Chapter 15 (p. 233))]
- **Encode-vs-generate criterion (methodological).** [INFERENCE] A construction *generates* (rather than *encodes*) geometry only if the emergent metric/causal structure is fixed by abstract (algebra, state, dynamics) data with no pre-chosen factorization, region, symmetric/Hadamard vacuum, or causal order. Graded axes $\Delta_{\text{geo}}$ (geometric input) and $\kappa_\Phi$, plus a 5-item smuggling checklist $g_1$–$g_5$. Across six surveyed constructions (CLPW, Connes–Rovelli, Leutheusser–Liu, Cao–Carroll, causal sets, tensor networks) none clears the bar — the frontier coincides with the Type III1 no-factorization fact. This makes the H0/H4 "encodes, not generates" verdict operational. → *[notes* (Working note: *Formalize: an encode-vs-generate criterion for emergent geometry (OP-46)* (repo: `notes/2026-06-08-iter3-encode-vs-generate-criterion.md`)), *OP-46* (Chapter 15 (p. 233))]
- **Streetlight caveat on the convergence claim.** [INFERENCE] The "Convergence" observation above (H0/H1/H3/H4/H6 wagering that causal/algebraic structure precedes geometry) is partly a *shared-input* artifact: H0/H4 share Killing-symmetric vacua + AdS/large-$N$; H1/H3/H6 share the Malament/HKM theorem + the unproven Hauptvermutung; H7 is AdS/large-$N$. Agreement among methods that share their assumptions is weaker evidence for the wager than genuine independent convergence. → *[notes* (Working note: *Red-team: the demotion-*

*and-derivation posture and its fixed cores* (repo: `notes/2026-06-08-iter3-redteam-demotion-and-derivation.md`))]

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## Iteration 4 refinements (2026-06-08)

*From the iteration-4 attacks + convergence audit (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter4-*). Net: the standing verdict is unchanged; the central diagnosis is harder.*

- **HYP-dS-NOMIN-R3 (dS first-law escape executed → near-no-go).** `[INFERENCE, high]` The one un-foreclosed iteration-3 escape — an $SO(d+1,1)$ family of static-patch modular Hamiltonians — was executed and **fails**: a finite-dimensional isometry orbit supplies $\leq \dim SO(d+1,1) = 10$ global Killing-charge constraints, which cannot Radon-invert to the infinite-dimensional local $\delta G_{\mu\nu}(x)$ (no "radius modulus" / local region continuum); the family is geometric in a *common* state only at Bunch–Davies (probe null there); its first-law content is second-order with $\delta^2 S_{\text{gen}} < 0$ (concave GSL, not an invertible equality). Corroborated by the published Chen–Xu execution (arXiv:2511.00622), which reaches entropy-matching + a 2nd-order quantum first law and **no** $\delta G_{\mu\nu}$. Scoped to the strict unitary single-static-patch isometry-orbit common-state route; non-unitary dS/CFT and Jacobson small-ball routes remain open. → *[notes* (Working note: *Execute the dS first-law escape: an SO(d+1,1) family of modular Hamiltonians (OP-41)* (repo: `notes/2026-06-08-iter4-ds-first-law-escape.md`))]
- **HYP-dS-CARDINALITY (reusable principle).** `[INFERENCE]` Einstein-from-first-law requires an *infinite-dimensional* local region family probing every bulk point (AdS: an RT surface through every point/orientation, Radon-invertible to $\delta G_{\mu\nu}(x)$). Isometry orbits of a maximally-symmetric background are *finite-dimensional* and yield only global Killing charges — so local dS dynamics-from-entanglement needs an independently

infinite local family (Jacobson small balls), which the orbit structurally lacks.

- **OP-48 disjunction + HYP-SPLIT-GEOM (GPT on Type III$_1$).** [ESTABLISHED/INFERENCE] No-go for exact composition / local-tomography on a *single* Type III$_1$ factor; split-property qualified *yes* for nested regions with a nonzero collar (intermediate Type I factor, genuine factorization). But the rescue is not clean — the canonical intermediate factor is singled out only by the vacuum's modular conjugation $J$, **relocating** the BW/CHM geometric-reference circularity to the factorization stage, and failing for adjacent regions. Full A+B merge needs a state-/collar-free composition surrogate (OPEN). → *[notes* (Working note: *Resolve OP-48: can GPT framework-axioms live on a Type III$_1$ algebra? (split-property test)* (repo: `notes/2026-06-08-iter4-gpt-type3-resolution.md`))]
- **HYP-CKV-VACUITY + GAP-NCG-LORENTZIAN-SIGNATURE (encode→generate near-vacuous).** [INFERENCE] Recovering geometry from a non-symmetric modular flow is squeezed by Borchers/Wiesbrock (half-sidedness ⟺ positivity — the emergent translation comes from a relocated positivity premise) and Caminiti et al. (arXiv:2502.02633: recovered generators are conformal Killing vectors, presupposing symmetry). Connes' reconstruction yields only a *Riemannian* manifold; a non-CKV readout (modular-Berry / Connes spectral distance) yielding a *Lorentzian* metric is the only unrealized move that would turn encode into generate. → *[notes* (Working note: *Break the Bisognano–Wichmann/CHM circularity: geometry from non-symmetric modular flow? (OP-46)* (repo: `notes/2026-06-08-iter4-modular-circularity.md`))]
- **Convergence note.** The iteration-4 audit finds the new-insight trend **diminishing** (third iteration at PARTIAL); analytical saturation ≈1–3 iterations away; object-level confirmation non-forecastable. See *Chapter 21 (p. 497)* § Iteration 4 update.

## Iteration 5 refinements (2026-06-08) — closing push, saturation

*From the closing analytical push (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter5-*). No track flipped the headline; the diagnosis is now readout-independent. Consolidated in Chapter 38 (p. 705).*

- **HYP-CKV-VACUITY — upgraded to a readout-independent effective no-go.** `[INFERENCE — HIGH that no current construction generates Lorentzian geometry from non-symmetric modular data; MEDIUM that it is a true no-go vs a limitation of known readouts]` The OP-46 non-CKV *Lorentzian* readout was executed via modular-Berry / kinematic space. It produced the **first emergent Lorentzian metric** in the survey (Huang–Ma $dS_2$, arXiv:2003.12252), crossing the signature barrier the Riemannian Cao–Carroll and Connes-spectral routes fail — **but the signature is inherited from the symmetric CFT vacuum, not generated.** The obstruction now closes at the *same signature-installation step* across all three known readouts (stress-tensor-flux/Sorce–CKV; modular-Berry/zero-mode; Lorentzian-NCG/η-foliation-twist). Formally open only for a hypothetical readout installing no symmetry/η/foliation/twist (none exists) — the single hedge keeping the headline from a theorem. → *[notes* (Working note: *OP-46: emergent LORENTZIAN geometry from a non-CKV modular flow? (the verdict-flipping move)* (repo: `notes/2026-06-08-iter5-op46-lorentzian-nonckv-readout.md`))]
- **OP-48(c) conditional no-go.** `[ESTABLISHED/INFERENCE]` On a single Type $III_1$ factor, "collar-free + state-independent" composition and "singles out QM" are mutually exclusive: the genuinely collar-free surrogate (Haagerup standard form + Connes fusion) is type-agnostic and a *universal embezzler* ( $\kappa_{\max} = 0$, van Luijk et al.); any QM-selecting axiom forces a Type I factor (split construction), re-importing the collar + reference state. The QM-selecting composition axiom **coincides** with the

geometry-importing input — tightening "encodes-not-generates." → *[notes* (Working note: *OP-48 point (c): a state-/collar-free composition surrogate on Type III1?* (repo: `notes/2026-06-08-iter5-op48c-collarfree-composition.md`))]

- **HYP-dS-CARDINALITY-R1 — the root obstruction.** `[INFERENCE, high]` In *every* known route to Λ>0 Einstein-from-entanglement — AdS-FGHMVR, Jacobson small-balls, CLPW Type II1 — **the area-entropy coefficient** $\eta = 1/4G$ **is an INPUT, not an output.** The two surviving dS routes fail on this: the pseudo-entropy first law inherits non-unitarity (complex central charge); Jacobson-in-CLPW re-imports η (the II1 trace encodes one area datum, not a local continuum). → *[notes* (Working note: *The two dS routes surviving the cardinality no-go: pseudo-entropy and Jacobson-in-CLPW* (repo: `notes/2026-06-08-iter5-surviving-dS-routes.md`))]
- **Saturation note.** Fourth consecutive iteration at PARTIAL / encodes-not-generates / not-yet-physics; the named decidable list is settled; the verdict-flipping lever did not flip. The analytical program is **saturated** — see *Chapter 38 (p. 705)*. The only genuinely-open analytical lever is OP-44/OP-CP1 (nonperturbative crossed-product fate).

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## Iteration 6 refinements (2026-06-10) — new-input iteration

*From the new-input iteration (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter6-*). No refinement moves the headline; the obstruction broadens (four readout families), a new named hedge is added (HYP-ENCODING-SCREEN), and one program hypothesis (H5) gains confidence inside a confirmed bound.*

- **HYP-CKV-VACUITY-R3 — four readout families; the installation/selection trichotomy; two new smuggle-list items.** `[INFERENCE — HIGH as a classification of the surveyed pool; MEDIUM that the trichotomy is exhaustive of` `possible mechanisms — the standing hedge]` Every known construction with a Lorentzian-signature output traces the Lorentzian datum to exactly one of three forms: **(i) a symmetry-distinguished state** (CKV/symmetric-vacuum inheritance — the iter-4/5 closures; also the symmetric saddles of Lee's demonstrations); **(ii) an indefinite-pairing axiom** (η/Krein/time-orientation/twist in NCG; the $(\leq n, \leq n)$-eigenvalue axiom + indefinite spin scalar product in causal fermion systems — the **fourth readout family**, closing at the same signature-installation step: the necessity lemma, derived and numerically re-checked by finder and referee, shows that for positive-semidefinite operator classes the CFS *lightlike* relation is empty — the indefiniteness axiom is a necessary carrier of the signature, the Krein η relocated into the first definition); **(iii) a signature-valued template with dynamical selection** (the $\mathcal{S}$-parametrized hypersurface-deformation template in Lee, JHEP 06 (2020) 070, arXiv:1912.12291; the signature-potential toy model of Bermudez 2025 `[SPECULATIVE]`). Forms (i)–(ii) *install* the Lorentzian datum; form (iii) installs the possibility space {Euclidean, Lorentzian} and generates only the *choice*. **Smuggle list extended by two named items:** (a) indefinite-pairing/(n,n)-flag axiom; (b) signature-valued parameter space / deformed-algebra template. CFS status: ENCODE as demonstrated (all worked examples are Dirac-sea constructions on pre-given Minkowski; minimizer existence proven only for finite-dimensional $\mathcal{H}$; no substantive published external critique found in a targeted search `[OPEN]`); candidate-GENERATE as a program, unproven. **The open target compresses to its minimal form: derive the indefinite pairing (a canonical (n,n) operator class, or equivalently a Krein η — e.g. from the modular conjugation $J$) from $(\mathcal{A}, \omega)$ modular data rather than positing it. No such derivation exists or is sketched `[OPEN]`.** Process matrices do not help: local causal arrows are presupposed at every node, and spacetime-embedded indefinite causal order provably reduces to definite acyclic order (Vilasini–Renner, PRA 110, 022227 (2024); companion arXiv:2408.13387). No post-2024 weakening of the Borchers–Wiesbrock positivity–translation linkage was found. → *[notes* (Working note: *OP-46 attack, iteration 6: the generation hunt — causal fermion systems, signature-selection models, and the broadened smuggling list* (repo: `notes/2026-06-10-iter6-generation-construction-survey.md`))]

- **HYP-dS-CARDINALITY-R1 — extended on four fronts; the pattern now also covers a phenomenological chain.** `[INFERENCE, high]` (i) **QRF observer-relativity:** De Vuyst–Eccles–Höhn–Kirklin prove PW=CLPW and that entropies assigned to one subregion relative to distinct observers "can drastically differ" (arXiv:2405.00114; follow-up 2412.15502, verified) — not only is the trace normalization observer-tied, the entropy *value* is QRF-relative. (ii) **HWZ-to-dS lift:** the Hollands–Wald–Zhang dynamical entropy extends to dynamical dS cosmological horizons (Zhao, arXiv:2503.16138, EPJC 85:750 (2025)) with η supplied by the Wald-type formalism, not derived `[INFERENCE, medium]`. (iii) **Maldacena's observer** (arXiv:2412.14014) cancels the imaginary phase of the dS sphere partition function — but the $A/4G$ *coefficient* still comes from the classical saddle: the observer fixes the phase, not η. (iv) **String side:** Zigdon (arXiv:2604.25918) — under the stated assumption that the QG sphere partition function receives a nonzero $1/G_N$ contribution, perturbative string theory admits no dS×compact solutions at any order in $\alpha'$, $g_s$ (tree-level action vanishes), in tension with Gibbons–Hawking: on the perturbative-string side no $A/4G$ saddle is available at all under those assumptions `[INFERENCE, medium — conditional, silent on nonperturbative dS]`. **η is input — or absent — never output.** (v) **Phenomenological extension:** the Verlinde–Zurek/GQuEST chain installs $\eta = 1/4G$ at the modular→metric step — the same coefficient, now in an *inference chain to an observable* rather than a derivation route; a GQuEST positive would, after exclusion of alternatives, be most naturally read as "modular fluctuations gravitate with the *assumed* η" `[INFERENCE]`. → *[notes* (Working note: *Iteration 6 — de Sitter and closed-universe* *quantum gravity: the 2023–2026 literature delta* (repo: `notes/2026-06-10-iter6-closed-universe-and-dS-holography.md`)), *notes* (Working note: *Quantum-gravity phenomenology 2022–2026: the distinguishing-prediction hunt, audited* (repo: `notes/2026-06-10-iter6-qg-phenomenology.md`))]

- **HYP-ENCODING-SCREEN (new named item — the third hedge).** `[INFERENCE – conditional schema; its application inherits the OP-46 encode-only classification's MEDIUM hedge]` **Proposition:** let Φ be encode-only in the OP-46 sense (consumes background data $g_1$–$g_5$; outputs no metric data beyond input) and let $T_A = (\mathcal{A}, \omega; \Phi)$ be a wager-true theory whose *entire* bridge to spacetime observables passes through Φ. Then $T_A$'s observational content equals that of the semiclassical EFT $E$ whose background Φ consumed — the prediction map factors $(\mathcal{A}, \omega) \to E \to \{P(O)\}$ — so no observable separates $T_A$ from any geometry-first theory with IR limit $E$. **Corollary (one-way — "iff" weakened to "only if" per referee):** given the screen, a class-level distinguishing test exists *only if* some wager-true theory has a non-factoring (generative) bridge; whether a generative bridge *suffices* for a class-separating observable is `[OPEN]`. The OP-46 verdict-flipper is a *prerequisite for*, not a guarantee of, the missing distinguishing experiment — *Chapter 38 (p. 705)* §7's no-test claim and §8's reopening condition coincide as obstructions. **Corollary (portability, survey-scoped):** every surveyed observable-statement conjecture (noise PSDs, decoherence rates, w(z) statistics) transplants between wager-true and geometry-first theory classes; confirmation yields abduction, never class separation. A distinguishing prediction would require a "reverse Weinberg–Witten" theorem ("observable X is realizable only without fundamental metric degrees of freedom"); none exists *and none has been attempted* — a literature fact, tag `[OPEN, as of 2026-06]`, NOT a structural impossibility (the impossibility argument only restates the screen under the unproven net-sharing assumption that every UV/global/topological observable is a net invariant — exactly the reopening lever). This reverse-WW target is **un-**

**registered** and carries no OP-ID (it is *not* OP-50, which is the closed-universe Hilbert-space problem); it lives here, in this corollary. Weinberg–Witten itself runs the other direction and is evaded by holography (e.g. composite massless spin-2: Carone–Claringbold–Vaman, arXiv:1710.09367), an evasion symmetric across both theory classes. **Positive-evidence note (2026-06-10 distinguishing-test hunt):** the screen now has a published, peer-reviewed worked instance — the Sharmila–Vermeulen–Datta three-class interferometer-noise taxonomy (*Nat. Commun.***17**, 701 (2026), arXiv:2505.22892) classifies by the mechanism-blind two-point correlator and co-classes emergent and fundamental-metric models (geontropic + Károlyházy in class (b); Oppenheim CQ + CSL in class (a)) — plus two author-side degeneracy concessions (Sidan A & Banks; Erlich). The screen is reinforced with evidence, its grade and condition unchanged. **Hedge accounting: the §7 no-test claim is exactly as strong as the encode-only classification — it carries the OP-46 MEDIUM hedge explicitly.**→ *[notes* (Working note: *Iteration 6, Track A1 — A Distinguishing Prediction for the Geometry-from-Algebra Wager: Found or Refused* (repo: `notes/2026-06-10-iter6-distinguishing-prediction-hunt.md`))]

- **H5 (amplitudes/positive geometry) — scope rewritten, confidence low → low-to-medium; key bound confirmed and sharpened.** `[SPECULATIVE, low-to-medium]` Two-tier established core (referee-approved): the amplituhedron core is *proven* at tree level for planar N=4 (BCFW tiling, Inventiones 2025; a second amplituhedron exists for ABJM); the surfaceology core (Arkani-Hamed–Frost–Plamondon–Salvatori–Thomas, arXiv:2309.15913, 2311.09284) covers **massless colored theories at all loop orders and all genus** — non-SUSY gluons in any D, pions, colored fermions — with loop-*integrated* formulas existing but formal (physical-kinematics continuation open: the integrand-only bound is only PARTIALLY dissolved; the massive-non-SUSY-matter clause of the old bound survives). **Rodina correction (binding, per referee):** Rodina (PRL 134, 031601 (2025)) and Zhou (arXiv:2604.07195) take locality (and, for Zhou, unitarity) as HYPOTHESES — so the result enters as "hidden zeros/UV-scaling uniquely determine the Tr(φ³) tree amplitude *within a local ansatz*, plus a 2-split factorization not implied by unitarity (extended to all topological orders and string amplitudes, arXiv:2405.09608)" — genuine and new, one rung below "locality as derived." Cosmohedra: a single polytope for the full Tr(φ³) cosmological wavefunction, combinatorics proven 2026; kinematic flow is a boundary reformulation of bulk time evolution on fixed FRW, not generation of time `[SPECULATIVE]`. **The decisive referee bound stands and now looks structural: no positive geometry / surface formalism for gravity (no color, no surface; June 2026)** `[OPEN]` — and every construction consumes flat-space/FRW kinematic data as input, consistent with encode-not-generate. Gravitational EFT bootstrap: consistency conditions remain rigid, with the new caveat that gravitational positivity is loop-corrected (calculable G-suppressed negativity; state dispersively; D=11 has no upper bound — one-sided). Five JHEP issue/page references from this track enter the Bibliography (p. 776)`[unverified]` (search-snippet provenance); the gravituhedron paper 2012.15780 is single-author Trnka. → *[notes* (Working note: *Amplitudes and Positive Geometry 2023–2026: Surfaceology, Cosmohedra, Hidden Zeros, and the Gravity Bound* (repo: `notes/2026-06-10-iter6-amplitudes-positive-geometry.md`))]

- **HYP-GIE-DISCRIM — complicated by a mass-independent entangling-phase claim.** `[SPECULATIVE/INFERENCE — recent preprint]` Braccini–Serafini–Bose (arXiv:2602.19306; note Bose is a BMV original author) claim analytically that in qubit-coupled Stern–Gerlach protocols the leading-order gravitational entangling phase between the two qubits is **mass-independent** (unitary and open dynamics; higher orders reintroduce mass dependence via imperfect recombination). If it holds: (a) the $m^2$-vs-$m^3$ mass-scaling discriminator is *protocol-dependent* and must be restated per-protocol; (b) the experimental mass requirement for QGEM-class tests may relax in qubit-phase designs. Related status (per referee correction on rebuttal independence): the Aziz–Howl loophole is under attack via three distinct arguments from **two** author groups — Marletto–Oppenheim–Vedral–Wilson (2511.07348; the no-go's own proponents) and Tang–Yang–Han–Kan–Liu–Xue (2512.13675 *and* 2604.16276 — same six authors; 2604.16276 verified real, arguing the Aziz–Howl effect is "essentially classical" semiclassical wavepacket motion and "negligibly small" under proper Gaussian-wavepacket analysis) — trending dead but `[CONTESTED]` absent a concession, and the rebuttal literature is not yet sociologically independent of the no-go's stakeholders. → *[notes* (Working note: *Quantum-gravity phenomenology 2022–2026: the distinguishing-prediction hunt, audited* (repo: `notes/2026-06-10-iter6-qg-phenomenology.md`))]

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## Iteration 7 refinements (2026-06-10)

*From the lever-execution iteration (4 attack tracks on the four named decidable levers, all adversarially refereed; see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter7-*). No refinement moves the headline (sixth consecutive iteration). Two new named items enter per the AGENTS.md §3 named-ID convention (HYP-MODULAR-TRICHOTOMY; HYP-FACTORIZATION-IS-GEOMETRY); HYP-CKV-VACUITY advances to -R4 with its hedge hardened; HYP-ENCODING-SCREEN inherits the hardening.*

- **HYP-MODULAR-TRICHOTOMY (new named item — the iteration-7 no-go lemma).** `[INFERENCE — ansatz-scoped on its sesquilinear leg; enumerative on its non-sesquilinear leg]` Under a flagged **modular-constructibility ansatz** (sesquilinear Hermitian pairings canonically built from $(\mathcal{M}, \Omega)$ have Gram operator a self-adjoint element of $\{\Delta\}''$), every canonical pairing from modular data is: **(i) positive** ( $f(\Delta) \geq 0$ — incl. the Hilbert form and, on its OS domain, the cone form), or **(iii) a sign-of-Δ Krein form** $\langle \cdot, \varepsilon(\Delta) \cdot \rangle$ for *any* sign function $\varepsilon$, $J$-odd or not (the referee's strengthening of prong (iii) beyond $J$-odd sign functions) — each genuinely indefinite but unitarily universal in the Borchers/HSMI class, generically algebra-incompatible ($\mathrm{Ad}\,\varepsilon(\Delta) \notin \mathrm{Aut}(\mathcal{M})$, numerically demonstrated finite-dim), and non-localizable without a posited carrier; the remaining canonical pairings **(ii)** — symplectic $\mathrm{Im}\langle\cdot,\cdot\rangle|_K$, complex-symmetric $\langle J\cdot,\cdot\rangle$, real $J$-split — are signature-free **by separate enumeration of the known form types, not by the ansatz**. Hence no geometry-bearing, algebra-compatible, localized Lorentzian $(n, n)$ pairing is derivable from modular data alone, *modulo the ansatz — the explicit residual*. Companion positive sub-result (the lemma's input, referee-corrected): $\eta_0 = \mathrm{sgn}(\ln\Delta)$ is a distinguished $J$-odd self-adjoint unitary in $\{\Delta\}''$ — **one member of the infinite family of $J$-odd sign functions $\varepsilon(\Delta)$,** $\varepsilon(1/\lambda) = -\varepsilon(\lambda)$ (e.g. $\mathrm{sgn}(\sin\ln\Delta)$, referee-constructed and numerically confirmed); it is singled out only by an *added* modular-rescaling-invariance axiom ($\Delta \mapsto \Delta^s$). The form $\langle\cdot, \eta_0\cdot\rangle$ is indefinite iff $\omega$ is non-tracial, with Krein signature $(\infty, \infty)$ on any type III factor (spec $\Delta = *0, \infty$) holds only for III1; the $(\infty, \infty)$ conclusion survives for all type III — referee correction). Non-uniqueness *strengthens* the negative verdict: every member of the family is equally canonical and equally geometry-void. The vacuity theorem ($\Delta_{\mathrm{geo}} = 0$, unitary universality by one multiplicity cardinal) substantially coincides with the known uniqueness of non-degenerate standard pairs, extended by "any fixed function of $\Delta$ rides along" — the geometric-vacuity corollary is the new-to-the-wiki content. Structural bonus, severity-none per referee (the strongest finding of the track): $J$ **carries reflection positivity abstractly** — $\langle a^*\Omega, Ja\Omega\rangle \geq 0$ for all $a \in \mathcal{M}$ by self-duality of the natural cone (consistent with the Neeb–Ólafsson standard-subspace ↔ reflection-positivity dictionary) — so the $g_3$ trap is upgraded from symmetric-state inheritance to a **structural fact**: the modular conjugation sits on the Euclidean/positive side, and no $J$-induced pairing can be secretly Lorentzian. Verified analytically and over 500 numerical trials, independently replicated by the referee (fresh code, fresh RNG). [→ [notes* (Working note: *OP-46 attack, iteration 7: η-from-modular-data — the modular sign η0 exists, is geometry-*

*void, and a trichotomy lemma closes the J-route* (repo: `notes/2026-06-10-iter7-indefinite-pairing-attempt.md`))]

- **HYP-CKV-VACUITY-R4 — the J-route is closed; the hedge hardens MEDIUM → MEDIUM-HIGH (conditional); the target re-compresses to the carrier problem.** `[INFERENCE — HIGH as a classification of the surveyed pool; MEDIUM-HIGH that the closure is a true no-go, conditional on the HYP-MODULAR-TRICHOTOMY constructibility ansatz]` The -R3 compressed target ("derive the indefinite pairing — Krein η / (n,n) operator class — from modular data; no such derivation exists or is sketched") receives its **mandatory precision repair**: the "no such derivation exists" sentence is now *literally false* — η0 achieves the literal target, smuggle-free per the house $g_1$–$g_7$ instrument — yet **nothing flips**, because the derived pairing is provably geometry-void (unitary universality $\Longrightarrow$ $\Delta_{\text{geo}} = 0$; algebra-incompatibility; non-localizability). The -R3 compression was therefore *lossy*. The faithful residual: **derive an algebra-compatible, LOCALIZED (fiberwise)** $(n, n)$ **pairing from modular data** — which is the carrier problem (see HYP-FACTORIZATION-IS-GEOMETRY below). Slogan, refereed: *the modular sign is to signature what thermal time is to time.* Literature direction-map `[OPEN — negative search, referee-corroborated, scoped to both sessions' queries]`: all prior art runs Krein → modular — Gottschalk (JMP 43 (2002) 4753) *assumes* Krein structure and derives Bisognano–Wichmann; Jakóbczyk's 1980s line builds modular theory *inside* a posited indefinite-metric space (1985, OSTI 5135323, record-verified); the NCG twist papers obtain the Krein space *from the twist* (Devastato–Farnsworth–Lizzi–Martinetti, JHEP 03 (2018) 089, arXiv:1710.04965 — author list corrected; Nieuviarts 2024–26). No publication derives a Krein fundamental symmetry from $(\mathcal{M}, \omega)$ modular data. → *[notes* (Working note: *OP-46 attack, iteration 7: η-from-modular-data — the modular sign η0 exists, is geometry-void, and a trichotomy lemma closes the J-route* (repo: `notes/2026-06-10-iter7-indefinite-pairing-attempt.md`))]

- **HYP-FACTORIZATION-IS-GEOMETRY (new named item — where all five routes close).** `[INFERENCE, high as a convergence observation; the underlying question OPEN]` The carrier/localization step — supplying a *local* carrier (a factorization, region net, fiber structure, or preparation primitive) on which an abstract algebraic datum could become geometric — is the single step at which **five independent routes** now close: (1) the Type $III_1$ no-factorization frontier (iter 3: the encode/generate boundary coincides with the hand-chosen factorization); (2) the split-property rescue (iter 4: the canonical Type I factor is fixed only by the vacuum's $J$ — the circularity relocated to the factorization stage, HYP-SPLIT-GEOM); (3) the CFS carrier posit (iter 6: the $(\leq n, \leq n)$ operator class is the Krein η relocated into the definition, on a posited carrier); (4) the OP-48c composition layer (iters 5–7: QM-selecting composition forces the Type I collar; iteration 7 relocates the failure further, to the normal-product-*preparation* primitive — the state postulate as installation choice); (5) **the $\eta_0$ localization step (iter 7): the canonical modular sign exists but is non-localizable — universal, algebra-incompatible, carrier-free.** Statement of the named problem: *exhibit (or refute the existence of) an algebra-compatible, localized* $(n, n)$ *pairing — equivalently a modular-canonical local carrier — derived from* $(\mathcal{M}, \omega)$ *data alone.* This is now simultaneously the CONCLUSION §8 reopening condition and (via HYP-ENCODING-SCREEN) the prerequisite for any distinguishing experiment. → *[notes* (Working note: *OP-46 attack, iteration 7: η-from-modular-data — the modular sign $\eta_0$ exists, is geometry-void, and a trichotomy lemma closes the J-route* (repo: `notes/2026-06-10-iter7-indefinite-pairing-attempt.md`)), *notes* (Chapter 27 (p. 628))]
- **HYP-ENCODING-SCREEN — hardening note (hedge accounting).** `[INFERENCE — conditional schema, unchanged]` The screen's application inherits the OP-46 encode-only classification's hedge, which iteration 7 hardened: **MEDIUM → MEDIUM-HIGH, conditional on the HYP-MODULAR-TRICHOTOMY constructibility ansatz.** The §7 no-test

claim is now exactly as strong as that hardened classification — same condition attached. (The A2 track mildly reinforces the screen from the other side: the P1a descent discharge works *by consuming* pre-installed $\mathfrak{so}(d+1,1)$ structure — the discharge mechanism is itself an encoding step.) → *[notes* (Working note: *OP-46 attack, iteration 7: η-from-modular-data — the modular sign η0 exists, is geometry-void, and a trichotomy lemma closes the J-route* (repo: `notes/2026-06-10-iter7-indefinite-pairing-attempt.md`)), *notes* (Working note: *OP-44 iteration-7 attack: the order-$G_N^2$ obligation for the dS static patch — quadratic descent discharged, residue located at footnote 12 and the cubic charge* (repo: `notes/2026-06-10-iter7-op44-orderG2-descent.md`))]

- **OP-48(c) horn C — upgraded to theorem-conditional-on-framework (registry note; full statement in *Chapter 15 (p. 233)*).** `[INFERENCE, high – the FP-forcing step is textually grounded but interpretive ("forced-modulo-rejecting-OP-48's-question", per referee)]` The full-pair reading of "recovers QM" is forced: all three reconstructions quantify composition over the designated pair and install product states at framework level; the separation half of local tomography *holds* on the III1 Haag-dual pair (own derivation, referee re-derived), so the failure locus is exactly the normal-product-preparation primitive (PS), with the uniform $1 - 1/\sqrt{2}$ gap; (PS) on the pair forces Type I. Residue = the normal-state postulate + the composition-primitive choice: (X1) singular product states (existence via the corrected Schlieder/Roos + Hahn–Banach route — the max-tensor universal-property route is invalid, referee MAJOR) or (X2) non-OPT III1-embezzlement composition; both exit the framework. Encodes-not-generates recurring at the state-postulate layer. → *[notes* (Working note: *OP-48(c) iteration-7 attack: the full-pair reading is forced — horn C loses its definitional residue; plus the OP-50 watch* (repo: `notes/2026-06-10-iter7-op48c-fullpair-forcing.md`))]
- **Byproduct (A3): $S_{\mathrm{SVD}}(\mathrm{cap}) = S_{\mathrm{dS}}/2$, region-independent.** `[SPECULATIVE]` Under the A3 phase bookkeeping AND the unproven junction additivity (OP-49 residual (c′)), the universal

part of the dS3 hemisphere SVD spectrum gives $S_{\mathrm{SVD}} = \pi\tilde{c}/6 = \mathrm{Re}\, S_A^{\mathrm{pseudo}} = S_{\mathrm{dS}}/2$ for every cap — the only intrinsically real two-state entropy retains exactly the horizon constant and zero local geometry (the ENCODES tie-in survives only at this conditional level). The numerical value is `[SPECULATIVE]` (in any trace-class realization $\mathrm{Tr}\sqrt{\mathcal{T}^\dagger\mathcal{T}} > 1$ strictly, since equality would force a real pseudo-spectrum contradicting the verified complex $S_A$ — referee correction); the region-*independence* is `[INFERENCE, conditional on (c′)]`. Retained as a falsifiable byproduct prediction for any future direct computation, with both flags. → *[notes* (Working note: *OP-49 SVD residual closed: the SVD first law fails on the dS$_3$ hemisphere because* $c + c^* = 0$ (repo: `notes/2026-06-10-iter7-op49-svd-computation.md`))]

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## Iteration 8 refinements (2026-06-10) — the net/family levers

*From the net/family lever-execution iteration (see notes/ (repo:* `notes/`*; see the notes index, repo:* `notes/README.md`*) iter8-*). No headline movement (7th consecutive); the principal hedge hardens toward HIGH on a narrowed condition.*

- **HYP-CKV-VACUITY-R5 — the carrier no-go upgrades from single-algebra to net/family.** `[INFERENCE – HIGH classification; conditional on the net-constructibility ansatz]` Iteration 7's single-algebra trichotomy left one structurally-richer input uncovered: net-modular data (relative modular operators, $\{J_O\}$, HSMI, modular intersection, modular-Berry over a net). Iteration 8 closes it via the **same boost-covariance engine**: a local indefinite multiplication form has a sign locus the modular flow $p \mapsto e^{-2\pi t}p$ boosts away, so local ⟹ non-covariant ⟹ non-canonical; every modular-covariant net-data form is positive-and-local, signature-free, or indefinite-but-non-local — never both indefinite and localized. Localization enters only through the net's index set, which **is** the geometry (BGL; Morinelli–Neeb–Ólafsson Euler element as input;

Lashkari–Leung–Moosa–Ouseph JHEP 10 (2025) 153). **Hedge: MEDIUM-HIGH → toward HIGH, conditional now on a net-constructibility ansatz** (narrower than the single-algebra ansatz), with two named obligations outstanding (naturality theorem; modular-Berry-holonomy escape). → *[notes* (Working note: *OP-46 / carrier-problem attack, iteration 8: the net upgrade — net-modular data adds operators, not a localized indefinite form; the parametrization is the geometry* (repo: `notes/2026-06-10-iter8-net-carrier-no-go.md`))]

- **HYP-FACTORIZATION-IS-GEOMETRY — restated; the carrier-convergence count STAYS at five.** `[INFERENCE]` The net/family no-go is the **net-level extension of route (5)'s boost-covariance engine**, not a sixth independent route (binding referee correction to the finder's "sixth convergence"). The carrier problem — derive an algebra-compatible, *localized* (fiberwise) $(n, n)$ pairing from modular data — remains the single compressed analytical reopening target, now with the net upgrade and its net-index-set smuggle folded into route (5). Five routes converge there.
- **HYP-ENCODING-SCREEN — inherits the hardened grade.** `[INFERENCE]` The §7 no-test corollary inherits the OP-46 hardening: **MEDIUM-HIGH → toward HIGH, conditional on the net-constructibility ansatz** (same condition). No new hedge; no experimental channel emerged.
- **OP-48c horn-C / the normal-state postulate — X1 is now a theorem-with-a-price-tag.** `[INFERENCE, high]` Singular product states across the $III_1$ Haag-dual pair provably exist (Schlieder ⟹ C*-independence ⟹ Roos ⟹ Hahn–Banach; non-normal by Takesaki), no reconstruction axiom *fails* on them (they are upstream of every axiom's $\sigma$-additive state space), and admitting them relinquishes density-matrix states + $\sigma$-additivity + finite tomographic dimension jointly — so the normal-state postulate is a load-bearing, non-derivable installation choice. Horn C stays theorem-conditional-on-framework; encodes-not-generates recurs at the state-postulate layer, now theorem-grade. → *[notes* (Working note: *OP-48(c) EXIT X1: singular*

*product states on the III₁ Haag-dual pair exist (theorem) and constitute a framework exit, not a closable residue* (repo: `notes/2026-06-10-iter8-op48c-X1-singular-states.md`))]

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## Iteration 9 refinements (2026-06-10) — the carrier no-go's naturality content

*From the carrier-naturality lever iteration (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter9-*). No headline movement (eighth consecutive confirmation); the principal hedge's grade is unchanged and its condition sharpened; one named obligation discharged.*

- **HYP-CKV-VACUITY-R6 — the net-constructibility ansatz replaced by a naturality formulation; the modular-Berry-holonomy obligation discharged.** `[INFERENCE — HIGH classification; conditional on the net-naturality theorem + a located commutant-to-generated gap]` (A1) The boost-covariance engine is promoted to a **naturality obstruction** (Lemma N2: a natural sign locus is flow-carried, hence boosted out of the open index set), and the constructibility *ansatz* is discharged to **Lemma N1** (the Gram operator is forced into the commutant of Δ — *referee correction:* the commutant, not the algebra generated by Δ, so a commutant-to-generated gap survives). The irreducible residual: "localized" presupposes the index-set order, which is the geometry. (A2) The **modular-Berry-holonomy escape closes negative** — the holonomy-natural form is the signature-free symplectic Kirillov–Kostant (Crofton) form; the only indefinite object, the kinematic-space $dS_2$ metric, is vacuum/symmetry-inherited (Huang–Ma 2110.08703; the iteration-5 closure). The larger-structure loophole (subfactor standard invariant / planar algebra) closes *against* the evader: a rigid C*-tensor category has positive hom-space inner products by axiom, so enlarging the symmetry shrinks the candidate pool. **Grade unchanged**

**(MEDIUM-HIGH→HIGH conditional); condition sharpened; carrier-convergence count stays at five.**

- **HYP-FACTORIZATION-IS-GEOMETRY ⇔ reverse-Weinberg–Witten (the iteration-9 identification).** `[INFERENCE, high]` (A3) The carrier problem and the reverse-WW theorem are **the same open problem**: A1-concrete (the GNS pair net + vacuum is a complete invariant) is a conditional theorem via Weiner's algebraic Haag theorem (arXiv:1006.4726), whose hypotheses are exactly the geometric inputs HYP-FACTORIZATION-IS-GEOMETRY isolates; so A1 unconditional ⇔ no positive carrier solution ⇔ no reverse-WW witness. The reverse-WW target is pinned to this corollary, unregistered, `[OPEN, as of 2026-06]`.
- **HYP-ENCODING-SCREEN — inherits the unchanged grade and sharpened condition; the no-test target now pinned to the carrier problem.** `[INFERENCE]` Grade MEDIUM-HIGH→HIGH, conditional on the net-constructibility/naturality content. The §7 no-test corollary is now structurally identified with the §8 reopening condition (via A3): the empirical and analytical blockers are one problem.

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## Iteration 10 refinements (2026-06-10) — the carrier no-go's two residual obligations

*From the carrier-gap lever iteration (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter10-*). No headline movement (ninth consecutive confirmation); the principal hedge's grade AND condition are both unchanged — the designated verdict-bearing lever (close the commutant-to-generated gap) was executed and did not close. Two submitted outcome labels were corrected by the referee (a condition-shedding overclaim struck; a re-hardening claim refuted).*

- **HYP-CKV-VACUITY-R6 — UNCHANGED; the commutant-to-generated gap did not close, so the hedge sheds no condition.** `[INFERENCE — HIGH` `classification; conditional on the net-naturality theorem + the located commutant-to-generated gap]` (A1) The designated verdict-flipper — close the iteration-9 commutant-to-generated gap and push the hedge to unconditional HIGH — was attacked head-on via the Bisognano–Wichmann ergodicity of the vacuum, and the claimed closure is a **binding major error**: ergodicity empties the *centralizer*$\mathcal{M}_\omega = \mathcal{M} \cap \{\Delta^{it}\}'$, but the iteration-9 gap is the representation-space multiplicity gap $\langle\Delta\rangle'' \subsetneq \{\Delta^{it}\}'$, which (because $\Delta \notin \mathcal{M}$) stays non-empty even in the BW case and contains the canonical indefinite $\eta_0 = \mathrm{sgn}(\ln \Delta)$ . The hedge stays **at -R6**, grade and condition both unchanged; it does **not** advance to "-R7 / $n_1$ alone." Two kept sub-results re-establish the standing obstruction: the Type III1 centralizer is genuinely large for degenerate spectrum (integrable weight ⟹ II $_\infty$ centralizer; Connes–Takesaki flow of weights; Popa realization math/0011084) and the state is a faithful normal trace on it (Takesaki, semifinite). (A2) **No carrier lives in the gap (RESOLVED-NEGATIVE):** the gap is empty in the geometric/Lebesgue sector (centralizer trivial; Shlyakhtenko free-quasi-free structure theory) and geometry-void in the almost-periodic sector — the strongest-ever candidate $\mathrm{sgn}(h) \in \mathcal{M}$ clears bars (i)/(iv) by an inner automorphism yet has $\Delta_{\mathrm{geo}} = 0$, its complete invariant the finer per-modular-eigenspace signature (referee correction: NOT the global $(p, q)$ pair; 4 toy conjugacy classes, Skolem–Noether), localization failing via the inherited $n_1$ presupposition (the boost-orbit/wedge-filling mechanism struck as sector-inconsistent). **Combined:** the commutant residual is substantially closed for algebra-internal (Ad-inner) carriers; a residual sliver — algebra-compatible carriers in $\{\Delta^{it}\}' \setminus \mathcal{M}$ — is NOT discharged; the hedge narrows substantially toward but not provably to $n_1$ alone. Carrier-convergence count **stays at five** (centralizer-completion of route (5)).

- **HYP-FACTORIZATION-IS-GEOMETRY — route (5) unchanged; the commutant-to-generated gap remains the located residual.** `[INFERENCE, high]` The carrier problem's residual after iteration 10 is exactly the iteration-9 residual: the net-naturality theorem plus the located commutant-to-generated gap (now sharpened to: the gap is real and wide for degenerate modular spectrum, empty in the geometric sector by centralizer-triviality, and contains no carrier — but is not provably *closed* for algebra-compatible carriers outside $\mathcal{M}$). Count stays five; no route added; no headline flip.

- **HYP-ENCODING-SCREEN — inherits the unchanged grade and unchanged condition.** `[INFERENCE]` Grade MEDIUM-HIGH→HIGH, conditional on the net-naturality theorem + the located commutant-to-generated gap (iteration-9 status). The §7 no-test corollary and the reverse-WW identification (pinned to the §8 reopening condition) are untouched; no experimental channel emerged.

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## Iteration 11 refinements (2026-06-10) — the carrier reduction, net-completeness handles, and the terminal-saturation audit

*From the carrier-reduction / net-completeness / terminal-saturation iteration (see notes/ (repo:* `notes/`*; see the notes index, repo:* `notes/README.md`*) iter11-*). No headline movement (tenth consecutive confirmation); the principal hedge's grade AND condition are both unchanged — the located commutant-gap residual re-grounds its vocabulary on a NAMED external open problem without closing to a theorem; A1's submitted logical-equivalence reduction was MAJOR-CORRECTED.*

- **HYP-CKV-VACUITY-R6 — UNCHANGED; the located commutant-gap residual re-grounds on Connes' bicentralizer problem (NAMED, OPEN in general as of 2026), it does NOT close to a proved theorem.** `[INFERENCE – HIGH classification; conditional on the net-naturality theorem + the located commutant-to-generated gap]` (A1) The algebra-internal (inner, $J \in \mathcal{M}$) carrier horn lives in the modular centralizer $M_\psi = \mathcal{M} \cap \{\Delta_\psi^{it}\}'$, so centralizer/bicentralizer structure theory names the iteration-9 commutant-gap residual. **Binding referee correction (centerpiece MAJOR-CORRECTED):** the submitted "carrier no-go REDUCES (logical equivalence) to Connes' bicentralizer problem" is **downgraded** to "is RELATED TO centralizer/bicentralizer structure theory" — the spine conflated trivial centralizer ($M_\psi = \mathbb{C}1$, killing a carrier, *unconditionally achievable on every $III_1$ factor*; Marrakchi–Vaes, Crelle 809 (2024) 247, Thm A) with Haagerup's bicentralizer-triviality (an *irreducible/large*-centralizer state in which a carrier *exists*) — opposite conditions. **Second binding correction:** the mission's "resolved-by-Houdayer–Marrakchi–Tomatsu" premise is FALSE; Connes' bicentralizer problem is **OPEN in general as of 2026** (only large classes and equivalent reformulations settled). **Grade AND registry-condition UNCHANGED; no R-advance; carrier-convergence count stays at five.** The full no-go stays two-bar: the bicentralizer route touches only the inner horn (as a loose structural connection, not a reduction); the $\{\Delta^{it}\}' \setminus \mathcal{M}$ sliver is disposed of separately by non-localizability + algebra-incompatibility, unchanged. *(Subregion-localized carriers would, if anything, live in the relative bicentralizer $BC_\varphi(\mathcal{N} \subset \mathcal{M})$ of the local inclusions — Ando–Haagerup–Houdayer–Marrakchi, Math. Ann. 376 (2020) 1145 — a suggestive structural connection, tagged `[SPECULATIVE]`, not an established reduction.)*

- **HYP-FACTORIZATION-IS-GEOMETRY — route (5) unchanged; the located residual now has a NAMED external connection.** `[INFERENCE, high]` The carrier problem's residual after iteration 11 is exactly the iteration-9/10 residual (the net-naturality theorem plus the located commutant-to-generated gap), with the gap's algebra-internal horn now connected to centralizer/bicentralizer structure theory and through it to Connes' (genuinely open) bicentralizer conjecture — an external resolution clock, not a program-internal condition. Count stays five; no route added; no headline flip. The three iteration-11 net-completeness handles (holography of information, T-duality, MIP*=RE) are independent *roads to* this gate, not new convergence routes.

- **HYP-ENCODING-SCREEN — inherits the unchanged grade and unchanged condition; reverse-WW re-grounded from a third direction.** `[INFERENCE]` Grade MEDIUM-HIGH→HIGH, conditional on the net-naturality theorem + the located commutant-to-generated gap. The §7 no-test corollary and the reverse-WW identification (pinned to the §8 reopening condition) are untouched in grade/condition; iteration 11 adds an independent re-grounding of the RWW≡carrier identity from the holographic (holography of information + its Dec-2025 arXiv:2512.18912 failure mode), geometric (T-duality class-symmetry), and operational (MIP*=RE/Tsirelson) directions. Reverse-WW stays `[OPEN, as of 2026-06]`, un-registered, NOT OP-50; no experimental channel emerged. New sub-question NQ-1 (does the holographic-completeness failure regime coincide with carrier existence?) recorded here, un-IDed.

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## Iteration 12 refinements (2026-06-19) — the physics-net lever retired, the indefinite-metric route closed, external channels dry

*From the iteration-12 tracks (B1/B2/B3, all UPHELD-WITH-CORRECTION) plus the subquestion-#3 frontier scout. No headline movement (ELEVENTH consecutive confirmation); the principal hedge's grade AND condition both unchanged; terminal saturation re-confirmed.*

- **HYP-CKV-VACUITY-R6 — UNCHANGED** (grade MEDIUM-HIGH→HIGH conditional AND condition: the net-naturality theorem + the located commutant-to-generated gap). `[INFERENCE — HIGH classification]` No R-advance, no R7. Iteration 12 minted no R-advance: the physics-net relative-bicentralizer lever is non-load-bearing (B1), no 2023–2026 external math touched the residual (B2/B3), and the indefinite-metric / Krein carrier route is closed-negative (scout).

- **HYP-FACTORIZATION-IS-GEOMETRY — route (5) annotated; subquestion #3 CLOSED-NEGATIVE; count stays FIVE.** `[INFERENCE, high]` (i) The relative-bicentralizer "externalized vocabulary" from iteration-11 A1 is confirmed **NAMING-ONLY / non-load-bearing** (B1): the relative-bicentralizer triviality question is genuinely OPEN as pure operator algebra (AHHM is discrete-inclusion-only) but cannot close the no-go (trivial bicentralizer is carrier-friendly). (ii) The **indefinite-metric / PT-symmetric / Krein carrier route (subquestion #3) is CLOSED-NEGATIVE** (scout): it collapses to encode-not-generate by the gauge-artifact / installed-not-derived / $\eta_0$-equivalent trichotomy — the *same* route-5 carrier problem in Krein vocabulary, **not a sixth route**. Two of iteration-11's three open subquestions are thereby discharged (the third, the bicentralizer-flow signed-obstruction, folds into B1's non-load-bearing bicentralizer territory). Carrier-convergence count **stays FIVE**.
- **HYP-ENCODING-SCREEN — UNCHANGED; two 2026 instances recorded.** `[INFERENCE]` Grade MEDIUM-HIGH→HIGH, condition unchanged; reverse-WW stays pinned to the §8 reopening condition, `[OPEN, as of 2026-06]`, un-registered, NOT OP-50. Two 2026 instances of the screen are logged, both INPUT-presupposing: the Morinelli–Neeb–Olafsson Euler-pair geometric-BW / Spin–Statistics theorems (symmetry = input; arXiv:2603.26390 / 2508.10960), and "GR from RG" (arXiv:2602.11806, a confirmed evade-not-invert Weinberg–Witten instance with a UV-Dirichlet-fixed installed metric — discharging the iteration-11 minor "WW gloss not confirmable" defect). Neither is a carrier, a reverse-WW inversion, or a sixth route.

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## Iteration 16 refinements (2026-07-03) — the multi-wedge closure; the hedge

## advances R6 → R7; the residual is pinned to (E_O)

*From the clock-(1)-triggered iteration (tracks G1–G4, referees R1/R2/R3 binding; see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter16-*). Headline UNCHANGED (15th consecutive confirmation) — but the principal hedge moves for the first time since iteration 9: the diffuse commutant-to-generated-gap condition is replaced by a single named open hypothesis.*

- **HYP-CKV-VACUITY-R7 — MINTED (replaces R6; grade unchanged, conditional HIGH; the first referee-verified hedge advance since iteration 9).** `[INFERENCE — HIGH classification; condition verbatim per referee R1]` Condition string: *"Conditional HIGH on: (i) the localization-template hypothesis **n1** (irreducible, per iter-9, unchanged); AND (ii) hypothesis **(E_O)** — vacuum ergodicity on massive double-cone algebras,* $M(O)_\omega = M(O) \cap \{\Delta_O^{it}\}' = \mathbb{C}1$ *for every double cone O. Direction: given the Lemma-G hypotheses (H, + weak additivity) and n1, (E_O) makes the net-naturality no-go a theorem (Lemma G, referee-verified); ¬(E_O) is necessary for any counterexample fiber but NOT sufficient (rotation equivariance, the Lemma-N2 cross-fiber web, and the inherited centralizer-sign vacuity remain as generativity walls). The commutant-to-generated gap is NOT closed: it is non-empty ((−1)^N, η0) and classified — localized branch dead (G-loc), fully-natural non-localized branch = gauge gradings with Poincaré-invariant eigenprojections, Δ_geo = 0 (G-glob)."* House history: two prior R7 mints (iteration 10) were struck — for a wrong-object error and an overstated discharge; this mint differs by referee-verified written proofs on the correct object ( $\{\Delta_W^{it}\}' \subset B(H)$) and by exhibiting, not denying, the gap's occupants. → *[notes* (Working note: *Track G1, iteration 16: direct assault on the net-naturality theorem — Lemma G formalized; the multi-wedge lever closes the non-localized sliver; the localized horn is a three-line theorem; the residual shrinks to double-cone* *vacuum ergodicity (E_O)* (repo: `notes/2026-07-03-iter16-G1-multiwedge-naturality.md`))]

- **LEM-NET-NATURALITY-G — new named item (route-5 closure lemma, two branches proof-grade).** On a Haag-dual split BW net with unique vacuum and weak additivity: **(G-loc)** every localized Aut($\mathcal{N}$,Ω)-natural carrier is ±1 `[INFERENCE, high — referee-verified proof, iteration 16]` (Borchers scaling $\Longrightarrow \mathrm{Fix}(\Delta_W^{it}) = \mathbb{C}\Omega$; vacuum pinning uses only bounded $\Delta^{it}$; Reeh–Schlieder kill); **(G-glob)** the wedge boosts of all translated wedges generate $\mathcal{P}_+^\uparrow$ (identity $\Lambda_{W+a}(t)\Lambda_W(t)^{-1} = T((1-\Lambda_W(t))a)$, verified by direct computation), so every fully-natural non-localized algebra-compatible carrier is a gauge-symmetry implementer with Poincaré-invariant eigenprojections, $\Delta_{\text{geo}} = 0$ `[INFERENCE, high — referee-verified]`; the sliver is **non-empty** (Fock parity $(-1)^N$) and exhaustively geometry-void; the DHR n1-inheritance reading is framing only `[INFERENCE, medium]`; the conformal-net closure is a transport sketch, not established `[INFERENCE, medium-high]`; **(G-field)** the reduction pins the sole residual to massive double-cone fibers `[INFERENCE, high]`. The multi-wedge lever (intersecting ALL wedge-flow constraints) was genuinely unused in iterations 9–10 (single-Δ throughout; referee-verified against the records).
- **(E_O) — new named open hypothesis (the residual, in one line).** *Vacuum ergodicity on massive double-cone algebras:* $M(O)_\omega = \mathbb{C}1$ for every double cone $O$ in a massive BW net. `[OPEN]` — not settled by Marrakchi–Vaes genericity (generic ≠ specific vacuum state), by III1-ness (a type statement, not ergodicity), or by any 2024–2026 result surveyed; the massive double-cone modular flow is non-geometric with unknown spectrum, and (E_O) is exactly a question about that unknown. Watch item (δ) of the re-sharpened clock (1). *(Iteration 17 attacked it directly: free-field reduction to one-particle continuous spectrum; three structure-theory routes and the disproof all fail; see the Iteration 17 refinements below.)*
- **HYP-FACTORIZATION-IS-GEOMETRY — route (5) hardened two-branches-to-proof-grade; count stays**

**FIVE.** [INFERENCE, high] (i) G2's three constructive angles (state-family naturality; DL split-inclusion canonical data; core-descent) all BLOCKED — the co-moving evasion of Lemma N2 is real off-vacuum but the family class closes (Lemma S2, sketch-grade, an *extension* of the posed conjecture — Lemma N2 and the dossier statement are untouched); the DL flip fails algebra-compatibility without an installed leg-identification η; LEM-CORE-DESCENT (proof-grade transport corollary) sends core-built carriers to the closed centralizer class or to dual-flow-annihilated trace data. (ii) G4's sweep: 7 of 8 untried frameworks REPRODUCE-SCREEN; the survivor is the **DCNG** watch item (definable-carrier no-go, continuous model theory — the gap as a *definability* question; no-go direction only; gated on an η0-definability lemma) [OPEN – new named target]. (iii) Morinelli–Neeb (Adv. Math. **458** (2024) 109960, arXiv:2312.12182) prove a two-directional, **BW-conditional** bridge: given BW, regularity forces Euler-generated modular groups, and conversely an Euler element plus BW yields regularity and localizability — theorem-level confirmation the two smuggle items (n1 ↔ η) are interchangeable *relative to an installed BW/wedge structure*, deriving neither from (algebra, state). (iv) G3: the Marrakchi transfer is RESOLVED-NEGATIVE at the level of the object (the expectation hypothesis is definitional; the flow is undefined for the physics inclusions); the bicentralizer route stays non-load-bearing, now with the AHHM structural output additionally without carrier interaction.

- **HYP-ENCODING-SCREEN — inherits the R7 condition.** [INFERENCE] Grade MEDIUM-HIGH→HIGH, now conditional on {n1 AND (E_O)}. The §7 no-test corollary and the reverse-WW identification are untouched in content; a proof of (E_O) would make both unconditional on the stated hypotheses. No experimental channel emerged; NOT-YET-PHYSICS unchanged.

## Iteration 17 refinements (2026-07-07) — the (E_O) assault: R7 UNCHANGED, the

## free-field reduction refines the (E_O) item

*The first direct attack on the residual (E_O) (five finder lenses E1–E5; three binding adversarial referees; watch-sweep #2 in the same pass). Verdict unchanged (16th consecutive); the hedge did NOT move — (E_O) was neither proved nor disproved.* *See Chapter 37 (p. 693).*

- **HYP-CKV-VACUITY-R7 — UNCHANGED (no R8, no retreat).** `[INFERENCE — HIGH classification; condition {n₁ AND (E_O)} unchanged]` A dedicated five-lens assault neither proved nor disproved (E_O) for any subclass, so the hedge sheds no condition and gains none; grade (conditional HIGH) and condition string unchanged. Carrier-convergence count stays **FIVE**; no numeric ID consumed. (Contrast iteration 16, which advanced R6 → R7.)
- **(E_O) — status append: the free-field REDUCTION (referee-verified PARTIAL).** For the free massive scalar, (E_O) is `[OPEN]` and reduces via the second-quantization functor (M(O) = R(K_O), Ω the Fock vacuum, Δ_O = Γ(δ_O), σ_t = Ad Γ(δ_O^{it}) gauge-invariant number-preserving) to a one-particle spectral question: `[INFERENCE, high]` **purely CONTINUOUS spectrum of the double-cone one-particle modular Hamiltonian ln δ_O (equivalently: the one-particle modular flow is weakly mixing / has no eigenvectors) ⇒ σ^ω weakly mixing on M(O) ⇒ M(O)_ω = ℂ1.** Binding correction: the naive "no eigenvalue 0 of ln δ_O ⇒ ergodic" reading is INSUFFICIENT — Γ(u)dΓ(A)Γ(u)* = dΓ(uAu*) makes N = dΓ(1) and every dΓ(E) modular-fixed regardless of point spectrum, so no-eigenvalue-0 kills only fixed Weyl operators; the continuous-spectrum condition is what discharges the bilinear/number-type elements. The load-bearing spectral fact is `[OPEN]` — numerical-only (Bostelmann–Cadamuro–Minz arXiv:2209.04681; Cadamuro arXiv:2312.08525), near-multiplication with an unresolved non-diagonal kernel; **no closed-form massive double-cone modular Hamiltonian exists** (Longo–Morsella

arXiv:2012.00565, v3 comment: the massive-ball result "contained a gap, and have been removed"). Numerics LEAN toward continuous spectrum (hence toward (E_O)) but do not prove it. **If that spectral fact is proved for the free field, R7 → R8 fires on the free-massive-scalar subclass** (the net-naturality no-go a theorem there, modulo n1); general class (H) stays conditional.

- **(E_O) — general (H): all three structure-theory routes provably FAIL (referee CONFIRMED).** [INFERENCE, high] (a) weak-mixing ⟹ trivial centralizer is [ESTABLISHED], but vacuum weak mixing on a BOUNDED double cone is not derivable from spatial cluster decomposition (the modular orbit carries no operator to spacelike infinity); (b) the wedge-to-double-cone HSMI/Wiesbrock reduction breaks at Δ_O-fixity ⇏ Δ_W-fixity and reconstructs only the ambient Poincaré flows (succeeds iff Δ_O geometric = the conformal case); (c) Marrakchi–Vaes genericity cannot be upgraded to the distinguished vacuum (a dense G_δ omits prescribed points; inner perturbations preserve ergodicity-status via M_{ω∘Ad u} = u*M_ω u; Poincaré fixes ω). Marrakchi arXiv:2606.23636 (relative bicentralizer flow ergodic *with expectation*) does not transfer — the double-cone gate is expectation-free.
- **(E_O) — adversarial disproof FAILED (referee CONFIRMED).** [INFERENCE, high] No non-scalar element of M(O)_ω is constructible from any symmetry class (H) guarantees; every candidate (localized conserved flux; SO(d−1) rotation invariants; hidden point spectrum; integrable/almost-periodic; and split-interpolating factors, central sequences, η0, (−1)^N) dies on the triple wall {bounded ∧ in M(O) ∧ exactly Δ_O-fixed}. Honest status is BARE OPEN (absence-of-witness, not absence-of-existence — the referee's binding down-calibration of the finder's "leaning true").

## References

For all cited works — Tomita–Takesaki and Connes' classification, Bisognano–Wichmann, Buchholz–D'Antoni–Fredenhagen, Ryu–Takayanagi/HRT, Lewkowycz–Maldacena, FGHMVR, Casini–Huerta / Ceyhan–Faulkner (QNEC), Maldacena's AdS/CFT, Maldacena–Susskind (ER=EPR), Witten and Chandrasekaran–Longo–Penington–Witten (crossed product / Type II), Jacobson (1995, 2015), Malament (1977) / Hawking–King–McCarthy (1976), Bombelli–Lee–Meyer–Sorkin and the causal-set literature, Page–Wootters, Connes–Rovelli, Strominger et al. (celestial/BMS/soft theorems), Arkani-Hamed–Trnka and the positive-geometry program, Almheiri–Dong–Harlow / PYHP / Harlow (holographic QEC), Penington / Almheiri et al. (Page curve / islands), Hardy (2001), Chiribella–D'Ariano–Perinotti (2011), Gleason, PBR, Tsirelson, Popescu–Rohrlich, Renou et al. (2021) — see the Bibliography (p. 776). No citation, number, or theorem on this page is invented; where a result's scope is uncertain it is flagged inline.

CHAPTER 20

# The Assumptions Ledger: Bedrock or Artifact?

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This page is the wiki's central registry of the **assumptions** on which physical theory rests. Its purpose is not merely to list them but to **classify** each one along an axis from *fundamental* to *historical artifact*, and to make explicit the consequence of relaxing it. The deliverable distinguishing question is: **which things treated as fundamental are not?** Throughout, every nontrivial claim carries an epistemic marker per *Chapter 2 (p. 8)*. The structural map of how these assumptions feed each theory lives in *Chapter 3 (p. 15)*; the conjectures that would replace or relax them live in *Chapter 19 (p. 392)*.

## Classification scheme

Each assumption is assigned exactly one classification. These are *not* epistemic markers (those tag individual claims); they describe the **status of the assumption as a load-bearing premise**:

- **fundamental** — appears irreducible: no accepted derivation exists, relaxing it breaks the theory's consistency, and it currently functions as a primitive law-like posit.
- **plausibly-fundamental** — deeply load-bearing and empirically robust, but either conditional on a framework, parasitic on a deeper assumption, or known to be only locally/approximately exact.
- **historical-artifact** — treated (often historically) as fundamental, but now understood to be derived, false at a

deeper level, or an inheritance of the regime in which it was discovered.

- **mathematical-convenience** — a representational or computational choice (possibly forced in *form* but not load-bearing in *content*), or an idealization whose rigorous status is incomplete.
- **untested-empirically** — a deep background posit with no decisive experimental test or theorem either way; often smuggled in implicitly.

`[INFERENCE]` The single most useful discipline this page enforces is separating a **true logical inconsistency** from a **domain-of-validity mismatch** from an **unsolved-but-consistent problem**. Most "assumptions that look shaky" are domain boundaries, not contradictions. See *Chapter 14 (p. 191)*.

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# A. Cross-cutting assumptions (the master table)

| ID | Assumption | Appears in | Classification | Reasoning (epistemic) |
|---|---|---|---|---|
| **A-1** | **Microcausality / locality**: $[\phi(x), \phi(y)] = 0$ for spacelike $(x-y)^2 < 0$ | *QFT* (Chapter 8 (p. 89)), *SR* (Chapter 4 (p. 25)), *info* (Chapter 12 (p. 155)), *GR* (Chapter 10 (p. 121)) | fundamental | `[ESTABLISHED as a defining axiom of rel` Drives spin–statistics, CPT, Reeh–Schlied no-signaling even with Bell-violating entar Caveat: microcausality the *axiom* is fundar *local realism* the intuition is precisely wha Schlieder undermine. Expected to be modi Planck scale. |

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|---|---|---|---|---|
| A-3 | **Exact linearity of QM** (superposition principle) | *QM* (Chapter 7 (p. 71)), *math* (Chapter 13 (p. 172)), *info* (Chapter 12 (p. 155)) | plausibly-fundamental | `[ESTABLISHED to extraordinary precision]` Underlies all interference; Weinberg-type and Bialynicki-Birula–Mycielski bounds find no deviation. Demoted from "fundamental" because (i) it is the explicit target of objective-collapse programs, and (ii) nonlinear QM permits superluminal signaling — so linearity may be *derivative* of no-signaling + Hilbert space rather than primitive. |
| A-4 | **Born rule as a separate axiom**: $\Pr = \mathrm{Tr}(\rho E)$ | *QM* (Chapter 7 (p. 71)), *math* (Chapter 13 (p. 172)), *info* (Chapter 12 (p. 155)) | mathematical-convenience | [Form ESTABLISHED/forced; *independence* CONTESTED] Gleason's theorem ($\dim \geq 3$, noncontextuality) makes the form *unique*. Whether it must be *postulated* (von Neumann packaging) or *derived* (Deutsch–Wallace, Zurek envariance, GPT reconstructions) is open; reconstructions suggest it is not logically independent. Treating it as free-standing overstates its independence. |
| A-5 | **Complex** (not real/quaternionic) Hilbert space | *QM* (Chapter 7 (p. 71)), *math* (Chapter 13 (p. 172)), *info* (Chapter 12 (p. 155)) | plausibly-fundamental | `[INFERENCE/CONTESTED]` Long thought conventional (Stueckelberg: real-QM-with-superselection reproduces QM). Recent multipartite Bell-type tests (Renou et al., 2021–2022) argue complex QM is *not* reproducible by a real-Hilbert-space theory under standard tensor-product/network-independence locality. Principled-but-conditional, not unconditional. |

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|---|---|---|---|---|
| A-6 | **Lorentz / Poincaré invariance** is exact (stable vacuum, energy bounded below) | *SR* (Chapter 4 (p. 25)), *QFT* (Chapter 8 (p. 89)), *particle* (Chapter 9 (p. 106)), *GR* (Chapter 10 (p. 121)), *cosmo* (Chapter 11 (p. 139)), *classical* (Chapter 4 (p. 25)) | plausibly-fundamental | `[ESTABLISHED locally to extreme precision; INFERE` Basis of Wigner classification, spin–statistics, CPT; sp needed for stability. Demoted because (1) globally it i spontaneously breaks Poincaré via the cosmic (CMB) Planck-scale Lorentz violation is OPEN, constrained |
| A-7 | **CPT symmetry** holds exactly | *QFT* (Chapter 8 (p. 89)), *particle* (Chapter 9 (p. 106)), *SR* (Chapter 4 (p. 25)) | plausibly-fundamental | `[ESTABLISHED as a theorem GIVEN its premises]` CPT (A-1) + Lorentz invariance (A-6) + unitarity (A-2) + H independent postulate — exactly as fundamental as th Tested to high precision (neutral-kaon mass differenc |

| | | | | |
|---|---|---|---|---|
| A-8 | **Spacetime is a smooth Lorentzian continuum** (differentiable manifold) | *GR* (Chapter 10 (p. 121)), *QFT* (Chapter 8 (p. 89)), *math* (Chapter 13 (p. 172)), *cosmo* (Chapter 11 (p. 139)), *info* (Chapter 12 (p. 155)) | mathematical-convenience | `[INFERENCE/OPEN — deliberately` Underwrites every confirmed GR/ prediction, so fundamental *withi* But convergent hints (Planck-scal $\sim \ell_P^{-2}$, UV divergences, area-law B holographic finiteness, lattice-as-l triviality, discrete QG programs) s continuum is an effective idealizat leaning, not a settled result. |
| A-9 | **Spacetime is fundamental** (not emergent from entanglement/information) | *GR* (Chapter 10 (p. 121)), *info* (Chapter 12 (p. 155)), *QFT* (Chapter 8 (p. 89)), *cosmo* (Chapter 11 (p. 139)) | untested-empirically | `[SPECULATIVE/OPEN]` Working assu GR/QFT. The "spacetime from entanglement" program (Ryu–Tak JLMS, entanglement-wedge recon ER=EPR, first-law → linearized E rigorous proof-of-concept — but *o* AdS/CFT. Generalization to our d universe is unknown and currentl untestable. The deepest place whe habitual "fundamental" label may artifact. |

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|---|---|---|---|---|
| **A-10** | **Quantum fields are the fundamental DOF** (particles emergent) | *QFT* (Chapter 8 (p. 89)), *particle* (Chapter 9 (p. 106)) | plausibly-fundamental | `[INFERENCE]` Locality + cluster decomposition + Q essentially force a field description (Weinberg's fol theorem); particles are frame-ambiguous (Unruh, global particle number in curved space). Demoted because fundamental only *within* QFT — whether fields emerge from strings/boundary/pre-geometr is OPEN, and condensed matter exhibits emergent gauge fields and Lorentz invariance. |
| **A-11** | **Spacetime has exactly** $3 + 1$ **dimensions** | *GR* (Chapter 10 (p. 121)), *QFT* (Chapter 8 (p. 89)), *particle* (Chapter 9 (p. 106)), *cosmo* (Chapter 11 (p. 139)) | plausibly-fundamental | `[ESTABLISHED at accessible scales; OPEN a prior` Inverse-square law holds from sub-mm to cosmological scales; dimension enters Lovelock's theorem and Wigner classification. Demoted: it is empirical *input*, not derived; KK/string scenarios allow small extra dimensions; exchange statistics i dimension-dependent (anyons in 2D). |

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| A-12 | **Fundamental constants are constant** in space and time | *particle* (Chapter 9 (p. 106)), *cosmo* (Chapter 11 (p. 139)), *QFT* (Chapter 8 (p. 89)), *GR* (Chapter 10 (p. 121)) | plausibly-fundamental | `[ESTABLISHED to tight bounds; OPEN at the ma` Constancy of *dimensionless* constants ($\alpha$, $m_p/m$ tested by quasar spectra, the Oklo reactor, atom clocks, BBN/CMB — no confirmed variation. Demoted: directly testable (occasional CONTES -variation claims have not survived); dynamical theories predict slow variation; only dimensionl ratios are meaningful (a units artifact to flag). S *Chapter 16 (p. 308)*. |
| A-13 | **Cosmological principle**: large-scale homogeneity + isotropy | *cosmo* (Chapter 11 (p. 139)), *GR* (Chapter 10 (p. 121)) | plausibly-fundamental | [ESTABLISHED above $\sim 100$ Mpc; INFERENC exact] CMB isotropic to $\sim 10^{-5}$ after dipole subtraction; homogeneity inferred via Copernic principle + isotropy. Demoted: holds *statisticall* only *above* a scale; FLRW is a statistical average backreaction is CONTESTED (believed small); claimed anomalies CONTESTED; inflationary o OPEN. |
| A-14 | **Past hypothesis**: the universe began in extraordinarily low (gravitational) entropy | *thermo* (Chapter 5 (p. 40)), *statmech* (Chapter 6 (p. 56)), *cosmo* (Chapter 11 (p. 139)) | fundamental | [INFERENCE bordering ESTABLISHED that it *needed*; OPEN *why*] Since microdynamics are C symmetric, an asymmetric arrow *must* be sourc an asymmetric boundary condition — a near-for inference. *Necessity* is fundamental; *explanatio* wide open (Penrose's Weyl-curvature quantifica inflation relocates rather than removes the puzz Kept fundamental: no accepted dynamical deriv exists. |

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| **A-15** | **A fundamental action principle exists**; dynamics from stationary $S$ | *classical* (Chapter 4 (p. 25)), *QFT* (Chapter 8 (p. 89)), *particle* (Chapter 9 (p. 106)), *GR* (Chapter 10 (p. 121)), *math* (Chapter 13 (p. 172)) | plausibly-fundamental | [INFERENCE — with a clear demotion of its *classical* status] Stationary action as a *classical* primitive is **derived** as the $\hbar \to 0$ stationary-phase limit of the path integral — its ground is quantum. The deeper claim (a fundamental quantum action) is plausibly effective: $e^{iS/\hbar}$ is heuristic in 4D (no rigorous Lorentzian measure), and S-matrix/bootstrap and operator-algebraic programs formulate physics without a manifest Lagrangian (non-Lagrangian CFTs). | Without a fundamental Lagrangian, dynamics is fixed by consistency conditions (unitarity, crossing, conformal/operator data). Noether-style symmetry–conservation links need reformulation. |

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| **A-16** | **The universe is computable** (physical Church–Turing thesis) | *info* (Chapter 12 (p. 155)), *math* (Chapter 13 (p. 172)) | untested-empirically | `[OPEN/CONTESTED]` Suggested by all known physics, but *not* a theorem. Continuous dynamics, real-valued data, and Malament–Hogarth spacetimes leave it open; the *efficient* (extended) thesis is separately challenged by quantum computing (BQP vs BPP). Often smuggled in implicitly; flagging it as untested is the honest move. | If hypercomputation is realizable, the law $\leftrightarrow$ simulability link breaks, undermining "it-from-bit"/digital physics. If only the *efficient* thesis fails (already indicated by QC), classical simulability is lost but computability survives. |
| **A-17** | **Equivalence principle**: matter couples universally and minimally to one metric | *GR* (Chapter 10 (p. 121)), *classical* (Chapter 4 (p. 25)), *cosmo* (Chapter 11 (p. 139)) | plausibly-fundamental | [ESTABLISHED to $\sim 10^{-15}$ for the weak EP] (torsion balances, MICROSCOPE). Einstein EP licenses geometrization. Demoted: a *single* universal metric is an assumption (multi-metric/scalar-tensor alternatives are constrained, not excluded); EP in the *quantum* regime (superposed/entangled masses) is OPEN; any new light scalar induces composition-dependent forces. | EP violation means inertial $\neq$ gravitational mass, forbidding clean geometrization and requiring extra fields/metrics; a discovery of a fifth force / new scalar. |

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| A-18 | **EFT decoupling**: UV decouples from IR; low-energy physics is renormalizable + suppressed higher-dim operators | *QFT* (Chapter 8 (p. 89)), *particle* (Chapter 9 (p. 106)), *statmech* (Chapter 6 (p. 56)), *info* (Chapter 12 (p. 155)) | plausibly-fundamental | `[ESTABLISHED as working principle; with one maj` Wilsonian RG makes renormalizability *emergent* ( operators $\sim (E/\Lambda)^n$); Appelquist–Carazzone + EF (ChPT, Fermi theory, gravity-as-EFT, SMEFT) con Demoted by a glaring exception: the **cosmologica constant problem** is a failure of naive decouplin vacuum energy fails to decouple, off by $\sim$ 120 orde hierarchy problem is a related relevant-operator se |
| A-19 | **Time is an external classical parameter** / a global preferred time exists | *QM* (Chapter 7 (p. 71)), *cosmo* (Chapter 11 (p. 139)), *GR* (Chapter 10 (p. 121)) | historical-artifact | `[INFERENCE — artifact verdict, sharpened]` In NI Pauli's theorem forbids a self-adjoint time operato to a bounded-below $H$, so $t$ is a c-number while $x$ i operator — an asymmetry from non-relativistic, fix background thinking. In cosmology, FLRW singles time / the CMB frame. Acute in QG: Wheeler–DeW H |

# B. Per-domain foundational assumptions

The cross-cutting table above captures assumptions shared across domains. Below are the **domain-specific** foundational posits, with the same five-way classification (here folding the source schema's `likely-fundamental` $\to$ *plausibly-fundamental*, and `conventional-choice` $\to$ *mathematical-convenience* unless otherwise noted). IDs continue the global sequence as `D-<domain>-n`.

## B.1 Quantum mechanics (non-relativistic) — see *Chapter 7 (p. 71)*

| ID | Assumption | Classification | Note |
|---|---|---|---|
| **QM-1** | States are rays in a complex Hilbert space; amplitudes superpose | plausibly-fundamental | Subsumes A-3, A-5. Linearity `[ESTABLISHED]`; *complexity* plausibly-fundamental (Renou et al.). |
| **QM-2** | Observables are self-adjoint operators; values lie in the spectrum | plausibly-fundamental | Self-adjointness `[ESTABLISHED]` forced (Stone, spectral theorem). *Exhaustiveness* of self-adjoint ↔ measurable is an idealization (superselection, POVMs). |
| **QM-3** | Born rule gives outcome probabilities | plausibly-fundamental | See A-4. Form forced by Gleason (dim $\geq 3$); derivability CONTESTED. |
| **QM-4** | Projection/collapse postulate is a real, distinct dynamical process | untested-empirically | `[CONTESTED]` The measurement problem. Cannot be derived from unitary evolution; replacement (GRW/CSL vs Everett/decoherence vs QBism) unknown. The bare postulate is very likely *not* fundamental as stated. See *Chapter 15 (p. 233)*. |
| **QM-5** | Composite systems combine by tensor product | plausibly-fundamental | `[ESTABLISHED operationally]` Encodes local tomography + entanglement; singled out in GPT reconstructions. Could be otherwise in exotic theories. |
| **QM-6** | Time is an external classical parameter (no time observable) | historical-artifact | See A-19. Pauli's theorem; relational reformulations exist. |
| **QM-7** | Global phase + overall normalization are unphysical (states are rays) | mathematical-convenience | `[ESTABLISHED]` Pure representational convention; only relative phases matter. (Inter-superselection-sector phases are unobservable for a *deeper, dynamical* reason — distinct from this convention.) |

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|---|---|---|---|
| **QM-8** | Hilbert space is separable | mathematical-convenience | `[ESTABLISHED for NRQM]` Adequate for all standard single-system QM; non-separable spaces appear only in special constructions. |
| **QM-9** | Spin–statistics (integer → boson, half-integer → fermion) | fundamental (within physics) / *imported* into NRQM | `[ESTABLISHED]` An *added postulate* in NRQM but a *theorem* in QFT (microcausality + positive energy + Lorentz). Its NRQM status as imported input is a domain-of-validity signal. In 2D: braid group → anyons. |
| **QM-10** | Measurement-independence / "free choice" of settings (Bell) | plausibly-fundamental | `[CONTESTED/OPEN]` Denying it (superdeterminism/retrocausality) preserves locality but is widely regarded as conspiratorial. A load-bearing assumption, not settled. |

## B.2 Quantum field theory — see *Chapter 8 (p. 89)*

| ID | ASSUMPTION | CLASSIFICATION | NOTE |
|---|---|---|---|
| **QFT-1** | Fields are the fundamental DOF | plausibly-fundamental | See A-10. Weinberg's folk theorem; deeper substrate OPEN. |
| **QFT-2** | Fixed smooth globally-Lorentzian background | plausibly-fundamental | See A-8. Spectacular but expected to fail at $\ell_P$; continuum limit existence is the constructive/triviality question. |
| **QFT-3** | Microcausality | fundamental | See A-1. Least negotiable assumption. |
| **QFT-4** | Poincaré invariance + unique stable vacuum + spectrum condition | plausibly-fundamental | See A-6. |

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|---|---|---|---|
| **QFT-5** | Renormalizability as a selection criterion | **historical-artifact** | `[ESTABLISHED reinterpretation]` Once "sacred"; Wilson's RG reinterpreted it as an emergent low-energy property (irrelevant operators suppressed). Non-renormalizable EFTs are predictive. A clear superseded artifact. |
| **QFT-6** | A well-defined functional measure $\mathcal{D}\phi$ exists | mathematical-convenience (status unclear) | `[CONTESTED/OPEN]` Lorentzian "measure" is not a measure in $d = 4$; Euclidean construction rigorous only for super-renormalizable/2D–3D models. 4D YM existence + mass gap is a Clay problem. A phenomenally successful heuristic. |
| **QFT-7** | Gauge "symmetry" is a fundamental interaction-generating principle | mathematical-convenience (heuristic framing) | `[CONTESTED interpretation]` Gauge symmetry is a *redundancy*, not a physical symmetry (Elitzur; observables are gauge-invariant). The physical core (massless spin-1 consistency, charge conservation) is fundamental; "derive interactions by gauging" is heuristic. See *Chapter 17 (p. 325)*. |
| **QFT-8** | Asymptotic in/out states + well-defined $S$-matrix (LSZ, asymptotic completeness) | plausibly-fundamental | `[INFERENCE]` LSZ/Haag–Ruelle justify it given a mass gap. Fails for confinement (no free quarks), massless particles (IR), and CFTs (no particles). Restricted domain. |
| **QFT-9** | Wick rotation / Osterwalder–Schrader reconstruction is faithful | mathematical-convenience (reach unclear) | `[ESTABLISHED where conditions hold]` A theorem given reflection positivity; fails at finite chemical potential (sign problem) and for real-time dynamics. |
| **QFT-10** | Unitarity of evolution / $S$-matrix | fundamental | See A-2. |

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| **QFT-11** | Finite field content + a single fixed Fock space suffice | mathematical-convenience (strictly false) | `[ESTABLISHED limitation]` **Haag's theorem**: the interacting theory does *not* live in the free Fock space; inequivalent representations are generic (contrast Stone–von Neumann in finite QM). AQFT replaces this with a net of algebras. |

## B.3 General relativity — see *Chapter 10 (p. 121)*

| ID | Assumption | Classification | Note |
|---|---|---|---|
| **GR-1** | Spacetime is a smooth 4D Lorentzian manifold, signature $(-,+,+,+)$ | plausibly-fundamental | See A-8, A-11. Signature is fundamental to causality; smoothness an idealization; $(-+++)$ vs $(+---)$ purely conventional. |
| **GR-2** | Levi-Civita connection: torsion-free + metric-compatible | mathematical-convenience (a choice) | `[ESTABLISHED as a choice]` Einstein–Cartan (torsion from spin) and teleparallel gravity (flat connection + torsion) agree with GR for spinless matter — the "Geometric Trinity." Metric compatibility is deeply tied to the EP. |
| **GR-3** | Second-order field equations, action linear in $R$ (plus $\Lambda$) | mathematical-convenience (leading EFT term) | `[ESTABLISHED structurally]` **Lovelock's theorem**: in 4D the only divergence-free symmetric 2-tensor from $g$ and its first two derivatives is $aG_{\mu\nu} + bg_{\mu\nu}$. Higher-curvature terms are generic EFT corrections; "linear in $R$ " is the leading term. $\Lambda$ is genuinely fundamental as an allowed term. |
| **GR-4** | Matter couples minimally/universally to one metric (EP) | plausibly-fundamental | See A-17. |

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|---|---|---|---|
| **GR-5** | Diffeomorphism invariance / background independence | **fundamental** | `[ESTABLISHED]` The deepest structural principle. The **hole argument** forces identifying $\varphi^* g$ with $g$: points have no physical identity apart from fields. Dictates the constraint structure, absence of preferred time, and most obstructions to quantization. Any QG theory must reproduce it. |
| **GR-6** | Time-orientable + globally hyperbolic (admits a Cauchy surface) | mathematical-convenience (an added restriction) | `[INFERENCE]` Imposed for a well-posed initial-value problem (Choquet-Bruhat–Geroch). Bare EFE permit CTCs (Gödel, Kerr interior). Whether nature enforces it = (strong) cosmic censorship + chronology protection, both OPEN/CONTESTED. |
| **GR-7** | Matter satisfies (pointwise) energy conditions | **historical-artifact** | `[CONTESTED]` Crucial hypotheses of the singularity/area/censorship theorems, but *not* laws of GR — constraints on matter. QFT violates the pointwise conditions (Casimir, Hawking, squeezed states); the SEC is violated by inflation and by $\Lambda > 0$. Replaced by averaged/quantum conditions (ANEC, QNEC). |
| **GR-8** | $\Lambda$ is a single constant of nature (not dynamical) | untested-empirically | `[OPEN/CONTESTED]` The unique Lovelock term beyond $G_{\mu\nu}$; $\Lambda$CDM fits well. But the value is $\sim 120$ orders below naive QFT vacuum energy (the $\Lambda$ problem). Whether constant, vacuum energy, anthropic, or quintessence is unresolved; recent surveys *hint* (not conclusively) at evolving dark energy. |

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| **GR-9** | Gravity is purely classical (metric is a c-number) | plausibly-fundamental | `[INFERENCE/OPEN]` Classical GR is fantastically successful, but semiclassical $G_{\mu\nu} = 8\pi G\langle T_{\mu\nu}\rangle$ is at best approximate and internally problematic. Near-universal expectation: the metric must be quantum. Fundamental within its domain, expected emergent/approximate at $\ell_P$. |

## B.4 Classical mechanics — see *Chapter 4 (p. 25)*

| ID | Assumption | Classification | Note |
|---|---|---|---|
| **CM-1** | Absolute universal time $t$ (Galilean structure) | **historical-artifact** | `[ESTABLISHED false fundamentally]` SR demolishes absolute simultaneity; Newtonian time is the $c \to \infty$ limit. |
| **CM-2** | Smooth finite-dim configuration manifold; deterministic ODE evolution | plausibly-fundamental | `[INFERENCE/CONTESTED at margins]` Picard–Lindelöf gives uniqueness for Lipschitz fields. Cracks: Lipschitz failure $\to$ non-uniqueness (Norton's dome, $\ddot{x} = \sqrt{x}$); finite-time non-collision singularities (Xia 1992, $N \geq 5$). Physical vs point-particle artifact is OPEN. |
| **CM-3** | Forces specified independently and superpose; $L = T - V$ | mathematical-convenience (modeling convention) | `[INFERENCE]` Works because classical forces derive from potentials; not logically forced (dissipative/nonholonomic systems need generalizations). The form $L = T - V$ is a low-energy shadow of the relativistic/path-integral origin. |
| **CM-4** | Stationary-action principle | plausibly-fundamental | See A-15. **Derived** as the $\hbar \to 0$ limit of the path integral; genuinely *stationary* (saddle points past caustics), not "least." |
| **CM-5** | Phase space carries a canonical symplectic structure; Poisson algebra | **fundamental** (within the framework) | `[ESTABLISHED]` The most robust structural feature — survives (deformed) into QM via Poisson $\to$ commutator; preserved by canonical transforms and the flow (Liouville). Darboux: all symplectic manifolds look locally identical (rigid, not conventional). |

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| **CM-6** | Euclidean $E^3$ space + existence of inertial frames | **historical-artifact** | `[ESTABLISHED false fundamentally]` GR replaces flat space + global inertial frames with dynamical curved spacetime; only *local* inertial frames exist. Smuggles in the origin-of-inertia (Mach) question, OPEN/CONTESTED. |
| **CM-7** | Observables have simultaneous arbitrarily-precise real values | **historical-artifact** | `[ESTABLISHED false fundamentally]` QM forbids it (uncertainty, noncommutativity); the classical phase-space point is the $S \gg \hbar$ coarse-grained limit. |
| **CM-8** | Positive-definite mass tensor; regular (nondegenerate) Lagrangian | mathematical-convenience (regularity assumption) | `[INFERENCE]` Makes the Legendre transform invertible. Deliberately violated by the most important theories (gauge fields, reparametrization-invariant systems, relativistic point particle) — *singular* Lagrangians needing Dirac constraint theory. |
| **CM-9** | Time-reversal invariance of microscopic mechanics | plausibly-fundamental | [ESTABLISHED for standard $L$; CONTESTED as universal] Holds for $L$ quadratic in velocities; broken by velocity-dependent/dissipative forces and *fundamentally* by CP-violation. Underlies the Loschmidt paradox. See *Chapter 5 (p. 40)*. |

## B.5 Thermodynamics — see *Chapter 5 (p. 40)*

| ID | ASSUMPTION | CLASSIFICATION | NOTE |
|---|---|---|---|
| **TH-1** | Equilibrium states exist, specified by few macrovariables (thermodynamic limit) | plausibly-fundamental | `[ESTABLISHED for short-range systems]` Precondition for the formalism. Fails for long-range/gravitating/small/nonequilibrium system — a domain-of-validity boundary, not a law. |
| **TH-2** | Entropy is extensive/additive | mathematical-convenience (domain-limited) | `[ESTABLISHED short-range; CONTESTED in genera` A consequence of short-range interactions; gravitating systems are non-extensive (area-law BH entropy). Tsallis/Rényi relax additivity (physical necessity contested). |

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| **TH-3** | Temperature well-defined and positive (zeroth law) | mathematical-convenience | Zeroth-law transitivity `[ESTABLISHED]`/structural; *positivity* is NOT fundamental — bounded-spectrum systems realize negative absolute temperature ($T = \partial U/\partial S$), experimentally confirmed and "hotter than $+\infty$." |
| **TH-4** | Second law is exact and exceptionless | **historical-artifact** (refined) | `[CONTESTED/refined]` StatMech + fluctuation theorems make it *statistical*: $P(+\Sigma)/P(-\Sigma) = e^{\Sigma/k_B}$; Jarzynski/Crooks promote the inequality to a trajectory-ensemble equality. The "absolute" reading is a macroscopic-limit artifact; the *probabilistic* statement is fundamental. |
| **TH-5** | Low-entropy past underlies the arrow of time | plausibly-fundamental (need) / OPEN (why) | See A-14. The *need* is near-forced; the *explanation* is unsolved. |
| **TH-6** | Third law ($S \to S_0$ as $T \to 0$; absolute zero unattainable) | **fundamental** | `[ESTABLISHED]` For non-degenerate (or sub-exponentially degenerate) ground states — a quantum fact (discrete spectrum). The *zero reference* is conventional (Planck); the vanishing of entropy *differences* and unattainability are fundamental. |
| **TH-7** | Equal a priori probability (microcanonical postulate) | plausibly-fundamental | See SM-1. Load-bearing micro$\to$macro bridge; full first-principles thermalization derivation OPEN. |
| **TH-8** | Heat/work cleanly separable; reversible quasi-static processes exist | mathematical-convenience (idealization) | `[ESTABLISHED as idealization]` Reversible = infinitely-slow zero-power limit, never exact; bounds real processes (Carnot, Clausius). |
| **TH-9** | $k_B$ is a fundamental conversion factor | mathematical-convenience (units) | `[ESTABLISHED]` Post-2019 SI, $k_B$ is a *fixed defined constant*: temperature is literally an energy unit. A units convention, though a fundamental energy-per-DOF scale is physical. |

## B.6 Statistical mechanics — see *Chapter 6 (p. 56)*

| ID | ASSUMPTION | CLASSIFICATION | NOTE |
|---|---|---|---|
| **SM-1** | Equal a priori probability (microcanonical) | mathematical-convenience (partly forced) | `[CONTESTED]` The natural Liouville-invariant measure (strong dynamical justification), but no theorem derives it for realistic systems; partly MaxEnt/indifference. Justified pragmatically. |
| **SM-2** | Ergodicity (time average = ensemble average) | **historical-artifact** | `[ESTABLISHED neither necessary nor sufficient]` KAM shows generic Hamiltonian systems are *not* ergodic, yet StatMech works. Superseded by typicality/concentration-of-measure/Khinchin for $\sim 10^{23}$ DOF. |
| **SM-3** | Thermodynamic limit ($N, V \to \infty$ , $N/V$ fixed) | mathematical-convenience (idealization) | `[ESTABLISHED]` True non-analyticities (phase transitions) exist only in the strict limit (Lee–Yang); finite systems have crossovers. Captures real behavior because rounding is negligible for macroscopic $N$. |
| **SM-4** | Molecular chaos (Stosszahlansatz) | **fundamental** (locus of time-asymmetry) | `[ESTABLISHED as where irreversibility enters]` Provably false for *post*-collision velocities; Lanford's theorem: holds for typical data for shor times. The precise point where a typicality/initial-condition assumption enters; not derivable from reversible mechanics. |
| **SM-5** | Coarse-graining (entropy relative to a macrostate partition) | **fundamental** (need) / conventional (specific choice) | `[CONTESTED interpretation]` Subjective (Jaynes) vs objectively-singled-out macrovariables (Boltzmann/Lebowitz/Goldstein). Either way irreversibility cannot be stated without a coarser-than-microstate description. |
| **SM-6** | Low-entropy initial condition / past hypothesis | **fundamental** | See A-14. Cannot be eliminated from StatMech. |
| **SM-7** | Gibbs $N!$ + cell size $h^{dN}$ | **historical-artifact** | `[ESTABLISHED]` Inserted by hand classically (Gibbs paradox, extensivity); QM *derives* both — indistinguishability gives $N!$, $h =$ Planck's constant sets the cell. Fundamental only once Q is recognized. |
| **SM-8** | Canonical $e^{-\beta H}$ is the universal equilibrium distribution | plausibly-fundamental | `[ESTABLISHED in domain; CONTESTED at margins]` Follows from reservoir trace / MaxEnt / ETH for short-range additive systems. Fails for long-range (gravitating, some plasmas); Tsallis proposals CONTESTED. |

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| **SM-9** | Eigenstate Thermalization Hypothesis (ETH) | plausibly-fundamental | `[INFERENCE, strong numerical support]` The quantum mechanism of thermalization (Deutsch, Srednicki, RMT). Not proven for any realistic $H$; provably fails for integrable (GGE) and many-body-localized systems. Domain: generic non-integrable, non-MBL. |

## B.7 Cosmology — see *Chapter 11 (p. 139)*

| ID | Assumption | Classification | Note |
|---|---|---|---|
| **CO-1** | Cosmological principle | plausibly-fundamental | See A-13. |
| **CO-2** | GR is correct on cosmological scales | plausibly-fundamental | `[ESTABLISHED in tested regimes; INFERENCE on horizo` Tested in the solar system / pulsars / GW mergers — sc $10^{30}$ smaller than the Hubble radius. Applying it unmod the whole universe is an extrapolation; modified-gravit alternatives constrained, not all excluded. |
| **CO-3** | Matter is a perfect fluid with simple equations of state | mathematical-convenience (replaceable modeling) | `[INFERENCE]` Perfect-fluid form is *forced* by FLRW sym background level; dust DM and $w = -1$ DE are simplif *choices* data has not yet required abandoning. |
| **CO-4** | Spatial flatness ( $\Omega_k = 0$) | mathematical-convenience (empirically-motivated choice) | `[INFERENCE]` An inflation prediction, consistent with Cl ($ |
| **CO-5** | Cold dark matter exists as a distinct dominant component | plausibly-fundamental (existence) / OPEN (identity) | `[ESTABLISHED that something beyond baryons+GR is ne` Multi-pillar concordance (rotation curves, lensing, Bull Cluster, CMB peaks, BBN). "Cold" favored over hot/wa *Particle* identity (WIMP/axion/sterile-$\nu$/PBH) OPEN a searches. See *Chapter 15 (p. 233)*. |
| **CO-6** | Dark energy is a cosmological constant ( $w = -1$ exactly) | mathematical-convenience (minimal choice) | [ESTABLISHED that acceleration occurs; OPEN/CON that it is exactly $\Lambda$] $\Lambda$ is simplest and fits almost all data exactly is a choice. Recent BAO read by some as hinting evolving DE (CONTESTED, not established). |

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| **CO-7** | Primordial fluctuations: adiabatic, Gaussian, near-scale-invariant | plausibly-fundamental (data) / INFERENCE (inflationary origin) | `[ESTABLISHED as data description]` $n_s \approx 0.965$, super-horizon correlations measured. Inflationary *origin* compelling but unproven; some alternatives reproduce the spectrum; B-mode tensors not detected. |
| **CO-8** | Inflation occurred | plausibly-fundamental (bordering SPECULATIVE) | `[INFERENCE]` Solves horizon/flatness/monopole and predicts the spectrum, but: no primordial GW detection; inflaton unidentified; eternal-inflation measure problem; needs its own low-entropy initial conditions (Penrose). Leading paradigm, not established fact. See *Chapter 19 (p. 392)*. |
| **CO-9** | Low-entropy initial macrostate (past hypothesis) | **fundamental** | See A-14. Near-perfect thermal uniformity = high thermal but very low *gravitational* entropy (Weyl $\approx 0$). |
| **CO-10** | Standard thermal-equilibrium reasoning + single arrow apply to the cosmos | mathematical-convenience (not rigorously founded for gravity) | `[INFERENCE/CONTESTED]` Self-gravitating systems have negative specific heat and no global max-entropy equilibrium; global $T/S$ for the universe is a useful idealization that breaks down conceptually for gravity. |
| **CO-11** | Exactly three light neutrino species ( $N_{\text{eff}} \approx 3.044$) | mathematical-convenience (empirically-grounded, easily freed) | `[ESTABLISHED to good precision]` BBN + CMB damping tail consistent with 3.044 (the excess over 3 is a real QFT prediction). Held fixed but easily promoted to a free parameter (dark radiation, $H_0$-tension probes). |
| **CO-12** | Cosmic time $t$ is a globally well-defined preferred coordinate | historical-artifact | See A-19. FLRW singles out the CMB frame — spontaneous, not law-breaking; routine treatment as fundamental is a convenience of the high-symmetry idealization. |

## B.8 Particle physics / Standard Model — see *Chapter 9 (p. 106)*

| ID | Assumption | Classification | Note |
|---|---|---|---|
| PP-1 | Nature is a local, Lorentz-invariant, unitary relativistic QFT | plausibly-fundamental | `[ESTABLISHED to extreme precision]` (electron $g-2$ to $\sim 12$ sig figs). Plausibly emergent from a deeper structure; QG likely modifies locality at $\ell_P$. |
| PP-2 | Gauge group is exactly $SU(3)_c \times SU(2)_L \times U(1)_Y$ | plausibly-fundamental | `[ESTABLISHED as low-energy description]` Product structure + hypercharge normalization look unifiable (SU(5), SO(10), Pati–Salam); coupling running suggests near-unification at $\sim 10^{16}$ GeV (better in MSSM). A possible low-energy shadow. See *Chapter 18 (p. 349)*. |
| PP-3 | Hypercharge/fermion representations are what they are (e.g. $Q_L \sim (3,2)_{1/6}$) | **fundamental** (within SM) | `[ESTABLISHED]` Fixed by **anomaly cancellation** (quantum consistency) + measured charges. Charge *quantization* (proton/electron equality to $\sim 10^{-21}$) is *not* explained in the SM — elegantly explained by GUT embedding. Fundamental to the SM but points beyond it. |
| PP-4 | Exactly three generations with identical gauge numbers | untested-empirically | `[ESTABLISHED empirically]` Invisible $Z$ width $\rightarrow$ exactly 3 light active neutrinos; no 4th chiral generation (Higgs data). *Why* three is completely OPEN. Could-be-otherwise. |
| PP-5 | EWSB by a single elementary scalar doublet (tachyonic mass term) | mathematical-convenience (minimal realization) | [ESTABLISHED that a $\sim 125$ GeV SM-like scalar exists] Minimal single-doublet is a *choice* (2HDM, composite Higgs, technicolor). The elementary-scalar assumption *creates* the hierarchy problem — theoretically suspect; possibly composite. CONTESTED whether elementary. |
| PP-6 | Neutrinos are massless (minimal SM) | **historical-artifact** | `[ESTABLISHED FALSE]` Oscillations prove nonzero mass. A minimality assumption baked into the original SM; fixed by the Weinberg dim-5 operator or $\nu_R$. The clearest case of a wrong original SM assumption. See *Chapter 14 (p. 191)*. |

| | | | |
|---|---|---|---|
| **PP-7** | Neutrino masses are Dirac (vs Majorana) | untested-empirically | `[OPEN/CONTESTED]` Genuinely unknown. Majorana (seesaw) explains lightness and is favored by many; $0\nu\beta\beta$ (the decisive test) not observed. |
| **PP-8** | The QCD $\bar{\theta}$ term is (or relaxes to) $\approx 0$ | untested-empirically | `[OPEN]` Strong-CP problem: nothing forces $\bar{\theta} \approx 0$; measured $< 10^{-10}$ vs naturalness $\mathcal{O}(1)$. Options: anthropic, massless up (disfavored by lattice), or a Peccei–Quinn axion (SPECULATIVE but well-motivated). |
| **PP-9** | The SM has $\sim$ 19 free parameters fixed only by experiment | plausibly-fundamental | `[ESTABLISHED counting]` Rises to $\sim$ 26–28 with neutrino masses. The strong hierarchies (e.g. top/electron Yukawa $\sim 10^5$) suggest an underlying flavor theory. The most glaring "descriptive, not explanatory" aspect. See *Chapter 16 (p. 308).* |
| **PP-10** | Fixed 4D Minkowski; gravity neglected | mathematical-convenience (domain restriction) | `[ESTABLISHED valid choice]` Gravitational coupling $\sim (E/M_{\mathrm{Pl}})^2$ negligible at colliders. A deliberate domain restriction, and exactly the boundary where the SM must be completed (gravity non-renormalizable; SM is an EFT below $M_{\mathrm{Pl}}$). |
| **PP-11** | $B$ and $L$ are (accidental, approximate) global symmetries | **historical-artifact** | `[ESTABLISHED]` Accidental symmetries of the renormalizable Lagrangian, not imposed; anomalous (only $B-L$ non-anomalous), violated by sphalerons; $B$-violation *required* for baryogenesis (Sakharov). An artifact of restricting to low-dimension operators. |

## B.9 Information theory & physics — see *Chapter 12 (p. 155)*

| ID | ASSUMPTION | CLASSIFICATION | NOTE |
|---|---|---|---|
| **IT-1** | Information has an observer-independent measure (Shannon $H$) | **fundamental** | `[ESTABLISHED]` Shannon/Khinchin uniqueness theorem; operational meaning (compression/communication limits) proven and validated. |

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|---|---|---|---|
| **IT-2** | The logarithm + base-2 "bit" are the natural units | mathematical-convenience (base) / fundamental (log form) | The log form is *forced* by additivity; base-2 vs nats vs base-$d$ is pure convention ($k_B \ln 2$ links to thermo). |
| **IT-3** | States are density operators; info measures are spectral functions (von Neumann $S$) | **fundamental** (conditional on QM) | `[ESTABLISHED within QM]` Follows from Hilbert-space postulates + Gleason/Born; $S$ singled out by Schumacher compression/quantum AEP. Conditional on QM being final (OPEN). |
| **IT-4** | Gravitational entropy/info bounds scale with AREA (holographic principle) | plausibly-fundamental | `[INFERENCE]` Bekenstein–Hawking area-extensivity on multiple independent derivations (Hawking, Euclidean action, string microstates, LQG area spectra); area-law entanglement + Bousso bound reinforce. Lacks a fully general background-independent proof. See *Chapter 17 (p. 325)*. |
| **IT-5** | Landauer's principle: erasure costs $k_B T \ln 2$ per bit | **fundamental** (inequality) / CONTESTED (independence) | `[ESTABLISHED]` Derivable from statmech + Jarzynski/Crooks; experimentally confirmed near the bound. Whether it is *independent* of the second law (Earman–Norton, Hemmo–Shenker) is CONTESTED, but the inequality is not. |
| **IT-6** | Spacetime geometry is emergent from entanglement (RT / "it-from-qubit") | untested-empirically | See A-9. Rigorous only in AdS/CFT (RT, JLMS, QEC reconstruction, first-law $\rightarrow$ linearized Einstein). Generalization unknown. |
| **IT-7** | AdS/CFT holds exactly (bulk QG = boundary CFT) | plausibly-fundamental (duality) / OPEN (exactness) | `[INFERENCE]` Enormous consistent evidence (symmetries, spectra, correlators, integrability, BH thermo); proven in some lower-dim/BPS cases; no general nonperturbative proof. Treat duality as plausibly-fundamental, exactness/genericity as open. |

| | | | |
|---|---|---|---|
| **IT-8** | ER=EPR (entanglement = geometric wormhole connection) | untested-empirically | `[SPECULATIVE]` Maldacena–Susskind conjecture; supported by traversable-wormhole + thermofield-double constructions. No first-principles derivation; status for generic entanglement unknown. See *Chapter 19 (p. 392)*. |
| **IT-9** | Physical processes are computable (physical Church–Turing) | untested-empirically | See A-16. |
| **IT-10** | Information is conserved (unitarity) | **fundamental** (within QM) | See A-2. Page curve reproduced via QES/islands `[INFERENCE]`; mechanism in our universe still being worked out. |
| **IT-11** | The continuum is the right substrate (vs discrete "it-from-bit") | historical-artifact (leaning; arguably unclear) | See A-8. UV divergences, area-law BH-entropy finiteness, holographic finiteness hint the continuum is effective (Wheeler's "it from bit"). Far from settled. |

## B.10 Mathematics for physics — see *Chapter 13 (p. 172)*

| ID | Assumption | Classification | Note |
|---|---|---|---|
| **MA-1** | Pure states are rays in a complex separable Hilbert space (field is $\mathbb{C}$) | plausibly-fundamental | See A-5. Separability a near-universal convenience; "why complex" genuinely open; quaternionic QM consistent but unmotivated. |
| **MA-2** | Observables are self-adjoint (not merely symmetric) | **fundamental** | `[ESTABLISHED]` Spectral + Stone theorems require self-adjointness; choosing a self-adjoint extension is a *physical* choice (boundary conditions). PT-symmetric QM is a similarity reformulation, not a refutation. |

| | | | |
|---|---|---|---|
| **MA-3** | Symmetries act by unitary/antiunitary reps; evolution is unitary | **fundamental** | `[ESTABLISHED]` Wigner's theorem + Stone; antiunitarity forced for time reversal. The measurement "collapse" is the one tension (CONTESTED interpretational issue, not a mathematical defect). |
| **MA-4** | Spacetime is a smooth finite-dim manifold; fields are smooth bundle sections | mathematical-convenience (effective) | See A-8. Distributional/weak solutions show smoothness is often relaxed even classically. |
| **MA-5** | Gauge fields ARE connections on principal $G$-bundles | **fundamental** | `[ESTABLISHED]` Wu–Yang/Trautman dictionary; Aharonov–Bohm + Dirac quantization confirm the bundle topology is physical. "Gauge = redundancy" is standard `[INFERENCE]`, refined (not refuted) by large-gauge/soft charges. The structure group is dictated by experiment. |
| **MA-6** | Born rule + recovery of Kolmogorov probability in decohered subalgebras | plausibly-fundamental | See A-4. Form near-inevitable via Gleason; derivability CONTESTED. |
| **MA-7** | Relativistic QFT is axiomatizable (Wightman/Haag–Kastler/OS) + a 4D interacting example exists | plausibly-fundamental | `[ESTABLISHED for the axioms + low-dim models]` Whether central 4D theories satisfy them is the **YM mass-gap** problem (OPEN). Minority view: perturbative 4D QED/$\phi^4$ are "trivial" (Aizenman–Duminil-Copin proved $\phi^4_4$ triviality) — a domain-of-validity matter, not an inconsistency. |
| **MA-8** | The Lorentzian path-integral "measure" $\mathcal{D}\phi\, e^{iS}$ is a legitimate measure | **historical-artifact** | `[ESTABLISHED]` No $\sigma$-additive translation-invariant measure exists in infinite dimensions; $e^{iS}$ is non-absolutely-convergent. Rigorous content is via Euclidean reflection-positive measures + continuation, oscillatory-integral methods, or perturbation theory. A productive abuse of notation. |
| **MA-9** | Renormalization is meaningful; the Wilsonian EFT picture organizes cutoff dependence | **fundamental** | `[ESTABLISHED]` Empirically (QED, EW precision) + BPHZ-rigorous order by order. The shift from "embarrassing infinity-subtraction" to "EFT + RG flow" is a genuine conceptual correction. Nonperturbative continuum existence for non-asymptotically-free theories is OPEN (triviality). |

| | | | |
|---|---|---|---|
| **MA-10** | Local algebras are type $III_1$ von Neumann factors | plausibly-fundamental | [ESTABLISHED in AQFT] Reeh–Schlieder, Bisognano–Wichmann; explains entanglement-area divergences and the need for *relative* (not absolute) entropy. Algebra type can change in limits (type II in some gravitational/large-$N$ crossovers). |
| **MA-11** | Symmetry groups are Lie groups; reps continuous/unitary; superselection organizes inequivalent reps | **fundamental** | [ESTABLISHED] Wigner particle classification, universal covers ($SL(2,\mathbb{C})$, spin), Doplicher–Haag–Roberts theory. Generalized/categorical symmetries and quantum groups *extend* but do not overturn this. |

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# C. Narrative: the artifact-suspects

The table's real payload is the set of assumptions **commonly treated as fundamental that are probably not**. These deserve scrutiny because mistaking an artifact for a law misdirects the unification program (*Chapter 18 (p. 349)*). I rank them by how consequential the misclassification would be.

## C.1 Spacetime fundamentality (A-9) — the deepest suspect

[SPECULATIVE/OPEN] The habit of treating the smooth manifold $(M, g)$ as the bedrock of physics is the most likely place where "fundamental" is an artifact of working inside background-dependent frameworks. Two distinct demotions stack: (i) the *continuum* may be effective (A-8, A-11, MA-4), and (ii) *geometry itself* may be emergent from entanglement (A-9, IT-6). The AdS/CFT results — $S_{\text{EE}} = \frac{\text{Area}(\gamma)}{4G_N}$ (Ryu–Takayanagi) and the derivation of linearized Einstein equations from the entanglement first law $\delta S = \delta\langle H\rangle$ — are a genuine proof-of-concept [INFERENCE] that *some* worlds reconstruct geometry from boundary information.

But they live in a special tractable corner (asymptotically AdS, large $N$, strong coupling). Whether our de Sitter-like, non-holographic universe works this way is untestable today. The honest verdict: do not assert spacetime is emergent (that overstates the evidence), but flag every "spacetime is fundamental" usage as **untested-empirically**, not fundamental.

## C.2 Renormalizability (QFT-5) — the cleanest *resolved* artifact

`[ESTABLISHED reinterpretation]` This is the model case of a successfully *demoted* assumption, and the template for the rest. For decades, renormalizability was treated as a non-negotiable selection rule on fundamental Lagrangians. Wilson's RG dissolved this: irrelevant operators are suppressed as $(E/\Lambda)^n$, so low-energy renormalizability is an **emergent consequence** of integrating out high scales, not an axiom. Non-renormalizable EFTs (chiral perturbation theory, Fermi theory, gravity-as-EFT, the SM-as-EFT with dimension-6 operators) are perfectly predictive. That a "sacred" requirement turned out to be a derived organizing fact is the strongest existence-proof that other "fundamental" assumptions can be artifacts.

## C.3 External / preferred time (A-19, QM-6, CO-12) — a regime inheritance

`[INFERENCE]` Time as an external c-number parameter is an inheritance of non-relativistic, fixed-background thinking. Pauli's theorem ($t$ cannot be a self-adjoint operator conjugate to a bounded-below $H$) shows the $x/t$ asymmetry is structural to NRQM, not to nature; the Wheeler–DeWitt constraint $\hat{H}|\Psi\rangle = 0$ is *timeless*; and relational schemes (Page–Wootters, partial observables) recover dynamics as correlations with a clock subsystem. Cosmic time (CO-12) is the same artifact wearing a cosmological hat: FLRW's high symmetry singles out the CMB frame, but in a realistic inhomogeneous universe no global cosmic time exists. This is a domain-of-validity issue, not a contradiction — but the routine "fundamental time" framing is artifactual.

### C.4 Energy conditions (GR-7) and the second law's absoluteness (TH-4) — laws that are really regime statements

`[CONTESTED]` Two classic "laws" that are actually regime-restricted. The *pointwise* energy conditions are hypotheses of the singularity, area, and censorship theorems — yet quantum fields violate them (Casimir, Hawking, squeezed states), and the strong energy condition is violated by both inflation and the observed $\Lambda > 0$. They survive only in averaged/quantum form (ANEC, QNEC). Likewise, the *exceptionless* second law is a macroscopic-limit shadow of a *statistical* law: the fluctuation theorem $P(+\Sigma)/P(-\Sigma) = e^{\Sigma/k_B}$ makes transient entropy decreases exponentially rare but nonzero, and Crooks/Jarzynski promote the inequality to an exact equality over trajectory ensembles. Neither is a logical inconsistency — each is a precise idealization mistaken for an absolute.

### C.5 The cosmological constant as a constant (GR-8, A-18) — the anomaly hiding in a "fundamental" label

`[OPEN/CONTESTED]` Treating $\Lambda$ as a fixed constant of nature is the minimal Lovelock-allowed choice, but it sits atop the worst quantitative mismatch in physics: the observed value is $\sim 120$ orders of magnitude below naive QFT vacuum-energy estimates, and this *is* the place where EFT decoupling (A-18) demonstrably fails. Calling $\Lambda$ "fundamental" papers over whether it is vacuum energy, anthropically selected, or a dynamical field (quintessence) — with recent surveys *hinting* (not establishing) at evolving dark energy. This is the artifact-suspect with the highest stakes for observational cosmology. See *Chapter 15 (p. 233)*.

### C.6 The classical least-action principle (A-15, CM-4) and the path-integral "measure" (MA-8)

`[INFERENCE]` The principle of *least* (really *stationary*) action looked teleological and fundamental for two centuries; its modern status is **derived** as the $\hbar \to 0$ stationary-phase limit of the path integral. Conversely, the path-integral "measure" $\mathcal{D}\phi\, e^{iS}$ that grounds it is itself not a measure in any rigorous sense in $d = 4$ (MA-8) — a productive abuse of notation whose rigorous content lives in Euclidean reflection-positive measures and analytic continuation. So a "fundamental" classical principle rests on a quantum object that is itself a heuristic: a two-layer reminder that organizing principles and their mathematical encodings can both be effective rather than primitive.

### C.7 The gauge "principle" (QFT-7, MA-5) — physical content vs heuristic framing

`[CONTESTED interpretation]` "Interactions arise by gauging a symmetry" is enormously productive but is a *heuristic*, not a law: gauge symmetry is a redundancy of description (Elitzur's theorem forbids its spontaneous breaking; observables are gauge-invariant), and the same physics is expressible via gauge-invariant Wilson loops or bundle/connection geometry (MA-5, which *is* fundamental as a mathematical identity). The artifact is the *framing as a symmetry principle*; the physical core (massless-spin-1 consistency, charge conservation, the physical bundle topology probed by Aharonov–Bohm) is real. Recent large-gauge/soft-charge work refines rather than refutes this.

### C.8 Honest non-suspects

`[INFERENCE]` Equally important is what is *not* an artifact, lest the page over-correct: **microcausality (A-1)**, **unitarity (A-2)**, **diffeomorphism invariance (GR-5)**, the **symplectic structure (CM-5)**, **self-adjointness (MA-2)**, **molecular chaos as the locus of irreversibility (SM-4)**, the **third law (TH-6)**, **anomaly cancellation (PP-3)**, and **Shannon/Landauer (IT-1, IT-5)** all survive scrutiny as genuinely fundamental within their domains. The discipline is to demote what deserves demotion without inflating skepticism into a blanket suspicion of structure.

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## D. How to use this ledger

- When a page on this wiki invokes an assumption, **cite its ID** (e.g. "assuming A-1 microcausality") so contradictions and dependencies stay traceable.
- A claim that *relaxing* an assumption resolves a tension belongs in *Chapter 19 (p. 392)*; a claim that two assumptions are *jointly inconsistent* belongs in *Chapter 14 (p. 191)*; a genuinely *unresolved* assumption belongs in *Chapter 15 (p. 233)*.
- The dependency chain A-1 + A-2 + A-6 ⇒ A-7 (CPT) is the canonical worked example of an assumption that is a *theorem given others* — use it as the template when assessing whether a "fundamental" posit is actually derivative.

## Iteration 2 additions and refinements (2026-06-08)

*New assumptions and sharpenings from the iteration-2 tracks (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`)), each refereed.*

- **A-20 — "Our universe is asymptotically eternal de Sitter with constant Λ."** Classification: **historical-artifact / modeling-choice** [CONTESTED]. DESI DR2 (2025) prefers dynamical $w_0 w_a$CDM dark energy at 2.8–4.2σ ($w_0 > -1$, $w_a < 0$, phantom crossing), and the swampland/TCC conjectures disfavor stable eternal dS. *If relaxed:* the "no unitary eternal-dS dual" obstruction partly dissolves and the holographic/emergence target shifts to metastable dS or rolling quintessence (raising the bar to reproducing a dynamical $w(z)$). → *OP-42, [notes* (Working note: *Can emergent-spacetime / holography graduate to our Λ>0 universe?* (repo: `notes/2026-06-08-de-sitter-and-emergent-spacetime.md`)), *Chapter 11 (p. 139)*]
- **A-21 — The horizon area-entropy law $S = A/4\ell_P^2$ is an INPUT to the entropic-gravity derivations.** Classification: **mathematical-convenience / load-bearing-input**

[INFERENCE]. Jacobson, Padmanabhan, Verlinde, and Jacobson's entanglement-equilibrium route all *assume* area-proportional entropy (directly, via screen bit-counts, or via vacuum entanglement); recovering Einstein's equations from it is a consistency / equation-of-state result, not a microscopic explanation. *Caveats:* Sakharov induced gravity instead *outputs* the area law; in AdS/CFT, Ryu–Takayanagi *derives* it. *If relaxed (microscopic origin found):* would convert the entropic programs from analogy to physics (OP-45). → *[notes* (Working note: *Entropic / emergent gravity: genuine derivation, or suggestive analogy?* (repo: `notes/2026-06-08-entropic-and-emergent-gravity.md`))]

- **A-22 — Measurement non-contextuality of probabilities (Gleason's premise).** Classification: **plausibly-fundamental, but substantive** `[ESTABLISHED as the premise; its necessity is interpretation-dependent]` . The probability assigned to a projector is independent of which orthogonal resolution completes it — distinct from Kochen–Specker non-contextuality of *values*, and *rejected* by contextual hidden-variable theories (Bohm). The "Born form is forced" conclusion is conditional on it. → *OP-43, [notes* (Working note: *Is the Born Rule Fundamental or Derivable, and Is Quantum Mechanics Exactly Linear?* (repo: `notes/2026-06-08-born-rule-and-quantum-linearity.md`))]
- **A-23 — Geometric modular flow imported at the interpretation stage of the crossed-product program.** Classification: **mathematical-convenience / hidden-input** `[CONTESTED diagnosis]`. The crossed product itself is purely algebraic (Connes–Takesaki; a Killing horizon is *not* required for its existence). But every *semiclassical* construction reproducing $S_{\text{gen}}$ uses a state with *geometric* modular flow (Bisognano–Wichmann boost; Hislop–Longo/CHM for special regions). For generic (algebra, state) pairs modular flow is non-local, and no derivation of geometry from genuinely non-geometric modular flow has been shown — the precise sense in which geometry is quietly imported into "algebra-before-geometry." → *OP-46,*

*[notes* (Working note: *Does the modular / crossed-product (von Neumann algebra) program deliver background independence?* (repo: `notes/2026-06-08-algebraic-background-independence.md`))]

**Refinements to existing assumptions:**

- **A-3 (exact linearity of QM) — sharpened.** Generic *deterministic* nonlinear modifications of the Schrödinger equation permit superluminal signalling (Gisin 1990; Polchinski 1991; reaffirmed Bielińska–Eckstein–Horodecki 2024), so for deterministic dynamics exact linearity is **derivative of no-signaling** `[INFERENCE, strong]`, not primitive. The live exception is *stochastic* objective collapse (GRW/CSL, Diósi–Penrose), which evades the deterministic no-go and is **untested-empirically** at the macroscopic-superposition scale. Reclassify as *plausibly-fundamental-but-derivative; empirically open at the collapse scale.* → *[notes* (Working note: *Is the Born Rule Fundamental or Derivable, and Is Quantum Mechanics Exactly Linear?* (repo: `notes/2026-06-08-born-rule-and-quantum-linearity.md`))]
- **A-4 / QM-3 (Born rule) — sharpened.** The *form* $p = \mathrm{Tr}(\rho E)$ is forced by Gleason (dim≥3) **and** Busch / Caves–Fuchs–Manne–Renes (dim=2, POVM version, at the price of additivity over all compatible effects) — the qubit case is *not* unconstrained. A 2026 rigidity theorem (Torres Alegre, arXiv:2602.09056 [preprint, verified real, not peer-reviewed]) extends this Born-*form* rigidity to **infinite-dimensional normal state spaces** via no-signaling + normal steering ($\sigma$-additive GHJW) + $\sigma$-affinity, but *presupposes* the von Neumann-algebra scaffold — relocating, not dissolving, the foundational question. The residual open content is strictly the *quantitative-probability* problem (OP-43), conditional on A-22. → *[notes* (Working note: *Is the Born Rule Fundamental or Derivable, and Is Quantum Mechanics Exactly Linear?* (repo: `notes/2026-06-08-born-rule-and-quantum-linearity.md`))]

## Iteration 3 note (2026-06-08)

- **Qualify the Type III₁ "fixed core."** Iteration 3's red-team flags that "local QFT algebras are Type III₁ for any Λ" — used as a fixed anchor of the demotion-and-derivation posture — holds `[ESTABLISHED]` only for QFT on a *fixed background*. Once gravity is dynamical, gravitational dressing alters the algebra at leading order in $G_N$ (Giddings arXiv:2505.22708, 2510.24833), so as a core of the *gravitational* theory it is `[OPEN]`/`[CONTESTED]` (see *OP-44* (Chapter 15 (p. 233)), A-23). Broader lesson: the posture's "fixed cores" are **heterogeneous** — each conditional on a stated background or premise (area law is an input only for the Jacobson–Padmanabhan–Verlinde lineage; no-signaling is shared but not quantum-forcing; unitarity is conditional on standard QM and violated by GRW/CSL). → *[notes* (Working note: *Red-team: the demotion-and-derivation posture and its fixed cores* (repo: `notes/2026-06-08-iter3-redteam-demotion-and-derivation.md`)), *Chapter 21 (p. 497)* § Iteration 3 update]

## 2026-07-02 note (Track R2 — measurement problem operationalized)

- **A-3 (exact linearity) — empirical status sharpened.** The stochastic-collapse exception's *canonical* parameter points are now excluded: GRW ($r_C = 10^{-7}$ m, $\lambda = 10^{-16}$ s$^{-1}$) at **9.1σ** and the parameter-free Diósi–Penrose model ($R_0 > 4.9$ Å, 90% CL) — XENONnT, PRL **136**, 120201 (2026), arXiv:2506.05507 `[ESTABLISHED]`. "Empirically open at the collapse scale" now means: open only for colored-noise/dissipative extensions and unnatural parameter corners, which evade radiation bounds by construction `[INFERENCE]`. Classification unchanged (*plausibly-fundamental-but-derivative*); the residual openness has narrowed materially.
- **QM-4 (projection postulate as real dynamics) — unchanged classification, updated pressure.** The only *testable* replacement family (objective collapse) lost its

motivated targets (above); the untestable families' cost ledger was repriced by the 2023–2026 Local-Friendliness results (*Quantum* 7, 1112 (2023); **9**, 1819 & 1851 (2025)) — which constrain assumption combinations, not interpretations. Still *untested-empirically* / `[CONTESTED]`. → *[notes* (Working note: *Track R2 — The Measurement Problem, Operationalized (2026-07 bound state)* (repo: `notes/2026-07-02-measurement-problem-operationalized.md`)), *Chapter 15 (p. 233)* OP-2/OP-26]

## References

See the Bibliography (p. 776) for canonical sources underlying the claims above — including standard treatments of microcausality, spin–statistics, and CPT (Streater–Wightman; Weinberg, *The Quantum Theory of Fields* Vol. I); Gleason's theorem and quantum foundations; Wilson's renormalization group; Lovelock's theorem; the fluctuation theorems (Jarzynski, Crooks); the Ryu–Takayanagi formula and AdS/CFT; the $\phi_4^4$ triviality result (Aizenman–Duminil-Copin); and the standard cosmological concordance analyses. Specific experimental bounds (e.g. weak-EP tests, $N_{\mathrm{eff}}$, $\alpha$-variation) are quoted at the precision stated in the primary literature; consult BIBLIOGRAPHY.md before citing exact figures.

PART V

# THE INVESTIGATION

This is the part where the book stops surveying and starts working. Over six iterations, the project attacked its own best candidate — the wager that causal, algebraic, and entanglement structure is more fundamental than metric geometry — with every tool it had, under standing adversarial referee.

Chapter 21, the Findings, is the narrative spine: the cumulative synthesis of all six iterations, recording what proved solid, what broke, and what changed its tag along the way. The five chapters that follow are the iteration syntheses in sequence. Iteration 2 stress-tested the wager against the recent literature and found two coherent programs joined by thin bridges. Iteration 3 shifted from surveying to *attempting* — and red-teamed the project's own posture. Iteration 4 measured its own diminishing returns. Iteration 5 executed the one designated verdict-flipping move; it did not flip, and the analytical phase was declared saturated. Iteration 6 injected two years of new literature in a deliberate attempt to overturn the result — and tightened it instead.

These syntheses distill roughly thirty long-form working notes — derivation attempts, literature sweeps, red-team reports — which are not reproduced here but are cited throughout and remain public in the repository (notes/, indexed in notes/README.md).

What this part establishes is the book's central result, and it is a negative one of unusual precision: the wager *encodes* geometry but does not *generate* it. Part VI states what follows.

CHAPTER 21

# Findings: The Running Synthesis (Iterations 1–11)

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This page distills the current state of the investigation into six questions: *what is solid; where do the frameworks genuinely clash; which deep principles look most fundamental; what candidate unifying directions survive scrutiny; is a universal theory even possible; and what should we work on next*. Every claim is tagged per *Chapter 2 (p. 8)*. Detail and derivations live on the linked pages; this page is the map, not the territory.

> **Headline (iteration 1).** The two pillars of physics — relativistic quantum field theory (*the Standard Model* (Chapter 9 (p. 106))) and *general relativity* (Chapter 10 (p. 121)) — are each confirmed to extraordinary precision and are *mutually consistent everywhere we can currently test*. Of the 17 catalogued inter-framework "contradictions" (*Chapter 14 (p. 191)*), **none is a demonstrated logical inconsistency**: under adversarial refereeing they resolve into domain-of-validity mismatches, unsolved-but-consistent problems, and conceptual tensions — and they cluster precisely where *both* theories are extrapolated past their tested regimes (the Planck scale, black-hole interiors, the initial singularity). Independently, several of the strongest results in mathematical physics converge on one wager: that **causal / algebraic / entanglement structure is more fundamental than metric geometry** (*Chapter 19 (p. 392)*, *Chapter 17 (p. 325)*). That wager is currently confined to the anti-de-Sitter corner and has no distinguishing experimental test — which is exactly why it is not yet physics.

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## 1. What is firmly established (the load-bearing core)

These are the parts any future theory must reproduce, not overturn.

- **Quantum mechanics works and is weird in a precise way.** The Hilbert-space formalism, unitary evolution, and the Born rule (whose *form* is forced by *Gleason's theorem* (the Glossary (p. 747)) for $\dim \geq 3$) are confirmed to high precision. Bell/CHSH inequalities are violated in loophole-free experiments, *and* no-signaling holds — so nature is non-locally correlated but not superluminally communicating. `[ESTABLISHED]` See *Quantum Mechanics* (Chapter 7 (p. 71)).
- **The Standard Model is an anomaly-free chiral gauge theory** $SU(3)_c \times SU(2)_L \times U(1)_Y$; renormalized perturbation theory predicts e.g. the electron anomalous moment to ~10+ significant figures. `[ESTABLISHED]` See *Particle Physics* (Chapter 9 (p. 106)), *Quantum Field Theory* (Chapter 8 (p. 89)).
- **General relativity is confirmed as a classical geometric theory** across perihelion precession, light bending, Shapiro delay, binary-pulsar decay, direct gravitational-wave detection, and black-hole shadows; below the Planck scale it is a predictive effective field theory. `[ESTABLISHED]` See *General Relativity* (Chapter 10 (p. 121)).
- **Thermodynamics, statistical mechanics, and their bridge are secure**, now sharpened by *exact* fluctuation theorems (Jarzynski, Crooks) and the renormalization-group account of universality. Black-hole mechanics maps exactly onto the laws of thermodynamics ($S_{\mathrm{BH}} = k_B c^3 A/4G\hbar$, $T_H = \hbar c^3/8\pi G k_B M$). `[ESTABLISHED]` See *Thermodynamics* (Chapter 5 (p. 40)), *Statistical Mechanics* (Chapter 6 (p. 56)).
- **$\Lambda$CDM is a 6-parameter concordance cosmology** fitting the CMB, BBN, large-scale structure, and the `[ESTABLISHED]` fact of cosmic acceleration. See *Cosmology* (Chapter 11 (p. 139)).
- **Structural theorems are bedrock:** Noether (symmetry $\Rightarrow$ conservation), Wigner's classification (a relativistic particle *is* a

unitary irrep of the Poincaré group), microcausality, unitarity ( $S^{\dagger}S = 1$), and Lovelock's rigidity (Einstein + $\Lambda$ is essentially the unique 4D second-order metric theory). `[ESTABLISHED]`

The honest takeaway: there is **far more established, mutually consistent physics than is usually implied** by "GR and QM are incompatible." The incompatibility is real but narrow and located.

## 2. The sharpest genuine tensions

Catalogued and refereed in *Chapter 14 (p. 191)* (IDs `GC-n`). The single most important finding of the audit: **classify before alarming.** Highlights, with their refereed kind:

- **[GC-1] Background-dependence vs background-independence** — *domain mismatch + OPEN foundational.* Minkowski QFT presupposes a fixed metric; GR makes the metric dynamical. The referee downgraded the popular "mutually exclusive in principle" framing: QFT on fixed *curved* backgrounds is rigorous, and perturbative graviton EFT places a dynamical metric inside QFT below $M_{\mathrm{Pl}}$. The genuine obstruction is a *superposed/fluctuating causal structure*, which deprives microcausality and the particle/vacuum concepts of their usual definitions. `[ESTABLISHED contrast / OPEN resolution]`
- **[GC-2] Perturbative non-renormalizability of gravity** — an *`[ESTABLISHED]` theorem* ($G$ has mass dimension $-2$; two-loop $C^3$ divergence, Goroff–Sagnotti 1985), with an *`[OPEN]`* UV completion. This is **not** an inconsistency: an EFT with a cutoff is fully consistent below it.
- **[GC-3] Cosmological-constant problem** — *`[OPEN]` fine-tuning*, not a falsified prediction. In strict EFT logic $\Lambda$ is a renormalized free parameter; what "fails" is naturalness. The famous "$10^{120}$" is a *regularization-dependent* ratio (an electroweak cutoff gives ~$10^{60}$); but finite QCD/electroweak condensates already overshoot the observed value by 40–60 orders. See *Chapter 16 (p. 308)*.

- **[GC-4] Black-hole information paradox** — *domain-of-validity tension*, **not** (per referee) a logical inconsistency. Unitarity is now widely favored: AdS/CFT is manifestly unitary, and island/replica-wormhole computations reproduce the unitary Page curve in controlled (largely AdS, large-$c$) settings `[INFERENCE]`; no first-principles mechanism exists for a realistic 4D black hole `[OPEN]`.
- **[GC-6] Measurement problem** — *structural incompleteness*, not a bare inconsistency: the unitary axiom and the Born/projection rule have disjoint stipulated domains, and contradiction appears only under the contested premise that unitary evolution is *universal*. Decoherence explains the pointer basis but yields an improper mixture; it does not select an outcome. `[ESTABLISHED tension / OPEN resolution]`
- **[GC-7] Arrow of time** — *`[ESTABLISHED]` structure* + *`[OPEN]` cosmological origin*. Time-symmetric microlaws cannot yield monotone entropy increase without a time-asymmetric input (Loschmidt, Zermelo); the residual mystery is *why the early universe had such low gravitational entropy*, exported to cosmology and quantum gravity.

**Net:** the "war" between QM and GR is, on inspection, a handful of located domain-mismatches plus deep-but-consistent unsolved problems. That reframing is itself a load-bearing result for the feasibility question (§5).

## 3. The most promising candidate deep principles

From the ranked synthesis in *Chapter 17 (p. 325)*:

- **Most likely genuinely fundamental:** (i) **Noether's theorem + representation theory** — theorem-level, regime-universal `[INFERENCE, high]`; (ii) **unitarity / information conservation** [INFERENCE, high, *conditional on QM being final*]; (iii) **microcausality / causal order** — least negotiable for relativistic QFT, with the sharp conjecture that *causal order outlives spatial locality* `[INFERENCE]`; (iv) the **Past Hypothesis**

— uniquely a *boundary condition* rather than a law `[OPEN-if-irreducible]`.

- **Deepest meta-principle:** the **renormalization group / effective field theory**. `[ESTABLISHED]` Its lesson may be *anti-foundationalist* — that "fundamental" and "emergent" are scale-relative, which directly bears on whether a final theory exists (§5).
- **The central live convergence.** Multiple independent results — the holographic area bound, Ryu–Takayanagi, Jacobson's derivation of the Einstein equation as an equation of state, Van Raamsdonk's disentangling argument, and the Type $\mathrm{III}_1$ structure of local QFT algebras — point the *same way*: **information/entanglement/entropy may be deeper than geometry.** [SPECULATIVE overall; rising to `[INFERENCE]` for the RT sub-claim *within AdS*]. The honest caveat (see §4) is that essentially every rigorous result here lives in the negatively-curved AdS corner, which has the *wrong sign of* $\Lambda$ for our accelerating universe.

# 4. Candidate framework directions and their honest status

Nine directions are developed and refereed in *Chapter 19 (p. 392)* (`H0`–`H8`). They are **research directions, not results.** Strikingly, they converge on one wager — *causal/algebraic/entanglement structure precedes metric geometry* — approached through four lenses:

- **Modular / algebraic** (`H0`, `H4`): take the Type $\mathrm{III}_1$ von Neumann algebra and its Tomita–Takesaki modular flow as primitive; the crossed product with an observer/clock yields a Type II trace, an emergent time, and an area-leading generalized entropy; the entanglement first law $\delta S_A = \delta\langle K_A\rangle$ gives the *linearized* Einstein equation.
- **Causal order** (`H1`, `H3`, `H6`): a partial order is primitive; a causal poset already fixes the conformal metric (Malament/HKM), so only one scalar (scale/volume, or a relational clock) need be

supplied; sharpened by causal-set theory and celestial/BMS boundary symmetry.

- **Amplitudes / positive geometry** (`H5`): amplitudes *are* the canonical form of a positive geometry (amplituhedron); locality and unitarity become derived boundary properties — established only for planar $\mathcal{N} = 4$ SYM.
- **Operational / quantum-error-correction / post-quantum** (`H7`, `H8`): bulk geometry as the continuum limit of a QECC; QM as the rigid limit of an operational information theory.

**The most epistemically defensible entry is the negative one.** `H2` argues that "spacetime from entanglement," judged as a *universal* theory, currently fails two bars — (1) all rigorous results are AdS/large-$N$, linearized, wrong-$\Lambda$-sign; (2) Type $\mathrm{III}_1$ algebras have no tensor factorization, so the "qubits sewing space" picture is *ill-defined* (though not logically inconsistent — modular theory is the rigorous reformulation). It supplies concrete **graduation criteria** (a de Sitter RT analog; full nonlinear Einstein from entanglement; a dynamically-formed asymptotically-flat black hole). `[INFERENCE]`

**Shared, possibly fatal obstacles** every direction must survive: the **Weinberg–Witten theorem** (an emergent massless spin-2 needs an emergent extra dimension), the **Type $\mathrm{III}_1$ obstruction**, **AdS-confinement** of the rigorous results, and the **absence of any distinguishing near-term experimental prediction**. On the wiki's own anti-crank standard (`AGENTS.md` (in the repository) §4), a direction with no distinguishing test is not yet physics. Confidence across `H0`–`H8` ranges from *very-low* to *medium*; none is asserted as established.

## 5. Is a single universal theory of physics possible?

The feasibility assessment (refereed) turns on **disambiguating three senses of "universal/final":**

- **(A) A consistent unified quantum-gravity *framework* reconciling GR and QFT — POSSIBLE, high confidence.** The clashes of §2 are domain-mismatches and unsolved-but-consistent problems, *not* proven contradictions; consistency conditions on a quantum theory of gravity are extraordinarily rigid; and AdS/CFT is a working **existence proof** `[INFERENCE]` that *some* consistent quantum gravity exists. Whether the framework *for our universe* is found and **empirically confirmed** is genuinely `[OPEN]` — plausibly attainable mathematically, but possibly never decided by experiment given the ~15-orders-of-magnitude "desert" between accessible energies and $M_{\mathrm{Pl}}$.
- **(B) A *final* theory — a true bottom to the tower of effective theories — possible but NOT demonstrably so.** The Wilsonian "tower with no bottom" reading (reinforced by $\phi_4^4$ triviality) and the emergence programs (geometry from thermodynamics/entanglement) make *"no fundamental bottom layer"* a serious live alternative; "fundamental" may be partly scale-relative. Lean: a final *framework* is more plausible than a final *substrate*. `[SPECULATIVE]`
- **(C) A *complete* theory that fixes all contingent parameters — the LEAST likely.** The string landscape + anthropic selection `[SPECULATIVE/CONTESTED]`, the cosmological-constant problem `[OPEN]`, and the contingent low-entropy Past Hypothesis `[OPEN]` each threaten parameter-fixing completeness even if the underlying framework is unique.

**Net assessment.** Modestly likely as a consistent unified *framework* (whose existence in principle is essentially uncontested); substantially less likely as a parameter-fixing, complete-and-final *bottom*; with **empirical confirmation, rather than logical possibility, as the most probable point of permanent failure.** We explicitly *decline* the common overreach that Gödel's incompleteness theorems forbid a final theory — undecidability of a theory's *consequences* is not incompleteness of its *laws*. `[INFERENCE]`

## 6. Highest-leverage next questions

Prioritized in *Chapter 40 (p. 739)*; registry in *Chapter 15 (p. 233)*.

1. **Get one real-universe ($\Lambda > 0$) result** out of the emergence program — a de Sitter RT/entanglement analog, or nonlinear Einstein from entanglement. This single deliverable would do more than anything else to move the central wager (§3–4) from AdS folklore toward physics. It is `H2`'s graduation criterion.
2. **Black-hole information as the decisive test bed** — push the Page curve beyond controlled AdS toy models toward a dynamically-formed, asymptotically-flat black hole.
3. **Operationalize the measurement problem** — tighten collapse-model (GRW/CSL) bounds via interferometry/optomechanics; this is the one corner of [GC-6] that is *experimentally* decidable.
4. **Sharpen the GR↔QFT interface** — for each item in *Chapter 14 (p. 191)*, state precisely which incompatibility is logical vs domain-of-validity (continue the refereeing already begun).
5. **Distinguishing-prediction audit** — for every program in *Chapter 18 (p. 349)*, record the single most plausible near-term *distinguishing* observation; flag the (many) programs that have none.
6. **Assumption-demotion program** — attack the highest-value "fundamental-suspects" in *Chapter 20 (p. 458)*: the smooth spacetime continuum, exact linearity of QM, the Born rule as a separate axiom, locality, and 3+1 dimensions.
7. **State the cosmological-constant problem without folklore** — a fully regularization-aware account separating the genuine puzzle from cutoff artifacts.

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## Iteration 2 update (2026-06-08)

*Five refereed deep-research tracks (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`)) pushed on the highest-*

*leverage questions. Net effect: the central wager is* ***sharper and more constrained, not confirmed****. The referee flagged 0/25 new claims unsound; several were corrected in place.*

**What moved.**

- **The Λ>0 graduation barrier is structural, not temporary.** `[INFERENCE, high]` There is no de Sitter RT surface, no uncontested unitary dS dual, and — decisively — **no dS analog of the AdS entanglement-first-law ⇒ Einstein chain**. The genuine positive: the CLPW Type $II_1$ static-patch algebra gives a *derived finite Λ>0 entropy* with a maximal-entropy state (= empty dS), a real foothold for H4 — but it is observer-relative, background-fixed *entropy bookkeeping*, not dynamics. H2's negative meta-conclusion stands and is reinforced. → *[notes* (Working note: *Can emergent-spacetime / holography graduate to our Λ>0 universe?* (repo: `notes/2026-06-08-de-sitter-and-emergent-spacetime.md`)); OP-41, OP-42, A-20]
- **The algebraic/holographic program *encodes* geometry; it does not yet *generate* it.** `[INFERENCE, high — conceded by its own architects]` The modular/crossed-product results (Witten; CLPW; KLS 2024–25, extending Type II to any Killing horizon) are a rigorous reorganization of semiclassical gravity on a *fixed* background with a Killing/conformal symmetry; "background independence" in Witten 2308.03663 means state-independence on a fixed worldline algebra. Whether any von Neumann-algebraic subregion description survives *nonperturbatively* is OPEN (Giddings 2505.22708, "algebraic spacetime disruption"). → *[notes* (Working note: *Does the modular / crossed-product (von Neumann algebra) program deliver background independence?* (repo: `notes/2026-06-08-algebraic-background-independence.md`)); OP-44, OP-46, A-23, H4-R2/R3]
- **The Born rule is forced in form, open in meaning.** `[ESTABLISHED form / OPEN meaning]` Gleason (dim≥3) + Busch/CFRS (dim=2) force $p = \mathrm{Tr}(\rho E)$ given non-contextuality of probabilities; but no no-collapse derivation of the

*quantitative* (unequal-amplitude) rule avoids re-importing a Born-equivalent premise. → *[notes* (Working note: *Is the Born Rule Fundamental or Derivable, and Is Quantum Mechanics Exactly Linear?* (repo: `notes/2026-06-08-born-rule-and-quantum-linearity.md`)); OP-43, A-22]

- **Exact linearity is plausibly derivative of causality — but untested where it matters.** `[INFERENCE strong / CONTESTED that it is fundamental]` Generic *deterministic* nonlinearity permits superluminal signalling (Gisin; Polchinski; Bielińska–Eckstein–Horodecki 2024); *stochastic* objective collapse (GRW/CSL, Diósi–Penrose) evades the no-go and is empirically open. [→ A-3 refinement]
- **Tabletop quantum gravity (BMV) is a class-falsifier, not a quantization proof.** `[INFERENCE for the exclusions; loophole status CONTESTED]` A positive result excludes mean-field semiclassical and classical-channel gravity and constrains stochastic-CQ — establishing non-classicality of the Newtonian sector — but **not** graviton quantization. The mediation-vs-locality premise is logically stronger than no-signaling and is actively disputed (Di Biagio; Aziz–Howl vs rebuttals, 2025). → *[notes* (Working note: *Would gravitationally-induced entanglement prove the gravitational field is quantum?* (repo: `notes/2026-06-08-tabletop-quantum-gravity.md`)); OP-40, HYP-GIE-DISCRIM]
- **Entropic gravity, disaggregated.** [mixed] Jacobson (1995) is a genuine theorem *given* the area law + Unruh $T$; Padmanabhan a structural rewriting; Sakharov a mechanism (which *outputs* the area law); Verlinde-2011 a heuristic; Verlinde-2016 dark matter testable but **disfavored at cluster/CMB scales**. *Iteration-1 correction:* galaxy weak-lensing (Brouwer 2021) *agrees with* EG — it does not falsify it. → *[notes* (Working note: *Entropic / emergent gravity: genuine derivation, or suggestive analogy?* (repo: `notes/2026-06-08-entropic-and-emergent-gravity.md`)); OP-45, OP-47, A-21, HYP-ENTROPIC-GRAVITY-SPLIT]

**Does it cohere into one program? — Partial.** [INFERENCE] The synthesis (*notes* (Chapter 22 (p. 571))) finds **two internally-coherent programs**: **(A) geometry-from-algebra** (de Sitter, algebraic, entropic tracks — one substrate, one method, one decisive deliverable: the dS entanglement first law, OP-41) and **(B) is-the-quantum-framework-final-and-does-it-cover-gravity** (Born/linearity + BMV — reconstruction theorems, no-go arguments, experiments). They are joined only by **thin bridges**: no-signaling/causal-order as a shared load-bearing constraint, and BMV non-classicality as a *logical prerequisite* for the whole emergence program. The tempting "it's all information" unifier is **refused** — "information" names non-interchangeable objects across the two clusters (von Neumann-algebraic relative entropy vs operational probability/LOCC monotones).

The most defensible posture is therefore a **program of demotion-and-derivation**: hold the established cores fixed (local QFT Type III₁ for any Λ (undressed, $G_N = 0$); the area law; no-signaling/causal-order; the Gleason–Busch-forced Born *form*; unitarity — but see the Iteration 3 update below: these "cores" are **heterogeneous** and three are conditional, Type III₁ chief among them) and prosecute as *suspected-derivative* the status of metric geometry, the local-subsystem picture, exact linearity, and the Born rule's axiom-status — each to a decisive yes/no test. **No track has yet graduated from *encoding* to *generating* geometry, and Programs A and B do not (yet) merge.** That is the honest iteration-2 verdict: a sharp, coherent strategy with a single highest-leverage target (a Λ>0 entanglement first law), not a unification.

# Iteration 3 update (2026-06-08)

*Iteration 3 shifted from surveying to **attempting** (three attemptive tracks + a red-team of our own posture; see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter3-*). Net: the coherence verdict is **unchanged at PARTIAL** — and iteration 3 confirms that is correct. Progress was real but **almost entirely negative/structural**: obstructions got sharper and the* *posture got more honestly hedged, while the central wager got no closer to confirmation or to a distinguishing test.*

**Did Program A move? Sharpened, not advanced.** `[INFERENCE, high]` The de Sitter first-law attempt (*notes* (Working note: *Attempt: a de Sitter entanglement first law (OP-41)* (repo: `notes/2026-06-08-iter3-de-sitter-first-law-attempt.md`)), *OP-41* (Chapter 15 (p. 233))) converted iteration-2's bare "no dS first-law chain exists" into a *located* obstruction: AdS's FGHMVR derivation inverts a *continuum* of boundary-anchored regions to a pointwise $\delta G_{\mu\nu}$, but the dS static patch supplies **one** region and **one** Killing vector, so $\delta S_{\rm gen} = \delta\langle K\rangle$ yields only the single $\xi$-integrated Hamiltonian constraint (= the dS generalized second law), not the tensor equation; and the variational principle is **sign-reversed** — empty dS *maximizes* $S_{\rm gen}$ (Banihashemi–Jacobson–Svesko–Visser, $dE = -T\,dS$), structurally opposite to AdS RT minimization. Separately, Jacobson 2015 *is* already a local entanglement-equilibrium derivation of the full Einstein equation valid at any $\Lambda$ — but non-holographic and presupposing the area-entropy coefficient and a fixed-volume maximal-entanglement hypothesis, so it answers a different question. One un-foreclosed escape remains (an $SO(d+1,1)$ family of static-patch modular Hamiltonians; Chen–Xu 2511.00622); the obstruction is tagged `[OPEN]`**, not a no-go**.

**Encode vs generate, made rigorous** (*notes* (Working note: *Formalize: an encode-vs-generate criterion for emergent geometry (OP-46)* (repo: `notes/2026-06-08-iter3-encode-vs-generate-criterion.md`)), *OP-46* (Chapter 15 (p. 233))). `[INFERENCE, high]` A decision-usable criterion (two graded axes — geometric-input $\Delta_{\rm geo}$ and post-processing $\kappa_\Phi$ — plus a five-item "smuggling" checklist) grades six emergence constructions. Finding: **no surveyed construction generates Lorentzian spacetime from factorization-free, region-free, symmetric-state-free data**; the encode/generate frontier coincides exactly with a hand-chosen tensor factorization or graph — i.e. with the Type III₁ no-factorization fact in disguise. Cao–Carroll "space from Hilbert space" comes closest but outputs only a *Riemannian spatial* metric after positing a factorization.

**The A+B merge: narrow success, deep failure** (*notes* (Working note: *Attempt: Merge Program A (geometry-from-algebra) and Program B (quantum-framework-final)* (repo: `notes/2026-06-08-iter3-merge-attempt-operational-substrate.md`))).
`[INFERENCE, medium]` On a shared von Neumann algebra with faithful normal state $(\mathcal{M}, \omega)$ one *can* non-circularly co-derive Pillar B's Born **form** (lifted to infinite dimensions by the 2026 causal-rigidity theorem, *arXiv:2602.09056* (the Bibliography (p. 776))) + linearity **and** Pillar A's KMS/thermal time + crossed-product Type II entropy — with the "observer" demoted to an internal quantum reference frame, genuinely dissolving the agent/no-agent clash *here*. But the GPT axioms that fix the Hilbert-space framework (tensor composition, purification, continuous reversibility) have **no Type III₁ formulation** and fail in infinite dimensions (candidate-no-go ***OP-48* (Chapter 15 (p. 233))**) — and fail precisely in the closed-universe, no-external-observer regime. So iteration-2's "thin bridge" upgrades to a **substantive partial merge at one sublayer, a located no-go at the deeper one** — *not* the "it's all information" fusion.

**Red-team of our own posture — four forced weakenings** (*notes* (Working note: *Red-team: the demotion-and-derivation posture and its fixed cores* (repo: `notes/2026-06-08-iter3-redteam-demotion-and-derivation.md`)); 0 fatal errors):

1. **The "fixed cores" are heterogeneous, and three are conditional.** Most importantly, **local QFT Type III₁ "for any Λ" is a fixed core only on a *fixed background*; once gravity is dynamical it is `[OPEN]`/`[CONTESTED]`** (gravitational dressing alters the algebra at leading order in $G_N$ — Giddings 2505.22708 / 2510.24833). This corrects the iteration-2 framing above, which listed it flatly as fixed. The area law is an *input* only for the Jacobson–Padmanabhan–Verlinde lineage (it is *output* by Sakharov, *derived* in AdS by RT); no-signaling is shared but does *not* single out QM (PR-boxes); unitarity is conditional on standard QM and is deliberately violated by GRW/CSL.
2. **"Demotion-and-derivation" is a research *heuristic*, not a falsifiable program** — its falsifiable content lives entirely in

the object-level deliverables (OP-41, OP-44, the BMV $m^2$-vs-$m^3$ test, collapse bounds) and in the bracketed `[SPECULATIVE]` wager it quarantines.

3. **The six-lens "convergence" is partly a streetlight effect** — the lenses agree partly because they share load-bearing inputs (Killing-symmetric vacua, AdS/large-$N$, the unproven causal-set Hauptvermutung), which is weaker evidence than genuine independent convergence.
4. **CLPW Type II₁ is a feature of the static patch's *bounded area*, not a Λ-diagnostic** — Schwarzschild–de Sitter (Λ>0) is II∞×II∞ (KLS), so the II₁-vs-II∞ distinction tracks the horizon/observer-Hamiltonian spectrum, not the sign of Λ (refines the iteration-2 H2-R1 correction further).

**What survived every attack and was strengthened:** the deflationary core (the catalogued GR/QM "contradictions" are not logical inconsistencies); the central diagnosis **"the algebraic/holographic program *encodes* geometry, it does not *generate* it"** (reinforced by the verified Giddings results); the Born form/meaning split; linearity-as-derivative-of-no-signaling; the refusal of the "it's all information" fusion; the dS negative meta-result; and BMV-as-class-falsifier-not-quantization-proof.

**Empirical falsifiability portfolio (the honest bottom line).** `[INFERENCE, high]` Every live channel tests the program's logical *gate* (is gravity non-classical? — BMV/QGEM, ~10–20+ yr), the status of its *cores* (is linearity/Lorentz exact? — objective-collapse bounds decisive ~2035; LIV/EP null anchors), or its correct *target* (is the universe really eternal dS? — DESI/Euclid $w(z)$, 2–7 yr; currently a 2.8–4.2σ dynamical-dark-energy hint). **None tests the geometry-from-algebra wager itself.** Matter-wave / levitated-optomechanics mass scale is the shared enabling technology gating both live channels. By the wiki's own anti-crank standard, the wager is therefore **still not yet physics** — a sharp, coherent, well-hedged research *strategy*, not a result.

## Iteration 4 update (2026-06-08)

*Three attack tracks + a web-verified literature refresh + a convergence audit (see notes/ (repo: `notes/`; see the notes index, repo: `notes/README.md`) iter4-*)* ***closed the last open escape route from iteration 3*** *and confirmed convergence. The standing verdict is* ***unchanged for the third consecutive iteration*** *— PARTIAL coherence; geometry-from-algebra* ***encodes but does not generate*** *geometry; the wager is* ***not-yet-physics*** *— but the central diagnosis now rests on a much harder argument.*

**The dS first-law escape was executed and failed — a scoped near-no-go.** `[INFERENCE, high]` Iteration 3 left one unforeclosed route to a Λ>0 Einstein-from-entanglement derivation: enlarge the single static-patch boost to the full $SO(d+1,1)$ family of observer/static-patch modular Hamiltonians. Executed, it fails on one interlocking obstruction whose decisive component is a **dimension count**: the isometry orbit is *finite-dimensional* ( $\dim = 2d$), supplying at most $\dim SO(d+1,1) = 10$ global Killing-charge constraints, which cannot Radon-invert to the *infinite-dimensional* local field $\delta G_{\mu\nu}(x)$ — it lacks the "radius modulus" / continuum of anchored regions that AdS's FGHMVR inverts; compounded by the family being geometric in a *common* state only at Bunch–Davies (where the probe vanishes) and a second-order, sign-definite ($\delta^2 S_{\text{gen}} < 0$) concave-GSL character. The **published execution of exactly this route** (Chen–Xu, arXiv:2511.00622) lands precisely where the count predicts: entropy-matching and a second-order quantum first law, **no** $\delta G_{\mu\nu}$. This promotes HYP-dS-NOMIN toward a near-no-go (*Chapter 19 (p. 392)* HYP-dS-NOMIN-R3) and yields a reusable principle (HYP-dS-CARDINALITY): *Einstein-from-first-law needs an infinite-dimensional local region family probing every bulk point; the finite isometry orbit of a maximally-symmetric background gives only global charges*. Two routes survive: the non-unitary dS/CFT

Type II₁ algebra. → *[notes* (Working note: *Execute the dS first-law escape: an SO(d+1,1) family of modular Hamiltonians (OP-41)* (repo: `notes/2026-06-08-iter4-ds-first-law-escape.md`))]

**OP-48 settled as a sharp disjunction.** `[ESTABLISHED/INFERENCE]` Can the GPT framework-fixing axioms live on a Type III₁ algebra? **(a)** Hard **no-go** for exact local tomography / tensor composition on a *single* Type III₁ factor; **(b)** but the **split property** (nested $\mathcal{O}_1 \subset \mathcal{O}_2$, nonzero collar) supplies an intermediate Type I factor with a genuine factorization and local density matrices — defeating the would-be hard no-go; **(c)** yet this is *not* a clean reconstruction: the canonical intermediate factor is singled out only by the vacuum's modular conjugation $J$, **relocating** the Bisognano–Wichmann/CHM geometric-reference-state circularity to the factorization stage (HYP-SPLIT-GEOM), and it fails for adjacent (zero-collar) regions. A full A+B merge still requires a *state-/collar-free* composition surrogate — **OPEN**. → *[notes* (Working note: *Resolve OP-48: can GPT framework-axioms live on a Type III₁ algebra? (split-property test)* (repo: `notes/2026-06-08-iter4-gpt-type3-resolution.md`))]

**The encode→generate frontier is near-vacuous under the standard readout.** `[INFERENCE]` The OP-46 clause-(iii) escape (recover geometry from a *non-symmetric* modular flow) is squeezed by two chained, web-verified theorems: Borchers/Wiesbrock (half-sidedness ⇔ positivity — the emergent translation group comes *from* a relocated positivity premise, not neutral data) and Caminiti et al. (arXiv:2502.02633: recovered boost generators are *conformal Killing vectors*, presupposing a symmetric background). Connes' reconstruction, moreover, yields only a **Riemannian** manifold, not a Lorentzian one (GAP-NCG-LORENTZIAN-SIGNATURE). The genuinely-live, unrealized target is a *non-CKV* readout (modular-Berry / Connes spectral distance) yielding a Lorentzian metric. → *[notes* (Working note: *Break the Bisognano–Wichmann/CHM circularity: geometry from non-symmetric modular flow? (OP-46)* (repo: `notes/2026-06-08-iter4-modular-circularity.md`))]

**Literature refresh (web-verified).** `[ESTABLISHED]` (i) **Muon $g-2$ is now SM-consistent**: the 2025 Theory-Initiative white paper (arXiv:2505.21476) adopts a lattice-QCD HVP average, giving $a_\mu^{\rm SM} = 116\,592\,033(62) \times 10^{-11}$ and $\Delta a_\mu = 38(63) \times 10^{-11}$ — **no tension** (a residual internal lattice-vs-$R$-ratio / CMD-3 HVP tension persists). The historic ~4–5σ anomaly is largely resolved; OP-21 and *Chapter 9 (p. 106)* are updated. Peripheral to the central wager. (ii) The three previously-`[unverified]` 2026 preprints are **all real / web-confirmed** (no fabrication): 2603.06211 (Zhang), 2601.18856 (Pronskikh), 2601.13012 (Pati — fixes the *inner product*, **not** linearity). (iii) Collapse-model bounds tightened but not closed; first Lorentzian-signature asymptotic-safety fixed-point evidence (incremental, `[CONTESTED]`); DESI still DR2.

**Convergence verdict — how far to a conclusion.** `[INFERENCE, high]` The audit (*notes* (Working note: *Convergence / completeness audit — how far from a conclusion?* (repo: `notes/2026-06-08-iter4-convergence-audit.md`))) finds the new-insight trend **diminishing** (three iterations at the same verdict; marginal yield now a sharper obstruction or a self-correction, not new positive structure) and distinguishes two distances:

- **(A) Analytical saturation — close (~1–3 more focused iterations).** "Is a universal theory possible?" is answered in disambiguated form (senses A/B/C); the central wager is named; *why* it's not-yet-physics has survived every attack. What remains is a *finite, named* list: OP-48 point (c) (collar-free composition surrogate); OP-44/OP-CP1 (nonperturbative crossed-product fate); the OP-46 non-CKV Lorentzian readout (the only move that would flip the headline verdict).
- **(B) Object-level confirmed theory — not forecastable.** Gated not just by the ~15-order-of-magnitude empirical desert but, more bindingly, by the fact that **the wager has no distinguishing experimental test even in principle** — every live channel tests the program's gate, cores, or target, none the wager itself.
- **Recommendation: HYBRID** — attack the three named decidable problems over the final ~1–3 iterations, do the cheap

documentation passes (done), and stand up lightweight tracking of the four empirical channels (BMV; collapse bounds ~2035; DESI/Euclid $w(z)$; LIV/EP), since those — not further analysis — gate the object-level verdict.

This is the honest standing conclusion: **a rigorous, stable map of exactly why a universal theory is open and what would close it — converging on analytical saturation, with the object-level question handed to experiment.**

## Iteration 5 update — analytical saturation reached (2026-06-08)

*The closing analytical push (four web-verified attack/route tracks + synthesis; see notes/ (repo:* `notes/`*; see the notes index, repo:* `notes/README.md`*) iter5-*) executed the one move iteration 4 designated as the only verdict-flipper — and **it did not flip**. Two of three named decidable problems closed as no-gos/bounds, the third sharpened to a located boundary; the program reached the **analytical-saturation** condition. Fourth consecutive iteration at PARTIAL / encodes-not-generates / not-yet-physics. The standing verdict is now consolidated in the capstone **Chapter 38** **(p. 705)**.*

- **The verdict-flipper was executed and did not flip.** `[INFERENCE, high]` The OP-46 non-CKV *Lorentzian* readout (modular-Berry / kinematic-space) produced a **genuine first** — the first emergent *Lorentzian* ($dS_2$) metric in the survey, crossing the signature barrier the Riemannian Cao–Carroll and Connes-spectral routes fail (Huang–Ma, arXiv:2003.12252) — **but its Lorentzian signature is inherited from the symmetric CFT vacuum, not generated** from non-symmetric data. "Encodes-not-generates" is now **readout-independent**: it closes at the *same signature-installation step* across all three known readouts (stress-tensor-flux/Sorce–CKV, modular-Berry/zero-mode, Lorentzian-NCG/η-foliation-twist). → *[notes* (Working note: *OP-46: emergent LORENTZIAN geometry from a non-CKV modular flow? (the verdict-flipping* *move)* (repo: `notes/2026-06-08-iter5-op46-lorentzian-nonckv-readout.md`))]

- **OP-48(c) settled as a clean conditional no-go.** `[ESTABLISHED/INFERENCE]` On a single Type III1 factor, "collar-free + state-independent" composition and "singles out QM" are mutually exclusive: the genuinely collar-free surrogate (Haagerup standard form + Connes fusion) is type-agnostic and a *universal embezzler* ($\kappa_{\max} = 0$, van Luijk et al.); any QM-selecting axiom forces a Type I factor (the split construction), re-importing the collar + reference state. The QM-selecting composition axiom **coincides** with the geometry-importing input. → *[notes* (Working note: *OP-48 point (c): a state-/collar-free composition surrogate on Type III1?* (repo: `notes/2026-06-08-iter5-op48c-collarfree-composition.md`))]
- **OP-44/OP-CP1 sharpened to a regime boundary.** `[INFERENCE, high]` The crossed-product Type II structure survives as an approximate, QRF-relational leading-order truncation **below** $N_* \sim R/2GE_f$; **at/above** it, Giddings' algebraic spacetime disruption degenerates the inclusion structure. So "Type III1 for any Λ" is a robust *kinematic/leading-order* input but **not a fixed nonperturbative core** — with no theorem on either side (the sole residual formally-open lever). → *[notes* (Working note: *OP-44/OP-CP1: nonperturbative fate of the crossed-product Type II structure* (repo: `notes/2026-06-08-iter5-op44-nonperturbative-crossed-product.md`))]
- **Both surviving dS routes fail.** `[INFERENCE, high]` The dS/CFT pseudo-entropy first law evades the cardinality facet but inherits **non-unitarity** (complex central charge $c = i\,3R/2G$); Jacobson-in-CLPW merely re-imports $\eta = 1/4G$ (the CLPW II1 trace encodes one area datum, not a local continuum). Unifying root (HYP-dS-CARDINALITY-R1): **the area-entropy coefficient is an INPUT, not an output, in every known route.** New decidable sub-problem: OP-dS-PSEUDO-REALITY. → *[notes* (Working note: *The two dS routes surviving the*

*cardinality no-go: pseudo-entropy and Jacobson-in-CLPW* (repo: `notes/2026-06-08-iter5-surviving-dS-routes.md`))]

**Saturation declared.** `[INFERENCE, high]` Iteration 5 produced zero new positive structure and zero verdict movement — only sharpening, closure, and one symmetry-bound positive. Per the iteration-4 criterion the analytical program is **saturated**. The standing conclusion, its two explicit hedges (OP-46 residual MEDIUM; OP-48c horn-C INFERENCE), and the residual watch-list now live in ***Chapter 38 (p. 705)***; the empirical channels gating the *object-level* question are tracked in ***Chapter 39 (p. 725)***. The object-level question remains **not forecastable** — the wager has no distinguishing test even in principle.

## Iteration 6 update (2026-06-10) — new-input iteration

*The first post-saturation iteration: not further reasoning over the saturated map, but a deliberate injection of* ***new input*** *— 6 web-verified research tracks + 4 adversarial attack tracks (22 agents, all refereed; ~90 findings kept, zero fatal; see notes/ (repo:* `notes/`*; see the notes index, repo:* `notes/README.md`*) iter6-*). The standing verdict is* ***UNCHANGED at the headline level for the fifth consecutive iteration*** *— PARTIAL coherence; Program A encodes-not-generates; the wager not-yet-physics; object-level not forecastable — but the hedge structure underneath it was genuinely restructured: one hedge shrank, one residual lever closed negative, one new named hedge was added, and the central obstruction broadened from three readout families to four.*

- **The no-test claim is now a corollary, not an observation (ENCODING-SCREEN) — and a third hedge is added.** `[INFERENCE]` A direct, literature-wide attempt to overturn CONCLUSION §7's sharpest sentence ("no distinguishing experimental test even in principle") failed and *strengthened* it: if Program A is encode-only in the OP-46 sense, its observational content equals that of the EFT whose background it consumed, so every candidate signal (VZ/GQuEST geontropic noise, DSW horizon decoherence, everpresent-Λ w(z), trapped-ion Bisognano–Wichmann) factors through physics equally available to geometry-first descriptions. A class-separating test would require a "reverse Weinberg–Witten" theorem ("observable X is realizable *only* without fundamental metric degrees of freedom") — no theorem of that shape exists. The §8 analytical reopening condition and the missing experiment are thereby **the same obstruction** (one-way: a generative bridge is a *prerequisite for*, not a guarantee of, a class-separating observable — the referee weakened the Corollary's "iff" to "only if"). Because the screen inherits the OP-46 encode-only classification, the no-test claim now carries the same MEDIUM hedge — recorded as HYP-ENCODING-SCREEN, the wiki's **third named hedge**. → *[notes* (Working note: *Iteration 6, Track A1 — A Distinguishing Prediction for the Geometry-from-Algebra Wager: Found or Refused* (repo: `notes/2026-06-10-iter6-distinguishing-prediction-hunt.md`))]

- **The obstruction broadens to a FOURTH readout family, and a new smuggling pattern is named.** `[INFERENCE, high]` Causal fermion systems — the most serious candidate previously absent from the wiki — install the Lorentzian datum at their *first axiom*: the $(\leq n, \leq n)$-eigenvalue condition on the operator class is a necessary carrier of the signature (an elementary spectral lemma, derived and numerically re-checked by finder and referee, shows the lightlike/null-cone relation is empty for positive-semidefinite operators) — the Krein η relocated into the definition; all worked examples are Dirac-sea encodes on a pre-given Minkowski background, with minimizer existence proven only for finite-dimensional $\mathcal{H}$. Separately, a genuinely new **selection-type** pattern surfaced (Sung-Sik Lee's matrix model, JHEP 06 (2020) 070, arXiv:1912.12291; a 2025 classical signature-potential toy model): signature appears as a state-dependent dynamical *output*, but on a posited carrier with a hand-installed signature-valued constraint-algebra template — these generate only the *choice*, not the possibility space. Result: HYP-CKV-VACUITY-R3 — every known Lorentzian-output route is installation-type (symmetric state; indefinite-pairing axiom) or selection-type (signature-valued template); the smuggle list gains two named items, and the OP-46 open target compresses to its minimal form: **derive the indefinite pairing (Krein η / (n,n) operator class) from modular data**. CONCLUSION §8's "none exists or has been sketched" gets a precision repair (Lee 2020 is the nearest sketch; it fails the full spec). → *[notes* (Working note: *OP-46 attack, iteration 6: the generation hunt — causal fermion systems, signature-selection models, and the broadened smuggling list* (repo: `notes/2026-06-10-iter6-generation-construction-survey.md`))]

- **OP-44's interior is restructured — still no theorem at $N_*$, but both sides hardened and a fourth option appeared.** `[ESTABLISHED (cited theorems) / INFERENCE (bearing)]` Jensen–Raju–Speranza (arXiv:2412.21185) prove the AdS boundary time-band algebra strictly has **no commutant** nonperturbatively (a theorem-grade boundary-anchored instance of the failure side) *and* a rigidity theorem (the modular crossed product is the only crossed product of a III1 algebra with a trace — the survival side has a unique landing spot within the crossed-product class). De Vuyst–Eccles–Höhn–Kirklin (arXiv:2411.19931) locate the order-$G_N^2$ proof obligation at the **quadratic (boost) linearization-instability constraints** — cleanest for spatially closed (dS-like) universes. KLS II (arXiv:2601.07910) and Chandrasekaran–Flanagan (arXiv:2601.07915) make the survival side theorem-grade at linearized order for the *complete* constraint family, including a GSL for non-stationary perturbations and a proof of the quantum focusing conjecture in the perturbative regime. Kudler-Flam–Witten (arXiv:2510.06376) add a published fourth verdict option: the nonperturbative algebra may exist **only after averaging over** $N$. Herderschee–Kudler-Flam (arXiv:2510.01556) realize the iter-5 "split-property downside" — but by stringy (Hagedorn) physics at sub-string-length separations, not by $O(\kappa)$ dressing; every published post-breakdown candidate remains *algebraic*, weakening (not refuting) Giddings' "beyond algebras" horn. The $N_*$ boundary itself is unmoved. → *[notes* (Working note: *OP-44 & OP-48(c) iteration-6 attack: theorem-shaped brackets on the crossed-product fate, and a hardened horn C* (repo: `notes/2026-06-10-iter6-op44-op48c-levers.md`)), *notes* (Working note: *OP-44 literature delta (2024–2026): the survival side hardens perturbatively; the failure side turns out algebraic; no theorem on either side of $N_*$* (repo: `notes/2026-06-10-iter6-algebraic-frontier.md`))]

- **OP-48(c) horn C is PARTIALLY DISCHARGED — the hedge shrinks but does not vanish.** `[INFERENCE, high – theorem-shaped under the full-pair reading]` Under the GPT-faithful (full-pair, normal-product-state) reading of "recovers QM," the "must interpose Type I" step is now theorem-shaped: a normal product state across a commuting pair forces a Type I interpolating factor (GNS factorization + the established split-inclusion equivalences, anchored to Doplicher–Longo 1984 / D'Antoni–Longo 1983 / Summers 1990), so on the Haag-dual pair $(\mathcal{M}, \mathcal{M}')$ exact QM-type composition holds iff $\mathcal{M}$ is Type I; in the approximate reading, every normal state of a III1 pair sits at uniform norm distance $\geq 1 - 1/\sqrt{2}$ from every product/separable state (Summers–Werner + Tsirelson). The residue relocates from operator algebra to (a) the GPT formalization choice (full-pair vs subsystem reading) and (b) the now-sharpened alternative: commuting III1 factors support their own consistent multipartite composition theory via universal embezzlement (Quantum 9, 1818 (2025)) — a proven *spectrum* of composition regimes with two physically motivated endpoints, which algebra classifies but cannot choose. The choice is a physical/axiomatic input — encodes-not-generates recurring at the composition layer. → *[notes* (Working note: *OP-44 & OP-48(c) iteration-6 attack: theorem-shaped brackets on the crossed-product fate, and a hardened horn C* (repo: `notes/2026-06-10-iter6-op44-op48c-levers.md`))]
- **OP-49 closes to NEAR-NO-GO (conditional exclusion, resolved-negative).**

`[INFERENCE, high, with named residuals]` In the dS/CFT pseudo-entropy first law (arXiv:2511.07915), the entire region- and state-dependent — hence the entire linearized-Einstein — content is carried by the **imaginary part** (the real part is the region-independent constant $S_{\text{dS}}/2$; Doi et al., PRL 130, 031601). Reality therefore forces triviality: any protocol making the pseudo-entropy real either nulls the probe ($\delta\langle T_{tt}\rangle \equiv 0$, no Einstein constraint) or deforms the states/dictionary back to a real-Euclidean/AdS-type (encodes) computation — grounded in the verified Guo–He–Zhang pseudo-Hermiticity reality theorem (JHEP 05 (2023) 021) and mirrored by the unitary-AdS timelike-EE first law (arXiv:2511.17098), where the roles of Re/Im invert exactly under $c \to ic$. Named residuals: exceptional-point (non-diagonalizable) transition matrices; an SVD-entropy first law (with a holomorphy-breaking argument against it — conjecture, decidable by a stated computation); openness of the horn-(ii) protocol class. House language mirrors OP-41: near-no-go. **Correction absorbed:** the iter-5 parenthetical had the dS Re/Im split backwards — the divergent, region-dependent part is *imaginary*; the finite universal part is *real* (R6A4-F2). → *[notes* (Working note: *OP-49 resolved-negative (conditional): reality of the dS/CFT pseudo-entropy excludes nontrivial first-law content* (repo: `notes/2026-06-10-iter6-op49-pseudo-entropy-reality.md`))]

- **The closed-universe Hilbert-space debate is mapped — new problem OP-50 `[CONTESTED]`.** `[ESTABLISHED that the debate exists; the premise CONTESTED]` A 2024–2026 literature (Usatyuk–Wang–Zhao, SciPost Phys. 17, 051 (2024); Harlow–Usatyuk–Zhao, arXiv:2501.02359; Nomura–Ugajin) argues dim $\mathcal{H}_{\text{closed}} = 1$ per α-sector, with standard QM recovered only relative to an internal observer of effective dimension $\sim e^{S_{Ob}}$ — an independent, path-integral route to the same "QM-selection requires an observer-anchored factorization" structure OP-48(c) derived algebraically. The one-dimensionality premise is actively contested (Antonini–Rath–Sasieta–Swingle–Vilar López; Hong Liu's observer-free subregion-subalgebra framing; Engelhardt–Gesteau) — a structural *analogy* that reinforces, but cannot harden, horn (C). QRF results (DEHK PW=CLPW) show even entropy *values* are observer-relative; Maldacena's observer fixes the dS sphere partition function's *phase* but not its coefficient; Zigdon's perturbative-string result suggests the candidate microscopic theory produces no $A/4G$ saddle at all — **η input, or absent, never output** (HYP-dS-CARDINALITY-R1 extended). → *[notes* (Working note: *Iteration 6 — de Sitter and closed-universe quantum gravity: the 2023–2026 literature delta* (repo: `notes/2026-06-10-iter6-closed-universe-and-dS-holography.md`))]

- **Amplitudes scope update — H5 confidence raised, the gravity bound confirmed structural.** `[ESTABLISHED/INFERENCE]` Surfaceology extends amplitudes-from-combinatorics beyond planarity (all genus), beyond SUSY (non-SUSY gluons in any D, pions, colored fermions), and partially beyond integrands — for colored, massless theories; Rodina (PRL 134, 031601 (2025)) proves hidden zeros uniquely fix $\mathrm{Tr}(\varphi^3)$ trees *within a local ansatz* (locality is a hypothesis, not derived — referee correction); cosmohedra package the full $\mathrm{Tr}(\varphi^3)$ cosmological wavefunction. The decisive bound stands: **no positive geometry / surface formalism for gravity** (no color, no surface), and every construction consumes flat-space/FRW kinematic data as input — encode-direction evidence. H5: SPECULATIVE, confidence low → low-to-medium, scope rewritten. → *[notes* (Working note: *Amplitudes and Positive Geometry 2023–2026: Surfaceology, Cosmohedra, Hidden Zeros, and the Gravity Bound* (repo: `notes/2026-06-10-iter6-amplitudes-positive-geometry.md`))]
- **Landscape: two additions and six stale-fixes.** `[ESTABLISHED as published programs]` New entries: **Oppenheim's postquantum classical–quantum gravity** (PRX 13, 041040 (2023); theorem-level decoherence–diffusion trade-off, Nat. Commun. 14, 7910 (2023); already squeezed ≥15 orders by tabletop data, PRResearch 6, 033076 (2024)) and the **swampland dark-dimension scenario** (micron-scale inverse-square-law falsifier; a claimed O(1)-parameter DESI fit

`[CONTESTED]`). Both *do* carry distinguishing tests — demonstrating the wiki's anti-crank standard is passable in this field, which sharpens by contrast the central wager's not-yet-physics demotion. Stale-fixes: causal-set Hauptvermutung (Müller 2025 preprint splits it into variants with claimed truth values); everpresent Λ CMB-disfavored in current implementations + no published DESI confrontation (an identified gap); GFT condensate mean-field-justified (PRL 130, 141501 (2023)); first CDT–FRG quantitative bridge; spin foams numerically Lorentzian; LQG quantum-OS phenomenology. → *[notes* (Working note: *Iteration 6 — R4: The Other Background-Independent Programs, 2023–2026 (Landscape Refresh)* (repo: `notes/2026-06-10-iter6-background-independent-programs.md`))]

- **Phenomenology: GQuEST is the first funded Program-A-adjacent experiment — and still not a wager test.** `[ESTABLISHED (chain) / INFERENCE (taxonomy)]` The Verlinde–Zurek geontropic program and GQuEST (PRX 15, 011034 (2025); under construction at Caltech) is the first experiment probing a conjecture *derived in* the modular/causal-diamond vocabulary the wager privileges — but its chain re-imports $\eta = 1/4G$ at the modular→metric step (extending HYP-dS-CARDINALITY-R1's pattern to a phenomenological chain), and it is decision-grade only for [minimal graviton-EFT IR noise $\delta L^2 \sim \ell_p^2$, Carney et al. PRD 113, 106002 (2026)] vs [exotic Planck-normalized UV-IR mixing $\delta L^2 \sim \ell_p L/4\pi$]. Meanwhile the CQ gate may close **from the classical side**: ultralocal-continuous dead, ultralocal-discrete below the experimental upper bound (with stated caveats), non-local continuous surviving by ~1 order — possibly before BMV ever runs (no funded BMV roadmap exists). DSW horizon decoherence is real but deflated (Biggs–Maldacena, arXiv:2405.02227: ordinary thermal matter does the same — thermality, not horizon-ness). → *[notes* (Working note: *Quantum-gravity phenomenology 2022–2026: the distinguishing-prediction hunt, audited* (repo: `notes/2026-06-10-iter6-qg-phenomenology.md`))]

- **Experiment refresh: every standing number re-verified; four rows change.** `[ESTABLISHED]` g–2 final + WP2025 hold; DESI DR2 (2.8–4.2σ) still current and the **w(z) decision point compresses to 2027** (DESI full 5-yr analysis; Euclid DR1 Oct 2026; Roman launch 2026-08-30; LSST underway); S8 substantially eased on the KiDS axis (KiDS-Legacy 0.73σ from Planck — DES Y6 still in tension `[CONTESTED]`) while H0 persists; a new ~4σ LHCb $B^0 \to K^{*0}\mu^+\mu^-$ anomaly (OP-21 updated); Σm_ν presses below the oscillation floor (~3σ effective-negative-mass tension, entangled with the $w_0 w_a$ degeneracy — one ΛCDM stress, not two); matter-wave record 1.7×10⁵ Da at μ=15.5 (strongest macrorealism exclusion to date); GW250114 + GWTC-4.0/5.0 deliver the strongest strong-field GR nulls (area law confirmed, arXiv:2509.08054, PRL 135, 111403; no echoes; Kerr-consistent ringdown); LHAASO pushes linear LIV to $E_{\mathrm{QG},1} > 10\, E_{\mathrm{Pl}}$. None of it touches the wager — exactly as the watchlist caveat predicts. → *[notes* (Working note: *Iteration 6 — Experiment & Anomaly Refresh (mid-2026)* (repo: `notes/2026-06-10-iter6-experiment-refresh.md`))]

**Saturation status after new input.** `[INFERENCE, high]` The saturation diagnosis survives its first genuine stress test: ~90 refereed findings of new 2023–2026 input produced **zero verdict movement at the headline level** (fifth consecutive iteration) — only hedge-level restructuring. The ledger now reads: **three named hedges** (OP-46 residual, MEDIUM, unchanged; OP-48c horn C, *shrunk* — theorem-shaped under the full-pair reading, residual = GPT-formalization choice + the III₁-embezzlement-composition alternative; HYP-ENCODING-SCREEN, *new* — the §7 no-test claim inherits the OP-46 MEDIUM hedge), one residual lever **closed negative** (OP-49 → near-no-go), one restructured but open (OP-44, with a located order-$G_N^2$ obligation and a four-option verdict space), one new contested problem (OP-50), and one compressed verdict-flipper target (derive the indefinite pairing from modular data). The object-level question remains handed to experiment, with the 2027 w(z) verdict the nearest-dated event that could reframe the target. See *Chapter 38 (p. 705)* (amended), *Chapter 39 (p. 725)* (rewritten caveat + four new rows), and *the iteration-6 synthesis* (Chapter 26 (p. 623)).

## Iteration 7 update (2026-06-10) — the decidable levers, attacked

*Iteration 7 executed the four named decidable levers iteration 6 left on the table (the compressed OP-46 target; OP-44's order-$G_N^2$ obligation; OP-49's SVD residual; OP-48c's formalization-choice residue + the OP-50 watch). 9 agents; 4 attack tracks, all adversarially refereed with independent re-derivation, numerical replication, and live citation audits; referee corrections and outcome checks are binding over the original claims. The standing verdict is **UNCHANGED at the headline level for the SIXTH consecutive iteration** — PARTIAL coherence; Program A encodes-not-generates; the wager not-yet-physics; object-level not forecastable. What moved is the hedge ledger: the principal hedge **hardened** (OP-46 MEDIUM → MEDIUM-HIGH, conditional), the last substantive OP-49 residual **closed negative** (conditional), OP-48c's definitional residue **resolved**, and OP-44's obligation was **split into three named theorem-shaped parts** — including the first identified perturbative foothold for the failure side.*

- **The designated verdict-flipper was executed — and resolved NEGATIVE in the most informative way possible: the compressed target is achievable but provably geometry-void.** `[ESTABLISHED (η₀ construction) / INFERENCE (vacuity + trichotomy)]` A canonical Krein fundamental symmetry **is** derivable from pure $(\mathcal{M}, \Omega)$ modular data: $\eta_0 = \mathrm{sgn}(\ln \Delta)$ is a smuggle-free ($g_1$–$g_6$ F, $g_7$ T; $\kappa_\Phi = 0$) $J$-odd self-adjoint unitary in $\{\Delta\}''$, indefinite iff $\omega$ is non-tracial, with Krein signature $(\infty, \infty)$ on any type III factor — *one member of an infinite family of $J$-odd sign functions* $\varepsilon(\Delta)$ (referee construction: $\mathrm{sgn}(\sin \ln \Delta)$; uniqueness requires an added modular-rescaling-invariance axiom — a correction that pushes *negative*). But three vacuity results show it carries **zero geometric information**: (i) in the Borchers/HSMI class — exactly where modular data has causal content — $(H, \Delta, J, \varepsilon(\Delta))$ is unitarily universal (one multiplicity cardinal; $\Delta_{\text{geo}} = 0$*proven*, not graded; substantially the known uniqueness of non-degenerate standard pairs, extended by "any fixed function of Δ rides along"); (ii) generic algebra-incompatibility ($\operatorname{Ad}\varepsilon(\Delta) \notin \operatorname{Aut}(\mathcal{M})$, numerically demonstrated against an $\operatorname{Ad}\Delta^{it}$ control); (iii) non-localizability without a posited carrier. A new **modular-form trichotomy lemma** (ansatz-scoped on its sesquilinear leg, enumerative on its non-sesquilinear leg) closes the J-route generally: every canonical pairing from modular data is positive, signature-free, or sign-of-Δ Krein (indefinite but universal). Bonus structural result (the strongest finding per referee): the $g_3$ trap is **structural** — $\langle a^*\Omega, Ja\Omega\rangle \geq 0$ by self-duality of the natural cone, so the modular conjugation $J$ carries reflection positivity *abstractly* (the Euclidean datum), not merely in Bisognano–Wichmann vacua; no $J$-induced pairing can be secretly Lorentzian. Literature direction-map (negative search, referee-corroborated): all prior art runs Krein → modular (Gottschalk, JMP 43 (2002) 4753; Jakóbczyk 1985, OSTI 5135323; the NCG twist line — Devastato–Farnsworth–Lizzi–Martinetti, JHEP 03 (2018) 089, arXiv:1710.04965), never modular → Krein. **Net:** the OP-46 hedge hardens **MEDIUM → MEDIUM-HIGH (conditional on the modular-constructibility ansatz)**; HYP-ENCODING-SCREEN inherits; the iteration-6 compressed sentence gets a mandatory precision repair — its literal reading is achieved yet flips nothing, so the faithful reopening target is **"derive an algebra-compatible, LOCALIZED (fiberwise) $(n, n)$ pairing from modular data"** — the carrier problem (new named item HYP-FACTORIZATION-IS-GEOMETRY), the **fifth independent route to converge there**. Slogan: *the modular sign is to signature what thermal time is to time.→ [notes* (Working note: *OP-46 attack, iteration 7: η-from-modular-data — the modular sign η0 exists, is geometry-void, and a trichotomy* *lemma closes the J-route* (repo: `notes/2026-06-10-iter7-indefinite-pairing-attempt.md`))]

- **OP-44's order-$G_N^2$ obligation splits three ways (P1a/P1b/P1c) — one part discharged, two located residues, and the first perturbative failure-side foothold.** [SHARPENED-OPEN] (P1a) Quadratic-order descent of the *complete* global-dS constraint set to the CLPW static-patch crossed product is **discharged at inference grade** (conditional on the inherited quantum-imposition hedge, with fixed-point factoriality not separately proven): decompose $\mathfrak{so}(d+1,1) = (\mathbb{R}_B \oplus \mathfrak{so}(d)) \oplus \mathfrak{m}$ relative to the pode observers — the boost constraint *is* the CLPW constraint, the compact $\mathfrak{so}(d)$ twirl preserves Type II (AKL centrality criterion), and the $2d$ coset constraints are solved by the observer worldline acting as a complete dynamical frame; confirmed at the literature level by DEHK §2.5.1 verbatim (citing FJLRW arXiv:2403.11973 and AKL arXiv:2407.01695). This strengthens the survival side below $N_*$ — but only by consuming pre-installed $\mathfrak{so}(d+1,1)$ structure, itself an encoding step (HYP-ENCODING-SCREEN mildly reinforced). (P1b = R1) DEHK footnote 12: classical sufficiency of the Taub conditions *with worldline clock sources* is unchecked, and distributional stress-energy exits the FMM/AMM Sobolev framework — located, decidable `[OPEN]`. (P1c = R2) The cubic constraint **does not split across the horizon**: the one-sided boost-weighted cubic charge ($\Delta_3 = (3d+1)/2$) has vacuum fluctuations $\sim A/\delta^{2(d-1)}$ (vs $A/\delta^{d-1}$ quadratic — the latter reproducing the known area law as a consistency check), so subdominance requires a **Planck-length dressing-collar floor $\delta \gtrsim \ell_P$ in every dimension** — the gravitational parallel of OP-48's stringy $\ell_s$ floor (the iter-4/iter-6 collar-tax leg sharpened from a second independent direction); the relative cocycle is not manifestly affiliated to the CLPW algebra (strictly weaker than provably outer; free-field scaling caveats binding — the exponent could soften). The obstruction horn of R2 is **the first identified perturbative foothold for the failure-side reading of OP-44, at order $\kappa^3$ rather**

**than at $N_*$**; both horns are theorem-shaped (discharge = cubic edge modes, KLS-II-style; obstruction = no affiliated renormalization ⟹ outer derivation). The FMM/AMM classical-exhaustion input (quadratic Taub conditions are the only classical obstruction; Moncrief I/II, FMM I, AMM II) carries `[INFERENCE, medium-high]` for its dS consequence — the $\Lambda > 0$ transfer of the cone theorem is unproven. Literature delta 2024–2026: the nearest discharge machinery exists (Kirklin's all-orders GSL, arXiv:2412.01903, JHEP 07 (2025) 192; Ali Ahmad–Jefferson 2501.01487) but nobody has executed the dS closed-slice cubic step. **The bulk $N_*$ statement remains theorem-free on both sides; nothing here moves it.**→ *[notes* (Working note: *OP-44 iteration-7 attack: the order-$G_N^2$ obligation for the dS static patch — quadratic descent discharged, residue located at footnote 12 and the cubic charge* (repo: `notes/2026-06-10-iter7-op44-orderG2-descent.md`))]

- **OP-49's last substantive residual closed: the SVD first law on the dS3 hemisphere FAILS (resolved-negative, conditional, on referee-narrowed grounds).** `[INFERENCE]` The iteration-6 stated decidable computation was performed. **Unconditional pillar:** $S_{\text{SVD}}$ is real by construction while the verified arXiv:2511.07915 carrier ($\delta\langle K_A\rangle$, imaginary $T_{tt}$) is purely imaginary — the literal first law $\delta S_{\text{SVD}} = \delta\langle K_A\rangle$ fails with no assumptions beyond the paper's own verified content; that alone closes the residual as the iter-6 holomorphy conjecture predicted, now upgraded to an exact operator identity ( $\delta\,\mathrm{Tr}(\mathcal{T}^\dagger\mathcal{T})^m = 2m\,\mathrm{Re}\,\mathrm{Tr}[(\mathcal{T}^\dagger\mathcal{T})^{m-1}\mathcal{T}^\dagger\delta\mathcal{T}]$ — the SVD response is the Re-projection of a $\mathbb{C}$-linear functional). **Conditional mechanism (genuinely novel, not proven):** the SVD replica object is an alternating $c/c^*$ branched cover whose sheetwise-additive twist dimension carries $c + c^* = 0$ — cap-size blindness at the universal level — but exact blindness additionally requires an unproven heterogeneous-cover **junction additivity** at the branch points, where $2m$ interfaces between conjugate theories meet; interface-CFT precedent (Sakai–Satoh-type transmission-dependent effective central charges) shows naive additivity generically fails (four referee majors, all kept with corrections). Either way no local Einstein constraint survives in the SVD channel: a θ0-independent response fails the wave equation and CHM inversion outright, and even a cap-dependent real response cannot equal the imaginary carrier. **OP-49 stays NEAR-NO-GO with the residual list rewritten: (a) exceptional points; (b) beyond-2511.07915-family generalization; (c′) heterogeneous-cover junction additivity** — three items, not two. Byproduct $S_{\mathrm{SVD}} = S_{\mathrm{dS}}/2$ (region-independent horizon constant) **drops to** `[SPECULATIVE]` (the $q = 1$ self-normalization is in tension with the trace-norm inequality given the verified complex pseudo-spectrum); region-independence `[INFERENCE, conditional on (c′)]` — consistent with the encodes pattern: the only intrinsically real two-state entropy retains exactly the horizon constant and zero local geometry. Absence re-confirmed (timestamped search-bounded, 2026-06-10): no SVD-in-dS computation or SVD first law exists in the literature; this computation is new. → *[notes* (Working note: *OP-49 SVD residual closed: the SVD first law fails on the dS*$_3$ *hemisphere because* $c + c^* = 0$ (repo: `notes/2026-06-10-iter7-op49-svd-computation.md`))]

- **OP-48(c)'s definitional residue is RESOLVED: the full-pair reading of "recovers QM" is forced (forced-modulo-rejecting-OP-48's-question, per referee) — and the failure locus relocates to the product-preparation primitive.** `[ESTABLISHED (verbatim axiom texts) / INFERENCE, high (the forcing)]` All three GPT reconstructions — Hardy quant-ph/0101012, CDP arXiv:1011.6451, Masanes–Müller arXiv:1004.1483, axioms checked verbatim against the primary PDFs — quantify composition universally over THE designated pair (no existential quantifier over embedded subfactor pairs exists) and install product states at *framework* level, upstream of local tomography; the subsystem reading is inexpressible as a weakening of any reconstruction axiom and vacuous where expressible — it can only be implemented by deleting the full pair from the system roster, i.e. abandoning OP-48's question. The sharpest new mathematics of the iteration (referee re-derived, correct): **the separation half of local tomography HOLDS on the Type III1 Haag-dual pair** (span$\{ab\colon a \in \mathcal{M}, b \in \mathcal{M}'\}$ is σ-weakly dense, so distinct normal states differ on some product experiment) — the failure locus is *not* tomography but the **normal-product-preparation primitive (PS)**, violated uniformly with the Summers–Werner gap $1 - 1/\sqrt{2}$. **Horn C upgrades to theorem-conditional-on-framework** (a condition shared by Hardy/CDP/MM at framework level, weaker than any single paper's axiom set), conditional exactly on (i) the carried Summers–Werner primary spot-check, (ii) the carried split-equivalence technicalities, (iii) the normal-state postulate. The residue relocates to two named exits: **(X1)** singular (non-normal) product states — existence holds, but by the referee-corrected Schlieder/Roos + min-norm-domination + Hahn–Banach route (the submitted max-tensor universal-property route is mathematically invalid); admitting them abandons the program's own normal-state postulate; **(X2)** the III1/embezzlement composition endpoint, which exits the OPT framework entirely. Both exits leave the framework rather than reinterpret its axioms — the state postulate is an installation choice; encodes-not-generates recurring at the state-postulate layer. **OP-50 watch:** three verified post-2026-01 developments — Harlow–Usatyuk–Zhao published (JHEP 02 (2026) 108); Nomura–Ugajin arXiv:2602.13387; Higginbotham arXiv:2512.17993 (JHEP 03 (2026) 183) — plus a **third position**: Miyata–Horikoshi–Tajima–Ugajin (arXiv:2606.04575), symmetric-orbifold closed-universe sectors whose *pre-gauge-constraint* dominant sector is exponentially large with hyperfinite Type II1 structure and whose *post-constraint* physical Hilbert space grows only polynomially in $N$ (neither strictly 1-dim nor $e^{S_{Ob}}$; cross-feeds OP-44's survival side as a model-specific construction, not a theorem). OP-50 stays `[CONTESTED]`, now three-cornered. Negative sweep (dated process note): no EGH/ARSSV/Liu direct responses post-2026-

01. Still NOT-YET-PHYSICS: no experimental channel. → *[notes* (Working note: *OP-48(c) iteration-7 attack: the full-pair reading is forced — horn C loses its definitional residue; plus the OP-50 watch* (repo: `notes/2026-06-10-iter7-op48c-fullpair-forcing.md`))]

**Saturation status and the hedge ledger after iteration 7.** `[INFERENCE, high]` The saturation diagnosis survives its second stress test — this time against the wiki's *own* designated decidable levers rather than new field input: four refereed tracks, every outcome label upheld by the referee (one conditionally, on narrower grounds), and **zero verdict movement at the headline level — the sixth consecutive iteration**. The ledger now reads: **hedge 1 (OP-46 residual) HARDENED, MEDIUM → MEDIUM-HIGH conditional on the modular-constructibility ansatz** — the J-route is closed by construction-plus-vacuity rather than by absence, and the reopening target re-compresses to the carrier problem, where **five independent routes now close** (the Type III1 no-factorization frontier; the split/collar relocation; the CFS carrier posit; the OP-49 protocol deformations; and now the $\eta_0$ localization step — recorded as HYP-FACTORIZATION-IS-GEOMETRY); **hedge 2 (OP-48c horn C) shrinks again and changes character** — no definitional freedom remains in the reading; the residue is the normal-state postulate plus the composition-primitive choice (X1/X2); **hedge 3 (HYP-ENCODING-SCREEN) inherits the hardened MEDIUM-HIGH grade** with the same ansatz condition attached. One residual lever closed negative-conditional (OP-49's SVD channel, residual list now a/b/c′); one restructured into three theorem-shaped parts with the first failure-side perturbative foothold at order $\kappa^3$ (OP-44, P1a discharged / R1, R2 located); $N_*$ untouched. No experimental channel emerged from any track — NOT-YET-PHYSICS unchanged; the experiment watchlist is untouched and the 2027 w(z) decision point remains the nearest-dated external event. See *Chapter 38 (p. 705)* (amended), *Chapter 15 (p. 233)*, *Chapter 19 (p. 392)*, and *the iteration-7 synthesis* (Chapter 27 (p. 628)).

# Iteration 8 update (2026-06-10)

*Iteration 8 re-attacked the four standing analytical levers from a structurally-richer angle than iteration 7: the carrier/localization no-go was tested against the net/family input it had not yet covered (A1); OP-44's R1/R2 residues were pushed to theorem-shape (A2); the OP-49 cap-blindness mechanism was scrutinized (A3); and the OP-48c X1 exit was promoted to a theorem (A4). 4 attack tracks, all adversarially refereed with independent re-derivation, numerical replication, and live citation audits; referee corrections and outcome checks are **binding over the original claims** (one submitted outcome label was overturned — see A3). The standing verdict is **UNCHANGED at the headline level for the SEVENTH consecutive iteration** — PARTIAL coherence; Program A encodes-not-generates; the wager not-yet-physics; object-level not forecastable. What moved: the principal hedge **hardened further toward HIGH and its condition narrowed**; OP-44's two residues sharpened to theorem-shape on both legs; OP-48c's X1 exit became a theorem-with-a-price-tag; OP-49's (c′) was reframed-and-narrowed but **did not close**.*

- **The carrier no-go is upgraded from a single-algebra to a net/family no-go — and the principal hedge hardens toward HIGH on a NARROWED condition.** [INFERENCE — HIGH as a classification; the closure conditional on the net-constructibility ansatz] The one structurally-richer input iteration 7's single-algebra trichotomy did not reach — net-modular data (relative modular operators $\Delta_{\sigma|\rho}$, families $\{J_O\}$, half-sided modular inclusions, modular intersection, the modular-Berry connection over a net $O \mapsto \mathcal{M}(O)$) — was shown to add strictly more **operators** but no new localized indefinite **form**. The engine is the **covariance–locality obstruction**: a local indefinite multiplication form has a sign locus that the modular flow $p \mapsto e^{-2\pi t}p$ boosts away, so local ⇒ non-covariant ⇒ non-canonical (verified numerically in the chiral standard pair; robust to the $x$-vs-$p$ locality-representation subtlety). Every modular-covariant Hermitian form from net data is therefore (P) positive-and-local, (S) signature-free, or (N) indefinite-but-non-local — never simultaneously indefinite and localized. Localization lives only in the net's parametrization, which **is** the geometry: BGL prove positivity of energy ⇔ isotony; Morinelli–Neeb–Ólafsson take the **Euler element as an input** defining wedge localization; Lashkari–Leung–Moosa–Ouseph (JHEP 10 (2025) 153) state that *"$AdS_2$ emerges from insisting on a local representation of the boundary algebras that satisfy Haag's duality."* **Hedge move:** OP-46 / HYP-CKV-VACUITY hardens from MEDIUM-HIGH (conditional on the single-algebra ansatz) **toward HIGH, now conditional on a single net-constructibility ansatz** — the condition is narrower, with two named obligations outstanding before HIGH is unconditional (replace the ansatz by a naturality theorem; rule out a modular-Berry-holonomy-natural localized indefinite form). **Referee-binding count correction:** this is the **net-level extension of the fifth route's boost-covariance engine to the family input** — a *strengthening of route (5),* **not** a sixth independent route. The carrier-convergence count **stays at five** (HYP-FACTORIZATION-IS-GEOMETRY); the net upgrade and its net-index-set smuggle are folded into route (5). HYP-ENCODING-SCREEN inherits the hardened grade with the same condition. → *[notes* (Working note: *OP-46 / carrier-problem attack, iteration 8: the net upgrade — net-modular data adds operators, not a localized indefinite form; the parametrization is the geometry* (repo: `notes/2026-06-10-iter8-net-carrier-no-go.md`))]

- **OP-44's R1 resolves to a theorem-shaped failure-side result with a renormalization escape; R2's obstruction horn is reinforced.** [SHARPENED-OPEN] **(R1)** For a distributional worldline-clock source the second-order Taub charge $Q_2$ **diverges** ($\varepsilon^{2-d}$, $\ln\varepsilon$ for $d=2$): the FMM/AMM cone theorems and Hintz's solvability theorem (arXiv:2306.07715, Thm 1.1: $f \in C^{\infty}_{\mathrm{sc}}$, divergence-free) are proven only for smooth, spatially compactly-supported sources, so DEHK footnote 12's clock-sufficiency **cannot be discharged in the naive**

**distributional form** (Pound 1506.06245, verbatim: $\delta^2 G[h_1, h_1] \sim 1/r^4$, non-integrable; "there exists no solution to the original, fully nonlinear equation with a point particle source"). `[ESTABLISHED computation]`**Escape (corrected mechanism):** the divergence is a pure self-energy absorbed by **Gralla–Wald / Pound puncture (effective-source) regularization** — *not* an EM-style scalar mass renormalization, which at second order is only one component of the singular-field subtraction — and the Taub charge of the finite **regular** field is read off; sufficiency holds **iff** the regularized charge satisfies the FMM cone condition (a conditional, not an instance; the regularization supplies the clock's structure as INPUT — HYP-ENCODING-SCREEN reinforced). `[INFERENCE, medium-high]`**(R2)** All three proposed cancellations of the cubic $\delta^{-2(d-1)}$ divergence fail at leading order: smearing only relocates the cutoff $\delta \to w$ (the fixed-dimension composite $\mathcal{O}_3$ keeps its intrinsic short-distance power; the explicit $w^{d-2\Delta_3}$ figure is a separate bulk-OPE power-counting estimate, not the term-by-term continuation of the shell exponent — referee flag); **KLS-II edge modes are provably quadratic-order only** (arXiv:2601.07910: $X = \delta^2 \mathcal{Q}^R / 4G_N\beta$; "addresses only second-order perturbations") and structurally cannot reach the cubic charge; normal-ordering touches only the lower-dimension self-contraction descendants, leaving the fully-connected $\langle :\mathcal{O}_3 :: \mathcal{O}_3 : \rangle_c$ channel invariant. The R2 obstruction horn (the first perturbative failure-side foothold, order $\kappa^3$) is reinforced as an **INFERENCE**, with the constrained-graviton tensor structure / boost-weighted transverse contraction the **sole surviving escape**. The $\delta \gtrsim \ell_P$ collar floor and the $N_*$ boundary are untouched. Verdict unchanged; encodes-not-generates reinforced. → *[notes* (Working note: *OP-44 iteration-8: the two located residues executed — R1 (worldline-clock Taub finiteness) resolves two-horn with a renormalization condition; R2 (cubic non-splittability) reinforced against smearing, edge modes, and normal-ordering* (repo: `notes/2026-06-10-iter8-op44-R1-R2.md`))]

- **OP-49's (c′) is reframed and narrowed — but does NOT close; OP-49 stays NEAR-NO-GO.** `[INFERENCE, medium]` The bra/ket "interface" of the dS SVD cover was argued to be the transmissive orientation-reversal (permutation-diagonal) defect — a dimension-zero junction operator giving twist dimension exactly $\propto m(c + c^*) = 0$, hence cap-blindness with no transmission-dependent correction. **The submitted RESOLVED-POSITIVE label was overturned by the referee** (UPHELD-IN-DIRECTION-WITH-MAJOR-CORRECTION): the transmissive-junction mechanism is valid at real-$c$ orientation reversal but **extrapolated to the non-unitary imaginary-$c$ regime** where transmission coefficients violate the unitary bounds and the Karch–Kusuki–Ooguri–Sun–Wang machinery (which assumes SSA, the $c$-theorem, positive $c$) does not apply — by the track's own admission "not rigorously excluded." So (c′) is **REFRAMED-AND-NARROWED to the complex-$c$ extension, NOT retired**: OP-49 keeps residuals **(a) exceptional points, (b) beyond-2511.07915-family, (c′) heterogeneous-cover junction additivity (narrowed)**. The **unconditional pillar is independently re-verified** and untouched by (c′): $S_{\mathrm{SVD}} \in \mathbb{R}_{\geq 0}$ by construction while the verified carrier $\delta\langle K_A\rangle \in i\mathbb{R} \setminus \{0\}$, so the literal SVD first law fails with no assumptions beyond the paper's own content. Cap-blindness remains a **conditional** secondary observation, not a near-unconditional second pillar; no CONCLUSION hedge loosens or tightens. (Citation fix carried: arXiv:0705.3129 is Fuchs–Gaberdiel–Runkel–Schweigert, not "Bachas et al."; the transmissive-defect criterion is Bachas–Brunner–Roggenkamp.) → *[notes* (Working note: *OP-49 residual (c′) closed: the dS SVD bra/ket junction is the transmissive orientation-reversal defect, so cap-blindness is exact* (repo: `notes/2026-06-10-iter8-op49-junction-additivity.md`))]
- **OP-48c's X1 exit is now a theorem-with-a-price-tag; horn C stays theorem-conditional-on-framework.** `[INFERENCE, high]` **X1-EXISTENCE (theorem):** a singular (non-normal) product state $\omega(ab) = \varphi_A(a)\varphi_B(b)$ across the type

III1 Haag-dual pair $(\mathcal{M}, \mathcal{M}')$ exists — chain (every link live-verified): $\mathcal{M}$ a factor ⇒ the Schlieder property (central-carrier fact) ⇒ C*-independence (Halvorson) ⇒ a product extension on $C^*(\mathcal{M} \cup \mathcal{M}')$ (Roos 1970, via Rédei–Summers) ⇒ a state on $B(\mathcal{H})$ by positive Hahn–Banach; and **every** such state is non-normal because $\mathcal{M} \vee \mathcal{M}' = B(\mathcal{H})$ is type I while $\mathcal{M} \overline{\otimes} \mathcal{M}'$ is type III, so the **Takesaki** W*-isomorphism characterizing normal product states fails (ref [30]=Takesaki 1958, **not** Florig–Summers — citation corrected). **X1-DISQUALIFICATION (sharper than posed):***no* reconstruction axiom "provably fails" on a singular state — Hardy/CDP/Masanes–Müller all quantify only over finite-dim, $\sigma$-additive (normal/density-matrix) states, so a singular state is **not an element of any framework's state space**; disqualification sits upstream of every axiom. Hence X1 does **not** collapse horn C to "theorem-conditional-only-on-X2"; horn C **stays theorem-conditional-on-framework**, and X1 is confirmed a **costed framework EXIT** whose price is now a theorem (**X1-COST**): admitting singular product states relinquishes density-matrix states + $\sigma$-additivity + finite tomographic dimension **jointly**, each constitutive of all three GPT reconstructions. The normal/density-matrix state postulate is thereby proven a load-bearing, non-derivable installation choice — encodes-not-generates recurring at the state-postulate layer, now theorem-grade. (Asymmetry recorded: the *separation* half of local tomography HOLDS on the III1 pair, so the exit is at the preparability/$\sigma$-additivity/finiteness layer, not at distinguishability. Citation fix: "normal states weak-* dense" is the standard density of vector states, not Fell's theorem.) X2 (the non-OPT III1-embezzlement composition endpoint) untouched. No experimental channel; NOT-YET-PHYSICS unaffected. → *[notes* (Working note: *OP-48(c) EXIT X1: singular product states on the III1 Haag-dual pair exist (theorem) and constitute a framework exit, not a closable residue* (repo: `notes/2026-06-10-iter8-op48c-X1-singular-states.md`))]

**Saturation status and the hedge ledger after iteration 8.** `[INFERENCE, high]` The saturation diagnosis survives its third stress test — this time against the structurally-richer net/family inputs the single-algebra levers had not covered: four refereed tracks, **zero verdict movement at the headline level — the seventh consecutive iteration** — with one submitted outcome label (A3 RESOLVED-POSITIVE) caught and downgraded by the referee, evidence the discipline is binding rather than rubber-stamping. The ledger now reads: **hedge 1 (OP-46 residual) HARDENS further toward HIGH on a NARROWED condition** — the no-go is upgraded from single-algebra to net/family, conditional now on a single net-constructibility ansatz (two named obligations outstanding); the carrier-convergence count **stays at five** (the net upgrade strengthens route (5), it is not a sixth route); **hedge 3 (HYP-ENCODING-SCREEN) inherits** the hardened grade and condition. OP-44 sharpened on both legs to theorem-shape (R1 a regularization-conditional failure-side result; R2's obstruction horn reinforced with one named escape) — $N_*$ and the $\ell_P$ floor untouched. OP-48c's X1 exit promoted to a theorem-with-a-price-tag; horn C unchanged (theorem-conditional-on-framework). OP-49's (c′) reframed-and-narrowed but **still standing**; OP-49 stays NEAR-NO-GO; no hedge moved from A3. No experimental channel emerged from any track — NOT-YET-PHYSICS unchanged; the watchlist is untouched and the 2027 w(z) decision point remains the nearest-dated external event. See *Chapter 38 (p. 705)* (amended), *Chapter 15 (p. 233)*, *Chapter 19 (p. 392)*, and *the iteration-8 synthesis* (Chapter 28 (p. 634)).

## Distinguishing-test hunt (2026-06-10) — the 2024–2026 experimental/phenomenology literature, swept

*A targeted hunt, not a full iteration: three adversarially-refereed tracks (H1 literature sweep; H2 reverse-Weinberg–Witten / net-invariant analysis; H3 deep-read of the verified lead). No headline movement (8th confirmation). The screen is reinforced with positive evidence rather than re-hedged.*

- **No in-principle distinguishing test exists in the 2024–2026 literature.** `[INFERENCE, high]` The verified lead — Sharmila–Vermeulen–Datta, *Nat. Commun.* **17**, 701 (2026), arXiv:2505.22892 — resolves *negatively*: its three-class fluctuation taxonomy classifies by the two-point correlation function (mechanism-blind by construction), and its own model table is many-to-one — geontropic Verlinde–Zurek (emergent) and Károlyházy (fundamental-metric stochastic) co-inhabit class (b); Oppenheim CQ and CSL co-inhabit class (a) — so no class is realizable only by a non-fundamental-metric model. Verbatim, the authors are *"agnostic to the microscopic origins of the SFs."* This is the cleanest published worked instance of HYP-ENCODING-SCREEN Corollary 2 (UV-portability). Two emergent-spacetime programs concede observational degeneracy with fundamental-metric physics in their own words (Sidan A & Banks, arXiv:2502.15108, predictions *"fit extant data equally well"*; Erlich, arXiv:2504.03084, low- $\ell$ effect *"not unique"*). → *[notes* (Working note: *Distinguishing-Test Hunt (2026-06-10) — An In-Principle Test of the Geometry-from-Algebra Wager in the 2024–2026 Literature: Found or Refused* (repo: `notes/2026-06-10-distinguishing-test-hunt.md`))]
- **Every global/topological/boundary observable that could carry priority information is a net invariant.** `[INFERENCE – conditional on net-sharing A1, the reopening lever]` Theta-angle / global form (Casini–Magán DHR-for-higher-form-symmetries, arXiv:2511.21810, verified), 't Hooft anomaly (IR-matched + net-invariant), soft hair / memory (IR sector of the graviton EFT), topological entanglement entropy (DHR total quantum dimension), modular Berry phase (Tomita–Takesaki-built), indefinite causal order (reduces to definite acyclic order, Vilasini–Renner) — each is shared between algebra-first and geometry-first descriptions of the same physics. **Binding scope:** the corollary "therefore no reverse-Weinberg–Witten observable exists" is a *literature-absence* claim, tag `[OPEN, as of 2026-06]`, NOT a structural impossibility — it restates the screen under the unproven assumption A1, so it cannot

upgrade absence to a theorem. The reverse-WW target is **unregistered** (it is *not* OP-50, which is the closed-universe Hilbert-space problem); it lives in HYP-ENCODING-SCREEN's portability corollary.

- **Net effect on the hedge ledger: zero.** No hedge created, none removed; the §7 no-test corollary keeps its OP-46 MEDIUM-HIGH grade and modular-constructibility condition unchanged. The hunt converted a survey-scoped portability inference into a *positively evidenced* one (a peer-reviewed mechanism-blind taxonomy + two author-side degeneracy concessions) without moving the verdict.

## Iteration 9 update (2026-06-10)

*Iteration 9 attacked the carrier no-go's residual naturality content directly and chased the one uncovered escape from the iteration-8 net/family no-go. Four adversarially-refereed tracks (A1: is the net-constructibility ansatz replaceable by a naturality theorem; A2: does the modular-Berry-holonomy escape produce a localized indefinite form; A3: is reverse-Weinberg–Witten independently provable or a free witness; A4: does the constrained-graviton TT tensor structure cancel OP-44 R2's cubic horizon divergence). Referee corrections and outcome checks are **binding over the original claims** — one submitted outcome label was overturned (A4: RESOLVED-NEGATIVE → SHARPENED-OPEN) and one verdict-bearing overclaim was struck (A2's "unconditional HIGH"). Every load-bearing citation re-verified live; key mathematics independently re-derived; numerics independently replicated. The standing verdict is **UNCHANGED at the headline level for the EIGHTH consecutive iteration** — PARTIAL coherence; Program A encodes-not-generates; the wager not-yet-physics; object-level not forecastable. What moved: the principal hedge's GRADE is unchanged (MEDIUM-HIGH→HIGH conditional) but its CONDITION is **sharpened** — the net-constructibility ansatz is replaced by a naturality formulation whose constructibility half is a theorem-modulo-a-located-gap and whose localization half is proven irreducible; one* *of the two iteration-8 named obligations (the modular-Berry-holonomy escape) is* ***discharged RESOLVED-NEGATIVE****; the reverse-WW target is pinned to the §8 reopening condition; and OP-44 R2's failure-side foothold is* ***de-hardened*** *after a fatal computational error in the submitted graviton cancellation.*

- **The carrier no-go's constructibility ansatz is replaced by a naturality THEOREM-modulo-a-located-gap; the localization half is proven irreducible; the hedge sharpens in precision, not grade.** `[INFERENCE — HIGH as a classification; closure conditional on the naturality formulation + a located commutant-to-generated gap]` Under the BFV/GMA naturality formulation, the Gram operator of an $\mathrm{Aut}(\mathcal{N}, \Omega)$-natural modular-covariant carrier is **forced** (not assumed) into the commutant of $\{\Delta_O^{it}\}$ by C2-equivariance, and the GMA reconstruction (Buchholz–Dreyer–Florig–Summers math-ph/9805026; Summers–White hep-th/0304179) makes the geometry a functor of the modular data — so iteration-8's flagged constructibility *ansatz* is discharged to **Lemma N1**. **Binding referee correction:** naturality forces $\eta$ into the *commutant* of $\Delta$, not into the *abelian algebra generated by* $\Delta$; these are unequal in the generic Type-III degenerate-spectrum case, so Lemma N1 **weakens but does not eliminate** the constructibility ansatz — a commutant-to-generated gap survives. The boost-covariance engine is promoted to a **naturality obstruction (Lemma N2)**: a localized indefinite natural form has a natural sign locus $\Lambda$ carried by the modular flow ($p \mapsto e^{-2\pi t}p$, Borchers), with no fixed locus in the open index set, so the only flow-invariant sign pattern is the global constant (definite) — localized + natural + indefinite is impossible (numerically reconfirmed: $\|m_{\mathrm{boost}} - m\|_\infty = 2$ for local sign forms; constant form boost-invariant; sgn(modular-frequency) natural but nonlocal). The **irreducible residual**: the predicate "localized" is defined relative to the index set $\mathcal I · *Chapter 39 (p. 725)*
- *Chapter 1 (p. 1)* — project overview and full index
- *Chapter 3 (p. 15)* — the map of frameworks and their reductions

- *Chapter 14 (p. 191) · Chapter 15 (p. 233) · Chapter 20 (p. 458)*
- *Chapter 17 (p. 325) · Chapter 18 (p. 349) · Chapter 19 (p. 392)*
- *Chapter 16 (p. 308) · Chapter 2 (p. 8) · Chapter 40 (p. 739)* · the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835))


## References

See the Bibliography (p. 776). This synthesis rests on the sources cited across the domain and analysis pages; no claim here is asserted beyond what those pages support. s region order, a base functor that is part of the SOURCE datum — so "localized" presupposes $n_1$, which (BGL; Neeb–Ólafsson Euler element; Lashkari–Leung–Moosa–Ouseph) IS the geometry. **Hedge move:** OP-46 / HYP-CKV-VACUITY-R5 → **-R6**, grade **unchanged at MEDIUM-HIGH→HIGH conditional**, condition **sharpened** from an unproven von-Neumann-membership ansatz to "conditional on the very index-set order the program seeks to derive" + the located commutant-to-generated gap. The larger-structure loophole closes in the wrong direction: a carrier natural under the subfactor standard invariant / planar algebra cannot generate signature (a rigid C*-tensor category has positive hom-space inner products by axiom; enlarging the symmetry shrinks the candidate pool). **Count correction (binding):** this is the promotion of route (5)'s boost-covariance engine from ansatz-scoped to naturality-scoped — the **same engine**, not a new route; the carrier-convergence count **stays at five**. (math 0304340 is Bisch, not Jones — citation fix.) → *[notes* (Working note: *OP-46 / carrier obligation (i), iteration 9: the naturality promotion — constructibility becomes a theorem, localization is the irreducible residual (and is the geometry)* (repo: `notes/2026-06-10-iter9-carrier-naturality.md`))]

- **The modular-Berry-holonomy escape closes RESOLVED-NEGATIVE — one of the two iteration-8 named obligations is discharged.** [INFERENCE] The single grounds. **(a)** The modular-Berry curvature $F = P_0([X_1, X_2])$ is **antisymmetric** — a commutator whose boundary expectation equals the Kirillov–Kostant–Souriau **symplectic** form on a coadjoint orbit of the conformal group $SO(d, 2)$ (the Crofton form), with symplectic-type holonomy that preserves a **signature-free** form, not a signature-$(n, n)$ symmetric pairing; it falls in the already-closed signature-free leg of the net-data dichotomy (verified in 2305.16384, 1812.02176, 2505.04682; robust to the Virasoro state-changing generalization 2111.05345). **(d)** The only indefinite (Lorentzian $dS_2$) object is the **kinematic-space metric**, which carries the Lorentzian signature via an embedded Minkowski $\eta_{\mu\nu}$ and whose signature comes from the conformal symmetry of the modular Hamiltonian for a spherical entangling surface, computed **on the vacuum** (Huang–Ma 2110.08703) — symmetry/vacuum-inherited from the $SO(d, 2)$ coadjoint orbit, not generated by the connection (the iteration-5 closure HYP-BERRY-LORENTZIAN-ONLY-FROM-SYMMETRY). **Binding referee corrections:** the indefinite object is the kinematic-space metric (distinct from the positive-definite QFI/Fubini–Study metric on state space — the original "symmetric part = positive-definite QFI" characterization is wrong); the claimed third ground (b: "type-$III_1$ centralizers trivial") is FALSE and contradicted by 2505.04682 (type-$III_1$ factors generically carry states with non-trivial modular zero modes) — but (b) is not needed, (a) and (d) each suffice. The closure is naturality-by-strong-argument, not naturality-by-theorem; the residual is the *general* program residual (the net-naturality theorem, obligation i), not Berry-specific — so the Berry-specific obligation (ii) is **discharged**. The carrier-convergence count **stays at five** (the Berry case folds into route (5) + the iteration-5 symmetry-inheritance closure, not a sixth route). The submission's "unconditional HIGH" was a **binding overclaim and is struck**: the hedge stays MEDIUM-HIGH→HIGH CONDITIONAL on the net-constructibility ansatz, exactly the iteration-8 status, with only obligation (i) now between HIGH and a theorem. → *[notes* (Working note: *Carrier obligation (ii): the modular-Berry-*

*holonomy escape closes — the holonomy-natural form is symplectic (signature-free) or symmetry-inherited, never a localized indefinite carrier* (repo: `notes/2026-06-10-iter9-modular-berry-holonomy.md`))]

- **Reverse-Weinberg–Witten is the empirical-side shadow of the carrier problem, not an independent target.** `[INFERENCE, high as a structural identification, conditional on the carrier ansatz]` A1 ("every observable is an invariant of the GNS pair, net + vacuum"), read in its load-bearing form, is a **CONDITIONAL theorem**: by Weiner's algebraic Haag theorem (arXiv:1006.4726, verified verbatim), under the split property and geometric modular action the unitary-equivalence class of the vacuum-sector net of double-cone algebras on a fixed Cauchy surface **completely determines the AQFT model** — without assuming the connecting unitary intertwines vacua or symmetry reps. A1-*abstract* (over single abstract algebras) is vacuous (Connes–Haagerup: every local algebra is *the* unique hyperfinite type $III_1$ factor; Buchholz's "monads") — the load-bearing object is A1-*concrete*, the GNS pair $(\{\mathcal{A}(O)\}, \omega_0)$ with modular data. Charged-sector and global/topological candidates are net invariants too (Doplicher–Roberts reconstructs the superselection category and field net from the vacuum-sector net in $D > 2$; theta-angle / global form / line operators are DHR / Haag-duality-violation invariants, Casini–Magán 2511.21810) — the H2 closure re-used, not relitigated. **The located gap:** Weiner's hypotheses ARE the geometry — GMA demands the modular data act geometrically on a pre-given causal index set (Euler element as input where posited; where *derived*, 2312.12182, it is from net + Bisognano–Wichmann + a regularity condition, all of which already encode the causal/Lie structure); the causal index set is $n_1$ (BGL: positivity-of-energy $\Leftrightarrow$ isotony); split is the OP-48 collar. So A1-concrete is proved only "relative to a net whose geometry is already installed." **Central identification:** A1 unconditional $\iff$ the carrier problem has no positive solution $\iff$ no reverse-WW witness — proving any one is proving all three. The reverse-WW *witness* is **strictly**

**harder** than a positive carrier solution (carrier necessary but not proven sufficient for a class-separating observable). This pins the standing reverse-WW target (`[OPEN, as of 2026-06]`) to the existing §8 reopening condition (HYP-FACTORIZATION-IS-GEOMETRY) rather than leaving it free-floating; it does not flip the headline, add a route (count stays FIVE), move a hedge, or invoke OP-50 (reverse-WW remains un-registered; OP-50 is canonically the closed-universe Hilbert-space problem). HYP-ENCODING-SCREEN keeps its MEDIUM-HIGH→HIGH (net-constructibility-conditional) grade. (GMA paper is the four-author Buchholz–Dreyer–Florig–Summers; "Buchholz–Summers" is acceptable shorthand but incomplete.) → *[notes* (Working note: *The Reverse-Weinberg–Witten Target Is the Carrier Problem Seen From the Empirical Side (2026-06-10)* (repo: `notes/2026-06-10-iter9-reverse-weinberg-witten.md`))]

- **OP-44 R2's graviton escape stays OPEN; the failure-side foothold is de-hardened after a fatal computational error.** `[SHARPENED-OPEN]` The submitted claim that the constrained-graviton (transverse-traceless) tensor structure leaves OP-44 R2's order-$\kappa^3$ cubic boost-charge horizon divergence robust (closing the escape on the FAILURE side) is **OVERTURNED**. Two findings survive: **F1** — the boost weight $K = \int X^u T_{uu}$ multiplies the $uu$-stress component by a *scalar* weight $X^u(y)$ carrying no transverse indices, so it cannot itself drive a polarization cancellation (scope-limited; does not by itself close the escape); **F3** — the requested $d = 2$ ($dS_2$/Rindler$_2$) case is **degenerate** (the 3D graviton has $D(D-3)/2 = 0$ propagating polarizations), so the cubic charge vanishes by *absence of modes*, not by tensor cancellation (uninformative for the escape). The load-bearing **F2/F4 are FATAL**: the decisive contraction $\Pi_{ij,kl}\hat{n}^i\hat{n}^j\hat{n}^k\hat{n}^l$ was claimed $= (d-2)/(d-1) > 0$ but is actually $= 0$ in every dimension — the transverse projector annihilates its own axis ($P_{ij}\hat{n}^j = 0$), so the TT structure **annihilates** the naive leading $\delta^{-2(d-1)}$ all-radial term rather than leaving it robust. The surviving same-power coefficient needs the full-Hessian contraction (not performed); it is nonzero for transverse dim $\geq 3$ but **zero for transverse dim 2** (the central $D = 4/d = 3$ codim-2 case). **R2's escape is therefore NOT closed on the failure side**; it stays OPEN with the residual coefficient uncomputed and dimension-dependent. Consequence: the $\ell_P$ collar floor / first perturbative failure-side foothold **reverts from** `[INFERENCE, medium-high]` **to** `[INFERENCE, medium]`. Headline untouched; carrier count stays FIVE; $N_*$ and the rest of OP-44 untouched; no headline flip (the submission's self-limiting posture was the one correct part). → *[notes* (Working note: *OP-44 R2 closed on the failure side: the constrained-graviton tensor structure does not cancel the cubic boost-charge horizon divergence* (repo: `notes/2026-06-10-iter9-op44-graviton-escape.md`))]

**Saturation status and the hedge ledger after iteration 9.** `[INFERENCE, high]` The saturation diagnosis survives its fourth lever-execution stress test — this time against the carrier no-go's *residual naturality content* and its single uncovered escape: four refereed tracks, **zero verdict movement at the headline level — the eighth consecutive iteration** — with one submitted outcome label overturned (A4) and one verdict-bearing overclaim struck (A2), evidence the discipline is binding. The ledger now reads: **hedge 1 (OP-46 residual) — grade UNCHANGED (MEDIUM-HIGH→HIGH conditional), condition SHARPENED** (the ansatz is replaced by a naturality formulation: constructibility a theorem-modulo-a-located-commutant-to-generated-gap; localization proven irreducible = the index-set order = $n_1$); one of its two iteration-8 named obligations (the modular-Berry-holonomy escape) **discharged RESOLVED-NEGATIVE**, leaving the net-naturality theorem (plus the new commutant-to-generated gap) as the residual between HIGH and a theorem; the carrier-convergence count **stays at five** (A1, A2, A3 all fold into route (5)). **Hedge 3 (HYP-ENCODING-SCREEN) inherits** the unchanged grade and sharpened condition; the reverse-WW target is now **pinned** to the §8 reopening condition rather than free-floating. OP-44 R2's failure-side foothold **de-hardened** (medium-high → medium) after the graviton cancellation was shown fatally wrong; the escape stays OPEN; $N_*$ and the $\ell_P$ floor's *in-principle* status untouched. No experimental channel emerged from any track — NOT-YET-PHYSICS unchanged; the watchlist is untouched and the 2027 $w(z)$ decision point remains the nearest-dated external event. See *Chapter 38 (p. 705)* (amended), *Chapter 15 (p. 233)*, *Chapter 19 (p. 392)*, and *the iteration-9 synthesis* (Chapter 29 (p. 636)).

## Iteration 10 update (2026-06-10)

*Iteration 10 attacked the two named obligations the iteration-9 hedge left between HIGH and a theorem — the* ***commutant-to-generated gap*** *that iteration 9 located inside Lemma N1 (A1, the designated verdict-bearing lever: close the gap and push the principal hedge to unconditional HIGH), and whether a carrier could live in that gap (A2) — plus the one decidable OP-44 R2 sub-computation iteration 9 left uncomputed (A3: the surviving same-power coefficient of the cubic boost-charge horizon divergence at the physical D=4). Three adversarially-refereed tracks; referee corrections and outcome_checks are* ***binding over the original claims****. Two of the three submitted outcome labels were corrected by the referee — A1's claimed condition-shedding was struck as a major error, and A3's submitted RESOLVED-NEGATIVE was refuted back to SHARPENED-OPEN — evidence the discipline remains binding rather than rubber-stamping. Every load-bearing citation re-verified live; key mathematics independently re-derived; numerics independently replicated (seed 20260610). The standing verdict is* ***UNCHANGED at the headline level for the NINTH consecutive iteration*** *— PARTIAL coherence; Program A encodes-not-generates; the wager not-yet-physics; object-level not forecastable. What moved this iteration:* ***nothing in grade or condition.*** *The principal hedge stays exactly at its iteration-9 R6 status; the designated verdict-flipper (close the commutant-to-generated gap) was attacked head-on and did* ***not*** *close.*

**hedge to unconditional HIGH — was executed and did NOT fire: the gap does not close, the hedge sheds no condition.** `[INFERENCE — verdict-neutral null result]` The attack's verdict-bearing tail is sound and uncontested (PARTIAL unchanged; carrier-convergence count FIVE; no localized indefinite centralizer element exists, so no carrier could live in the gap; the reverse-Weinberg–Witten ⇔ carrier-no-go identification and the encoding-screen inherit unchanged). But its **centerpiece is a major error and is struck**: the claim that in the Bisognano–Wichmann case — the only case where the boost engine operates — the vacuum's ergodicity (Borchers' Lebesgue boost spectrum; Longo 1978) makes the modular centralizer trivial, $\mathcal{M}_\omega = \mathbb{C}$, and thereby **empties** the iteration-9 commutant-to-generated gap, repairing the Lemma-N1 deficiency. The centralizer $\mathcal{M}_\omega = \mathcal{M} \cap \{\Delta^{it}\}'$ is the **wrong object**: the iteration-9 gap is the *representation-space* multiplicity gap $\langle\Delta\rangle'' \subsetneq \{\Delta^{it}\}'$ driven by spectral degeneracy of $\Delta$, and since $\Delta \notin \mathcal{M}$, ergodicity of $\mathcal{M}_\omega$ does not empty it — the gap remains non-empty even in the BW case and contains the canonical indefinite $\eta_0 = \mathrm{sgn}(\ln\Delta)$. **Net:** the commutant-to-generated gap does **not** close; the iteration-9 Lemma-N1 deficiency is **not** repaired; HYP-CKV-VACUITY stays at **R6** — conditional HIGH on $n_1$**and** the located commutant-to-generated gap — it does **not** advance to "$n_1$ alone." Two genuine sub-results are kept and re-establish (not advance) the standing picture: the modular centralizer of a Type III1 factor is genuinely large and the gap is real and wide for degenerate spectrum (integrable weight ⟹ Type $\mathrm{II}_\infty$ centralizer, Connes–Takesaki flow of weights; Popa-type realization: any finite $(N,\tau)$ occurs as a centralizer, arXiv:math/0011084), and the state restricts to a faithful normal **trace** on its centralizer (Takesaki), so the centralizer is semifinite and any indefinite element it carries is trace-balanced — signed dimension, no localization. The geometric-vs-degenerate exclusion (a geometric modular flow keeps a non-trivial centralizer only via an internal gauge multiplet, which by DHR commutes with the boost and carries zero spacetime localization) re-establishes the inherited Lemma-N2/$n_1$ obstruction. → *[notes* (Working note: *OP-46 / carrier obligation (i), iteration 10: the net-naturality theorem (negative horn) — the closure attempt was STRUCK (the gap does NOT close); hedge stays at R6* (repo: `notes/2026-06-10-iter10-naturality-commutant-gap.md`))]

- **No carrier lives in the gap: the gap is geometry-void on both sides, but by the inherited $n_1$/algebra-incompatibility obstruction, not by ergodicity — RESOLVED-NEGATIVE.** [INFERENCE] Where geometry lives (geometric/Bisognano–Wichmann vacua: homogeneous **Lebesgue** modular spectrum, no modular eigenvectors), the *centralizer* is trivial $\mathcal{M} \cap \{\Delta^{it}\}' = \mathbb{C}$, so no algebra-internal carrier exists in the gap (this is the centralizer-triviality fact for Lebesgue-spectrum free Araki–Woods factors, traced to Shlyakhtenko's free-quasi-free-states structure theory; Houdayer arXiv:0809.3827 proves the *bicentralizer* trivial and Boutonnet–Houdayer arXiv:1602.01741 place amenable modular-invariant subalgebras in the almost-periodic summand — both corroborate but neither *is* the biconditional). Where the gap is non-empty (almost-periodic / discrete-spectrum states) a self-adjoint centralizer element $h \notin W^*(\Delta)$ exists whose sign is genuinely indefinite (signature $(8, 8, 0)$ in a $4 \times 4$ toy, $\Delta$ multiplicities $(1, 4, 6, 4, 1)$) and, because $\mathrm{sgn}(h) \in \mathcal{M}$, algebra-compatible by an **inner** automorphism — clearing the two bars ((i) indefinite and (iv) algebra-compatible) that iteration-7's $\eta_0$ failed, the strongest carrier candidate the program has produced — yet it remains **geometry-void**: $\Delta_{\text{geo}} = 0$. **Two binding referee corrections to the load-bearing sub-arguments** (the vacuity survives both): the complete $\mathrm{Aut}(\mathcal{M}, \omega)$-invariant of a centralizer sign-involution is **not** the global signature pair $(p, q)$ (the toy has 4 distinct conjugacy classes of signature-$(8, 8)$ involutions, not 1; $M_n$ has only inner automorphisms by Skolem–Noether) but the **finer signature resolved per modular eigenspace** — still a discrete, input-indexed, choice-dependent sign pattern carrying zero metric/causal content; and the auxiliary "centralizer element is boost-orbit-constant hence wedge-filling" mechanism is **struck** as sector-inconsistent (boost-sweeps-the-wedge requires geometric flow, but $h$ exists only in the almost-periodic sector where $\sigma_t$ is not a boost), the localization-failure surviving via the inherited mechanism that localization itself presupposes the net's index-set order $n_1$ (iteration-9 localization-irreducibility). **Combined with A1:** the commutant-to-generated residual is **substantially closed for algebra-internal (centralizer, Ad-inner) carriers** but a residual sliver — algebra-compatible carriers with $\mathrm{Ad}\,\eta \in \mathrm{Aut}(\mathcal{M})$ yet $\eta \notin \mathcal{M}$, living in $\{\Delta^{it}\}' \setminus \mathcal{M}$ — is **not** discharged, so the hedge narrows substantially toward but **not provably to** $n_1$ **alone**; grade and condition unchanged. Carrier-convergence count stays FIVE (centralizer-completion of route (5), not a sixth route). → *[notes* (Working note: *OP-46 / carrier obligation, iteration 10: the positive dual executed — the commutant-to-generated gap is geometry-void too (centralizer carrier sgn(h) is empty where geometric, universal-and-non-local where it exists)* (repo: `notes/2026-06-10-iter10-carrier-in-the-gap.md`))]

- **OP-44 R2's decidable D=4 coefficient does NOT resolve negative: the cubic horizon coefficient VANISHES at the physical D=4 under the projector the coincidence limit selects — R2 stays SHARPENED-OPEN, the iteration-9 posture upheld.** `[SHARPENED-OPEN]` The submitted attempt to re-harden the R2 obstruction at the physical $D = 4$ ( $m = D - 2 = 2$) by replacing iteration-9's radial-subtracting graviton projector with a symmetric-traceless-on-full-cut projector $\Pi^{ST}$ (no within-cut radial subtraction) is **refuted on both the physics and the arithmetic**. (Physics) The standard graviton TT projector subtracts the propagation direction $\hat{k}$; the leading horizon divergence is a coincidence-limit object dominated by $\hat{k} \parallel \hat{n}$ (the within-cut separation), so the physically-correct projector **is** radial-subtracting (the iteration-9 $\Pi^{\mathrm{rad}}$), under which the all-radial term is annihilated in every $D$ and the full Hessian vanishes at $m = 2$; the submission's reductio ("$\Pi^{\mathrm{rad}}$ gives 0 polarizations at $D = 4$, forbidding gravitational waves") is a category error — the trace of the cut-restricted, coincidence-limit contraction is not the free-graviton polarization count. (Arithmetic, projector-independent) Even granting $\Pi^{ST}$, the submitted Hessian closed form is wrong; the exact contraction's genuine leading $s^2$ coefficient is $(m-1)(m-2)/2$, which **vanishes at** $m=2$, reproducing — not refuting — the iteration-9 finding (only an $s^4/2$ subleading remnant survives at $m=2$). The $\ell_P$ collar floor in 4D is therefore **not** re-established; OP-44 R2's failure-side foothold stays `[INFERENCE, medium]` and SHARPENED-OPEN; the carrier-convergence count stays FIVE; the OP-46 principal hedge is untouched by this track. → *[notes* (Working note: *OP-44 R2 D4 coefficient: the surviving cubic-divergence coefficient is nonzero at the physical D=4 — the iteration-9 transverse-dim-2 vanishing used the wrong graviton projector; the ell_P collar floor stands* (repo: `notes/2026-06-10-iter10-op44-R2-D4-coefficient.md`))]

**Saturation status and the hedge ledger after iteration 10.** `[INFERENCE, high]` The saturation diagnosis survives its fifth lever-execution stress test — this time against the **designated verdict-bearing lever itself** (close the commutant-to-generated gap and push the principal hedge to unconditional HIGH): three refereed tracks, **zero verdict movement at the headline level — the ninth consecutive iteration** — with two of three submitted outcome labels corrected by the referee (A1's condition-shedding struck as a major error; A3's RESOLVED-NEGATIVE refuted to SHARPENED-OPEN), evidence the discipline is binding. The ledger now reads: **hedge 1 (OP-46 residual) — grade AND condition UNCHANGED** at the iteration-9 R6 status (MEDIUM-HIGH→HIGH conditional on the net-naturality theorem **plus** the located commutant-to-generated gap); the gap did not close, the hedge shed no condition, no carrier lives in the gap. The one positive movement available — narrowing toward $n_1$ alone — is real but partial: the commutant residual is substantially closed for algebra-internal (Ad-inner) carriers, with a residual sliver (carriers in $\{\Delta^{it}\}' \setminus \mathcal{M}$) undischarged. **Hedge 3 (HYP-ENCODING-SCREEN) inherits** the unchanged grade and condition; the reverse-WW identification stays pinned to the §8 reopening condition. OP-44 R2's foothold stays `[INFERENCE, medium]` and SHARPENED-OPEN; the $\ell_P$ floor in 4D is not re-established; $N_*$ untouched. No experimental channel emerged from any track — NOT-YET-PHYSICS unchanged; the watchlist is untouched and the 2027 $w(z)$ decision point remains the nearest-dated external event. See *Chapter 38 (p. 705)* (amended), *Chapter 15 (p. 233)*, *Chapter 19 (p. 392)*, and *the iteration-10 synthesis* (Chapter 30 (p. 641)).

## Iteration 11 update (2026-06-10)

*Iteration 11 tested whether the standing verdict could move from three directions: A1 — does the carrier no-go reduce to a named, externally-tracked theorem (closing it) or re-ground on a named open problem (cleaning the hedge); A2 — do three independent handles on "is net+vacuum a complete invariant" produce a referee-UPHELD reverse-Weinberg–Witten witness (a headline FLIP); A3 — a terminal-saturation meta-audit asking whether any purely-reasoning lever capable of moving the verdict or the hedge still remains. Three adversarially-refereed tracks; referee corrections and outcome_checks are* ***binding over the original claims****. A1's centerpiece was MAJOR-CORRECTED (the claimed logical-equivalence reduction downgraded). Every load-bearing citation re-verified live; the keystone new datum (arXiv:2512.18912, Geng–Jafferis–Shrivastava–Tata, 21 Dec 2025) is genuine and its quoted phrases exact. The standing verdict is* ***UNCHANGED at the headline level for the TENTH consecutive iteration*** *— PARTIAL coherence; Program A encodes-not-generates; the wager not-yet-physics; object-level not forecastable. What moved this iteration:* ***nothing in grade, condition, or count.*** *The principal hedge's principal residual re-grounds its vocabulary on a named open problem (cleaner, not closed); the carrier-convergence count stays at FIVE.*

- **A1 — the carrier no-go's inner horn RE-GROUNDS on Connes' bicentralizer problem (a NAMED open problem), it does NOT close to a proved theorem; the**

**mission's "resolved-by-Houdayer–Marrakchi–Tomatsu" premise is FALSE.**

`[INFERENCE — HIGH classification; the re-grounding cleaner-not-closed]`

The algebra-internal (inner, J∈M) carrier horn lives in the modular centralizer $M_\psi = \mathcal{M} \cap \{\Delta_\psi^{it}\}'$, so centralizer/bicentralizer structure theory is the correct vocabulary for the located commutant-gap residual that iterations 9–10 carried. **Binding referee correction (the centerpiece is MAJOR-CORRECTED):** the submitted headline "the carrier no-go REDUCES (logical equivalence) to Connes' bicentralizer problem" does **not** hold and is **downgraded from "REDUCES / logical equivalence" to "is RELATED TO centralizer/bicentralizer structure theory."** The spine conflated two *opposite* conditions — trivial centralizer ($M_\psi = \mathbb{C}1$, which kills a carrier and is **unconditionally achievable on every III$_1$ factor**, Marrakchi–Vaes, Crelle 809 (2024) 247, Thm A) vs Haagerup's bicentralizer-triviality condition, which yields an *irreducible/large*-centralizer state in which a carrier **exists** (the opposite of what closing the no-go needs). So the relation to Connes' (genuinely open) problem is a loose structural connection, not an equivalence; the inner horn stays closed by the inherited geometry-void/non-localizability facts, not by the bicentralizer conjecture. **Second binding correction:** the mission's premise that Connes' bicentralizer problem was "RESOLVED in general by Houdayer–Marrakchi–Tomatsu" is **FALSE** — the conjecture is **OPEN in general as of 2026** (only large classes — amenable, free Araki–Woods/no-eigenvector quasi-free, Cartan, free products, semisolid, *q*-deformed, tensor-III$_1$ — and equivalent reformulations are settled; Marrakchi 2308.15163 solves Kadison's problem only for III$_1$ factors *satisfying* the conjecture). The open status cites those sources, **not** Marrakchi–Vaes (whose Thm A is unconditional). **Net:** the principal hedge's commutant-gap residual re-grounds on a named, externally-tracked open problem — strictly cleaner and more honest, with an external resolution clock — but the grade is unchanged (MEDIUM-

HIGH→HIGH conditional), the registry condition string is unchanged, and the no-go does **not** close to a proved theorem. Carrier-convergence count stays at **FIVE** (this is route (5)'s reduction-to-known-problem framing, not a sixth route). → *[notes* (Working note: *OP-46 / carrier obligation, iteration 11 (track A1): the carrier no-go's inner horn is RELATED TO Connes' bicentralizer problem (structural connect...* (repo: `notes/2026-06-10-iter11-carrier-bicentralizer.md`))]

- **A2 — three independent handles on net-completeness all land back on the carrier gate; NO reverse-WW witness, NO headline FLIP.** `[INFERENCE, high]` (i) **Holography of information** (AdS/Wheeler–DeWitt arXiv:2107.14802; null infinity arXiv:2002.02448; de Sitter arXiv:2303.16316) is a STATE-completeness theorem *within one fixed gravitational theory*, conditional on the gravitational Gauss law / diffeomorphism constraints — i.e. it presupposes the geometry; it does not address the theory-level reverse-WW question and corroborates rather than circumvents the iteration-9 RWW≡carrier identity. (ii) Its **Dec-2025 failure mode** (Geng–Jafferis–Shrivastava–Tata, *The Fate of Information Localizability and Holography in Quantum Gravity*, arXiv:2512.18912, 21 Dec 2025): *"holography cannot be proven at the perturbative level in Newton's constant $G_N$ via the non-localizability of information,"* via *"the emergence of a perturbatively localized observer"* — the mechanism (per the paper's body, binding referee correction) is information **re-localizing** when bulk operators are gravitationally dressed to a *nontrivial background distribution* or to an emergent localized observer/clock (the Hamiltonian constraint satisfied *by construction via dressing*, NOT evaded by geometry going away). An installed localized observer is exactly the kind of carrier the wiki's carrier problem concerns — so completeness co-locates with the carrier gate. (iii) **T-duality/mirror symmetry** is an exact CFT isomorphism between geometrically distinct targets, establishing that net+vacuum does NOT determine the target *geometry* — but class-symmetrically (the two geometries are the *same physical theory*, no class-separating observable), so it embarrasses geometry-first Program G (two operationally identical metrics) and *supports* the emergent-geometry ontology while producing no reverse-WW witness. (iv) *MIP=RE / Connes embedding** (arXiv:2001.04383, $C_{qa} \subsetneq C_{qc}$) underdetermines the algebra in the *wrong place* — the split/tensor (Type-I) structure, the OP-48 collar smuggle-item — not net-completeness; it confirms A1-abstract vacuity and sharpens OP-48 but supplies no independent handle on reverse-WW. **Net:** the iteration-9 RWW≡carrier identity is independently re-grounded from a third direction; no half yields a witness; reverse-WW stays a literature-absence `[OPEN, as of 2026-06]`, NOT an impossibility, NOT OP-50; count stays FIVE; no hedge moves. One new sub-question is recorded (NQ-1: does the holographic-completeness *failure* regime coincide with carrier existence?), un-IDed, inside HYP-ENCODING-SCREEN. → *[notes* (Working note: *Net + Vacuum as a Complete Invariant: Three Independent Handles, One Carrier Gate (2026-06-10)* (repo: `notes/2026-06-10-iter11-net-completeness-handles.md`))]

- **A3 — terminal analytical saturation CONFIRMED (referee-UPHELD); every verdict-bearing residual is decidable only by external input.** `[INFERENCE, high]` Of the seven residuals the wiki lists as OPEN/decidable (carrier net-naturality theorem; commutant-to-generated gap; OP-44 R2 D=4 coefficient/$N_*$ boundary; OP-49 a/b/c′; OP-48c X2; OP-50; reverse-WW), **none is decidable by further reasoning in a way that could move the verdict or the hedge.** Two (OP-44 R2 higher-D coefficient; OP-49 residuals) are reasoning-decidable but **headline-inert by their own scoping**; the other five are decidable only by external input — four by the *same* math breakthrough (the carrier net-naturality theorem ⇔ commutant-gap closure ⇔ reverse-WW witness ⇔ HYP-CKV-VACUITY-as-unconditional-HIGH), and OP-50 by a community resolution that even settled cannot harden the headline. The one dated experimental input (2027 DESI five-year $w(z)$) bears on the target, not the wager. The marginal-yield trend is **monotone to zero**: new positive structure zero since iteration 2; hedge-*grade* movement present at iterations 7–8, absent at 9–11; hedge-*condition* (registry) movement present at 7–9, absent at 10–11. The first derivative (grade) reached zero at iteration 9, the second (condition) at iteration 10; iteration 11 added a *precision* re-grounding of the hedge's vocabulary onto a named external problem but minted **no R-advance**. A targeted live web search (2026-06) for a modular-data carrier, a reverse-WW distinguishing observable, and a 2025–2026 net-naturality/GMA theorem returned **no decidable external lever**; the nearest emergent-metric constructions (arXiv:2504.06698; arXiv:2602.11806) produce AdS/Riemannian-corner output from symmetric/background-presupposing input — the exact encode-pattern HYP-ENCODING-SCREEN predicts. **Recommendation: TERMINAL SATURATION — consolidate + watch-mode; no genuine remaining decidable verdict-moving iteration exists.**→ *[notes* (Working note: *Terminal-saturation audit — is the analytical program done?* (repo: `notes/2026-06-10-iter11-terminal-saturation-audit.md`))]

**Saturation status and the hedge ledger after iteration 11.** `[INFERENCE, high]` The saturation diagnosis survives its sixth lever-execution stress test and, for the first time, a dedicated *meta-audit* asking whether anything could still move the verdict by reasoning: three refereed tracks, **zero verdict movement at the headline level — the tenth consecutive confirmation** — with A1's centerpiece MAJOR-CORRECTED (the claimed bicentralizer reduction downgraded), evidence the discipline remains binding. The ledger now reads: **hedge 1 (OP-46 residual) — grade AND registry-condition UNCHANGED** at the iteration-9/10 R6 status (MEDIUM-HIGH→HIGH conditional on the net-naturality theorem **plus** the located commutant-to-generated gap); the gap's residual now has a cleaner *external* name — it relates to centralizer/bicentralizer structure theory and, through it, to Connes' bicentralizer problem (OPEN in general as of 2026) — but the no-go does **not** close to a theorem and no R-advance is minted. **Hedge 3 (HYP-ENCODING-SCREEN) inherits** the unchanged grade and condition; the reverse-WW identification stays pinned to the §8 reopening condition, re-grounded from a third (holography/T-duality/Tsirelson) direction, still `[OPEN, as of 2026-06]`, still NOT OP-50. OP-44 R2's foothold and $N_*$ are untouched (no A3-track re-attack). No experimental channel emerged from any track — NOT-YET-PHYSICS unchanged; the watchlist is untouched and the 2027 $w(z)$ decision point remains the nearest-dated external event. **The A3 audit's verdict — terminal analytical saturation, only external input can now move the headline — is the README-sanctioned obtained-outcome and governs the consolidation decision.** See *Chapter 38 (p. 705)* (amended, capstone note added), *Chapter 15 (p. 233)*, *Chapter 19 (p. 392)*, and *the iteration-11 synthesis* (Chapter 31 (p. 645)).

## Iteration 12 update (2026-06-19)

**Verdict unchanged — ELEVENTH consecutive confirmation.** PARTIAL coherence / Program A encodes-not-generates geometry / wager not-yet-physics / object-level not forecastable. Principal hedge HYP-CKV-VACUITY stays at -R6 in grade (MEDIUM-HIGH->HIGH conditional) AND condition (net-naturality theorem + located commutant-to-generated gap). Carrier-convergence count stays **FIVE**. No new numeric registry ID. Three adversarially-refereed tracks (B1, B2, B3), all UPHELD-WITH-CORRECTION, all verdict-neutral.

- **B1 (physics-net bicentralizer lever) — RESOLVED-NEGATIVE-AS-A-LEVER / OA-QUESTION-OPEN.** `[INFERENCE, high]` The actual gating object for a physics net is the **relative** bicentralizer of the irreducible, infinite-index, **non-discrete** local inclusion M(O) subset M(W) — and AHHM's relative-bicentralizer theorem (Ando-Haagerup-Houdayer-Marrakchi, Math. Ann. 376 (2020) 1145) is restricted to **discrete** inclusions with N amenable, so the triviality question is GENUINELY OPEN as pure operator algebra. The injective-III_1 single-algebra fact (B(M(O),phi) = C1 by Haagerup) is real but addresses the wrong object. Decisively, the lever is NON-LOAD-BEARING: closing the inner carrier horn needs a TRIVIAL CENTRALIZER M_psi = C1, which is unconditional on every III_1 factor (Marrakchi-Vaes, Crelle 809 (2024) 247, Thm A, verbatim-verified, no bicentralizer hypothesis) — so the inner horn does not wait on any (relative-)bicentralizer fact; and TRIVIAL BICENTRALIZER <=> exists a state with LARGE/irreducible centralizer (Haagerup biconditional, Acta Math. 158 (1987) 95), which is carrier-ADMITTING. Trivial (relative) bicentralizer is therefore the carrier-FRIENDLY condition and points the WRONG way for the no-go. The no-go for physics nets stands on the inherited bars (iter-7 eta_0 vacuity; **iter-9 boost-no-fixed-locus engine — the actual BW-net-closing mechanism**; iter-10 commutant-gap null). Iter-11 A1 §7.1 subquestion #1 is thereby **retired as a lever**, one of iteration 11's three open subquestions discharged. Citation-precision repair recorded: the iter-11 claim that Marrakchi-Vaes Thm A is "conditional on the bicentralizer conjecture" is corrected — Thm A is UNCONDITIONAL.

- **B2 (2023-2026 external-math sweep) — CONFIRMS-SATURATION.** `[INFERENCE, high]` A web-verified sweep found no result closing, refuting, or directly addressing the net-naturality theorem or the located commutant-to-generated gap. Every cited paper either presupposes geometry as input (Morinelli-Neeb-Olafsson Euler-element program arXiv:2508.10960 / 2412.20410 [single-author Morinelli] / 2511.09360 / 2603.26390; Liu lecture notes arXiv:2510.07017; Weiner arXiv:1006.4726 conditional on split + GMA), or is orthogonal (Naaijkens-Penneys-Wallick arXiv:2605.10693 — an LTO-internal Euclidean-side reflection-positivity axiom; it constructs no Lorentzian metric, no indefinite pairing, no carrier witness, and makes **no** claim that the modular conjugation J carries the Euclidean datum [the iter-7 J-as-RP-carrier gloss is trimmed]), or re-confirms an open external problem without bearing on the carrier reduction (Ando-Goldbring arXiv:2605.12776 — Connes' bicentralizer OPEN in general). The residual stays OPEN as iterations 10 and 11 left it.

- **B3 (mid-2026 external-input sweep) — CONFIRMS-SATURATION.** `[INFERENCE, high]` No new decidable verdict-moving lever. The one genuinely-new 2026 carrier-adjacent mathematics — the Morinelli-Neeb-Olafsson Euler-pair geometric-BW / Spin-Statistics theorems (arXiv:2603.26390 Part II / 2508.10960) — is, by verbatim abstract read, a cleaner statement of HYP-ENCODING-SCREEN (symmetry = INPUT; BW / Spin-Statistics derived FROM the Euler element; NO construction of causal/metric structure from non-symmetric data): NOT a carrier, NOT a reverse-WW inversion, **NOT a sixth route — count stays FIVE**. Connes' bicentralizer confirmed OPEN in general (Ando-Goldbring, arXiv:2605.12776). New objective-collapse bounds (VIP/XENONnT: CSL ~2 orders, DP ~5x; primary arXiv:2506.05507, PRL March 2026) tighten the A-3 CORE, not the wager. DESI five-year w(z) verdict still 2027, low-z-SN-calibration-dependent (DR2 dynamical-dark-energy preference reduced to < 2 sigma; Huang-Cai-Wang arXiv:2502.04212), bears on TARGET not wager. "GR from RG" (arXiv:2602.11806) confirmed an evade-not-invert WW instance (it explicitly engages Weinberg-Witten and unfreezes a UV-Dirichlet-fixed installed metric) — discharging the iter-11 minor defect that the WW gloss was "not confirmable."

**Net:** terminal analytical saturation UPHELD as of 2026-06-19. One of iteration 11's three open subquestions retired as a non-load-bearing lever (a sharpening of the saturation diagnosis); external channels still dry. No carrier closes; no reverse-WW witness; the extraordinary-claim gate does not fire.

## Iteration 13 update (2026-06-19)

*A new-angle structural/synthesis meta-audit (3 refereed tracks: C1 unique-gate / bottleneck-completeness; C2 cross-program triangulation; C3 how-close calibration) — explicitly NOT a carrier re-attack (saturated). Net: the audit CONFIRMS the bottleneck map is complete over the tracked set and triangulation yields an ontological leaning but no testable invariant — re-confirming the consolidation. Verdict unchanged for the* ***12th consecutive iteration****; principal hedge unmoved in grade and condition; carrier-convergence count stays FIVE; no new registry IDs.*

**The bottleneck map is complete over the tracked set** `[INFERENCE]`. The carrier problem (≡ reverse-Weinberg–Witten ≡ the net-naturality theorem) is the unique *analytical reasoning-decidable* gate to the Program-A *wager*. Every candidate independent obstruction fails the two-part test "simultaneously a wager-gate AND a neglected reasoning-decidable lever": the measurement problem (GC-6), Λ (GC-3), OP-50 closed-universe, BH-information (GC-4/5), and the GC-1..17 clashes are each either downstream of or orthogonal to the wager AND themselves external-gated (collapse experiments / experiment+fine-tuning / community resolution / deeper-QG input). The only wager-touching clashes (GC-1, GC-13, GC-16) obstruct exactly what the carrier problem already encodes [INFERENCE — they are *targeted by*, not *identical to*, the carrier formulation]. This is an absence-over-the-enumerated-set claim, not a theorem of completeness. One scope clarification (already implicit in CONCLUSION §3/§7): the gate is unique *to the wager*, not *to unification* — AdS/CFT is an existence proof for §3A independently of the carrier. One recorded contingency: if the Programs-A/B fusion-refusal (§2) were ever overturned, the measurement problem could become a wager-gate and the audit would need re-running.

**Triangulation yields one shared invariant, and it discriminates nothing** `[INFERENCE]`. Across the five programs the unique referee-survivable shared structural invariant is the area law S=A/4G — but it is a semiclassical retrodiction every program AND every geometry-first description reproduces, hence screen-PASSING, not class-SEPARATING; it re-confirms the encode screen rather than breaking it. The shared "geometry-is-emergent" leaning is ontological only: it fractures into incompatible substrates (string/loop/causal-set/algebraic) and is rejected by asymptotic safety. A shared vocabulary is not a shared prediction.

**"How close are we" is axis-dependent and collapses to "how close is the carrier theorem"** [CALIBRATION]. Framework-consistency is largely won (no proven GR-QFT inconsistency; AdS/CFT a strongly-evidenced worked example, NOT a theorem — read "existence proof" as "modulo the AdS/CFT conjecture"). A derived candidate theory is math-gated at the single carrier bottleneck. Experimental traction on the wager is exactly zero with no in-principle test (BMV bears on the gate, is many orders — "~6-8 orders" — below decisive). The nearest dated decision point — DESI five-year (DR3) w(z) analysis ~2027 (survey separately extended through Dec 2028) — bears on the TARGET, not the wager. The three axes are not independent: experimental traction is downstream of the derived theory, which is the chokepoint, so the binding bottleneck is EXTERNAL INPUT, not reasoning. A confirmed universal theory remains NOT FORECASTABLE.

**Consolidation HOLDS.** No genuine independent reasoning-decidable bottleneck and no referee-survivable testable shared constraint emerged; terminal analytical saturation is re-confirmed from the frame/synthesis side. Active mode stays consolidation + lightweight watch on the two clocks (the carrier/net-naturality/bicentralizer/GMA mathematics, and the 2027 DESI w(z) verdict).

## Iteration 14 update (2026-06-19) — the verdict surviving its maximal adversarial assault

*Iteration 14 was structurally different from all thirteen before it. Instead of testing one more lever and watching the verdict survive, it INVERTED the burden of proof: each of three adversarial tracks (D1, D2, D3) was instructed to assume the standing verdict is WRONG and to build — from all web-verifiable 2024-2026 literature — the single strongest possible falsification of its assigned pillar, with no hedging toward the verdict. A referee governed each track under the binding anti-crank protocol (a plausibility hope is a failed break; an unverifiable break-attempt is treated as failed). Net result:* ***all three pillars survived their strongest assault; no prong broke.*** *This is the 13th consecutive non-falsification, but earned by surviving a maximal attack rather than by another passive lever-test.*

**Pillar 1 — "encodes but does not GENERATE geometry" (the carrier no-go). HELD.** The strongest 2026 generative candidate is Fullwood–Vedral–Guzman-Gonzalez, "On Lorentzian symmetries of quantum information" (arXiv:2604.07471, Apr 2026) — the most aggressive claim in the surveyed literature because it asserts the Lorentzian **signature** (1,3) is an *output*, deriving the Minkowski form and the full Lorentz group SO+(1,3) from "preservation of linear entropy" on qubit observables (via S_L(rho)=2 det(rho), det = the SL(2,C)-invariant on Herm_2(C)). [The construction is web-verified real; the spin-homomorphism SL(2,C)→SO+(1,3) mathematics it rests on is independently web-verified correct.] **The attack collapses at one named step:** the (1,3) signature is NOT generated — it is a fixed structural fact of the chosen carrier Herm_2(C), installed the moment one picks a *complex* (rather than real/quaternionic) two-level Hilbert space; the determinant on 2×2 complex-Hermitian matrices IS a quadratic form of signature exactly (1,3), unconditionally. So step (b)'s "derivation" is a tautology of the carrier and step (c) reads off a signature already fixed by the C-structure. This is precisely the indefinite-pairing-in-carrier smuggle named in HYP-CKV-VACUITY-R3 item (ii) and the iter-7 modular-form trichotomy. The takedown does not rest on the authors' framing: they *claim* the relativistic form emerges (abstract), yet their conclusion concedes no explicit spacetime-emergence *mechanism* ("...remains an open and profound challenge"). Structurally it is an **internal** Lorentz symmetry on qubit-observable space — no locality, no net, no causal order, no map to spacetime points — not generated spacetime. Cross-check candidates confirm the same closure (emergent-modular-Virasoro arXiv:2606.11984 presupposes a fixed Euclidean CFT cylinder + conformal vacuum; the Neeb–Olafsson Euler-element net framework arXiv:2412.20410 uses Euler wedges + a Reeh–Schlieder vacuum + a posited Lie-group representation = the iter-12 "cleaner statement of the encoding screen, not a carrier"; the companion arXiv:2502.13293 outputs only EUCLIDEAN 3-space). **The exact installation step that defeats every candidate: the Lorentzian signature enters through the choice of carrier (complex -structure / determinant form / Euler element / conformal vacuum), never as an output of non-symmetric (algebra, state) dynamics*. Carrier count unchanged at FIVE.

**Pillar 2 — the wager has no in-principle distinguishing test. HELD.** Two-pronged maximal attack. **Prong 1 (geontropic/GQuEST as a candidate reverse-Weinberg–Witten observable):** the EFT-baseline paper (arXiv:2409.03894) concedes a geontropic detection "would signal a SEVERE BREAKDOWN of effective quantum field theory" — standard graviton-EFT noise is ΔL/L ~ 10^-38, while the Verlinde–Zurek signal targeted by GQuEST (arXiv:2404.07524, PRX 15, 011034 (2025)) is ~10^-21 and size-scaling, prima facie a signal the encoded EFT cannot produce. **Collapses on a class-vs-axis error:** the geontropic signal separates EFT-violating from EFT-respecting quantum gravity, which is ORTHOGONAL to the wager's emergent-vs-fundamental axis. The killing fact is in the GQuEST authors' own words — "diverse approaches to quantum gravity yield α=O(1)... conformal field theory, dilaton theory, and hydrodynamics" all "predict metric fluctuations of the same scale" — so non-emergent frameworks give the identical signal, and the amplitude $\langle\varphi^2\rangle\propto\alpha$ re-imports the holographic $\eta=1/4G$ at the modular→metric step. Screen-PASSING, a primary-sourced sharpening of the iter-6 Sharmila–Vermeulen–Datta co-classing. Three fresh 2026 sources independently confirm (all web-verified): arXiv:2602.06091 ("Entanglement Before Spacetime") concedes the Newtonian phase "arises only after... introducing the infinity twistor" (same phase both pictures); arXiv:2503.04886 (PRD) reproduces standard cosmological perturbations and concedes the VZ mode's existence "remains an open question"; the non-Gaussianity line (arXiv:2606.09119) discriminates the quantumness/gate, "NOT its ontological status (fundamental vs emergent)." **Prong 2 (Schneider 2024, global content is non-effective — the more serious attack):** establishes, referee-survivably, that "field observables in GR are only defined after fixing the global manifold structure" and global content "acts like a superselection rule that must be independently specified" — which is EXACTLY the wiki's own named opening (the un-proven net-sharing assumption). But Schneider constructs **no observable**, "remains ambiguous on whether global content differences produce empirically distinguishable predictions," and ends on an explicit unresolved fork. Per the iter-9 identity (reverse-WW witness strictly harder than the reverse-WW theorem ≡ the OPEN carrier problem, via Weiner arXiv:1006.4726) and the anti-crank protocol, a "there may be a seam" argument is a research program, not an in-principle discriminator. The seam is corroborated; it is not breached.

**Pillar 3 — terminal analytical saturation / not forecastable. HELD (FAILED-TO-FALSIFY + NO-NEW-INPUT).** Strongest reasoning-lever candidate: the non-Hermitian/Krein cMERA route (arXiv:2606.17983, Jun 2026, "Emergent de Sitter Space and Non-Unitary Tensor Networks from Non-Hermitian Quantum Criticality") — the most dangerous-looking object because it couples a genuinely indefinite (Krein-type) operator structure to a Lorentzian (dS) output via an explicit non-unitary continuous-MERA network. **Collapses for the iter-12**

**reason, now web-instantiated:** verified verbatim, it "formulat[es] a non-unitary cMERA for a CONCRETE non-Hermitian CRITICAL fermion chain" — the critical point is PRE-SELECTED input, the dS geometry emerges only RELATIVE to that installed criticality, and the construction REPRODUCES (does not derive) the log entanglement coefficient. This is the iter-12 "Krein route = route-5 carrier problem in Krein vocabulary, not a sixth route" closure now with a worked example CONFIRMING it. No carrier; no sixth route; count stays FIVE. Strongest external-input candidate: DESI's April-2026 survey completion (verified-real, genuinely postdating the iter-13 framing) — but survey-done is not w(z)-verdict (clock advanced, not ticked), and DESI bears on the TARGET (eternal dS vs quintessence), never the wager. The bicentralizer channel (arXiv:2605.12776 reformulates; arXiv:2511.11409 Houdayer–Marrakchi establishes an equivalence) leaves the problem OPEN, and per the iter-11 binding fact a trivial bicentralizer is the carrier-FRIENDLY condition, so even its resolution cannot close the no-go. **Honest meta-verdict: no reasoning-decidable lever the iter-11/12/13 audits missed could be constructed; the non-Hermitian-criticality angle was the best new shot and it lands squarely inside the encoding screen.**

**Net.** The verdict held against the strongest assault three independent adversarial tracks could mount, each refereed to ATTACK-FAILS-VERDICT-HOLDS. No carrier closes; no reverse-WW witness; no generative-vs-encoding construction; no signature smuggle defeated; no sixth route. Positive value added (all web-verified): six fresh 2026 primary sources logged as new published instances of the encoding screen (2604.07471, 2602.06091, 2503.04886, 2606.09119, 2606.17983, 2412.20410), and Schneider 2024 located the screen's sole seam from outside the wiki, exactly where the wiki places it, without breaching it. **Registry effect: NONE — carrier count FIVE, hedge HYP-CKV-VACUITY-R6 / HYP-ENCODING-SCREEN unmoved, no new numeric ID. Verdict unchanged, 13th consecutive; terminal analytical saturation re-confirmed against post-iter-13**

**mid-2026 literature.** Citation-precision bind carried from the D3 referee: state "No Houdayer–Marrakchi (or any) RESOLUTION of the bicentralizer problem exists; 2511.11409 (selfless-equivalence) and 2605.12776 (model-theoretic reformulation) both leave it OPEN" — verdict-bearing content untouched.

## Iteration 15 update — the net-naturality theorem attacked from three genuinely-new mathematical frameworks; all three reproduce the encoding screen

**Outcome: NO HANDLE FOUND. 14th consecutive confirmation. The carrier no-go is now robust across every modern mathematical framework tried.**

Three track agents brought frameworks NEVER previously applied to the carrier problem in this wiki — disjoint from the exhausted Tomita–Takesaki / centralizer / bicentralizer machinery of iters 5–14 — and asked whether any gives a new *decidable* handle on the net-naturality theorem. All three reported **REPRODUCES-SCREEN**; all three referees upheld (none sustained a refutation; corrections were attribution/precision only, none verdict-bearing).

- **E1 — Factorization-algebra / functorial-QFT** (Costello–Gwilliam prefactorization algebras; Atiyah–Segal–Lurie cobordism QFT; the Benini–Carmona–Grant-Stuart–Schenkel categorical **equivalence** `AQFT ≅ time-orderable prefactorization algebra`, arXiv:2412.07318 / 1903.03396, web-verified real). The equivalence is a genuinely-new precise object but **not a handle**: both sides are functors over the SAME fixed orthogonality / causal-disjointness relation ⊥ (= smuggle-item $n_1$), plus the O(d)-structured bordism site carrying the signature. It transports the carrier question verbatim — a **fourth independent witness** to the obstruction already logged in iter-9 §4 (Benini–Schenkel–Woike arXiv:1709.08657), not a novel point. Referee BC: re-attribute arXiv:2212.08175 to Gwilliam–Rejzner; count stays FIVE (categorical presentation of route 5, not a sixth route).

- **E2 — Connes–Atiyah–Singer index theory** (spectral triples / unbounded KK; bounded Kasparov K-homology; coarse geometry / Roe algebras). Every index pairing factors through the **Dirac operator** `D` (the metric datum) or its phase `F=sign(D)`; in the Lorentzian case `D` is Krein-self-adjoint on a **pre-installed Krein space** whose indefinite form IS the `(n,n)` carrier (arXiv:1611.07062, web-verified). A single local $III_1$ factor is K-theoretically structureless — but that is only the single-algebra shadow; the carrier content survives in the relations + vacuum. Referee BC: spectral-localizer attribution split (Loring–Schulz-Baldes finite-volume vs Kaad arXiv:2508.08668 KK); type-$III_1$ vacuity is a single-algebra result only.
- **E3 — Higher-categorical / homotopical / non-unitary tensor-categorical AQFT** (operadic AQFT on orthogonal categories; Lorentzian bordism FFT, Bunk–MacManus–Schenkel arXiv:2308.01026; cobordism hypothesis; logarithmic/Hermitian tensor categories beyond iter-9's rigid C*-point). Causal/localization structure is the orthogonality relation of the base; an indefinite signature exists categorically only outside the unitary world and there only as **installed reflection/dagger data** (Müller–Stehouwer arXiv:2301.06664) — the iter-12 Krein closure in categorical vocabulary. Does NOT close the iter-9 commutant-to-generated gap. Referee BC: attribution fixes; cobordism hypothesis carries no load beyond "the bordism site is the geometric input."

**The common smuggle, stated once:** across all three frameworks (and the prior Tomita–Takesaki and Krein routes), the geometric input enters as one of two interchangeable objects — the causal-disjointness/orthogonality/index-set order $n_1$ (localization template), or a metric/Dirac/Krein datum $\eta$ (signature carrier). None makes $n_1$ or $\eta$ an *output* of algebra+state.

**Registry: nothing moved.** Verdict UNCHANGED (14th consecutive); HYP-CKV-VACUITY-R6 grade (MEDIUM-HIGH→HIGH) and condition (net-naturality theorem + located commutant-to-generated gap) UNMOVED; HYP-ENCODING-

SCREEN no-test corollary UNMOVED; carrier-convergence count **FIVE**; no new ID consumed (next free OP-51 / A-24 / GC-18 / H9). The deliverable is the dossier — the carrier problem posed for external resolution, now demonstrated robust across factorization-algebra, index/K-theory, and higher-categorical mathematics.

## Iteration 16 update (2026-07-03) — the multi-wedge closure: the first hedge advance since iteration 9; the residual pinned to (E_O)

**Outcome: verdict UNCHANGED — FIFTEENTH consecutive confirmation — but the principal hedge MOVES: HYP-CKV-VACUITY advances R6 → R7 on a referee-verified proof package.** The iteration was clock-gated and legitimate: watch-sweep #1's clock-(1) input (Marrakchi arXiv:2606.23636, 2026-06-22) triggered it per the ROADMAP saturation rule. Four tracks (G1–G4), three binding referees (R1/R2/R3), every load-bearing citation web-verified; one pipeline fabrication caught and discarded (process record in the G3 note).

- **G1 (the verdict-bearing move) — the commutant-to-generated gap is CLASSIFIED, not closed** `[INFERENCE, high — referee-verified proofs]`. The multi-wedge naturality lever — intersecting the constraints of ALL wedge modular flows, verified genuinely unused in iterations 9–10 — yields **LEM-NET-NATURALITY-G**: **(G-loc)** every *localized* natural carrier is ±1, a written proof independently re-derived by the referee at every step (Borchers/BW scaling $\Longrightarrow \mathrm{Fix}(\Delta_W^{it}) = \mathbb{C}\Omega$ via the Cauchy–Schwarz equality case; vacuum pinning using only bounded $\Delta^{it}$ — no domain trap; Reeh–Schlieder kill); **(G-glob)** the boosts of all translated wedges generate $\mathcal{P}_+^\uparrow$ (group identity verified by direct computation), so every *fully-natural non-localized* algebra-compatible carrier is a **gauge-symmetry implementer** with Poincaré-invariant eigenprojections — $\Delta_{\mathrm{geo}} = 0$, proof-grade. Decisively honest: the gap is **NON-empty** — Fock parity $(-1)^N$ certifies occupants — but every occupant is a charge/statistics grading, exhaustively geometry-void. The sole surviving branch (massive double-cone fibers) is pinned to one named hypothesis, **(E_O)**: vacuum ergodicity on massive double-cone algebras, $M(O)_\omega = \mathbb{C}1$, [OPEN] — genuinely open (Marrakchi–Vaes genericity cannot reach the specific vacuum state; the massive double-cone modular flow is non-geometric with unknown spectrum). Direction (binding): (E_O) ⇒ the net-naturality no-go is a **theorem** on the stated hypotheses (+ weak additivity); ¬(E_O) is necessary but NOT sufficient for a counterexample fiber. Referee-bound downgrades: the DHR n1-inheritance reading is framing only; the conformal-net closure is a transport sketch. Two prior R7 mints were struck (iteration 10); this one differs by working on the correct object ($\{\Delta_W^{it}\}' \subset B(H)$), exhibiting rather than denying the gap's occupants, and carrying written proofs that survived full adversarial re-derivation.

- **G2 — three untried constructions BLOCKED; the state-family extension closed as a class** [INFERENCE, high]. The co-moving evasion of Lemma N2 (the boost moves the state, so a fixed-sign-locus argument fails) is *mathematically real off-vacuum* — and closed anyway by the section-rigidity trichotomy (Lemma S2, sketch-grade; recorded as an **extension** of the posed conjecture — Lemma N2 and the dossier statement stand verbatim). The Doplicher–Longo split-inclusion flip Θ is canonical, genuinely (∞,∞), not a function of Δ — and fails algebra-compatibility precisely because identifying the tensor legs requires a non-canonical anti-isomorphism (an installed η); where canonical it degenerates to PCT (Lemma PC, sketch-grade: standard-form canonicity is cone-covariant transport). LEM-CORE-DESCENT (proof-grade transport corollary): core-built carriers descend to the closed centralizer class or carry only trace data annihilated by the dual flow's scaling $\tau \circ \theta_s = e^{-s}\tau$.

- **G3 — the Marrakchi lever RESOLVED-NEGATIVE at the level of the object; watch target re-sharpened** [INFERENCE, high]. From the full TeX source: the "with expectation" hypothesis is **pervasive and definitional** — the relative bicentralizer flow is built against φ = φ∘E_N (Takesaki) before any proof step; the author flags it "essential". For M(O) ⊂ M(W) no ambient state's modular flow globally preserves M(O) (the BW flows are geometric and move O — smuggle-item $n_1$ again), so the flow is undefined and the sweep-#1 watch target was **ill-posed as stated** — retired and replaced (synthesis §6). The AHHM structural output has no carrier interaction; iter-12's non-load-bearing grade is strengthened. Sole transferable residue: the expectation-free relative T-invariant (adjacent, non-load-bearing conjecture T(M(O) ⊂ M(W)) = {0}).

- **G4 — 7 of 8 untried frameworks REPRODUCE-SCREEN; one conditional survivor** `[INFERENCE, high]`. Noncommutative Dirichlet forms (constitutive positivity g6), free Araki–Woods (the input orthogonal representation IS the boost), infinite-index subfactor invariants, entropic order parameters (region-indexed detectors), QRF/gravity crossed products (installed flow/observer), net cohomology (poset as axiom zero), and the 2024–26 web sweep all close at a located smuggle step. Survivor: **DCNG** — in the 2025 continuous-logic W*-axiomatizations the gap becomes a *definability* question; the definable-carrier no-go is a named watch item / iter-17 lever (no-go direction only; gated on an $\eta_0$-definability lemma; the *localized* half is not even posable without installing $n_1$ as a logical sort — the smuggle reproduces at the level of the logic's signature). Dossier §6 hardening: **Morinelli–Neeb** (Adv. Math. **458** (2024) 109960, arXiv:2312.12182) prove a two-directional, **BW-conditional** bridge between the two smuggle items — theorem-level confirmation that $n_1$ and η are interchangeable relative to an installed BW/wedge structure, deriving neither from (algebra, state).

**Standing-record repairs (binding, propagated):** Doplicher–Longo = Invent. Math. **75** (1984) 493–536 (iter-12 note fixed in place); "irreducible" for M(O) ⊂ M(W) is FALSE (collar algebra in the relative commutant; conclusions survive via Takesaki's criterion — dated addenda on the iter-12 note and watch-sweep #1);

Houdayer–Marrakchi 2511.11409 states an implication, not an equivalence.

**Saturation status and the hedge ledger after iteration 16.** `[INFERENCE, high]` The headline stands a fifteenth time — no carrier, no reverse-WW witness, no generative construction, no distinguishing test; NOT-YET-PHYSICS unchanged; the 2027 DESI $w(z)$ verdict remains the nearest dated external event. What moved: **hedge 1 (OP-46 residual) advances R6 → R7** — grade unchanged (conditional HIGH), condition restructured from the diffuse {n1 + commutant-to-generated gap} to the sharp **{n1 AND (E_O)}**; **hedge 3 (HYP-ENCODING-SCREEN) inherits** the R7 condition — a proof of (E_O) would make the no-go and its no-test corollary theorems on the stated hypotheses. Carrier-convergence count stays **FIVE**; no numeric ID consumed (next free OP-51 / A-24 / GC-18 / H9); three named items enter (LEM-NET-NATURALITY-G; (E_O); DCNG). Watch-mode resumes on the re-sharpened five-item clock-(1) state with (E_O) as the principal item. See the *iteration-16 synthesis* (Chapter 36 (p. 684)).

## Iteration 17 update (2026-07-07) — the (E_O) assault: free-field reduction, three route-closures, disproof failure; hedge stays R7

**Outcome: verdict UNCHANGED — SIXTEENTH consecutive confirmation; the principal hedge STAYS at R7 — (E_O) neither proved nor disproved.** The first direct attack on the residual (E_O) (five finder lenses E1–E5; three binding referees; watch-sweep #2 in the same pass, dry). Clock-legitimate — (E_O) is the designated principal watch item (δ), and the assault is the sanctioned next step. Every load-bearing 2020–2026 citation re-verified live.

- **Free field (referee PARTIAL): (E_O) OPEN, sharply reduced.**

  `[INFERENCE, high on the reduction; OPEN on the spectral input]` Via second quantization, M(O)_ω = ℂ1 is IMPLIED by purely continuous one-particle spectrum of ln δ_O (weak mixing). The naive "no-eigenvalue-0 ⟹ ergodic" reading is REFUTED (bosonic dΓ(A) are modular-fixed regardless of point spectrum). Spectral fact numerical-only (BCM arXiv:2209.04681; Cadamuro arXiv:2312.08525); no closed-form massive double-cone modular Hamiltonian (Longo–Morsella arXiv:2012.00565, massive-ball result "contained a gap, and have been removed"). Leaning-true, unproven.

- **General (H) (referee CONFIRMED): all three structure-theory routes FAIL.** `[INFERENCE, high]` clustering not transportable to bounded-region modular mixing; Δ_O-fixity ⇏ Δ_W-fixity; genericity not pinnable to the vacuum. Marrakchi arXiv:2606.23636 does not transfer (expectation-free gate).
- **Disproof FAILED (referee CONFIRMED): no carrier.** `[INFERENCE, high]` eight candidate mechanisms die on {bounded ∧ in M(O) ∧ exactly Δ_O-fixed}; honest status BARE OPEN (referee down-calibrated the finder's "leaning true").
- **Hedge ledger:** HYP-CKV-VACUITY R7 UNCHANGED (grade conditional HIGH, condition {n1 AND (E_O)}); the (E_O) item gains a free-field internal reduction. Carrier-convergence count stays FIVE; no numeric ID consumed. NOT-YET-PHYSICS unchanged; 2027 DESI w(z) remains the nearest dated external event. See the *iteration-17 synthesis* (Chapter 37 (p. 693)).

# References

See the Bibliography (p. 776). This synthesis rests on the sources cited across the domain and analysis pages; no claim here is asserted beyond what those pages support.

CHAPTER 22

# Iteration 2 Synthesis: Does the Unified Program Cohere?

Iteration: 2

## Question

Across the five iteration-2 investigations (de Sitter / emergent spacetime, the Born rule and quantum linearity, tabletop quantum gravity, entropic / emergent gravity, and the algebraic / crossed-product program), is there a single defensible *unified research program*? If so, what does it commit to, what does it explain, what is its sharpest internal tension, and what near-term deliverables would settle it — and, stated honestly, does it actually cohere or are we manufacturing unity with a slogan?

The discipline of this page is to resist the most seductive false unifier ("it's all information-theoretic") and to report the *truthful* coherence verdict, even when that verdict is "partial."

## State of the art (the fixed structural cores)

The synthesis takes as **fixed** the following rigorous results, and treats everything else as suspected-derivative.

**Operational quantum core.** The Born-rule *functional form* $p = \mathrm{Tr}(\rho E)$ is forced — Gleason's theorem for $\dim \mathcal{H} \geq 3$, and Busch / Caves–Fuchs–Manne–Renes for $\dim = 2$ (POVM version) — given the Hilbert-space structure and non-contextuality of probabilities

`[ESTABLISHED]`. This is the rigidity the operational reconstructions (no-signaling + local tomography + purification + reversibility) point to.

**Algebraic QFT.** Local relativistic QFT subregions carry Type $\mathrm{III}_1$ von Neumann algebras (Connes-Haag-Araki property of the vacuum): no trace, no tensor factorization, no local density matrix `[ESTABLISHED]`. Crucially this holds for *all* signs of $\Lambda$ — Minkowski, de Sitter, AdS — before gravitational dressing [ESTABLISHED; see [synthVerdict refinement]].

**Horizon entropy.** A horizon obeys the Bekenstein-Hawking area law $S = A/4G$, with renormalized generalized-entropy form

$$S_{\text{gen}} = \frac{\langle A \rangle}{4G} + S_{\text{out}}.$$

This is an **input** to every emergence derivation, not an output; the microscopic origin of $A/4G$ is not supplied `[INFERENCE, strong]`.

**Crossed-product Type II structure.** Adjoining an observer/clock and taking the crossed product by the modular automorphism group converts the $\mathrm{III}_1$ algebra into a Type II algebra with a semifinite trace, density matrices, finite $S_{\text{gen}}$, and a derived Generalized Second Law (Witten 2022; Chandrasekaran-Penington-Witten 2022; Chandrasekaran-Longo-Penington-Witten / CLPW 2022). Kudler-Flam-Leutheusser-Satishchandran (2024-25) extend this to *any* bifurcate Killing horizon `[ESTABLISHED, given a background]`.

**Causal order.** A causal order fixes the conformal metric (Malament; Hawking-King-McCarthy), and no-signaling / microcausality survive where naive locality fails `[ESTABLISHED]`. Causal order — not the full metric — is the robust survivor.

**Linearity bound to causality.** Generic *deterministic* nonlinear modifications of QM permit superluminal signalling (Gisin 1989; Polchinski 1991; reaffirmed Bielińska-Eckstein-Horodecki 2024) `[ESTABLISHED for the deterministic exclusion]`.

**Load-bearing positive results that presuppose geometry.** Jacobson 1995 (Einstein equation as equation of state from $\delta Q =$

$T\,dS$ on local Rindler horizons) assumes causal/metric structure plus the area law and Unruh $T$ `[ESTABLISHED]`. Faulkner-Guica-Hartman-Myers-Van Raamsdonk 2013 / Lashkari et al. derive only the *linearized* Einstein equations from the entanglement first law $\delta S = \delta\langle K\rangle$ on a *fixed* AdS background `[ESTABLISHED]`. RT/HRT is a holographic dictionary entry, not a from-scratch derivation of bulk geometry `[ESTABLISHED]`.

## Key arguments

`[INFERENCE]` **The defensible program is not a theory-of-everything but a *program of demotion-and-derivation*.** Hold the fixed cores above; treat metric geometry, the local-subsystem/qubit picture, exact QM linearity, and the Born rule's axiom-status as **suspected-derivative**; prosecute each suspicion to a yes/no verdict via a short list of decisive deliverables. Stated positively: "Causal/algebraic/entanglement structure plus a small set of forced operational axioms is more fundamental than metric geometry and than the postulate-list packaging of QM; geometry and the apparent independence of the Born rule are reorganizations of that structure."

`[INFERENCE]` **The program's defensibility comes entirely from its scoping.** Every load-bearing positive result is *conceded* to presuppose rather than generate geometry, so "spacetime from entanglement" is retained only as a *constrained program* (H2) with explicit graduation criteria — never as a framework.

`[SPECULATIVE]` **The central wager — algebra/entanglement before geometry — is true of our $\Lambda > 0$ universe.** This is the speculative heart. In every place the program is rigorous it is demonstrably a *re-encoding* of geometry; in every place it would be a genuine derivation it is unachieved (see Strongest internal tension).

`[ESTABLISHED form / OPEN meaning]` **The Born rule is both forced and mysterious.** Its *form* is forced (Gleason/Busch); its *operation as chance over single outcomes* in a no-collapse setting is unresolved. No Everettian derivation (frequency operators, Zurek envariance, Deutsch-Wallace, Carroll-Sebens) closes the quantitative (unequal-amplitude) step without re-importing a Born-equivalent symmetry / rationality / typicality premise, and all require a decoherence preferred basis whose own selection invokes Born `[OPEN for quantitative meaning]`.

`[INFERENCE, strong]` **Exact linearity is plausibly derivative of no-signaling**, not primitive — but it remains empirically untested at the collapse scale, where stochastic objective-collapse (GRW/CSL) and gravitationally-induced (Diósi-Penrose) models evade the deterministic no-go `[CONTESTED that exact linearity is fundamental]`.

`[INFERENCE]` **BMV / gravitationally-induced entanglement is a falsifier of a *specific class* of classical gravity theories**, not a proof that "gravity is quantized." A positive result excludes mean-field semiclassical gravity and Kafri-Taylor-Milburn classical-channel models and constrains Oppenheim stochastic-CQ gravity, establishing non-classicality of the Newtonian (off-shell, longitudinal) sector — but does NOT establish graviton quantization, renormalizability, or any UV/Fock structure `[INFERENCE for the exclusions; the loophole status is CONTESTED]`.

`[CONTESTED]` **"It's all information-theoretic" must be refused as the unifier.** "Information" denotes non-interchangeable objects across the clusters — von Neumann-algebraic relative entropy and crossed-product traces in the geometry-from-algebra cluster vs operational probability and LOCC monotones in the quantum-framework cluster. Gluing them with the word "information" is exactly the false unity this audit exists to prevent.

## Sharpest obstructions

**Reorganization vs generation (the single deepest tension).** `[INFERENCE]` The entire algebraic/holographic cluster is now *conceded by its own architects* to presuppose geometry: a fixed background with a Killing/conformal symmetry is an input at every step (it is what makes the modular flow geometric, via Bisognano-

Wichmann / Hislop-Longo / CHM); the constructions are perturbative in $G_N$ (or $1/N$) about a fixed saddle; Witten's own "A Background-Independent Algebra in Quantum Gravity" (arXiv:2308.03663) defines "background independent" as independence of the *state* on a *fixed* worldline algebra — with the spacetime $M$ re-entering as a choice of state; Jacobson/Padmanabhan take the area law as input. So the wager is, wherever rigorous, a re-encoding of geometry; wherever it would be a genuine derivation, it is unachieved. This is not a contradiction but a precisely-located gap.

**The Bisognano-Wichmann / CHM circularity.** `[INFERENCE]` The modular flow is "geometric" only because a symmetric vacuum on a given spacetime is assumed. To break the circularity one must recover an emergent boost/Killing structure and horizon from an (algebra, state) pair whose modular flow is *not* a known symmetric vacuum. Until done, the algebraic program must be cited as reorganization.

**No $\Lambda > 0$ dynamics-from-entanglement.** `[INFERENCE]` CLPW's Type $\mathrm{II}_1$ static-patch algebra (with maximal-entropy state = empty dS) is a genuine $\Lambda > 0$ *entropy* result, but supplies no RT surface, no dual, and no dynamical equation. There is no de Sitter analog of the AdS first-law $\Rightarrow$ linearized-Einstein chain. dS/CFT yields a *non-unitary* Euclidean boundary functional (complex conformal weights for $m^2\ell^2 > d^2/4$), not a unitary boundary Hilbert space. The deepest target — nonlinear, background-independent Einstein-from-entanglement — is unachieved even in AdS, so dS is at least two unsolved steps away.

**Nonperturbative fate of the algebra.** `[OPEN/CONTESTED]` Giddings (arXiv:2505.22708) argues the III $\to$ II story is a leading-order special case of gravitational dressing, and nonperturbatively the algebraic structure itself may fail ("algebraic spacetime disruption"). If so, the von Neumann framework is only a leading-order reorganization.

**The quantitative-Born wall.** `[OPEN]` No no-collapse derivation of unequal-amplitude probabilities uses premises provably inequivalent to Born; the GPT reconstruction *relocates* rather than dissolves the question.

**BMV loopholes.** `[CONTESTED]` Classical-but-nonlocal mediators (Hall-Reginatto, at the price of nonlocal signalling); the Di Biagio "mediation $\neq$ relativistic locality" objection; and the Aziz-Howl claim that classical gravity + QFT matter entangles via virtual matter ($m^3$ scaling) — all actively disputed (rebuttals argue an ultra-local propagator kills the effect and any residual entanglement is matter-mediated). The decisive inference rests on the General Witness Theorem, which is strong but has documented locality assumptions.

**Wrong-idealization risk.** `[CONTESTED]` DESI's 2.8-4.2$\sigma$ dynamical-dark-energy hint plus swampland/TCC may move the target from eternal dS to metastable dS or rolling quintessence — the program may be aiming at the wrong idealization (`ASSUME-ETERNAL-dS`).

## What would count as decisive progress

A direction with no distinguishing near-term test is a research program, not a theory. The two empirically live channels are gravitationally-induced entanglement (BMV) and objective-collapse / nonlinearity bounds; the cosmological channel is DESI/Euclid/Roman dynamical dark energy. The decisive deliverables, in leverage order:

1. **dS entanglement first law [OP-dS-FIRSTLAW]** — show in a controlled $\Lambda > 0$ setting that $\delta S = \delta\langle K\rangle$ implies the *linearized* Einstein equations (the genuine dS analog of FGHMVR). Decides whether emergence graduates from $\Lambda > 0$ entropy bookkeeping (exists) to $\Lambda > 0$ dynamics-from-entanglement (does not). Single highest-leverage item.
2. **Nonperturbative fate of crossed-product Type II [OP-CP1]** — survival beyond all-orders-in-$1/N$, or replacement by a non-algebraic structure (Giddings). Either answer is decisive.

3. **Emergent geometry from a non-symmetric modular flow** — recover a Killing/boost structure and horizon as an *output* of an (algebra, state) pair with no prior Reeh-Schlieder vacuum. Directly decides reorganization-vs-generation.
4. **Controlled BMV/QGEM experiment [OP-40, HYP-GIE-DISCRIM]** — joint GIE detection + gravitational-decoherence bound; resolve Aziz-Howl ($m^3$) vs ultra-local-propagator, and Di Biagio mediation-vs-locality. The $m^2$-vs-$m^3$ scaling discriminates genuine quantum gravity from the classical-gravity loophole.
5. **Detect/exclude objective-collapse nonlinearity [A-3-REF]** — GRW/CSL, Diósi-Penrose at macroscopic-superposition scale; and prove/refute a theorem strengthening Gisin/Polchinski to "any causal, state-resolving, deterministic dynamics on a tensor-product Hilbert space is exactly linear."
6. **Born-free quantitative derivation [OP-BORN-QUANT]** — operational Born-free preferred-basis definition, then show whether any no-collapse derivation of unequal-amplitude Born succeeds with premises provably not Born-equivalent.
7. **Dynamical dark energy + landscape verdict [ASSUME-ETERNAL-dS]** — confirm/refute $w(z) \neq -1$ via DESI DR3+/Euclid/Roman/LSST; theorem-level statement on whether metastable dS is in the landscape. Decides the correct *target*.

## Proposed registry items (with referee verdict)

**[REG-SYNTH-1] Unified program = demotion-and-derivation, not theory-of-everything.**

- Referee verdict: **keep**, severity **minor**.
- Refined statement: The most defensible unified program is a research **heuristic** (not itself a falsifiable program; per the iteration-3 red-team, its falsifiable content lives only in the object-level deliverables — OP-41, OP-44, the BMV $m^2$-vs-$m^3$ test, collapse bounds) of demotion-and-derivation: hold fixed

the established cores (local QFT Type $\mathrm{III}_1$ for any $\Lambda$ sign (undressed, $G_N = 0$); the area law; no-signaling/causal-order; the Gleason-Busch-forced Born form; unitarity), and prosecute as suspected-derivative the status of metric geometry, the local-subsystem picture, exact linearity, and Born's axiom-status — each to a yes/no verdict. Every load-bearing positive result (RT/HRT, linearized-Einstein-from-first-law, crossed-product Type II entropy, Jacobson's Clausius derivation) is conceded to *presuppose* rather than *generate* geometry. [INFERENCE for coherence as strategy; SPECULATIVE for the wager being true of our $\Lambda > 0$ universe; OPEN for whether any track graduates to genuine geometry-generation.]

**[REG-SYNTH-2] Algebra-type structural signal.**

- Referee verdict: **keep with correction**, severity **minor** (overclaim flagged).
- Refined statement: The original "algebra-type-encodes-$\Lambda$-sign as the cleanest new structural framing" **overstates novelty and cleanliness** and is downgraded. The accurate content: local QFT subregions are Type $\mathrm{III}_1$ *independent of the sign of* $\Lambda$ (Minkowski, dS, AdS alike, before dressing) `[ESTABLISHED]`. Gravitational dressing via the crossed product converts these to Type II, and it is the $\mathrm{II}_1$-vs-$\mathrm{II}_\infty$ distinction — existence/normalizability of a tracial maximal-entropy state ($\mathrm{II}_1$ = dS static patch, admits empty-dS maximal-entropy state; $\mathrm{II}_\infty$ = black-hole/AdS-thermal, no maximal-entropy state) — that carries the dS-specific information, **not a faithful $\Lambda$-sign-to-type bijection** (CLPW 2022; CPW 2022). CLPW's $\mathrm{II}_1$ is the genuine $\Lambda > 0$ algebraic foothold.

**[REG-SYNTH-3] No-signaling as load-bearing core axiom.**

- Referee verdict: **keep with correction**, severity **minor** (mis-scoping flagged).
- Refined statement: No-signaling is a genuine, load-bearing shared constraint (it grounds exact linearity in cluster two and causal-order primacy in cluster one), but it is **not a stand-alone quantum-forcing axiom on par with**

**Gleason/Busch**: PR-box / box-world is no-signaling but non-quantum. Treat it as *one of several* GPT axioms (with local tomography + purification + reversibility), not as singling out quantum theory by itself. `[CONTESTED as sufficient.]`

**[REG-SYNTH-4] BMV as logical prerequisite ("gate").**

- Referee verdict: **keep with correction**, severity **minor**.
- Refined statement: BMV non-classicality is *logically prior* to the emergence program (if gravity were fundamentally classical/stochastic-CQ, the algebra-before-geometry wager would be misdirected), but the gate is **strong, not airtight**: it rests on the General Witness Theorem inference (entanglement via a local mediator ⇒ non-classical mediator), which assumes LOCC-type locality and excludes non-mediated channels `[CONTESTED]`.

**[REG-SYNTH-5] Meta-verdict: PARTIAL coherence (refuse the "information" fusion).**

- Referee verdict: **keep**, severity **minor**.
- Refined statement: One program (geometry-from-algebra) genuinely unifies three tracks via the modular/crossed-product + area-law substrate; verdict = rigorous reorganization of semiclassical gravity on a fixed background, not yet derivation of geometry. A second, methodologically distinct program (is-the-quantum-framework-final-and-does-it-cover-gravity) unifies two tracks via reconstruction theorems, no-go arguments, and tabletop experiments. The two are joined only thinly — by shared tools (no-signaling/causal-order) and by BMV as a logical prerequisite — and the "it's all information" fusion is correctly refused. `[INFERENCE.]`

## Verdict

**Partial coherence — and that is the truthful answer, not a fragmented "no" nor a forced "yes."**

There are **two internally-unified programs**, joined by genuine but slender bridges:

- **Program A — geometry-from-algebra** (three tracks: de Sitter-emergence, algebraic-background-independence, entropic/emergent-gravity). One substrate (modular/crossed-product algebra + the area-entropy law), one method (derive geometry/entropy from algebra+entanglement), one binding obstacle set (AdS-confinement, Type $III_1$, no nonlinear/background-independent Einstein, the area law as unexplained input), one decisive deliverable (the dS entanglement first law). Iteration-2 verdict, mutually reinforcing across all three tracks: **rigorous reorganization of semiclassical gravity on a fixed background, NOT yet a derivation of geometry**, with CLPW's Type $II_1$ result as the strongest $\Lambda > 0$ foothold. This cluster genuinely coheres. `[INFERENCE, confidence high.]`
- **Program B — is-the-quantum-framework-final-and-does-it-cover-gravity** (two tracks: Born-rule/linearity reconstruction; tabletop BMV). Method: reconstruction theorems, no-go arguments, and *experiments* — not holographic mathematics. This cluster also coheres internally. `[INFERENCE, confidence high.]`
- **The bridges are real but thin and mostly methodological:** (i) both deploy no-signaling as a load-bearing constraint; (ii) the GPT operational reconstruction is the axiom set that would make "physics is information" substantive, linking Born/linearity to the algebraic wager; (iii) BMV's bearing on whether gravity is non-classical is *logically prior* to the whole emergence program. These are bridges of shared tools and logical prerequisite, not a shared substrate or a shared derivation.

Net: **one coherent program + one coherent adjacent program, joined by no-signaling/causal-order as a genuine but slender shared principle and by BMV as a logical gate.** Coherence = PARTIAL. [INFERENCE for coherence as a research strategy; SPECULATIVE for the central wager being true of our $\Lambda > 0$ universe; OPEN for graduation to actual geometry-generation. Overall confidence: high.] *(Iteration-3 red-team relabel, propagated here: "demotion-and-derivation" is a research* ***heuristic****, not itself a falsifiable program — its only falsifiable content is the object-level deliverables. Consistent with Chapter 21 (p. 497) and Chapter 23 (p. 582).)*

## Open subquestions

- `[OPEN]` Can a de Sitter entanglement first law yield linearized Einstein at $\Lambda > 0$? (OP-dS-FIRSTLAW)
- `[OPEN]` Does the crossed-product Type II structure survive nonperturbatively in $G_N$, or does algebraic spacetime disruption replace it? (OP-CP1)
- `[OPEN]` Can semiclassical geometry be recovered as an *output* of a non-symmetric modular flow, breaking the Bisognano-Wichmann/CHM circularity?
- `[OPEN]` Is there an operational, Born-free derivation of the *quantitative* Born rule in a no-collapse theory? (OP-BORN-QUANT)
- `[CONTESTED]` Will BMV/QGEM resolve the $m^2$-vs-$m^3$ scaling and the Di Biagio / Aziz-Howl disputes decisively? (HYP-GIE-DISCRIM)
- `[OPEN]` Can Gisin/Polchinski be strengthened to a theorem forcing exact linearity for *any* causal, state-resolving, deterministic dynamics? (A-3-REF)
- `[CONTESTED]` Is the correct target eternal dS, metastable dS, or rolling quintessence? (ASSUME-ETERNAL-dS)
- `[OPEN]` Do "encode geometry" and "generate geometry" differ by any agreed operational criterion?

CHAPTER 23

# Iteration 3 Synthesis: From Surveying to Attempting

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Iteration: 3

This note records the iteration-3 investigation and folds in the synthesis-verdict review (one substantive technical correction was forced; the conclusion is unchanged). The honest one-line headline: **the program got more rigorously understood and more honestly hedged, the obstructions got sharper, and the central bet got no closer to being either confirmed or empirically decidable.** `[INFERENCE, high confidence]`

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## Updated coherence verdict (still-partial)

The coherence verdict is **PARTIAL**, unchanged from iteration 2, and the iteration-3 work confirms this is the *correct* verdict rather than a placeholder. `[INFERENCE, high]`

Iteration 3 ran four moves: (1) a de Sitter entanglement first-law attempt (OP-41); (2) a formal encode-vs-generate criterion (OP-46); (3) a Program-A + Program-B merge attempt; (4) a red-team pass over the synthesis page. Three of the four sharpened obstructions rather than removing them; the fourth forced the synthesis page to retract over-claims back to what its own source notes already supported.

What survived every attack — the project's best work — and was *strengthened*, not weakened:

- The **deflationary core** `[ESTABLISHED-as-argued]`: the catalogued GR/QM "contradictions" are domain-mismatches or unsolved-but-consistent problems, not logical inconsistencies (referee verdicts in *Chapter 14 (p. 191)* are sound).
- The **central diagnosis** `[INFERENCE, high]`: the algebraic/holographic program **ENCODES geometry, it does not GENERATE it** — strengthened by the verified Giddings results (arXiv:2505.22708, 2510.24833 [VERIFIED]: dressed operators generically fail to commute spacelike at leading order in $G_N$).
- The **Born form/meaning split**, **linearity-as-a-derivative-of-no-signaling**, the refusal of an "it's all information" fusion, the dS negative meta-result, and **BMV-as-class-falsifier-not-quantization-proof**.

The wager — *algebra/entanglement before geometry is true of our* $\Lambda > 0$ *universe* — remains `[SPECULATIVE]` and, by the wiki's own anti-crank standard, still **not yet physics**: it has no distinguishing near-term test (see *Empirical falsifiability portfolio*).

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# Did Program A move?

**SHARPENED, NOT ADVANCED — and that distinction is the honest headline.** `[INFERENCE, high]`

Program A (geometry-from-algebra) did not graduate from "encoding" to "generating" geometry at $\Lambda > 0$. The dS-first-law attempt (OP-41, see *the attempt note* (Working note: *Attempt: a de Sitter entanglement first law (OP-41)* (repo: `notes/2026-06-08-iter3-de-sitter-first-law-attempt.md`))) produced a precisely-located *obstruction*, not a derivation.

### (1) Jacobson 2015 re-read: a real but different result

Jacobson 2015 (arXiv:1505.04753, PRL 116 201101) [VERIFIED] is correctly re-read as already a local, dS-valid derivation of the **full Einstein field equation** from entanglement equilibrium.

> **Synthesis-verdict correction (the one substantive technical fix).** Iteration-3's draft claimed Jacobson is "the NONLINEAR semiclassical Einstein equation ... in one respect (nonlinearity) STRONGER than the AdS FGHMVR linearized result." This conflates two senses of "nonlinear" and is `[CONTESTED-as-stated]`. The corrected characterization:

Jacobson derives, from a **first-order** entanglement-equilibrium (stationarity at fixed volume) condition imposed *pointwise* in small geodesic balls, the full Einstein equation

$$G_{\mu\nu} + \Lambda\, g_{\mu\nu} = 8\pi G\, \langle T_{\mu\nu} \rangle$$

as a **local tensor equation, including** $\Lambda$, for conformal matter (nonconformal matter requires the unproven $\delta X$ / entanglement-entropy-variation conjecture). The derivation is first-order in the state variation — *exactly like FGHMVR*. The genuine contrast is **not** nonlinear-vs-linearized but: Jacobson reconstructs a *complete local tensor equation pointwise* because Einstein gravity is algebraic in curvature (no derivatives of curvature), so a small-ball first-order condition at every point reconstructs the full tensor equation. Bueno–Min–Speranza–Visser (arXiv:1612.04374) show the *same recipe* yields only the **linearized** equations for higher-curvature gravity — isolating that the "full equation" is a property of Einstein gravity's local curvature-algebraic structure, not of the derivation's perturbative order. FGHMVR instead reconstructs the bulk equation *explicitly linearized about pure AdS*. `[ESTABLISHED, corrected]`

Crucially, Jacobson **presupposes geometry**: the area–entropy density $\eta = 1/(4G\hbar)$ is [ASSUMED, not derived]; the *maximal* (not merely stationary) vacuum-entanglement hypothesis at **fixed volume** is required (fixed-radius spoils the coefficient by $(d+1)/3$ ); and the matter is conformal. It is **non-holographic / local-thermodynamic**, answering a *different* question than the holographic/algebraic OP-41. This is a **reconciliation, not closure** — iteration 2 already conceded Jacobson "presupposes geometry." `[ESTABLISHED]`

## (2) The CLPW Type II$_1$ route breaks — and we now know exactly where

The CLPW static-patch route (arXiv:2206.10780 [VERIFIED]: Type II$_1$, empty dS = maximum-entropy state) was pushed toward linearized Einstein and **broke at a newly-identified structural step**, not merely "stalled." The break is at the conversion-to-tensor step (Step 5 of the attempt), and it breaks for a *structurally different reason* than in AdS:

- **(i) One Killing vector, not a family of anchoring regions [decisive].** AdS FGHMVR obtains a local equation $\delta G_{\mu\nu}(x) = 8\pi G\, \delta T_{\mu\nu}(x)$ because the timelike boundary supplies a *continuum* of ball regions (all positions, sizes, boosts) $\Rightarrow$ a continuum of independent first-law constraints inverting to a pointwise field equation. The dS static patch supplies **exactly one** distinguished region and **one** Killing vector $\xi$ (the static-patch boost). So $\delta S_{\text{gen}} = \delta\langle K\rangle$ yields only the **single scalar constraint**

  $$\int_\Sigma \xi \cdot \delta C = 0$$

  — the $\xi$-contracted, slice-integrated linearized Hamiltonian constraint = horizon-area stationarity = the dS GSL — **not** the tensor $\delta G_{\mu\nu}$. There is no boundary to slide the region along.
- **(ii) No timelike conformal boundary** $\Rightarrow$ no boundary stress tensor, and the "surface" is the cosmological *horizon* (a bifurcate Killing horizon), fixed by background isometry, not an extremized RT min-cut. There is no min-cut/extremization to linearize. `[ESTABLISHED]`
- **(iii) The variational principle is sign-REVERSED.** Banihashemi–Jacobson–Svesko–Visser (arXiv:2208.11706, JHEP 01(2023)054 [VERIFIED]) establish $dE_\xi = -T_{\text{dS}}\, dS_{\text{hor}}$: adding Killing energy *decreases* horizon area/entropy. With empty dS *maximizing* $S_{\text{gen}}$, the dS variational principle is a concave **maximization** ($\delta^2 S_{\text{gen}} < 0$), structurally opposite to AdS RT **minimization** (Hollands–Wald positive-canonical-

energy). The AdS machinery cannot be transcribed. `[INFERENCE, high — rests on established facts]`

- **(iv) The modular Hamiltonian is geometric only for Bunch–Davies.** $K_{\mathrm{BD}} = 2\pi H_\xi$ holds for Bunch–Davies; for a perturbed state, modular flow is non-geometric (no Bisognano–Wichmann), so $\delta\langle K\rangle$ is not a clean boost energy. Geometry is presupposed, not generated (the iteration-2 algebraic-background-independence verdict, now in explicit dS dress; see Working note: *Does the modular / crossed-product (von Neumann algebra) program deliver background independence?* (repo: `notes/2026-06-08-algebraic-background-independence.md`)). `[INFERENCE, high]`

## Net on Program A

Genuine technical progress of the **negative** kind. Iteration 2 said "no CLPW-rooted dS first-law ⇒ Einstein chain exists." Iteration 3 says **why** (one region / one Killing vector vs AdS's continuum; sign-reversed maximization) **and** identifies one concrete, citable escape: a **family of static-patch modular Hamiltonians over the** $SO(d+1,1)$ **observer orbit** (Chen–Xu covariant observers, arXiv:2511.00622 [VERIFIED], which builds exactly this $L^2(SO(1,d))$ QRF family) or **sub-patch CHM causal diamonds** (concretely, the dS-diamond modular Hamiltonian of Fröb, arXiv:2308.14797, JHEP 12(2023)074), to manufacture the missing continuum of constraints.

> **Tagging correction (forced by the verdict review).** The draft tagged the obstruction "near-no-go (HYP-dS-NOMIN)." This **overstates**. No theorem forbids manufacturing the missing continuum; and Fröb cautions that the dS-diamond modular Hamiltonian is geometric *only* in the full-static-patch limit and is non-local for finite diamonds. The route is non-trivial but **UN-foreclosed**. Correct tag: **a sharpened-but-OPEN structural obstruction with one identified, unexecuted escape route** — `[OPEN]`, not [near-no-go]. Whether the $SO(d+1,1)$ family yields a pointwise $\delta G_{\mu\nu} = 8\pi G\,\delta T_{\mu\nu}$ is **untested**; calling it "the missing continuum of constraints" should be hedged to "a candidate mechanism

whose ability to produce a pointwise field equation is untested." `[OPEN]`

**Verdict: Program A advanced in *understanding* but did not advance toward *geometry-generation.***

This is reinforced by the encode-vs-generate criterion (OP-46, see *the criterion note* (Working note: *Formalize: an encode-vs-generate criterion for emergent geometry (OP-46)* (repo: `notes/2026-06-08-iter3-encode-vs-generate-criterion.md`))): the encode/generate frontier coincides *exactly* with the presence of a hand-chosen tensor-factorization $\mathcal{H} = \bigotimes_i \mathcal{H}_i$ or graph $G = (V, E)$ — which is the Type $\mathrm{III}_1$ no-factorization fact in disguise. No surveyed construction generates Lorentzian spacetime from factorization-free, region-free, symmetric-state-free data. Cao–Carroll(–Michalakis) "space from Hilbert space" comes closest, but outputs only a *Riemannian spatial* metric and only after positing a factorization (a surrogate for input-geometry $g_5$). [INFERENCE, high on the classification; medium on the criterion being uniquely right]

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## The A+B merge attempt

**SUBSTANTIVE PARTIAL MERGE at one sublayer; thin methodological bridge at the deeper sublayer; NOT empty re-description, NOT a full merge.** A real but bounded upgrade to iteration-2's "thin bridge." `[INFERENCE, medium]` (See *the merge note* (Working note: *Attempt: Merge Program A (geometry-from-algebra) and Program B (quantum-framework-final)* (repo: `notes/2026-06-08-iter3-merge-attempt-operational-substrate.md`)).)

The shared object is a von Neumann algebra with faithful normal state $(\mathcal{M}, \omega)$ under the no-signaling / normal-steering (GHJW) constraint.

### What merged — the Born-form + thermal/entropic sublayer

On $(\mathcal{M}, \omega)$ one can **non-circularly co-derive**:

- **(a) Pillar B's Born-rule FORM** + exact linearity, lifted to infinite-dim normal state spaces by the 2026 causal-rigidity theorem (arXiv:2602.09056 [VERIFIED], published 7 Feb 2026: recovers Born on vN-algebra normal state spaces via GNS, under no-signaling + normal steering + $\sigma$-affinity); and
- **(b) Pillar A's KMS/thermal-time** (Tomita–Takesaki) + emergent modular/causal flow + (via an *internal* crossed-product clock) a Type II trace, finite generalized entropy, GSL.

Neither pillar presupposes the other on this sublayer, and the "observer" is demoted to an internal gauge-invariant **quantum reference frame** (CLPW + covariant-observer 2511.00622 [VERIFIED]), genuinely dissolving the agent/no-agent category clash **here**.

### What did NOT merge — the Hilbert-space-FRAMEWORK sublayer

The GPT axioms that single out Hilbert space / the complex field / tensor-product composition have **no Type III formulation** and provably **fail in infinite dimensions**. Three located breaks (candidate-no-go **OP-GPT-TYPE3**):

- **BREAK 1 — composition not even statable.** Local tomography presupposes $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$; Type $\text{III}_1$ has no tensor factorization and no density matrices (Connes–Haag–Araki) `[ESTABLISHED]`. The 2026 theorem is honest: it lifts only the Born-rule *form*, not composition.
- **BREAK 2 — load-bearing axioms false in $\infty$-dim.** Hardy continuous-reversibility and CDP purification fail (the shift operator is an isometry, not unitary; orthogonal projections are not sequentially compact) `[ESTABLISHED]`.
- **BREAK 3 — the crossed-product trace re-imports geometry.** The III $\to$ II crossed-product trace exists only with a clock Hamiltonian bounded below in a fixed-background semiclassical perturbative window — re-importing the preferred-state/fixed-geometry that Pillar A should *generate* (Bisognano–Wichmann/CHM circularity), and may fail

nonperturbatively (Giddings 2505.22708/2510.24833 [VERIFIED]). `[INFERENCE, high]`

### Net on the merge

**YES** for a single non-circular substrate on the Born-form + thermal/entropic sublayer; **NO** (with the precisely-located candidate-no-go OP-GPT-TYPE3) for the Hilbert-space-framework sublayer. The shared no-signaling principle does real non-tautological work on both sides (so *not* empty re-description), but composition/factorization cannot be posed on Type III (so *not* a full merge). The merge **"succeeded narrowly, failed deeply"** — and crucially it did **not** merge in the cosmological closed-universe / no-external-observer regime that was the prompt's crux, because that is exactly where the BREAK 3 perturbative window closes. `[INFERENCE, medium]`

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## Red-team impact (what we weakened)

The red-team found **0 fatal errors, 4 serious cracks, 2 cosmetic**; all load-bearing 2025–2026 citations re-verified exactly. The cracks are the synthesis page **over-compressing better-hedged source notes**. Four forced weakenings: `[INFERENCE/CONTESTED, high that the weakenings are warranted]`

1. **"Fixed cores" re-tagged from flat to HETEROGENEOUS.** The five "fixed cores" are not the same kind of object, and three are not actually fixed:
   - **(i) Type III$_1$ "for any $\Lambda$"** is `[ESTABLISHED]` for QFT on a *fixed background* but `[CONTESTED/OPEN]` as a fixed core of the *gravitational* theory — Giddings shows gravitational dressing alters the algebra at leading order in $G_N$. The synthesis page listing it as ESTABLISHED-fixed **directly contradicts** the wiki's own algebraic-background-independence note (which tags it OPEN/CONTESTED, and Giddings backs the note). **This is a genuine internal**

  **inconsistency** that must be resolved by propagating the note's qualification up.

  - **(ii) Area law** is NOT a universal input — derived in AdS (RT), output in Sakharov; load-bearing only for the Jacobson–Padmanabhan–Verlinde lineage specifically.
  - **(iii) No-signaling** is genuinely shared but NOT quantum-forcing (PR-boxes are no-signaling yet non-quantum) — demote from "core that forces structure" to "one of several GPT axioms."
  - **(iv) Born form** is forced only conditional on non-contextuality + effect-additivity.
  - **(v) Unitarity** is conditional on standard QM being final and is deliberately violated by live GRW/CSL.
  - Honest header: *"a small set of robust-but-non-homogeneous anchors, each conditional on a stated background or premise."*

2. **"Demotion-and-derivation" demoted from "falsifiable program" to "research HEURISTIC."** As a meta-strategy it forbids nothing and is unfalsifiable *by* scoping; its only falsifiable content lives in the object-level deliverables (OP-dS-FIRSTLAW, OP-CP1, BMV $m^2$-vs-$m^3$, collapse bounds) and in the bracketed `[SPECULATIVE]` wager it quarantines. "Falsifiable" was a category error applied to a procedure.
3. **"Independent convergence of six lenses" weakened to "convergence of methods sharing load-bearing inputs"** (a streetlight artifact). H0/H4 share Killing-symmetric vacua + AdS/large-$N$; H1/H3/H6 share the continuum Malament/HKM theorem + the unproven Hauptvermutung; H7 is AdS/large-$N$. Agreement of methods sharing their assumptions is weaker evidence for the wager than genuine independent convergence.
4. **CLPW Type $\mathrm{II}_1$ kept as a $\Lambda > 0$ entropy foothold but stripped of "$\Lambda$-DIAGNOSTIC" billing.** Trace normalizability ($\mathrm{II}_1$ vs $\mathrm{II}_\infty$) is set by the horizon/observer-Hamiltonian spectrum, not by $\Lambda$: Schwarzschild–de Sitter ($\Lambda > 0$) is $\mathrm{II}_\infty \times \mathrm{II}_\infty$ (KLS, PRD 111 025013 [VERIFIED]). So $\mathrm{II}_1$ is a feature of the

static patch's *bounded area*, not of $\Lambda > 0$ as such — a persistent emphasis error.

**What survived every attack** (the wiki's best work): the deflationary core; the "encodes-not-generates geometry" verdict (STRENGTHENED by Giddings); the Born form/meaning split; linearity-as-derivative-of-no-signaling; the PARTIAL-coherence verdict; the refusal of the "it's all information" fusion; the dS negative meta-result; BMV-as-class-falsifier-not-quantization-proof.

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## Empirical falsifiability portfolio

Every live channel tests the program's logical **GATE** (is gravity non-classical?), its correct **TARGET** (is the universe really dS?), or the status of its **FIXED CORES** (is linearity/Lorentz exact?). **None tests the geometry-from-algebra wager itself.** `[INFERENCE, high]`

| Experiment | Decides which claim | Current status | Timeline (yrs) |
|---|---|---|---|
| **BMV / QGEM** gravitationally-induced entanglement: two $\sim 10^{-14}$ kg masses in superposition; measure $m^2$ vs $m^3$ witness scaling | Program A's logical **GATE** (REG-SYNTH-4 / OP-40). Positive result excludes mean-field semiclassical gravity + Kafri–Taylor–Milburn classical-channel models; $m^2 =$ quantum mediator, $m^3 =$ Aziz–Howl virtual-matter loophole. Does NOT prove graviton quantization (HYP-GIE-DISCRIM). Refutes the wager's gate if gravity is shown fundamentally classical | `[OPEN, not yet performed]` Mass/coherence $\sim$ 6–10 orders beyond current matter-wave records; General Witness Theorem locality premise actively `[CONTESTED]` (Di Biagio; Aziz–Howl $m^3$ rebuttals, 2025) | 10–20+ (2030s–2040s; not imminent) |

| | | | |
|---|---|---|---|
| **Objective-collapse** (CSL / Diósi–Penrose) bounds: matter-wave interferometry, optomechanics, ultracold cantilevers, MAQRO-class space interferometry, spontaneous-X-ray/heating searches | Directly tests A-3-REF / linearity-as-derivative-of-no-signaling. Stochastic collapse EVADES the deterministic Gisin–Polchinski no-go and is the empirically live way exact linearity could be FALSE. Detection refutes "exact unitarity as a fixed core" (weakening (1)(v)); null results push $\lambda$ down. Also bounds DP gravitational self-collapse competing with BMV | `[OPEN, actively bounding]` X-ray/LISA-Pathfinder limits already exclude the parameter-free Diósi–Penrose model + large swaths of GRW/CSL space `[INFERENCE]`; no positive detection | 0–10 (decisive CSL coverage by $\sim$2035) |

| | | | |
|---|---|---|---|
| **Dynamical dark energy** $w(z)$: DESI DR2/DR3+, Euclid, Roman, LSST/Vera Rubin; CPL $w_0, w_a$ vs $-1$ | The correct **TARGET** for Program A (ASSUME-ETERNAL-dS / wrong-idealization risk). $w(z) \neq -1 \Rightarrow$ aim at metastable dS / rolling quintessence, not eternal dS, weakening motivation for a strict $\Lambda > 0$ first law (OP-dS-FIRSTLAW). Does NOT test the wager directly — tests which background it must reproduce | [CONTESTED, \$\sim 3–4\sigma\$ hint] DESI DR2 (arXiv:2503.14738 [VERIFIED]) 2.8–4.2\$\sigma\$ for dynamical DE (3.1\$\sigma\$ DESI+CMB); 2025–26 joint analyses (e.g. arXiv:2511.22512) report persistence. Not yet 5\$\sigma\$; SNe-systematics-dependent | 2–7 (DR3+ / first Euclid+Roman through \$\sim\$2030) |

| | | | |
|---|---|---|---|
| **LIV / EP** tests: GRB photon time-of-flight (Fermi-LAT, CTA), birefringence, clock-comparison; torsion-balance (Eöt-Wash), LLR, MICROSCOPE $\eta <\sim 10^{-15}$ | Robustness of the surviving **FIXED CORES**: causal order / microcausality and the metric GR presupposes. Confirmed LIV refutes "exact local Lorentz + single causal order is held-fixed" (and threatens emergent-causal-order tracks H1/H3/H6 if discreteness breaks Lorentz). EP violation ⇒ new gravity-sector physics constraining entropic/emergent models (Verlinde-2016 already disfavored) | `[ESTABLISHED null to high precision]` No confirmed LIV; some Planck-scale dispersion excluded by GRBs `[INFERENCE]`. MICROSCOPE confirms EP to $\sim 10^{-15}$. Most likely a null-result anchor, not a discovery channel | 0–10 (C next-ger through $\sim$2 |
| **Matter-wave interferometry** at increasing mass: Talbot–Lau, large-molecule/nanoparticle interferometers, levitated optomechanics | Foundational **enabler** shared by Program B's two live channels: builds toward BMV mass scale AND bounds objective collapse (mass-dependent interference loss = collapse signal). Empirical content behind demoting "unitarity" from core to conditional | `[OPEN, advancing]` Superposition verified up to $\sim 10^4$–$10^5$ amu + nanoparticle regimes; orders below BMV/collapse-decisive masses `[ESTABLISHED for current records]` | 5–15 (th bridge technolo gating b BMV an decisive collapse tests) |

## Net progress

Iteration 3 produced **real but almost entirely NEGATIVE / STRUCTURAL progress, and the coherence verdict is UNCHANGED at PARTIAL — correctly so.** `[INFERENCE, high]`

Three of the four moves sharpened obstructions; the fourth forced retractions back to the source notes:

1. **The dS-first-law attempt did NOT advance Program A toward geometry-generation.** It converted iteration-2's bare "no dS first-law chain exists" into a located structural obstruction — one region / one Killing vector vs AdS's continuum, plus a sign-reversed concave maximization — leaving exactly **one un-foreclosed escape** (an $SO(d+1,1)$ family of modular Hamiltonians, concretely instantiated by Chen–Xu covariant observers and dS-diamond CHM constructions). Tagged `[OPEN]`, not "near-no-go."
2. **The encode-vs-generate criterion is a genuine deliverable.** It makes the central distinction decision-usable (two graded axes $\Delta_{\text{geo}}$, $\kappa_\Phi$ + a 5-item smuggling checklist $g_1$–$g_5$), classifies six constructions, and isolates that the encode/generate frontier coincides *exactly* with a hand-chosen factorization/graph — the Type $\text{III}_1$ no-factorization fact in disguise. Net finding: **NO surveyed construction generates Lorentzian spacetime from factorization-free, region-free, symmetric-state-free data.**
3. **The A+B merge succeeded only on the Born-form + thermal sublayer and failed** (candidate-no-go OP-GPT-TYPE3) on the Hilbert-space-framework sublayer — notably failing in the cosmological no-observer regime that was the crux.
4. **The red-team found no fatal errors but forced four weakenings**, including resolving a genuine internal contradiction (Type $\text{III}_1$ as "fixed core" once gravity is on).

The **deflationary core** and the **central diagnosis** ("encodes, does not generate") survived every attack and were STRENGHTENED by the verified Giddings results. The **wager** remains `[SPECULATIVE]` and, crucially, **without any distinguishing near-term test**: every live channel tests the gate, the target, or the cores — none tests geometry-from-algebra. By the wiki's anti-crank standard, the wager is therefore **still not yet physics**.

**Honest one-line summary:** more rigorously understood, more honestly hedged, obstructions sharper, central bet no closer to confirmation or empirical decidability.

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## Next highest-leverage targets

1. **[Re-aimed OP-dS-FIRSTLAW-R2 — single highest-leverage theory target.]** Attempt the one un-foreclosed escape: build a FAMILY of static-patch modular Hamiltonians over the $SO(d+1,1)$ observer orbit (Chen–Xu, arXiv:2511.00622, $L^2(SO(1,d))$ QRF) or sub-patch CHM diamonds (Fröb, arXiv:2308.14797), and test whether the family supplies enough **independent** first-law constraints — each with geometric modular flow in a **common** state — to invert to a local $\delta G_{\mu\nu}$. If it cannot (constraints not independent, or modular flow geometric only in Bunch–Davies), HYP-dS-NOMIN is promoted from conjecture to a true no-go and the holographic dS route is closed. `[OPEN]`
2. **[Settle OP-GPT-TYPE3 — converts the partial merge into a full merge or a true no-go.]** Decide whether the GPT framework-fixing axioms (local tomography/tensor-composition, purification, continuous reversibility) admit ANY formulation on a Type $\mathrm{III}_1$ factor or its normal state space. Concretely: can the 2026 causal-rigidity theorem's $\sigma$-affinity + normal-steering be strengthened to fix **composition**, not just the probability rule? If composition provably needs factorization

(which $III_1$ lacks), this is a no-go for the Hilbert-space-framework half. `[OPEN]`

3. **[Break the Bisognano–Wichmann/CHM circularity — deepest reorganization-vs-generation test.]** Recover a Killing/boost structure and a horizon as the **OUTPUT** of an $(\text{algebra}, \text{state})$ pair whose modular flow is NOT a known symmetric (BW/Hadamard) vacuum. This is the criterion's clause (iii), satisfied by no surveyed construction; succeeding here is the only thing that would turn "encode" into "generate." `[OPEN]`
4. **[Resolve the nonperturbative fate of the crossed product — OP-CP1.]** Determine whether the III $\rightarrow$ II crossed-product Type II structure survives beyond all-orders-in-$1/N$, or is replaced by Giddings' non-algebraic structure (arXiv:2510.24833). Bears on BREAK 3 of the merge and on whether "Type $III_1$ for any $\Lambda$" can be a fixed core at all once gravity is on. Either answer is decisive. `[OPEN/CONTESTED]`
5. **[Propagate the red-team weakenings into the synthesis page — documentation, not research.]** Re-tag the "fixed cores" list as heterogeneous with per-item status; demote "falsifiable program" to "research heuristic"; add the shared-input/streetlight caveat to the convergence claim; strip "$\Lambda$-diagnostic" billing from CLPW $II_1$. Required for the wiki to stop contradicting its own better-hedged notes.
6. **[Mature the shared enabling technology — highest-leverage EMPIRICAL target.]** Push matter-wave / levitated-optomechanics mass scale upward: it simultaneously gates the BMV gate-test of Program A AND tightens objective-collapse bounds on the linearity assumption — the two live channels of Program B — and is the prerequisite both ultimately depend on. **No theory target is empirically decidable without it.**

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CHAPTER 24

# Iteration 4 Synthesis: Diminishing Returns, Measured

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Iteration: 4

This note folds the iteration-4 attacks and the synthesis-verdict review into one record. The honest one-line headline: **iteration 4 moved no headline claim — it reinforced all three — by closing the last open escape route from iteration 3, settling one open problem as a sharp disjunction, and promoting one obstruction toward a near-no-go; the marginal yield is now a sharper obstruction or a self-correction, not new positive structure.** `[INFERENCE, high]`

The three headline claims are unchanged: (1) coherence is **PARTIAL**; (2) the geometry-from-algebra program **ENCODES** but does not **GENERATE** geometry; (3) the central wager is **not-yet-physics** — it has no distinguishing near-term test. What changed is *precision* and the *closing of escape routes* — convergence behavior, the third consecutive iteration at this verdict. `[INFERENCE, high]`

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## Did the standing verdict change?

**NO.** Unchanged at the headline level, **sharpened** at the structural level. `[INFERENCE, high]`

The standing verdict holds across iterations 2 $\rightarrow$ 3 $\rightarrow$ 4: PARTIAL coherence; geometry-from-algebra encodes, does not generate, geometry; the wager is not-yet-physics. Iteration 4 did not move any of these — it reinforced all three. The central diagnosis ("encodes, does not generate") is now backed by a *much harder* argument than iteration 3 supplied.

The de Sitter first-law obstruction is upgraded from iteration-3's hedged `[OPEN]` **sharpened obstruction with one unforeclosed escape** to a **scoped near-no-go**: the one escape (the $SO(d+1,1)$ static-patch modular-Hamiltonian family) was actually *executed* and fails on three independent, individually-sufficient grounds. The decisive one is a hard **cardinality mismatch** — a finite-dimensional isometry orbit ($\dim \mathcal{O} = 2d = 6$ in 4d) supplying at most $\dim SO(d+1,1) = 10$ scalar Killing-charge constraints, which cannot Radon-invert to the infinite-dimensional local field $\delta G_{\mu\nu}(x)$ (6 independent functions on $M^4$ after Bianchi). *INFERENCE, high — rests on the established FGHMVR Radon-inversion mechanism and the verified finite orbit dimension; see the [attempt note* (Working note: *Iteration 4: the SO(d+1,1)-family escape from the dS first-law obstruction (OP-41 / HYP-dS-NOMIN)* (repo: `notes/2026-06-08-iter4-dS-firstlaw-SO-family-attempt.md`)).]

This is **NOT** an unconditional theorem. It is explicitly scoped to the strict **unitary, single-static-patch, isometry-orbit, common-state** route. Two routes remain genuinely open: the non-unitary dS/CFT pseudo-entropy route (arXiv:2511.07915), and Jacobson's non-algebraic small-ball continuum (which already works, but is non-holographic). `[INFERENCE, high]`

The load-bearing citation is verified. **Chen–Xu, arXiv:2511.00622, "An algebra for covariant observers in de Sitter space"** (Bin Chen, Jie Xu) is the *published execution* of exactly this route — an averaged modular crossed product over static patches, the $L^2(SO(1,d))$ QRF, all $SO(1,d)$ constraints — and it reaches entropy-matching and a (second-order) quantum first law but derives **no** local Einstein equation / $\delta G_{\mu\nu}$, landing precisely where the cardinality argument predicts. `[ESTABLISHED that the paper exists and is so characterized]`

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## Program A status

**OBSTRUCTION SHARPENED / one target promoted toward no-go — NOT advanced toward geometry-generation.** `[INFERENCE, high]`

Program A (geometry-from-algebra) made **zero** progress toward *generating* geometry at $\Lambda > 0$ and lost its single highest-leverage open theory target. Specifically:

- **HYP-dS-NOMIN promoted** from iteration-3's `[INFERENCE-conjecture + one un-foreclosed escape]` to a scoped near-no-go (new registry item **HYP-dS-NOMIN-R3**):

  > There is no derivation of the local tensor equation $\delta G_{\mu\nu} = 8\pi G\,\delta T_{\mu\nu}$ from a dS static-patch generalized-entropy first law $\delta S_{\text{gen}} = \delta\langle K\rangle$ via the FGHMVR mechanism, even after enlarging to the full $SO(d+1,1)$ family, for three independent reasons: **(i)** finite-dimensional orbit $\Rightarrow \leq 10$ scalar Killing-charge constraints, cannot invert to the infinite-dim local field; **(ii)** the family is geometric in a *common* state only at Bunch–Davies, where the probing perturbation $\delta\Psi$ is null; **(iii)** the non-trivial first-law content is second-order, where $\delta^2 S_{\text{gen}} < 0$ renders it a concave GSL inequality, not an invertible equality. `[INFERENCE, high; near-no-go for the scoped route]`

- **A reusable generalization extracted** (new registry item **HYP-dS-CARDINALITY**): Einstein-from-first-law requires an **infinite-dimensional local region family probing every bulk point** (in AdS: an RT surface through every bulk point in every orientation, Radon-invertible to $\delta G_{\mu\nu}(x)$). Isometry orbits of a maximally-symmetric background are **finite-dimensional** and yield only global Killing charges. Therefore local dS dynamics from entanglement requires an *independently infinite local family* (Jacobson's small geodesic balls — one at every bulk point in every frame), which the isometry orbit structurally lacks. This converts a soft "finiteness worry" into a hard dimension count, and is the most substantive product of iteration 4. `[INFERENCE, high]`

The escape routes that survive — dS/CFT pseudo-entropy with complex central charge $c = i\,3R_{\mathrm{dS}}/2G_N$ (2511.07915); Jacobson's continuum realized *inside* a single CLPW Type $\mathrm{II}_1$ algebra (importing the local continuum without a boundary) — are exactly the iteration-3-listed verdict-movers, now isolated as the **only** remaining ways Program A could move from encode to generate at $\Lambda > 0$.

**Status verb: the diagnosis is reinforced and the live frontier is narrower, not closer to a positive result.**

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## Merge status

**UNCHANGED in kind, SHARPENED in detail.** `[INFERENCE, medium]`

The A+B merge remains **"narrow success / deep failure"**: the Born-form + thermal/entropic sublayer merges non-circularly on a shared $(\mathcal{M}, \omega)$; the Hilbert-space-FRAMEWORK sublayer (GPT composition / local-tomography) does not. Iteration 4 did **not** convert the partial merge into a full merge, nor into a clean no-go — but it **settled OP-48 / OP-GPT-TYPE3 as a sharp DISJUNCTION** with a precise dividing line, replacing iteration-3's vaguer "candidate-no-go / missing-theorem gap":

- **(a) NO-GO** for exact GPT local tomography / tensor composition on a **single Type $\mathrm{III}_1$ factor** (no factorization, no density matrix). `[ESTABLISHED]`
- **(b)** State-/collar-dependent **YES** for nested regions $\mathcal{O}_1 \subset \mathcal{O}_2$ under the **split property** — an exact tensor factorization $\mathcal{H} \cong \mathcal{H}_1 \otimes \mathcal{H}_2$ with local density matrices and statistical independence *does* exist, defeating the would-be hard no-go **as a no-go**. `[ESTABLISHED]`
- **(c)** But it is **NOT** a clean operational reconstruction, for two live-web-verified reasons:

1. The canonical intermediate Type I factor $\mathcal{F} = \mathcal{R}(K) \vee J\,\mathcal{R}(K)\,J$ is made canonical **only** by the vacuum $\Omega$'s modular conjugation $J$ (infinitely many non-canonical factors otherwise) — re-importing the BW/CHM geometric reference state Program A was meant to *generate* (new registry item **HYP-SPLIT-GEOM**: the split-property rescue **relocates** the circularity to the factorization stage, it does not escape it).
2. It requires a **nonzero collar** and fails for touching/adjacent regions, so the operationally natural "compose adjacent subsystems" has no exact tensor structure.

The split-property facts (canonical factor fixed by $J$ relative to the vacuum; nontrivial relative commutant) are web-verified. Confirmed (via the literature track) that the 2026 Born-form rigidity result **arXiv:2602.09056** and its finite-dim predecessor (both Torres Alegre) treat only the single-system probability **RULE** and do **not** fix composition — so the merge's deeper sublayer stays unmerged.

**Net:** a full merge still requires solving (c) — a state-/collar-free composition surrogate on a Type III$_1$ factor — which remains **OPEN**. (See the forthcoming attack note `2026-06-08-iter4-gpt-type3-resolution.md`.)

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## Literature-refresh impact

**ONE genuinely claim-moving result; ONE citation defect to scrub; the rest re-confirmation.** `[ESTABLISHED]` Every citation independently checked on the live web verifies exactly; **NO fabrication anywhere** — there are no fabricated IDs to scrub, only one *misattribution* and assorted residue.

### Claim-moving (action = update): Muon $g-2$

**arXiv:2505.21476** (Muon $g-2$ Theory Initiative WP2025), verified via WebSearch against the Theory-Initiative and CERN Courier pages, switches the leading-order HVP input to a **lattice-QCD average**, shifting the SM prediction to

$a_\mu^{\rm SM} = 116\,592\,033(62) \times 10^{-11}$,

so that the difference with the experimental world average is $\Delta a_\mu = 38(63) \times 10^{-11}$ — **no tension**. The historic $\sim 4$–$5\sigma$ muon $g-2$ anomaly is thereby largely **resolved into SM-consistency** (an internal lattice-vs-$R$-ratio / CMD-3 tension persists). Any wiki page still listing muon $g-2$ as a *live BSM anomaly* is now **STALE** and must be updated. [VERIFIED]

> Scope note: muon $g-2$ is **peripheral** to the central wager — it touches the Standard-Model-precision core, not geometry-from-algebra — so it does **not** move the headline verdict, only a domain page. `[INFERENCE, high]`

## Citation to scrub (defect, not fabrication)

**arXiv:2308.14797 is REAL but was MISATTRIBUTED** in the iteration-3 synthesis note as "Forste et al." Verified on the live arXiv abstract: it is **"Modular Hamiltonian for de Sitter diamonds," single-authored by Markus B. Fröb, JHEP 12 (2023) 074.** Iteration 4 correctly flags and fixes this (new item **GAP-FROEB-CITATION**); a cleanup task to propagate the fix has been spawned. [VERIFIED]

## Three previously-flagged IDs cleared ([unverified] → [web-confirmed])

| arXiv ID | Author / venue | Content | Promotion note |
|---|---|---|---|
| **2603.06211** | J. Zhang (Oxford) | "Summing to Uncertainty": additivity is the load-bearing premise across five Born-rule derivations (Gleason, Busch, Deutsch–Wallace, Zurek, Finkelstein–Hartle) | REAL; promote to web-confirmed |
| **2601.18856** | V. Pronskikh (Fermilab) | "Operationally induced preferred basis in unitary quantum mechanics"; exact-title match | REAL; promote to web-confirmed |

| | | | |
|---|---|---|---|
| **2601.13012** | A. K. Pati | "No-Signalling Fixes the Hilbert-Space **Inner Product**" | REAL; **trim the "/linearity" rider** on promotion — the paper fixes the inner product, **not** linearity (it *assumes* linear/unitary dynamics) |

All three were genuinely flagged with prudence; the papers are genuine. The one required edit is the inner-product-not-linearity trim on 2601.13012. [VERIFIED]

### Non-moving refreshes (action = add / none)

- Newer collapse-model bounds (2401.04665, 2411.17588, 2512.02838): tighten but do **not** close CSL/DP space; KA-4 / O-4 conclusion unchanged.
- First Lorentzian-signature asymptotic-safety fixed-point evidence (D'Angelo, arXiv:2310.20603, PRD 109 066012): incremental, still `[CONTESTED]`, no distinguishing test.
- DESI $w(z)$ still DR2; the 2.8–4.2$\sigma$ hint stands; DR3 forthcoming — no claim moves.
- Island-beyond-AdS and KLS / crossed-product follow-ups already current.

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## How far from a conclusion

**Two DIFFERENT distances, and they must not be conflated.** `[INFERENCE, high]`

### Analytical saturation — CLOSE (≈ 1–3 more focused iterations)

The analytical conclusion is already largely in hand and stable across iterations 2 $\rightarrow$ 3 $\rightarrow$ 4:

1. **"Is a universal theory possible?"** is answered in disambiguated form:
   - **(A)** a consistent unified GR+QFT framework is **POSSIBLE** at high confidence (AdS/CFT is an existence proof; the 17 catalogued clashes are domain-mismatches, not contradictions);

   - **(B)** a final substrate is possible but **not demonstrably so** (a Wilsonian tower-with-no-bottom is live);
   - **(C)** a parameter-fixing complete theory is **least likely**.
2. The one central wager is named precisely: *causal/algebraic/entanglement structure precedes metric geometry, true of our* $\Lambda > 0$ *universe*.
3. **Why** it is not-yet-physics is established and has survived every attack across three iterations: it encodes but does not generate geometry, and it has no distinguishing near-term test.
4. Iteration 4 **shrank** the remaining analytical to-do list by closing the single highest-leverage open theory target (OP-41 via its only un-foreclosed route).

What remains is a **finite, named** list:

- settle **OP-48 / OP-GPT-TYPE3 point (c)** — the state-/collar-free composition surrogate (full merge vs clean no-go);
- **OP-44 / OP-CP1** — the nonperturbative fate of the crossed product once gravity is dynamical (whether "Type $\mathrm{III}_1$ for any $\Lambda$" can even be a fixed core);
- the **OP-46 clause-(iii)** construction — recover Killing/horizon as the OUTPUT of a genuinely non-symmetric, non-CKV modular flow (the *only* move that turns encode into generate).

The OP-46 escape is itself now **near-vacuous** under the standard geometric readout (new registry item **HYP-CKV-VACUITY**). Two chained theorems, web-verified, force this: **(a)** Borchers/Wiesbrock — half-sidedness $\Leftrightarrow$ positivity, so the $ax + b$ group is generated *from the relocated positivity premise*, not from neutral data; **(b)** Caminiti et al. (arXiv:2502.02633) — recovered boost generators are *conformal Killing vectors*, which presuppose a symmetric background. The vacuity is **conditional on the standard readout** — a non-CKV readout (modular Berry / Connes spectral distance) is the genuinely live, currently-unrealized target. After these are settled, marginal structural insight goes to near-zero and the program becomes a watch-list. [INFERENCE, high on the classification; medium that the vacuity is a true no-go rather than a limitation of the standard readout]

### Object-level confirmed theory — NOT FORECASTABLE

This is a *stronger* obstruction than mere distance. It is gated by two things the program cannot schedule:

- **(i)** the $\sim 15$-order-of-magnitude empirical desert between accessible energies ($\sim$ TeV) and $M_{\mathrm{Pl}} \approx 1.22 \times 10^{19}$ GeV;
- **(ii)** more bindingly, the central wager currently has **NO distinguishing experimental test even in principle.** Every live channel tests the program's logical **GATE** (is gravity non-classical? — BMV/QGEM, gated by matter-wave/optomechanics mass scale, $\sim 5$–$15$ yr), the status of its fixed **CORES** (is linearity/Lorentz exact? — objective-collapse bounds decisive $\sim 2035$; LIV/EP nulls), or its correct **TARGET** (eternal dS? — DESI DR3+/Euclid $w(z)$, 2–7 yr). **None tests geometry-from-algebra itself.**

So even optimistic enabling-technology timelines would open the **gate**, not confirm the theory. The object-level conclusion lacks a defined experimental decision procedure — it is non-forecastable, not merely far. `[INFERENCE, high]`

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## Recommendation

**HYBRID.** [confidence: high]

Iteration 4 mostly sharpened obstructions and confirmed convergence — say so plainly. It produced one genuinely new high-value **NEGATIVE** result (the dS cardinality no-go), settled one open problem as a sharp **disjunction** (OP-48), promoted one open obstruction toward a near-no-go (the clause-(iii) vacuity), found **ZERO** new positive structure, and stabilized the PARTIAL / not-yet-physics verdict for the **third consecutive iteration**. The new-insight trend is **DIMINISHING** — each iteration's marginal yield is increasingly a sharper obstruction or a self-correction, not new positive structure or a new degree of freedom in the problem.

Recommended allocation for the remaining $\sim$ 1–3 iterations:

1. **ATTACK the three named, genuinely-decidable analytical problems.** OP-48 point (c) and OP-44/OP-CP1 each saturate a gap either way; the OP-46 non-CKV-readout construction (modular Berry / Connes spectral distance on a non-symmetric state, yielding a **Lorentzian** metric) is the **only** move that would change the headline verdict — note that Connes reconstruction gives only a **Riemannian** manifold (new registry item **GAP-NCG-LORENTZIAN-SIGNATURE**, web-verified).
2. **DO the two cheap documentation passes:** the spawned propagation/citation-scrub task, and the muon $g-2$ staleness update.
3. **SIMULTANEOUSLY stand up lightweight tracking** for the four empirical channels (BMV $m^2$-vs-$m^3$; collapse bounds $\sim$ 2035; $w(z)$ DESI DR3+/Euclid; LIV/EP), since those — not further analysis — gate the object-level verdict.

Pure continue-attacking risks diminishing returns; pure pivot-to-tracking prematurely abandons 2–3 genuinely decidable named problems.

**Accept all new registry items** — each is web-grounded and correctly tagged: **HYP-dS-NOMIN-R3**, **HYP-dS-CARDINALITY**, **HYP-CKV-VACUITY**, the **OP-GPT-TYPE3** refinement, **HYP-SPLIT-GEOM**, **GAP-FROEB-CITATION**, **GAP-LL-ENERGY-POSITIVITY-R2**, **GAP-NCG-LORENTZIAN-SIGNATURE**.

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## Next targets

1. **Attack OP-48 point (c).** Exhibit (or prove impossible) a **state-independent, collar-free** composition surrogate that singles out QM from split-composition axioms on a Type $\mathrm{III}_1$

factor. A constructive *yes* converts the partial A+B merge into a full merge; a proven *no* is a clean no-go fixing the boundary of the analytical question. Either answer saturates a named gap. `[OPEN]`

2. **Resolve OP-44 / OP-CP1.** The nonperturbative fate of the crossed-product Type II structure once gravity is dynamical — does a von Neumann-algebraic subregion description survive beyond all-orders-in-$1/N$, or does Giddings' algebraic spacetime disruption (arXiv:2505.22708 / 2510.24833) replace it? Decides whether "Type $III_1$ for any $\Lambda$" can be a fixed core at all. `[OPEN/CONTESTED]`
3. **Attempt the OP-46 clause-(iii) escape under a NON-CKV readout.** Recover a Killing/boost structure + horizon as the OUTPUT of an $(\text{algebra}, \text{state})$ pair whose modular flow is genuinely non-symmetric (non-Bunch–Davies, non-Hadamard), using modular-Berry information-metric or Connes spectral distance, **crucially yielding a Lorentzian (not merely Riemannian) metric.** This is the only move that would turn "encode" into "generate" and change the headline verdict. `[OPEN]`
4. **Attempt the two un-foreclosed dS routes** that survive the cardinality no-go: (a) the non-unitary dS/CFT pseudo-entropy route (2511.07915) accepted as physics despite a complex central charge, or (b) Jacobson's infinite small-ball family realized **inside** a single CLPW Type $II_1$ algebra (importing the local continuum without a boundary). Either would move Program A from encodes toward generates at $\Lambda > 0$. `[OPEN]`
5. **Documentation/hygiene (cheap, required).** Update any wiki page treating muon $g-2$ as a live BSM anomaly to SM-consistent per WP2025 (arXiv:2505.21476); propagate the spawned iteration-3 red-team fixes (heuristic-not-program relabel; Caves–Fuchs–Manne–Renes spelling; Fröb-not-Forste byline) into the iteration-2 synthesis note; promote 2603.06211 / 2601.18856 / 2601.13012 to web-confirmed, trimming the "/linearity" rider on 2601.13012.

6. **Stand up lightweight experiment-tracking** for the four empirical channels (BMV $m^2/m^3$ $\sim2030s–40s; objective-collapse decisive $\sim 2035; $w(z)$ DESI DR3+/Euclid 2–7 yr; LIV/EP nulls), since the object-level verdict is gated by these, not by further analysis — and none currently tests the geometry-from-algebra wager directly.

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## Iteration-4 notes

- Working note: *Iteration 4: the SO(d+1,1)-family escape from the dS first-law obstruction (OP-41 / HYP-dS-NOMIN)* (repo: `notes/2026-06-08-iter4-dS-firstlaw-SO-family-attempt.md`) — the executed $SO(d+1,1)$-family escape; the cardinality no-go (HYP-dS-NOMIN-R3) and its three failure counts.
- `2026-06-08-iter4-gpt-type3-resolution.md` — OP-48 resolved as a sharp disjunction (Type $\mathrm{III}_1$ no-go vs split-property qualified yes); HYP-SPLIT-GEOM. [companion attack note]
- `2026-06-08-iter4-modular-circularity.md` — the OP-46 Bisognano–Wichmann/CHM circularity test; HYP-CKV-VACUITY; GAP-NCG-LORENTZIAN-SIGNATURE. [companion attack note]

## Prior synthesis and registries

- *Chapter 23 (p. 582)* — the iteration-3 synthesis this one updates (and whose "Forste et al." mis-citation is corrected to Fröb).
- *Chapter 22 (p. 571)* — the iteration-2 synthesis (target of the pending propagation/citation-scrub task).
- Working note: *Does the modular / crossed-product (von Neumann algebra) program deliver background independence?* (repo: `notes/2026-06-08-algebraic-background-independence.md`) — the encode-not-generate / Type $\mathrm{III}_1$ background note.
- *Chapter 21 (p. 497)* — consolidated findings register.
- *Chapter 40 (p. 739)* — iteration roadmap and open-problem queue.
- *Chapter 15 (p. 233)* — OP-41, OP-44/OP-CP1, OP-46, OP-48/OP-GPT-TYPE3.

- *Chapter 19 (p. 392)* — H0–H7; HYP-dS-NOMIN-R3, HYP-dS-CARDINALITY, HYP-CKV-VACUITY, HYP-SPLIT-GEOM.

CHAPTER 25

# Iteration 5 Synthesis: Saturation

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Iteration: 5

This note folds the four iteration-5 attacks and the synthesis-verdict review into one closing record. The one-line headline: **iteration 5 executed the verdict-flipping move that iteration 4 named, and it did not flip the verdict; it closed two of the three named decidable problems as no-gos/bounds, sharpened the third to a located boundary, produced exactly one genuine-but-symmetry-bound positive result (the first emergent Lorentzian metric in the survey), and thereby reached the saturation condition iteration 4 forecast.** `[INFERENCE, high]`

The three headline claims are **unchanged for the fourth consecutive iteration**: (1) coherence is **PARTIAL**; (2) the geometry-from-algebra program (Program A) **ENCODES** geometry, it does not **GENERATE** it; (3) the central wager — that causal/algebraic/entanglement structure precedes metric geometry in our $\Lambda > 0$ universe — is **not-yet-physics**, with no distinguishing test even in principle. What iteration 5 changed is *precision* and the *closing of the named to-do list*, not direction. `[INFERENCE, high]`

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## Did any track flip the headline verdict?

**NO.** No track flipped the headline. The standing verdict holds across iterations $2 \to 3 \to 4 \to 5$. `[INFERENCE, high]`

Iteration 4 made an unusually sharp falsifiable bet: it designated the **OP-46 clause-(iii) non-CKV Lorentzian readout on a non-symmetric state** as *the only move that could flip the headline verdict*. Iteration 5 executed exactly that move — via the modular-Berry / Huang–Ma kinematic-space readout — and the verdict did not flip. *INFERENCE, high; see [iter-4 synthesis* (Chapter 24 (p. 599)) for the designation and *the OP-46 attack note* (Working note: *OP-46: emergent LORENTZIAN geometry from a non-CKV modular flow? (the verdict-flipping move)* (repo: `notes/2026-06-08-iter5-op46-lorentzian-nonckv-readout.md`)) for the execution.]

The execution produced a **genuine positive**: the modular-Berry / kinematic-space route is the **first surveyed construction to output a genuinely Lorentzian** (de Sitter, $dS_2$) emergent metric, crossing the **N4 signature barrier** that both the Riemannian Cao–Carroll route and the Connes spectral route fail. `[ESTABLISHED, web-verified — Huang-Ma, arXiv:2003.12252.]` So the signature barrier is **not absolute**.

But the positive **confirms rather than refutes** "encodes, not generates": the Lorentzian metric appears **only on the symmetric CFT vacuum with geometric CHM modular flow** (spherical entangling surfaces), where the canonical Berry connection is fixed by conformal symmetry. For a non-symmetric state $\omega$ the zero-mode block has no canonical complement and the connection is gauge-ambiguous. Hence the Lorentzian signature is **inherited from a pre-installed symmetry (the g3 input)**, not generated from non-symmetric algebraic data. `[INFERENCE, high.]`

The decisive structural upgrade is that the diagnosis is now **readout-independent**. The "encodes-not-generates" obstruction closes at the *same signature-installation step* across all three known readouts:

1. **Stress-tensor-flux / Sorce–CKV gate** — the conformal Killing vector is supplied by hand.

2. **Modular-Berry / zero-mode projector** — the zero-mode complement is supplied by conformal symmetry.
3. **Lorentzian NCG / $\eta$-foliation-twist** — Connes reconstruction is provably *Riemannian* `[ESTABLISHED — arXiv:0810.2088]`; there is **no Lorentzian reconstruction theorem**, and every known Lorentzian spectral triple installs the signature via a fundamental symmetry ($\eta$ form / time-orientation form / foliation / twist) that must commute with the algebra — a re-imported g4 causal input. [ESTABLISHED — van den Dungen, arXiv:1711.07299; Nieuviarts twisted triple, arXiv:2512.15450, web-verified.]

In all three, geometry (and specifically *Lorentzian* signature) appears only where a conformal/Killing symmetry or an equivalent causal datum is **pre-installed**. That common closing point is what makes the obstruction structural rather than a property of one formalism. `[INFERENCE, high.]`

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## Status of the three named decidable problems and the two dS routes

Iteration 4 reduced the path-to-saturation to a finite, named list: **OP-48(c)**, **OP-44/OP-CP1**, and **OP-46 clause-(iii)**. Iteration 5 settled all three. The two surviving dS routes were also sharpened. Each is summarized below; full arguments live in the per-attack notes.

### OP-48(c) — state-independent, collar-free composition surrogate singling out QM on a Type III$_1$ factor

**SETTLED as a clean conditional no-go** (medium-high confidence); `didItMove = advanced` — a genuine upgrade from OPEN, not a re-description. *INFERENCE, medium-high; see [the OP-48c attack note* (Working note: *OP-48 point (c): a state-/collar-free composition surrogate on Type III₁?* (repo: `notes/2026-06-08-iter5-op48c-collarfree-composition.md`)).]

- **(A)** A state-independent, collar-free surrogate **does exist**: the commuting pair $(\mathcal{M}, \mathcal{M}')$ from the Haagerup standard form, with **Connes fusion** as its weight-independent composition law. This **refutes** the iter-3/4 presumption that canonical composition must cost a collar plus a reference state. [ESTABLISHED, web-verified — van Luijk et al., axioms A1–A5 / Theorem A, arXiv:2401.07299, 2401.07292; review 2510.07563.]
- **(B)** But it is **type-agnostic and does not single out QM**: there is no local-tomography clause (Haag duality is strictly weaker), and on the physical Type $\mathrm{III}_1$ factor it is a **universal embezzler**, $\kappa_{\max} = 0$ — the operational antipode of finite-dimensional QM (Type I, $\kappa_{\max} = 2$). `[ESTABLISHED, web-verified.]`
- **(C)** Any surrogate that *does* single out QM-type composition (bounded embezzlement / genuine tensor factorization / local tomography) must interpose a Type I factor — the **split construction** — re-importing the collar (BREAK 2) and reference-state canonicity (BREAK 1). `[INFERENCE, high.]`

**Net:** "collar-free + state-independent" and "singles out QM" are **mutually exclusive on a single $\mathrm{III}_1$ factor**. The named problem is **closed**. This did not flip the headline; it *tightened* "encodes-not-generates" by showing the QM-selecting axiom **coincides** with the geometry/state-importing input rather than being orthogonal to it. The 2026 rigidity results do not rescue selection: Pati (arXiv:2601.13012) **presupposes** the bipartite tensor product; Torres Alegre (arXiv:2602.09056) fixes only the Born *form*, not composition. `[ESTABLISHED that the papers are so characterized.]`

## OP-44 / OP-CP1 — nonperturbative fate of the crossed-product Type II structure

**SHARPENED to regime-dependent with a located boundary; remains formally OPEN** (the one residual open lever); `didItMove = sharpened-obstruction`. *INFERENCE, high; see [the OP-44 attack note* (Working note: *OP-44/OP-CP1:*

*nonperturbative fate of the crossed-product Type II structure* (repo: `notes/2026-06-08-iter5-op44-nonperturbative-crossed-product.md`)).]

Let $N_* \sim R/(2GE_f)$ mark where the dressed operator's Schwarzschild radius $r_s$ reaches the static-patch scale $R$:

- **Below** $N_*$ ($r_s \ll R$): the von Neumann algebraic subregion description **survives** as an approximate, coarse-grained, QRF-relational Type II structure. The CLPW/KLS $\mathrm{III}_1 \to \mathrm{II}$ crossed product is the **faithful leading-order** $\mathcal{O}(\kappa)$ **truncation** of Giddings' gravitational dressing — "a small piece of a more general algebraic structure." `[ESTABLISHED, web-verified — Giddings, arXiv:2510.24833.]`
- **At/above** $N_*$ ($r_s \gtrsim R$): the spacelike complement is consumed, and the inclusion/commutant structure degenerates ("algebraic spacetime disruption"), with a "network of Hilbert-space inclusions" the preliminary proposed replacement. `[ESTABLISHED that arXiv:2505.22708 is so characterized.]`

**Net:** "Type $\mathrm{III}_1$ for any $\Lambda$" is a robust **kinematic input** and a controlled **leading-order structure**, **not** a fixed nonperturbative core. Crucially, there is **no no-go theorem and no all-orders survival proof on either side of** $N_*$ — this is genuine regime-dependence, not a settled no-go. The two literatures do not contradict (the crossed-product results make no nonperturbative claim; Giddings makes no claim the leading-order truncation is wrong). The framing question — "can Type $\mathrm{III}_1$-for-any-$\Lambda$ be a fixed nonperturbative core?" — is answered effectively **"no at the nonperturbative level"** while its kinematic/leading-order status is preserved. `[INFERENCE, high.]`

## OP-46 clause (iii) / HYP-CKV-VACUITY — the designated verdict-flipper

**ATTEMPTED and did NOT flip; promoted toward a readout-independent no-go**; `didItMove = sharpened-obstruction`. [INFERENCE — HIGH that no current construction generates Lorentzian geometry from non-symmetric modular data;

MEDIUM that this is a true no-go versus a limitation of all currently-known readouts.]

The substance is given in the section above. The summary: HYP-CKV-VACUITY is now **effectively a no-go on every readout currently known** (stress-tensor flux, modular-Berry, Lorentzian-NCG). It remains **formally open** (not a theorem) only for a hypothetical readout that installs **no** fundamental symmetry / no $\eta$ / no foliation / no twist — and none such exists or has been sketched. This is the single epistemic hedge that keeps the headline from being a theorem. `[INFERENCE, medium-high.]`

## Surviving dS routes — neither moves Program A from ENCODES to GENERATES at $\Lambda > 0$

**Both SHARPENED; PARTIAL reinforced**; `didItMove = sharpened-obstruction`. The two un-foreclosed escapes after the iter-4 cardinality no-go fail for **distinct, now-identified** reasons. *INFERENCE, high; see [the surviving-dS-routes note* (Working note: *The two dS routes surviving the cardinality no-go: pseudo-entropy and Jacobson-in-CLPW* (repo: `notes/2026-06-08-iter5-surviving-dS-routes.md`)).]

- **Route (a) — dS/CFT pseudo-entropy first law** (arXiv:2511.07915; web-verified: $c = i\,3R/2G$, non-Hermitian $\mathcal{T}_A$ , complex pseudo-entropy, $\mathcal{I}^+$, complexified geodesics, $\mathrm{dS}_2$ KG equation). This **genuinely evades** the iter-4 cardinality / radius-modulus facet — CFT regions of every size on $\mathcal{I}^+$ give a true radius continuum, and it **is** a real first-law $\Rightarrow$ linearized-$\mathrm{dS}_3$ -Einstein implication. But it inherits a **different disqualifying assumption: non-unitarity**. Its real geometry is the real section of an analytic continuation of the AdS/CHM computation (Maldacena $\ell_{\mathrm{AdS}} \to i\,\ell_{\mathrm{dS}}$), with reality *imposed* on complexified geodesics, not generated; pseudo-entropy becomes physical only when post-selection removes the non-unitary content. `[INFERENCE, high.]`
- **Route (b) — Jacobson's infinite small-ball family inside a single CLPW Type $\mathrm{II}_1$ algebra.** This **fails the import**: the CLPW Type $\mathrm{II}_1$ algebra has a *unique finite trace*, $\mathrm{Tr}(1) = 1$,

encoding exactly **one** area datum (the global horizon). Internalizing Jacobson's infinite *local* small-ball family requires re-supplying the fixed area-entropy coefficient $\eta = 1/4G$ at every bulk point — it merely **re-is** Jacobson non-algebraically, confirming HYP-dS-CARDINALITY. `[INFERENCE, high.]`

**Unifying refinement (HYP-dS-CARDINALITY-R1):** in **every** known route — AdS-FGHMVR, Jacobson, CLPW — the **area-entropy coefficient is an INPUT, not an output**. That is the true root obstruction, untouched by either surviving route. One new genuinely-decidable sub-problem was spun off: **OP-dS-PSEUDO-REALITY** (can $\operatorname{Im} S(\mathcal{T}_A)$ be physically nulled while retaining nontrivial $\delta G$?). `[INFERENCE, high.]`

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## Is analytical saturation reached?

**YES.** `[INFERENCE, high.]`

Iteration 4 forecast saturation "$\sim$ 1–3 iterations away," contingent on settling a finite named list, and stated the criterion explicitly: *after these are settled, marginal structural insight goes to near-zero and the program becomes a watch-list*. Iteration 5 settled the list:

- **OP-48(c)** — SETTLED (clean conditional no-go).
- **OP-46 clause (iii)** — the designated verdict-flipper — executed and **foreclosed on all known readouts** (effective readout-independent no-go).
- **OP-44 / OP-CP1** — sharpened to a located regime boundary $N_*$ ; the sole residual formally-open item, but no longer headline-moving and with no theorem possible on either side without genuinely new input.

Two of three named problems closed as no-gos/bounds, the third sharpened to a boundary, and the **one verdict-flipping lever was attempted and did not flip**. Iteration 5 produced **zero new positive structure and zero verdict movement** — only sharpening, closure, and one genuine-but-symmetry-bound positive (the first Lorentzian output, which confirms the diagnosis). This is the **fourth consecutive iteration** at PARTIAL / encodes-not-generates / not-yet-physics. By the iter-4 criterion, the precondition for saturation is now met. `[INFERENCE, high.]`

**What saturation is and is not.** "Analytically saturated" is a statement about the *marginal yield of further surveying-and-sharpening*, not a proof of any headline claim. The headline rests on one MEDIUM-confidence hedge (OP-46 residual) and one INFERENCE leg (OP-48c horn (C)); it is **not** a theorem. The **object-level** question — a confirmed universal theory — remains **NOT FORECASTABLE**, unchanged: it is gated by the $\sim$ 15-order Planck desert and, more bindingly, by the wager having **no distinguishing experimental test even in principle**. Saturation concerns the analytical program, not the physics. `[INFERENCE, high.]`

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## Updated standing

- **Coherence: PARTIAL** — unchanged across iterations 2–5, now analytically saturated. `[INFERENCE, high.]`
- **Program A (geometry-from-algebra): ENCODES, does not GENERATE.** Now backed by a **readout-independent** obstruction (closes identically across the stress-tensor-flux/Sorce-CKV, modular-Berry/zero-mode, and Lorentzian-NCG/$\eta$-foliation-twist readouts) and **tightened** by a web-verified operational invariant (van Luijk embezzlement $\kappa_{\max}$) showing the QM-selecting composition axiom **coincides** with the geometry/state-importing input rather than being orthogonal to it. `[INFERENCE, high.]`
- **Central wager: NOT-YET-PHYSICS** — no distinguishing test even in principle; unchanged. `[INFERENCE, high.]`
- **Confidence per track:** OP-48c **medium-high**; OP-46 **HIGH** that no current construction generates Lorentzian geometry from non-symmetric modular data, **MEDIUM** that it is a true no-go rather than a limitation of all currently-known

readouts; OP-44 **regime-dependent** (no theorem on either side); surviving-dS **PARTIAL** reinforced. `[INFERENCE, high.]`

The one thing that changed at the object level is the discovery that the signature barrier (N4) is **not absolute** — a real, citable positive (Huang–Ma $dS_2$). It is recorded as a positive that *confirms* the diagnosis, because the Lorentzian signature is inherited from the symmetric vacuum, not generated. [ESTABLISHED that the construction outputs a Lorentzian metric; INFERENCE, high that the signature is inherited.]

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## Residual open levers

Four items remain formally open; none is headline-moving, and each is now a watch-list or hardening item rather than an analytical lever. `[INFERENCE, high.]`

1. **OP-44 / OP-CP1 (the genuinely-open item).** The nonperturbative crossed-product fate is regime-dependent with boundary $N_* \sim R/(2GE_f)$ but formally open: **no no-go theorem and no all-orders survival proof on either side**. It cannot flip "encodes-not-generates" either way, and closing it further would require new mathematics (quantitative all-orders dressing control, or a proof that the algebra fails past $N_*$) that is not currently on the table. **Watch-list, not lever.**
2. **OP-46 clause (iii) residual.** HYP-CKV-VACUITY is a no-go only over **currently-known** readouts; it remains formally open for a hypothetical readout installing **no** fundamental symmetry / no $\eta$ / no foliation / no twist. None such exists or is sketched. Confidence is **MEDIUM** (not high) that this is a true no-go rather than a limitation of all presently-conceived readouts — **the single epistemic hedge that keeps the headline from being a theorem.**
3. **OP-dS-PSEUDO-REALITY (newly spun off).** Whether the imaginary pseudo-entropy of dS/CFT (arXiv:2511.07915) can be reduced to a real, physical quantity (genuine post-selection / real reduced density matrix) **without** trivializing the first-law ⇒

Einstein content. A physical nulling of $\operatorname{Im} S$ with nontrivial $\delta G$ would promote route (a) toward GENERATES. A sharp, decidable, currently-unanswered test isolating the single step on which route (a)'s verdict turns.

4. **OP-48(c) horn (C).** The "must interpose Type I" step is **INFERENCE** — strongly supported by the van Luijk $\kappa_{\max}$ classification but not a cited theorem in the exact "no QM-selection without Type I" phrasing. A hardenable-but-currently-uncited link; the only non-ESTABLISHED leg of an otherwise settled conditional no-go.

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## Recommendation

**Declare saturation and write the capstone.** `[INFERENCE, high.]`

The precondition the iter-4 criterion set is met: the named list is settled, the verdict-flipping lever was attempted and did not flip, and marginal structural yield has gone to near-zero. Concretely:

1. **Write `CONCLUSION.md`** recording the converged verdict (PARTIAL / encodes-not-generates / not-yet-physics) with its two explicit hedges (OP-46 residual MEDIUM; OP-48c horn (C) INFERENCE), the one object-level positive (first Lorentzian output, signature inherited), and the four residual open levers as a watch-list.
2. **Switch remaining effort to lightweight experiment-tracking** of the four channels that bear on the broader landscape — **BMV** ($m^2$-vs-$m^3$ entanglement-witness scaling); **objective-collapse bounds** (next decisive window $\sim 2035$); $w(z)$ dark-energy equation of state (**DESI DR3+ / Euclid**, 2–7 yr horizon); and **LIV / EP** null tests. Note explicitly that **none of these tests the geometry-from-algebra wager itself** — they constrain the surrounding physics, not the central wager, which remains untestable even in principle.

The honest reading the prompt anticipated holds: iteration 5 closed the named problems as no-gos/bounds and produced no verdict-flip, so **saturation is reached**, and the right next action is a capstone plus a watch-list, not a sixth analytical iteration. `[INFERENCE, high.]`

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CHAPTER 26

# Iteration 6 Synthesis: The Verdict Meets New Input

## 1. What was attempted

Iteration 5 declared analytical saturation: further *reasoning* would re-derive the same map, and only **new input** — an experiment, or a genuinely new mathematical readout — could move the verdict. Iteration 6 was the deliberate test of that claim: six research tracks swept the 2023–2026 literature the wiki had not yet ingested (closed-universe/dS holography; the algebraic frontier; amplitudes/positive geometry; the background-independent program landscape; QG phenomenology; a full experiment refresh), and four attack tracks tried to break the standing conclusion at its strongest points (hunt a distinguishing prediction; find a generating construction; move the OP-44/OP-48c residual levers; decide OP-49). Every track was adversarially refereed with citation audits; corrected claims are binding over originals.

## 2. Per-track verdicts

| Track | Verdict (one line) |
|---|---|
| A1 distinguishing-prediction hunt | No channel distinguishes the wager even in principle; the no-test fact upgraded to a *corollary* of encode-not-generate (ENCODING-SCREEN, "only if" form), inheriting the OP-46 MEDIUM hedge; GQuEST added as an EFT-breakdown (not wager) channel. |

| | |
|---|---|
| A2 generation-construction survey | No generation: CFS = fourth readout family (the (≤n,≤n) axiom is the relocated Krein η); new selection-type pattern (Lee 2020 — nearest sketch, fails the §8 spec); HYP-CKV-VACUITY-R3; target compresses to deriving the indefinite pairing from modular data. |
| A3 OP-44/OP-48c levers | Both levers moved, neither flips: JRS no-commutant + crossed-product rigidity; DEHK order-$G\_N^2$ obligation located; horn C partially discharged (theorem-shaped under the full-pair reading; residue = formalization choice + $III_1$-embezzlement alternative). |
| A4 OP-49 pseudo-entropy | NEAR-NO-GO, resolved-negative (conditional): the Einstein content lives entirely in Im S; reality forces triviality; three named residuals; iter-5 Re/Im parenthetical corrected. |
| R1 closed universe / dS | Debate mapped → OP-50 `[CONTESTED]`; observer-rescue = path-integral analogue of OP-48(c) (reinforces, cannot harden); HYP-dS-CARDINALITY-R1 extended: η input or absent, never output. |
| R2 algebraic frontier | Survival side theorem-hardened at linearized order (full constraint family; non-stationary GSL; QFC proof); failure-side replacements all still algebraic; survives-after-averaging option; N* unmoved; encode-direction evidence. |
| R3 amplitudes | H5 scope expands (low → low-to-medium) inside a confirmed structural bound: no positive geometry for gravity; all constructions consume flat-space/FRW kinematic data — encodes. |
| R4 other programs | Two new landscape entries (Oppenheim CQ; dark dimension) — both genuinely testable, sharpening the wager's demotion by contrast; six stale-fixes. |
| R5 QG phenomenology | GQuEST/VZ is Program-A-adjacent, not a wager test (η=1/4G re-imported); CQ gate may close from the classical side before BMV; DSW deflated (thermality, not horizon-ness). |
| R6 experiment refresh | All standing numbers hold; w(z) decision point compresses to 2027; four rows change; nothing touches the wager — exactly as the caveat predicts. |

## 3. Hedge accounting (before → after)

| # | Hedge (before iteration 6) | After iteration 6 |
|---|---|---|
| 1 | **OP-46 residual** — HYP-CKV-VACUITY a no-go only over known readouts (three families); MEDIUM that it is a true no-go | **Unchanged grade, broader base**: four readout families (CFS added); smuggle list extended (indefinite-pairing/(n,n) axiom; signature-valued template); slightly higher confidence (the designated most-serious absent candidate did not pass); still MEDIUM. Target compressed: derive the indefinite pairing from modular data. |
| 2 | **OP-48(c) horn C** — "must interpose Type I" is INFERENCE, not a cited theorem | **SHRUNK (partially discharged)**: theorem-shaped under the full-pair (normal-product-state) reading; uniform Bell gap $1-1/\sqrt{2}$ in the approximate reading. Residue: the GPT formalization choice + the consistent multipartite $III_1$-embezzlement composition alternative (a proven spectrum of regimes; algebra cannot choose the endpoint). |
| 3 | *(did not exist)* | **NEW — HYP-ENCODING-SCREEN**: the §7 no-distinguishing-test claim is a corollary of encode-not-generate ("only if" direction) and is exactly as strong as the encode-only classification — it inherits hedge 1's MEDIUM grade explicitly. |
| — | **OP-49** (residual lever, not a hedge) — open decidable test | **CLOSED-NEGATIVE** at near-no-go grade (conditional; three named residuals — exceptional points; SVD-entropy first law; horn-(ii) protocol-class openness). |

## 4. Why the verdict stands

`[INFERENCE, high]` Three independent reasons, each refereed:

1. **The attack tracks converged on the standing diagnosis from new directions.** The strongest absent candidate (causal fermion systems) closes at the *same* signature-installation step as the three known readouts; the genuinely novel pattern (dynamical signature selection) still installs the possibility space

by hand; the closed-universe debate independently re-derives the "QM only relative to an observer-anchored factorization" structure by a path-integral route. New input did not open new escape routes — it multiplied instances of the same closure.

2. **The no-test obstacle deepened from fact to corollary.** The two standing obstacles (no generation; no distinguishing test) are now one fact: an encode-only program's observables factor through the EFT it encodes, so the §8 reopening condition and the missing experiment coincide. This is *stronger* support for not-yet-physics, honestly hedged (it inherits the MEDIUM encode-only classification — hedge 3).
3. **The experiment side behaved exactly as the watchlist caveat predicted.** Every 2022–2026 result — including the first funded Program-A-adjacent experiment (GQuEST) — lands in the gate/core/target/program-adjacent taxonomy; the chains that look wager-flavored re-import η=1/4G at the decisive step. Two surveyed competitor programs (CQ gravity; dark dimension) *do* carry distinguishing tests, so the demotion of the wager cannot be excused by field-wide untestability.

Nothing found bears against the closed items (OP-41, OP-46 closures, OP-48a/b); zero findings were fatal; the few corrections ran *against* overclaims in the wiki's favor as often as against the tracks (Rodina hypotheses; "iff"→"only if"; binary→spectrum), which is what a healthy adversarial process looks like.

## 5. What remains (residual levers after iteration 6)

1. **OP-46 compressed target** — derive the indefinite pairing (Krein η / canonical (n,n) operator class, e.g. from the modular conjugation J) from (algebra, state) modular data. The single analytical verdict-flipper, now also the prerequisite for any distinguishing experiment (hedge 3).
2. **OP-44 / OP-CP1** — the order-$G\_N^2$ obligation is now *named* (quadratic boost constraints; DEHK): solving or obstructing that sector is the next decidable step; the four-option verdict space

(survives / fails / regime-dependent / survives-after-averaging) awaits a theorem at N*.

3. **OP-48(c) residue** — a principled argument for (or against) the full-pair formalization of "recovers QM", or any physical axiom selecting between the Type I and $III_1$-embezzlement composition endpoints.
4. **OP-49 residuals** — one stated decidable computation (δS_SVD for the $dS_3$ hemisphere under the 2511.07915 perturbation family); exceptional-point protocols.
5. **OP-50** `[CONTESTED]` — the closed-universe premise; watch the ARSSV / Liu / Engelhardt–Gesteau branches for a theorem.

## 6. What could move things next (only new input)

- **The 2027 w(z) decision point** (DESI full 5-yr; Euclid DR1 2026-10-21; Roman launch 2026-08-30) — the nearest-dated event; a confirmed $w(z)\neq-1$ reframes Program A's eternal-dS target (A-20) and would demand an everpresent-Λ confrontation that does not yet exist in print.
- **GQuEST results (~2–5 yr)** — either outcome is informative (EFT-breakdown discriminator); a detection would reopen the LANDSCAPE without flipping the wager.
- **CQ falsification (running now)** — would close the gate from the classical side, possibly before BMV; the asymmetric gate-closure route.
- **OP-44's order-$G_N^2$ obligation** — the one named, mathematically decidable analytical step left on the survival/failure frontier.
- **The compressed OP-46 target** — "derive the indefinite pairing from modular data": any sketch meeting it reopens CONCLUSION §8 *and*, via ENCODING-SCREEN, the experimental question simultaneously.

CHAPTER 27

# Iteration 7 Synthesis: The Decidable Levers, Executed

## 1. What was attempted

Iteration 6 closed with four named, decidable levers and an explicit prediction: only their execution (or new physics input) could move anything. Iteration 7 executed all four. A1 attacked the compressed OP-46 target — *derive the indefinite pairing (Krein $\eta$) from modular data* — the designated verdict-flipper, by attempting the construction rather than surveying for it. A2 attacked OP-44's named order-$G_N^2$ obligation for the dS static patch (discharge it or locate the obstruction). A3 performed the iteration-6 *stated decidable computation* — the SVD-entropy first law on the dS3 hemisphere under the arXiv:2511.07915 perturbation family. A4 settled OP-48c's remaining definitional residue (full-pair vs subsystem reading of "recovers QM") against the primary reconstruction texts, and ran the OP-50 watch. Every track was adversarially refereed; corrected claims and referee outcome-checks are binding over originals.

## 2. Per-track verdicts

| TRACK | OUTCOME | VERDICT (ONE LINE) |
|---|---|---|
| A1 indefinite pairing from modular data | **RESOLVED-NEGATIVE** | The literal target is *achievable* — $\eta_0$ = sgn(ln Δ) is a canonical, smuggle-free Krein fundamental symmetry (one of an infinite J-odd family) — but provably geometry-void (unitary universality ⟹ Δ_geo = 0; algebra-incompatibility; non-localizability); the trichotomy lemma closes the J-route under a flagged ansatz; g3 upgraded structural; hedge hardens MEDIUM → MEDIUM-HIGH; target re-compresses to the carrier problem. |
| A2 OP-44 order-$G_N^2$ obligation | **SHARPENED-OPEN** | P1 splits P1a/P1b/P1c: quadratic descent to the CLPW crossed product discharged at inference grade (conditional); residues R1 (footnote-12 sufficiency with worldline clocks) and R2 (cubic horizon non-splittability; δ ≳ ℓ_P collar floor) located, theorem-shaped on both horns; R2's obstruction horn = the first perturbative failure-side foothold, at order $\kappa^3$; N* untouched. |
| A3 OP-49 SVD first law | **RESOLVED-NEGATIVE (conditional)** | The SVD first law fails on the dS3 hemisphere: unconditionally, S_SVD is real by construction vs the purely-imaginary verified carrier; the c+c* = 0 cap-blindness mechanism is conditional on heterogeneous-cover junction additivity (new residual c′); no Einstein content survives; byproduct S_SVD = S_dS/2 → `[SPECULATIVE]`. |
| A4 OP-48c formalization + OP-50 | **RESOLVED-POSITIVE** | The full-pair reading is forced (modulo rejecting OP-48's question); LT-separation HOLDS on the III1 pair, so the failure locus is the normal-product-preparation primitive; horn C → theorem-conditional-on-framework; residue = the normal-state postulate + exits X1/X2; OP-50 three-cornered (arXiv:2606.04575). |

# 3. Hedge ledger (before → after)

| # | Hedge (before iteration 7) | After iteration 7 |
|---|---|---|
| 1 | **OP-46 residual** — HYP-CKV-VACUITY-R3, MEDIUM that the installation/selection classification is a true no-go; compressed target "derive the indefinite pairing from modular data" | **HARDENED: MEDIUM → MEDIUM-HIGH, conditional on the modular-constructibility ansatz.** The route is now closed by *construction-plus-vacuity*, not by absence: the literal target is achieved ($\eta_0$) and provably geometry-void; the trichotomy lemma (HYP-MODULAR-TRICHOTOMY) covers the general case modulo the flagged ansatz — the explicit residual. Target re-compressed (mandatory precision repair): derive an **algebra-compatible, localized** (n,n) pairing — the carrier problem (HYP-FACTORIZATION-IS-GEOMETRY). |
| 2 | **OP-48(c) horn C** — theorem-shaped under the full-pair reading; residue = the GPT formalization choice + the $III_1$-embezzlement alternative | **SHRUNK AGAIN, character changed: theorem-conditional-on-framework.** The definitional residue is resolved (full-pair forced; subsystem reading inexpressible); the failure locus relocates from tomography to the normal-product-preparation primitive. Residue = the normal-state postulate + the composition-primitive choice (X1 singular states / X2 non-OPT composition) — exits from the framework, not readings of it. |
| 3 | **HYP-ENCODING-SCREEN** — the §7 no-test claim inherits the OP-46 MEDIUM hedge | **Inherits the hardened grade: MEDIUM-HIGH, same ansatz condition attached.** Mildly reinforced from the A2 side (the P1a discharge consumes pre-installed so(d+1,1) structure — itself an encoding step). |
| — | **OP-49** (residual lever) — one stated decidable computation outstanding (SVD) | **CLOSED-NEGATIVE (conditional, narrowed grounds).** Residual list rewritten: (a) exceptional points; (b) beyond-family; (c′) junction additivity. Still NEAR-NO-GO. |

| | |
|---|---|
| — **OP-44** (residual lever) — a named but vague order-$G_N^2$ obligation | **Restructured: P1a discharged (inference grade, conditional); R1 + R2 located, decidable; first failure-side perturbative foothold (κ³); ℓ_P collar floor cross-links to the stringy ℓ_s floor.** N* still theorem-free on both sides. |

# 4. Why the verdict stands (sixth consecutive iteration)

`[INFERENCE, high]` Three refereed reasons:

1. **The verdict-flipper was executed, succeeded literally, and flipped nothing — the strongest possible form of a negative.** Iterations 4–6 progressively compressed the one analytical move that would turn "encodes" into "generates" down to a single sentence; iteration 7 *carried out* that sentence. The result is a genuine first (a canonical Krein fundamental symmetry from pure modular data, smuggle-free by the house instrument) that is simultaneously a proof of its own irrelevance: every member of the η-family is unitarily universal where modular data has causal content, incompatible with the algebra, and localizable on nothing. The compressed target was a lossy compression; the *faithful* residual — an algebra-compatible, localized (n,n) pairing — is the carrier problem, where all five known routes now close (HYP-FACTORIZATION-IS-GEOMETRY). The obstruction did not merely survive its designated attack; it absorbed the attack as a fifth converging instance.
2. **The other three levers moved exactly as the standing diagnosis predicts.** OP-49's only intrinsically real two-state entropy retains the region-independent horizon constant and zero local geometry (the probe-vanishes pattern, again). OP-48c's failure locus relocated to the *state postulate* — the product-preparation primitive is an installation choice (encodes-not-generates at yet another layer). OP-44's discharge works by consuming pre-installed isometry structure, and its new obstruction horn prices the dressing collar at ℓ_P — the collar

tax, now from a second independent direction. Four attacks, four instances of the same closure geometry.

3. **The adversarial process ran against the tracks as hard as for them.** Five majors (A3's mechanism downgraded to conditional; the byproduct to SPECULATIVE; A4's X1 derivation replaced; the 2606.04575 scope corrected; A1's uniqueness claim refuted by referee counterexample) — and every correction *narrowed* a claim without flipping an outcome label. One outcome (A3) was upheld only conditionally and on narrower grounds; the unconditional pillar (real vs imaginary) suffices for the registry action. That is what a healthy verdict-preserving iteration looks like: sharper statements, harder hedges, zero motion at the headline.

Nothing found bears against any closed item; N* was not moved and must not be read as moved (A2 is entirely perturbative); no track produced an experimental channel — NOT-YET-PHYSICS stands unmodified.

## 5. What remains (the residual map after iteration 7)

1. **THE compressed analytical target — the carrier problem (HYP-FACTORIZATION-IS-GEOMETRY).** Derive an algebra-compatible, **localized** (fiberwise) (n,n) pairing from $(\mathcal{M}, \omega)$ modular data. Five independent routes close at this step (no-factorization frontier; split/collar relocation; CFS carrier posit; OP-49 protocol deformations; $\eta_0$ non-localizability). Via HYP-ENCODING-SCREEN it is simultaneously the §8 reopening condition and the prerequisite for any distinguishing experiment. The one remaining analytical verdict-flipper.
2. **OP-44's R1 and R2** — both located and decidable: R1 = distributional/smeared-source extension of linearization-stability theory (the footnote-12 check); R2 = cubic edge modes (discharge horn) vs an affiliation/outerness theorem for the cubic cocycle (obstruction horn — the $\kappa^3$ failure-side foothold).

The nearest theorem-shaped work anywhere on the survival/failure frontier.

3. **OP-49's residuals (a)/(b)/(c′)** — exceptional-point transition matrices; beyond-2511.07915-family generalization; heterogeneous-cover junction additivity (an interface-CFT computation: the junction contribution at a branch point where 2m conjugate-theory interfaces meet). None headline-moving.
4. **OP-48c's X1/X2 + the normal-state postulate** — a physical argument for or against admitting singular product states as preparations (X1), or for the non-OPT III1 composition endpoint (X2); the algebra classifies, it cannot choose.
5. **OP-50** `[CONTESTED]`**, three-cornered** — strict 1-dim / prediction-framework-on-1-dim / polynomial-dim II1 sector (arXiv:2606.04575); watch for a theorem on any corner (and fold in the pre-window no-go arXiv:2509.14338 on next rewrite).
6. **The experiment watchlist — unchanged.** No iteration-7 track touched any channel; the 2027 w(z) decision point (DESI 5-yr; Euclid DR1; Roman), GQuEST, and the CQ gate remain the only dated external events. The object-level question stays with experiment.

CHAPTER 28

# Iteration 8 Synthesis: The Net/Family Carrier No-Go

**Iteration-8 synthesis note (for the synthesizer's record; the human-facing capstone stays CONCLUSION.md).**

The single most important judgment this iteration demanded — flagged in the task and re-checked by me independently — is that the carrier track is **NOT** a clause- or verdict-level RESOLVED-POSITIVE. The A1 outcome is RESOLVED-**NEGATIVE**: the net/family upgrade does not evade the iteration-7 carrier no-go; it relocates the carrier into the net's index set. I re-derived the engine (covariance–locality obstruction) and confirmed the net-index-set smuggle is genuine against the extended smuggle list: net-modular data (relative modular operators, $\{J_O\}$, HSMI, modular intersection, modular-Berry) adds operators but every modular-covariant pairing remains positive-local or indefinite-nonlocal, and locality enters only through the net's parametrization, which is the geometry (BGL isotony⇔positivity; the Euler element is an input in Morinelli–Neeb–Ólafsson; Lashkari et al. "insisting on a local representation"). So the net no-go *strengthens* the hedge toward HIGH; it does not flip it.

Second-most-important: I am **not** incrementing the carrier-convergence route count. The A1 finder wrote "sixth independent convergence"; the referee corrected this to "net-level extension of the fifth route's boost-covariance engine." That correction is binding and I have propagated it everywhere — the wiki count stays **five** (HYP-FACTORIZATION-IS-GEOMETRY). Any future agent who reads only the finder's headline would over-count; the registry deltas and CONCLUSION amendments lock the five-route count and fold the net upgrade into route (5). This is the kind of subtle inflation the referee discipline exists to catch.

Third: the A3 overclaim. The finder submitted RESOLVED-POSITIVE on OP-49 (c′) ("cap-blindness EXACT", "unconditional"); the referee overturned it to VERDICT-UNCHANGED with (c′) reframed-and-narrowed, NOT closed, because the transmissive-junction mechanism is an INFERENCE-medium extrapolation into the non-unitary imaginary-*c* regime where the unitary transmissivity bounds fail. I have encoded the *referee* label, not the finder label — OP-49 keeps three residuals (a)/(b)/(c′), and no CONCLUSION hedge moves from A3. The unconditional reality-mismatch pillar (re-verified) is what keeps OP-49 NEAR-NO-GO regardless.

Net hedge movement: exactly one hedge moves this iteration — the principal one (OP-46 / HYP-CKV-VACUITY / inherited HYP-ENCODING-SCREEN), hardening from MEDIUM-HIGH-conditional-on-single-algebra-ansatz to MEDIUM-HIGH→HIGH-conditional-on-net-constructibility-ansatz. I kept it inside the MEDIUM-HIGH band (worded "MEDIUM-HIGH→HIGH") rather than declaring a clean HIGH, because two named proof obligations remain open (naturality theorem; modular-Berry-holonomy escape). This is the conservative reading and matches the seven-iteration pattern of tightening-without-flipping.

The other three tracks are verdict-neutral sharpenings: A2 (OP-44 R1 theorem-shaped failure-side result with a regularization escape that itself supplies input; R2 obstruction horn reinforced with one named escape), A4 (OP-48c X1 promoted to a theorem-with-a-price-tag, horn C unchanged). No experimental channel emerged from any track; object-level stays not forecastable; the 2027 w(z) point remains the nearest external event.

Seventh consecutive headline-unchanged iteration. The saturation diagnosis survives its third stress test, and the referee's overturn of A3 is positive evidence the process is binding rather than self-confirming.

CHAPTER 29

# Iteration 9 Synthesis: The Carrier No-Go Naturality Content; Reverse-WW = the Carrier Problem

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**Date:** 2026-06-10 · **Type:** lever-execution iteration (4 adversarially-refereed attack tracks) · **Headline movement:** none (8th consecutive confirmation) · **Hedge movement:** principal hedge sharpened in *condition*, not grade; one of its two outstanding obligations discharged negative.

# Per-track verdict table

| TRACK | TARGET | SUBMITTED | REFEREE | NET EFFECT |
|---|---|---|---|---|
| **A1** carrier-naturality | Replace the net-constructibility ansatz by a naturality theorem | SHARPENED-OPEN | UPHELD-WITH-CORRECTION | Constructibility half → naturality theorem (Lemma N1) **modulo a commutant-to-generated gap**; boost engine → naturality obstruction (Lemma N2); localization half **irreducible** (= index-set order = $n_1$). Hedge **grade unchanged**, condition sharpened. Count stays 5. |
| **A2** modular-Berry-holonomy | Does the holonomy escape yield a localized indefinite form? | RESOLVED-NEGATIVE (+ "unconditional HIGH") | UPHOLD-OUTCOME / DOWNGRADE-HEDGE | Escape closes: (a) holonomy form is signature-free symplectic (KKS/Crofton); (d) the indefinite kinematic metric is vacuum-inherited from the $SO(d, 2)$ orbit. **"Unconditional HIGH" struck** — stays CONDITIONAL. Obligation (ii) **discharged**. Ground (b) false (not needed). Count stays 5. |

| | | | | |
|---|---|---|---|---|
| **A3** reverse-Weinberg–Witten | Is reverse-WW independently provable or a free witness? | SHARPENED-OPEN | KEEP-ALL, UPHELD | A1-concrete is a *conditional* theorem (Weiner 1006.4726); its hypotheses ARE the geometry. reverse-WW-theorem ≡ carrier no-go; witness strictly harder. Target **pinned** to §8 reopening condition. No witness; no flip. Count stays 5. |
| **A4** OP-44 R2 graviton escape | Does the TT tensor structure cancel the cubic horizon divergence? | RESOLVED-NEGATIVE | **OVERTURNED → SHARPENED-OPEN** | Step-3 contraction $\Pi : \hat{n}\hat{n}\hat{n}\hat{n} = 0$ (not $(d-2)/(d-1)$): TT structure **annihilates** the naive leading term; residual coefficient uncomputed, zero in $D = 4$. Escape stays OPEN; foothold de-hardened to INFERENCE-medium. |

## Hedge ledger: before → after

- **Hedge 1 (OP-46 residual / carrier no-go).** Before: MEDIUM-HIGH→HIGH, conditional on the *net-*

*constructibility ansatz*, two named obligations outstanding (net-naturality theorem; modular-Berry-holonomy escape). After: MEDIUM-HIGH→HIGH (**grade unchanged**), condition **sharpened** — the ansatz replaced by a naturality formulation (constructibility a theorem modulo a located commutant-to-generated gap; localization proven irreducible), and the **modular-Berry-holonomy obligation discharged RESOLVED-NEGATIVE**. Sole residual between HIGH and a theorem: the net-naturality theorem itself plus the new commutant-to-generated gap.

- **Hedge 3 (HYP-ENCODING-SCREEN / §7 no-test).** Inherits unchanged grade and the sharpened condition. The reverse-WW target is now pinned to the §8 reopening condition (A3) rather than free-floating.
- **OP-44 R2 (not a CONCLUSION hedge).** Failure-side foothold de-hardened (medium-high → medium); escape OPEN.

## Why the verdict stands an eighth time

The two named obligations iteration 8 placed between the hedge and a theorem were the entire iteration-9 agenda. One (the Berry escape) closed *against* the evader; the other (the net-naturality theorem) was advanced to a theorem-modulo-a-located-gap but **not** completed — and A3 independently showed *why* it cannot be free: the naturality/A1 theorem is Weiner's algebraic Haag theorem, whose hypotheses are exactly the geometric inputs the carrier problem isolates. Every route to closing the hedge unconditionally runs back through the same $n_1$ smuggle (the index-set order). No new degree of freedom entered the problem; the A4 overturn even *removed* a previously-claimed failure-side foothold. This is the signature of saturation: the marginal yield of further reasoning is sharper obstructions and self-corrections, not new structure.

## The residual map after iteration 9

- **The carrier problem (HYP-FACTORIZATION-IS-GEOMETRY)** — still THE compressed analytical reopening target; five routes converge there; the net-naturality theorem (plus the commutant-to-generated gap) is what stands between the hedge and a theorem.

- **reverse-WW** — `[OPEN, as of 2026-06]`, pinned to the carrier problem; un-registered; strictly harder than a positive carrier solution.
- **OP-44** — R2 escape OPEN (residual same-power coefficient uncomputed, dimension-dependent); $N_*$ theorem-free both sides; R1 a regularization-conditional failure-side result; $\ell_P$ floor in-principle status untouched.
- **OP-48c** — X1 a theorem-with-a-price-tag; horn C theorem-conditional-on-framework. **OP-49** — NEAR-NO-GO, residuals (a)/(b)/(c′). Both untouched this iteration.
- **Experiment watchlist** — unchanged; 2027 $w(z)$ the nearest-dated external event.

## Binding discipline notes (carried)

One submitted outcome label overturned (A4) and one verdict-bearing overclaim struck (A2) — the referee discipline is binding, not rubber-stamping. Carrier-convergence count is **five** (A1/A2/A3 all fold into route (5); none is a sixth route). reverse-WW is **not OP-50** (OP-50 is the closed-universe Hilbert-space problem). math/0304340 is **Bisch**, not Jones. No new OP/GC/A/H numeric IDs consumed (next free: OP-51, A-24, GC-18, H9).

CHAPTER 30

# Iteration 10 Synthesis: The Commutant-to-Generated Gap, Attacked From Both Sides

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**One-line outcome.** The designated verdict-bearing lever — close the iteration-9 commutant-to-generated gap and push the principal hedge to unconditional HIGH — was executed and **did not fire**. Two of three submitted outcome labels were corrected by the referee (A1's condition-shedding centerpiece struck as a major error; A3's RESOLVED-NEGATIVE refuted to SHARPENED-OPEN). The standing verdict is **UNCHANGED, ninth consecutive confirmation**; the principal hedge moves in **neither grade nor condition**; the carrier-convergence count stays **five**; no carrier lives in the gap.

**Per-track verdict table.**

| TRACK | SUBMITTED | REFEREE OUTCOME | VERDICT BEARING |
|---|---|---|---|
| A1 — close the commutant-to-generated gap → unconditional HIGH | SHARPENED-OPEN (gap closes via BW ergodicity; hedge sheds the commutant condition) | **DOWNGRADE** — verdict-bearing tail upheld, centerpiece STRUCK (major error); verdict-neutral null result | Designated verdict-flipper; did NOT flip. Hedge stays -R6. |

| | | | |
|---|---|---|---|
| A2 — carrier-in-the-gap | RESOLVED-NEGATIVE | **UPHELD with two corrections** (invariant finer than $(p, q)$; wedge-filling mechanism struck; hedge-narrowing softened) | Strengthens; no flip; no carrier in the gap. |
| A3 — OP-44 R2 D=4 cubic coefficient | RESOLVED-NEGATIVE (R2 re-hardens; ℓ_P floor in 4D) | **OVERTURNED → SHARPENED-OPEN** (REFUTE SUSTAINED; iter-9 posture upheld) | Neutral; R2 stays open; foothold `[INFERENCE, medium]`. |

**The decisive correction (A1).** The attack's verdict-bearing tail is sound and uncontested — PARTIAL unchanged, count FIVE, no localized indefinite centralizer element, the reverse-WW ⇔ carrier-no-go identification and the encoding-screen inheriting unchanged. But the centerpiece is a category error on *which* gap iteration 9 located. Ergodicity of the Bisognano–Wichmann vacuum (Borchers' Lebesgue boost spectrum; Longo 1978) makes the modular **centralizer** $\mathcal{M}_\omega = \mathcal{M} \cap \{\Delta^{it}\}'$ trivial. But the iteration-9 gap is the **representation-space multiplicity gap** $\langle \Delta \rangle'' \subsetneq \{\Delta^{it}\}'$, driven by spectral degeneracy of $\Delta$ on $\mathcal{H}$. Because $\Delta \notin \mathcal{M}$, triviality of the centralizer says nothing about this gap, which stays non-empty even in the BW case and contains the canonical indefinite $\eta_0 = \mathrm{sgn}(\ln \Delta)$. So the gap does not close, the Lemma-N1 deficiency is not repaired, and **HYP-CKV-VACUITY stays at -R6** — conditional HIGH on $n_1$ AND the located commutant-to-generated gap.

**What is genuinely kept (re-establishing, not advancing).** Two ESTABLISHED sub-results survive A1: (1) the III1 modular centralizer is genuinely large for degenerate spectrum — integrable weight ⟹ Type $\mathrm{II}_\infty$ centralizer (Connes–Takesaki flow of weights), and any finite $(N, \tau)$ is realizable as a centralizer (Popa, math/0011084) — so the gap is real and wide, not closable on signature alone; (2) the state restricts to a faithful normal trace on its centralizer (Takesaki), so the centralizer is semifinite and the indefiniteness it admits is trace-balanced — signed dimension, no localization. The geometric-vs-degenerate exclusion (a geometric modular flow keeps a non-trivial centralizer only via an internal gauge multiplet, which by DHR commutes with the boost and carries zero spacetime localization) re-establishes the inherited Lemma-N2/$n_1$ obstruction.

**No carrier in the gap (A2), with the vacuity surviving two corrected sub-arguments.** Where geometry lives (Lebesgue modular spectrum), the centralizer is trivial — no algebra-internal carrier (Shlyakhtenko free-quasi-free structure theory; Houdayer 0809.3827 / Boutonnet–Houdayer 1602.01741 corroborate but are not the biconditional). Where the gap is non-empty (almost-periodic sector), the strongest carrier candidate the program has produced — $\mathrm{sgn}(h)$ for a centralizer $h \notin W^*(\Delta)$, indefinite signature $(8, 8, 0)$, and algebra-compatible by an **inner** automorphism since $\mathrm{sgn}(h) \in \mathcal{M}$ — clears the two bars ((i) indefinite, (iv) algebra-compatible) that iteration-7's $\eta_0$ failed, yet remains geometry-void ( $\Delta_{\mathrm{geo}} = 0$). The two corrections: its complete $\mathrm{Aut}(\mathcal{M}, \omega)$-invariant is the finer **per-modular-eigenspace signature** (not the global $(p, q)$ pair — 4 toy conjugacy classes, Skolem–Noether), still discrete/input-indexed/geometry-void; and the "boost-orbit-constant ⟹ wedge-filling" localization argument is struck as sector-inconsistent, the localization-failure surviving via the inherited $n_1$ presupposition.

**The one available positive movement, and why it is only partial.** Combining A1 and A2, the commutant-to-generated residual is **substantially closed for algebra-internal (centralizer, Ad-inner) carriers**. But a residual sliver — algebra-compatible carriers with $\mathrm{Ad}\,\eta \in \mathrm{Aut}(\mathcal{M})$ yet $\eta \notin \mathcal{M}$, living in $\{\Delta^{it}\}' \setminus \mathcal{M}$ — is not discharged. So the hedge narrows substantially toward but **not provably to $n_1$ alone**; grade and condition both unchanged.

**A3 (OP-44 R2) is verdict-neutral.** The decidable D=4 cubic-coefficient sub-computation does not resolve negative: under the radial-subtracting TT projector the coincidence limit selects, the leading boost-charge coefficient vanishes at the physical $D = 4$ ( $m = 2$), and even the submission's own $\Pi^{ST}$ Hessian has leading $s^2$-coefficient $(m-1)(m-2)/2 = 0$ at $m = 2$. R2 stays SHARPENED-

OPEN; the $\ell_P$ floor in 4D is not re-established; the foothold stays `[INFERENCE, medium]`.

**Why the verdict stands a ninth time.** The verdict-flipping condition (a referee-upheld generative, localized, algebra-compatible carrier surviving the encoding screen and the localization-is-geometry result) is exactly what A1/A2 sought in the one place it could still live — the commutant-to-generated gap — and it is not there: the gap does not close, and no carrier lives in it. The residual between HIGH and a theorem is unchanged (the net-naturality theorem + the still-open commutant-to-generated gap). Object-level still handed to experiment; the 2027 $w(z)$ decision point remains the nearest-dated external event.

CHAPTER 31

# Iteration 11 Synthesis: The Hedge Re-Grounded on a Named Open Problem; Terminal Saturation

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**Iteration 11 synthesis — the hedge re-grounded on a named open problem; saturation goes terminal.**

The single most important thing to get right this iteration is what did NOT happen. The mission framed A1 as potentially closing the carrier no-go to a proved theorem (its premise: Connes' bicentralizer problem was "resolved in general by Houdayer–Marrakchi–Tomatsu"). That premise is FALSE — verified by the referee against live sources: Connes' bicentralizer conjecture is OPEN in general as of 2026. So there was never a closure available, and the VERDICT-DELTA "re-verify three times before endorsing" gate (which triggers only on a closure-to-proved-theorem or a referee-UPHELD reverse-WW witness) does not fire. Neither extraordinary outcome occurred.

What did happen is the cleanest possible version of "nothing moved." For three iterations (9, 10) the principal hedge has carried a residual the program defined internally: a "commutant-to-generated gap." Iteration 11's genuine, surviving content is that this residual's algebra-internal horn lives in the modular centralizer $M_\psi$, so the *right vocabulary* for it is centralizer/bicentralizer structure theory — a body of mathematics with an external resolution clock not under the program's control. That is strictly cleaner and more honest. But the referee was emphatic that this is a re-grounding of vocabulary, NOT a reduction: the submission's "logical equivalence to Connes' bicentralizer problem" was MAJOR-CORRECTED, because it conflated two opposite conditions — a *trivial* centralizer (which kills a carrier, and which Marrakchi–Vaes make unconditionally achievable on every $III_1$ factor) versus Haagerup's *bicentralizer-triviality* (which yields an irreducible, *large*-centralizer state in which a carrier exists). Those are opposites; there is no equivalence. The no-go's inner horn stays closed by the inherited geometry-void and non-localizability facts, exactly as before. Grade unchanged, registry condition unchanged, count unchanged at FIVE.

A2 is the corroborating "no flip." Three genuinely independent handles on the reverse-WW question — the holographic (holography of information, plus its real Dec-2025 failure mode in arXiv:2512.18912), the geometric (T-duality/mirror symmetry), and the operational (MIP*=RE) — all reach the *same* carrier gate without producing a class-separating observable. The T-duality result is quietly the most interesting: it positively *supports* the wager's emergent-geometry ontology (two operationally identical metrics embarrass geometry-first Program G), but symmetrically across both theory classes, so it yields no witness. The holographic failure mode is the sharpest co-location: where holographic completeness fails, an emergent localized observer has been installed by gravitational dressing — which is precisely a carrier. These are roads to the gate, not new convergence routes; count stays FIVE.

A3 is the track that should govern what happens next, and it is referee-upheld: terminal analytical saturation. The marginal-yield curve is now flat to second order — grade-derivative zero since iteration 9, condition-derivative zero since iteration 10, and iteration 11 added only a precision re-grounding with no registry advance. Every residual that could move the verdict is gated on external input. This is the README-sanctioned obtained-outcome: a precise, reasoned account of why the verdict is not-yet-movable, which is itself the deliverable. The honest recommendation is TERMINAL-SATURATION: consolidate the standing deliverable and shift to lightweight watch-mode on (a) the carrier problem / any 2025–2027 net-naturality or bicentralizer development, and (b) the 2027 DESI five-year $w(z)$ verdict.

Net effect on the wiki: append-only iteration-11 blocks to FINDINGS / OPEN_PROBLEMS / HYPOTHESES; two conservative CONCLUSION amendments plus a §6 terminal-saturation capstone note; a CHANGELOG entry; verified-only BIBLIOGRAPHY additions. No file deletions, no retractions of prior text, no new numeric IDs. The tenth consecutive confirmation is recorded; the hedge is cleaner; the program is honestly done reasoning.

CHAPTER 32

# Iteration 12: Terminal Saturation Re-Confirmed — the Physics-Net Lever Retired and the Indefinite-Metric Route Closed

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> **One-line verdict.** A fresh mid-2026 sweep of the carrier/net-naturality math, the Connes bicentralizer problem, reverse-WW phenomenology, the 2027 DESI w(z) program, and the empirical channels finds **no new decidable lever that could move the verdict or the principal hedge by reasoning.** The iter-11 A3 terminal-saturation diagnosis (referee-upheld) is **UPHELD as of 2026-06**. The one genuinely-new 2026 mathematics in the carrier circle — the Morinelli–Neeb–Olafsson Euler-pair Bisognano–Wichmann/Spin–Statistics theorems — is a *cleaner statement of the encoding screen* (the symmetry group is INPUT), not a carrier solution. Standing verdict unchanged, **11th consecutive confirmation**; hedge unmoved in grade and condition; carrier-convergence count stays **FIVE**. `[INFERENCE, high]`

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## 1. Scope and method

This note re-runs the single external obligation the saturation diagnosis leaves open: *has a NEW (2026) external development appeared that constitutes a decidable verdict-moving lever?* It does not re-attack any gap. Every load-bearing reference below was web-verified live this session against the actual abstract/source. The sweep covers the four channels the mission named: (a) carrier / net-naturality / GMA-reconstruction math; (b) the Connes bicentralizer problem; (c) reverse-WW / distinguishing-test phenomenology; (d) the 2027 DESI five-year w(z) program + adjacent dark-energy probes (Euclid, Roman) and the empirical channels (BMV/QGEM, objective collapse, GQuEST/VZ, LIV/EP).

## 2. Sweep results — channel by channel

### 2a. Carrier / net-naturality / GMA-reconstruction math — one new 2026 result, encode-pattern

The genuinely-new 2026 item in the net-naturality/Euler-element/geometric-BW circle is the **Morinelli–Neeb–Olafsson "Orthogonal pairs of Euler elements"** program: Part I (arXiv:2508.10960, 14 Aug 2025 — classification, fundamental groups, twisted duality) and **Part II (arXiv:2603.26390, 27 Mar 2026 — Geometric Bisognano–Wichmann and Spin–Statistics Theorems)**. Abstract read verbatim: AQFT models are *generalized including Lie groups of symmetries whose Lie algebras admit an Euler element h*, with ad h diagonalizable, eigenvalues in {−1,0,1}; from an orthogonal Euler pair the authors *derive both a Bisognano–Wichmann Theorem and a Spin–Statistics Theorem for nets of standard subspaces and von Neumann algebras*.

**Adjudication — this is the encoding screen, not a carrier.** The Euler element is *the generator of the modular group* ***in the symmetry Lie algebra*** (verified verbatim from the program's surrounding literature): the Lie group of symmetries is the **INPUT**, and the causal/wedge/Lorentzian structure is "output" only relative to that pre-installed symmetry. This is exactly the object the iter-11 A3 audit named as the irreducible smuggle — *"forced to be signature-inherited from a pre-installed Euler-element/causal-index-set."* The result is a *stronger, cleaner* statement of HYP-ENCODING-SCREEN (geometry recovered precisely where the symmetry is installed by hand), **not** a construction of Lorentzian structure from genuinely non-symmetric (algebra, state) data, and therefore **not** a carrier solution, **not** a reverse-WW inversion, and **not** a sixth convergence route. Count stays FIVE. [INFERENCE, high]

The broader GMA-reconstruction literature returns only the standing Buchholz–Summers / Brunetti–Guido–Longo / Neeb–Morinelli corpus already underwriting Lemma N1/N2. No net-naturality theorem (the verdict-bearing residual) has appeared.

## 2b. Connes' bicentralizer problem — confirmed OPEN in general (2026)

Verified live: the conjecture B(M,φ)=C1 for every type III_1 factor (separable predual) **remains open in general as of 2026**; settled only for special classes (amenable, Cartan, free products, semisolid, q-deformed Araki–Woods, tensor-III_1). The May-2026 model-theory paper **arXiv:2605.12776** restates it as open via a "bicentralizer functor zeroset" reformulation — it is the same paper the iter-11 referee cited. **No Houdayer–Marrakchi–Tomatsu general-resolution paper exists.** This upholds the binding iter-11 referee correction in full. [OPEN]

## 2c. Reverse-WW / distinguishing-test phenomenology — still EVADE, not invert

The only 2026 item touching Weinberg–Witten is the **"GR from RG" essay (Sheikh-Jabbari–Taghiloo, arXiv:2602.11806)**, which proposes to **evade** WW — boundary conditions flowing from rigid Dirichlet (UV) to mixed Dirichlet–Neumann (IR) to "unfreeze" the metric, or the energy-momentum tensor vanishing/non-conserved by construction. The background metric is **Dirichlet-fixed in the UV** (installed), confirming the iter-11 minor-correction (it confronts/evades, does not invert, WW). Evading WW by zeroing the T_munu matrix element is precisely the handle-erasing move HYP-ENCODING-SCREEN predicts; it produces **no class-separating observable**. The **reverse-WW witness remains unattempted /** `[OPEN, as of 2026-06]`, the empirical-side shadow of the carrier problem (iter-9 identity), carrying no registry ID. `[OPEN]`

## 2d. The 2027 DESI w(z) program + adjacent probes — unchanged, target not wager

- **DESI**: the dynamical-dark-energy preference persists in DR2 (~3–4σ for w0waCDM, w0>−1, wa<0, possible phantom crossing) but is **dominated by low-redshift SN systematics** — correcting low-z SN calibration reduces it to **< 2σ** (arXiv:2502.04212; cf. calibration-independent SN-systematics tests). The decisive **five-year (DR3) w(z) verdict is still expected 2027** (DESI 3D map completed Apr 2026). `[CONTESTED]`
- **Roman / Euclid**: Roman launches 2026; Roman + Euclid joint dark-energy investigation by ~May 2027 (NASA/JPL). Adds multi-probe confrontation but no earlier decision point.
- All bear on Program A's correct **TARGET** (eternal dS vs metastable dS vs rolling quintessence), **never the wager** — status unchanged from the watchlist.

## 2e. Empirical channels — gate/core/target only, no wager bearing

- **Objective collapse**: NEW 2026 bounds — **VIP collaboration analysis of XENONnT** spontaneous-radiation data (Gran Sasso) improves **CSL by ~2 orders of magnitude** and tightens **Diósi–Penrose by ~factor 5**, no excess detected. Constrains a fixed **CORE** (exact linearity, A-3), not the wager. `[ESTABLISHED]`
- **BMV / QGEM**: ~6–8 orders below decisive mass/coherence; no funded milestone roadmap; 2026 work is

decoherence/tripartite-entanglement *modelling* (Sci. Rep. s41598-026-44184-2), not a run. Bears on the **gate**.

- **GQuEST / VZ**: design in PRX 15 (2025) 011034; preliminary build at Caltech; no science data as of 2026-06. The Sharmila–Vermeulen–Datta mechanism-blind correlator taxonomy still governs any signal. Program-adjacent, not a wager test.
- **GW strong-field nulls, LIV/EP nulls, matter-wave macroscopicity**: continuing nulls/records harden cores and enablers; none bears on the wager.

## 3. Residual enumeration with decidability classification (refreshed)

| RESIDUAL | STATUS (2026-06) | ROUTE | MOVES VERDICT/HEDGE? |
|---|---|---|---|
| Carrier net-naturality theorem (HYP-CKV-VACUITY-R6 / HYP-FACTORIZATION-IS-GEOMETRY) | Unchanged; 2026 Euler-pair BW/Spin–Statistics theorems are encode-pattern (symmetry = input), not a net-naturality theorem | **external-math-gated** | YES if solved — but no 2026 progress toward it |
| Commutant-to-generated gap | Unchanged (iter-10 null); no 2026 operator-algebra development closes it | **external-math-gated** | NO reasoning-move |
| Connes bicentralizer (related vocabulary, not a reduction — iter-11 binding) | Confirmed OPEN in general (arXiv:2605.12776) | **external-math-gated** | NO — related vocabulary only; not equivalent to the carrier no-go |
| Reverse-WW witness | `[OPEN, 2026-06]`; 2026 'GR from RG' evades not inverts | **external-math-gated / external-experiment-gated** | NO independent move — it *is* the carrier problem in empirical clothes |

| | | | |
|---|---|---|---|
| 2027 DESI five-year w(z) | Verdict still 2027; DR2 ~3–4σ → < 2σ under SN systematics | **external-experiment-gated** | NO — bears on TARGET not wager |
| Objective collapse (CSL/DP) | NEW 2026 VIP/XENONnT bounds tighten the core | **external-experiment-gated** | NO — CORE, not wager |
| OP-44 R2 higher-D coeff; OP-49 a/b/c′; OP-48c X1/X2; OP-50 | Unchanged | (i) reasoning-decidable but **headline-inert** / (ii) external | NO — headline-inert by own scoping |

**Result.** Of the verdict-bearing residuals, **zero** are reasoning-decidable in a way that could move the verdict or hedge. The two reasoning-decidable residuals stay headline-inert by their own scoping. Every verdict-bearing residual is **external-math-gated** (four collapse to the single carrier net-naturality breakthrough ⇔ commutant-gap closure ⇔ reverse-WW witness ⇔ HYP-CKV-VACUITY→unconditional HIGH) or **external-experiment-gated** (2027 DESI w(z), which bears on the target).

## 4. Adjudication — is the saturation diagnosis still correct?

**YES — terminal saturation is upheld as of 2026-06.** No purely-reasoning lever can move the verdict; only external input can. The classification:

- **Reasoning-decidable-but-headline-inert:** OP-44 R2 (higher-D / s^4 remnant coefficient), OP-49 (a/b/c′). Unchanged.
- **External-math-gated:** carrier net-naturality theorem; commutant-to-generated gap; reverse-WW witness; Connes bicentralizer (related vocabulary). The 2026 Euler-pair theorems and the model-theory bicentralizer paper are *adjacent activity in the right circle* but neither inverts the encoding screen nor closes the gap — they reconfirm it.

- **External-experiment-gated:** 2027 DESI five-year w(z) (target); objective-collapse coverage (core); BMV/QGEM gate; GQuEST/VZ program-adjacent. New 2026 results tighten cores/enablers, none touches the wager.

No genuine new decidable verdict-moving lever has appeared. The marginal-yield curve stays flat to second order (grade-derivative zero since iter 9; condition-derivative zero since iter 10; iter 11 added precision re-grounding only; iter 12 adds external confirmation only).

## 5. Conservative registry implication

NO new numeric ID; NO grade/condition change to any hedge; count stays FIVE; verdict unchanged (11th consecutive). Append-only watchlist/watch-item notes recording: Connes OPEN confirmed; the 2026 Euler-pair theorems flagged as an encode-pattern instance (explicitly NOT a sixth route, NOT a carrier); the 2026 objective-collapse core-tightening; the reaffirmed-2027 DESI w(z) verdict with strengthened SN-calibration caveat; the 'GR from RG' evade-not-invert instance.

## Referee verdict — track B3 (external-input + experiment sweep / terminal-saturation refresh), iteration 12 (2026-06-19)

*Adversarial referee, default stance REFUTE. Every load-bearing citation web-verified live this session against the actual abstract/source; the one genuinely-new mathematical result (Euler-pair Part II) fetched and read verbatim; the no-go red-team run against the binding iter-9/10/11 corrections and the standing five-route convergence count.* ***Stance NOT sustained as a refutation — the submission survives with citation-precision corrections only; no fatal, major, or verdict-bearing defect.***

**Outcome: CONFIRMS-SATURATION — UPHELD WITH CITATION-PRECISION CORRECTION. Standing verdict**

**UNCHANGED (11th consecutive). Carrier-convergence count FIVE. Hedge HYP-CKV-VACUITY stays -R6 (grade and condition). No new registry ID.**

## What is sound and verified (citations clean / faithful)

- **Euler-pair Part II (arXiv:2603.26390) is the encoding screen, not a carrier — confirmed verbatim.** I fetched the abstract directly. The Lie groups of symmetries / Euler element h are **pre-installed as input**; the Bisognano–Wichmann and Spin–Statistics theorems are **derived FROM** the Euler element; the paper makes **no claim** to construct causal/Lorentzian/metric structure from non-symmetric (algebra, state) data. This is exactly the pre-installed Euler-element/causal-index-set the iter-11 A3 audit named as the irreducible smuggle. The finder's adjudication — stronger/cleaner HYP-ENCODING-SCREEN instance, NOT a carrier, NOT a reverse-WW inversion, NOT a sixth route — is correct and survives the no-go red-team. **Count stays FIVE.** Part I (arXiv:2508.10960, Forum Math. 38(3) 2026) also verified.
- **Connes' bicentralizer problem OPEN in general — confirmed (arXiv:2605.12776).** Real, posted 12 May 2026; the 'bicentralizer functor zeroset' reformulation matches the finder's gloss. Upholds the binding iter-11 correction in full; no Houdayer–Marrakchi–Tomatsu general-resolution paper exists. The finder does **not** lean on the bicentralizer route to close the no-go — it correctly treats it as confirmed-open external vocabulary, respecting every binding iter-11 correction (trivial-centralizer ≠ trivial-bicentralizer; the no-go is not a reduction to Connes).
- **'GR from RG' (arXiv:2602.11806) — evade-not-invert, now CONFIRMABLE.** Live search confirms the paper explicitly engages Weinberg–Witten and unfreezes a UV-Dirichlet-fixed (installed) metric (Dirichlet→mixed Dirichlet/Neumann under RG flow). This **discharges** the iter-11 A3 minor defect (where the WW gloss was flagged 'not

confirmable'). Produces no class-separating observable; reverse-WW stays `[OPEN, 2026-06]`, the empirical-side shadow of the carrier problem, no registry ID. Correct.

- **VIP/XENONnT objective-collapse bounds — numbers exact.** CSL ~2 orders, DP ~factor 5, no excess; verified against the PRL March 2026 result. Constrains the A-3 CORE, not the wager. Correctly classified.
- **QGEM tripartite-entanglement (Sci. Rep. 2026, s41598-026-44184-2) — verified;** decoherence modelling, not a run; bears on the gate.
- **Marginal-yield curve flat to second order; terminal saturation UPHELD.** Every verdict-bearing residual is external-math-gated or external-experiment-gated; the two reasoning-decidable residuals (OP-44 R2; OP-49 a/b/c′) are headline-inert by their own scoping. No new decidable verdict-moving lever has appeared.

## Citation-precision corrections (binding over the finder's claims; none verdict-bearing)

1. **arXiv:2605.12776 authorship.** Authors are **Ando–Goldbring**, not "I. Goldbring et al." Correct in any append.
2. **arXiv:2502.04212 authorship. Huang–Cai–Wang** (Efstathiou in acknowledgments only). Strike "Efstathiou-type ... analysis"; attribute correctly.
3. **DESI residual magnitude.** Source abstract states the preference reduces to **"less than 2σ"**; the finder's "~0.5–1.5σ" is sharper than the cited source supports. Soften to "< 2σ".
4. **Primary VIP/XENONnT source.** Name **arXiv:2506.05507** (PRL March 2026) as the primary bound in any watchlist append; arXiv:2512.15393 and arXiv:2604.21705 (both verified real) are adjacent/auxiliary, not the primary result.

## No-go / extraordinary-claim red-team — clean

No carrier solution, no metric smuggle, no generative (vs encoding) construction, no reverse-WW witness anywhere in the sweep. The one new mathematical result presupposes the symmetry group and outputs geometry only relative to it — encode-pattern, verified verbatim. The extraordinary-claim gate does not fire. No conflict with dS-cardinality, the iter-7 trichotomy / η_0 vacuity, the iter-8 net no-go, the iter-9 naturality obstruction / reverse-WW = carrier identity, the iter-10 commutant-gap null, or the binding iter-11 bicentralizer correction. Count stays FIVE; no RESOLVED-POSITIVE offered.

## Registry implication (conservative)

NO new numeric ID; NO grade/condition change to any hedge; count FIVE; verdict unchanged (11th consecutive). Append-only watchlist/watch-item notes as the finder proposes, with corrections (1)–(4) applied: Connes OPEN confirmed (Ando–Goldbring, arXiv:2605.12776); the 2026 Euler-pair theorems flagged as an encode-pattern instance (explicitly NOT a sixth route, NOT a carrier); the objective-collapse core-tightening (primary arXiv:2506.05507); the reaffirmed-2027 DESI w(z) verdict with the SN-calibration caveat (Huang–Cai–Wang arXiv:2502.04212, preference < 2σ); the 'GR from RG' evade-not-invert instance (now confirmable, discharging the iter-11 minor defect).

## Net

A verdict-neutral external sweep whose substance is correct and whose load-bearing citations are all real and faithfully characterized. The single genuinely-new mathematical result is correctly adjudicated as an encoding-screen instance rather than a carrier. The CONFIRMS-SATURATION outcome and every verdict-bearing claim are **sustained**; the only defects are four citation-precision items, none of which touches the verdict, the hedge, or the count. Terminal saturation upheld; the wager remains not-yet-physics; the object-level forecast unchanged.

CHAPTER 33

# Iteration 13: The Frame Audited From Outside — Is the Carrier the Unique Gate, and How Close Are We?

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cat# Track C1 — Bottleneck-completeness audit: is the carrier problem the UNIQUE analytical gate to the Program-A wager?

> **One-line verdict.** The carrier problem is the unique analytical *reasoning-decidable* gate to the Program-A **wager** (geometry-from-algebra). The other obstructions the wiki tracks — the measurement problem, the cosmological-constant problem, OP-50 (closed-universe Hilbert space), the black-hole-information problem, and the 17 GR-QFT clashes — are **genuine independent problems of physics**, but for each one, it is *either* not a gate of the central wager *or* itself external-gated (decidable only by experiment or community resolution), never both a gate AND a neglected reasoning-lever. So the unique-gate claim survives, the terminal-saturation diagnosis is **strengthened**, and the only correction is a scope clarification the wiki's own text already implies. **No verdict, hedge, condition, or count moves.** `[INFERENCE, high]`

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## 1. What was audited, and against what

The mission: is the carrier problem (= reverse-Weinberg-Witten = the net-naturality theorem; HYP-FACTORIZATION-IS-

GEOMETRY) genuinely the UNIQUE analytical gate to the Program-A wager, or has the wiki under-weighted INDEPENDENT obstructions to a final theory? For each candidate obstruction I asked three questions:

- **(a) downstream?** Is it reducible to / a consequence of the carrier problem?
- **(b) independent?** Is it genuinely independent of the carrier problem?
- **(c) decidable?** Is it itself reasoning-decidable, or external-gated (experiment / community resolution)?

And then the *load-bearing* question the mission turns on: is it a **gate of the wager** (the encode→generate step), or a problem of physics that sits beside the wager?

The frame I am auditing is set by CONCLUSION §3/§4/§7/§8: the wager is *"causal/algebraic/entanglement structure precedes metric geometry, true of our Λ>0 universe"*; the gate is *"derive an algebra-compatible, localized (n,n) pairing from modular data"*; this gate is proven (iter-9) equivalent to the reverse-WW witness and to the net-naturality theorem.

## 2. The decisive distinction — gate of the *wager* vs problem of *physics*

The wiki's "single gate" language (ROADMAP: *"the single open analytical gate"*; CONCLUSION §8) is scoped, and the scope is load-bearing. CONCLUSION §3 already separates three senses of "is a universal theory possible":

- **§3A — a consistent unified GR+QFT framework: POSSIBLE, high confidence.** AdS/CFT is an existence proof. This is **NOT gated by the carrier problem** — it holds whether or not geometry is generated from algebra.
- **§3B — a final substrate: possible but not demonstrably so.** Wilsonian tower / emergence.
- **§3C — a parameter-fixing complete theory: least likely.** Threatened independently by Λ, the landscape, the Past

Hypothesis.

The carrier problem gates only the **wager** — the specific bet that geometry is *generated* (not merely *encoded*) by causal/algebraic structure. Conflating "the gate to the wager" with "the gate to unification" would overstate the claim. This is the one scope finding of the audit, and it *sharpens* rather than weakens the unique-gate claim: the gate is unique precisely because the wager is the wiki's narrowly-defined central object.

## 3. The five candidate independent obstructions — per-obstruction adjudication

| Obstruction | (a) downstream of carrier? | (b) independent? | (c) reasoning-decidable? | Gate of the wager? | Verdic |
|---|---|---|---|---|---|
| **Measurement / collapse** (OP-2, OP-26, GC-6) | No | **YES** (Program B; fusion refused, §2) | **No** — external-gated (collapse experiments) | **No** (tests a CORE / Program B, not encode→generate) | indepen problem physics, external gated, n neglecte lever |
| **Cosmological constant** (OP-3, GC-3) | No | **YES** | **No** — fine-tuning + experiment; Weinberg no-go targets Λ→0, a different object | **No** (bears on TARGET §3A/ §3C, not the gate) | indepen external gated, n neglecte lever |
| **OP-50 closed-universe Hilbert space** | No | **YES** (structural analogue of OP-48c horn-C) | **No** — external community resolution; `[CONTESTED]`, no theorem either side | **No** (structural analogy only; "cannot harden the headline", iter-11 A3) | indepen external gated, n neglecte lever |

| | | | | | |
|---|---|---|---|---|---|
| **BH information / Page curve / islands** (OP-15, GC-4, GC-5) | No | **YES** | **No** — gated by deeper QG input; unresolved for realistic BHs | **No** (tests whether semiclassical QFT is used out-of-domain) | independent, external-gated, not a neglected lever |
| **The 17 GR-QFT clashes** (GC-1..GC-17) | mixed | mixed | mostly domain-mismatch, no reasoning content bearing on the wager | only GC-1/13/16 touch the wager — and they ARE the carrier problem | none is a load-bearing independent gate |

## 3.1 Measurement / collapse (OP-2, OP-26, GC-6) — INDEPENDENT, external-gated, NOT a wager-gate

The measurement problem lives in **Program B** (is-the-quantum-framework-final), which CONCLUSION §2 keeps strictly separate from Program A: the two share only no-signaling/causal-order, and *"the it's-all-information fusion is REFUSED."* Resolving the carrier would not touch measurement; resolving measurement would not touch the carrier. It is genuinely independent — and it is **not reasoning-decidable**: the only interpretation-distinguishing models are objective-collapse, decided by experiment. **Web-verified (2026):** the collapse class is *not* falsified — XENONnT/Gran Sasso/VIP exclude specific CSL and squeeze Diósi-Penrose parameterizations, but models without spontaneous radiation remain viable (arXiv:2209.15082; Donadi et al.; PMC10138035 review: *"broader classes … have not been definitively falsified as a class"*). So it is an independent problem of physics that is external-gated, not a neglected reasoning-lever, and not a gate of the geometry-from-algebra wager.

## 3.2 Cosmological constant (OP-3, GC-3) — INDEPENDENT, external-gated, NOT a wager-gate

A fine-tuning / domain-mismatch, **not** a logical inconsistency (GC-3 referee: in EFT, Λ is a renormalized free parameter; *"no failed prediction and no inconsistency"*; vacuum energy gravitates as Λ by Lovelock uniqueness). **Web-verified:** Weinberg's no-go theorem is a genuine no-go but targets a *different object* than the wager — it forbids a local dynamical mechanism setting $\Lambda \to 0$ without fine-tuning (reduces to "minimum of a potential is zero"). The 2025 nonlocal-gravity route (arXiv:2502.07321, infinite-derivative gravity) *invalidates* Weinberg's no-go by lifting locality — a fine-tuning-evasion proposal, not a wager-bearing result. Λ bears on the wager's **TARGET** (our $\Lambda>0$ universe; the dS-cardinality no-go, §4) and on parameter-fixing completeness (§3C), never on the encode→generate gate.

### 3.3 OP-50 closed-universe Hilbert space — INDEPENDENT, external-gated, NOT a wager-gate

The wiki already classifies OP-50 as the **path-integral face of the OP-48(c) horn-C residue** — a structural *analogue* of the algebraic observer-relational no-go, not a downstream consequence of the carrier. **Web-verified (2025-2026):** genuinely `[CONTESTED]`, no theorem on either side — Usatyuk-Wang-Zhao / Harlow-Usatyuk-Zhao (1-dim per α-sector) vs Antonini et al. (arXiv:2507.10649, *"the baby universe is fine"*) vs Hong Liu (arXiv:2509.14327, observer-free subalgebra framing) vs Engelhardt-Gesteau (no semiclassical baby universe) vs the symmetric-orbifold polynomial-dim $II_1$ sector (arXiv:2606.04575). Per iter-11 A3 it *"cannot harden the headline"* — so it is correctly NOT folded into the carrier gate, and it could not be a neglected reasoning-lever because its premise is itself contested and externally decided.

### 3.4 BH information / Page curve / islands (OP-15, GC-4, GC-5) — INDEPENDENT, external-gated, NOT a wager-gate

**Web-verified (2024-2026):** the island/QES/Page-curve resolution is established only in AdS / lower-dimensional / semiclassical-plus-replica **toy models**; for asymptotically-flat, dynamically-formed 4D black holes the bulk mechanism is NOT settled (PMC12191829 review: *"unitary evolution in solvable models"*; the late-time radiation *"is necessarily a macroscopic superposition state"*; *"perhaps there is still much more to say"*).

This matches OP-15/GC-4 verbatim and confirms the wiki did not overclaim a resolution. The problem tests whether **semiclassical QFT is pushed outside its domain** near the horizon — it does not bear on whether modular data *generates* Lorentzian geometry. Gated by deeper QG input, not reasoning-decidable from the wiki's algebra.

### 3.5 The 17 clashes (GC-1..GC-17) — none a load-bearing independent gate

By the wiki's own referee verdicts: **zero are proven logical inconsistencies**; they decompose into domain-mismatches (GC-1,2,3,9,11,14,17), unsolved-but-consistent problems (GC-4,7,10,13), and conceptual tensions (GC-5,6,8,12,15,16). The only clashes that touch the wager directly — GC-1 (background-independence), GC-13 (area-law vs volume-extensive Hilbert space), GC-16 (spacetime-from-entanglement generality) — are **exactly what the carrier problem already encodes**: GC-1 is the encode-not-generate frontier; GC-13/GC-16 are the Type-III1/no-factorization root the carrier problem confronts. The rest are downstream of the already-audited independent obstructions (measurement, Λ, BH-info) or are domain-mismatches with no reasoning-decidable content bearing on geometry-from-algebra.

## 4. Adversarial test of the unique-gate claim

I tried to find an INDEPENDENT, reasoning-decidable bottleneck that the carrier-focus neglected. The candidates and why each fails the "neglected reasoning-lever" bar:

- **Measurement** — not a wager-gate (Program B), and external-gated. Fails (ii).
- **Λ** — not a wager-gate (target/completeness), and the one no-go in the area (Weinberg) is about a different object and is itself being evaded by external proposals. Fails (i) and (ii).
- **OP-50 / BH-info** — not wager-gates, and external-gated (community / deeper QG). Fail (ii).

- **The clashes** — the wager-relevant ones ARE the carrier problem; the rest are downstream/domain-mismatch. Fail (i).

No candidate is simultaneously (i) a gate of the wager and (ii) a neglected reasoning-decidable lever. The closest thing to an "independent decidable bottleneck" would be a clash that is a *proven logical inconsistency* forcing the wager false — and the wiki's central result is that **none of the 17 is a proven inconsistency.** So there is no INDEPENDENT-BOTTLENECK-FOUND.

## 5. What this adds to the terminal-saturation diagnosis

The iter-11/iter-12 audits established that every verdict-bearing *residual* is external-gated. This audit closes a complementary gap they did not explicitly address: that no *independent obstruction outside the residual list* is a neglected reasoning-bottleneck. The bottleneck map is **complete** — the carrier problem is not just the last open residual on a list, it is the unique reasoning-decidable gate to the wager, with the other great problems of physics sitting beside the wager (independent) rather than gating it. This **strengthens** the saturation diagnosis rather than perturbing it.

## 6. Conservative registry implication

NONE that consumes an ID. The verdict, the principal hedge (HYP-CKV-VACUITY-R6, grade and condition), the HYP-ENCODING-SCREEN no-test corollary, and the carrier-convergence count **FIVE** are all unmoved (this would be the 12th consecutive confirmation if logged). Optional append-only annotation: record the bottleneck-completeness result and the scope clarification (gate is unique *to the wager*, not *to unification*) under CONCLUSION §6/§8 or HYP-ENCODING-SCREEN. No grade, condition, or count change.

## 7. Honesty notes / limits of this audit

- This is a **conservative meta-audit**, not a new construction. It re-uses the wiki's own classifications and verifies the 2024-2026 status claims it leans on; it does not re-derive the carrier no-go.
- The "independence" judgments for measurement and Λ are `[ESTABLISHED]` (they follow from the §2 Programs-A/B split and the GC-3/GC-6 referee verdicts). The "completeness of the bottleneck map" judgment is `[INFERENCE]` — it is an absence-of-neglected-lever claim, defensible but not a theorem.
- A genuine adversary could argue the §2 Programs-A/B split is itself a modeling choice, and that a future *fusion* of the two programs could make measurement a wager-gate. The wiki explicitly refuses that fusion (§2, "information names non-interchangeable objects"); if that refusal were ever overturned, this audit would need re-running. I record that as the single contingency, not a current lever.

# C1 Referee — Bottleneck-completeness audit (iteration 13)

**Track:** C1 — is the carrier problem the unique analytical gate to the Program-A wager, or has the wiki under-weighted independent obstructions? **Default stance:** REFUTE. **Stance NOT sustained as a refutation. Outcome: UPHELD-WITH-CORRECTION (verdict-neutral meta-audit).**

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# 1. What the finder claimed, and the standard I held it to

The finder submitted **UNIQUE-GATE-CONFIRMED**: the carrier problem is the unique *reasoning-decidable* analytical gate to the Program-A **wager** (geometry-from-algebra); the measurement problem, the cosmological-constant problem, OP-50, the BH-information problem, and the 17 GR-QFT clashes are genuine independent problems of physics but none is *both* a gate of the wager *and* a neglected reasoning-decidable lever. One scope correction: the gate is unique to the *wager*, not to *unification*.

I held this to the iteration-13 mission constraints (meta-audit, NOT a carrier re-attack; conservative reading; web-verify every program-status claim; do not move verdict/hedge/count; do not consume new IDs) and to the binding state in CONCLUSION §2-§8, OP-2/3/15/26/50, GC-1..17, and the iter-11/iter-12 terminal-saturation audits.

# 2. Frame audit — is the unique-gate claim even well-posed?

The finder's decisive move is the **gate-of-the-wager vs problem-of-physics distinction**, and it is correct and load-bearing. CONCLUSION §3 already separates three senses of "possible": §3A (consistent unified framework — POSSIBLE, high-confidence, AdS/CFT an existence proof), §3B (final substrate), §3C (parameter-fixing completeness). The carrier gates only the **wager** (the encode→generate step), never §3A. The wiki's "single analytical gate" language (ROADMAP; §8) is therefore scoped, and the finder is right that conflating it with "the gate to unification" would overstate it. This is a real frame-sharpening, and it sharpens *toward* the standing verdict rather than against it — exactly what a conservative meta-audit should produce.

## 3. Per-obstruction adjudication — did the finder miss a neglected lever?

I ran the adversarial test the mission demands: find an obstruction that is simultaneously (i) a gate of the wager and (ii) a neglected *reasoning-decidable* lever. Each candidate fails, and the finder's reasons are grounded in the wiki's own referee verdicts:

- **Measurement (OP-2/26, GC-6):** Program B; §2 refuses the A/B fusion. External-gated (collapse experiments; only objective-collapse models are interpretation-distinguishing, matching GC-6 verbatim). Not a wager-gate. Web-verified: collapse class not falsified (PMC10138035). Fails (i) and (ii).
- **Λ (OP-3, GC-3):** fine-tuning/domain-mismatch, "no failed prediction and no inconsistency" (GC-3 referee). Weinberg's no-go targets $\Lambda \to 0$ dynamical adjustment — a *different object* than geometry-from-algebra — and is itself being evaded by external proposals (web-verified 2502.07321, nonlocal/infinite-derivative gravity). Bears on §3A target / §3C completeness, not the gate. Fails (i) and (ii).
- **OP-50:** structural analogue of OP-48c horn-C; `[CONTESTED]` with no theorem either side; external-community-gated (web-verified live dispute: Antonini 2507.10649 vs Usatyuk-Wang-Zhao vs Liu vs Engelhardt-Gesteau). iter-11 A3: "cannot harden the headline." Fails (ii).
- **BH-info (OP-15, GC-4/5):** web-verified unsettled for realistic asymptotically-flat dynamically-formed 4D black holes (islands established only in AdS/lower-D/semiclassical toy models); tests whether semiclassical QFT is used out-of-domain, not whether modular data generates Lorentzian geometry. External-gated by deeper QG input. Fails (ii).
- **The 17 clashes:** zero proven logical inconsistencies (the wiki's central GC result). Only GC-1/13/16 touch the wager — and those are what the carrier already targets. The rest are domain-mismatches or downstream of the already-audited independent obstructions. None is a load-bearing independent gate.

**No candidate is simultaneously (i) and (ii).** The closest possible independent decidable bottleneck — a clash that is a *proven inconsistency* forcing the wager false — does not exist, because none of the 17 is a proven inconsistency. So there is no INDEPENDENT-BOTTLENECK-FOUND, and the unique-gate claim survives. This is the correct, conservative result.

## 4. Overclaim red-team

- **No triangulation overclaim.** The finder uses *independence* to DENY neglected levers, not to assert convergence. The obstructions are explicitly stated to sit *beside* the wager (independent), not to *converge on* the carrier. This is the conservative direction; the anti-crank "shared vocabulary ≠ shared prediction" trap is avoided.
- **No carrier re-attack.** The finder explicitly does not re-derive or re-attack the carrier no-go; count stays FIVE; hedge untouched. Compliant.
- **The completeness judgment is honestly hedged.** §7 of the note concedes it is `[INFERENCE]`, an absence-of-neglected-lever claim "defensible but not a theorem." Calibration is correct. I bind this hedge into BC-1: the phrase "the bottleneck map is complete" must be softened to "no *tracked* obstruction is a neglected reasoning-decidable lever."
- **The single contingency (§2 fusion-refusal overturn) is correctly logged as a contingency, not a current lever.** Upheld.

## 5. Citation audit

All load-bearing program-status/literature claims web-verified this session or inherited from the binding iter-12 track-B3 referee's verbatim verification. Real, faithfully characterized, no fabrication: 2502.07321 (Weinberg-no-go/nonlocal-gravity), 2507.10649 (Antonini closed-universe), PMC10138035 (collapse class), BH-info 2025 status, the OP-50 dispute corpus, and the inherited Euler-pair / Connes-OPEN / GR-from-RG / VIP-XENONnT chain. See the citation_audit array. Only precision note: name arXiv:2506.05507 as the *primary* 2026 collapse bound (BC-4).

## 6. Outcome and binding corrections

**UPHELD-WITH-CORRECTION.** The submission is a sound, conservative, verdict-neutral meta-audit that strengthens the terminal-saturation diagnosis without moving any registry quantity. Corrections **BC-1 through BC-6** (binding, none verdict-bearing): keep the completeness claim at `[INFERENCE]` and soften "complete"; the scope clarification is an annotation, not a registry advance (no new ID); GC-1/13/16 "ARE the carrier" stays `[INFERENCE]`; name 2506.05507 as primary collapse source; strip the `cat#` typo; the §2-fusion contingency is the sole standing caveat.

**Verdict / hedge / count: UNMOVED.** Standing verdict confirmed (12th consecutive if logged); HYP-CKV-VACUITY-R6 unmoved in grade and condition; HYP-ENCODING-SCREEN no-test corollary unmoved; carrier-convergence count **FIVE**; next free IDs OP-51 / A-24 / GC-18 / H9. Terminal analytical saturation re-confirmed; the bottleneck map carries no hidden independent reasoning-bottleneck.

CHAPTER 34

# Iteration 14: The Verdict Survives Its Maximal Adversarial Assault

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> **One-line.** For the first time the burden was inverted: rather than test one more lever, three adversarial tracks were told to assume the standing verdict is WRONG and to build the single strongest web-verifiable 2024–2026 falsification of each load-bearing pillar, with no hedging. **All three pillars survived.** The verdict is confirmed a 13th consecutive time — but earned by surviving the strongest attack we could mount, not by another passive lever-test. `[INFERENCE, high]`

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## What was attacked, and how each attack collapsed

The standing verdict SURVIVED its strongest possible assault on all three pillars — no prong broke. Iteration 14 was deliberately different: instead of testing one more lever, three adversarial tracks were told to assume the verdict is wrong and to build the single strongest web-verifiable 2024-2026 falsification of each pillar, with no hedging. Pillar 1 (geometry is encoded, not generated) was attacked with the most aggressive 2026 generation claim — Fullwood–Vedral's assertion that the Lorentzian (1,3) signature is an OUTPUT of an information principle on qubits (arXiv:2604.07471) — and it collapsed because that signature is a fixed fact of the chosen complex carrier (the determinant on 2×2 complex-Hermitian matrices is unconditionally signature (1,3)), installed at the choice of Hilbert space, with the authors themselves conceding no emergence mechanism exists. Pillar 2 (no in-principle test) was attacked with GQuEST/geontropic interferometry as a candidate reverse-Weinberg–Witten observable, which collapsed on the authors' own admission that conformal-field-theory, dilaton, and hydrodynamic models — none emergent in the wager's sense — yield the identical signal, so it cannot separate emergent from fundamental geometry. Pillar 3 (terminal saturation) was attacked with a brand-new non-Hermitian/Krein cMERA de Sitter construction (arXiv:2606.17983) and DESI's April-2026 survey completion; both collapsed — the Krein construction pre-selects criticality as input (confirming, not breaking, the iter-12 closure) and DESI bears on the target, not the wager. The one thing the assault produced that the referee could not fully dismiss is philosopher Schneider's 2024 argument that global spacetime content is non-effective: it independently locates the screen's sole structural seam exactly where the wiki already places it, but it constructs no distinguishing observable and ends on an unresolved fork, so it corroborates the opening without breaching it. Net: carrier-convergence count stays FIVE, the principal hedge is unmoved, no new registry ID is consumed, and the verdict is confirmed for the 13th consecutive time — but this time earned by surviving the strongest falsification we could mount rather than by another passive lever-test. Terminal analytical saturation holds; only a new theorem (the carrier/reverse-Weinberg–Witten problem) or a new experiment can move it, and neither exists as of mid-2026.

## The one thing the assault could not fully dismiss (corroboration, not a breach)

NOT a break — but the single thing the assault produced that the referee could not fully dismiss is Schneider's global-content argument ("Spacetime Emergence: An (In)effective Story," Phil. of Physics 2024, arXiv:2410.00932), the strongest sub-result of track D2 prong 2. The referee explicitly graded it "the more serious attack" because it independently, from the philosophy-of-physics side and entirely outside this wiki's thread, LOCATES the screen's sole structural seam exactly where the wiki already places it: global / topological / superselection content that "acts like a superselection rule that must be independently specified" and "is not effective," directly targeting the verdict's own named un-proven premise (the net-sharing assumption that every UV/global/topological observable is a net invariant). What the referee CAN dismiss — and did — is the claim that this seam yields a witness: Schneider constructs no observable, develops no phenomenology, and ends on an explicit unresolved fork ("either... a topic of import within QG phenomenology... or else we ought to radically reorient"). So the crack is real as a CORROBORATION of where the only opening lives, not as a breach through it. Per the iter-9 identity (reverse-WW witness is strictly harder than the reverse-WW theorem, which equals the OPEN carrier problem via Weiner's algebraic Haag theorem), and per the binding anti-crank protocol (a plausibility hope is a failed break, not a falsification), this sharpens the verdict's self-described soft spot without moving it. Everything else the assault produced (Fullwood-Vedral signature-as-output; GQuEST/geontropic; non-Hermitian/Krein cMERA dS; DESI completion; the bicentralizer reformulation) the referees dismissed fully and on the merits.

## Verdict movement

HELD — 13th consecutive confirmation, but earned differently from all prior twelve. Iterations 1-13 each tested a lever and watched the verdict survive; iteration 14 inverted the burden and assaulted the standing verdict directly on all three pillars with the strongest 2024-2026 constructions, counterexamples, and arguments that could be web-verified, instructed NOT to hedge toward the verdict. All three pillars survived: Pillar 1 (encode-not-generate / carrier no-go) — FAILED-TO-FALSIFY, ATTACK-FAILS-VERDICT-HOLDS; Pillar 2 (no in-principle distinguishing test) — FAILED-TO-FALSIFY, ATTACK-FAILS-VERDICT-HOLDS; Pillar 3 (terminal saturation / not forecastable) — FAILED-TO-FALSIFY + NO-NEW-INPUT, ATTACK-FAILS-VERDICT-HOLDS. NO REGISTRY

MOVEMENT: carrier-convergence count stays FIVE; principal hedge HYP-CKV-VACUITY-R6 / OP-46 unmoved in grade (MEDIUM-HIGH→HIGH conditional) and condition (net-naturality theorem + located commutant-to-generated gap); HYP-ENCODING-SCREEN and the §7 no-test corollary inherit unchanged grade; no new numeric ID consumed. The qualitative gain is that the verdict is now stress-tested not just by "no lever moved it" but by "its strongest possible falsification was attempted and collapsed for stateable reasons" — which is a stronger epistemic posture than passive non-falsification. Terminal analytical saturation re-confirmed against the freshest (post-iter-13, mid-2026) literature.

CHAPTER 35

# The Carrier Problem, Posed: The Open Gate Distilled for External Solvers

> **What this is.** The single open analytical gate of the geometry-from-algebra program, distilled from fifteen adversarially-refereed iterations into its sharpest, most-attackable form — a self-contained statement an operator-algebraist, AQFT theorist, or noncommutative-geometer can pick up cold. At terminal analytical saturation, this is what reasoning can still usefully do: pose the problem so the external mathematics that *could* close it has the best possible target. Every reduction below is web-verified and epistemically tagged; the binding corrections from iterations 9–14 are folded in.

## The Net-Naturality Theorem (The Carrier Problem): A Self-Contained Open-Problem Statement for External Solvers

*Distilled from 15 adversarially-refereed iterations. This is the single most attackable form of the one open gate in the "geometry-from-algebra" program. An operator-algebraist / AQFT-theorist / noncommutative-geometer can pick this up cold.*

## 0. One-sentence statement

On a Haag-dual, split, Bisognano–Wichmann local net of von Neumann algebras with vacuum, is an `Aut(N,Ω)`-natural, modular-covariant, **localized**, indefinite `(n,n)` (Krein) Hermitian form **forced to be signature-inherited** from a pre-installed Euler element / causal index-set order (⇒ carrier no-go is unconditional; "Program A encodes but does not generate geometry" is a **theorem**), or can such a form exist that is **not** so forced (⇒ a *carrier*; geometry is **generated**; the verdict flips)?

## 1. Setup and definitions

- A **local net** `N = {M(O)}_{O∈I}` indexed by a partially ordered set `I` of spacetime regions, with a cyclic-separating vacuum `Ω`, isotony, locality (commuting algebras for causally disjoint regions), Haag duality, the split property, and the **Bisognano–Wichmann (BW)** property (the modular group `Δ_W^{it}` of a wedge acts as the boost). Each local algebra is the unique hyperfinite type III₁ factor.
- The **modular package** `m(N) = ({Δ_O^{it}}, {J_O}, {Δ_{O'|O}}, {P_{O',O}})`: modular flows, conjugations, relative modular operators, the modular-Berry/Connes connection over the net.
- **`Aut(N,Ω)`** is the group of vacuum-preserving net automorphisms: invertible maps `α` sending `M(O) → M(αO)` compatibly with isotony and the index-set order on `I`, with `α(Ω)=Ω`. By Tomita–Takesaki + BW it **contains the modular/boost flow** `{Δ_W^{it}}` and the geometric symmetries. "**`Aut(N,Ω)`-natural**" (used throughout, incl. §0 and the Conjecture) means: in the locally-covariant (Brunetti–Fredenhagen–Verch) idiom the carrier is a *natural transformation* of the carrier functor `Φ: Net → Krein` — i.e. for every `α ∈ Aut(N,Ω)` the form transforms equivariantly, `η_{αx} = α_*η_x` (equivalently `Ad U_α` commutes with the family `{η_x}`). Naturality under the modular subgroup is exactly the `[η, Δ_O^{it}]=0` condition below; naturality under the full `Aut(N,Ω)` is strictly stronger. `[INFERENCE — definitional, after iter-9 §1]`

- A **carrier** is a field of Hermitian forms `{η_x}` (equivalently a fundamental symmetry `J=J*`, `J²=1`, signature `(n,n)`) that is: **(modular-natural)** `[η, Δ_O^{it}]=0`; **(algebra-compatible)** `Ad J ∈ Aut(M)`; **(localized, "(loc)")** affiliated to `M(O) ⊊ M(W)` with vanishing cross-coupling between causally disjoint regions; **(natural)** equivariant under `Aut(N,Ω)`; and **(generative)** carrying NEW geometric content `Δ_geo ≠ 0` — i.e. *not* a function of `Δ` and *not* vacuum/symmetry-inherited.

## 2. The precise conjecture (the net-naturality theorem)

> **Conjecture (carrier no-go, net form).** Every `Aut(N,Ω)`-natural, modular-covariant carrier on a Haag-dual split BW net is positive-and-localized, signature-free, or indefinite-and-nonlocalized — **never both indefinite and localized**. Equivalently: any localized indefinite `(n,n)` form on such a net has its signature **inherited** from the net's causal index-set order `I` (the pre-installed Euler element), not generated by `(algebra, state)` data.

A proof makes "Program A encodes geometry, does not generate it" a theorem and the no-distinguishing-test corollary unconditional. A **counterexample** (a generative carrier) flips the verdict: geometry would be an *output* of algebra+state.

## 3. Established reductions and lemmas (the spine)

**Lemma N2 (the engine — `[INFERENCE, high; numerically demonstrated twice]`).** A localized indefinite natural form has a natural sign-locus `Λ ⊂ I` carried by every automorphism, hence by the modular flow. By the Borchers relation the modular flow acts as a free translation `p ↦ e^{-2πt} p`, which has **no fixed locus in the open index set**; the only flow-invariant sign-pattern is a global constant (definite). The half-line dilation's fixed points `p=0,∞` are the **wedge edge** — a geometric boundary outside the open `I`, i.e. smuggle-item `n₁`. Hence localized + natural + indefinite is impossible at finite locus. The only invariant indefinite object is `sgn(modular frequency)` — natural but **nonlocal** (Hilbert tail `~1/sep`).

**Lemma N1 (constructibility — `[downgraded to weakening, NOT a theorem]`).** Modular-naturality forces `η ∈ {Δ_0}'` (commutant), *not* into the abelian `W*({Δ_0})`; for type III₁ the modular spectrum is degenerate so these differ. **The iter-9 referee struck the claimed discharge to a theorem.** What survives is the **commutant-to-generated gap**: `η` lands in the commutant of the modular flow, and whether it is forced to be a *function of* `Δ` (geometry-void, $\eta_0$-class) or can be a genuinely new generated object is exactly the open residual.

**The exact residual (where the whole problem lives).** Two-fold and located: (i) the **localization-template hypothesis** $\mathbf{n_1}$ — "(loc)" is natural only relative to `I`'s causal order, and *that order is the geometry* (BGL: positivity ⟺ isotony; Neeb–Ólafsson: the order needs a pre-installed Euler element); removing it removes the meaning of "localized"; (ii) the **commutant-to-generated gap** — the surviving sliver of algebra-compatible carriers in `{Δ}' \ W*(Δ)` not provably reducible to a function of `Δ`. iter-10 confirmed the gap does **not** close: ergodicity of the BW vacuum empties only the *centralizer*, not the representation-space multiplicity gap, which retains the indefinite $\eta_0$`=sgn(ln Δ)`; the strongest candidate carrier clears the indefinite + algebra-compatible bars by an inner automorphism yet is **geometry-void**.

**The baseline obstruction (iter-7).** The canonical Krein fundamental symmetry $\eta_0$ `= sgn(ln Δ)` **does** exist, smuggle-free, from pure `(M,Ω)` data, is genuinely indefinite — but is **provably geometry-void**: unitarily universal across the entire Borchers/HSMI class (`Δ_geo = 0`, proven), generically algebra-incompatible, and non-localizable. Any carrier must be genuinely *not* of this class.

## 4. The FIVE converging routes (count is FIVE, never six)

All five are facets of ONE problem; new framings are re-expressions, not new routes:

1. stress-tensor-flux / Sorce–CKV readout;
2. modular-Berry / zero-mode readout;
3. Lorentzian-NCG / η-foliation-twist readout;
4. causal-fermion-system operator-spectral dichotomy (the `(≤n,≤n)` eigenvalue axiom IS the signature carrier);
5. the modular-sign / net-naturality route (this one), of which the following are **equivalent forms** (each a re-expression of the SAME route, not a sixth):
   - the **reverse-Weinberg–Witten theorem** (a class-separating observable realizable *only* without fundamental metric d.o.f.) — its *theorem* direction is co-extensive with the carrier no-go; a *witness* is strictly harder than a positive carrier solution. `[INFERENCE — the equivalence is this program's reduction; the reverse-WW witness itself is OPEN as of 2026-06, no construction exists]`;
   - the **completeness of (net + vacuum) as an invariant** = **Weiner's algebraic Haag theorem** (arXiv:1006.4726, M. Weiner, *An algebraic Haag's theorem*, Comm. Math. Phys. 2011) `[ESTABLISHED]`: under split + geometric modular action of the wedge algebras, the vacuum-sector net (double cones based on a fixed spacelike hyperplane) is a complete invariant — *no* assumption that the connecting unitary intertwines vacua or symmetry representations — but its hypotheses *are* the geometric inputs ($n_1$, the collar);
   - the **indefinite-metric / Krein-modular route** (iter-12, CLOSED-NEGATIVE): every Krein modular structure either installs `J` by hand (runs Krein→modular), is a gauge artifact modded out on passing to observables, or is $\eta_0$-equivalent / complex-pseudo-entropy.

## 5. Adjacent named open problems and the exact geometric input each framework smuggles

- **Connes' bicentralizer problem** (type III₁ factors; OPEN in general as of 2026, Ando–Goldbring arXiv:2605.12776). The located commutant-gap residual's *vocabulary* is RELATED TO (not logically reducible to — the iter-11 equivalence claim was major-corrected) centralizer/bicentralizer structure theory. A *trivial* (relative) bicentralizer is the carrier-**friendly** condition, so resolving it either way cannot by itself close the no-go (iter-12, non-load-bearing).

- **Factorization-algebra / functorial-QFT** (iter-15, E1, REPRODUCES-SCREEN). The Benini–Carmona–Grant-Stuart–Schenkel categorical **equivalence** `AQFT ≅ time-orderable prefactorization algebra` on globally hyperbolic Lorentzian manifolds (arXiv:2412.07318, Lett. Math. Phys. 2025; building on arXiv:1903.03396; the Lorentzian pAQFT factorization-algebra result is Gwilliam–Rejzner arXiv:2212.08175) is a NEW precise object but **not a handle**: both its sides are functors over the SAME fixed **orthogonality / causal-disjointness relation** `⊥` (= smuggle-item $n_1$), plus the Lorentzian time-orientation, and on the functorial side the **O(d)-structured bordism category carries the signature** (cobordism hypothesis installs geometric structure as a *source* parameter). [Published as Lett. Math. Phys. **116**, 13 (2026), DOI 10.1007/s11005-025-02035-7; the "2025" above is the acceptance-year DOI stamp, not the publication volume.] It transports the carrier question verbatim across the equivalence — a fourth independent witness to the same obstruction.
- **Connes–Atiyah–Singer index / K-theory / coarse geometry** (iter-15, E2, REPRODUCES-SCREEN). Every index pairing factors through the **Dirac operator** `D` (the metric datum; in spectral triples `D` *is* the Connes distance) or its bounded phase `F=sign(D)`. In the Lorentzian/indefinite case `D` is Krein-self-adjoint on a **pre-installed Krein space** whose indefinite inner product IS the `(n,n)` carrier `η` — *installed, not generated* (arXiv:1611.07062, "indefinite spectral triples over Krein spaces"). On a single local hyperfinite III₁ factor K-theory is structureless (no finite projections, no trace) — but this is only the single-algebra shadow; all carrier content survives in the **relations between algebras + the vacuum**. The coarse/Roe route smuggles the **metric coarse structure** and discards fine-scale localization. The spectral localizer (Loring–Schulz-

Baldes / Kaad arXiv:2508.08668) still requires a Dirac/first-order operator before yielding a numerical index.

- **Higher-categorical / homotopical / non-unitary tensor-categorical AQFT** (iter-15, E3, REPRODUCES-SCREEN). The orthogonality relation $\perp$ = causal disjointness is the **base of operadic AQFT** (Benini–Schenkel–Woike arXiv:1709.08657); the **Lorentzian bordism category** is the site (Bunk–MacManus–Schenkel arXiv:2308.01026); the free symmetric monoidal **bordism source** of the cobordism hypothesis is geometry-in-the-site. An indefinite $(n,n)$ signature can exist categorically only OUTSIDE the positivity-axiomatic unitary world, and there only as **installed reflection/dagger data** (Müller–Stehouwer arXiv:2301.06664) — the iter-12 Krein problem in categorical vocabulary. No natural functor out of the modular data supplies a localized indefinite carrier; the reformulation does **not** close the commutant-to-generated gap.

## 6. The exact common smuggle-item, stated once

Across Tomita–Takesaki, Krein/indefinite-metric, factorization-algebra, index/K-theory, and higher-categorical mathematics, the geometric input enters as **one of two interchangeable objects**: the **causal-disjointness / orthogonality / index-set order $n_1$** (the localization template), OR a **metric/Dirac/Krein datum $\eta$** (the signature carrier installed on a pre-given space). The open problem is precisely: **exhibit, or rule out, a structure in which $n_1$ (the causal order) and the signature $\eta$ are OUTPUTS of the algebra+state rather than indexing data of the source category / installed data on the carrier space.**

## 7. What a proof either way would imply for a universal theory of physics

- **No-go proven** (the honest expected outcome, now robust across all modern frameworks): "causal/algebraic/entanglement structure precedes metric geometry" is, for our $\Lambda>0$ universe, an **encoding strategy, not a generative mechanism** — a definitive theorem. By the HYP-ENCODING-SCREEN corollary it follows *unconditionally* that the central wager has **no in-principle distinguishing experimental test** (an encode-only program's observational content equals that of the EFT it encodes; the reverse-WW theorem holds). The program is permanently "not yet physics" absent a different idea; a unified GR+QFT framework remains POSSIBLE (AdS/CFT existence proof) but geometry-from-algebra is *not* the road to deriving it.

- **Carrier exhibited** (a generative counterexample): geometry — causal order and Lorentzian signature — is **generated** from `(algebra, state)` data. This would be the first construction outputting a Lorentzian causal/metric structure installing no fundamental symmetry, η-form, foliation, twist, `(n,n)`-flag axiom, or signature-valued template by hand. It would make the reverse-Weinberg–Witten witness *possible in principle* (a necessary, not sufficient, prerequisite for a distinguishing test), reopen the entire verdict, and convert the central wager from a research strategy into a candidate *result* — the first step toward a derived (not merely consistent) universal theory.

## 8. Status

Standing verdict unchanged 14 consecutive iterations (terminal analytical saturation, grade-derivative zero since iter-9, condition-derivative zero since iter-10). Every verdict-bearing residual is now decidable only by **external input**: a resolution of the net-naturality theorem (equivalently the carrier no-go / commutant-gap closure / a reverse-WW witness), an external resolution of Connes' bicentralizer problem, or a genuinely new mathematical readout not yet tried. The problem is **posed for external resolution.**

## Status append (2026-07-03, iteration 16 — the residual restructured; the posed problem above stands verbatim)

The §3 "exact residual" is superseded in its second component by the iteration-16 multi-wedge closure (LEM-NET-NATURALITY-G, referee-verified; Working note: *Track G1, iteration 16: direct assault on the net-naturality theorem — Lemma G formalized; the multi-wedge lever closes the non-localized sliver; the localized horn is a* *three-line theorem; the residual shrinks to double-cone vacuum ergodicity (E_O)* (repo: `notes/2026-07-03-iter16-G1-multiwedge-naturality.md`)): **(G-loc)** every localized natural carrier is ±1 (proof-grade); **(G-glob)** every fully-natural non-localized algebra-compatible carrier is a gauge-symmetry implementer with Poincaré-invariant eigenprojections, Δ_geo = 0 (proof-grade) — the commutant-to-generated gap is **non-empty ((−1)^N, η0) and classified, NOT closed**. The residual is now: (i) the localization-template hypothesis **n1** (unchanged, irreducible); (ii) hypothesis **(E_O)** — vacuum ergodicity on massive double-cone algebras, $M(O)_\omega = \mathbb{C}1$, [OPEN]. Direction: (E_O) ⟹ the conjecture in §2 is a theorem on the stated hypotheses (+ weak additivity); ¬(E_O) is necessary but NOT sufficient for a counterexample fiber. §6's two-item interchangeability now has a published theorem-level statement, BW-conditionally: Morinelli–Neeb, Adv. Math. **458** (2024) 109960 (arXiv:2312.12182). For external solvers the sharpest open questions are now **(E_O)** and the **DCNG** definability lever — see *Chapter 36 (p. 684)* §6. `[INFERENCE, high — referee-verified package]`

## Status append (2026-07-07, iteration 17 — (E_O) attacked directly; the posed problem stands verbatim)

The §3 residual (E_O) received a dedicated five-lens assault (three binding referees; *Chapter 37 (p. 693)*). It stands [OPEN] — neither proved nor disproved; **HYP-CKV-VACUITY stays R7.** For external solvers the sharpest decidable sub-target is now concrete: **for the free massive scalar, (E_O) reduces to purely continuous spectrum (no eigenvectors) of the double-cone one-particle modular Hamiltonian** $\ln \delta_O$ — via second quantization, continuous spectrum ⟹ weak mixing ⟹ $M(O)_\omega = \mathbb{C}1$. Correction folded in: the naive "no eigenvalue 0 ⟹ ergodic" reading is INSUFFICIENT (bosonic $d\Gamma(A)$ are modular-fixed regardless of point spectrum); the correct sufficient condition is continuous spectrum. The spectral fact is numerical-only (Bostelmann–Cadamuro–Minz arXiv:2209.04681; Cadamuro arXiv:2312.08525; no closed form — Longo–Morsella arXiv:2012.00565, massive-ball result "contained a gap, and have been removed"), leaning-true, unproven. For general class (H), all three structure-theory routes (weak-mixing-from-clustering; wedge→double-cone HSMI; Marrakchi–Vaes genericity) provably fail, and the adversarial disproof fails on eight candidate mechanisms — so the posed problem stands verbatim, sharpened not solved. `[INFERENCE, high — referee-verified package]`

CHAPTER 36

# Iteration 16: The Multi-Wedge Closure — the Gap Classified and the Hedge Advanced to R7

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**Outcome: Verdict UNCHANGED — 15th consecutive confirmation** (PARTIAL coherence / Program A encodes-not-generates / not-yet-physics / not forecastable). **But this is the first iteration since iteration 9 in which the principal hedge MOVED:** HYP-CKV-VACUITY advances **R6** → **R7** on a referee-verified proof package. The diffuse "commutant-to-generated gap" condition is replaced by a single named open hypothesis, **(E_O)**. **Iteration:** 16

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## 1. Why this iteration was legitimate (the saturation rule)

The ROADMAP terminal-saturation rule forbids new analytical iterations "unless one of those clocks delivers a genuinely new input." Clock (1) — the carrier / net-naturality / bicentralizer / GMA mathematics — delivered on 2026-06-22: Marrakchi's unconditional ergodicity theorem for the relative bicentralizer flow (arXiv:2606.23636), flagged by watch-sweep #1 as a near-miss with a sharpened watch target. Iteration 16 executed that target (G3), and used the occasion to run the two never-tried levers the new input suggested (G1's multi-wedge intersection; G2's state-family naturality) plus a broad untried-readout sweep (G4).

## 2. Per-track verdicts (all refereed; corrections binding)

| Track | Question | Outcome | Referee |
|---|---|---|---|
| **G1** — multi-wedge naturality assault (*note* (Working note: *Track G1, iteration 16: direct assault on the net-naturality theorem — Lemma G formalized; the multi-wedge lever closes the non-localized sliver; the localized horn is a three-line theorem; the residual shrinks to double-cone vacuum ergodicity (E_O)* (repo: `notes/2026-07-03-iter16-G1-multiwedge-naturality.md` ))) | Does intersecting the naturality constraints of ALL wedge modular flows (genuinely unused in iters 9–10, which were single-Δ) close the commutant-to-generated gap? | **PARTIAL — major negative-side hardening, proof-grade.** (G-loc): every *localized* natural carrier is ±1 — a written proof (Borchers scaling → Fix(Δ_W^{it}) = ℂΩ → vacuum pinning → Reeh–Schlieder kill), independently re-derived by the referee at every step. (G-glob): the wedge boosts of all translated wedges generate $\mathcal{P}_+$^↑ (group identity verified by direct computation), so every fully-natural non-localized algebra-compatible carrier is a **gauge-symmetry implementer** with Poincaré-invariant eigenprojections — Δ_geo = 0, proof-grade. The gap is **non-empty ((−1)^N certifies it) and classified, NOT closed**. Residual = exactly hypothesis **(E_O)**: vacuum ergodicity on massive double-cone algebras, M(O)_ω = ℂ1, [OPEN]. | **R1: UPHELD-WITH-CORRECTIONS.** R7 minted (§4 below). Bound down: DHR n1-inheritance = framing only; conformal-net closure = transport sketch, not established; (E_O) direction is (E_O) ⟹ closure / ¬(E_O) necessary-not-sufficient. |

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| **G2** — constructive carrier attempt (*note* (Working note: *Iteration 16 (track G2): the constructive direction — three untried carrier constructions (state-family naturality, split-inclusion canonical data, core-descent), executed and graded* (repo: `notes/2026-07-03-iter16-G2-constructive-carrier.md`))) | Do three untried constructions (state-family naturality over the Marrakchi–Vaes dense G_δ; Doplicher–Longo split-inclusion canonical data; continuous-core descent) yield a generative carrier? | **All three BLOCKED.** (A) The co-moving evasion of Lemma N2 is mathematically real off-vacuum, but the proposed section-rigidity trichotomy (Lemma S2, sketch-grade) closes the family class; the physical vacuum-wedge fiber is stabilizer-fixed and dies by Lemma N2 directly. (B) The pulled-back DL flip Θ fails algebra-compatibility unless a non-canonical anti-isomorphism (an installed η) identifies the tensor legs; where canonical it degenerates to PCT. (C) LEM-CORE-DESCENT (proof-grade transport corollary): core-built carriers either descend to the closed centralizer class or carry only trace data the dual flow scales away (θ-invariant values {0,∞}). | **R2: UPHELD-WITH-CORRECTIONS.** Scope fix: state-family naturality is an **extension** of the posed conjecture, not inside it — Lemma N2 and the dossier are not edited; S2 recorded as the extension's engine, completeness conditional on G1's (G-glob) class. MV genericity = corroboration, not the kill. |

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| **G3** — Marrakchi transfer audit (*note* (Working note: *Track G3 (iteration 16): full technical audit of Marrakchi arXiv:2606.23636 — the expectation hypothesis is pervasive-and-definitional, the physics-net transfer fails at the level of the object, and the AHHM hyperfinite-subfactor output is VACUOUS for local inclusions* (repo: `notes/2026-07-03-iter16-G3-marrakchi-transfer.md`))) | Does arXiv:2606.23636 (or its method) transfer to the expectation-free physics-net inclusions M(O) ⊂ M(W)? | **TRANSFER RESOLVED-NEGATIVE; watch target re-sharpened.** From the full TeX source: the "with expectation" hypothesis is pervasive AND definitional (the flow β^φ is built against φ = φ∘E_N via Takesaki's theorem, before any proof step; the author flags it "essential"). The transfer is impossible **as posed**: no state on M(W) has a modular flow globally preserving M(O) (else Takesaki gives the excluded expectation), so the sweep-#1 target "ergodicity for expectation-free inclusions" was **ill-posed** and is retired/re-sharpened (§6). The AHHM hyperfinite-subfactor re-check: no interaction with the carrier problem (the iter-12 non-load-bearing grade is strengthened; see |

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| §5 scope note). Sole transferable residue: the expectation-free relative T-invariant → adjacent non-load-bearing conjecture $T(M(O) \subset M(W)) = \{0\}$. | **R3: SUSTAINED** (all eight expectation-usage census points verbatim-verified against an independently re-downloaded source; a pipeline PDF-summarization fabrication was caught by the finder and corroborated by the referee — process record kept). |

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| **G4** — untried-readout sweep (*note* (Working note: *Iter-16 Track G4: the untried-readouts sweep — eight candidate frameworks graded against the carrier screen, and a definable-carrier no-go as the surviving lever* (repo: `notes/2026-07-03-iter16-G4-untried-readouts.md`))) | Does any framework not tried in 15 iterations give a decidable handle? | **7 of 8 REPRODUCE-SCREEN** (NC Dirichlet forms — constitutive positivity g6; free Araki–Woods — input orthogonal representation IS the boost; infinite-index subfactor invariants — borderline irrelevant; entropic order parameters — region-indexed detectors; QRF/crossed products — installed flow/observer; net cohomology — poset as axiom zero; 2024–26 web sweep — nothing passes). **1 conditional survivor:** continuous model theory of W*-algebras — the gap becomes a *definability* question, yielding the named target **DCNG** (definable-carrier no-go), a watch-item/iter-17 lever gated on an η0-definability lemma, no-go-direction-only. Bonus: **Morinelli–Neeb** (Adv. Math. 458 (2024) 109960, arXiv:2312.12182) proves a two-directional, **BW-conditional** bridge between the two smuggle items — theorem-level confirmation that n1 and η are interchangeable *relative to an installed BW/wedge structure*, deriving neither from (algebra, state). | **R3: SUSTAINED-WITH-CORRECTIONS.** All three high-risk screen-grades verbatim-verified; DCNG's honest grade = watch item, not result; the Morinelli–Neeb sentence bound to the BW-conditional wording above. |

## 3. Why the verdict stands a 15th time

No track produced a carrier, a reverse-Weinberg–Witten witness, a generative construction, or a distinguishing test. Every positive object exhibited this iteration ($(-1)^N$, the $\eta_0$-section, the DL flip Θ) was *found and classified as geometry-void or installed*, not offered as a carrier. The iteration's entire yield is negative-side: the no-go is now **proof-grade on two of its three branches** (localized; fully-natural non-localized) with the third branch reduced to one named open hypothesis. The encoding screen survives its sixth vocabulary (definability logic) and gains a published theorem-level statement of its central smuggle-item interchangeability (Morinelli–Neeb, BW-conditional). NOT-YET-PHYSICS is unchanged; no experimental channel emerged; the 2027 DESI w(z) verdict remains the nearest dated external event.

## 4. The hedge ledger — the first movement since iteration 9

**HYP-CKV-VACUITY: R6 → R7 (minted by referee R1, the first hedge-moving submission in the carrier track to survive full adversarial re-derivation).** Grade unchanged (conditional HIGH). Condition string (R1-verbatim, binding):

> Conditional HIGH on: (i) the localization-template hypothesis **n1** (irreducible, per iter-9, unchanged); AND (ii) hypothesis **(E_O)** — vacuum ergodicity on massive double-cone algebras, $M(O)_\omega = M(O) \cap \{\Delta_O^{it}\}' = \mathbb{C}1$ for every double cone O. Direction: given the Lemma-G hypotheses (H, + weak additivity) and n1, (E_O) makes the net-naturality no-go a theorem (Lemma G, referee-verified); ¬(E_O) is necessary for any counterexample fiber but NOT sufficient (rotation equivariance, the Lemma-N2 cross-fiber web, and the inherited centralizer-sign vacuity remain as generativity walls). The commutant-to-generated gap is NOT closed: it is non-empty ($(-1)^N$, $\eta_0$) and classified — localized branch dead (G-loc),

fully-natural non-localized branch = gauge gradings with Poincaré-invariant eigenprojections, Δ_geo = 0 (G-glob).

House history honored: two prior R7 mints (iteration 10) were struck — for a wrong-object error and an overstated discharge respectively. This mint commits neither (correct object {Δ_W^{it}}′ ⊂ B(H); gap explicitly non-empty; written proofs, independently re-derived). HYP-ENCODING-SCREEN inherits the R7 condition. Carrier-convergence count stays **FIVE**; no numeric ID consumed; LEM-NET-NATURALITY-G, (E_O), and DCNG enter as named items per the AGENTS.md §3 convention.

# 5. Standing-record repairs found this iteration (binding, propagated)

1. **Doplicher–Longo volume:** *Standard and split inclusions of von Neumann algebras* is Invent. Math. **75** (1984) 493–536, not "73" — web-verified (Springer DOI 10.1007/BF01388641). Fixed in the iter-12 note (4 occurrences); a correct BIBLIOGRAPHY entry added.
2. **"Irreducible" for M(O) ⊂ M(W) is FALSE** (R2, sustained): by Haag duality the relative commutant M(O)′ ∩ M(W) ⊇ M(O′ ∩ W) — the collar algebra — is enormous. The iter-12 note and watch-sweep #1 carried "irreducible"; the *conclusions* survive unchanged (the no-normal-expectation datum follows independently via Takesaki's criterion — the BW-geometric ambient flows move O), but the wording is repaired by dated addenda. Consequently the G3 AHHM-vacuity argument is scoped to what survives the repair: the AHHM relation's output remains without carrier interaction on iter-12's independent grounds.
3. **Houdayer–Marrakchi arXiv:2511.11409 states an implication, not an "equivalence"** (watch-sweep #1 wording repaired by addendum).
4. **The watch-sweep #1 sharpened watch target was ill-posed** ("ergodicity/rigidity for expectation-free irreducible inclusions" — the flow is undefined without the expectation, and

the inclusions are not irreducible). Replaced by the three-prong target in §6.

## 6. The re-sharpened watch state (supersedes the sweep-#1 target)

Clock (1) watch items after iteration 16:

- **(α)** any new *definition* of a relative-bicentralizer-type invariant for expectation-free inclusions, plus any theorem about it;
- **(β)** movement on bicentralizer *triviality* itself (unchanged from sweep #1);
- **(γ)** any computation of the expectation-free relative T-invariant for QFT-type inclusions, including the adjacent conjecture $T(M(O) \subset M(W)) = \{0\}$;
- **(δ) — NEW, the principal item:** any resolution of **(E_O)** (vacuum ergodicity on massive double-cone algebras) — the single hypothesis to which the carrier no-go's last open branch is now pinned. Adjacent technology: Marrakchi's Theorem-E-type averaging; Houdayer–Marrakchi selflessness (implication direction only).
- **(ε) — NEW:** the **DCNG** lever (definable-carrier no-go in continuous model theory), gated on the $\eta_0$-definability lemma; suitable for an operator-algebra/continuous-logic collaborator; no-go direction only.

## 7. What could move the verdict from here

Unchanged in kind, sharpened in address: (i) a proof of (E_O) → the net-naturality no-go becomes a **theorem** on the stated hypotheses (the honest expected outcome; makes the no-test corollary unconditional); (ii) a massive double-cone counterexample fiber surviving ALL walls (¬(E_O) is only the entry ticket) → the first generative carrier and a verdict flip; (iii) the external clocks (2027 DESI w(z) — target, not wager; Euclid mid-2027). The dossier stands as posed, with its §3 residual now carrying the iteration-16 status append.

CHAPTER 37

# Iteration 17: The (E_O) Assault — the Free-Field Reduction and Why the Hedge Holds at R7

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**Outcome: Verdict UNCHANGED — 16th consecutive confirmation** (PARTIAL coherence / Program A encodes-not-generates / not-yet-physics / not forecastable). **The principal hedge STAYS at R7 — (E_O) was neither proved nor disproved.** Unlike iteration 16 (which advanced R6 → R7), no hedge movement: no R-advance, no R-retreat, no numeric ID consumed, carrier-convergence count stays **FIVE**. What *is* genuinely new and refereed: (1) the free-field case of (E_O) is sharply **reduced** to a concrete one-particle spectral question, and the naive "no-eigenvalue-0 ⇒ ergodic" reading is **refuted**; (2) all three structure-theory routes to (E_O) for general nets provably **fail**; (3) the adversarial disproof **fails** on eight candidate mechanisms — no carrier, the 16-iteration streak is intact. **Iteration:** 17

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## 1. Why this iteration was legitimate (the clock rule)

The ROADMAP terminal-saturation rule forbids new analytical iterations "unless one of those clocks delivers a genuinely new input." Iteration 16 pinned the entire remaining no-go to a single named open hypothesis and designated it the **principal watch**

**item (δ)**: **(E_O)** — vacuum ergodicity on massive double-cone algebras. It never attacked (E_O) as a mathematical object; the iter-16 synthesis §7 named "a proof of (E_O)" as "the honest expected outcome" of the next legitimate step. Iteration 17 *is* that step: a dedicated, five-lens, direct assault on (E_O). This is new work, not a re-run of a closed lever. (Watch-sweep #2, run in the same pass, also re-surfaced the (E_O)-adjacent Marrakchi arXiv:2606.23636 — see *watch-sweep #2* (Working note: *Watch-mode sweep #2 — external-input channels, window 2026-07-02 → 2026-07-07* (repo: `notes/2026-07-07-watch-sweep-02.md`)).)

## 2. The exact object

> **Hypothesis (E_O)** [OPEN]**.** For the vacuum ω on the double-cone algebra M(O) of a massive theory in the standing class (H) (isotony, locality, Haag duality, split, Poincaré covariance with **unique** vacuum, Reeh–Schlieder, Bisognano–Wichmann for wedges, local algebras hyperfinite type $III_1$), the modular flow σ^ω_t = Ad Δ_O^{it} is **ergodic**: M(O)_ω := M(O) ∩ {Δ_O^{it}}′ = ℂ1 for every double cone O.

The wedge analogue M(W)_ω = ℂ1 is a theorem (Borchers scaling: the boost is the modular flow, the two lightlike translation semigroups are scaled to infinity, a boost-fixed vector is translation-fixed by a Cauchy–Schwarz equality hence = cΩ, and Reeh–Schlieder promotes this to the algebra). **That engine has no analogue for a bounded double cone** — no translation semigroup maps O into itself — which is exactly why (E_O) is the residual. The massless/conformal double cone *is* geometric (Hislop–Longo: Δ_O is a dilation), so (E_O) is expected to close there; the open case is specifically **massive** double cones, where Δ_O is non-geometric.

# 3. Per-lens verdicts (all refereed; corrections binding)

| LENS | TECHNOLOGY | THESIS | REFEREE |
|---|---|---|---|
| **E1** — free-field modular spectrum (*reduction* (Chapter 37 (p. 693))) | Second-quantization functor: $\Delta\_O = \Gamma(\delta\_O)$, $\sigma\_t = \mathrm{Ad}\,\Gamma(\delta\_O^{it})$; reduce (E_O) to a one-particle question | **E_O_OPEN — reduced.** Correctly identifies that $\sigma\_t$ is a *gauge-invariant, number-preserving* Bogoliubov flow, so $M(O)\_\omega$ is **not** controlled by fixed one-particle *vectors* alone: $\Gamma(u)d\Gamma(A)\Gamma(u)^* = d\Gamma(uAu^*)$ makes $N = d\Gamma(1)$ and every $d\Gamma(E)$ modular-fixed **regardless of point spectrum**. Hence "no eigenvalue 0 of $\ln \delta\_O \Rightarrow$ ergodic" is NECESSARY but PROVABLY NOT SUFFICIENT — the bosonic number-type operators survive. The residual is an *affiliation* clause: which functions of $\ln \delta\_O$ are affiliated to $M(O)$. | **Target A: PARTIAL.** Reduction confirmed; the naive slogan refuted; the correct sufficient condition is E2's (below). |

| | | | |
|---|---|---|---|
| **E2** — III₁ centralizer / ergodicity structure theory | Connes spectrum Γ(σ)=ℝ; Marrakchi–Vaes genericity; weak-mixing ⇔ trivial centralizer | **E_O_OPEN.** The clean sufficient condition: **weak mixing of σ^ω on M(O) ⟹ M(O)_ω = ℂ1** (elementary Tomita–Takesaki), and for the quasi-free vacuum this reduces to **purely continuous one-particle spectrum of ln δ_O** (no eigenvectors). But Γ(σ)=ℝ and III₁-ness are outer-structure invariants strictly weaker than centralizer-triviality; Marrakchi–Vaes genericity (dense G_δ) cannot pin the distinguished symmetric vacuum; and modular mixing on a *bounded* region is not a corollary of spacelike cluster decomposition (the modular orbit carries nothing to spacelike infinity). | **Target B: CONFIRMED** (route does not settle it; fatal gap identified). |

| | | | |
|---|---|---|---|
| **E3** — wedge→double-cone reduction | Half-sided modular inclusion (Wiesbrock/Borchers), modular intersection | **E_O_OPEN — clean impossibility for this route.** O = $W_1 \cap W_2$ and $M(O) \subset M(W_1)$, but Δ_O-fixity does NOT imply Δ_W-fixity (different algebras, same Ω, non-commuting modular operators); HSMI reconstructs the *ambient* boost/translations (scaled by Δ_W, sliding O out of itself), never the non-geometric Δ_O. The route closes (E_O) **iff Δ_O is geometric = the conformal case already conceded**. | **Target B: CONFIRMED.** Two demerits (below), conclusion survives. |

| | | | |
|---|---|---|---|
| **E4** — adversarial disproof | Construct a non-scalar element of M(O)_ω | **E_O_OPEN — disproof FAILS.** Four candidate mechanisms (localized conserved flux; SO(d−1) rotation invariants; hidden point spectrum of ln Δ_O; integrable/almost-periodic modular data) plus four the referee added (split-interpolating type-I factors; central sequences; $\eta_0$ = sgn(ln Δ_O); (−1)^N parity) **all die on the triple wall {bounded ∧ in M(O) ∧ exactly Δ_O-fixed}**. The SO(d−1) stabilizer supplies only unbounded, global, non-local generators; the only bounded modular-fixed vector is Ω (⟹ scalars via Reeh–Schlieder). No entry-ticket carrier. | **Target C: CONFIRMED**, with a binding calibration (below). |
| **E5** — is it already known? | Exhaustive AQFT / operator-algebra literature sweep | **E_O_OPEN in the literature.** No paper states, proves, or disproves trivial vacuum centralizer on a bounded double cone (massive OR even massless). The massive double-cone modular operator is *explicitly characterized* (numerically) yet the ergodicity question is **never posed**. (E_O) is a genuine open problem, not folklore — and, because the modular operator is in hand, an unusually **tractable** one for the free field. | Sustained (cross-checked against E1/E2). |

# 4. The one thing this iteration actually establishes (the free-field reduction)

The deliverable is a referee-verified reduction of the free-field case to a clean, decidable spectral statement:

> **Free-field reduction** [INFERENCE, high]. For the free massive scalar, M(O) = R(K_O), Ω the Fock vacuum, Δ_O = Γ(δ_O), σ_t = Ad Γ(δ_O^{it}) a gauge-invariant number-preserving flow. If the one-particle modular Hamiltonian **ln δ_O has purely continuous spectrum on the one-particle space** (equivalently: the one-particle modular flow is weakly mixing / has no eigenvectors), **then σ^ω is weakly mixing on M(O), hence M(O)_ω = ℂ1** — (E_O) holds for the free field, and the net-naturality no-go becomes a theorem on the free-massive-scalar subclass (modulo n1).

Two binding corrections make this the *right* statement:

1. **The naive reading is refuted.** "No eigenvalue 0 of ln δ_O ⟹ ergodic" is only necessary — it kills fixed *Weyl* operators W(f) but not the number-type dΓ(A), A ∈ {δ_O^{it}}′, which are modular-fixed for any point spectrum. Absolute continuity is not enough by fixed-vector counting; one needs the full weak-mixing / purely-continuous-spectrum condition, which *does* discharge the dΓ(A) obstruction (weak mixing forces the fixed-point *algebra* to ℂ1, bypassing any need to characterize affiliation directly). This corrects the too-quick "trivial kernel ⟹ (E_O)" intuition (bosonic Fock always carries the σ-fixed number operators).
2. **The spectral premise is unproven** [OPEN]. The massive double-cone one-particle modular Hamiltonian is known only **numerically** (Bostelmann–Cadamuro–Minz, *On the mass dependence of the modular operator for a double cone*, arXiv:2209.04681, Ann. Henri Poincaré 2023 — "close to a multiplication operator, but ... dependent on mass and angular momentum"; Cadamuro, arXiv:2312.08525) — a near-multiplication diagonal part plus an unresolved non-diagonal kernel "orders of magnitude smaller ... cannot confirm whether zero." **No closed-form massive double-cone modular Hamiltonian exists**: Longo–Morsella, *The massless modular Hamiltonian* (arXiv:2012.00565), v3 comment verbatim — "the results on the massive modular hamiltonian of a ball **contained a gap, and have been removed**." The numerics *lean* toward

continuous spectrum (a multiplication operator has purely continuous spectrum) — hence toward (E_O) holding — but this is not a proof.

**Net:** the reduction is proof-grade; the spectral fact it needs is `[OPEN]`, numerical-only, leaning-true. So even the decidable base case is genuinely open — but now pinned to a single concrete spectral computation rather than an abstract centralizer question.

## 5. Why the verdict stands a 16th time, and why the hedge does NOT move

No lens produced a carrier, a reverse-Weinberg–Witten witness, a generative construction, or a distinguishing test. (E_O) was neither proved (no spectral input; no criterion shown to hold for the vacuum on a bounded region) nor disproved (no non-scalar affiliated modular-fixed element exhibited). Per the house rule, **a hedge advances only when a condition is discharged or added** — neither happened, so **HYP-CKV-VACUITY stays at R7**, grade (conditional HIGH) and condition string {n1 AND (E_O)} both unchanged. This is the disciplined outcome: the iteration produced real refereed mathematics, but its *verdict-level* result is confirmation, not movement.

The verdict is nonetheless **modestly hardened at framing level** (not proof level): (i) the disproof was attempted head-on and failed cleanly, which strengthens — without proving — the case that (E_O) holds; (ii) the no-go is now shown to be **un-closable by every reasoning-decidable structure-theory route** (weak-mixing-from-clustering, wedge-to-double-cone HSMI, and genericity-upgrade all carry identified fatal gaps), which sharpens "only external input can move it"; (iii) the literature sweep confirms (E_O) has never even been *posed* for the bounded massive case, so the wiki's isolation of it is a genuine contribution.

## 6. Standing-record notes (binding; do NOT propagate the errors)

1. **Citation attribution:** arXiv:2209.04681 is **Bostelmann–Cadamuro–Minz**, NOT "Figliolini–Guido" (one finder mislabeled it; the referee caught it). Figliolini–Guido is the separate 1989 Ann. IHP paper *The Tomita operator for the free scalar field* and the 1994 *On the type of second quantization factors*. The mislabel is **not** written into any registry page.
2. **Overstated sub-claim, not propagated:** E3's "for massive nets the wedge pair is not a Wiesbrock modular intersection" is overstated and likely wrong — massive wedge modular flows *are* geometric (Bisognano–Wichmann; Mund's massive BW theorem), and modular intersections of wedges *do* reconstruct the Poincaré group for massive nets. E3's *conclusion* survives on the correct reasoning (the reconstructed flow is the ambient boost/translation, not Δ_O), so only the sub-claim is dropped.
3. **Calibration (binding):** E4's "net evidence FOR (E_O) / leaning true" is down-graded by the referee to **bare OPEN** — an adversary's failure to *construct* a witness from guaranteed symmetries is absence-of-witness, not absence-of-existence, since the massive double-cone modular structure is spectrally uncharted with no closed form.
4. **Unverified sub-claim:** the Figliolini–Guido 1989 "1 is not an eigenvalue of $(\Delta\_B+1)(\Delta\_B-1)^{-1}$" point could not be verbatim-confirmed (numdam metadata only) and stays `[unverified]`; it is not load-bearing.

## 7. The re-sharpened watch state (clock 1)

Unchanged in kind from iteration 16; sharpened at (δ):

- **(α)** new definitions/theorems for expectation-free relative-bicentralizer-type invariants — STATIC;
- **(β)** bicentralizer triviality — STATIC (Marrakchi arXiv:2606.23636 is *with expectation*; the double-cone gate is

expectation-free, so it does not transfer);

- **(γ)** expectation-free relative T-invariant computations (adjacent conjecture T(M(O) ⊂ M(W)) = {0}) — STATIC;
- **(δ) — PRINCIPAL, now sharpened:** any resolution of (E_O). The sharpest decidable sub-target is now **"prove purely continuous spectrum (no eigenvectors) of the massive double-cone one-particle modular Hamiltonian ln δ_O for the free scalar"** — if proved, (E_O) holds on the free subclass (via weak mixing) and **R7 → R8 would fire there** (the no-go a theorem on the free-massive-scalar subclass, modulo n1); general (H) stays conditional. Adjacent technology: Bostelmann–Cadamuro–Minz / Cadamuro numerics (arXiv:2209.04681, 2312.08525); the second-quantization functor (Figliolini–Guido; Conti–Morsella arXiv:2305.07606).
- **(ε)** the DCNG definability lever — STATIC (needs an operator-algebra / continuous-logic collaborator).

Clock (2) unchanged: 2027 DESI five-year w(z) (Euclid cosmology co-timed mid-2027) is the nearest dated external event.

## 8. What could move the verdict from here

Unchanged in kind, sharper in address: (i) a proof of purely-continuous massive double-cone one-particle modular spectrum → (E_O) on the free subclass → R7 → R8 there → the no-test corollary unconditional on that subclass; (ii) a proof (or general-net analogue) of (E_O) → the net-naturality no-go a theorem on class (H); (iii) a massive double-cone counterexample fiber surviving all walls (¬(E_O) is only the entry ticket) → the first generative carrier and a verdict flip; (iv) the external clocks (2027 DESI w(z) — target, not wager). The dossier stands as posed, its §3 residual now carrying the iteration-17 free-field reduction.

PART VI

# THE VERDICT

Six iterations of mapping, attacking, and refereeing converge here, on a conclusion that is neither triumphant nor defeatist — and is, for that reason, worth stating carefully.

Chapter 27 is the capstone. The verdict, in brief: a universal theory of physics is not forbidden and not delivered. A consistent unified framework is very plausibly possible — none of the seventeen catalogued clashes is a genuine contradiction, and an existence proof that *some* consistent quantum gravity exists is in hand. But the most developed candidate direction encodes geometry without generating it, and has no distinguishing experimental test even in principle; by this project's own standard, it is a sharp research strategy, not yet physics. The chapter states the verdict's three explicit hedges, explains why "saturation" is a claim about the marginal yield of further reasoning rather than a proof, and — most importantly — specifies the exact analytical construction that would reopen the analysis.

Chapter 28 hands the question to experiment: the watchlist of empirical channels — gravitationally induced entanglement, objective-collapse bounds, dynamical dark energy, and the rest —

with what each can and, just as carefully, cannot decide. Chapter 29 closes with the roadmap for whoever continues the work.

What this part establishes is the book's deliverable: not a theory, but a precise, adversarially tested account of why the question is open and what would close it.

CHAPTER 38

# Conclusion: The Converged Verdict

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**THE ONE-PARAGRAPH VERDICT.** A single universal theory of physics is **not forbidden and not delivered.** The two pillars — quantum field theory and general relativity — are mutually consistent everywhere we can test, and of the 17 catalogued inter-framework clashes **none is a proven logical inconsistency**; they are domain-mismatches and unsolved-but-consistent problems concentrated where both theories are extrapolated past their tested regimes. A *consistent unified framework* is therefore very plausibly **possible** (AdS/CFT is an existence proof that *some* consistent quantum gravity exists). But the most developed candidate direction — that **causal/algebraic/entanglement structure precedes metric geometry** — *encodes* geometry without yet *generating* it, and, decisively, **has no distinguishing experimental test even in principle.** By this wiki's own anti-crank standard it is therefore **not yet physics**: a sharp, coherent research *strategy*, not a result. The object-level question — *is there a final theory, and is it this one?* — is **not forecastable**, gated less by the ~15-order-of-magnitude energy desert than by the absence of any experiment that bears on the central wager. This verdict has been stable across iterations and has now survived, in sequence, a direct attempt to overturn it, a dedicated new-input iteration (6 research + 4 attack tracks), an iteration that executed its own four named decidable levers (iteration 7; all adversarially refereed), and a targeted hunt of the 2024–2026 experimental/phenomenology literature for an in-principle distinguishing test (2026-06-10; 3 refereed tracks; none found,

> the screen reinforced with positive mechanism-blindness evidence) — whose net effect was again to *tighten* it: the designated verdict-flipper was achieved *literally* and proved geometry-void, hardening the principal hedge (OP-46, MEDIUM → MEDIUM-HIGH conditional; the no-test claim inherits it — see §7); a second residual lever closed negative (OP-49's SVD channel); OP-48c's definitional residue was resolved; and the reopening target compressed to the carrier problem (§8); and an eighth iteration upgraded that carrier no-go from a single-algebra to a net/family result — net-modular data (relative modular operators, half-sided modular inclusions, modular intersection, the modular-Berry connection over a net) supplies more operators but no localized indefinite form, so the principal hedge hardens further toward HIGH on a narrowed (net-constructibility) condition while the carrier-convergence count stays at five; and a ninth iteration replaced the net-constructibility ansatz by a naturality formulation — the constructibility half discharged to a naturality theorem modulo a located commutant-to-generated-algebra gap, the localization half proven irreducible (it presupposes the net's index-set order, which is the geometry) — and discharged the modular-Berry-holonomy escape negative (the holonomy-natural form is the signature-free symplectic Kirillov–Kostant form; the only indefinite object, the kinematic-space metric, is symmetry-inherited from the conformal vacuum), leaving the net-naturality theorem as the sole residual between HIGH and a theorem, the hedge's grade unchanged and the count still five.

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## 1. What this is

A five-iteration, adversarially-refereed, web-verified investigation: map the best current physics, find where it breaks, separate fundamental assumptions from historical artifacts, search for deeper unifying principles, and evaluate — honestly — whether a universal theory is possible. The deliverable was never going to be a theory of everything from an armchair; it was to produce *either* a candidate framework with real backing *or* a precise account of why the evidence does not support one. We reached the second, in a strong and specific form. See *Chapter 1 (p. 1)* for navigation, *Chapter 21 (p. 497)* for the running synthesis, and *notes/* (repo: `notes/`; see the notes index, repo: `notes/README.md`) for the iteration-by-iteration work.

## 2. The converged headline (stable across iterations 2–7)

1. **Coherence is PARTIAL.** `[INFERENCE, high]` There is not one unified program but **two internally-coherent ones** joined by thin bridges: **(A) geometry-from-algebra** (modular/crossed-product, holography, entropic gravity) and **(B) is-the-quantum-framework-final** (Born-rule/linearity reconstructions + tabletop quantum gravity). They share only no-signaling/causal-order and the logical prerequisite that gravity be non-classical. The tempting "it's all information" fusion is **refused** — "information" names non-interchangeable objects across the two clusters.
2. **Program A ENCODES geometry; it does not GENERATE it.** `[INFERENCE, high]` Every load-bearing result (Ryu–Takayanagi, the entanglement-first-law⇒linearized-Einstein chain, the crossed-product Type II entropy, Jacobson's Clausius derivation) is *conceded by its own architects* to presuppose a fixed background with a Killing/conformal symmetry. This is now a **readout-independent** obstruction (see §4).
3. **The central wager is NOT-YET-PHYSICS.** `[INFERENCE, high]` It has no distinguishing near-term test; every live experiment tests the program's *gate*, *cores*, or *target*, none the wager itself.

## 3. Is a universal theory possible? — three senses

- **(A) A consistent unified GR+QFT framework — POSSIBLE, high confidence.** `[INFERENCE]` The clashes are domain-mismatches, not contradictions; consistency conditions on quantum gravity are extraordinarily rigid; AdS/CFT is a working existence proof that *some* consistent quantum gravity exists *(iteration-13 calibration: read this as a strongly-evidenced worked example* ***modulo the AdS/CFT conjecture****, which is itself unproven — a demonstration-by-example, not a theorem)*. Whether the framework *for our universe* is found and **empirically confirmed** is `[OPEN]`.
- **(B) A final substrate — possible but NOT demonstrably so.** `[SPECULATIVE]` The Wilsonian "tower with no bottom" (reinforced by $\phi^4_4$ triviality) and the emergence programs make *"no fundamental bottom layer"* a serious live alternative; "fundamental" may be scale-relative.
- **(C) A parameter-fixing complete theory — LEAST likely.** `[SPECULATIVE/CONTESTED]` The landscape + anthropics, the cosmological-constant problem, and the contingent low-entropy Past Hypothesis each threaten parameter-fixing completeness even if the framework is unique.

We explicitly **decline** the common overreach that Gödel's theorems forbid a final theory: undecidability of a theory's *consequences* is not incompleteness of its *laws*.

## 4. The central wager and why it did not graduate

The wager — *causal/algebraic/entanglement structure precedes metric geometry, true of our* $\Lambda > 0$ *universe* — was attacked at its strongest points and held the line as a *bet*, never graduating to a *result*:

- **de Sitter cardinality no-go** (iter 4–5). The one route to a Λ>0 Einstein-from-entanglement derivation — enlarging the static-patch boost to the full $SO(d+1,1)$ family — **fails on a dimension count**: a finite-dimensional isometry orbit gives ≤10 global Killing-charge constraints, which cannot Radon-invert to the infinite-dimensional local $\delta G_{\mu\nu}(x)$. The reusable principle (HYP-dS-CARDINALITY, refined -R1): in *every* known route — AdS-FGHMVR, Jacobson, CLPW — **the area-entropy coefficient $\eta = 1/4G$ is an INPUT, not an output.** That is the true root obstruction.
- **The encode→generate frontier is readout-independent** (iter 5). The designated verdict-flipping move — recover a *Lorentzian* metric from a *non-symmetric* modular flow — was executed via the modular-Berry / kinematic-space route. It produced a **genuine first**: the first emergent *Lorentzian* ($dS_2$) metric in the survey, crossing the signature barrier that the Riemannian Cao–Carroll and Connes-spectral routes fail (Huang–Ma, arXiv:2003.12252). `[ESTABLISHED]` **But the Lorentzian signature is inherited from the symmetric CFT vacuum, not generated** — it appears only where conformal symmetry fixes the connection. The obstruction closes at the *same signature-installation step* across all **four** known readout families (stress-tensor-flux/Sorce–CKV; modular-Berry/zero-mode; Lorentzian-NCG/η-foliation-twist; and — added in iteration 6 — the causal-fermion-system operator-spectral-dichotomy readout, whose first axiom, the $(\leq n, \leq n)$-eigenvalue condition on the operator class, is a necessary carrier of the signature: the Krein η relocated into the definition, per an elementary spectral lemma re-derived and numerically checked twice). Iteration 6 also identified a second pattern, **selection-type** constructions (Lee 2020; a 2025 classical toy model), which install a signature-valued template and generate only the *choice* of signature. That common closing point is what makes the obstruction structural. Iteration 7 then *executed* the compressed target — and it resolved negative in the most informative way: a canonical Krein fundamental symmetry

$\eta_0 = \mathrm{sgn}(\ln\Delta)$**is** derivable, smuggle-free, from pure $(\mathcal{M},\Omega)$ modular data (one member of an infinite $J$-odd family, per the referee's non-uniqueness correction) — but it is **provably geometry-void**: unitarily universal across the entire Borchers/HSMI class ($\Delta_{\mathrm{geo}} = 0$ proven, not graded), generically algebra-incompatible, and non-localizable without a posited carrier; a new modular-form trichotomy lemma (ansatz-scoped) shows every canonical modular pairing is positive, signature-free, or universal-indefinite; and the modular conjugation $J$ *structurally* carries reflection positivity — the Euclidean datum — via cone self-duality, so no $J$-induced pairing can be secretly Lorentzian. The open target therefore re-compresses to: derive an algebra-compatible, **localized** (fiberwise) $(n,n)$ pairing from modular data — the carrier problem (HYP-FACTORIZATION-IS-GEOMETRY), where five independent routes now converge — a convergence iteration 8 strengthened (not extended) by upgrading the modular-sign route to a net/family no-go via the same boost-covariance engine, and iteration 9 sharpened again by replacing the net-constructibility ansatz with a naturality formulation (the same engine promoted to a naturality obstruction) and discharging the modular-Berry-holonomy escape negative — the count staying at five throughout. `[INFERENCE, high]`

- **Composition cannot single out QM without re-importing geometry** (iter 5). On a single Type III1 factor, "collar-free + state-independent" composition and "singles out quantum theory" are **mutually exclusive**: the genuinely collar-free surrogate (Haagerup standard form + Connes fusion) is type-agnostic and on III1 is a *universal embezzler* ($\kappa_{\max} = 0$), the operational antipode of finite-dim QM; any QM-selecting axiom forces a Type I factor (the split construction), re-importing the collar and a reference-state. The QM-selecting composition axiom **coincides** with the geometry-importing input. [ESTABLISHED (A,B horns) / INFERENCE, high — the "must interpose Type I" step is now *theorem-shaped under the full-pair (normal-product-state) reading* (iteration 6: GNS factorization + the Doplicher–Longo/D'Antoni–Longo split-inclusion equivalences; plus a uniform Bell gap $1 - 1/\sqrt{2}$ in the approximate reading); the residue is the GPT formalization choice (full-pair vs subsystem) and the consistent III1 universal-embezzlement composition alternative]

The net of three iterations of attacks: the diagnosis "encodes, does not generate" was not merely defended — it was **tightened into a readout-independent structural obstruction**.

# 5. What is established, what is ruled out, what is open

**Firmly established** (the load-bearing core any theory must reproduce): the Hilbert-space + Born formalism (Born *form* forced by Gleason + Busch/CFMR); the Standard Model as an anomaly-free chiral gauge theory; GR as a classical geometric theory confirmed to GW/EHT precision; thermodynamics + the fluctuation theorems + RG/universality; ΛCDM; Noether, Wigner, microcausality, unitarity, Lovelock rigidity. See *Chapter 3 (p. 15)*, the *the domain chapters (Part II, Chapters 4–13)* pages.

**Ruled out or bounded** (this project's negative results): a holographic dS first law in the strict AdS sense (cardinality no-go, scoped); a QM-selecting collar-free composition on a single III1 factor (conditional no-go); generation of Lorentzian geometry from non-symmetric modular data on any *currently-known* readout; the muon $g - 2$ "anomaly" (resolved to SM-consistency by WP2025). The two surviving dS routes fail for distinct identified reasons (pseudo-entropy inherits non-unitarity; Jacobson-in-CLPW re-imports $\eta$); reality-with-nontrivial-$\delta G$ for the dS/CFT pseudo-entropy first law is likewise ruled out at near-no-go grade (OP-49, iteration 6).

**Genuinely open** — the residual levers, none headline-moving:

1. **OP-44/OP-CP1** — the nonperturbative crossed-product fate is regime-dependent (boundary $N_* \sim R/2GE_f$) with *no theorem on either side of the bulk $N_*$ boundary* — but the interior is now structured (iteration 6): the AdS time-band model has a proven

no-commutant statement on the failure side and a proven crossed-product rigidity/uniqueness statement on the survival side (Jensen–Raju–Speranza); the order-$G_N^2$ obligation is located at the quadratic (boost) linearization-instability constraints (De Vuyst–Eccles–Höhn–Kirklin), cleanest for spatially closed (dS-like) universes; the survival side is theorem-grade at linearized order for the complete constraint family incl. a non-stationary GSL and the QFC (KLS II; Chandrasekaran–Flanagan); and a fourth verdict option exists — survives-only-after-averaging over $N$ (Kudler-Flam–Witten). Iteration 7 split the order-$G_N^2$ obligation into three named parts (P1a/P1b/P1c): quadratic-order descent of the *complete* global-dS constraint set to the CLPW crossed product is discharged at inference grade (conditional on the quantum-imposition hedge; the discharge consumes pre-installed $\mathfrak{so}(d+1,1)$ structure, itself an encoding step), leaving two located, theorem-shaped residues — DEHK footnote-12 classical sufficiency with worldline clock sources (distributional sources exit the FMM/AMM framework), and cubic-constraint horizon non-splittability with a one-sided $\delta^{-2(d-1)}$ divergence whose subleading coefficient would force a Planck-length dressing-collar floor $\delta \gtrsim \ell_P$ (the gravitational parallel of OP-48's stringy $\ell_s$ floor) — proposed as a perturbative failure-side foothold at order $\kappa^3$, but **de-hardened in iteration 9 to `[INFERENCE, medium]`**: the constrained-graviton (transverse-traceless) escape stays open because the TT projector annihilates the naive leading all-radial $\delta^{-2(d-1)}$ term ( $P_{ij}\hat{n}^j = 0$), leaving the surviving same-power coefficient uncomputed and dimension-dependent (zero in the central $D = 4$ codim-2 case). The bulk $N_*$ boundary itself remains theorem-free on both sides. Watch-list.

2. **OP-46 residual** — HYP-CKV-VACUITY (now -R3) is a no-go only over *currently-known* readouts — four families after iteration 6 — and the smuggle list is extended by two named items: an **indefinite-pairing/(n,n)-flag axiom** (the CFS form of η) and a **signature-valued parameter space / deformed-algebra template** (the Lee/Bermudez selection-

type form). Iteration 7 executed the compressed target: the literal derivation *succeeds* ($\eta_0 = \mathrm{sgn}(\ln \Delta)$, smuggle-free) and is provably geometry-void, and the modular-form trichotomy lemma closes the J-route generally under a flagged modular-constructibility ansatz — so the hedge **hardens MEDIUM → MEDIUM-HIGH (conditional on that ansatz, the explicit residual)** — and iteration 8 hardens it **further toward HIGH on a narrowed (net-constructibility) condition** by upgrading the no-go from single-algebra to net/family modular data (relative modular operators, HSMI, modular intersection, modular-Berry over a net add operators but no localized indefinite form; localization lives only in the net's index set, which is the geometry), with two named obligations outstanding (a naturality theorem replacing the ansatz; ruling out a modular-Berry-holonomy-natural localized indefinite form) — and iteration 9 acted on both: it replaced the ansatz with a naturality formulation whose constructibility half is a theorem modulo a located commutant-to-generated-algebra gap (Lemma N1/N2) and whose localization half is proven irreducible (it presupposes the index-set order, which is the geometry), and it discharged the modular-Berry-holonomy obligation **negative** (the holonomy-natural form is the signature-free symplectic Kirillov–Kostant form; the only indefinite object, the kinematic-space metric, is vacuum/symmetry-inherited), leaving the net-naturality theorem (plus the new commutant-to-generated gap) as the sole residual between HIGH and a theorem — a residual iteration 10 attacked directly (close the commutant-to-generated gap to reach unconditional HIGH) and **did not move**: the gap does not close (ergodicity empties only the centralizer, not the representation-space multiplicity gap) and no carrier lives in it, so the hedge stays at its iteration-9 grade and condition, narrowing substantially toward but not provably to $n_1$ alone (a residual sliver of algebra-compatible carriers outside the algebra survives); the compressed target keeps its mandatory precision repair: derive an algebra-compatible **localized** (fiberwise) $(n, n)$

pairing from modular data — the carrier problem (HYP-FACTORIZATION-IS-GEOMETRY). **This MEDIUM-HIGH→HIGH-confidence hedge (hardened in iteration 8 by the net/family carrier no-go and sharpened-in-condition by iteration 9's naturality formulation, conditional on the net-naturality theorem plus a located commutant-to-generated gap) is the principal reason the headline is not a theorem — and, via HYP-ENCODING-SCREEN, it is also the hedge on the §7 no-test claim.**

3. **OP-dS-PSEUDO-REALITY (OP-49) — CLOSED-NEGATIVE at near-no-go grade** (iteration 6, mirroring OP-41's house language): in the dS/CFT first law the entire linearized-Einstein content is carried by the imaginary part of the pseudo-entropy (the real part is the region-independent constant $S_{\mathrm{dS}}/2$), so reality protocols either null the probe ($\delta G$ unconstrained) or deform back to a real-Euclidean/AdS-type encodes-computation. Conditional on named residuals — iteration 7 performed the stated SVD computation and the SVD first law *fails* on the $dS_3$ hemisphere (unconditional core: $S_{\mathrm{SVD}}$ is real by construction while the verified carrier is purely imaginary; the $c + c^* = 0$ cap-blindness mechanism is conditional on heterogeneous-cover junction additivity), so the list is now (a) exceptional-point transition matrices, (b) beyond-2511.07915-family generalization, (c′) junction additivity — none headline-moving.
4. **OP-48(c) horn (C)** — partially discharged (iteration 6): theorem-shaped under the full-pair (normal-product-state) reading, with a uniform Bell gap in the approximate reading. Iteration 7 resolved the definitional half: the full-pair reading of "recovers QM" is **forced** (all three reconstructions — Hardy, CDP, Masanes–Müller, axioms read verbatim — quantify composition over the designated pair; the subsystem reading is inexpressible as a weakening of any axiom), the separation half of local tomography even *holds* on the $III_1$ pair, and horn C upgrades to **theorem-conditional-on-framework**. The

shrunken residual: the normal-state postulate plus the composition-primitive choice — singular product states (X1) or the non-OPT $III_1$-embezzlement composition endpoint (X2), each an *exit from* the framework rather than a reading of it; no definitional freedom remains in the reading. Iteration 8 promoted the X1 exit to a theorem-with-a-price-tag: singular product states across the $III_1$ Haag-dual pair provably exist (factor⇒Schlieder [by the central-carrier fact, Kadison–Ringrose]⇒C*-independence⇒Roos product extension on plain C*-independence⇒Hahn–Banach; non-normal by Takesaki 1958 [via Rédei–Summers Prop. 2]), no reconstruction axiom *fails* on them (they are upstream of every axiom's finite-dim/σ-additive state space), and admitting them relinquishes density-matrix states + σ-additivity + finite tomographic dimension jointly — so horn C stays theorem-conditional-on-framework and the normal-state postulate is proven a load-bearing, non-derivable installation choice.

See *Chapter 15 (p. 233)*, *Chapter 19 (p. 392)*, *Chapter 14 (p. 191)*, *Chapter 20 (p. 458)*.

## 6. Why "analytical saturation," not "a final theory"

"Saturated" is a statement about the **marginal yield of further reasoning**, not a proof of any headline claim. Iterations 3, 4, and 5 each produced *sharper obstructions, closures, and self-corrections* rather than new positive structure or new degrees of freedom in the problem. Iteration 4 set the criterion ("after the named list is settled, marginal insight → near-zero"); iteration 5 settled the list and the one verdict-flipping lever did not flip. Continuing to *reason* will now mostly re-derive the same map. The headline rests on three explicit hedges (the OP-46 residual, **hardened to MEDIUM-HIGH in iteration 7 and further toward HIGH in iteration 8, its condition sharpened in iteration 9 to the net-naturality theorem plus a located commutant-to-generated gap, left unmoved in grade and condition by**

**iteration 10 — whose attempt to close that gap was struck as a major error — and left unmoved again by iteration 11, which re-grounded the located commutant-gap residual's vocabulary on the named (and genuinely open) Connes bicentralizer problem without closing it to a theorem (the submitted logical-equivalence reduction was major-corrected to a structural connection)**, after the literal compressed target was achieved and proved geometry-void and the no-go was upgraded from single-algebra to net/family; the OP-48c horn-C inference, now theorem-conditional-on-framework with the residue relocated to the normal-state postulate plus the composition-primitive choice; and the HYP-ENCODING-SCREEN hedge on the §7 no-test corollary, which inherits the OP-46 MEDIUM-HIGH grade with the same condition attached) and is honestly **not a theorem** — but it is a stable, well-supported, adversarially-tested verdict that has now survived a dedicated new-input iteration *and* the execution of its own four named decidable levers.

> **Capstone note — terminal analytical saturation (iteration 11).** A dedicated meta-audit (iteration 11, track A3, referee-upheld) asked whether *any* purely-reasoning lever capable of moving the verdict or the principal hedge still remains, and concluded **no**: every verdict-bearing residual is now decidable only by external input — the carrier net-naturality theorem (equivalently the commutant-gap closure / a reverse-Weinberg–Witten witness / HYP-CKV-VACUITY-as-unconditional-HIGH), an external resolution of Connes' bicentralizer problem (on which the located commutant-gap residual was re-grounded, and which is itself open in general as of 2026), or the 2027 DESI $w(z)$ experiment (which bears on the *target*, not the wager). The two genuinely reasoning-decidable residuals (OP-44 R2's higher-dimension coefficient; OP-49's a/b/c′) are headline-inert by their own scoping. The marginal-yield trend is monotone to zero: the grade-derivative reached zero at iteration 9, the condition-derivative at iteration 10, and iteration 11 produced only a precision re-grounding of the hedge's vocabulary with no registry advance. **Only new**

**physics input — an experiment, or a genuinely new mathematical readout — can now move the verdict**, which is exactly the README-sanctioned obtained-outcome (a precise reasoned account of why-not-yet). Active mode shifts to consolidation plus lightweight watch on the carrier problem and the 2027 DESI verdict.

**Capstone note — terminal saturation re-confirmed (iteration 12, 2026-06-19).** A three-track iteration (physics-net bicentralizer-lever examination; a web-verified 2023-2026 external-math sweep; a mid-2026 external-input sweep) left the verdict unmoved for the **eleventh** consecutive time, moving neither grade nor condition nor count. Its one substantive structural result is a sharpening of the saturation diagnosis: iteration-11's named "sharp next lever" — the relative-bicentralizer triviality of the physics BW-net local inclusions — is **retired as non-load-bearing**, because (a) it is currently OPEN as pure operator algebra (the AHHM relative-bicentralizer theorem is discrete-inclusion-only; physics local inclusions are non-discrete) and (b) trivial (relative) bicentralizer is the carrier-FRIENDLY condition (Marrakchi-Vaes Thm A unconditional trivial centralizer; Haagerup's large-centralizer biconditional), so it cannot close the no-go in either resolution. One of iteration 11's three open subquestions is thereby discharged as a lever, with the no-go for physics nets resting on the inherited geometry-void / non-localizability bars. No carrier closes; no reverse-WW witness; the extraordinary-claim gate does not fire. External channels remain dry: Connes' bicentralizer problem re-confirmed OPEN in general (arXiv:2605.12776); the new 2026 Euler-pair BW / Spin-Statistics mathematics is a cleaner statement of the encoding screen, not a carrier; new objective-collapse bounds (arXiv:2506.05507) tighten a CORE not the wager; the 2027 DESI w(z) verdict (low-z-SN-calibration-dependent, preference < 2 sigma) bears on the target not the wager.

**Capstone note — bottleneck-completeness and the calibrated "how close" (iteration 13, 2026-06-19).** A new-angle meta-audit of the *frame* (not a carrier re-attack) confirmed, from outside the saturated thread, that the carrier problem is the **unique analytical reasoning-decidable gate to the wager** — every other tracked obstruction (the measurement problem, the cosmological constant, the closed-universe Hilbert space OP-50, black-hole information, the seventeen clashes) is either downstream of or orthogonal to the geometry-from-algebra wager *and* itself external-gated, so none is a neglected reasoning-lever (an absence-over-the-tracked-set claim, [INFERENCE], not a theorem of completeness; the gate is unique *to the wager*, not to unification — AdS/CFT independently evidences that *a* unified framework exists). A cross-program triangulation found exactly one shared structural invariant — the area law $S = A/4G$ — but it is a semiclassical retrodiction every rival *and* every geometry-first description reproduces, hence screen-*passing*, not class-*separating*: it re-confirms the encode screen rather than breaking it. And the calibrated answer to "how close are we to a universal theory of physics" is axis-dependent and collapses, on the two binding axes, to "how close is the carrier theorem": a consistent framework is plausibly possible, a *derived* theory is one located (and currently unreachable-by-argument) theorem away, and no experiment yet conceived can referee the central wager — so the honest stance is a sharp, conservative *not-yet*, neither imminent "theory of everything" nor closed door. **Verdict unchanged a twelfth consecutive time; consolidation holds.**

**Capstone note — the verdict survives its maximal adversarial assault (iteration 14, 2026-06-19).** For the first time the burden was inverted: rather than test another lever, three adversarial tracks were commissioned to *falsify* the verdict — to assume it wrong and build the single strongest web-verifiable 2024–2026 case against each of its three pillars (encode-not-generate; no in-principle distinguishing test; terminal saturation), with the referee's role inverted to defend the standing verdict. **All three pillars held.** The strongest generative claim of 2026 — Fullwood–Vedral's assertion that the Lorentzian (1,3) signature is an *output* of an information principle on qubits (arXiv:2604.07471) — collapses because that signature is the determinant on $2 \times 2$ complex-Hermitian matrices, an unconditional fact of the chosen *complex* carrier, installed at the choice of Hilbert space and carrying no locality, net, or causal order (an internal symmetry, not generated spacetime); it is a fresh instance of the encoding screen, not a sixth route. The strongest candidate test — GQuEST/geontropic interferometry as a reverse-Weinberg–Witten observable — collapses on the proponents' own admission that conformal-field-theory, dilaton, and hydrodynamic models (none emergent in the wager's sense) yield the *identical* signal, so it co-classes rather than separates. The strongest "missed lever" — a non-Hermitian / Krein cMERA de Sitter construction (arXiv:2606.17983) — pre-selects criticality as input, confirming rather than breaching the iteration-12 Krein closure. The honest find: the verdict is now confirmed not merely by passive non-falsification (twelve times) but by surviving the strongest attack three adversarial tracks could mount — the 13th consecutive confirmation, and a stronger epistemic posture. The one thing the assault could not dismiss is a *corroboration*, not a breach: Schneider's independent philosophy-of-physics argument (*Spacetime Emergence: An (In)effective Story*, arXiv:2410.00932) locates the encode-screen's **sole** structural seam — the global / topological / superselection content treated as a net invariant (the §7/§8 un-proven net-sharing premise) — exactly where this wiki already places its own soft spot, but constructs no observable through it (a reverse-Weinberg–Witten witness remains strictly harder than the still-open carrier problem). The seam is sharpened, not crossed; the verdict, the principal hedge (HYP-CKV-VACUITY-R6), and the carrier-convergence count (FIVE) are all unmoved.

**Capstone note — the open problem posed; the no-go robust across all modern mathematics (iteration 15, 2026-06-19).** The final analytical iteration brought three frameworks *never before applied* to the gate — factorization algebras / functorial (cobordism) QFT, Connes–Atiyah–Singer index theory / K-theory / coarse geometry, and higher-categorical / homotopical / non-unitary tensor-categorical AQFT — to bear on the net-naturality theorem (the carrier problem), the external-mathematics channel the saturation audits named as the only thing that could move it. **All three reproduced the encode-not-generate screen from new directions**, each smuggling the identical geometric input: the causal-disjointness / orthogonality / index-set order $n_1$ (the localization template — the base of operadic AQFT, the bordism site), or a metric/Dirac/Krein datum installed on a pre-given space (a Lorentzian spectral triple's Dirac operator *is* the metric, on a pre-installed Krein space whose indefinite form *is* the carrier — Bizi–Brouder–Besnard, arXiv:1611.07062). The carrier no-go is therefore robust not merely across modular theory but across every modern mathematical lens tried — the **14th consecutive confirmation**. No referee-survivable new handle emerged; the categorical equivalence AQFT $\simeq$ time-orderable prefactorization algebra is a translation tool — a fourth independent witness to the same obstruction — not a sixth route (count stays FIVE). The iteration's deliverable is the one thing reasoning can still do at terminal saturation: the single open gate, distilled into a self-contained, formally-precise **open-problem statement an external operator-algebraist / AQFT-theorist / noncommutative-geometer can attack cold** — *the carrier-problem dossier* (Chapter 35 (p. 674)). The problem is now posed for external resolution; the verdict, the principal hedge (HYP-CKV-VACUITY-R6), and the carrier-convergence count (FIVE) are unmoved.

## 7. The object-level question is handed to experiment

A *confirmed* universal theory is **not forecastable**, and the binding reason is not the energy desert but that **the central wager has no distinguishing experimental test even in principle** — a claim that iteration 6 upgraded from an observation about the channel landscape to a *corollary* of the §4 encode-not-generate obstruction (HYP-ENCODING-SCREEN): an encode-only program's observational content equals that of the EFT it encodes, so every candidate signal factors through physics equally available to geometry-first descriptions, and a class-separating test would require a "reverse Weinberg–Witten" theorem, of which none exists. As a corollary it is exactly as strong as the encode-only classification and **inherits the OP-46 hedge — MEDIUM-HIGH→HIGH after iteration 8, conditional on the net-constructibility ansatz** (the third explicit hedge); and the implication is one-way — a generative readout is a *prerequisite for*, not a guarantee of, a distinguishing test, so the §8 reopening condition and the missing experiment coincide as obstructions. The empirical channels that *do* exist — gravitationally-induced entanglement (BMV/QGEM), objective-collapse bounds, dynamical dark energy $w(z)$, Lorentz-invariance/equivalence-principle nulls, and now the GQuEST/Verlinde–Zurek interferometer channel — test the program's *gate*, *cores*, and *target*, **none the wager itself**. No experiment decides the wager; exactly one funded experiment (GQuEST) probes a conjecture *inspired by* the wager's modular vocabulary, and its inference chain re-imports $\eta = 1/4G$ at the modular→metric step while its design papers concede the amplitude is not derived from any UV-complete theory. A dedicated 2024–2026 literature hunt (2026-06-10) confirms this with fresh, positive evidence: the verified lead — the Sharmila–Vermeulen–Datta three-class fluctuation taxonomy (*Nat. Commun.***17**, 701 (2026), arXiv:2505.22892), the most natural place a wager-discriminating interferometer signal would live — is explicitly mechanism-agnostic (*"Our approach is agnostic to the microscopic origins of the SFs"*) and places emergent and fundamental-metric models in the *same* correlation class, a published worked instance of the encoding screen rather than a counterexample to it; and the theorem-shaped escape, a *reverse Weinberg–Witten* observable (one realizable only without fundamental metric degrees of freedom), remains **unattempted and** `[OPEN, as of 2026-06]` — its absence is a literature fact, not a proven structural impossibility, so the no-test corollary keeps exactly its OP-46 MEDIUM-HIGH→HIGH grade and the net-constructibility condition. Iteration 9 pinned this escape to the §8 reopening condition rather than leaving it free-floating: A1 ("every observable is an invariant of the GNS pair, net + vacuum") is a *conditional* theorem (Weiner's algebraic Haag theorem, arXiv:1006.4726: split + geometric modular action $\Rightarrow$ the vacuum-sector net is a complete invariant), but its hypotheses are exactly the geometric inputs HYP-FACTORIZATION-IS-GEOMETRY isolates — so the reverse-WW *theorem* is equivalent to the carrier no-go, the reverse-WW *witness* is strictly harder than a positive carrier solution, and reverse-WW is the empirical-side shadow of the carrier problem, not an independent target. They are tracked in *Chapter 39 (p. 725)*. Beyond this point, only new physics input — an experiment, or a genuinely new mathematical readout — can move the verdict.

# 8. What would reopen the analysis

Precisely one analytical development would flip the headline from "encodes" to "generates": **a construction that outputs a Lorentzian causal/metric structure from genuinely non-symmetric (algebra, state) data, installing no fundamental symmetry, η-form, foliation, twist — nor (added iteration 6) an indefinite-pairing/(n,n) operator-class axiom or a signature-valued parameter space / deformed-algebra template — by hand.** No construction meeting the full spec exists; the nearest sketch (Lee, JHEP 06 (2020) 070, arXiv:1912.12291) achieves *dynamical signature selection* — a state-dependent signature output — but on a posited carrier plus a hand-installed signature-parametrized constraint-algebra template, with only translation-invariant (symmetric) saddles demonstrated, so it fails the spec. Equivalently (the compressed form, repaired in iteration 7 — the iteration-6 wording "no one has derived the indefinite pairing from modular data" is now *literally false*: $\eta_0 = \mathrm{sgn}(\ln \Delta)$ achieves it, smuggle-free, and is provably geometry-void): no one has derived an **algebra-compatible, localized** (fiberwise) $(n, n)$ pairing from modular data — the carrier problem (HYP-FACTORIZATION-IS-GEOMETRY), the step at which all five known routes close — a closure iteration 8 hardened by showing the net/family upgrade (relative modular operators, HSMI, modular intersection, modular-Berry over a net) relocates the carrier into the net's index set rather than evading it, and iteration 9 sharpened by replacing the net-constructibility ansatz with a naturality formulation — the constructibility half a theorem modulo a located commutant-to-generated-algebra gap, the localization half proven irreducible (it presupposes the index-set order, which is the geometry) — and by closing the modular-Berry-holonomy escape negative (signature-free symplectic curvature; vacuum-inherited kinematic metric); and iteration 10 attacked that located commutant-to-generated gap head-on and confirmed it does **not** close — the gap is real and wide for degenerate modular spectrum, empty in the geometric sector only at the level of the *centralizer* (not the representation-space gap, which keeps the indefinite $\eta_0$), and carries no carrier (the strongest-ever candidate clears the indefinite and algebra-compatible bars by an inner automorphism but is geometry-void) — so the residual between HIGH and a theorem is unchanged and the route count stays at five. Iterations 11–16 re-grounded that residual (Connes' bicentralizer problem, OPEN in general as of 2026; the indefinite-metric/Krein route closed-negative; every un-tried modern mathematical framework reproducing the encoding screen) and finally, via iteration 16's multi-wedge closure (LEM-NET-NATURALITY-G, referee-verified), **classified** the commutant-to-generated gap and pinned the entire remaining no-go to a single named open hypothesis — **(E_O)**, vacuum ergodicity on massive double-cone algebras, $M(O)_\omega = \mathbb{C}1$. Iteration 17 (2026-07-07) then subjected (E_O) to a dedicated five-lens assault (three binding referees): it stands [OPEN] — for the free massive scalar it is sharply **reduced** to a one-particle spectral question ($M(O)_\omega = \mathbb{C}1$ is implied by purely continuous spectrum of the double-cone one-particle modular Hamiltonian $\ln \delta_O$ / weak mixing; the naive "no-eigenvalue-0" reading refuted; the spectral fact numerical-only and unproven, no closed-form massive modular Hamiltonian existing), and for general class (H) all three structure-theory routes (weak-mixing-from-clustering; wedge-to-double-cone HSMI; Marrakchi–Vaes genericity) provably fail while the adversarial disproof also fails on eight candidate mechanisms — so the reopening target is sharpened but neither closed nor flipped, the hedge holds at R7, and the headline stands a **sixteenth** consecutive time. Empirically, a confirmed *positive* in any watch-list channel would resolve the program's *gate* or *target* (e.g. BMV confirming gravity is non-classical), but — to be clear — would still not confirm the wager. If either occurs, this conclusion is reopened and the relevant page is re-tagged with the change logged in the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)).

> **The open problem, posed (iteration 15).** The single analytical development above — the carrier problem / net-naturality theorem — is now distilled into a self-contained, formally-precise open-problem statement for external solvers: the precise conjecture, the established reductions and lemmas, the exact residual, the adjacent named open problems (Connes' bicentralizer; the factorization-algebra / index-theory / higher-categorical framings), and the exact geometric input every modern framework tried so far smuggles. See ***the carrier-problem dossier*** **(Chapter 35 (p. 674))**. Reasoning has taken the gate as far as argument can; it now awaits external mathematics.

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CHAPTER 39

# The Experiment Watchlist

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**Read this caveat first. NO channel on this page decides the central wager** (causal/algebraic/entanglement structure precedes metric geometry) — a fact that, as of iteration 6, is a *corollary* of the encode-not-generate obstruction (HYP-ENCODING-SCREEN, *Chapter 19 (p. 392)*) and inherits its MEDIUM hedge, not merely an observation about the 2026 landscape. The channels constrain the program's logical **gate** (is gravity non-classical?), its fixed **cores** (is linearity/Lorentz/strong-field GR exact?), its correct **target** (is the universe eternal dS?), or — a fourth category added this iteration — **program-adjacent phenomenology**: GQuEST is the first *funded* experiment probing a conjecture *inspired by* Program A's modular/causal-diamond vocabulary, but its inference chain re-imports the area-entropy coefficient $\eta = 1/4G$ at the modular→metric step (extending HYP-dS-CARDINALITY-R1's pattern from derivation routes to a phenomenological chain), so it is decision-grade only for **minimal-graviton-EFT IR noise vs exotic Planck-normalized UV-IR mixing** — on no reading a wager test. A positive anywhere yields *abductive* support at most, never class-level separation. Note also the **asymmetric gate-closure route**: falsification of the explicit classical-quantum (CQ) diffusion kernels would close the gate *from the classical side* — by eliminating the principal live local stochastic classical alternative, within that model-class scope — possibly years before any BMV-style entanglement witness runs. That the wager remains untestable even in principle until a *distinguishing* prediction is found is the standing obstacle, not a temporary gap.

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# Empirical channels

| CHANNEL | WHAT IT DECIDES | CURRENT STATUS |
|---|---|---|
| **BMV / QGEM** gravitationally-induced entanglement | Program A's logical **gate**: is gravity non-classical? ($m^2$ vs $m^3$ witness scaling) | Not yet performed; ~6–10 orders below required mass/coherence; **n funded experimental roadmap with milestones exists publi as of mid-2026**; mediation-vs-locality premise `[CONTESTED]`; a 202 claim that the leading-order qubit entangling phase is mass-INDEPENDENT in Stern–Gerlach protocols (Braccini–Serafini–Bos arXiv:2602.19306) would make the $m^2$-vs-$m^3$ discriminator protoco dependent `[SPECULATIVE/INFERENCE]`; witness family widened beyon the Newtonian phase: first genuinely post-Newtonian GIE channel – entanglement via frame dragging (Lense–Thirring), derived two independent ways (Wakakuwa–Petruzziello–Lantaño–Huelga–Plen arXiv:2606.31678) `[ESTABLISHED as derivation, un-refereed v1; same gate/class sta` |

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| **Objective-collapse** bounds (CSL / Diósi–Penrose) | A fixed **core**: is exact linearity true at the macroscopic-superposition scale? | Parameter-free DP excluded; DP $R_0 > 4.9$ Å (90% CL); canonical white-noise CSL points (GRW, Adler) excluded — GRW point 9.1σ from best fit (XENONnT, PRL 136, 120201 (2026), arXiv:2506.05507; $\lambda/r_C^2 < 3.0 \times 10^{-3}$ s$^{-1}$m$^{-2}$, most stringent for $r_C \lesssim 10^{-5}$ m); colored/dissipative extensions + large-$r_C$ corners survive; no detection | decisive coverage ~2035 | Detection falsifies "exact unitarity as a fixed core" (A-3); null results push $\lambda$ down. The empirically live way exact linearity could be false. |

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| **Dynamical dark energy** $w(z)$ (DESI DR3+, Euclid, Roman, LSST) | Program A's correct **target**: eternal dS, metastable dS, or rolling quintessence? | DESI DR2 (PRD 112, 083515 (2025)): 2.8–4.2σ preference for $w_0w_a$CDM, $w_0 > -1$, $w_a < 0$; sample- and SN-calibration-dependent `[CONTESTED]`; DESI 5-yr survey complete (Apr 2026, >47M objects); **full five-year analysis expected 2027 — the decision point**; Euclid DR1 restructured (ESA announcement 2026-06-15, verified): DR1-Foundation Nov 2026 (raw+calibrated only), complete DR1 with GC+WL cosmology products **mid-2027**; Roman at KSC since 2026-06-21, launch NET 2026-08-30 (Falcon Heavy); Rubin LSST 10-yr survey underway | ~1–3 yr (2027 first decisive verdict; multi-survey confrontation by ~2028–2030) | $w(z) \neq -1$ confirmed ⇒ the holographic/emergence target is *not* eternal dS, weakening the motivation for a strict Λ>0 first law (A-20). Tests the target, not the wager. |

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| **LIV / EP** nulls (GRB time-of-flight, CTA; torsion-balance, MICROSCOPE, LLR) | Robustness of the surviving **cores**: exact local Lorentz invariance + a single causal order | No confirmed LIV; LHAASO GRB 221009A (arXiv:2402.06009, PRL): $E_{\rm QG,1} > 10\,E_{\rm Pl}$ (linear) and $E_{\rm QG,2} > 6 \times 10^{-8}\,E_{\rm Pl}$ (quadratic) — first-order Planck-suppressed LIV excluded an order beyond the Planck scale; WEP: MICROSCOPE final $\eta(\mathrm{Ti}, \mathrm{Pt}) \sim 10^{-15}$ (PRL 129, 121102 (2022)); torsion balances $\sim 10^{-13}$ | 0–10 yr (mostly a null-result anchor) | Confirmed LIV would refute "exact Lorentz + single causal order held fixed" and threaten emergent-causal-order tracks (H1/H3/H6). |
| **Matter-wave / levitated optomechanics** (enabling tech) | Shared **enabler**: gates BMV *and* tightens collapse bounds | Record: sodium nanoclusters >7,000 atoms, $> 1.7 \times 10^5$ Da interfered in a time-domain Talbot–Lau interferometer (arXiv:2507.21211; Nature 649, 866 (2026)); macroscopicity $\mu = 15.5$ — the most stringent generic-macrorealism exclusion to date; still ~7–8 orders below BMV-decisive masses | 5–15 yr | The bridge technology both live channels depend on; no theory target is empirically decidable without it. |

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| **GQuEST / Verlinde–Zurek geontropic noise** (program-adjacent phenomenology) | EFT-breakdown discriminator: minimal graviton-EFT IR noise $\delta L^2 \sim \ell_p^2$ (Carney–Karydas–Sivaramakrishnan, PRD 113, 106002 (2026)) vs Planck-normalized correlated noise $\delta L^2 \simeq \ell_p L/4\pi$ (pixellon; amplitude $\alpha$ not derived from any UV-complete theory). **Not the wager** — the chain inputs $\eta = 1/4G$ | Design PRX 15, 011034 (2025); under construction at Caltech (DOE-funded); no science data as of 2026-06; existing bounds $\alpha \lesssim 0.6$–3 (IR-cutoff-dependent); $\alpha \gtrsim 0.1$ without IR cutoff already dead (LIGO) | ~2–5 yr ($5\sigma$ on $\alpha = 1$ in $O(10^5)$ s once running) | Detection ⇒ either severe IR breakdown of graviton EFT or a non-vacuum graviton state (Freidel–Oberfrank confound) — enormous; *abductive* support for holographic modular counting; reopens *Chapter 18 (p. 349)*; still does not flip the wager from not-yet-physics. Null ⇒ kills the $\alpha \sim O(1)$ pixellon normalization only. A signal would be classified by the **Sharmila–Vermeulen–Datta** three-class correlator taxonomy (*Nat. Commun.* **17**, 701 (2026), arXiv:2505.22892), whose classes are **mechanism-blind by construction** — geontropic (emergent) and Károlyházy (fundamental-metric stochastic) co-inhabit class (b); Oppenheim CQ and CSL co-inhabit class (a) — so no class is realizable |

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| | | | | metric model: a published worked instance of the encoding screen, not a wager discriminator (2026-06-10 distinguishing-test hunt). |
| **CQ decoherence–diffusion trade-off** (gate, from the classical side) | Is gravity non-classical? — by squeezing the consistent classical-quantum hybrid class (Oppenheim PRX 13, 041040 (2023); trade-off theorem Nat. Commun. 14, 7910 (2023)) | Data-driven (Janse et al., PRResearch 6, 033076 (2024)): ultralocal-continuous **dead**; ultralocal-discrete window below the experimental upper bound (modulo stated caveats); non-local continuous survives by ~1 order. Counter-pressure: Hotta–Murk–Terno "QG in disguise" embedding claim `[CONTESTED]` | Running now (analysis of existing force-noise + coherence data); years before BMV | Complete falsification of the explicit CQ kernels closes the gate by elimination of the principal classical alternative (model-class-scoped) — the **asymmetric gate-closure route**; a detection of anomalous diffusion would upend the gate entirely. |

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| **GW strong-field nulls** (core robustness anchor) | Is GR exact in the strong field; are remnants Kerr? (Recorded as a core *claim under test*, not a pre-existing wiki core) | GWTC-4.0 (2025-08-26) + tests-of-GR companions: no deviations, no echoes, Kerr-consistent; GW250114 (SNR 80): two ringdown modes, $(2, 2, 0)$ + overtone Kerr-consistent (tens of percent); **area law confirmed** (arXiv:2509.08054, PRL 135, 111403 (2025)); GWTC-5.0 (2026-05-26): 390 detections total | Continuous (O4/O5; next data release Dec 2026) | A confirmed echo or non-Kerr QNM would reopen near-horizon structure assumptions some Program-A constructions lean on; continuing nulls harden them. |
| **Neutrino-mass cosmology** (target-adjacent ΛCDM stressor) | Whether ΛCDM's neutrino sector is consistent — entangled with the $w_0w_a$ degeneracy (one underlying ΛCDM stress with the w(z) row, not two) | DESI DR2+CMB: $\Sigma m_\nu < 0.0642$ eV (95%, ΛCDM) — below the inverted-ordering floor; effective-mass parameter prefers negative values at ~3σ vs the oscillation floor (arXiv:2503.14744); relaxes to $< 0.163$ eV in $w_0w_a$CDM; direction dataset-dependent `[CONTESTED]` | ~1–4 yr (DESI 5-yr 2027; Euclid DR1; JUNO ordering; KATRIN) | Persistence after the 2027 w(z) verdict would indicate a real ΛCDM crack on the target side; dissolution in $w_0w_a$CDM would fold it into the w(z) decision. |

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## Channel notes

- **BMV / QGEM.** The Bose–Marletto–Vedral proposal: if two ~ $10^{-14}$ kg masses in spatial superposition become entangled purely via gravity, then — *given* the LOCC no-go and a local-mediation premise — the gravitational mediator is non-classical. The inferential weight sits on the mediation premise, which is logically stronger than no-signaling and is actively disputed (Di

Biagio 2025; Aziz–Howl 2025 $m^3$ loophole vs rebuttals). A decisive run needs a joint **entanglement detection + decoherence bound** and an $m^2$-vs-$m^3$ scaling test. See *Chapter 14 (p. 191)* GC-15, OP-40, HYP-GIE-DISCRIM. *2026 status (iteration 6):* the Aziz–Howl $m^3$ loophole is trending dead under three rebuttal arguments from two author groups (Marletto–Oppenheim–Vedral–Wilson 2511.07348; Tang–Yang–Han–Kan–Liu–Xue 2512.13675 + 2604.16276 — the latter now verified real, upgrading the tabletop note's [unverified] flag), but the rebuttal literature is not yet sociologically independent of the no-go's stakeholders `[CONTESTED]`. The mass-independent-entangling-phase claim (Braccini–Serafini–Bose, arXiv:2602.19306 — note Bose is a BMV original author) complicates HYP-GIE-DISCRIM: mass-scaling discriminators become protocol-dependent.

- **GQuEST / Verlinde–Zurek.** The VZ chain: holographic modular-fluctuation area law $\langle \Delta K^2 \rangle = \langle K \rangle = A/4G$ `[INFERENCE within AdS/CFT]` → AdS-to-flat lightsheet transfer `[SPECULATIVE]` → fluctuations gravitate as a coherent s-wave breathing mode (pixellon) [SPECULATIVE, load-bearing; Aalsma–Bak: whether QG vacua contain the required mode "remains unresolved"] → interferometer PSD with free $\alpha = O(1)$ . **The wager contributes no link.** Hogan's holographic noise (the Holometer's original target) is dead, and its null does NOT transfer to VZ. See HYP-dS-CARDINALITY-R1, Working note: *Quantum-gravity phenomenology 2022–2026: the distinguishing-prediction hunt, audited* (repo: `notes/2026-06-10-iter6-qg-phenomenology.md`), Working note: *Iteration 6, Track A1 — A Distinguishing Prediction for the Geometry-from-Algebra Wager: Found or Refused* (repo: `notes/2026-06-10-iter6-distinguishing-prediction-hunt.md`). *2026-06-10 distinguishing-test hunt:* the Sharmila–Vermeulen–Datta correlator taxonomy (verified *Nat. Commun.* **17**, 701 (2026)) is the sharpest published statement of what a signal pins down — the symmetry/decay class of the noise two-point function — and is itself mechanism-blind: its model table is many-to-one (CSL, Diósi–Penrose, Oppenheim, Károlyházy → class (a); geontropic + EFTs → class (b)), so the model→class map is non-invertible and cannot recover whether the metric is fundamental. See Working note: *Distinguishing-Test Hunt (2026-06-10) — An In-Principle Test of the Geometry-from-Algebra Wager in the 2024–2026 Literature: Found or Refused* (repo: `notes/2026-06-10-distinguishing-test-hunt.md`).

- **Objective collapse.** GRW/CSL and Diósi–Penrose evade the Gisin–Polchinski deterministic-nonlinearity no-go by being stochastic. X-ray/heating searches (Donadi et al. 2021) and LISA-Pathfinder already exclude the parameter-free DP model. See *Chapter 20 (p. 458)* A-3, Working note: *Is the Born Rule Fundamental or Derivable, and Is Quantum Mechanics Exactly Linear?* (repo: `notes/2026-06-08-born-rule-and-quantum-linearity.md`). *Iteration-6 note:* the "~decisive coverage by 2035" date has no external anchor — flag as an internal extrapolation; the interferometric frontier is advancing faster than older projections while the $10^5$–$10^{18}$ amu bulk remains untested. *2026-07 (Track R2):* the arXiv:2506.05507 bound is the **XENONnT Collaboration's** (the iteration-12 sweep's "VIP/XENONnT" conflated two programs — VIP-3 is that group's *Pauli-exclusion* arm, data from 2025, *Entropy* **26**, 752 (2024)); LISA-Pathfinder updates: PRA **111**, L020203 (2025) (rotational, ~2× at large $r_C$) and arXiv:2411.17588 [preprint] ( $\lambda \lesssim 8.3 \times 10^{-11}$ s$^{-1}$ at $r_C = 10^{-7}$ m). With GRW and Adler points both dead, the surviving Markovian space has no independently motivated targets; colored-noise/dissipative escapes evade radiation bounds *by construction*, so complete closure is not guaranteed by any funded roadmap — the "~2035 decisive coverage" flag stands. Bound table: Working note: *Track R2 — The Measurement Problem, Operationalized (2026-07 bound state)* (repo: `notes/2026-07-02-measurement-problem-operationalized.md`).
- **Dark energy $w(z)$.** DESI DR2 (arXiv:2503.14738) prefers $w_0w_a$ CDM over ΛCDM at 2.8–4.2σ. If confirmed, it reframes the dS-emergence target (A-20). See *Chapter 11 (p. 139), Chapter 16 (p. 308)*. *Iteration-6 note:* everpresent Λ — the nearest existing *template* of a wager-class prediction — is CMB-disfavored in its

current implementations (Das–Nasiri–Yazdi 2307.13743, preprint) and, as of 2026-06-10, no published confrontation with the DESI $w_0w_a$ posterior exists: an identified gap, watch for it.

- **LIV / EP.** Mostly a robustness anchor for the established cores; a *discovery* here would be the surprise that reopens the locality/causal-order assumptions.

## Theory watch-items (would reopen the *analysis*, not just the landscape)

These are the residual open levers from *Chapter 38 (p. 705)* §5. None is currently movable without genuinely new input:

1. **The verdict-flipper (OP-46 residual; compressed form).** A construction outputting a **Lorentzian** causal/metric structure from genuinely *non-symmetric* (algebra, state) data, installing **no** fundamental symmetry / η-form / foliation / twist — **nor an indefinite-pairing/(n,n) axiom, nor a signature-valued template** (HYP-CKV-VACUITY-R3). Compressed iteration-6 form: **derive the indefinite pairing from modular data.** Nearest sketch (Lee 2020) is selection-type and fails the spec. *Still the single development that would flip "encodes" to "generates" — and now, via HYP-ENCODING-SCREEN, also the prerequisite for any distinguishing experiment*. See *Chapter 19 (p. 392)*.
2. **OP-44 / OP-CP1.** Restructured (iteration 6): no-commutant theorem + crossed-product rigidity on the AdS time-band model (Jensen–Raju–Speranza); order-$G_N^2$ obligation located at the quadratic boost constraints (De Vuyst–Eccles–Höhn–Kirklin); survival side theorem-grade at linearized order incl. the QFC (KLS-II; Chandrasekaran–Flanagan); fourth option "survives-after-averaging over $N$" (Kudler-Flam–Witten). The bulk $N_*$ boundary is still theorem-free on both sides.
3. **OP-49 (OP-dS-PSEUDO-REALITY) — CLOSED (near-no-go, resolved-negative, conditional; iteration 6).** Removed from the open watch-items; residuals (exceptional-

point transition matrices; an SVD-entropy first law — one stated decidable computation remains) tracked in *Chapter 15 (p. 233)*.

4. **OP-48(c) horn (C).** Hedge shrunk (iteration 6): theorem-shaped under the full-pair reading; residual = the GPT formalization choice + the III1-embezzlement composition alternative.
5. **OP-50 (new, `[CONTESTED]`).** The closed-universe Hilbert-space problem: is $\dim \mathcal{H}_{\text{closed}} = 1$ per α-sector, with QM recovered only observer-relationally? An independent path-integral route to OP-48(c)'s conditional structure; watch the ARSSV / Hong Liu / Engelhardt–Gesteau counter-positions. See *Chapter 15 (p. 233)*.
6. **Reverse-Weinberg–Witten (un-registered; HYP-ENCODING-SCREEN portability corollary; `[OPEN, as of 2026-06]`).** Exhibit an observable $X$ realizable in some theory with **no** fundamental metric degrees of freedom and by **no** core-satisfying fundamental-metric theory — or prove no such observable exists. Its existence would be the wager's first in-principle distinguishing test; its impossibility would make the §7 not-yet-physics verdict permanent-in-principle rather than contingent. The 2024–2026 distinguishing-test hunt confirms it is **unattempted**: emergent-gravity constructions only *evade* Weinberg–Witten (remove the background; zero $T_{\mu\nu}$ matrix elements) — exactly the moves that erase observational handles. **NB: this is NOT "OP-50"** (a working-note misnomer); reverse-WW carries no registry ID. See *Chapter 19 (p. 392)* HYP-ENCODING-SCREEN, Working note: *Distinguishing-Test Hunt (2026-06-10) — An In-Principle Test of the Geometry-from-Algebra Wager in the 2024–2026 Literature: Found or Refused* (repo: `notes/2026-06-10-distinguishing-test-hunt.md`).

## 2026-06 external-input sweep note (iteration 12)

*Append-only; no new row, no registry ID. From the iteration-12 B3 track (referee-verified live).*

- **Connes' bicentralizer problem** re-confirmed **OPEN in general** (Ando–Goldbring, arXiv:2605.12776) — re-validates the binding iteration-11 correction; the carrier no-go is NOT equivalent to it.
- **Objective collapse** — new 2026 bounds tighten the A-3 **core**, not the wager: VIP/XENONnT (CSL ~2 orders, Diósi–Penrose ~5×; no excess; primary arXiv:2506.05507, PRL March 2026).
- **DESI five-year** $w(z)$ verdict reaffirmed **2027**; the DR2 dynamical-dark-energy preference reduces to **< 2σ** under low-z SN-calibration systematics (Huang–Cai–Wang, arXiv:2502.04212); Roman launch 2026, Roman+Euclid joint dark-energy ~2027. Bears on Program A's **target**, **never the wager**.
- **Theory-watch-item 6 (reverse-WW):** "GR from RG" (arXiv:2602.11806) confirmed an **evade-not-invert** Weinberg–Witten instance (UV-Dirichlet-fixed installed metric) — discharges the iteration-11 minor "WW gloss not confirmable" defect. Reverse-WW stays `[OPEN, as of 2026-06]`.
- **New 2026 carrier-adjacent math:** the Morinelli–Neeb–Olafsson Euler-pair geometric-BW / Spin–Statistics theorems (arXiv:2603.26390 Part II / 2508.10960 Part I) are a cleaner statement of HYP-ENCODING-SCREEN (symmetry = input) — **NOT** a carrier, **NOT** a sixth route.

*Watch-sweep #2 (2026-07-07) — no dated clock movement. From the iteration-17 pass (referee-verified live).*

- **DESI five-year** $w(z)$ analysis ~2027 UNCHANGED (2026-07-03 DESI post = outreach cosmic-web visualization, no science; observing into 2028). Euclid DR1 mid-2027 UNCHANGED. Roman launch NET **2026-08-30** on Falcon Heavy UNCHANGED (NASA blog 2026-07-06 vertical-positioning prelaunch milestone at KSC). Tabletop-QG cores (Oppenheim CQ; QGEM/BMV; GQuEST/Verlinde–Zurek): no in-window dated movement. The 2027 DESI $w(z)$ decision point holds as the nearest dated external event; none of these tests the wager.

- **Theory-watch-item (δ) — (E_O), the principal analytical clock:** STATIC as to resolution, but iteration 17 sharpened its free-field branch — the sharpest decidable sub-target is now "prove purely continuous spectrum (no eigenvectors) of the massive double-cone one-particle modular Hamiltonian $\ln \delta_O$ for the free scalar" (⟹ (E_O) on the free subclass via weak mixing ⟹ R7 → R8 there). Adjacent numerics: Bostelmann–Cadamuro–Minz arXiv:2209.04681; Cadamuro arXiv:2312.08525 (leaning-true, no proof/closed-form). See *Chapter 37 (p. 693)*.

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## How to use this page

Update a row when a channel reports new bounds or a detection; log the change and any re-tag in the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)). If a **theory watch-item** is resolved, reopen *Chapter 38 (p. 705)* and re-run the relevant analytical track. A confirmed positive in any empirical channel resolves a *gate/core/target* — record it here and propagate to the affected pages, but do **not** mistake it for confirmation of the central wager.

CHAPTER 40

# Roadmap: What Comes Next

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**Update (post-iteration-5):** the analytical program is **saturated** — five iterations converged on a stable PARTIAL / encodes-not-generates / not-yet-physics verdict, and the one designated verdict-flipping lever (OP-46 Lorentzian non-CKV readout) was executed and did not flip. The standing verdict is consolidated in *Chapter 38 (p. 705)*. Active effort now shifts from analysis to **lightweight experiment-tracking** (*Chapter 39 (p. 725)*). The Tier-1/2 analytical items below are retained as a *record of what was attacked and closed*; the only genuinely-open analytical lever is OP-44/OP-CP1 (nonperturbative crossed-product fate, regime boundary $N_*$), plus the standing "what would reopen the analysis" trigger in *Chapter 38 (p. 705)* §8.

**Update (post-iterations-6–7) — saturation held under new input and lever execution.** Iteration 6 injected two years of new external literature (10 refereed tracks); iteration 7 *executed* the four named decidable levers. The verdict is unchanged for the **sixth consecutive iteration**, but the frontier sharpened to a single object. The compressed verdict-flipper has been re-compressed twice: from "derive geometry from algebra" → "derive an indefinite pairing from modular data" (iter 6) → **executed and found geometry-void** (iter 7: $\eta_0 = \mathrm{sgn}(\ln \Delta)$ exists, smuggle-free, but is unitarily universal / algebra-incompatible / non-localizable) → the now-current target: **derive an algebra-compatible, *localized* (n,n) pairing from modular data — the carrier problem (*Chapter 19 (p. 392)* HYP-FACTORIZATION-IS-GEOMETRY), where five independent closure routes**

**converge.** The OP-46 hedge hardened MEDIUM → MEDIUM-HIGH. Iteration 8 (in flight) attacks the carrier problem via net/relative modular data plus three named residues (OP-44 R1/R2; OP-49 junction additivity (c′); OP-48c exit X1).

**The current live agenda, by leverage:**

1. **The carrier problem** (HYP-FACTORIZATION-IS-GEOMETRY) — the single analytical development that would flip "encodes" to "generates". Attacked in iteration 8; if it survives there, the next inputs are net-cohomological / modular-Berry-curvature constructions and any 2024–2026 result deriving *localization* (not just a pairing) from modular data.
2. **The distinguishing-test sweep** — the *binding* blocker per *Chapter 38 (p. 705)* §7 is not the algebra but the absence of any experiment that separates the wager from a geometry-first description (HYP-ENCODING-SCREEN). A standing iteration should sweep the live experimental literature for a candidate distinguishing test, refereed against the screen (write the chain wager → observable, mark each link). *Seeded lead (held, unrefereed):* Sharmila–Vermeulen–Datta, "Signatures of correlation of spacetime fluctuations in laser interferometers," *Nat. Commun.* 2025, arXiv:2505.22892 — a mechanism-agnostic, correlator-based fluctuation classification; expected to *reinforce* §7 (a correlator cannot separate fundamental from emergent metric) rather than threaten it.
3. **The named residues** — OP-44 R1 (distributional clock sources vs FMM/AMM linearization stability) and R2 ($\kappa^3$ cubic-charge horizon non-splittability); OP-49 (c′) junction additivity; OP-48c X1/X2 framework exits. Each is decidable by a stated computation.
4. **Empirical anchors** (*Chapter 39 (p. 725)*) — the 2027 DESI 5-yr $w(z)$ verdict (nearest dated event), GQuEST results, the CQ-gravity squeeze. None tests the wager; all test its gate/cores/target.

**Update (post-iterations-8–12) — TERMINAL ANALYTICAL SATURATION; the agenda above is now a record, not a live list.** Iterations 8–12 (plus a distinguishing-test hunt and a dedicated frontier scout) executed every named lever above and a dedicated terminal-saturation meta-audit. The verdict is unchanged for the **eleventh consecutive iteration** (iteration 12, 2026-06-19); the principal hedge has not moved in grade or condition since iteration 9. The carrier problem (≡ reverse-Weinberg–Witten ≡ the net-naturality theorem; *Chapter 19 (p. 392)* HYP-FACTORIZATION-IS-GEOMETRY) is the single open analytical gate, and every *reasoning-decidable* approach to it is now closed or retired:

- the **commutant-to-generated gap** (iter 10) does not close, and contains no carrier;
- the **bicentralizer re-grounding** (iter 11) is a *loose structural connection*, not a reduction, to Connes' (genuinely open) bicentralizer problem; the **physics-net relative-bicentralizer lever** (iter 12) is *non-load-bearing* (trivial bicentralizer is the carrier-friendly condition);
- the **indefinite-metric / PT-symmetric / Krein carrier route** (iter 12 scout; iter-11 A1 subquestion #3) is **closed-negative** — it collapses to encode-not-generate by a gauge-artifact / installed-not-derived / $\eta_0$-equivalent trichotomy;
- all three of iteration 11's open subquestions are discharged; a web-verified 2023–2026 external-math sweep found no result touching the residual.

**Conclusion (referee-upheld A3 audit, iteration 11; re-confirmed iteration 12):** every verdict-bearing residual is now decidable only by **external input** — a genuinely new operator-algebra / AQFT theorem (the net-naturality theorem / a reverse-WW witness / an external resolution of Connes' bicentralizer problem), or the **2027 DESI five-year** $w(z)$ experiment (which bears on the *target*, not the wager). **No purely-reasoning iteration can move the verdict.** Active mode is therefore **consolidation + lightweight watch**, on exactly two clocks: (1) the carrier / net-naturality / bicentralizer / GMA-reconstruction mathematics (2025–2027), and (2) the 2027 DESI $w(z)$ verdict (nearest dated external event; the DR2 dynamical-dark-energy preference reduces to < 2σ under low-z SN-calibration systematics). New analytical iterations are **not** launched unless one of those clocks delivers a genuinely new input. The standing deliverable — the maximally-sharp reasoned account of *why-not-yet* — is the README-sanctioned obtained-outcome.

**Update (2026-07-03, iteration 16) — clock (1) fired; the iteration ran; the hedge advanced; watch-mode resumes on a sharper state.** Watch-sweep #1's clock-(1) input (Marrakchi arXiv:2606.23636) sanctioned the first post-saturation analytical iteration (G1–G4, referees binding; *synthesis* (Chapter 36 (p. 684))). Outcome: verdict unchanged (**15th consecutive**), but **HYP-CKV-VACUITY advances R6 → R7** — the first hedge movement since iteration 9. The commutant-to-generated gap is now **classified (not closed)** by the referee-verified LEM-NET-NATURALITY-G (localized branch: dead, proof-grade; fully-natural non-localized branch: gauge gradings, Δ_geo = 0, proof-grade); the entire no-go now rests on {n1 AND **(E_O)**} where (E_O) = vacuum ergodicity on massive double-cone algebras [OPEN]. **The active watch state (supersedes sweep-#1's target, which iteration 16 showed ill-posed):** clock (1) = (α) new definitions/theorems for expectation-free relative-bicentralizer-type invariants; (β) bicentralizer triviality; (γ) expectation-free relative T-invariant computations (adjacent conjecture T(M(O) ⊂ M(W)) = {0}); **(δ) any resolution of (E_O) — the principal item**; (ε) the DCNG definability lever (needs an operator-algebra/continuous-logic collaborator). Clock (2) unchanged: 2027 DESI five-year w(z) (Euclid cosmology co-timed mid-2027). New analytical iterations remain gated on a genuinely new clock input.

**Update (2026-07-07, iteration 17) — the (E_O) assault ran; verdict unchanged 16th; hedge STAYS R7; watch-mode resumes.** The principal watch item (δ) = (E_O) received a dedicated five-lens assault (E1–E5, three binding

> referees; *synthesis* (Chapter 37 (p. 693))), and *watch-sweep #2* (Working note: *Watch-mode sweep #2 — external-input channels, window 2026-07-02 → 2026-07-07* (repo: `notes/2026-07-07-watch-sweep-02.md`)) confirmed zero in-window strong findings. Outcome: (E_O) neither proved nor disproved — **HYP-CKV-VACUITY stays at R7** (no R8, no retreat). New refereed content: (i) free-field (E_O) sharply **reduced** to a one-particle spectral question (purely continuous spectrum of ln δ_O / weak mixing ⟹ M(O)_ω = ℂ1; the naive "no-eigenvalue-0" reading refuted; the spectral fact numerical-only, leaning-true — Bostelmann–Cadamuro–Minz arXiv:2209.04681, Cadamuro arXiv:2312.08525, no closed form per Longo–Morsella arXiv:2012.00565); (ii) all three structure-theory routes for general (H) provably **fail**; (iii) the adversarial disproof **fails** on eight candidate mechanisms (no carrier; bare OPEN). The five-item clock-(1) watch state (α)–(ε) persists with **(δ) (E_O) principal**, now carrying the free-field spectral sub-target as the sharpest decidable lever (a proof of purely-continuous one-particle massive double-cone modular spectrum ⟹ (E_O) on the free subclass ⟹ R7 → R8 there). Clock (2) unchanged: 2027 DESI five-year w(z). New analytical iterations remain gated on a genuinely new clock input.

What to work on next, and why. Items are framed as *questions to resolve* or *artifacts to produce*, prioritized by leverage — how much a credible answer would move the central question of whether a universal theory is possible. Cross-references point to where the work lands.

> Selection principle: prefer questions that are (a) load-bearing for unification, (b) at least partly decidable by existing or near-term evidence or by rigorous math, and (c) currently under-served in this wiki. De-prioritize questions whose answers would change nothing downstream.

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## Tier 0 — Maintenance & integrity (every iteration)

- **Lint pass** per `AGENTS.md` (in the repository): contradictions across pages, stale numbers, orphan pages, missing links, overclaiming, ID integrity.
- **Bibliography verification**: replace every `[unverified]` in the Bibliography (p. 776) with a checked reference, or remove it.
- **Quantitative refresh**: re-check measured values and tensions (e.g. $H_0$, $S_8$, muon $g-2$ after lattice/experiment updates) and re-tag in *Chapter 16 (p. 308)*.

## Tier 1 — Highest leverage (drive the central question)

1. **Sharpen the GR↔QFT interface.** Make precise, in *Chapter 14 (p. 191)*, exactly which incompatibilities are *logical* vs *domain-of-validity*. Treat gravity as an effective field theory and document what is actually known to fail (and at what scale) vs assumed to fail.
2. **The black-hole information problem as a decisive test bed.** Track the status of unitarity, the Page curve, island/replica-wormhole results, and what they do and do not establish. This is the cleanest arena where quantum theory and gravity must be reconciled.
3. **The measurement problem, operationalized.** Catalogue what each resolution (decoherence + branching, objective collapse, hidden variables, epistemic/QBist) *costs* and which make distinguishable predictions (e.g. collapse-model bounds from interferometry). Promote anything testable from `[CONTESTED]` toward decidability.
4. **Cosmological-constant problem, stated without folklore.** A careful, regularization-aware account (see *Chapter 16 (p. 308)*) of what is genuinely puzzling vs an artifact of naive cutoff estimates.

## Tier 2 — Structural / mathematical

5. **Assumption demotion program.** For each `fundamental/plausibly-fundamental` entry in *Chapter 20 (p. 458)*, ask: is there a derivation that would demote it? Highest-value targets: smooth-continuum spacetime, exact linearity of QM, the Born rule, locality, 3+1 dimensions.
6. **Rigor gap in interacting QFT.** Document the constructive-QFT frontier and the Yang–Mills mass-gap problem; clarify what "the Standard Model exists mathematically" would even require.
7. **Information-theoretic reconstructions of QM.** Evaluate how far axiomatic reconstructions (Hardy; entropic; CBH) actually *explain* the Hilbert-space formalism vs re-encode it.

## Tier 3 — Candidate-framework development (gated by the verification protocol)

8. **Emergent-spacetime program.** Stress-test the entanglement→geometry claim (Ryu–Takayanagi, Jacobson's derivation, tensor networks): what would count as evidence *outside* AdS, and is there a falsifiable prediction? See *Chapter 19 (p. 392)*, *Chapter 18 (p. 349)*.
9. **Distinguishing predictions audit.** For every program in *Chapter 18 (p. 349)*, record the single most plausible near-term *distinguishing* observation. Programs with none are flagged accordingly.

## Tier 4 — Empirical anchors to monitor

- Gravitational-wave catalogues (strong-field GR, possible deviations).
- CMB / large-scale-structure surveys (dark sector, inflationary predictions, tensions).

- Precision tests: equivalence-principle, Lorentz-invariance, and constancy-of-constants bounds (each tightens an entry in *Chapter 20 (p. 458)*).
- Collider / flavor / neutrino results bearing on beyond-Standard-Model structure.
- Tabletop quantum-gravity proposals (gravitationally-induced entanglement) and collapse-model bounds.

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## Method for each iteration

1. Pick a small number of Tier-1/2 items.
2. Fan out reading; write structured findings; **verify adversarially** before promoting any claim (see `AGENTS.md` (in the repository) §4).
3. Update the affected pages in place; reconcile *Chapter 21 (p. 497)*; log in the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)).
4. Re-ask the central question: did anything move the case for/against a universal theory? Record the delta in *Chapter 21 (p. 497)*.

# Glossary

This page collects precise, cross-linked definitions for the mathematical objects, physical concepts, and named results used throughout this wiki. Entries are alphabetical. Each definition is meant to be *correct as stated* within its domain of validity; where a term carries genuine subtlety or dispute, an epistemic marker (`[ESTABLISHED]`, `[INFERENCE]`, `[SPECULATIVE]`, `[OPEN]`, `[CONTESTED]`) flags it (see *Chapter 2 (p. 8)* for the calibration scheme). For the map of how domains relate, see *Chapter 3 (p. 15)*; for the canonical-source pointers, see the Bibliography (p. 776).

## A

**Action** — The functional $S[\gamma] = \int L\,dt$ (or $\int \mathcal{L}\,d^4x$ in field theory) assigning a number to each history $\gamma$. Physical trajectories make $S$ *stationary*, not necessarily least `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

**Action–angle variables** — Canonical coordinates $(I, \phi)$ for an integrable system in which $\dot{I} = 0$ and $\dot{\phi} = \omega(I)$; motion is linear winding on invariant tori. The cleanest picture of integrable order, against which chaos is the failure `[ESTABLISHED]`.

**Adiabatic (cosmological) perturbations** — Primordial fluctuations in which all species share a common local time-shift, so number-density ratios are spatially constant. Observed to high accuracy and predicted by single-field inflation `[ESTABLISHED as description]`. See *Chapter 11 (p. 139)*.

**ADM formalism** — Arnowitt–Deser–Misner $3+1$ decomposition of the metric into spatial metric $h_{ij}$, lapse $N$, and shift $N^i$, recasting general relativity as constrained Hamiltonian evolution of geometry `[ESTABLISHED]`. Foundation of numerical relativity and canonical quantization attempts. See *Chapter 10 (p. 121)*.

**AdS/CFT correspondence** — Conjectured exact duality between a quantum-gravity theory in asymptotically anti-de Sitter spacetime and a conformal field theory on its boundary `[INFERENCE: overwhelming evidence, no general nonperturbative proof]`. See *Chapter 12 (p. 155)*, *Chapter 18 (p. 349)*.

**Algebraic QFT (Haag–Kastler)** — Formulation of QFT as a *net* of local operator algebras $\mathcal{O} \mapsto \mathfrak{A}(\mathcal{O})$ satisfying isotony, microcausality, and covariance, avoiding any privileged Hilbert-space representation `[ESTABLISHED structural framework]`. See *Chapter 8 (p. 89)*.

**Anomaly (quantum/chiral)** — A classical symmetry broken by quantization; e.g. the Adler–Bell–Jackiw anomaly $\partial_\mu j^{5\mu} = \frac{e^2}{16\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$. Physical for global currents ($\pi^0 \to 2\gamma$); must cancel for gauged currents `[ESTABLISHED]`. See *Chapter 9 (p. 106)*.

**Asymptotic freedom** — The decrease of a gauge coupling at high energy due to a negative beta function; in QCD $\beta_0 = 11 - \frac{2}{3}n_f > 0$ makes quarks weakly coupled at short distance `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Asymptotic safety** — Conjecture that gravity (or another theory) is UV-completed by a nontrivial fixed point of the renormalization-group flow `[SPECULATIVE/OPEN]`. See *Chapter 15 (p. 233)*.

**Atiyah–Singer index theorem** — Equates the analytic index of an elliptic operator (e.g. net chiral zero modes of the Dirac operator) with a topological integral, ind(\slashed$D$) $=$ $\int_M \hat{A}(M)\,\mathrm{ch}(F)$ `[ESTABLISHED]`. Underlies anomalies and instanton fermion counting. See *Chapter 13 (p. 172)*.

# B

**Baryon acoustic oscillations (BAO)** — Frozen-in sound-horizon scale in the matter distribution, a standard ruler probing cosmic geometry and expansion history `[ESTABLISHED]`. See *Chapter 11 (p. 139)*.

**Bekenstein bound** — Upper bound on the entropy in a weakly gravitating region of radius $R$ and energy $E$: $S \leq 2\pi k_B RE/\hbar c$ `[INFERENCE; sharp only in weak gravity]`. See *Chapter 12 (p. 155)*.

**Bekenstein–Hawking entropy** — Black-hole entropy proportional to horizon *area*: $S_{\text{BH}} = k_B c^3 A/(4G\hbar) = k_B A/4\ell_P^2$ [ESTABLISHED theory; microstate counting `[OPEN]` for generic black holes]. See *Chapter 5 (p. 40)*.

**Bell's theorem** — No local-hidden-variable theory satisfying measurement independence can reproduce all quantum correlations; the CHSH inequality $|S| \leq 2$ for LHV is violated up to the Tsirelson bound $2\sqrt{2}$ by quantum systems, confirmed loophole-free `[ESTABLISHED]`. The escape via denying measurement independence (superdeterminism) is `[OPEN/CONTESTED]`. See *Chapter 7 (p. 71)*.

**Beta function** — $\beta(g) = \mu\, dg/d\mu$, governing the renormalization-group running of a coupling; its sign and zeros determine asymptotic freedom, Landau poles, and fixed points `[ESTABLISHED]`. See *renormalization group*.

**Bianchi identity (contracted)** — The geometric identity $\nabla^\mu G_{\mu\nu} \equiv 0$, which forces local energy–momentum conservation $\nabla^\mu T_{\mu\nu} = 0$ and reduces the Einstein equations to 6 independent dynamical equations `[ESTABLISHED]`. See *Chapter 10 (p. 121)*.

**Black-hole information paradox** — Apparent conflict between Hawking's thermal (pure→mixed) evaporation and quantum unitarity. Recent island/replica-wormhole computations reproduce a unitary Page curve `[INFERENCE within their assumptions]`; a complete mechanism for realistic black holes is `[OPEN/CONTESTED]`. See *Chapter 14 (p. 191)*.

**Bohmian mechanics (de Broglie–Bohm)** — Interpretation restoring definite particle positions guided by the wavefunction; empirically equivalent to standard QM, paying with explicit nonlocality and contextuality

`[ESTABLISHED equivalence; CONTESTED as preferred]`. See *Chapter 7 (p. 71)*.

**Boltzmann entropy** — $S_B = k_B \ln W$, the logarithm of the number of microstates compatible with a macrostate; the surface-entropy definition. Contrast the Gibbs (volume) entropy; their inequivalence for small/bounded systems is `[CONTESTED]`. See *Chapter 6 (p. 56)*.

**Boltzmann equation** — Kinetic evolution equation for the one-particle distribution $f$ with a collision integral; valid in the dilute-gas (Grad) limit. Its $H$-theorem yields irreversibility only after the molecular-chaos assumption `[ESTABLISHED]`. See *Chapter 6 (p. 56)*.

**Born rule** — Outcome probability $\Pr(A \in \Delta) = \mathrm{Tr}(\rho\, E_A(\Delta))$. Gleason's theorem makes it the unique noncontextual probability measure on projections for dim $\geq 3$ `[ESTABLISHED as form]`; whether it is *derivable* from unitary dynamics is `[CONTESTED]`. See *Chapter 7 (p. 71)*, *Chapter 2 (p. 8)*.

**Bousso (covariant entropy) bound** — Holographic bound $S_{\text{light-sheet}} \leq A/4G\hbar$ on the entropy crossing a light-sheet of a surface of area $A$; the covariant, generally applicable form of holography `[INFERENCE]`. See *Chapter 12 (p. 155)*.

**BRST symmetry** — A global fermionic symmetry of the gauge-fixed action encoding gauge redundancy, used to prove unitarity and renormalizability of gauge theories `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

# C

**C*-algebra** — A complex Banach $*$-algebra satisfying the C*-identity $\|a^*a\| = \|a\|^2$. The Gelfand–Naimark theorem represents any C*-algebra as bounded operators on a Hilbert space `[ESTABLISHED]`. The natural setting for algebraic QM/QFT and superselection. See *Chapter 13 (p. 172)*.

**Canonical commutation relation (CCR)** — $[\hat{x}, \hat{p}] = i\hbar 1$. Has no bounded or finite-dimensional realization; its exponentiated (Weyl) form is essentially unique for finitely many degrees of freedom by Stone–von Neumann but admits inequivalent representations for infinitely many `[ESTABLISHED]`. See *Stone–von Neumann theorem*.

**Canonical ensemble** — Probability distribution $\rho = Z^{-1}e^{-\beta H}$ for a system in thermal contact with a reservoir at temperature $T = 1/k_B\beta$; $F = -k_BT \ln Z$ `[ESTABLISHED for short-range interactions]`. See *Chapter 6 (p. 56)*.

**Carnot bound** — Maximum efficiency of a heat engine between reservoirs, $\eta = 1 - T_c/T_h$, a direct consequence of the second law `[ESTABLISHED]`. See *Chapter 5 (p. 40)*.

**Chern number / characteristic class** — Topological invariant of a bundle; e.g. the second Chern number $\frac{1}{8\pi^2}\int \mathrm{tr}(F \wedge F) \in \mathbb{Z}$ classifies instanton sectors, and the first Chern number gives the quantized Hall conductance (TKNN) `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

**CHSH inequality** — See *Bell's theorem*. The four-correlator form $S = |E(a,b) - E(a,b') + E(a',b) + E(a',b')| \leq 2$ for LHV, with quantum maximum $2\sqrt{2}$.

**CKM matrix** — Unitary mixing matrix relating quark mass and weak-interaction eigenstates; its single irreducible phase is the sole source of Standard-Model quark CP violation (Jarlskog $J \approx 3 \times 10^{-5}$) `[ESTABLISHED]`. See *Chapter 9 (p. 106)*.

**Coarse-graining** — Partition of microstate space into macroscopically distinguishable cells; thermodynamic entropy and irreversibility are defined relative to such a partition. The *need* for it is structurally fundamental; the specific choice has a conventional element `[CONTESTED interpretation]`. See *Chapter 6 (p. 56)*.

**Coleman–Mandula theorem** — Under reasonable assumptions (nontrivial S-matrix, mass gap, finitely many particle types), the only Lie-algebra symmetries of the S-matrix that combine spacetime and internal symmetries are direct products of the Poincaré group with internal symmetries — forbidding nontrivial

**Conformal field theory (CFT)** — A QFT invariant under the conformal group; sits at renormalization-group fixed points and anchors the UV/IR endpoints of RG flows. Most rigorously controlled interacting QFTs (especially in 2D) `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Connection** — On a fiber bundle, a rule for parallel transport / covariant differentiation; for a principal $G$-bundle it is a $\mathfrak{g}$-valued one-form whose curvature is the gauge field strength $F = dA + A \wedge A$ `[ESTABLISHED]`. The geometric content of the gauge principle. See *Chapter 13 (p. 172)*.

**Constraint (Hamiltonian)** — A phase-space relation that must vanish (weakly) on physical states, e.g. the GR momentum and Hamiltonian constraints. First-class constraints generate gauge transformations; their quantization yields the Wheeler–DeWitt equation $H\Psi = 0$ `[ESTABLISHED]`. See *problem of time*.

**Contextuality (Kochen–Specker)** — No noncontextual assignment of definite values to all observables (for $\dim \geq 3$) reproduces quantum predictions; measured values can depend on which compatible observables are measured alongside `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Cosmological constant ($\Lambda$)** — The unique term Lovelock's theorem permits beyond the Einstein tensor; drives late-time accelerated expansion. Its observed value lies $\sim 10^{120}$ below naive QFT vacuum-energy estimates — the cosmological-constant problem `[OPEN]`. See *Chapter 14 (p. 191)*, *Chapter 11 (p. 139)*.

**Cosmological principle** — Spatial homogeneity and isotropy on large scales ($\gtrsim 100$ Mpc); the symmetry assumption underlying FLRW `[ESTABLISHED statistically; INFERENCE as exact]`. See *Chapter 20 (p. 458)*.

**CPT theorem** — Any local, Lorentz-invariant QFT with a Hermitian Hamiltonian is invariant under the combined operation of charge conjugation, parity, and time reversal `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Critical exponents / universality** — Power-law behavior of thermodynamic quantities near a continuous phase transition (e.g. $\xi \sim |t|^{-\nu}$), with exponents depending only on dimension and symmetry (universality classes), explained by RG fixed points `[ESTABLISHED]`. See *Chapter 6 (p. 56)*.

**Crooks fluctuation theorem** — Detailed symmetry $P_F(+W)/P_R(-W) = e^{\beta(W-\Delta F)}$ between forward and reverse driving protocols; quantifies the exponential rarity of transient second-law violations `[ESTABLISHED]`. See *Chapter 5 (p. 40)*.

# D

**Dark energy** — The component (constant or dynamical) driving cosmic acceleration; modeled as $\Lambda$ with equation of state $w = -1$. Acceleration is `[ESTABLISHED]`; whether dark energy is exactly $\Lambda$ is `[OPEN/CONTESTED]`. See *Chapter 11 (p. 139)*.

**Dark matter** — A non-baryonic, cold, gravitating component required by rotation curves, lensing, the CMB peaks, and structure formation. Its existence (beyond baryons + GR) is near-`[ESTABLISHED]`; its particle identity is `[OPEN]`. See *Chapter 11 (p. 139)*, *Chapter 9 (p. 106)*.

**Decoherence** — Suppression of interference in a subsystem through entanglement with an uncontrolled environment, selecting a quasi-classical pointer basis and yielding effective classicality. The dynamics is `[ESTABLISHED]`; its foundational significance for the measurement problem is `[CONTESTED]`. See *Chapter 7 (p. 71)*.

**Density matrix** — Positive, unit-trace operator $\rho$ encoding pure ( $\rho^2 = \rho$) or mixed states; the reduced state $\rho_A = \mathrm{Tr}_B\, \rho_{AB}$ captures all locally accessible statistics `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Diffeomorphism invariance** — Invariance under smooth coordinate changes; in GR physical states are equivalence classes under diffeomorphisms (background independence), making spacetime points have no physical identity apart from fields (hole argument) `[ESTABLISHED]`. See *Chapter 10 (p. 121)*, *Chapter 17 (p. 325)*.

**Dirac–Bergmann constraint analysis** — Hamiltonian treatment of singular (gauge) Lagrangians via first- and second-class constraints and Dirac brackets; the classical skeleton of gauge theory `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

**Domain of validity** — The regime (energies, scales, couplings, idealizations) in which a theory is empirically reliable. Distinguishing a true inconsistency from a domain-of-validity mismatch is central to this wiki's epistemics. See *Chapter 2 (p. 8)*, *Chapter 14 (p. 191)*.

# E

**Effective field theory (EFT)** — A theory valid below a cutoff $\Lambda$, organizing all symmetry-allowed operators by mass dimension into relevant/marginal/irrelevant, with $\mathcal{L}_{\text{eff}} = \sum_i c_i \Lambda^{4-d_i} \mathcal{O}_i$. Renormalizability becomes an emergent low-energy property `[ESTABLISHED]`. See *Chapter 8 (p. 89)*, *Chapter 17 (p. 325)*.

**Ehrenfest's theorem** — $\frac{d}{dt}\langle\hat{x}\rangle = \langle\hat{p}\rangle/m$, $\frac{d}{dt}\langle\hat{p}\rangle = -\langle\nabla V\rangle$; recovers Newtonian motion for narrow packets but fails for chaotic systems beyond the (logarithmic) Ehrenfest time `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Einstein field equations (EFE)** — $G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$; ten coupled quasilinear second-order PDEs relating curvature to stress-energy `[ESTABLISHED]`. See *Chapter 10 (p. 121)*.

**Eigenstate thermalization hypothesis (ETH)** — Ansatz that few-body observables in individual energy eigenstates of a generic non-integrable many-body Hamiltonian already take thermal values, explaining thermalization of isolated quantum systems [INFERENCE: strong numerical support, unproven; fails for integrable and many-body-localized systems]. See *Chapter 6 (p. 56)*.

**Energy conditions** — Pointwise positivity constraints on $T_{\mu\nu}$ (weak/null/strong/dominant) used in singularity and censorship theorems; violated by quantum fields and by a positive $\Lambda$, so the pointwise forms are now seen as too strong `[CONTESTED]`, replaced by averaged/quantum versions (ANEC, QNEC). See *Chapter 10 (p. 121)*.

**Entanglement** — Nonseparability of a composite quantum state; the source of Bell violations and a quantifiable resource (teleportation, dense coding). Pure-state bipartite entanglement is measured by the entanglement entropy $S(\rho_A)$ `[ESTABLISHED]`. See *Chapter 7 (p. 71), Chapter 12 (p. 155)*.

**Entanglement entropy** — Von Neumann entropy $S(\rho_A) = -\operatorname{Tr}\rho_A \ln \rho_A$ of a subsystem. In continuum QFT it is UV-divergent (area law); only universal/finite pieces are cutoff-independent, and local QFT algebras are Type III (no density matrices) `[ESTABLISHED]`. See *Chapter 12 (p. 155)*.

**Envariance** — Environment-assisted invariance; Zurek's symmetry argument attempting to derive the Born rule within a no-collapse (Everettian) framework `[CONTESTED]`. See *Chapter 7 (p. 71)*.

**Equivalence principle** — Universality of free fall (weak EP, tested to $\sim 10^{-15}$) and local Lorentz/position invariance (Einstein EP); the basis for geometrizing gravity `[ESTABLISHED]`. See *Chapter 10 (p. 121)*.

**ER=EPR** — Conjecture (Maldacena–Susskind) identifying entanglement (EPR) with geometric wormhole connections (Einstein–Rosen) `[SPECULATIVE]`. See *Chapter 18 (p. 349)*.

**Ergodicity** — Equality of long-time and phase-space averages on the energy surface. Generic Hamiltonian systems are *not* ergodic (KAM); statistical mechanics is now justified more by typicality/measure concentration than by strict ergodicity [ESTABLISHED that ergodicity is neither necessary nor sufficient as usually invoked]. See *Chapter 6 (p. 56)*.

**Euler–Lagrange equations** — $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = 0$; coordinate-invariant equations of motion from stationary action `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

**Everett (many-worlds) interpretation** — Retains only unitary evolution; measurement is branching into decoherent worlds. Pays with branching multiplicity and an unsettled account of probability `[ESTABLISHED equivalence to QM predictions; CONTESTED foundations]`. See *Chapter 7 (p. 71).*

## F

**Faddeev–Popov / Gribov ambiguity** — Gauge-fixing procedure introducing ghost fields; globally obstructed in non-abelian theories (Gribov copies; Singer's theorem shows no global gauge section exists), so it is only locally valid nonperturbatively `[ESTABLISHED]`. See *Chapter 13 (p. 172).*

**Fermi–Dirac / Bose–Einstein statistics** — Occupation numbers $\langle n_k \rangle = [e^{\beta(\epsilon_k - \mu)} \pm 1]^{-1}$ (+ fermions, – bosons), following from identical-particle (anti)symmetrization `[ESTABLISHED]`. See *Chapter 6 (p. 56)*, *spin–statistics theorem*.

**Feynman path integral** — Propagator as a sum over histories $\int \mathcal{D}[q]\, e^{iS/\hbar}$; the classical equations arise by stationary phase as $\hbar \to 0$. Rigorous via Euclidean (Wiener) measure in QM and some low-dimensional QFTs; the literal Lorentzian measure does not exist `[ESTABLISHED]`. See *Chapter 13 (p. 172).*

**Fiber bundle** — A space locally a product $U \times F$ of base and fiber but possibly globally twisted; principal $G$-bundles host gauge connections, and the bundle topology is physical (Dirac quantization, Aharonov–Bohm) `[ESTABLISHED]`. See *Chapter 13 (p. 172).*

**First law of entanglement** — Perturbative equality $\delta S_A = \delta\langle K_A \rangle$ (modular Hamiltonian $K_A$); in holographic setups it yields the *linearized* Einstein equations — a concrete "gravity from entanglement" result `[INFERENCE within AdS/CFT; SPECULATIVE as a route to full GR]`. See *Chapter 12 (p. 155).*

**Fluctuation–dissipation theorem** — Relates a system's linear response to its equilibrium fluctuation spectrum; Einstein, Nyquist, and Green–Kubo relations are special cases `[ESTABLISHED]`. See *Chapter 6 (p. 56).*

**Fluctuation theorems** — Exact relations (Jarzynski equality, Crooks theorem, Gallavotti–Cohen) constraining nonequilibrium work and entropy-production distributions, sharpening the second law into an equality over trajectory ensembles `[ESTABLISHED]`. See *Chapter 5 (p. 40)*.

**FLRW metric** — Friedmann–Lemaître–Robertson–Walker line element, the unique homogeneous-isotropic spacetime, with scale factor $a(t)$ and curvature $k$ `[ESTABLISHED]`. See *Chapter 11 (p. 139)*.

**Free energy** — Thermodynamic potentials obtained as Legendre transforms of internal energy: Helmholtz $F = U - TS$, Gibbs $G = U - TS + pV$, grand $\Omega = U - TS - \mu N$, each minimized at equilibrium under its control variables `[ESTABLISHED]`. See *Chapter 5 (p. 40)*.

**Friedmann equations** — Dynamical equations for $a(t)$ from the EFE under FLRW symmetry: $H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3}$ and the acceleration equation $\ddot{a}/a = -\frac{4\pi G}{3}(\rho + 3p) + \Lambda/3$ `[ESTABLISHED]`. See *Chapter 11 (p. 139)*.

## G

**Gauge invariance / gauge principle** — Invariance under local transformations of internal (or spacetime) symmetry; interactions follow from the covariant derivative $D_\mu = \partial_\mu - igA_\mu^a T^a$. Gauge symmetry is a *redundancy* of description, not a physical symmetry (Elitzur's theorem forbids its spontaneous breaking) `[ESTABLISHED]`; framing it as a fundamental "principle" is `[CONTESTED]`. See *Chapter 8 (p. 89)*, *Chapter 17 (p. 325)*.

**Generalized probabilistic theories (GPT)** — Operational framework reconstructing quantum theory from information-theoretic axioms (local tomography, purification, continuous reversibility); illuminating but with multiple inequivalent axiom sets `[OPEN]`. See *Chapter 7 (p. 71)*.

**Generalized second law (GSL)** — Conjecture that ordinary entropy plus black-hole horizon entropy never decreases `[INFERENCE, strong support]`. See *Chapter 5 (p. 40)*.

**Geodesic** — Extremal-proper-time worldline of a free-falling body: $\ddot{x}^\mu + \Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$; "gravity = inertial motion in curved spacetime" `[ESTABLISHED]`. See *Chapter 10 (p. 121)*.

**Geodesic deviation** — Relative acceleration of nearby geodesics governed by curvature: $D^2\xi^\mu/d\tau^2 = -R^\mu{}_{\alpha\nu\beta}u^\alpha\xi^\nu u^\beta$; the operationally measurable tidal signature of gravity `[ESTABLISHED]`. See *Chapter 10 (p. 121)*.

**Gibbs entropy** — $S_G = -k_B \int \rho \ln \rho \, d\Gamma$ (classical) or $-k_B \, \mathrm{Tr}(\rho \ln \rho)$ (von Neumann, quantum); the volume-entropy definition. Constant under exact dynamics; coarse-graining is required for increase `[ESTABLISHED]`. See *Chapter 6 (p. 56)*.

**Gleason's theorem** — For dim $\geq 3$, the only probability measures on the projection lattice of a Hilbert space are of the form $\mathrm{Tr}(\rho\,\cdot)$, fixing the Born rule's form given noncontextuality `[ESTABLISHED]`. See *Born rule*.

**Goldstone bosons** — Massless (or pseudo-) excitations accompanying spontaneous breaking of a continuous global symmetry (Goldstone's theorem); when the symmetry is gauged they are "eaten" via the Higgs mechanism `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Grand unified theory (GUT)** — Embedding of $SU(3) \times SU(2) \times U(1)$ into a simple group (SU(5), SO(10), Pati–Salam), explaining charge quantization and predicting proton decay `[INFERENCE/SPECULATIVE; minimal SU(5) excluded by proton-decay bounds]`. See *Chapter 18 (p. 349)*.

**Groenewold–van Hove theorem** — No quantization map exists from the full classical Poisson algebra to operators respecting $\{\cdot,\cdot\} \mapsto [\cdot,\cdot]/i\hbar$ on all observables; operator ordering is a genuine ambiguity `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

# H

**Haag's theorem** — In QFT the interaction picture does not exist: the free and interacting field representations are unitarily inequivalent, so the naive perturbative starting point is mathematically inconsistent (yet renormalized perturbation theory works) `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Hamilton–Jacobi equation** — $H(q, \partial S/\partial q, t) + \partial S/\partial t = 0$; a single first-order PDE whose complete integral solves the dynamics, and the classical limit of the Schrödinger equation via $S = -i\hbar \ln \psi$ `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

**Hamiltonian mechanics** — Phase-space dynamics $\dot{q}^i = \partial H/\partial p_i$, $\dot{p}_i = -\partial H/\partial q^i$ on a symplectic manifold `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

**Hawking radiation** — Thermal emission from a black-hole horizon at temperature $T_H = \hbar c^3/(8\pi G k_B M)$, derived from QFT on a curved background `[INFERENCE: not directly observed, but theoretically robust]`. See *Chapter 10 (p. 121)*.

**Higgs (Brout–Englert–Higgs) mechanism** — Spontaneous breaking of a gauge symmetry by a scalar vacuum expectation value, giving mass to gauge bosons ($M_W = \frac{1}{2}gv$, $v \approx 246$ GeV) without breaking gauge invariance explicitly `[ESTABLISHED]`. See *Chapter 9 (p. 106)*.

**Hilbert space** — Complete complex inner-product space; states of a quantum system are rays (unit vectors up to phase), and observables are self-adjoint operators on it `[ESTABLISHED]`. See *Chapter 7 (p. 71), Chapter 13 (p. 172)*.

**Holographic principle** — Conjecture that the degrees of freedom in a region are bounded by its boundary area in Planck units, so a gravitational theory is encoded on a lower-dimensional boundary `[INFERENCE; realized concretely in AdS/CFT]`. See *Chapter 12 (p. 155), Chapter 17 (p. 325)*.

**Hubble tension** — $\sim 4$–$6\sigma$ discrepancy between the early-universe ($H_0 \approx 67$–$68$) and local distance-ladder ($H_0 \approx 73$ km/s/Mpc) determinations of the expansion rate `[CONTESTED: systematics vs new physics]`. See *Chapter 11 (p. 139), Chapter 14 (p. 191)*.

# I

**Identical particles** — Indistinguishable quanta whose multiparticle states must be symmetric (bosons) or antisymmetric (fermions); in 2D the exchange structure is richer (anyons, braid group) `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Inflation** — A hypothesized epoch of slow-roll accelerated expansion solving the horizon, flatness, and monopole problems and seeding near-scale-invariant fluctuations [INFERENCE bordering SPECULATIVE; no primordial B-mode detection, inflaton unidentified]. See *Chapter 11 (p. 139)*.

**Instanton** — A finite-action solution of the Euclidean field equations representing tunneling between topologically distinct gauge vacua, labeled by an integer winding (second Chern) number `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

**Integrability (Liouville–Arnold)** — A Hamiltonian system with as many independent Poisson-commuting conserved quantities as degrees of freedom; motion confines to invariant tori `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

# J

**Jacobson's derivation** — Demonstration that the Einstein equations follow as an equation of state from horizon thermodynamics ($\delta Q = T\delta S$ on local Rindler horizons) `[SPECULATIVE/INFERENCE: suggestive of emergent gravity]`. See *Chapter 17 (p. 325)*.

**Jarzynski equality** — Exact nonequilibrium identity $\langle e^{-\beta W}\rangle = e^{-\beta \Delta F}$, valid arbitrarily far from equilibrium, implying $\langle W\rangle \geq \Delta F$ `[ESTABLISHED]`. See *Chapter 5 (p. 40)*.

# K

**KAM theorem** — Kolmogorov–Arnold–Moser: most invariant tori of an integrable system survive small perturbations, provided their frequencies satisfy a Diophantine (non-resonance) condition; resonances seed chaos `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

**KMS condition** — Kubo–Martin–Schwinger characterization of thermal equilibrium states via analyticity of correlation functions in imaginary time; via Tomita–Takesaki it identifies "temperature" and modular "time" with a common algebraic origin `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

**Kochen–Specker theorem** — See *contextuality*.

**Kolmogorov complexity** — Length $K(x)$ of the shortest program generating a string $x$; machine-independent up to an additive constant but provably uncomputable `[ESTABLISHED]`. See *Chapter 12 (p. 155)*.

**Kraus (operator-sum) representation** — General quantum channel $\rho \mapsto \sum_k K_k \rho K_k^\dagger$ with $\sum_k K_k^\dagger K_k = 1$; via Stinespring/Naimark every channel and POVM is a unitary/projective measurement on a dilated space `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

# L

**Lagrangian** — Function $L = T - V$ (or density $\mathcal{L}$) whose action governs the dynamics; the specific $T - V$ form is a low-energy shadow of deeper (relativistic/path-integral) structure `[ESTABLISHED, INFERENCE as to origin]`. See *Chapter 4 (p. 25)*.

**Landauer's principle** — Erasing one bit of information dissipates at least $k_B T \ln 2$ of heat, linking logical irreversibility to thermodynamic cost; experimentally confirmed near the bound `[ESTABLISHED; its independent status vs the second law is CONTESTED]`. See *Chapter 12 (p. 155)*.

**Landau pole** — Energy at which a coupling (e.g. in QED or $\phi^4$) formally diverges, signaling that the theory is at most an EFT with a finite UV cutoff (triviality) `[INFERENCE, strong]`. See *Chapter 8 (p. 89)*.

**ΛCDM** — The concordance cosmological model: GR with a cosmological constant and cold dark matter, specified by six base parameters, fitting the CMB, BBN, BAO, and supernovae [ESTABLISHED as a fit; constituents `[OPEN]`]. See *Chapter 11 (p. 139)*.

**Legendre transform** — Map exchanging a variable for its conjugate (e.g. $L \to H, U \to F$); underlies the Hamiltonian formalism and thermodynamic potentials `[ESTABLISHED]`. See *Chapter 4 (p. 25)*, *Chapter 5 (p. 40)*.

**Levi-Civita connection** — The unique torsion-free, metric-compatible connection on a (pseudo-)Riemannian manifold, with Christoffel symbols $\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$ [ESTABLISHED; the torsion-free choice is conventional, cf. Einstein–Cartan]. See *Chapter 10 (p. 121)*.

**Lindblad (GKSL) equation** — Most general generator of completely-positive, trace-preserving Markovian dynamics: $\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \sum_k (L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\})$; governs open-system evolution and decoherence `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Liouville's theorem** — Hamiltonian phase-space flow preserves volume ($d\rho/dt = 0$), so fine-grained entropy is conserved — the technical reason coarse-graining is needed for irreversibility `[ESTABLISHED]`. See *Chapter 6 (p. 56)*.

**Lovelock's theorem** — In 4D, the only divergence-free symmetric 2-tensor built from the metric and its first two derivatives (with second-order field equations) is $aG_{\mu\nu} + bg_{\mu\nu}$, making the Einstein equations plus Λ essentially unique `[ESTABLISHED]`. See *Chapter 10 (p. 121)*.

**LSZ reduction** — Lehmann–Symanzik–Zimmermann formula relating S-matrix elements to amputated correlation functions, justified for isolated mass shells with a mass gap; fails for confined or massless asymptotic states `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Lyapunov exponent** — Rate $\lambda$ of exponential divergence of nearby trajectories, $|\delta x(t)| \sim |\delta x(0)|e^{\lambda t}$; $\lambda > 0$ is the signature of deterministic chaos `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

# M

**Majorana mass / neutrino** — A fermion identical to its own antiparticle; Majorana neutrino masses (via the seesaw or Weinberg operator) would violate lepton number, testable by neutrinoless double-beta decay. Dirac vs Majorana nature of neutrinos is `[OPEN]`. See *Chapter 9 (p. 106)*.

**Mandelstam–Tamm bound** — Time–energy inequality bounding the rate of state evolution by the energy spread, reinterpreting $\Delta E\, \Delta t \gtrsim \hbar$ (which is *not* a Robertson relation, since there is no time operator) `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Margolus–Levitin theorem** — Quantum speed limit: a state of mean energy $E$ above the ground state needs at least $t_\perp \geq \pi\hbar/2E$ to reach an orthogonal state `[ESTABLISHED]`. See *Chapter 12 (p. 155)*.

**Maxwell relations** — Equalities among partial derivatives of thermodynamic potentials (e.g. $(\partial S/\partial V)_T = (\partial p/\partial T)_V$) following from exactness of the differentials `[ESTABLISHED]`. See *Chapter 5 (p. 40)*.

**Maxwell's demon** — Thought experiment apparently violating the second law by sorting molecules; resolved by the entropy cost of information acquisition and especially erasure (Szilard, Landauer, Bennett) `[ESTABLISHED]`. See *Chapter 5 (p. 40)*.

**Measurement problem** — The structural incompatibility between linear unitary evolution (which never selects a single outcome) and the stochastic projection postulate, with no rule specifying what triggers collapse ("the shifty split") `[OPEN/CONTESTED]`. The deepest internal problem of QM. See *Chapter 7 (p. 71), Chapter 14 (p. 191)*.

**Microcausality** — Vanishing of commutators of local observables at spacelike separation, $[\phi(x), \phi(y)] = 0$ for $(x-y)^2 < 0$; the QFT encoding of relativistic causality and no-signaling `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Microcanonical ensemble** — Uniform (Liouville) measure on the energy shell; $S_B = k_B \ln \Omega(E)$. The equal-a-priori-probability postulate bridges micro to macro `[INFERENCE/OPEN as to first-principles derivation]`. See *Chapter 6 (p. 56)*.

**Modular Hamiltonian** — Generator $K_A = -\ln \rho_A$ of the Tomita–Takesaki modular flow of a region's state; central to relative entropy, energy conditions, and the entanglement first law `[ESTABLISHED]`. See *Chapter 12 (p. 155)*.

**Molecular chaos (Stosszahlansatz)** — Assumption that pre-collision velocities are uncorrelated; the precise time-asymmetric input that lets the Boltzmann $H$-theorem produce irreversibility `[ESTABLISHED as the locus of time-asymmetry]`. See *Chapter 6 (p. 56)*.

**Moyal star product** — Noncommutative product $f \star g = fg + \frac{i\hbar}{2}\{f, g\} + O(\hbar^2)$ deforming the Poisson algebra; the phase-space (deformation-quantization) route from classical to quantum mechanics `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

**Mutual information** — $I(X;Y) = H(X) + H(Y) - H(X,Y) = D(p(x,y) \| p(x)p(y)) \geq 0$; the shared information between two systems, finite even when individual entropies diverge `[ESTABLISHED]`. See *Chapter 12 (p. 155)*.

# N

**Negative temperature** — For systems with a bounded energy spectrum, population inversion gives $dS/dU < 0$, hence $T < 0$, which is "hotter" than any positive $T$; experimentally realized [ESTABLISHED; admissibility tied to the CONTESTED Boltzmann-vs-Gibbs entropy debate]. See *Chapter 5 (p. 40)*.

**No-cloning theorem** — Linearity of quantum mechanics forbids a universal device that copies an unknown state; underlies quantum cryptography and sharpens the information paradox `[ESTABLISHED]`. See *Chapter 12 (p. 155)*.

**No-signaling theorem** — Local marginal statistics are independent of a distant party's measurement choice, so entanglement cannot transmit usable information faster than light `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Noether's theorem** — Every continuous symmetry of an action yields a conserved current/charge; in Hamiltonian form, $\dot{J} = 0 \iff \{J, H\} = 0$ `[ESTABLISHED]`. See *Chapter 4 (p. 25), Chapter 17 (p. 325)*.

# O

**Objective-collapse models (GRW/CSL, Diósi–Penrose)** — Modifications of QM adding stochastic nonlinear terms that spontaneously localize macroscopic superpositions, predicting tiny, *testable* deviations from unitary QM `[OPEN, empirically constrained]`. See *Chapter 7 (p. 71), Chapter 19 (p. 392)*.

**Osterwalder–Schrader reconstruction** — Theorem relating Euclidean (reflection-positive) correlation functions to a Lorentzian Wightman QFT; underlies rigorous and lattice work but requires reflection positivity (which can fail at finite density — the sign problem) `[ESTABLISHED where conditions hold]`. See *Chapter 8 (p. 89)*.

# P

**Page curve** — The entanglement entropy of Hawking radiation as a function of time, rising then falling for a unitarily evaporating black hole; recently reproduced via quantum-extremal-surface/island computations `[INFERENCE within assumptions]`. See *Chapter 12 (p. 155)*.

**Partition function** — Generating functional $Z = \mathrm{Tr}\, e^{-\beta \hat{H}}$ (statistical mechanics) or $\int \mathcal{D}\phi\, e^{-S_E}$ (Euclidean QFT); all equilibrium thermodynamics follows by differentiating $\ln Z$, and the Euclidean path integral *is* a partition function (Wick-rotation isomorphism) `[ESTABLISHED]`. See *Chapter 6 (p. 56)*, *Chapter 8 (p. 89)*.

**Past hypothesis** — The posit of an extraordinarily low-entropy initial cosmological state, required to ground the thermodynamic arrow of time given time-symmetric microdynamics `[INFERENCE that it is necessary; OPEN why it holds]`. See *Chapter 6 (p. 56)*, *Chapter 11 (p. 139)*.

**Pauli's theorem (no time operator)** — No self-adjoint operator is canonically conjugate to a Hamiltonian bounded below; time is treated as a c-number parameter, not an observable `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**PBR theorem** — Pusey–Barrett–Rudolph: under the assumption that independently prepared systems have independent physical states, the quantum state cannot be a mere probability distribution over deeper "ontic" states (a $\psi$-epistemic model) — the wavefunction is "real" `[INFERENCE, conditional on the preparation-independence assumption]`. See *Chapter 7 (p. 71)*.

**Penrose–Hawking singularity theorems** — Under energy conditions and a trapped surface (or expansion), spacetime is geodesically incomplete, proving singularities are generic in gravitational collapse and cosmology `[ESTABLISHED]`. See *Chapter 10 (p. 121)*.

**Phase transition** — Non-analyticity of a thermodynamic potential, occurring only in the thermodynamic limit (Lee–Yang); classified by order and universality class `[ESTABLISHED]`. See *Chapter 6 (p. 56)*.

**PMNS matrix** — Pontecorvo–Maki–Nakagawa–Sakata lepton mixing matrix governing neutrino oscillations; its large mixing angles and undetermined CP phase contrast with the nearly

diagonal CKM matrix
`[ESTABLISHED that mixing is large; CP phase OPEN]`. See *Chapter 9 (p. 106)*.

**Poincaré group** — The isometry group of Minkowski spacetime (Lorentz transformations + translations); its unitary irreducible representations classify particles by mass and spin (Wigner) via the Casimirs $P^2 = m^2$, $W^2 = -m^2 s(s+1)$ `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Poisson bracket** — Lie-algebraic structure $\{f, g\} = \sum_i (\partial_{q^i} f \, \partial_{p_i} g - \partial_{p_i} f \, \partial_{q^i} g)$ on phase space; evolution is $df/dt = \{f, H\} + \partial_t f$. Survives (deformed) into the quantum commutator `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

**POVM** — Positive operator-valued measure $\{E_i \geq 0, \sum_i E_i = 1\}$; the most general quantum measurement, generalizing projective (von Neumann) measurements `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Problem of time** — In canonical gravity the Hamiltonian is a sum of constraints, so $H\Psi = 0$ (Wheeler–DeWitt) describes a "timeless" quantum universe; reconciling this with observed time evolution is unresolved `[OPEN/CONTESTED]`. See *Chapter 10 (p. 121)*, *Chapter 14 (p. 191)*.

# Q

**QBism** — Quantum Bayesianism: reads the wavefunction and Born rule as an agent's subjective degrees of belief, dissolving the measurement problem at the cost of a non-objective $\psi$ `[CONTESTED interpretation]`. See *Chapter 7 (p. 71)*.

**Quantum error correction (QEC)** — Encoding logical information redundantly across physical degrees of freedom to protect against noise; in AdS/CFT it models how bulk operators are reconstructed from boundary subregions (entanglement-wedge reconstruction)
`[ESTABLISHED as code theory; INFERENCE as a model of holography]`. See *Chapter 12 (p. 155)*.

# R

**Reduced density matrix** — See *density matrix*.

**Reeh–Schlieder theorem** — The vacuum is cyclic and separating for every local algebra: acting with operators in an arbitrarily small region densely spans the whole Hilbert space, exhibiting ubiquitous vacuum entanglement (no contradiction with no-signaling) `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Relational quantum mechanics** — Rovelli's interpretation in which physical facts are relative to the systems interacting, with no observer-independent absolute state `[CONTESTED]`. See *Chapter 7 (p. 71)*.

**Renormalization** — Procedure absorbing cutoff dependence into a finite set of measured parameters; modern Wilsonian view treats it as RG flow of an effective theory rather than infinity-subtraction `[ESTABLISHED]`. See *renormalization group*, *Chapter 8 (p. 89)*.

**Renormalization group (RG)** — The flow of couplings with energy scale, $\beta(g) = \mu\, dg/d\mu$; fixed points define scale-invariant (conformal) theories, and irrelevant operators are suppressed at low energy, explaining universality and emergent renormalizability `[ESTABLISHED]`. See *Chapter 6 (p. 56)*, *Chapter 17 (p. 325)*.

**Ricci / Riemann curvature** — $R^\rho{}_{\sigma\mu\nu}$ is the tensorial measure of gravity ($[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma$); its contraction (Ricci) sources the Einstein tensor `[ESTABLISHED]`. See *Chapter 10 (p. 121)*.

**Rigged Hilbert space** — Gelfand triple $\Phi \subset \mathcal{H} \subset \Phi'$ making rigorous the Dirac formalism of continuous-spectrum "eigenstates" (e.g. position, momentum) that lie outside $\mathcal{H}$ `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

**Robertson uncertainty relation** — $\Delta A\, \Delta B \geq \frac{1}{2}|\langle[A, B]\rangle|$, a statement about ensemble spreads in a fixed state (not measurement disturbance), specializing to $\Delta x\, \Delta p \geq \hbar/2$ `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Ryu–Takayanagi formula** — In holography, the entanglement entropy of a boundary region equals the area of the minimal homologous bulk surface, $S(A) = \mathrm{Area}(\gamma_A)/4G_N$; covariant (HRT) and quantum-corrected (QES) versions extend it `[INFERENCE within AdS/CFT]`. See *Chapter 12 (p. 155)*.

# S

**Sakharov conditions** — The three requirements for dynamically generating a baryon asymmetry: baryon-number violation, C and CP violation, and departure from thermal equilibrium `[ESTABLISHED as necessary conditions]`. See *Chapter 9 (p. 106)*.

**Schmidt decomposition** — Canonical form $|\psi\rangle_{AB} = \sum_i \sqrt{p_i}\,|i\rangle_A|i\rangle_B$ of a bipartite pure state, with the $p_i$ the eigenvalues of either reduced density matrix `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Schrödinger equation** — $i\hbar\,\partial_t|\psi\rangle = H|\psi\rangle$; deterministic, linear, unitary evolution of a closed quantum system, with $U(t) = e^{-iHt/\hbar}$ `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Second law of thermodynamics** — Entropy of an isolated system does not decrease, $\Delta S \geq 0$ (Clausius $dS \geq \delta Q/T$); phenomenologically absolute, but statistically a probabilistic statement refined by fluctuation theorems `[ESTABLISHED at macroscale; refined by statistical mechanics]`. See *Chapter 5 (p. 40)*.

**Self-adjoint operator** — A symmetric operator equal to its adjoint *including domains*; required (not mere symmetry) for a real spectrum, a spectral measure, and unitary dynamics (Stone). Inequivalent self-adjoint extensions encode different physics (boundary conditions) `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

**Shannon entropy** — $H(X) = -\sum_x p(x)\log_2 p(x)$, the unique measure of average uncertainty (up to base), equal to the optimal lossless compression rate `[ESTABLISHED]`. See *Chapter 12 (p. 155)*.

**Spectral theorem** — Every self-adjoint operator decomposes as $A = \int_{\sigma(A)} \lambda \, dE_A(\lambda)$ against a projection-valued measure on its spectrum; underwrites both the value set and the Born rule `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

**Spin–statistics theorem** — In relativistic QFT, integer-spin fields are bosonic (commuting) and half-integer-spin fields are fermionic (anticommuting), forced by microcausality, positive energy, and Lorentz invariance; an *added postulate* in non-relativistic QM `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Spontaneous symmetry breaking (SSB)** — A symmetric dynamics with a non-symmetric ground state; the symmetry is realized nonlinearly (Goldstone modes) rather than on the vacuum. Requires infinitely many degrees of freedom (inequivalent vacua) `[ESTABLISHED]`. See *Chapter 8 (p. 89)*, *Chapter 17 (p. 325)*.

**Standard Model** — The renormalizable, anomaly-free chiral gauge theory with group $SU(3)_c \times SU(2)_L \times U(1)_Y$ spontaneously broken to $SU(3)_c \times U(1)_{\text{em}}$; the most precisely tested theory in science `[ESTABLISHED, with confirmed incompleteness via neutrino mass]`. See *Chapter 9 (p. 106)*.

**Stinespring dilation** — Theorem that every completely-positive map is a compression of a $*$-homomorphism (unitary) on a larger space; the structural basis for Kraus/POVM purification `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

**Stone's theorem** — A strongly continuous one-parameter unitary group has a unique self-adjoint generator, tying unitary dynamics to self-adjointness of the Hamiltonian `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

**Stone–von Neumann theorem** — For finitely many canonical pairs, all irreducible representations of the (Weyl form of the) CCR are unitarily equivalent; uniqueness *fails* for infinitely many degrees of freedom, the technical root of inequivalent vacua, Haag's theorem, and SSB `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

**Strong CP problem** — The unexplained smallness ($\bar{\theta} \lesssim 10^{-10}$) of the CP-violating QCD theta angle, naturally $O(1)$; the Peccei–Quinn axion is the leading proposed resolution `[OPEN; axion SPECULATIVE]`. See *Chapter 9 (p. 106)*.

**Strong subadditivity** — The deep entropy inequality $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$, equivalent to nonnegativity of conditional mutual information; source of most holographic entropy constraints `[ESTABLISHED]`. See *Chapter 12 (p. 155)*.

**Superselection sector** — A subspace between which coherent superpositions are unobservable (e.g. different charge or, for infinite systems, distinct phases), arising from inequivalent representations `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Supersymmetry (SUSY)** — Proposed graded symmetry relating bosons and fermions (evading Coleman–Mandula via the Haag–Łopuszański–Sohnius extension), addressing the hierarchy problem and improving gauge unification `[SPECULATIVE; no superpartners observed]`. See *Chapter 18 (p. 349)*.

**Symplectic form** — Closed, nondegenerate two-form $\omega = dq^i \wedge dp_i$ on phase space defining Hamiltonian vector fields via $\iota_{X_H}\omega = dH$; the rigid geometric core of classical mechanics (Darboux's theorem) `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

# T

**Tensor product** — The composition rule $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$ for quantum subsystems, encoding local tomography and entanglement; singled out among operationally reasonable rules in GPT reconstructions `[ESTABLISHED operationally; INFERENCE as necessity]`. See *Chapter 7 (p. 71)*.

**Thermodynamic limit** — The idealization $N, V \to \infty$ at fixed density, in which extensivity holds, ensembles are equivalent, and genuine phase transitions exist

**Third law of thermodynamics** — Entropy approaches a constant (conventionally zero) as $T \to 0$ for a non-degenerate ground state, and absolute zero is unattainable; a quantum consequence of discrete spectra `[ESTABLISHED]`. See *Chapter 5 (p. 40)*.

**Tomita–Takesaki modular theory** — For a cyclic separating state, the operator $S = J\Delta^{1/2}$ generates a canonical modular automorphism flow $\sigma_t(a) = \Delta^{it} a \Delta^{-it}$ and identifies the commutant ($JMJ = M'$); thermal (KMS) states are modular states `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

**Topological quantum field theory (TQFT)** — A QFT whose observables depend only on topology; formalized as a symmetric monoidal functor $Z : \mathrm{Bord}_n \to \mathcal{C}^{\otimes}$ (Atiyah–Segal), with the cobordism hypothesis classifying fully extended TQFTs `[ESTABLISHED for topological theories]`. See *Chapter 13 (p. 172)*.

**Triviality** — The conjecture/theorem that the continuum limit of a theory (e.g. $\phi^4$ in 4D, proven by Aizenman–Duminil-Copin) is the free theory, so it survives only as a cutoff EFT [ESTABLISHED for $\phi^4_4$; INFERENCE for QED]. See *Chapter 8 (p. 89)*.

**Tsirelson bound** — The quantum maximum $2\sqrt{2}$ of the CHSH expression, exceeding the classical 2 but below the no-signaling (PR-box) maximum 4 `[ESTABLISHED]`. See *Chapter 7 (p. 71)*.

# U

**Unitarity** — Conservation of total probability under time evolution / the S-matrix; non-negotiable within quantum mechanics, enforced order-by-order via Ward/BRST identities and the optical theorem `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Universality** — See *critical exponents / universality*.

**Unruh effect** — A uniformly accelerated observer perceives the Minkowski vacuum as a thermal bath at temperature $T = \hbar a / 2\pi k_B c$ ; demonstrates the observer-dependence of "particle" in QFT

`[INFERENCE: theoretically robust, not directly observed]`. See *Chapter 8 (p. 89)*.

# V

**Vacuum (QFT)** — The lowest-energy state; for interacting theories it is not the free Fock vacuum (Haag's theorem), is entangled across regions (Reeh–Schlieder), and carries energy whose gravitational weight is the cosmological-constant problem `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Von Neumann entropy** — $S(\rho) = -\operatorname{Tr}(\rho \ln \rho)$; the quantum analogue of Shannon entropy, zero for pure states and maximal for the maximally mixed state, and the entanglement entropy of a bipartite pure state `[ESTABLISHED]`. See *Chapter 12 (p. 155)*.

**Von Neumann algebra** — A $*$-subalgebra of bounded operators closed in the weak operator topology; classified into types I, II, III. Local QFT algebras are Type $\mathrm{III}_1$ (no trace, no local density matrix) `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

# W

**Ward identities** — Relations among correlation functions enforcing a (gauge) symmetry at the quantum level; ensure renormalizability and unitarity `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**Weinberg operator** — The unique dimension-5 Standard-Model EFT operator, $\frac{c_{ij}}{\Lambda}(\bar{L}^c_{Li}\tilde{\phi}^*)(\tilde{\phi}^\dagger L_{Lj})$, giving Majorana neutrino masses $m_\nu \sim v^2/\Lambda$ — the leading window to physics beyond the SM `[ESTABLISHED as the leading operator]`. See *Chapter 9 (p. 106)*.

**Weinberg–Witten theorem** — Forbids, under broad assumptions, massless particles of spin $> 1$ carrying a Lorentz-covariant conserved current, and spin $> 2$ carrying a covariant stress-energy tensor — constraining how gravitons or composite gauge bosons can be *emergent* (the graviton must evade it, e.g. via the absence of a covariant boundary stress tensor) `[ESTABLISHED]`. See *Chapter 18 (p. 349)*.

**Wheeler–DeWitt equation** — The quantum Hamiltonian constraint $H\Psi = 0$ of canonical gravity, formally timeless; see *problem of time* `[ESTABLISHED as the constraint; interpretation OPEN]`. See *Chapter 10 (p. 121)*.

**Wick rotation** — Analytic continuation $t \to -i\tau$ relating Lorentzian QFT to Euclidean statistical field theory; rigorous under reflection positivity (Osterwalder–Schrader) but problematic in generic curved/Lorentzian backgrounds `[ESTABLISHED where conditions hold]`. See *Chapter 13 (p. 172)*.

**Wightman axioms** — Axiomatic characterization of a relativistic QFT in terms of operator-valued distributions, Poincaré covariance, the spectrum condition, microcausality, and a cyclic vacuum `[ESTABLISHED framework; 4D interacting realizations OPEN]`. See *Chapter 8 (p. 89)*.

**Wigner's theorem** — Every symmetry preserving transition probabilities is implemented by a unitary or antiunitary operator (antiunitary for time reversal) `[ESTABLISHED]`. See *Chapter 13 (p. 172)*.

**Wilson loop** — Gauge-invariant trace of the holonomy of a connection around a closed loop; an order parameter for confinement and the natural gauge-invariant observable `[ESTABLISHED]`. See *Chapter 8 (p. 89)*.

**WKB approximation** — Semiclassical expansion in $\hbar$ with phase $e^{iS/\hbar}$; recovers classical mechanics by stationary phase and captures tunneling ($\sim e^{-S/\hbar}$), but the $\hbar \to 0$ limit is singular and non-uniform `[ESTABLISHED]`. See *Chapter 4 (p. 25)*.

# Y

**Yang–Mills theory** — Non-abelian gauge theory with field strength $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu$ and self-interacting gauge bosons; the basis of QCD and the electroweak sector. Rigorous 4D existence with a mass gap is a Clay Millennium Problem `[OPEN]`. See *Chapter 8 (p. 89)*, *Chapter 15 (p. 233)*.

## References

For canonical textbooks and landmark papers underlying these definitions, see the Bibliography (p. 776).

BACK MATTER

# Bibliography

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This page is the consolidated source list for the wiki. It contains **only real, standard references** that the author is confident exist: canonical textbooks (author + title) and genuinely landmark papers (author, year). Where a precise detail (volume, page, exact year) is uncertain, the entry carries an inline `[unverified]` flag and gives author + title + approximate year only — per the *Chapter 2 (p. 8)* rule that we never fabricate citations, numbers, or theorems.

**Scope of the epistemic tags here.** A reference's *existence* is treated as `[ESTABLISHED]` unless flagged `[unverified]`. The tag attached to an entry instead grades the *epistemic status of the physics the source reports*: a textbook of confirmed theory is `[ESTABLISHED]`; a foundational-interpretation paper is `[CONTESTED]` or `[SPECULATIVE]`; a research-frontier program is `[SPECULATIVE]` or `[OPEN]`. This lets the bibliography double as a quick map of *how firm the underlying claim is*, consistent with the conventions in *Chapter 2 (p. 8)*.

> **How to cite from wiki pages.** Domain pages should link here rather than re-listing full citations. Cite as, e.g., "Weinberg, *QTF* I–III (see the Bibliography (p. 776))." Numerical values should always be sourced to the Particle Data Group or Planck (below), never stated from memory — see *Chapter 16 (p. 308)*.

A handful of sources are load-bearing in more than one domain (MTW, Wald, Reed–Simon, Nielsen–Chuang, Haag, Glimm–Jaffe, Peskin–Schroeder, Weinberg's *QTF*, Landau–Lifshitz, the Hawking 1975 and Bekenstein 1973 papers, Jarzynski/Crooks, Wilson–Kogut). Each is listed once under its **primary** domain and cross-referenced from the others to avoid duplication.

# 1. Quantum mechanics and foundations

Primary support for *Chapter 7 (p. 71)*.

## Textbooks and monographs

**P. A. M. Dirac**, *The Principles of Quantum Mechanics* (Oxford Univ. Press, 1930; 4th ed. 1958). `[ESTABLISHED]` — Origin of the bra–ket formalism, representation-independent state/operator picture, and transformation theory underlying the Hilbert-space axioms.

**J. von Neumann**, *Mathematical Foundations of Quantum Mechanics* (Springer, 1932; Princeton Univ. Press transl. 1955). `[ESTABLISHED]` — The rigorous Hilbert-space axiomatization: self-adjoint operators, the spectral theorem, density matrices, the projection postulate, and the first serious analysis of measurement.

**A. Peres**, *Quantum Theory: Concepts and Methods* (Kluwer, 1995). `[ESTABLISHED]` — Operationally rigorous text emphasizing measurement, Kochen–Specker, Bell, and careful statement of the formalism.

**J. S. Bell**, *Speakable and Unspeakable in Quantum Mechanics* (Cambridge Univ. Press, 1987; 2nd ed. 2004). `[ESTABLISHED]` (results) / `[CONTESTED]` (interpretive morals) — Collected papers including the Bell inequality, the critique of the measurement "shifty split," and incisive analyses of nonlocality and hidden variables.

**M. A. Nielsen and I. L. Chuang**, *Quantum Computation and Quantum Information* (Cambridge Univ. Press, 2000; 10th anniversary ed. 2010). `[ESTABLISHED]` — Standard reference for density matrices, POVMs, Kraus/CPTP maps, entanglement measures, Tsirelson's bound, and no-cloning. *(Also primary for §7; cross-listed there.)*

## Landmark papers

**A. M. Gleason**, "Measures on the closed subspaces of a Hilbert space," *J. Math. Mech.* **6**, 885 (1957). `[ESTABLISHED]` — Derives the Born-rule form as the unique noncontextual probability measure on projections for $\dim \mathcal{H} \geq 3$.

**S. Kochen and E. P. Specker**, "The problem of hidden variables in quantum mechanics," *J. Math. Mech.* **17**, 59 (1967). `[ESTABLISHED]` — The Kochen–Specker theorem establishing quantum contextuality for $\dim \mathcal{H} \geq 3$: no noncontextual value assignment $v : \mathcal{P}(\mathcal{H}) \to \{0, 1\}$ exists.

**J. F. Clauser, M. A. Horne, A. Shimony, R. A. Holt**, "Proposed experiment to test local hidden-variable theories," *Phys. Rev. Lett.* **23**, 880 (1969). `[ESTABLISHED]` — The CHSH inequality $|\langle S\rangle| \leq 2$ classically (vs. Tsirelson's $2\sqrt{2}$ quantum bound); foundation of all subsequent Bell tests.

**V. Gorini, A. Kossakowski, E. C. G. Sudarshan** (1976) and **G. Lindblad**, "On the generators of quantum dynamical semigroups," *Commun. Math. Phys.* **48**, 119 (1976). `[ESTABLISHED]` — Independent derivations of the GKSL/Lindblad generator, the most general Markovian (CPTP-semigroup) open-system master equation.

**P. Busch**, "Quantum states and generalized observables: a simple proof of Gleason's theorem," *Phys. Rev. Lett.* **91**, 120403 (2003); and **C. M. Caves, C. A. Fuchs, K. Manne, J. M. Renes**, "Gleason-type derivations of the quantum probability rule for generalized measurements," *Found. Phys.* **34**, 193 (2004; arXiv:quant-ph/0306179). `[ESTABLISHED]` — The Gleason–Busch (effect/POVM) theorem, extending the Born-form uniqueness to $\dim \mathcal{H} = 2$ at the price of additivity over all compatible effects. *(Fourth author is Manne, not Schack.)*

## Recent preprints (2026, not yet peer-reviewed)

**E. O. Torres Alegre**, "Causal Rigidity of Born-Type Probability Rules in Infinite-Dimensional Operational Theories," arXiv:2602.09056 (Pontifical Catholic Univ. of Chile, submitted

7 Feb 2026). [arXiv preprint — **verified real**, not peer-reviewed] — A **rigidity** theorem (Thm 1, "Causal Rigidity"): under (i) no-signaling, (ii) normal steering via $\sigma$-additive purification — an infinite-dimensional generalization of the Hughston–Jozsa–Wootters/GHJW theorem — and (iii) $\sigma$-affinity of probability assignments under countable preparation mixtures (plus implicit continuity/monotonicity of $\Phi$), the post-processing map in $P = \Phi(\tau)$ must be the identity, making the Born rule the **unique** no-signaling-compatible probability assignment on **infinite-dimensional normal operational state spaces** (lifting the finite-dimensional Gleason–Busch form-rigidity). **Caveat:** §6 *presupposes* a von Neumann-algebra representation and then recovers $\tau = |\langle\phi|\psi\rangle|^2$ — it does NOT derive the Hilbert space or the algebraic scaffold, so it sharpens the Born-*form* rigidity rather than grounding the formalism (relocates, not dissolves; cf. the GPT-reconstruction verdict KA-5 in Working note: *Is the Born Rule Fundamental or Derivable, and Is Quantum Mechanics Exactly Linear?* (repo: `notes/2026-06-08-born-rule-and-quantum-linearity.md`)). Authors flag normal steering as "the strongest assumption." — *Note (iteration-4, web-confirmed):* the other 2026 preprint IDs cited in the Born-rule note are **all real** (promoted from `[unverified]`): arXiv:2603.06211 (J. Zhang, *Summing to Uncertainty* — additivity as the shared load-bearing premise across Born-rule derivations); arXiv:2601.18856 (V. Pronskikh, *Operationally induced preferred basis in unitary quantum mechanics*); arXiv:2601.13012 (A. K. Pati, *No-Signalling Fixes the Hilbert-Space Inner Product* — fixes the **inner product**, *not* linearity, which it assumes). No fabrication was found anywhere in the wiki's citations.

## Iteration-4 verified additions

**M. B. Fröb**, "Modular Hamiltonian for de Sitter diamonds," *JHEP* **12** (2023) 074 (arXiv:2308.14797). `[ESTABLISHED]` — *Single-authored.* The dS-diamond modular Hamiltonian is the Casini–Huerta–Myers conformal-Killing-vector flow, geometric only in the conformal vacuum, reducing to the static-patch boost in the large-diamond limit. (Corrects an iteration-3 "Forste et al." misattribution.) Bears on the OP-41 dS first-law analysis (see §9, *Chapter 19 (p. 392)*).

**Muon $g-2$ Theory Initiative (WP2025)**, "The anomalous magnetic moment of the muon in the Standard Model" (arXiv:2505.21476, 2025). `[ESTABLISHED]` — Lattice-QCD HVP-based SM prediction $a_\mu^{\rm SM} = 116\,592\,033(62) \times 10^{-11}$, $\Delta a_\mu = 38(63) \times 10^{-11}$ (SM-consistent); supersedes the WP2020 data-driven value for the $g-2$ anomaly status. See *Chapter 15 (p. 233)* OP-21 and *Chapter 9 (p. 106)*.

## Iteration-6 verified additions (2026-06-10)

*All arXiv IDs below were re-resolved by the iteration-6 referees against arXiv/journal listings on 2026-06-10. Journal references flagged `[unverified]` rest on search-snippet or Springer-sighting provenance only and must not be promoted until a DOI is checked. Per EPISTEMICS.md §4, no citation here is invented; titles are quoted only where verified.*

**Closed-universe / dS holography (OP-50, HYP-dS-CARDINALITY-R1):**

**M. Usatyuk, Z. Wang, Y. Zhao**, *SciPost Phys.* **17**, 051 (2024). `[ESTABLISHED, within 2D models]` — Including spacetime wormholes in the inner product degenerates the Gram matrix of perturbatively-distinct closed-universe states to rank 1 in JT gravity + matter and a topological model: a unique closed-universe state per theory. The proven core of the OP-50 debate.

**D. Harlow, M. Usatyuk, Y. Zhao**, arXiv:2501.02359 [JHEP ref **unverified** — Springer sighting only]. `[CONTESTED premise]` — The "observer rescue": an internal observer with $S_{Ob}$ degrees of freedom is described by an effective QM on a Hilbert space of dim ~$e^{S_{Ob}}$ (gravitational-path-integral argument, not a theorem). Counter-positions: Antonini–Rath–Sasieta–Swingle–Vilar López (arXiv:2507.10649); H. Liu (arXiv:2509.14327); Engelhardt–Gesteau (arXiv:2504.14586).

**J. Maldacena**, arXiv:2412.14014 (2024). `[ESTABLISHED as the paper's claim]` — An observer mostly cancels the dimension-dependent phase $i^{D+2}$ of the dS sphere partition function; the $A/4G$ coefficient still comes from the classical saddle (η input).

**Algebraic frontier (OP-44, OP-48c):**

**K. Jensen, S. Raju, A. J. Speranza**, arXiv:2412.21185; *JHEP* **06** (2025) 242. `[ESTABLISHED]` — The AdS boundary time-band algebra strictly has no commutant nonperturbatively; coarse-grained observer version reduces to the modular crossed product at leading order; uniqueness: the modular crossed product is the only crossed product of a III1 algebra with a trace.

**J. De Vuyst, S. Eccles, P. A. Höhn, J. Kirklin**, "Linearization (in)stabilities and crossed products," arXiv:2411.19931; *JHEP* (2025) 211. `[ESTABLISHED]` — Locates the order-$G_N^2$ obligation at the quadratic (boost) linearization-instability constraints; "essentially unambiguous" for spatially closed spacetimes. Companion: PW=CLPW + QRF-relative entropies, arXiv:2405.00114, *JHEP* (2025) 146.

**M. S. Klinger, J. Kudler-Flam, G. Satishchandran**, "Generalized Entropy is von Neumann Entropy II: The complete symmetry group and edge modes," arXiv:2601.07910 (2026, 104 pp). `[ESTABLISHED]` — Perturbative Type II structure survives the complete constraint family with boost-supertranslation edge modes; entropy = $S_{\text{gen}}$ + edge-mode + constant term.

**V. Chandrasekaran, É. É. Flanagan**, "Subregion algebras in classical and quantum gravity," arXiv:2601.07915 (2026, 111 pp). `[ESTABLISHED]` — GSL for non-stationary linearized perturbations of Killing horizons; quantum focusing conjecture proven within the linearized/perturbative regime.

**J. Kudler-Flam, E. Witten**, "Emergent Mixed States for Baby Universes and Black Holes," arXiv:2510.06376. `[ESTABLISHED]` — Baby-universe state sequences fail to converge at large N;

appropriate averaging over N yields mixed-state convergence with nontrivial commutants (the OP-44 "survives-after-averaging" option). Companion: Mellin averaging over N, arXiv:2605.15180.

**A. Herderschee, J. Kudler-Flam**, "Stringy algebras, stretched horizons, and quantum-connected wormholes," arXiv:2510.01556. `[ESTABLISHED]` — Hagedorn spectrum breaks the split property for regions within a string length; algebraic definition of stretched horizons; Type III ER=EPR factors.

**H. Liu**, "Lectures on entanglement, von Neumann algebras, and emergence of spacetime," arXiv:2510.07017 (2025, 119 pp). `[ESTABLISHED, review]` — The actual 2025 synthesis of the algebraic program (corrects the garbled "KLS-trio review" lead).

**L. van Luijk, A. Stottmeister, H. Wilming**, "Multipartite Embezzlement of Entanglement," *Quantum* **9**, 1818 (2025); arXiv:2409.07646. `[ESTABLISHED]` — A multipartite system of commuting $III_1$ factors on which every state is embezzling: the consistent non-Type-I composition endpoint in the OP-48(c) horn-C residue. Synthesis: van Luijk, arXiv:2510.07563 (type classification ↔ operational entanglement properties).

**Pseudo-entropy / OP-49:**

**W.-z. Guo, S. He, Y.-X. Zhang**, arXiv:2209.07308; *JHEP* **05** (2023) 021. `[ESTABLISHED]` — Reality of pseudo-(Rényi-)entropy trace moments is necessary-and-sufficiently characterized by η-pseudo-Hermiticity of the transition matrix (diagonalizable case), with a Tomita–Takesaki reality class. The reality theorem grounding the OP-49 near-no-go.

**K. Doi, J. Harper, A. Mollabashi, T. Takayanagi, Y. Taki**, *Phys. Rev. Lett.* **130**, 031601 (2023). `[ESTABLISHED]` — $dS_3/CFT_2$ pseudo-entropy: Re S = $S_{\mathrm{dS}}/2$, finite, region-independent; all region-dependence and divergence imaginary. (Source of the iter-5 Re/Im inversion correction.)

**Li–Xiao–He–Sun**, arXiv:2511.17098 (2025). [ESTABLISHED re paper content, equation-level verified; no verbatim cancellation quote — paraphrase only] — Timelike-EE first law in AdS/CFT

⇔ linearized Einstein equations; the imaginary part is a state- and interval-independent constant dropping out of variations — the unitary mirror of the dS case under $c \to ic$.

**A. J. Parzygnat, T. Takayanagi, Y. Taki, Z. Wei**, arXiv:2307.06531; *JHEP* **12** (2023) 123. `[ESTABLISHED]` — SVD entanglement entropy: the only known intrinsically real, non-negative two-state entropy; the OP-49 residual protocol (no first law exists as of 2026-06).

**Phenomenology (watchlist):**

**E. Verlinde, K. Zurek**, *JHEP* **04** (2020) 209; and *Phys. Rev. D* **106**, 106011 (2022). `[INFERENCE within AdS/CFT]` — Modular-fluctuation area law $\langle \Delta K^2 \rangle = \langle K \rangle = A/4G$ and its shockwave reproduction: the anchor of the geontropic chain (η input at the modular→metric step).

**S. M. Vermeulen et al. (GQuEST)**, *Phys. Rev. X* **15**, 011034 (2025). `[ESTABLISHED design]` — Photon-counting Michelson design; 5σ on the pixellon benchmark α=1 in $O(10^5)$ s; pixellon derivation from a UV-complete theory "still underway" (stated admission). Sibling bound: QUEST, *PRL* **135**, 101402 (2025), arXiv:2410.09175.

**D. Carney, S. Karydas, A. Sivaramakrishnan**, *Phys. Rev. D* **113**, 106002 (2026). `[ESTABLISHED]` — Minimal graviton-EFT null prediction: arm-length variance ~$\ell_p^2$, no IR divergences; a VZ-magnitude detection would "signal a severe breakdown of effective quantum field theory in low energy quantum gravity" (verbatim verified). Confound: Freidel–Oberfrank, arXiv:2601.17849 (PSD depends on the graviton quantum state).

**J. Oppenheim**, "A postquantum theory of classical gravity?," arXiv:1811.03116; *Phys. Rev. X* **13**, 041040 (2023). `[ESTABLISHED framework / CONTESTED physics]` — CPTP classical–quantum gravity; stochastic metric + matter, no measurement postulate.

**J. Oppenheim, C. Sparaciari, B. Šoda, Z. Weller-Davies**, arXiv:2203.01982; *Nat. Commun.* **14**, 7910 (2023).

`[ESTABLISHED, theorem within CQ]` — The decoherence–diffusion trade-off: coherence forces classical-sector diffusion observable as force noise.

**T. Janse, D. G. Uitenbroek, J. van Everdingen, S. Plugge, B. Hensen, T. H. Oosterkamp**, arXiv:2403.08912; *Phys. Rev. Research* **6**, 033076 (2024). `[ESTABLISHED]` — Force-noise compilation lowering the spacetime-diffusion bound ≥15 orders; ultralocal-continuous dead, ultralocal-discrete below the upper bound (with stated caveats), non-local continuous within ~1 order.

**A. Biggs, J. Maldacena**, arXiv:2405.02227. `[ESTABLISHED as published counter-reading]` — Ordinary finite-temperature matter reproduces the qualitative Danielson–Satishchandran–Wald horizon-decoherence effect: thermality, not horizon-ness, does the work (the DSW deflation). (Bibliographic fix from the A1 referee: arXiv:2505.17450 is a Lifshitz moving-mirror holographic analogue by Kawamoto–Lee–Yeh, NOT an independent reproduction of the DSW rate.)

**V. Vilasini, R. Renner**, *Phys. Rev. A* **110**, 022227 (2024); companion arXiv:2408.13387. `[ESTABLISHED]` — Indefinite-causal-order processes embedded in acyclic spacetime admit a definite, acyclic fine-grained causal-order explanation: process matrices do not generate causal order.

**Generation-construction survey (OP-46 / HYP-CKV-VACUITY-R3):**

**S.-S. Lee**, *JHEP* **06** (2020) 070; arXiv:1912.12291 (full text checked). `[SPECULATIVE as physics; ESTABLISHED that the paper exhibits it]` — Matrix model in which signature is a state-dependent dynamical output (dS-like Lorentzian regions bridged by Euclidean ones) on a posited carrier + signature-parametrized hypersurface-deformation template. The nearest sketch to CONCLUSION §8's spec; fails it (selection-type).

**F. Finster et al. (causal fermion systems)** — program review arXiv:1709.04781 (full text checked); CUP textbook listing arXiv:2411.06450 (2024/25); status paper arXiv:2603.05018 (Farnsworth, Finster, Paganini, Singh — future-dated ID verified real). `[ESTABLISHED definitions; INFERENCE on the η-identification]` — The $(\leq n, \leq n)$-eigenvalue axiom + indefinite spin scalar product as the relocated Krein η (fourth readout family); minimizer existence only finite-dimensional; all worked examples Dirac-sea encodes.

**L. Rodina**, *Phys. Rev. Lett.* **134**, 031601 (2025). `[ESTABLISHED — with the binding hypotheses caveat]` — Hidden zeros ⟺ "subset" enhanced BCFW UV scaling; zeros uniquely fix Tr(φ³) tree amplitudes **assuming an ordered local propagator structure and trivial numerators** (locality is a hypothesis, not derived — must not be entered as "locality as derived"). Companion conjecture line: Zhou, arXiv:2604.07195 (also assumes locality+unitarity).

**O. Müller**, "On the Hauptvermutung of Causal Set Theory," arXiv:2503.01719 (v2 Dec 2025; preprint, not peer-reviewed, proofs not independently checked). [web-confirmed real] — Two precise inequivalent formulations: one false, one true; countable-set version holds.

**Experiment refresh:**

**LIGO–Virgo–KAGRA**, GW250114 area-law companion: arXiv:2509.08054; *Phys. Rev. Lett.* **135**, 111403 (2025). `[ESTABLISHED]` — Hawking's area law $A_f > A_1 + A_2$ confirmed at high credibility (the correct citation — NOT the ringdown-spectroscopy paper 2509.08099, which carries the $(2, 2, 0)$ +overtone Kerr-consistency at tens-of-percent level; SNR 80 discovery / 76.9 catalog).

**LHCb**, $B^0 \to K^{*0}\mu^+\mu^-$ comprehensive amplitude analysis: arXiv:2512.18053; *PRL* 2026, DOI 10.1103/24g9-yn9d. `[ESTABLISHED measurement / CONTESTED interpretation]` — ~4σ

SM tension [σ-figure from press coverage of the PRL] on full Run 1+2; the flagship surviving flavor anomaly (OP-21).

**KiDS-Legacy**, arXiv:2503.19441 (cosmic shear: $S_8 = 0.815^{+0.016}_{-0.021}$, 0.73σ from Planck) and arXiv:2503.19442 (joint analysis, Planck-consistent). `[ESTABLISHED]` — The S8 easing on the KiDS axis; DES Y6 remains in tension per arXiv:2602.12238 `[CONTESTED]`.

**S. Pedalino, A. Ramírez-Galindo, S. Ferstl, K. Hornberger, M. Arndt, S. Gerlich (Vienna)**, arXiv:2507.21211; *Nature* **649**, 866 (2026). `[ESTABLISHED]` — Interference of sodium nanoclusters >7,000 atoms / >1.7×10$^5$ Da; macroscopicity μ=15.5 — the most stringent generic-macrorealism exclusion to date.

**LHAASO**, GRB 221009A LIV bounds: arXiv:2402.06009, *PRL* (v3 Feb 2026). `[ESTABLISHED]` — $E_{\mathrm{QG},1} > 10\, E_{\mathrm{Pl}}$, $E_{\mathrm{QG},2} > 6 \times 10^{-8}\, E_{\mathrm{Pl}}$.

`[unverified]` **journal-reference ledger (binding; do not promote without a DOI check):**

From R1 (Springer-sighting-only "JHEP" refs, six): Harlow–Usatyuk–Zhao 2501.02359; Higginbotham 2512.17993 (arXiv page: "Submitted to JHEP"); MSS 2506.18054; Nomura–Vafa-camp NV 2310.16994; ARTW 2511.04511; CCGPR 2303.16316.

From R3 (search-snippet JHEP issue/page refs, five): the colored-Yukawa loop-integrated paper arXiv:2406.04411 is now **promoted** — De–Pokraka–Skowronek–Spradlin–Volovich, *JHEP* **09** (2024) 160, DOI 10.1007/JHEP09(2024)160 (web-verified against OSTI/arXiv listings 2026-06-19); enter the remaining four flagged R3 journal refs as arXiv-ID-only with `[unverified]` journal fields. Attribution fix: the gravituhedron paper arXiv:2012.15780 is single-author **Trnka**.

Other riders: Nimmrichter et al. arXiv:2502.13821 "accepted PRL [journal ref **still unverified** — arXiv carries no journal-ref/DOI as of 2026-06-19; leave flagged]"; diamagnetic microchip trap arXiv:2411.02325 ["Accepted for publication in *Phys. Rev. A*" per arXiv, **no volume/page yet** — leave

flagged]. **Promoted (web-verified 2026-06-19):** Gonzalo–Montero–Obied–Vafa arXiv:2209.09249 = *JHEP* **11** (2023) 109, DOI 10.1007/JHEP11(2023)109 (ADS 2023JHEP...11..109G); the Barrett–Crane entry **PRD 111, 026014 (2025)** is Dekhil–Jercher–Pithis, *"Phase transitions in tensorial group field theory: Landau-Ginzburg analysis of the causally complete Lorentzian Barrett-Crane model,"* DOI 10.1103/PhysRevD.111.026014. **Over-attribution removed:** the "~$10^{-13}$ d displacement" figure is **not** in arXiv:2604.16276 (Xue et al., verified real) — its abstract/body argue the Aziz–Howl effect is "essentially classical" / "negligibly small"; the unsourced number has been struck wiki-wide. Bibliographic fixes carried from referees: Sharmila et al. is *Nat. Commun.* **17**, 701 (2026); Bub et al. arXiv:2408.11094 is published as *PRL* **134**, 121501 (2025).

## Iteration-7 verified additions (2026-06-10)

*All references below were re-verified live by the iteration-7 referees (arXiv/journal pages fetched; verbatim quotes checked against primary PDFs where load-bearing). No* `[unverified]` *riders were introduced by this iteration.*

**Modular theory ↔ Krein structures (A1, the verdict-flipper track):**

**H. Gottschalk**, "Die Momente gefalteter Gauß-Maße und ihre Anwendung in der Quantenfeldtheorie" / related Krein-space QFT work, *J. Math. Phys.* **43**, 4753 (2002). `[ESTABLISHED]` — Prior art running **Krein → modular**: modular structure built atop a posited indefinite-metric space; the reverse direction (modular → Krein) is absent from the literature (the A1 direction-map finding).

**L. Jakóbczyk**, on Tomita–Takesaki theory in indefinite-metric (Krein-space) QFT, 1985 (OSTI 5135323). `[ESTABLISHED]` — Same direction: Krein structure assumed, modular objects derived.

**A. Devastato, S. Farnsworth, F. Lizzi, P. Martinetti**, "Lorentz signature and twisted spectral triples," *JHEP* **03** (2018) 089 (arXiv:1710.04965). `[ESTABLISHED]` — The twisted-spectral-triple route: the Krein/η datum installed via the twist (author list corrected by the iteration-7 referee from a truncated form).

**Crossed products / constraints (A2):**

**J. Kirklin**, all-orders generalized second law, arXiv:2412.01903; *JHEP* **07** (2025) 192. `[ESTABLISHED]` — The nearest existing discharge machinery for higher-order constraint imposition; does not execute the dS closed-slice cubic step.

**S. Ali Ahmad, R. Jefferson** (AAJ), perturbative crossed products beyond subleading order, arXiv:2501.01487. `[ESTABLISHED]` — First controlled step past leading order (carried from iteration 6; re-anchored here for the A2 descent argument).

**Closed-universe / OP-50 watch (A4):**

**D. Harlow, M. Usatyuk, Y. Zhao**, *JHEP* **02** (2026) 108 (arXiv:2501.02359). `[CONTESTED premise]` — Now journal-published; upgrades the iteration-6 `[unverified]` JHEP flag for this entry.

**K. Higginbotham**, observer complementarity, *JHEP* **03** (2026) 183 (arXiv:2512.17993). `[CONTESTED]` — Now journal-published; upgrades the iteration-6 `[unverified]` flag.

**Y. Nomura, T. Ugajin**, partial-observability follow-up, arXiv:2602.13387. `[ESTABLISHED as claims]`

**Miyata–Horikoshi–Tajima–Ugajin**, arXiv:2606.04575. `[ESTABLISHED as the paper's construction]` — The third OP-50 position: a pre-gauge-constraint hyperfinite $II_1$ dominant sector with a post-constraint polynomial-dimension physical Hilbert space (scope per the iteration-7 referee).

## Distinguishing-test-hunt verified additions (2026-06-10)

*The 2024–2026 experimental/phenomenology sweep + the reverse-Weinberg–Witten / net-invariant analysis. Every arXiv ID below resolves to the stated authors and title; decisive primary-source quotes verified verbatim where load-bearing.*

**The verified lead (mechanism-blind taxonomy — positive ENCODING-SCREEN evidence):**

**B. Sharmila, S. M. Vermeulen, A. Datta**, "Signatures of correlation of spacetime fluctuations in laser interferometers," *Nat. Commun.* **17**, 701 (2026) (online 23 Dec 2025); arXiv:2505.22892. `[ESTABLISHED]` — Three-class taxonomy of spacetime-fluctuation two-point functions: class (a) factorised (Oppenheim CQ, CSL, Diósi–Penrose), (b) inverse/polynomial (Károlyházy, Verlinde–Zurek geontropic, EFTs), (c) exponential (entanglement-holographic/mesoscopic). Load-bearing verbatim quote confirmed present: *"Our approach is agnostic to the microscopic origins of the SFs."* The model→class map is many-to-one (fundamental and emergent models co-classed), so the taxonomy classifies *gravity models*, not fundamental-vs-emergent ontology — a published worked instance of HYP-ENCODING-SCREEN Corollary 2. (Bibliographic note already on record in the [unverified]-ledger fixes; entered here as a standalone source.)

**Author-side concessions of observational degeneracy (independent screen confirmations):**

**Sidan A, T. Banks**, holographic-spacetime inflation / CMB, arXiv:2502.15108 (2025); *Phys. Rev. D* **112** (2025) [volume per submission, not independently re-checked this session]. `[ESTABLISHED re content]` — Inflationary scalar+tensor predictions *"fit extant data equally well"* (verbatim) as field-theoretic inflation; the differences are *"radically different"* conceptually (no transplanckian problem), not observational. **Quotation-hygiene correction (binding):** the string "No

claim is made that the CMB could observationally distinguish emergent/holographic spacetime from conventional metric inflation" is a PARAPHRASE, not a verbatim quote — record it as such (EPISTEMICS §4: no fabricated quotes).

**J. Erlich**, stochastic composite gravity (GRF 2025 essay), arXiv:2504.03084 (2025). `[ESTABLISHED]` — Predictions (BH entropy; low-ℓ CMB modification; a stochastic-vs-QFT out-of-equilibrium disagreement) are explicitly portable: the low-ℓ effect is *"not unique."* Nuance on record (not verdict-changing): the out-of-equilibrium stochastic-vs-QFT test is framed as a positive experimental test of the framework, but it bears on exact linearity of QM (the gate / objective-collapse axis), NOT on geometry-priority — so it is not a reverse-WW.

**Other 2024–2026 sweep channels (all screened — dead/gate/member-level for the wager):**

**R. Schützhold**, "Stimulated Emission or Absorption of Gravitons by Light," *Phys. Rev. Lett.* (2025), HZDR. `[ESTABLISHED, verified]` — ~$10^6$ km effective-path interferometer probing stimulated graviton absorption/emission; tests the radiative-sector **gate** (is the gravitational field quantized / do gravitons exist), not derivational priority. Self-stated: *"would not directly prove the existence of gravitons ... strong supporting evidence."*

**L. Aalsma, S.-E. Bak**, "Modular Fluctuations in Cosmology," arXiv:2503.04886; *Phys. Rev. D* **112**, 026017 (2025) [journal ref carried from iteration 6, arXiv identity + quote re-verified]. `[ESTABLISHED]` — Causal-diamond modular fluctuations → near-scale-invariant CMB spectrum, requires quantizing an s-wave mode by hand; model-agnostic. Verbatim: *"whether such a mode exists in the vacuum of any theory of quantum gravity remains an open question."*

**G. Calcagni, L. Modesto**, primordial GWs from Weyl-invariant QG, arXiv:2206.07066; *JHEP* **12** (2024) 024. `[ESTABLISHED]` — $r_{0.05} \approx 0.01$ (BICEP/LiteBIRD) + blue-tilted SGWB (DECIGO) from a single UV-complete construction; a member-level

parameter prediction, not class-level — the quasi-scale-invariant spectrum is also produced by fundamental-metric quantized gravity.

**K. Hashimoto, K. Matsuo, K. Murata, G. Ogiwara, D. Takeda**, ML emergent-spacetime from linear response, arXiv:2411.16052. `[ESTABLISHED]` — A neural net *assuming AdS/CFT* reconstructs a bulk metric as *"interpretable weights"*; emergence is presupposed by training on the holographic dictionary, not tested (the encode-only architecture made literal).

**D. Jafferis et al.**, traversable-wormhole quantum simulation (Google Sycamore), *Nature* **612**, 51 (2022), DOI 10.1038/s41586-022-05424-3; criticism **G. Kobrin, T. Schuster, N. Y. Yao**, arXiv:2302.07897. `[ESTABLISHED]` — The gravitational dual is an interpretive dictionary over an engineered system; a simulation cannot distinguish emergence-in-nature from simulation by construction.

**Reverse-Weinberg–Witten / net-invariant analysis (H2 track):**

**C. Casini, J. M. Magán**, "A generalization of the DHR theorem for higher form symmetries," arXiv:2511.21810 (submitted 26 Nov 2025). `[ESTABLISHED, verified live]` — Generalized order/disorder parameters in $D > 2$ (Wilson/'t Hooft loops) are labeled by groups attached to Haag-duality-violation data of the net — i.e. the global form / line-operator spectrum is a **net (DHR/HDV) invariant**, hence EFT-shared between algebra-first and geometry-first descriptions (load-bearing only under the net-sharing assumption A1).

**C. D. Carone, T. V. B. Claringbold, D. Vaman**, "Composite graviton self-interactions in a model of emergent gravity," arXiv:1710.09367. [ESTABLISHED — author list and physics confirmed this session; corrects a prior author-less citation] — A worked Weinberg–Witten evasion (composite massless spin-2 via emergent gravity); the evasion is symmetric across both theory classes, so it supplies no class-separating observable.

**J. J. Heckman, M. Hübner, C. Murdia**, "On the Holographic Dual of a Topological Symmetry Operator," *Phys. Rev. D* **110**, 046007 (2024); arXiv:2401.09538. `[ESTABLISHED]` — Supports "no global $p$-form symmetry in AdS" and the SymTFT-inside-holography picture; a no-global-symmetry result under A1/holography does NOT upgrade reverse-WW's literature-absence to a structural impossibility.

**A. Parnachev, N. Poovuttikul**, "Topological Entanglement Entropy, Ground State Degeneracy and Holography," arXiv:1504.08244 (2015). `[ESTABLISHED]` — TEE ↔ GSD ↔ holography; TEE $\gamma = \log \mathcal{D}$ is the total quantum dimension of the DHR superselection category, a net+state invariant — diagnoses topological order, not derivational order.

**B. Czech, J. de Boer, D. Espíndola, B. Najian, J. van der Heijden, J. Zukowski**, "Changing states in holography: From modular Berry curvature to the bulk symplectic form," arXiv:2305.16384; and **W.-z. Huang, M.-z. Ma**, "Emergence of Kinematic Space from Quantum Modular Geometric Tensor," arXiv:2110.08703. `[ESTABLISHED]` — Modular Berry / kinematic-space data are net+state invariants (Tomita–Takesaki–built); the one emergent Lorentzian readout (Huang–Ma dS2) inherits its signature from the symmetric vacuum (iter-5 closure).

*(Vilasini–Renner, PRA **110**, 022227 (2024), already on record at BIBLIOGRAPHY line 85 — indefinite causal order reduces to definite acyclic order — is reused by the H2 track; no new entry needed.)*

**`[unverified]` riders introduced by this hunt (do not promote without a DOI/page check):**

Banks–Sidan A *Phys. Rev. D* **112** volume number (arXiv identity + the "fit extant data equally well" quote ARE verified; the journal volume is not independently re-checked this session).

Schützhold PRL (2025) volume/page (APS Physics + HZDR coverage confirm the result; the exact PRL coordinate is not pinned this session).

Delhom–Giacomelli analog-gravity lecture notes arXiv:2512.14209 and the Steinhauer-line analog-Hawking PRL **126**, 111301 (2021) — cited in the analog-gravity dead-by-construction note; loose but immaterial (the verdict does not depend on either resolving precisely).

## Iteration-8 verified additions (2026-06-10)

*All re-verified live by the iteration-8 referees (arXiv/journal pages fetched; load-bearing quotes checked against primary PDFs). No `[unverified]` riders introduced.*

**Net/family carrier no-go (A1):**

**Y. Lashkari (M. Lashkari), C. Leung, M. Moosa, A. Ouseph** (as listed), "...local representation of the boundary algebras / $AdS_2$ emergence," *JHEP* **10** (2025) 153; arXiv (verified). `[ESTABLISHED]` — Verbatim: *"$AdS_2$ emerges from insisting on a local representation of the boundary algebras that satisfy Haag's duality"* — localization is an imposed input, the load-bearing "net-index-set IS the geometry" datum.

**V. Morinelli, K.-H. Neeb, G. Ólafsson**, Euler-element / Bisognano–Wichmann for standard-subspace nets. `[ESTABLISHED]` — The Euler element (wedge localization) is an *input*, not derived from modular data.

(Borchers–Buchholz–Longo positivity⟺isotony, and the chiral standard-pair machinery, already on record.)

**OP-44 R1/R2 (A2):**

**P. Hintz**, "...solvability / linearization," arXiv:2306.07715. `[ESTABLISHED]` — Thm 1.1 solvability for smooth, divergence-free, spatially-compact sources; Remark 1.2 extends the *linear* theorem to $\Lambda > 0$ "with purely notational modifications" (verbatim, verified).

**A. Pound**, second-order self-force / puncture regularization, arXiv:1506.06245. `[ESTABLISHED]` — $\delta^2 G[h_1, h_1] \sim 1/r^4$ non-integrable; no solution to the fully nonlinear point-particle-sourced equation; the puncture/effective-source scheme.

(KLS-II, arXiv:2601.07910, already on record — edge modes "address only second-order perturbations.")

**OP-49 junction (A3):**

**C. Bachas, I. Brunner, D. Roggenkamp**, conformal-interface transmissivity (transmissive-defect criterion $T^1 = T^2$). `[ESTABLISHED]`

**J. Fuchs, M. R. Gaberdiel, I. Runkel, C. Schweigert**, arXiv:0705.3129. `[ESTABLISHED]` — Topological-defect / orientation-reversal machinery (corrects a prior "Bachas et al." attribution of this ID).

**A. Karch, Y. Kusuki, H. Ooguri, H.-Y. Sun, M. Wang**, interface effective central charge / transmission-dependent log coefficient (SSA, $c$-theorem, positive $c$ assumed — hence inapplicable at imaginary $c$). `[ESTABLISHED, unitary regime]`

**OP-48c X1 (A4):**

**M. Takesaki**, *On the direct product of $W^*$-factors*, Tôhoku Math. J. **10**, 116–119 (1958). `[ESTABLISHED]` — The normal-product-state $\Leftrightarrow$ $W^*$-isomorphism theorem (Rédei–Summers Prop. 2): the obstruction that forces singular product states across a Type III1 Haag-dual pair to be **non-normal** ($\mathcal{M} \vee \mathcal{M}' = B(\mathcal{H})$ is Type I while $\mathcal{M} \overline{\otimes} \mathcal{M}'$ is Type III). This is the non-normality obstruction — *not* Florig–Summers; the iteration-8 submission's joint "Takesaki/Florig–Summers" attribution is corrected here.

**M. Florig, S. J. Summers**, *On the statistical independence of algebras of observables*, J. Math. Phys. **38**, 1318–1328 (1997). `[ESTABLISHED]` — The C*-side **faithful**-product-state ⇔ C*-independence-in-the-product-sense characterization (Rédei–Summers Prop. 4 = ref [14]); a separate, stronger result, *not* the non-normality obstruction and not load-bearing for X1-existence.

**H. Roos**, *Independence of local algebras in quantum field theory*, Commun. Math. Phys. **16**, 238–246 (1970). `[ESTABLISHED]` — On commuting C*-algebras the C*-independence extension

state may be chosen a **product state** (the existence input for X1, via the Rédei–Summers post-Def-1 sentence). **Plain** C*-independence suffices — no product-sense isomorphism needed (Rédei–Summers Prop. 1: plain $\not\Rightarrow$ product-sense in general).

**M. Rédei, S. J. Summers**, *When are quantum systems operationally independent?*, arXiv:0810.5294 (2009). `[ESTABLISHED]` — Assembles the Roos / Takesaki / Florig–Summers independence results (Props 1/2/4); the primary carrier for the X1 existence theorem. **H. Halvorson** (math-ph/0602036) supplies the Schlieder ⇔ C*-independence implication box (Def 3.4 is the *definition* of Schlieder, not the proof of Step 1).

The **central-carrier fact** — every nonzero projection in a factor has central carrier $I$ — is **Kadison–Ringrose**, *Fundamentals of the Theory of Operator Algebras* (1983/1986); it (not Halvorson Def 3.4) is the proof of Step 1, factor ⟹ Schlieder.

## Iteration-9 verified additions (2026-06-10)

*Referee-audited live this session; no `[unverified]` riders introduced.*

**Carrier naturality + reverse-WW (A1, A3):**

**Z. Weiner**, "An algebraic version of Haag's theorem," arXiv:1006.4726. `[ESTABLISHED]` — Under the split property + geometric modular action, the unitary-equivalence class of the vacuum-sector net of double-cone algebras on a fixed Cauchy surface **completely determines the AQFT model**: the load-bearing "net + vacuum is a complete invariant" theorem behind the reverse-WW ⇔ carrier-problem identification.

**D. Buchholz, O. Dreyer, M. Florig, S. J. Summers**, "Geometric modular action and spacetime symmetry groups," arXiv:math-ph/9805026. `[ESTABLISHED]` — GMA: the geometry is a functor of the modular data (the constructibility-naturality input; "Buchholz–Summers" is incomplete shorthand for this four-author paper).

**S. J. Summers, R. White**, arXiv:hep-th/0304179. [ESTABLISHED] — GMA reconstruction of the spacetime symmetry group from modular data.

**D. Bisch**, arXiv:math/0304340. [ESTABLISHED] — Subfactor / planar-algebra standard-invariant reference (attribution corrected: **Bisch, not Jones**).

**K.-H. Neeb, G. Ólafsson**, arXiv:1704.01336. [ESTABLISHED] — Antiunitary representations / Euler element as the localization datum (the "fiber bundles over ordered symmetric spaces" reading: geometry in the base, no signature in the fibers).

arXiv:2312.12182 — partial-converse Euler-element generation (Euler element *derived* from net + Bisognano–Wichmann + regularity, which already encode the causal/Lie structure). [ESTABLISHED]

**Modular-Berry holonomy (A2):**

**B. Czech et al.**, arXiv:2305.16384 (modular Berry curvature ↔ bulk symplectic form); **W.-z. Huang, M.-z. Ma**, arXiv:2110.08703 (kinematic space from the modular geometric tensor — the Lorentzian $dS_2$ metric, vacuum-inherited); supporting: arXiv:1812.02176, arXiv:2505.04682, arXiv:2111.05345. [ESTABLISHED] — Establish the modular-Berry curvature as the signature-free symplectic Crofton form and the kinematic-space metric's signature as symmetry-inherited.

## Iteration-10 verified additions (2026-06-10)

**Connes, A.; Takesaki, M.** — "The flow of weights on factors of type III." *Tôhoku Mathematical Journal* **29** (1977), 473–575. — Cited for: the modular centralizer of an integrable weight on a Type III1 factor is a Type $II_\infty$ factor (trivial flow of weights ⟺ III1), so the commutant-to-generated gap is real and wide for degenerate modular spectrum (iteration-10 A1-F1).

**Popa, S.** — "On a class of type $II_1$ factors with Betti numbers invariants." arXiv:math/0011084. — Cited for: a free-product/Popa-type realization in which any prescribed finite von Neumann algebra $(N, \tau)$ occurs as the centralizer of a state on a Type III1 factor (the centralizer can be a Type $\mathrm{II}_1$ factor or any prescribed large algebra). *Replaces the digest-flagged miscitation arXiv:1804.05706 for this point* (iteration-10 A1-F1).

**Shlyakhtenko, D.** — free-quasi-free-states structure theory of free Araki–Woods factors. — Cited for: the centralizer-triviality fact for Lebesgue-spectrum (geometric/BW) modular flow — $\mathcal{M} \cap \{\Delta^{it}\}' = \mathbb{C}1$ when $\Delta$ has no eigenvectors. This, not Connes–Størmer / Houdayer / Boutonnet–Houdayer, is the source of the biconditional (iteration-10 A2-F2 referee correction). *(Attribution per the iteration-10 referee; pin the precise Shlyakhtenko reference — "Free quasi-free states," Pacific J. Math. 177 (1997) 329–368 — on the next bibliography pass if confirming the exact paper.)*

**Houdayer, C.** — bicentralizer triviality for type III1 factors. arXiv:0809.3827. — Cited as a *corroborating* (not biconditional) result: proves the *bicentralizer* trivial, distinct from the centralizer-triviality used in A2-F2. Attribution-precision entry to forestall the same over-attribution the iteration-8/9 referees flagged.

**Boutonnet, R.; Houdayer, C.** — amenable modular-invariant subalgebras lie in the almost-periodic summand. arXiv:1602.01741. — Cited as *corroborating* the fork in A2-F2 (amenable modular-invariant subalgebras / almost-periodic summand), again explicitly NOT the source of the centralizer biconditional.

## Iteration-11 verified additions (2026-06-10)

### Operator algebras — Connes' bicentralizer problem (the named open problem the hedge re-grounds on):

**Haagerup, U.** — "Connes' bicentralizer problem and uniqueness of the injective factor of type $\mathrm{III}_1$." *Acta Math.* **158** (1987) 95–148. — The biconditional linking bicentralizer-triviality to the existence of a state with an irreducible (large) modular centralizer. Cited for the genuine content of the iteration-11 A1 re-grounding (NOT for a reduction of the carrier no-go).

**Marrakchi, A.** — "Kadison's similarity problem and the bicentralizer conjecture" (arXiv:2308.15163). — Solves Kadison's problem *only for* type $\mathrm{III}_1$ factors that *satisfy* the bicentralizer conjecture; treats the conjecture itself as OPEN. Load-bearing source for "Connes' bicentralizer problem is open in general as of 2026" — the correction to the mission's false HMT-resolved premise.

**Marrakchi, A. & Vaes, S.** — *Crelle* (J. reine angew. Math.) **809** (2024) 247. — Theorem A: the modular centralizer $M_\psi = \mathbb{C}1$ is achievable on *every* type $\mathrm{III}_1$ factor, **unconditionally** (does not invoke the bicentralizer conjecture). The fact that breaks the submitted logical-equivalence reduction (killing a carrier needs trivial centralizer, achievable for free; the bicentralizer condition is the opposite).

**Ando, H., Haagerup, U., Houdayer, C. & Marrakchi, A.** — "Structure of bicentralizer algebras and inclusions of type III factors." *Math. Ann.* **376** (2020) 1145–1194 (arXiv:1804.05706). — The relative bicentralizer $BC_\varphi(\mathcal{N} \subset \mathcal{M})$ of an inclusion with expectation. Cited only as the *suggestive structural connection* for subregion-localized carriers (tagged SPECULATIVE — not an established reduction).

**Holography of information and its failure mode (A2):**

**Geng, H., Jafferis, D. L., Shrivastava, A. & Tata, S.** — "The Fate of Information Localizability and Holography in Quantum Gravity" (arXiv:2512.18912, 21 Dec 2025). — Keystone new datum. Verbatim-verified: *"holography cannot be proven at the perturbative level in Newton's constant $G_N$ via the non-localizability of information,"* via *"the emergence of a perturbatively localized observer."* The failure mechanism is re-localization via dressing to a nontrivial background / emergent observer-clock — co-locating holographic completeness with the carrier gap.

**Laddha, A., Prabhu, S., Raju, S. & Shrivastava, P.** — holography of information at null infinity (arXiv:2002.02448). — Massless information localized in an infinitesimal neighbourhood of the past boundary of future null infinity. (Self-correction from the digest: fourth author is Shrivastava, not "Sasmal.")

**Raju, S.** — holography of information in AdS / Wheeler–DeWitt (arXiv:2107.14802). — Two states agreeing at the boundary for an infinitesimal time agree everywhere in the bulk (perturbative in $G_N$).

**Chakraborty et al. / de Sitter holography of information** (arXiv:2303.16316). — Cosmological correlators in an arbitrarily small region completely specify the state. (Cited for the de Sitter instance of the conditional state-completeness result.)

**Operational / Tsirelson (A2):**

**Ji, Z., Natarajan, A., Vidick, T., Wright, J. & Yuen, H.** — "MIP*=RE" (arXiv:2001.04383). — $C_{qa} \subsetneq C_{qc}$ (negative answer to Tsirelson's problem; Connes embedding refuted). Cited for: operational correlation data does not pin down the operator algebra — but the underdetermination is in the split/tensor (Type-I) structure (the OP-48 collar), NOT in net-completeness.

*(T-duality/mirror symmetry in A2-F3 rests on standard textbook/established results — exact CFT isomorphism between geometrically distinct targets — and on the nLab "no complete construction of these SCFTs" status note for the full mirror-SCFT-isomorphism caveat; no new primary arXiv citation is load-bearing there, so none is added beyond the inline note.)*

## Iteration-12 verified additions (2026-06-19)

**Operator algebras (carrier-lever / bicentralizer).**

Ando, Haagerup, Houdayer, Marrakchi, *Structure of bicentralizer algebras and inclusions of type III factors*, Math. Ann. 376 (2020) 1145-1194 (arXiv:1804.05706). [VERIFIED] — defines/studies the **relative** bicentralizer of inclusions; its relative-bicentralizer theorem is restricted to **discrete** inclusions with N amenable (the discrete-only scope is the load-bearing fact for B1; physics local inclusions are non-discrete).

Already in the standing iter-11 corpus; re-cited here for the discrete-only scope.

Marrakchi, Vaes, *Ergodic states on type III_1 factors and ergodic actions*, Crelle (J. reine angew. Math.) 809 (2024) 247 (arXiv:2305.14217). [VERIFIED verbatim] — Theorem A: ergodic states (trivial centralizer M_psi = C1) form a dense G_delta on **any** III_1 factor, **UNCONDITIONAL**, no bicentralizer hypothesis. (Corrects the iter-11 A1 misattribution that called Thm A "conditional on the bicentralizer conjecture.")

Haagerup, *Connes' bicentralizer problem and uniqueness of the injective factor of type III_1*, Acta Math. 158 (1987) 95-148. [VERIFIED] — the large-centralizer biconditional (trivial bicentralizer <=> exists a faithful normal state with (M_psi)' cap M = C1) — the carrier-admitting direction. Already in the standing corpus.

Ando, Goldbring, *(model-theory / Connes bicentralizer)*, arXiv:2605.12776 (May 2026). [VERIFIED] — re-confirms Connes' bicentralizer problem is OPEN in general as of 2026. (Authors are Ando-Goldbring, not "I. Goldbring et al.")

**AQFT / encoding-screen (external-math sweep).**

Morinelli, Neeb, Olafsson, *(Euler-element / geometric Bisognano-Wichmann / Spin-Statistics program)*, arXiv:2508.10960; and Part II arXiv:2603.26390. [VERIFIED by abstract read] — BW / Spin-Statistics derived FROM the Euler element (symmetry is INPUT); a cleaner statement of the encoding screen, NOT a carrier, NOT a reverse-WW inversion, NOT a sixth route.

Morinelli (single-author), arXiv:2412.20410. [VERIFIED — single-author] — Euler-program; part of the same encoding-screen family.

arXiv:2511.09360 (Euler-program). [VERIFIED real per sweep] — per-paper authorship to be confirmed before bibliography insertion (not all three-author); mark [authorship unverified] until checked.

Liu, *(lecture notes, modular/AQFT)*, arXiv:2510.07017. [VERIFIED real] — presupposes geometry as input; fails the encode-not-generate screen.

Naaijkens, Penneys, Wallick, arXiv:2605.10693. [VERIFIED real] — LTO-internal Euclidean-side reflection-positivity axiom + Tomita-Takesaki for Haag duality on cone regions; ORTHOGONAL — constructs no Lorentzian metric, no indefinite pairing, no carrier witness; makes NO claim that J carries the Euclidean datum.

Weiner, *(algebraic Haag theorem)*, arXiv:1006.4726. [VERIFIED verbatim, standing corpus] — split + geometric modular action => the vacuum-sector net is a complete invariant (conditional on the geometric inputs).

Sheikh-Jabbari, Taghiloo, *"GR from RG"*, arXiv:2602.11806. [VERIFIED real] — RG-induced Einstein-Hilbert term with a UV-Dirichlet-fixed metric; explicitly engages Weinberg-Witten; an EVADE-not-invert WW instance (discharges the iter-11 "WW gloss not confirmable" minor defect).

**Experiment-watchlist (target/core, not wager).**

VIP / XENONnT objective-collapse bounds, primary arXiv:2506.05507 (PRL, March 2026). [VERIFIED real] — CSL ~2 orders, DP ~5x; tighten the A-3 core. Adjacent/auxiliary: arXiv:2512.15393, arXiv:2604.21705 [both VERIFIED real].

Huang, Cai, Wang, *(low-z SN calibration-bias analysis for DESI w(z))*, arXiv:2502.04212. [VERIFIED — abstract states dynamical-dark-energy preference "less than 2 sigma"] — Efstathiou in acknowledgments only (not an author).

## Iteration-14 verified additions (2026-06-19)

*From the maximal adversarial assault (the strongest 2024–2026 falsification attempts on the three pillars). The four load-bearing entries below were independently web-verified this session; the remaining encoding-screen instances surfaced by the assault are catalogued in the iteration-14 D-track notes (all referee-verified live).*

**Fullwood, J., Vedral, V. & Guzmán-González, E.** — "On Lorentzian symmetries of quantum information."

arXiv:2604.07471 (Apr 2026). [VERIFIED] — Derives the Minkowski form and $SO^{+}(1,3)$ from "preservation of linear entropy" on qubit observables (linear entropy $S_L = 2\,\mathrm{det}$; $\mathrm{det}$ = the SL(2,$\mathbb{C}$)-invariant on $\mathrm{Herm}_2(\mathbb{C})$). The strongest 2026 "signature-as-output" claim — but the (1,3) signature is intrinsic to the *complex* carrier (the determinant on $2 \times 2$ complex-Hermitian matrices has signature exactly (1,3), unconditionally), with no locality/net/causal order. Recorded as a 2026 instance of HYP-ENCODING-SCREEN (the indefinite-pairing-in-carrier smuggle), **not** a carrier and **not** a sixth route.

**Schneider, M. D.** — "Spacetime emergence: an (in)effective story." arXiv:2410.00932 (Philosophy of Physics, 2024). [VERIFIED] — Argues GR's spacetime "irreducibly include[s] global physical content, which is not effective" and that this global content "acts like a superselection rule that must be independently specified." An *external* (philosophy-of-physics) corroboration that locates the encode-screen's sole structural seam — the global/topological net-sharing premise — exactly where this wiki names its own soft spot; constructs no distinguishing observable.

**Chou, K.-H. & Chang, P.-Y.** — "Emergent de Sitter Space and Non-Unitary Tensor Networks from Non-Hermitian Quantum Criticality." arXiv:2606.17983 (Jun 2026). [VERIFIED] — A non-unitary cMERA built on a *pre-selected* non-Hermitian critical fermion chain; the de Sitter output emerges relative to the installed criticality (the coefficient is reproduced, not derived). Confirms the iteration-12 closure that the indefinite-metric / Krein route is route-5 in Krein vocabulary, not a sixth route.

**Verlinde, E. & Zurek, K. (GQuEST / geontropic):** "Modular fluctuations from shockwave geometry" / GQuEST design, arXiv:2404.07524 (PRX 15, 011034 (2025)); EFT baseline "Response of interferometers to the vacuum of quantum gravity," arXiv:2409.03894. [VERIFIED] — The geontropic interferometer signal; its proponents concede "conformal field theory, dilaton theory, and hydrodynamics" yield the same $\alpha =$

$O(1)$ amplitude, so it co-classes emergent and fundamental descriptions (screen-passing) and its amplitude re-imports $\eta = 1/4G$ at the modular→metric step. The freshest worked instance of the §7 no-test corollary.

### Iteration-15 verified additions (2026-06-19) — the un-tried mathematical frameworks

*From the final analytical iteration (factorization algebras / functorial QFT; index theory / K-theory; higher-categorical AQFT) — all reproduce the encode-not-generate screen. The Krein-spectral-triple entry was independently web-verified this session; the framework references are referee-verified, with attributions corrected per the referees.*

**Bizi, N., Brouder, C. & Besnard, F.** — "Space and time dimensions of algebras with applications to Lorentzian noncommutative geometry and quantum electrodynamics." arXiv:1611.07062. [VERIFIED] — Defines indefinite (pseudo-Riemannian) spectral triples "over Krein spaces instead of Hilbert spaces." The index-theory confirmation of the iteration-12 Krein closure: a Lorentzian spectral triple installs a Krein space whose indefinite inner product **is** the $(n, n)$ carrier $\eta$ — installed, not generated.

**Benini, M., Carmona, V., Grant-Stuart, A. & Schenkel, A.** — the categorical equivalence AQFT $\simeq$ time-orderable prefactorization algebra on globally hyperbolic Lorentzian manifolds (arXiv:2412.07318; building on arXiv:1903.03396). [referee-verified] — A new precise object but **not** a handle: both sides are functors over the same causal-disjointness / orthogonality relation (= the localization template $n_1$). A fourth independent witness to the same obstruction.

**Gwilliam, O. & Rejzner, K.** — "The observables of a perturbative algebraic quantum field theory form a factorization algebra." arXiv:2212.08175. [referee-verified] — pAQFT observables form a factorization algebra on a *fixed* Lorentzian manifold (the geometry is the base).

**Bunk, S., MacManus, J. & Schenkel, A.** — "Lorentzian bordisms in algebraic quantum field theory." arXiv:2308.01026 (Lett. Math. Phys. 115 (2025)). [referee-verified] — the Lorentzian bordism *site* for functorial AQFT (geometry in the site).

**Müller, L. & Stehouwer, L.** — reflection / dagger structures inducing an indefinite signature in (non-unitary) categorical field theory. arXiv:2301.06664. [referee-verified] — the categorical-vocabulary form of the iteration-12 installed-Krein problem.

## Watch-sweep #1 verified additions (2026-07-02) — post-saturation watch-mode

*From the first watch-mode sweep (window 2026-06-15 → 2026-07-02; Working note: Watch-mode sweep #1 — external-input channels, window 2026-06-15 → 2026-07-02 (repo: `notes/2026-07-02-watch-sweep-01.md`)). Zero referee-surviving strong findings; these entries are watch-channel records, not verdict movers. All web-verified live this session (un-refereed v1 preprints unless noted).*

**Marrakchi, A.** — "Ergodicity of the bicentralizer flow and Kadison's problem." arXiv:2606.23636 (v1 2026-06-22, math.OA). [VERIFIED] — Unconditional: the relative bicentralizer flow of a type $III_1$ irreducible subfactor **with expectation** is always ergodic; consequence: such subfactors contain a maximal abelian subalgebra, completing Kadison's 1967 problem. NOT bicentralizer triviality (Connes' problem stays OPEN); no carrier content (output is abelian/definite); does not transfer to the expectation-free physics-net inclusions (the iter-12 blocker). Watch value: the tracked adjacent cluster is actively moving.

**Bateman, S. & Turok, N.** — "Escape from Ostrogradsky via Hidden Ghost Parity." arXiv:2607.00096 (v1 2026-06-30, hep-th). [VERIFIED] — Four-derivative scalar QFT on a Krein space with a Krein Born rule and tree-level positivity via a "hidden ghost parity" κ from an O(1,1) embedding; the paper itself: "no preferred choice of ghost parity." A new instance of the iteration-12 Krein closure, channel (B) installed-not-derived ($n_1$ = fixed Minkowski commutator; κ global and non-unique); useful probabilistic-consistency machinery for indefinite carriers, no naturality content.

**Wakakuwa, E., Petruzziello, L., Lantaño, T. B., Huelga, S. F. & Plenio, M. B.** — "Relativistic Gravity-Induced Entanglement via Frame Dragging." arXiv:2606.31678 (v1 2026-06-30). [VERIFIED] — First genuinely post-Newtonian GIE channel (Lense–Thirring), derived two independent ways (quantized frame-dragging Hamiltonian; on-shell linearized-QG action). Gate-channel item (BMV/QGEM row); η and the causal order installed by design, openly.

**External clocks (news, verified live):** ESA Euclid DR1 restructured (published 2026-06-15) — DR1-Foundation Nov 2026 (no cosmology products), complete DR1 with GC+WL **mid-2027** (cosmos.esa.int/web/euclid/dr1-timeline); NASA Roman at KSC 2026-06-21, launch NET 2026-08-30, Falcon Heavy (science.nasa.gov/blogs/roman/2026/06/21/).

**Dated in-window markers, inside the screen (not load-bearing):** arXiv:2606.17951 (one-parameter phantom-crossing w(z) form); arXiv:2606.19427 (AI-in-the-loop w(z) parametrization search); arXiv:2606.21826 (DESI-era dark-energy review); arXiv:2606.30928 (Cardy-type open/closed CFT from conformal nets — the net is input); arXiv:2606.28868 (finite-dim Krein metric-congruence — η installed per representation).

## Iteration-16 verified additions (2026-07-03) — the multi-wedge iteration

*From the clock-(1)-triggered iteration (tracks G1–G4 + referees R1/R2/R3; Chapter 36 (p. 684)). All web-verified live this session.*

**Doplicher, S. & Longo, R.** — "Standard and split inclusions of von Neumann algebras." *Invent. Math.* **75** (1984) 493–536, DOI 10.1007/BF01388641. [VERIFIED — Springer/ADS/EuDML] — Standard split inclusions, the canonical interpolating type I factor, and the canonical implementation machinery. **Volume repair:** the iter-12 ledger's "Invent. Math. 73" was a typo, corrected in place this iteration.

**Morinelli, V. & Neeb, K.-H.** — "From local nets to Euler elements." *Adv. Math.* **458** (2024) 109960 (arXiv:2312.12182). [VERIFIED — published] — The two-directional, **BW-conditional** bridge between the two smuggle items: given BW, regularity forces Euler-generated modular groups; conversely an Euler element plus BW yields regularity and localizability. Theorem-level confirmation that n1 and η are interchangeable *relative to an installed BW/wedge structure* — deriving neither from (algebra, state). Hardens dossier §6.

**Borchers, H.-J.** — "The CPT-theorem in two-dimensional theories of local observables." *Commun. Math. Phys.* **143** (1992) 315–332, DOI 10.1007/BF02099011. [VERIFIED] — The Borchers commutation relations (spectrum condition + half-sided translations ⇒ modular scaling), the engine of LEM-NET-NATURALITY-G's (G-loc) branch.

**Marrakchi, A.** — arXiv:2606.23636 **full-source audit** (this iteration; abstract entry above under watch-sweep #1). [VERIFIED — full TeX source, twice independently downloaded] — The "with expectation" hypothesis is pervasive AND definitional (the relative bicentralizer flow is constructed against φ = φ∘E_N via Takesaki's theorem; the author flags the hypothesis "essential"); toolkit = averaging + ultrapower-implemented binormal states + approximate eigenstates of log Δ. Internal-bibliography caution: the preprint's "[Ha86] Acta Math. 69 (1986)" is a misprint (correct: Acta Math. **158** (1987) 95–148); its [Ma18] year is also suspect — import nothing from its bibliography unverified.

**Houdayer, C. & Marrakchi, A.** — "Selfless W*-probability spaces and Connes' bicentralizer problem." arXiv:2511.11409 (v2 2026-05-01; to appear J. Math. Soc. Japan per comments field). [VERIFIED — abstract verbatim] — States an **implication** (selflessness ⇒ bicentralizer-related structure), not an equivalence; the watch-sweep-#1 "equivalence" wording is repaired this iteration. Single-factor, no inclusion content; LOW relevance to expectation-free inclusions.

*AGHS / Arulseelan (continuous model theory of W-algebras, 2025 axiomatizations)** — arXiv:2508.17241 and companions. [VERIFIED — abstract level] — The axiomatizability infrastructure making the commutant-to-generated gap posable as a *definability* question (the DCNG watch item). Includes the II∞-undecidability sentence verified verbatim.

**Cipriani, F. & Sauvageot, J.-L.** — derivations from completely Dirichlet forms (the first-order differential calculus). [VERIFIED at listing level per the G4 referee] — The constitutive-positivity (g6) closure reason for the NC-Dirichlet-form readout: the form is completely *positive*-definite by construction; no indefinite version exists in the theory's foundations.

**Longo, R. & Xu, F.** — "Von Neumann Entropy in QFT." arXiv:1911.09390. [VERIFIED — attribution repaired per referee R2: Longo–Xu, not single-author] — The F-formula used in the G2 split-inclusion angle.

## Interpretation, measurement, and dynamical-collapse models (mostly `[CONTESTED]` / `[SPECULATIVE]`)

**H. Everett III**, "Relative state formulation of quantum mechanics," *Rev. Mod. Phys.* **29**, 454 (1957). `[CONTESTED]` — The no-collapse (relative-state / many-worlds) interpretation retaining only unitary evolution.

**D. Bohm**, "A suggested interpretation of the quantum theory in terms of hidden variables, I & II," *Phys. Rev.* **85**, 166 & 180 (1952). `[CONTESTED]` — The de Broglie–Bohm pilot-wave theory: an explicit deterministic, contextual, nonlocal hidden-variable completion empirically equivalent to standard QM for the usual observables.

**G. C. Ghirardi, A. Rimini, T. Weber**, "Unified dynamics for microscopic and macroscopic systems," *Phys. Rev. D* **34**, 470 (1986). `[SPECULATIVE]` (empirically testable, not yet decided) —

The GRW objective-collapse model: an explicit modification of unitary dynamics with experimentally bounded parameters.

**W. H. Zurek**, "Decoherence, einselection, and the quantum origins of the classical," *Rev. Mod. Phys.* **75**, 715 (2003). `[ESTABLISHED]` (decoherence mechanism) / `[CONTESTED]` (envariance derivation of the Born rule) — Authoritative review of decoherence, pointer-basis einselection, and the emergence of classicality.

*Functional-analysis underpinnings (Reed–Simon) are listed under §8; rigorous operator-algebra results (Gelfand–Naimark, Bargmann, Gleason) are cross-referenced from §8.*

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## 2. Quantum field theory and particle physics

Primary support for *Chapter 8 (p. 89)* and *Chapter 9 (p. 106)*.

### Textbooks

**S. Weinberg**, *The Quantum Theory of Fields*, Vols. I–III (Cambridge Univ. Press, 1995–2000). `[ESTABLISHED]` — The deepest first-principles treatment: derives QFT from Poincaré invariance, unitarity, and cluster decomposition; authoritative on Wigner's classification, renormalization, gauge theories, electroweak symmetry breaking, and anomalies.

**M. E. Peskin and D. V. Schroeder**, *An Introduction to Quantum Field Theory* (Addison-Wesley, 1995). `[ESTABLISHED]` — Standard graduate text for canonical + path-integral methods, renormalization, RG, gauge theories, and the chiral anomaly; the working calculational reference (and the QFT backbone for the Standard Model).

**M. D. Schwartz**, *Quantum Field Theory and the Standard Model* (Cambridge Univ. Press, 2014). `[ESTABLISHED]` — Modern text tying QFT formalism to SM phenomenology, EFT, and precision calculations.

**A. Zee**, *Quantum Field Theory in a Nutshell*, 2nd ed. (Princeton Univ. Press, 2010). `[ESTABLISHED]` — Path-integral-first and physically motivated; strong on the Wilsonian/EFT picture and conceptual structure.

**C. Itzykson and J.-B. Zuber**, *Quantum Field Theory* (McGraw-Hill, 1980). `[ESTABLISHED]` — Comprehensive classic on perturbation theory, renormalization, functional methods, and QED precision diagrammatics.

**C. Quigg**, *Gauge Theories of the Strong, Weak, and Electromagnetic Interactions*, 2nd ed. (Princeton Univ. Press, 2013). `[ESTABLISHED]` — Focused, physically motivated exposition of the SM gauge structure, electroweak theory, and the Higgs mechanism.

**R. F. Streater and A. S. Wightman**, *PCT, Spin and Statistics, and All That* (Benjamin, 1964; Princeton reissue 2000). `[ESTABLISHED]` — Canonical reference for the Wightman axioms, the reconstruction theorem, spin–statistics, and CPT — the rigorous operator/correlation foundation.

**R. Haag**, *Local Quantum Physics: Fields, Particles, Algebras*, 2nd ed. (Springer, 1996). `[ESTABLISHED]` — Definitive text on algebraic QFT (Haag–Kastler nets), DHR superselection, modular theory, Reeh–Schlieder, Bisognano–Wichmann, and Haag's theorem. *(Also primary for §8.)*

**J. Glimm and A. Jaffe**, *Quantum Physics: A Functional Integral Point of View*, 2nd ed. (Springer, 1987). `[ESTABLISHED]` — Authoritative account of constructive QFT: rigorous construction of $\phi^4$ in 2 and 3 dimensions, the Osterwalder–Schrader axioms, and the methods/limits of the program. *(Also primary for §8.)*

## Data reference

**Particle Data Group** (R. L. Workman *et al.*), *Review of Particle Physics*, *Prog. Theor. Exp. Phys.* (updated regularly; latest edition supersedes prior ones). `[ESTABLISHED]` — The standard source for measured SM parameters, masses, mixing matrices, and review articles. **Use for any numerical value** (see

*Chapter 16 (p. 308)*). [unverified] exact lead-author/year for the current edition; cite the edition in hand.

## Landmark papers — structure of QFT

**E. P. Wigner**, "On unitary representations of the inhomogeneous Lorentz group," *Annals of Mathematics* **40**, 149 (1939). [ESTABLISHED] — Classifies particles as unitary irreducible representations of the Poincaré group (labeled by mass $m$ and spin/helicity).

**K. G. Wilson and J. Kogut**, "The renormalization group and the $\varepsilon$ expansion," *Physics Reports* **12**, 75 (1974). [ESTABLISHED] — The Wilsonian RG/EFT reframing of renormalization; foundational for relevant/marginal/irrelevant operators and universality. *(Also primary for §5; cross-listed there.)*

**H. Reeh and S. Schlieder**, *Nuovo Cimento* **22**, 1051 (1961); **J. J. Bisognano and E. H. Wichmann**, *J. Math. Phys.* **16**, 985 (1975). [ESTABLISHED] — Vacuum cyclicity/entanglement (Reeh–Schlieder) and modular-flow-equals-boost (Bisognano–Wichmann); structural theorems linking locality, modular theory, and the Unruh effect.

**R. D. Sorkin**, "Impossible measurements on quantum fields," in *Directions in General Relativity*, Vol. 2 (1993); **C. J. Fewster and R. Verch**, "Quantum fields and local measurements," *Commun. Math. Phys.* **378**, 851 (2020). [OPEN] (problem) / [INFERENCE] (proposed framework) — The relativistic measurement problem (impossible measurements / superluminal signaling) and a modern consistent local-measurement framework.

## Landmark papers — the Standard Model

**S. L. Glashow**, *Nucl. Phys.* **22**, 579 (1961); **S. Weinberg**, *Phys. Rev. Lett.* **19**, 1264 (1967); **A. Salam** (1968). [ESTABLISHED] — Founding papers unifying the electroweak interactions.

**G. 't Hooft and M. Veltman**, *Nucl. Phys. B* **44**, 189 (1972). [ESTABLISHED] — Renormalizability of spontaneously broken non-Abelian gauge theories, making the electroweak theory predictive (Nobel 1999).

**P. W. Higgs**, *Phys. Rev. Lett.* **13**, 508 (1964); **F. Englert and R. Brout**, *Phys. Rev. Lett.* **13**, 321 (1964); **G. Guralnik, C. R. Hagen, T. W. B. Kibble**, *Phys. Rev. Lett.* **13**, 585 (1964). `[ESTABLISHED]` — The Brout–Englert–Higgs mechanism; the boson was discovered by ATLAS and CMS in 2012 (Nobel 2013).

**D. J. Gross and F. Wilczek**, *Phys. Rev. Lett.* **30**, 1343 (1973); **H. D. Politzer**, *Phys. Rev. Lett.* **30**, 1346 (1973). `[ESTABLISHED]` — Asymptotic freedom in non-Abelian gauge theory, the foundation of perturbative QCD (Nobel 2004).

**S. L. Adler**, *Phys. Rev.* **177**, 2426 (1969); **J. S. Bell and R. Jackiw**, *Nuovo Cimento A* **60**, 47 (1969). `[ESTABLISHED]` — The chiral (ABJ) anomaly: a classical symmetry broken by quantization, with physical consequences and gauge-anomaly-cancellation constraints.

**M. Kobayashi and T. Maskawa**, *Prog. Theor. Phys.* **49**, 652 (1973). `[ESTABLISHED]` — Three quark generations permit CP violation via a single CKM phase (Nobel 2008).

**R. D. Peccei and H. R. Quinn**, *Phys. Rev. Lett.* **38**, 1440 (1977); **S. Weinberg** (1978); **F. Wilczek** (1978). `[ESTABLISHED]` (mechanism) / `[OPEN]` (axion existence) — The Peccei–Quinn solution to the strong-CP problem and the prediction of the axion.

**A. D. Sakharov**, *JETP Lett.* **5**, 24 (1967). `[ESTABLISHED]` (necessary conditions) / `[OPEN]` (realized mechanism) — The three conditions (B violation, C/CP violation, departure from equilibrium) for dynamically generating the baryon asymmetry. See *Chapter 15 (p. 233)*.

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## 3. General relativity and gravitation

Primary support for *Chapter 10 (p. 121)*.

### Textbooks and treatises

**C. W. Misner, K. S. Thorne, J. A. Wheeler**, *Gravitation* (Freeman, 1973; reissued Princeton Univ. Press, 2017). `[ESTABLISHED]` — The canonical comprehensive reference ("MTW"); definitive on geometry, the equivalence principle, ADM, and gravitational waves. Source for sign-convention tables (see *Chapter 10 (p. 121)*).

**R. M. Wald**, *General Relativity* (Univ. of Chicago Press, 1984). `[ESTABLISHED]` — The standard rigorous graduate text; authoritative on causal structure, singularity theorems, energy, asymptotic flatness (ADM/Bondi), and QFT in curved spacetime.

**S. M. Carroll**, *Spacetime and Geometry: An Introduction to General Relativity* (Addison-Wesley, 2004). `[ESTABLISHED]` — Modern, clear graduate text; good on the Einstein–Hilbert variational derivation, diffeomorphism invariance, and observational connections.

**S. W. Hawking and G. F. R. Ellis**, *The Large Scale Structure of Space-Time* (Cambridge Univ. Press, 1973). `[ESTABLISHED]` — Definitive treatment of global causal structure, energy conditions, and the Hawking–Penrose singularity theorems. *(Also primary for §6; cross-listed there.)*

### Landmark papers and theorems

**R. Penrose**, "Gravitational collapse and space-time singularities," *Phys. Rev. Lett.* **14**, 57 (1965). `[ESTABLISHED]` — The original trapped-surface singularity theorem; foundational for black-hole and cosmological singularity results (2020 Nobel).

**R. Arnowitt, S. Deser, C. W. Misner**, "The dynamics of general relativity" (1962; reprinted as arXiv:gr-qc/0405109). `[ESTABLISHED]` — The original ADM $3 + 1$ Hamiltonian formulation, constraints, and definition of ADM energy.

**Y. Choquet-Bruhat and R. Geroch**, "Global aspects of the Cauchy problem in general relativity," *Commun. Math. Phys.*

**14**, 329 (1969). `[ESTABLISHED]` — Existence and uniqueness of the maximal globally hyperbolic development: the rigorous statement of GR's determinism.

**R. Schoen and S.-T. Yau**, "On the proof of the positive mass conjecture in general relativity," *Commun. Math. Phys.* **65**, 45 (1979); **E. Witten**, *Commun. Math. Phys.* **80**, 381 (1981). `[ESTABLISHED]` — Proofs of the positive-energy theorem ( $M_{\mathrm{ADM}} \geq 0$), foundational for the consistency of GR's notion of total energy.

**D. Christodoulou and S. Klainerman**, *The Global Nonlinear Stability of the Minkowski Space* (Princeton Univ. Press, 1993). `[ESTABLISHED]` — Landmark proof that Minkowski space is nonlinearly stable; a benchmark global-existence result in GR PDE theory.

**B. P. Abbott *et al.* (LIGO Scientific & Virgo Collaborations)**, "Observation of gravitational waves from a binary black hole merger," *Phys. Rev. Lett.* **116**, 061102 (2016). `[ESTABLISHED]` — First direct detection (GW150914); confirmation of GR's dynamical, strong-field, radiative regime.

**S. W. Hawking**, "Particle creation by black holes," *Commun. Math. Phys.* **43**, 199 (1975). `[ESTABLISHED]` (radiation) / `[OPEN]` (information fate) — Derivation of Hawking radiation; origin of the information paradox at the GR/QFT/thermodynamics interface. *(Load-bearing across §5, §6, §7; primary listing here.)*

**C. M. Will**, "The confrontation between general relativity and experiment," *Living Reviews in Relativity* **17**, 4 (2014). `[ESTABLISHED]` — Authoritative, regularly updated review of the PPN formalism and the tests defining GR's domain of validity. `[unverified]` exact article number of the most recent revision; cite the current *Living Reviews* version.

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# 4. Classical mechanics

Primary support for *Chapter 4 (p. 25)*.

### Textbooks

**H. Goldstein, C. Poole, J. Safko**, *Classical Mechanics*, 3rd ed. (Addison-Wesley, 2002). [ESTABLISHED] — Standard graduate text; canonical treatment of Lagrangian/Hamiltonian formalism, canonical transformations, Hamilton–Jacobi theory, rigid bodies, and chaos.

**L. D. Landau and E. M. Lifshitz**, *Mechanics* (Course of Theoretical Physics, Vol. 1), 3rd ed. (Pergamon, 1976). [ESTABLISHED] — Develops all of mechanics from the principle of least action with exceptional economy; authoritative on conservation laws, small oscillations, and the rigid body.

**V. I. Arnold**, *Mathematical Methods of Classical Mechanics*, 2nd ed. (Springer, GTM 60, 1989). [ESTABLISHED] — The definitive geometric/symplectic treatment: differential forms, symplectic manifolds, Liouville–Arnold integrability, action–angle variables, KAM; the bridge to modern mathematics (*Chapter 13 (p. 172)*).

**J. E. Marsden and T. S. Ratiu**, *Introduction to Mechanics and Symmetry*, 2nd ed. (Springer, 1999). [ESTABLISHED] — Authoritative on the geometry of symmetry: moment maps, Marsden–Weinstein symplectic reduction, Lie-group actions, and the modern form of Noether's theorem.

**V. I. Arnold, V. V. Kozlov, A. I. Neishtadt**, *Mathematical Aspects of Classical and Celestial Mechanics*, 3rd ed. (Springer, 2006). [ESTABLISHED] — Comprehensive modern survey of integrable systems, perturbation theory, KAM, Arnold diffusion, and the rigorous state of celestial mechanics.

**R. P. Feynman and A. R. Hibbs**, *Quantum Mechanics and Path Integrals* (McGraw-Hill, 1965). [ESTABLISHED] — The path-integral formulation $\propto e^{iS/\hbar}$ and the derivation of the classical least-action principle as its stationary-phase limit.

## Landmark papers and original sources

**E. Noether**, "Invariante Variationsprobleme," *Nachr. Ges. Wiss. Göttingen* (1918). `[ESTABLISHED]` — The original paper establishing the symmetry–conservation correspondence (Noether's first and second theorems).

**H. Poincaré**, *Les Méthodes Nouvelles de la Mécanique Céleste*, 3 vols. (Gauthier-Villars, 1892–1899). `[ESTABLISHED]` — Landmark origin of qualitative dynamics, the three-body problem, homoclinic tangles, and deterministic chaos / nonintegrability.

**A. N. Kolmogorov** (1954), **V. I. Arnold** (1963), **J. Moser** (1962) — the original KAM papers; synthesized in **J. Moser**, *Stable and Random Motions in Dynamical Systems* (Princeton Univ. Press, 1973). `[ESTABLISHED]` — Persistence of invariant tori under perturbation; foundational for the integrable–chaotic transition.

**P. A. M. Dirac**, *Lectures on Quantum Mechanics* (Belfer Graduate School, 1964); and "Generalized Hamiltonian dynamics" (1950). `[ESTABLISHED]` — Constraint analysis of singular Lagrangians (Dirac–Bergmann), Dirac brackets, and the bracket→commutator correspondence.

**F. Bayen, M. Flato, C. Fronsdal, A. Lichnerowicz, D. Sternheimer**, "Deformation theory and quantization, I & II," *Ann. Phys. (NY)* **111** (1978). `[ESTABLISHED]` (math) — Founding papers of deformation quantization and the star product, formalizing quantization as a deformation of the classical Poisson algebra.

**Z. Xia**, "The existence of noncollision singularities in Newtonian systems," *Annals of Mathematics* **135** (1992), 411–468. `[ESTABLISHED]` — Rigorous proof that finite-time noncollision singularities exist in the Newtonian $N$-body problem ($N \geq 5$), bounding global determinism. See *Chapter 14 (p. 191)*.

**J. D. Norton**, "The dome: an unexpectedly simple failure of determinism," *Philosophy of Science* **75** (2008). `[CONTESTED]` —

Sharp articulation of the non-Lipschitz indeterminism debate internal to Newtonian mechanics.

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# 5. Thermodynamics and statistical mechanics

Primary support for *Chapter 5 (p. 40)* and *Chapter 6 (p. 56)*.

## Textbooks

**H. B. Callen**, *Thermodynamics and an Introduction to Thermostatistics*, 2nd ed. (Wiley, 1985). `[ESTABLISHED]` — The canonical axiomatic (postulatory) treatment: fundamental relation, extremum principles, Legendre transforms, Maxwell relations, stability.

**E. Fermi**, *Thermodynamics* (Dover, 1956); **M. W. Zemansky and R. H. Dittman**, *Heat and Thermodynamics* (McGraw-Hill; multiple eds.). `[ESTABLISHED]` — Classic clear expositions of the four laws, Carnot's theorem, and the operational definition of absolute temperature. `[unverified]` exact edition/year of the Zemansky–Dittman volume in use.

**L. D. Landau and E. M. Lifshitz**, *Statistical Physics, Part 1* (Course of Theoretical Physics, Vol. 5), 3rd ed. (Pergamon). `[ESTABLISHED]` — Authoritative bridge from statistical mechanics to thermodynamics: equilibrium ensembles, fluctuations, phase transitions, and the third law from quantum statistics.

**K. Huang**, *Statistical Mechanics*, 2nd ed. (Wiley, 1987). `[ESTABLISHED]` — Standard pedagogical text: Boltzmann equation and H-theorem, the ensembles, quantum gases, and the Ising model via the transfer matrix.

**R. K. Pathria and P. D. Beale**, *Statistical Mechanics*, 3rd ed. (Elsevier/Academic Press). `[ESTABLISHED]` — Comprehensive coverage of quantum statistics, the grand canonical ensemble, BEC, fluctuation–dissipation, and an introduction to critical phenomena and the RG.

**L. E. Reichl**, *A Modern Course in Statistical Physics* (Wiley; multiple eds.). `[ESTABLISHED]` — Strong on nonequilibrium statistical mechanics, kinetic theory, linear response, and the fluctuation–dissipation theorem.

**N. Goldenfeld**, *Lectures on Phase Transitions and the Renormalization Group* (Addison-Wesley, 1992). `[ESTABLISHED]` — Conceptually careful treatment of scaling, universality, and Wilsonian RG, including the $\varepsilon$-expansion.

**A. I. Khinchin**, *Mathematical Foundations of Statistical Mechanics* (Dover, 1949). `[ESTABLISHED]` — Rigorous demonstration that the equivalence of time and phase averages for macroscopic sum-functions rests on high dimensionality, not strict ergodicity — a cornerstone of the typicality viewpoint.

## Historical origins

**S. Carnot**, *Réflexions sur la puissance motrice du feu* (1824); **R. Clausius** (1865, introduction of entropy); **L. Boltzmann**, *Lectures on Gas Theory* (1896–98). `[ESTABLISHED]` — Origins of the Carnot bound, the entropy concept, and the kinetic foundation of the second law.

## Landmark papers — equilibrium, critical phenomena, foundations

**L. Onsager**, "Crystal statistics. I. A two-dimensional model with an order–disorder transition," *Phys. Rev.* **65**, 117 (1944). `[ESTABLISHED]` — Exact solution of the 2D Ising model; the landmark demonstration of a genuine phase transition with non-mean-field critical exponents.

**K. G. Wilson and J. Kogut**, "The renormalization group and the $\varepsilon$ expansion," *Phys. Rep.* **12**, 75 (1974). `[ESTABLISHED]` — RG explanation of critical phenomena and universality (Wilson, Nobel 1982). *(Primary entry; also load-bearing in §2.)*

**E. T. Jaynes**, "Information theory and statistical mechanics," *Phys. Rev.* **106**, 620 (1957). `[CONTESTED]` — The maximum-entropy / inferential reinterpretation of statistical mechanics;

central to the disputed information-theoretic foundations. See *Chapter 12 (p. 155)*.

**J. L. Lebowitz**, "Boltzmann's entropy and time's arrow," *Physics Today* **46**(9), 32 (1993). `[INFERENCE]` — Influential modern statement of the typicality/Boltzmann-entropy resolution of the irreversibility paradoxes (Loschmidt, Zermelo) and the role of the low-entropy past.

**O. E. Lanford III**, "Time evolution of large classical systems," in *Dynamical Systems, Theory and Applications*, Lecture Notes in Physics **38** (Springer, 1975). `[ESTABLISHED]` — Rigorous derivation of the Boltzmann equation from Hamiltonian dynamics (Boltzmann–Grad limit) for short times, clarifying the status of molecular chaos.

## Landmark papers — nonequilibrium and fluctuation theorems

**L. Onsager**, *Phys. Rev.* **37**, 405 and **38**, 2265 (1931); **S. R. de Groot and P. Mazur**, *Non-Equilibrium Thermodynamics* (Dover, 1984). `[ESTABLISHED]` — Onsager reciprocal relations and linear irreversible thermodynamics; the near-equilibrium extension and connection to transport.

**R. Kubo**, "Statistical-mechanical theory of irreversible processes I," *J. Phys. Soc. Jpn.* **12**, 570 (1957). `[ESTABLISHED]` — Linear-response theory and the fluctuation–dissipation theorem (Green–Kubo relations for transport coefficients).

**C. Jarzynski**, "Nonequilibrium equality for free energy differences," *Phys. Rev. Lett.* **78**, 2690 (1997). `[ESTABLISHED]` — The Jarzynski equality $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$, an exact nonequilibrium identity refining the second law.

**G. E. Crooks**, "Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences," *Phys. Rev. E* **60**, 2721 (1999). `[ESTABLISHED]` — The Crooks fluctuation theorem relating forward/reverse work distributions; the detailed result from which Jarzynski follows.

**U. Seifert**, "Stochastic thermodynamics, fluctuation theorems and molecular machines," *Rep. Prog. Phys.* **75**, 126001 (2012). `[ESTABLISHED]` — Comprehensive review of stochastic thermodynamics and trajectory-level entropy production.

## Landmark papers — quantum thermalization

**J. M. Deutsch**, *Phys. Rev. A* **43**, 2046 (1991); **M. Srednicki**, "Chaos and quantum thermalization," *Phys. Rev. E* **50**, 888 (1994). `[INFERENCE]` (hypothesis, strongly supported) — The original formulations of the Eigenstate Thermalization Hypothesis (ETH) explaining thermalization in isolated quantum systems.

**L. D'Alessio, Y. Kafri, A. Polkovnikov, M. Rigol**, "From quantum chaos and eigenstate thermalization to statistical mechanics and thermodynamics," *Adv. Phys.* **65**, 239 (2016). `[INFERENCE]` — Comprehensive modern review tying ETH, quantum chaos, random-matrix theory, integrability/GGE, and the foundations of quantum statistical mechanics.

## Black-hole thermodynamics (interface with §3, §6, §7)

**J. D. Bekenstein**, "Black holes and entropy," *Phys. Rev. D* 7, 2333 (1973). `[ESTABLISHED]` (within semiclassical gravity) — Black-hole entropy $S = A/4$ (in $G = \hbar = c = k_B = 1$ units). *(Primary entry; companion to Hawking 1975, §3.)*

**J. M. Bardeen, B. Carter, S. W. Hawking**, "The four laws of black hole mechanics," *Commun. Math. Phys.* **31**, 161 (1973). `[ESTABLISHED]` — The isomorphism between black-hole mechanics and the four laws of thermodynamics.

**A. Strominger and C. Vafa**, "Microscopic origin of the Bekenstein–Hawking entropy," *Phys. Lett. B* **379**, 99 (1996). `[ESTABLISHED]` (for the specific BPS cases) / `[OPEN]` (general case) — String-theoretic microstate counting reproducing $S = A/4$ for certain extremal black holes. See *Chapter 15 (p. 233)*.

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# 6. Cosmology

Primary support for *Chapter 11 (p. 139)*.

## Textbooks

**S. Weinberg**, *Cosmology* (Oxford Univ. Press, 2008). `[ESTABLISHED]` — Authoritative graduate text; rigorous FLRW dynamics, the CMB and perturbation theory, and the anthropic bound on Λ.

**S. Dodelson and F. Schmidt**, *Modern Cosmology*, 2nd ed. (Academic Press, 2020). `[ESTABLISHED]` — Standard reference for the Einstein–Boltzmann hierarchy, CMB and matter power spectra, and the formalism behind CAMB/CLASS.

**V. Mukhanov**, *Physical Foundations of Cosmology* (Cambridge Univ. Press, 2005). `[ESTABLISHED]` (formalism) — Rigorous cosmological perturbation theory and the inflationary generation of primordial fluctuations.

**D. Baumann**, *Cosmology* (Cambridge Univ. Press, 2022). `[ESTABLISHED]` — Modern pedagogical synthesis of background dynamics, inflation, and perturbations with current observational framing.

**E. W. Kolb and M. S. Turner**, *The Early Universe* (Addison-Wesley, 1990). `[ESTABLISHED]` — Classic reference on thermal history, freeze-out, BBN, and relics; standard source for the hot-Big-Bang formalism.

**G. F. R. Ellis, R. Maartens, M. A. H. MacCallum**, *Relativistic Cosmology* (Cambridge Univ. Press, 2012). `[ESTABLISHED]` — Careful treatment of the cosmological principle, averaging/backreaction (the fitting problem), and the foundations of the FLRW idealization.

## Data reference

**Planck Collaboration**, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020). `[ESTABLISHED]` — The definitive CMB determination of the six Λ

CDM parameters; baseline for the $H_0$-tension and curvature discussions. **Use for numerical values** (see *Chapter 16 (p. 308)*); the wiki states such values approximately by design.

## Landmark papers and theorems

**A. H. Guth**, "Inflationary universe: a possible solution to the horizon and flatness problems," *Phys. Rev. D* **23**, 347 (1981). `[INFERENCE]` (paradigm, strongly favored) — Foundational inflation paper articulating the horizon, flatness, and monopole problems.

**A. Borde, A. H. Guth, A. Vilenkin**, "Inflationary spacetimes are not past-complete," *Phys. Rev. Lett.* **90**, 151301 (2003). `[ESTABLISHED]` (theorem) / `[SPECULATIVE]` (multiverse reading) — The BGV theorem: even eternally inflating spacetimes have a past boundary.

**S. Weinberg**, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1 (1989). `[OPEN]` — The canonical statement of the cosmological-constant problem and survey of attempted solutions; still the reference framing. See *Chapter 15 (p. 233)*.

**J. Martin**, "Everything you always wanted to know about the cosmological constant problem (but were afraid to ask)," arXiv:1205.3365 *C. R. Physique* **13**, 566–665 (2012), DOI 10.1016/j.crhy.2012.04.008. `[ESTABLISHED]` (pedagogy) — The standard regularization-aware treatment: dimensional regularization gives $m^4 \ln \mu$ structure, not $M_{\text{cut}}^4$.

**C. P. Burgess**, "The Cosmological Constant Problem: Why it's hard to get Dark Energy from Micro-physics," Les Houches lectures, arXiv:1309.4133. `[ESTABLISHED]` (pedagogy) — Naturalness / radiative-instability framing across EFT thresholds.

**A. Padilla**, "Lectures on the Cosmological Constant Problem," arXiv:1502.05296. `[ESTABLISHED]` (pedagogy) — Works Weinberg's no-go theorem through in detail; radiative instability; surveys sequestering and other exits.

**N. Kaloper, A. Padilla**, "Sequestering the Standard Model Vacuum Energy," *Phys. Rev. Lett.* **112**, 091304 (2014) (arXiv:1309.6562); **N. Kaloper, A. Padilla, D. Stefanyszyn, G. Zahariade**, "A Manifestly Local Theory of Vacuum Energy Sequestering," *Phys. Rev. Lett.* **116**, 051302 (2016) (arXiv:1505.01492). `[SPECULATIVE]` — Matter-loop vacuum-energy cancellation; the global version predicts transient dark energy and future collapse; graviton loops not covered by these versions.

**R. Carballo-Rubio, L. J. Garay, G. García-Moreno**, "Unimodular gravity vs general relativity: a status report," *Class. Quantum Grav.* **39**, 243001 (2022) (arXiv:2207.08499). `[ESTABLISHED]` (review) — UG ≡ GR except Λ-as-integration-constant; no further classical/semiclassical/quantum differences found; the "technically natural" reading is `[CONTESTED]`.

**A. Salvio**, "Unimodular Quadratic Gravity and the Cosmological Constant," *Phys. Lett. B* **856** (2024) 138920 (arXiv:2406.12958). `[ESTABLISHED]` (as literature verdict) — States explicitly that unimodular gravity does not solve the "new" CC problem (the observed value).

**DESI Collaboration**, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints," *Phys. Rev. D* **112**, 083515 (2025) (arXiv:2503.14738); **K. Lodha et al. (DESI)**, "Extended Dark Energy analysis using DESI DR2 BAO measurements," arXiv:2503.14743. `[ESTABLISHED]` (measurements) / `[CONTESTED]` (dynamical-DE interpretation) — The 3.1σ (BAO+CMB) / 2.8–4.2σ (+SNe) $w_0 w_a$ preference; parameterization-robustness and phantom-crossing preference.

**G. Efstathiou**, "Evolving Dark Energy or Supernovae Systematics?," arXiv:2408.07175; **D. D. Y. Ong, D. Yallup, W. Handley**, "A Bayesian Perspective on Evidence for Evolving Dark Energy," arXiv:2511.10631. `[CONTESTED]` — The counter-analyses: the preference is SN-sample-driven; in Bayes-factor terms DESI DR2+CMB alone modestly favor Λ

CDM, and the signal largely vanishes with recalibrated DES-Dovekie SNe (ln B −0.57±0.26 → −0.30±0.19).

**J. de Cruz Pérez, A. Gómez-Valent, J. Solà Peracaula**, "Dynamical Dark Energy models in light of the latest observations," arXiv:2512.20616 (accepted 2026). `[CONTESTED]` — Running-vacuum variants vs DESI DR2 + Planck PR4 + SNe; evidence comparable to $w_0 w_a$CDM, strongest with DES-Y5.

**L. A. Anchordoqui, I. Antoniadis, D. Lüst**, "S-dual Quintessence, the Swampland, and the DESI DR2 Results," arXiv:2503.19428. `[SPECULATIVE]` — Swampland-consistent (distance/dS/TCC) quintessence matching the DESI-fitted potential.

**R. Penrose**, "Singularities and time-asymmetry," in *General Relativity: An Einstein Centenary Survey*, eds. S. W. Hawking and W. Israel (Cambridge Univ. Press, 1979); and *The Road to Reality* (Jonathan Cape, 2004). `[SPECULATIVE]` — Origin of the Weyl curvature hypothesis and the gravitational-entropy framing of the low-entropy past and the arrow of time. See *Chapter 19 (p. 392)*.

**S. W. Hawking and G. F. R. Ellis**, *The Large Scale Structure of Space-Time* (Cambridge Univ. Press, 1973). `[ESTABLISHED]` — Singularity theorems and the rigorous geometric basis for the breakdown of FLRW at the initial singularity. *(Primary listing in §3.)*

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## 7. Information theory and quantum information

Primary support for *Chapter 12 (p. 155)*.

### Textbooks

**T. M. Cover and J. A. Thomas**, *Elements of Information Theory*, 2nd ed. (Wiley, 2006). `[ESTABLISHED]` — The canonical classical reference: AEP, data-processing inequality, capacity, rate–distortion, and the Kolmogorov-complexity chapter.

**M. Li and P. Vitányi**, *An Introduction to Kolmogorov Complexity and Its Applications* (Springer; multiple eds.). `[ESTABLISHED]` — Definitive treatment of algorithmic information theory: the invariance theorem, incomputability, relation to Shannon entropy, and Chaitin's results.

**M. A. Nielsen and I. L. Chuang**, *Quantum Computation and Quantum Information* (Cambridge Univ. Press, 2000/2010). `[ESTABLISHED]` — The standard quantum-information text: von Neumann entropy, no-cloning, teleportation, entanglement measures, channel capacities, error correction. *(Primary listing in §1.)*

**M. M. Wilde**, *Quantum Information Theory*, 2nd ed. (Cambridge Univ. Press, 2017). `[ESTABLISHED]` — Modern rigorous treatment of channels, entropies, and the information-theoretic structure underlying the operational generalization of the QM axioms.

## Foundational papers — classical and quantum information

**C. E. Shannon**, "A mathematical theory of communication," *Bell System Technical Journal* **27**, 379 & 623 (1948). `[ESTABLISHED]` — Founding paper: entropy, mutual information, source and channel coding theorems, channel capacity.

**E. H. Lieb and M. B. Ruskai**, "Proof of the strong subadditivity of quantum-mechanical entropy," *J. Math. Phys.* **14**, 1938 (1973). `[ESTABLISHED]` — Proof of strong subadditivity, the single most important entropy inequality, underlying both quantum information and holographic entropy constraints.

**J. A. Wheeler**, "Information, physics, quantum: the search for links," in *Complexity, Entropy and the Physics of Information* (Addison-Wesley, 1990). `[SPECULATIVE]` — The "it from bit" manifesto framing physics as fundamentally informational; the conceptual lodestar of the domain. See *Chapter 17 (p. 325)*.

## Thermodynamics of computation (interface with §5)

**R. Landauer**, "Irreversibility and heat generation in the computing process," *IBM J. Res. Dev.* **5**, 183 (1961). `[ESTABLISHED]` — Landauer's principle: erasing one bit dissipates at least $k_B T \ln 2$ of heat.

**C. H. Bennett**, "The thermodynamics of computation — a review," *Int. J. Theor. Phys.* **21**, 905 (1982). `[ESTABLISHED]` — Reversible computation and the resolution of the Maxwell-demon paradox via the cost of erasure.

**N. Margolus and L. B. Levitin**, "The maximum speed of dynamical evolution," *Physica D* **120**, 188 (1998); **S. Lloyd**, "Ultimate physical limits to computation," *Nature* **406**, 1047 (2000). `[ESTABLISHED]` (bounds) — The quantum speed limit / maximum operation rate and Lloyd's synthesis of physical bounds into computational ceilings.

## Holography and "spacetime from entanglement"

**J. D. Bekenstein**, "Universal upper bound on the entropy-to-energy ratio for bounded systems," *Phys. Rev. D* **23**, 287 (1981); **G. 't Hooft** (1993) and **L. Susskind**, "The world as a hologram," *J. Math. Phys.* **36**, 6377 (1995); **R. Bousso**, "A covariant entropy conjecture," *JHEP* 07 (1999) 004, and "The holographic principle," *Rev. Mod. Phys.* **74**, 825 (2002). `[INFERENCE]` (Bekenstein/Bousso bounds) / `[SPECULATIVE]` (holographic principle as fundamental) — The information-bound backbone of holography. `[unverified]` exact JHEP article number of Bousso 1999.

**S. Ryu and T. Takayanagi**, "Holographic derivation of entanglement entropy from AdS/CFT," *Phys. Rev. Lett.* **96**, 181602 (2006); **V. E. Hubeny, M. Rangamani, T. Takayanagi**, *JHEP* 07 (2007) 062. `[INFERENCE]` (within AdS/CFT) — The RT formula and its covariant (HRT) generalization: entanglement entropy as a minimal/extremal bulk area.

**A. Almheiri, X. Dong, D. Harlow**, "Bulk locality and quantum error correction in AdS/CFT," *JHEP* 04 (2015) 163; **F. Pastawski, B. Yoshida, D. Harlow, J. Preskill**, "Holographic quantum error-correcting codes," *JHEP* 06 (2015) 149. `[SPECULATIVE]` — AdS/CFT as a quantum error-correcting code and the HaPPY tensor-network model.

**T. Jacobson**, "Thermodynamics of spacetime: the Einstein equation of state," *Phys. Rev. Lett.* **75**, 1260 (1995); **M. Van Raamsdonk**, "Building up spacetime with quantum entanglement," *Gen. Rel. Grav.* **42**, 2323 (2010); **J. Maldacena and L. Susskind**, "Cool horizons for entangled black holes," *Fortsch. Phys.* **61**, 781 (2013). `[SPECULATIVE]` — Jacobson's thermodynamic derivation of Einstein's equations, the "spacetime from entanglement" thesis, and the ER=EPR conjecture. See *Chapter 18 (p. 349)*.

**D. N. Page**, "Information in black hole radiation," *Phys. Rev. Lett.* **71**, 3743 (1993). `[ESTABLISHED]` — The Page curve. **Island/QES core (coordinates verified 2026-07-02, arXiv abs pages + journal records fetched): G. Penington**, "Entanglement wedge reconstruction and the information paradox," arXiv:1905.08255, *JHEP* **09** (2020) 002; **A. Almheiri, N. Engelhardt, D. Marolf, H. Maxfield**, "The entropy of bulk quantum fields and the entanglement wedge of an evaporating black hole," arXiv:1905.08762, *JHEP* **12** (2019) 063; **G. Penington, S. H. Shenker, D. Stanford, Z. Yang**, "Replica wormholes and the black hole interior," arXiv:1911.11977, *JHEP* **03** (2022) 205; **A. Almheiri, T. Hartman, J. Maldacena, E. Shaghoulian, A. Tajdini**, "Replica wormholes and the entropy of Hawking radiation," arXiv:1911.12333, *JHEP* **05** (2020) 013. `[OPEN]` (paradox mechanism) / `[INFERENCE]` (island Page curve within controlled settings).

**Islands — standing objections (all verified 2026-07-02): H. Geng, A. Karch**, "Massive islands," arXiv:2006.02438, *JHEP* **09** (2020) 121; **H. Geng, A. Karch, C. Perez-Pardavila, S. Raju, L. Randall, M. Riojas, S. Shashi**, "Information transfer with a gravitating bath," arXiv:2012.04671, *SciPost Phys.***10**, 103 (2021), and "Inconsistency of islands in theories with long-range gravity," arXiv:2107.03390, *JHEP***01** (2022) 182; **H. Geng**, "Graviton mass and entanglement islands in low spacetime dimensions," arXiv:2312.13336, *SciPost Phys.***19**, 146 (2025); "Replica wormholes and entanglement islands in the Karch–Randall braneworld," arXiv:2405.14872, *JHEP***01** (2025) 063; "Making the case for massive islands," arXiv:2509.22775 (no journal ref); **S. Demulder, A. Gnecchi, I. Lavdas, D. Lüst**, "Islands and light gravitons in type IIB string theory," arXiv:2204.03669, *JHEP***02** (2023) 016. `[CONTESTED]` — the massive-graviton / gravitational-Gauss-law critique and its string-embedded refinements.

**Ensemble-vs-single-theory (verified 2026-07-02): D. Marolf, H. Maxfield**, "Transcending the ensemble: baby universes, spacetime wormholes, and the order and disorder of black hole information," arXiv:2002.08950, *JHEP* **08** (2020) 044; **J.-M. Schlenker, E. Witten**, "No ensemble averaging below the black hole threshold," arXiv:2202.01372, *JHEP* **07** (2022) 143. `[CONTESTED]` — whether replica-wormhole entropies are single-theory statements; see also arXiv:2510.06376 / 2605.15180 (OP-44 fourth option) and the OP-50 cluster.

**Crossed-product founding trio (verified 2026-07-02): E. Witten**, "Gravity and the crossed product," arXiv:2112.12828, *JHEP* **10** (2022) 008; **V. Chandrasekaran, R. Longo, G. Penington, E. Witten**, "An algebra of observables for de Sitter space," arXiv:2206.10780, *JHEP* **02** (2023) 082; **V. Chandrasekaran, G. Penington, E. Witten**, "Large N algebras and generalized entropy," arXiv:2209.10454, *JHEP* **04** (2023) 009; **E. Witten**, "A background independent algebra in quantum gravity," arXiv:2308.03663 (no journal ref listed). `[ESTABLISHED as mathematics / INFERENCE as physics]` — Type $\mathrm{III}_1 \to \mathrm{II}$ via the modular crossed product; vN entropy $= S_{\mathrm{gen}}$ up to an additive constant; the observer/clock is installed. (2024–2026 follow-ups on record in FINDINGS and the iteration-6 note: 2309.15897; 2306.01837; 2406.02116; 2406.01669; 2407.01695; 2501.01487; 2411.19931; 2601.07910; 2601.07915.)

**C. Akers, N. Engelhardt, D. Harlow, G. Penington, S. Vardhan**, "The black hole interior from non-isometric codes and complexity," arXiv:2207.06536 (no journal ref listed on arXiv). `[SPECULATIVE]` (leading proposal) — leading post-island account of interior reconstruction.

**S. Raju**, "Lessons from the information paradox," arXiv:2012.05770 [unverified journal-ref: *Phys. Rept.* (2022)]. — The holography-of-information counterpoint: exponentially small correlations + boundary completeness in massless gravity; companion cluster already on record at the A2 entries.

*(Hawking 1975 — the origin of the information paradox — is listed in §3.)*

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# 8. Mathematical physics

Primary support for *Chapter 13 (p. 172)*.

## Functional analysis and operator algebras

**M. Reed and B. Simon**, *Methods of Modern Mathematical Physics*, Vols. I–IV (Academic Press, 1972–1979). `[ESTABLISHED]` — The canonical rigorous reference for functional analysis in physics: Hilbert spaces, self-adjointness and self-adjoint extensions (deficiency indices), the spectral theorem, Stone's theorem, and scattering. Vol. I (Functional Analysis), II (Self-Adjointness), III (Scattering), IV (Analysis of Operators). *(Load-bearing for §1.)*

**M. Takesaki**, *Theory of Operator Algebras*, Vols. I–III (Springer, 1979/2003). `[ESTABLISHED]` — The definitive treatise on $C^*$- and von Neumann algebras, the Murray–von Neumann/Connes type classification, and Tomita–Takesaki modular theory with the KMS condition.

**R. Haag**, *Local Quantum Physics: Fields, Particles, Algebras*, 2nd ed. (Springer, 1996). `[ESTABLISHED]` — Algebraic QFT, the type-

III nature of local algebras, modular theory in physics, DHR superselection. *(Primary listing in §2.)*

## Geometry, topology, and representation theory

**M. Nakahara**, *Geometry, Topology and Physics*, 2nd ed. (IOP, 2003). `[ESTABLISHED]` — Widely used physicist's reference for manifolds, differential forms, fiber bundles, connections and curvature, characteristic classes, the Atiyah–Singer index theorem, and applications to gauge theory, anomalies, and instantons.

**A. O. Barut and R. Rączka**, *Theory of Group Representations and Applications* (PWN, 1977); **M. E. Taylor**, *Noncommutative Harmonic Analysis* (AMS, 1986). `[ESTABLISHED]` — Standard references for Lie groups, Lie algebras, and unitary representation theory in physics, including induced representations and Wigner's classification of Poincaré irreps. `[unverified]` exact year of the Taylor volume.

## Rigorous QFT, constructive results, and modern frameworks

**J. Glimm and A. Jaffe**, *Quantum Physics: A Functional Integral Point of View*, 2nd ed. (Springer, 1987). `[ESTABLISHED]` — Constructive QFT and the rigorous Euclidean path integral, including the Osterwalder–Schrader axioms and the construction of $P(\phi)_2$ and $\phi^4_3$. *(Primary listing in §2.)*

**A. Jaffe and E. Witten**, "Quantum Yang–Mills theory" (Clay Mathematics Institute Millennium Problem description, 2000). `[OPEN]` — The official statement of the Yang–Mills existence and mass-gap problem. See *Chapter 15 (p. 233)* and *Chapter 14 (p. 191)*.

**M. Aizenman** (*Phys. Rev. Lett.* 1981; *Commun. Math. Phys.* 1982) and **M. Aizenman and H. Duminil-Copin**, "Marginal triviality of the scaling limits of critical 4D Ising and $\phi^4_4$ models," *Annals of Mathematics* **194** (2021), 163–235. `[ESTABLISHED]` — Rigorous proof of triviality of the continuum limit of $\phi^4$ in four dimensions: some 4D theories exist only as cutoff effective theories. Central to the domain-of-validity discussion in *Chapter 8 (p. 89)*.

**J. Lurie**, "On the classification of topological field theories," *Current Developments in Mathematics 2008* (2009), 129–280. `[INFERENCE]` (proof program/sketch) — The cobordism hypothesis; central result on fully extended TQFTs and higher-categorical structures.

**K. Costello and O. Gwilliam**, *Factorization Algebras in Quantum Field Theory*, Vols. 1–2 (Cambridge Univ. Press, 2017/2021). `[INFERENCE]` (perturbative scope) — Rigorous formulation of perturbative QFT via factorization algebras and the Batalin–Vilkovisky formalism; a leading bridge between physicists' QFT and rigorous mathematics.

### Structural theorems and the meta-question

**V. Bargmann**, "On unitary ray representations of continuous groups," *Ann. Math.* **59** (1954); **A. M. Gleason** (1957, see §1); **I. M. Gelfand and M. A. Naimark** (1943) and **I. E. Segal** foundational $C^*$-algebra papers. `[ESTABLISHED]` — Projective/ray representations (Bargmann–Wigner), the near-uniqueness of the Born rule (Gleason), and the representability of $C^*$-algebras (Gelfand–Naimark–Segal).

**E. P. Wigner**, "The unreasonable effectiveness of mathematics in the natural sciences," *Comm. Pure Appl. Math.* **13** (1960), 1–14. `[OPEN]` (philosophical) — The original essay framing why mathematics is so effective in physics. See *Chapter 17 (p. 325)*.

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## 9. Quantum gravity and unification

There is currently **no experimentally confirmed theory of quantum gravity** `[ESTABLISHED]`; accordingly every entry below is `[SPECULATIVE]` or `[OPEN]` as physics, even where the *mathematics* is rigorous. Many key results in this area are also load-bearing in §3 (Hawking 1975, Bardeen–Carter–Hawking), §5 (Bekenstein, Strominger–Vafa), and §7 (holography, RT/HRT, ER=EPR, islands) and are not repeated here. This section collects the additional canonical *unification-program* references. See *Chapter 18 (p. 349)*, *Chapter 19 (p. 392)*, and *Chapter 15 (p. 233)*.

**J. M. Maldacena**, "The large-$N$ limit of superconformal field theories and supergravity," *Adv. Theor. Math. Phys.* **2**, 231 (1998). `[SPECULATIVE]` (conjecture, very strongly supported within its regime) — The AdS/CFT correspondence; the most concrete realization of holography and the engine behind much of §7. `[unverified]` exact page; widely cited as *Adv. Theor. Math. Phys.* 2 (1998) 231 / *Int. J. Theor. Phys.* 38 (1999) 1113.

**J. Polchinski**, *String Theory*, Vols. I–II (Cambridge Univ. Press, 1998). `[ESTABLISHED]` (as exposition of the framework) — The standard graduate reference for perturbative string theory, D-branes, and dualities. The *physical correctness* of string theory as the theory of nature is `[OPEN]`.

**C. Rovelli**, *Quantum Gravity* (Cambridge Univ. Press, 2004); **T. Thiemann**, *Modern Canonical Quantum General Relativity* (Cambridge Univ. Press, 2007). `[SPECULATIVE]` — Canonical references for loop quantum gravity (spin networks, the area/volume operators, spin foams). `[unverified]` exact year of the Thiemann volume.

**Caution (per *Chapter 2 (p. 8)*).** The maturity of the *mathematics* in these programs must never be conflated with empirical confirmation of the *physics*. None of the candidate quantum-gravity frameworks has a confirmed prediction distinguishing it from the others or from semiclassical gravity `[ESTABLISHED]`. Treat all unification claims sourced from this section as `[SPECULATIVE]` unless a domain page demonstrates otherwise.

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## Maintenance conventions

**Dedup rule.** A source load-bearing in multiple domains is listed once under its primary section and cross-referenced (not re-quoted) elsewhere. When adding a reference, search this page first.

**No fabrication.** Add only sources you are confident exist. If unsure of a detail, supply author + title + approximate year and append `[unverified]`. Never invent a DOI, volume, page, or theorem name. This mirrors the cardinal rule of *Chapter 2 (p. 8)*.

**Numbers.** All numerical constants and measured parameters trace to the Particle Data Group (§2) or Planck (§6); see *Chapter 16 (p. 308)*. Do not cite numbers from textbooks of memory.

**Provenance.** New entries should record which domain dossier or page motivated them; log additions in the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 835)).

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## References

This *is* the references page. Other wiki pages should cite back here rather than maintaining parallel source lists. Cross-domain meta-pages — *Chapter 18 (p. 349)*, *Chapter 17 (p. 325)*, *Chapter 21 (p. 497)*, *Chapter 40 (p. 739)* — should link individual entries above by author + short title.

# About This Book

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*Universal Physics* began as a research wiki, not a manuscript. It was built by a multi-agent process: a human author, Harshit Khemani, directing teams of AI research agents (Claude, built by Anthropic) across six iterations between June 8 and June 10, 2026 — dozens of agents per iteration, each assigned one focused question, each new claim passed through an adversarial referee whose explicit job was to break it.

The process had three standing rules. First, the epistemic-tag discipline (Chapter 2): every nontrivial claim carries exactly one of `[ESTABLISHED]`, `[INFERENCE]`, `[SPECULATIVE]`, `[OPEN]`, or `[CONTESTED]`, and no speculation may wear the clothes of fact. Second, web verification: citations were checked against the actual papers, and the bibliography contains real references only. Third, the append-only log: every change, correction, and demotion is recorded in the repository's changelog — including the places where the investigation reversed itself.

The iterations had distinct characters: 1 built the map; 2 stress-tested the central wager; 3 shifted from surveying to attempting (and red-teamed the project's own posture); 4 measured diminishing returns; 5 executed the final verdict-flipping lever and declared analytical saturation; 6 injected two years of new literature in a deliberate attempt to overturn the verdict, and failed.

The book's chapters reproduce the wiki's pages verbatim; only navigation chrome was trimmed. Roughly thirty long-form working notes behind the Part V syntheses are not reprinted here but remain public, with the full change history, at **github.com/HKTITAN/universal-physics** and **universal-**

# Publication History

The wiki's changelog is append-only; this history condenses it to one dated entry per iteration. Per-page micro-edits are omitted — the full log remains in the repository (`CHANGELOG.md`).

**2026-06-08 — Iteration 1: initial build.** First full pass, bootstrapped by a multi-agent workflow: ten parallel domain dossiers; cross-cutting analyses (contradictions, open-problems registry, assumptions ledger, unification-landscape survey, candidate principles); hypothesis generation through three independent lenses; adversarial verification of every contradiction and hypothesis. Conventions established: the five epistemic tags, the assumption-classification scheme, the anti-crank verification gate, and stable identifiers (OP-n, GC-n, Hn = H0–H8, A-n). Post-build structural QC was clean.

**2026-06-08 — Iteration 2: five refereed research tracks + synthesis.** First forward-research iteration (38 agents; 0/25 new claims flagged unsound). Tracks: de Sitter / emergent spacetime, Born rule & linearity, tabletop quantum gravity (BMV), entropic / emergent gravity, algebraic background-independence. Synthesis verdict: PARTIAL coherence — two internally-coherent programs joined by thin bridges; the "it's all information" fusion explicitly refused. Registries extended (OP-40…47, A-20…23); one iteration-1 mischaracterization of Verlinde-EG lensing corrected.

**2026-06-08 — Iteration 3: attempts + red-team.** Shifted from surveying to attempting (31 agents): located the exact obstruction to a de Sitter entanglement first law; delivered the encode-vs-generate criterion; attempted the A+B program merge (narrow success, deep failure — OP-48 added); red-teamed the project's own posture (0 fatal, 4 serious weakenings accepted). Coherence verdict unchanged at PARTIAL; progress negative/structural; the central wager still not yet physics.

**2026-06-08 — Iteration 4: attacks + literature refresh + convergence audit.** Executed and closed the last open dS escape (a scoped cardinality near-no-go); settled OP-48 as a sharp disjunction; found the encode→generate escape near-vacuous under the standard readout (HYP-CKV-VACUITY). Web-verified literature refresh: muon g−2 now SM-consistent (WP2025). Convergence audit: trend diminishing, analytical saturation ≈1–3 iterations away, object-level confirmation not forecastable. Verdict unchanged for the third consecutive iteration.

**2026-06-08 — Iteration 5: closing analytical push → analytical saturation; capstone.** The designated verdict-flipping move (the OP-46 Lorentzian non-CKV readout) was executed and did not flip — producing the survey's first emergent Lorentzian metric, with its signature inherited from a symmetric vacuum. OP-48(c) settled as a clean conditional no-go; OP-44 sharpened to a regime boundary; both surviving dS routes failed for identified reasons. Analytical saturation declared per the iteration-4 criterion; the capstone Conclusion (Chapter 27) and the Experiment Watchlist (Chapter 28) were written. A same-day documentation pass closed three propagation gaps (no physics conclusions changed).

**2026-06-10 — Iteration 6: new-input iteration; verdict unchanged a fifth consecutive time.** First post-saturation iteration: deliberate injection of new 2023–2026 literature (22 agents; 10 adversarially-refereed tracks — 6 research, 4 attack; ~90 findings, all kept, zero fatal). Headline unchanged; hedges restructured: the no-test claim upgraded to a corollary of encode-not-generate (HYP-ENCODING-SCREEN, new third hedge); causal fermion systems added as a fourth readout family closing at the same signature-installation step; OP-48(c) horn C partially discharged; OP-49 closed negative at near-no-go grade; OP-50 added `[CONTESTED]`; watchlist refreshed (w(z) decision point compresses to 2027).